

Problem Books in Mathematics

Pavle Mladenović

Combinatorics

A Problem-Based Approach



Springer

Problem Books in Mathematics

Series Editor:

Peter Winkler
Department of Mathematics
Dartmouth College
Hanover, NH 03755
USA

More information about this series at <http://www.springer.com/series/714>

Pavle Mladenović

Combinatorics

A Problem-Based Approach



Springer

Pavle Mladenović
Faculty of Mathematics
University of Belgrade
Belgrade, Serbia

ISSN 0941-3502

ISSN 2197-8506 (electronic)

Problem Books in Mathematics

ISBN 978-3-030-00830-7

ISBN 978-3-030-00831-4 (eBook)

<https://doi.org/10.1007/978-3-030-00831-4>

Library of Congress Control Number: 2018963721

© Springer Nature Switzerland AG 2019

This work is subject to copyright. All rights are reserved by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.

The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.

The publisher, the authors, and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, express or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Springer imprint is published by the registered company Springer Nature Switzerland AG.
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

Preface

This book is an extended version of the latest edition of my book written in Serbian under the title *Combinatorics*. I started gathering and selecting material and combinatorial problems for the book in the late 1980s while giving lectures to young talented mathematicians preparing for national and international mathematical competitions and while working as a jury member at the national mathematical olympiad.

In the 1990s and at the beginning of this century, till 2006, I was also involved in organizing national mathematical competitions. In two five-year periods, I was leader of the team of FR Yugoslavia (Serbia and Montenegro), first at the Balkan Mathematical Olympiad (BMO 1992–1996) and then at the International Mathematical Olympiad (IMO 1997–2001) and also was a jury member at these mathematical olympiads. This engagement along with my work at the University of Belgrade had a great influence on the form and content of this book.

The first Serbian edition was published in 1989, with two subsequent editions and one more printing in 1992, 2001, and 2013.

In Chapters 2–4, we introduce the basic combinatorial configurations that appear as a solution to many combinatorial problems, the binomial and multinomial theorems, and the inclusion-exclusion principle. The content of these chapters is standard in every textbook in enumerative combinatorics. The sources that were partially used here are the following: [5, 13, 15–17, 23, 27].

Chapter 5 introduces the notion of generating functions of a sequence of real numbers, which is a powerful tool in enumerative combinatorics. The sources that were used for this chapter are [4, 16, 19, 24, 27]. Chapter 6 is devoted to partitions of finite sets and partitions of positive integers. For more details, see [1, 5]. Chapter 7 deals with Burnside’s Lemma that gives a method of counting equivalence classes determined by an equivalence relation

on a finite set. For this topic and the more general Pólya enumeration theorem, see [26, 27, 30].

In Chapters 8 and 9, we introduce the basic notions and results of graph theory. The reader can find more details in [6, 14, 18, 20, 21, 30].

Several topics related to the existence of combinatorial configurations are presented in Chapter 10, and the sources used are books [2, 5] and journals [33, 34]. Several mathematical games are given in Chapter 11.

Topics from elementary probability are presented in Chapter 12. The interested reader can find a detailed presentation of discrete probabilistic models in Volume I of Feller's book [3].

In addition to elementary exercises, all the chapters also contain several problems of medium difficulty and some challenging problems often selected from materials that were suggested at national mathematical competitions in different countries. Chapter 13 contains challenging problems that are classified in seven sections. Several combinatorial problems that were suggested at BMO and IMO are included here. The sources that were used for the exercises and more challenging problems are the following books and journals: [7–12, 18, 22, 27–34].

Solutions to the exercises and problems from all the chapters are given at the end of the book. For a small number of problems that are easier or similar to previous problems, only answers or hints are provided.

I believe that the book may be useful not only to young talented mathematicians interested in mathematical olympiads and their teachers but also to researchers who apply combinatorial methods in different areas.

I would like to thank the anonymous referees for their useful comments and suggestion to include topics related to graph theory and elementary probability. I thank many colleagues from the Mathematical Society of Serbia and the members of the juries of BMO and IMO for sharing their experience and literature related to mathematical olympiads.

Finally, I am also very grateful to Mrs Alice Coople-Tošić for her language assistance during the final preparation of the book.

Belgrade, Serbia
2018

Pavle Mladenović

Contents

Preface	V
1 Introduction	1
1.1 Sets, Functions, and Relations	1
1.2 Basic Combinatorial Rules	4
1.3 On the Subject of Combinatorics	6
Exercises	8
2 Arrangements, Permutations, and Combinations	9
2.1 Arrangements	9
2.2 Arrangements Without Repetitions	10
2.3 Permutations	11
2.4 Combinations	11
2.5 Arrangements of a Given Type	13
2.6 Combinations with Repetitions Allowed	15
2.7 Some More Examples	16
2.8 A Geometric Method of Counting Arrangements	21
2.9 Combinatorial Identities	24
Exercises	27
3 Binomial and Multinomial Theorems	35
3.1 The Binomial Theorem	35
3.2 Properties of Binomial Coefficients	37
3.3 The Multinomial Theorem	42
Exercises	43
4 Inclusion-Exclusion Principle	49
4.1 The Basic Formula	49
4.2 The Special Case	51

4.3	Some More Examples	52
4.4	Generalized Inclusion-Exclusion Principle	58
	Exercises	58
5	Generating Functions	63
5.1	Definition and Examples	63
5.2	Operations with Generating Functions	65
5.3	The Fibonacci Sequence	67
5.4	The Recursive Equations	69
5.5	The Catalan Numbers	70
5.6	Exponential Generating Functions	72
	Exercises	72
6	Partitions	75
6.1	Partitions of Positive Integers	75
6.2	Ordered Partitions of Positive Integers	78
6.3	Graphical Representation of Partitions	80
6.4	Partitions of Sets	84
	Exercises	87
7	Burnside's Lemma	91
7.1	Introduction	91
7.2	On Permutations	92
7.3	Orbits and Cycles	95
7.4	Permutation Groups	99
7.5	Burnside's Lemma	102
	Exercises	104
8	Graph Theory: Part 1	107
8.1	The Königsberg Bridge Problem	107
8.2	Basic Notions	109
8.3	Complement Graphs and Subgraphs	113
8.4	Paths and Connected Graphs	115
8.5	Isomorphic Graphs	117
8.6	Euler's Graphs	119
8.7	Hamiltonian Graphs	120
8.8	Regular Graphs	121
8.9	Bipartite Graphs	122
	Exercises	123
9	Graph Theory: Part 2	127
9.1	Trees and Forests	127
9.2	Planar Graphs	130

9.3	Euler's Theorem	133
9.4	Dual Graphs	135
9.5	Graph Coloring	137
	Exercises	140
10	Existence of Combinatorial Configurations	141
10.1	Magic Squares	141
10.2	Latin Squares	147
10.3	System of Distinct Representatives	148
10.4	The Pigeonhole Principle	152
10.5	Ramsey's Theorem	153
10.6	Arrow's Theorem	156
	Exercises	160
11	Mathematical Games	165
11.1	The Nim Game	165
11.2	Golden Ratio in a Mathematical Game	167
11.3	Game of Fifteen	168
11.4	Conway's Game of Reaching a Level	170
11.5	Two More Games	174
	Exercises	175
12	Elementary Probability	177
12.1	Discrete Probability Space	177
12.2	Conditional Probability and Independence	183
12.3	Discrete Random Variables	185
12.4	Mathematical Expectation	187
12.5	Law of Large Numbers	191
	Exercises	193
13	Additional Problems	199
13.1	Basic Combinatorial Configurations	199
13.2	Square Tables	201
13.3	Combinatorics on a Chessboard	203
13.4	The Counterfeit Coin Problem	205
13.5	Extremal Problems on Finite Sets	206
13.6	Combinatorics at Mathematical Olympiads	209
13.7	Elementary Probability	217
14	Solutions	221
14.1	Solutions for Chapter 1	221
14.2	Solutions for Chapter 2	221
14.3	Solutions for Chapter 3	235

14.4 Solutions for Chapter 4	244
14.5 Solutions for Chapter 5	257
14.6 Solutions for Chapter 6	260
14.7 Solutions for Chapter 7	268
14.8 Solutions for Chapter 8	271
14.9 Solutions for Chapter 9	277
14.10 Solutions for Chapter 10	278
14.11 Solutions for Chapter 11	290
14.12 Solutions for Chapter 12	293
14.13 Solutions for Chapter 13	298
Bibliography	359
Index	363

Chapter 1



Introduction

1.1 Sets, Functions, and Relations

Sets. In this section, we introduce necessary notions and notation. A set is one of the basic notions in mathematics and is determined by its elements. We shall use the following notation:

$a \in A$ – element a belongs to set A ;

$a \notin A$ – element a does not belong to set A ;

$A = \{a \mid a \text{ has a property } \vartheta\}$ – the set of all elements with the property ϑ ;

$A \subset B$ – set A is a subset of set B , i.e., for any element $a \in A$, the relation $a \in B$ also holds;

$A = B$ – if $A \subset B$ and $B \subset A$;

$\emptyset$ – an empty set, i.e., a set without elements;

$A_1 \cup A_2 \cup \dots \cup A_n = \{a \mid a \text{ belongs to at least one of the sets } A_1, A_2, \dots, A_n\}$;

$A_1 \cap A_2 \cap \dots \cap A_n = \{a \mid a \text{ belongs to each of the sets } A_1, A_2, \dots, A_n\}$;

$A \setminus B = \{a \mid a \in A \text{ and } a \notin B\}$;

$\mathbb{P}(A)$ – the set of all subsets of set A ;

$\mathbb{N}$ – the set of positive integers $1, 2, 3, \dots$;

$\mathbb{N}_0$ – the set of nonnegative integers;

n -set – a set consisting of n elements;

$|A|$ – the number of elements of finite set A .

For some integers we introduce special notation.

$n! = 1 \cdot 2 \cdot \dots \cdot n$ for $n \in \mathbb{N}$, and $0! = 1$;

$$(2n)!! = 2 \cdot 4 \cdot \dots \cdot 2n; \quad (2n-1)!! = 1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1);$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \quad \text{for } k \in \{0, 1, \dots, n\}, \quad n \in \mathbf{N}_0; \quad \binom{n}{k} = 0 \quad \text{for } n < k.$$

We shall also use the following notation: $\forall$ (for all) is a universal quantifier; $\exists$ (there exists) is an existential quantifier; $\square$ is notation for the end of the proof; $\triangle$ is notation for the end of an example.

Cartesian product. For given sets A and B , the Cartesian product $A \times B$ is the set of all pairs (a, b) , such that $a \in A$ and $b \in B$. For example, if $A = \{1, 2\}$ and $B = \{p, q, r\}$, then $A \times B = \{(1, p), (2, p), (1, q), (2, q), (1, r), (2, r)\}$.

Similarly, for given sets $A_1, \dots, A_n$, where $n \geq 2$ is a positive integer, the Cartesian product $A_1 \times \dots \times A_n$ is the set of all n -tuples $(a_1, \dots, a_n)$, such that $a_1 \in A_1, \dots, a_n \in A_n$, i.e.,

$$A_1 \times \dots \times A_n = \{(a_1, \dots, a_n) \mid a_1 \in A_1, \dots, a_n \in A_n\}.$$

If $A_1 = \dots = A_n = A$, then the Cartesian product $A_1 \times \dots \times A_n$ is denoted by A^n and is called the Cartesian n -th power of the set A .

Functions. Let A and B be nonempty sets. A *function* $f : A \rightarrow B$ is defined if there is a rule that determines the unique output $b = f(a) \in B$ for any input $a \in A$. An element $a \in A$ is also called an *original*, and element $b = f(a) \in B$ is called the *value of function* f at point a , or the *image* of the original a . The set A is called the *domain* of function f , and the set B is the *codomain* or the set of outputs (the set of images). Two functions $f : A \rightarrow B$ and $g : A \rightarrow B$ are considered equal if $f(a) = g(a)$ for any $a \in A$.

Injective and surjective functions. A function $f : A \rightarrow B$ is called *injective* (or an *injection*, or *one-to-one*) if $f(a_1) \neq f(a_2)$ for all elements $a_1 \neq a_2$, $a_1, a_2 \in A$. A function $f : A \rightarrow B$ is *surjective* (or *surjection*, or *onto*) if for any element $b \in B$, there exists an element $a \in A$ such that $b = f(a)$. Finally, a function is *bijective* (or a *bijection*, or *one-to-one and onto*) if it is both injective and surjective.

If a function $f : A \rightarrow B$ is a bijection, then there exists the *inverse function* $f^{-1} : B \rightarrow A$ determined by $f^{-1}(b) = a$, for any $b = f(a) \in B$.

Fixed and moving points. Let us consider a function $f : A \rightarrow A$, i.e., a function for which the codomain coincides with the domain. An element $a \in A$ is a *fixed point* of function f if $f(a) = a$. Function $f : A \rightarrow A$ is an *identity function* if $f(a) = a$ for all $a \in A$. Element $a \in A$ is a *moving point* of function f if $f(a) \neq a$.

Function composition. Let us consider two functions $f : A \rightarrow B$ and $g : B \rightarrow C$. The *composition* $g \circ f$ of functions g and f is the function determined by $g \circ f(a) = g(f(a))$, for any $a \in A$. The composition $g \circ f$ is determined if the codomain of function f coincides with the domain of function g .

If $f : A \rightarrow B$ and $g : B \rightarrow C$ are both bijective functions, then the composition $g \circ f$ is also bijective, and there exist inverse functions $f^{-1} : B \rightarrow A$, $g^{-1} : C \rightarrow B$, and $(g \circ f)^{-1}$. Moreover, in that case the equality $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ holds.

Let us now consider two functions $f : A \rightarrow A$ and $g : A \rightarrow A$. For such functions both compositions $f \circ g$ and $g \circ f$ are defined, but $f \circ g$ is not necessarily equal to $g \circ f$, i.e., the operation $\circ$ is not commutative.

For any function $f : A \rightarrow A$, the n -th power of f can be defined for any $n \in \mathbb{N}$ as follows: $f^1 = f$ and $f^{n+1} = f^n \circ f$ for $n \in \mathbb{N}$.

A *sequence of the elements of set A* is function $f : \mathbb{N} \rightarrow A$. If we denote $f(n) = a_n \in A$ for any $n \in \mathbb{N}$, then the sequence is denoted by $(a_1, a_2, a_3, \dots)$, or simply by $(a_n)_{n \geq 1}$ or (a_n) .

Relations. A *binary relation* ρ on a set A is a collection of ordered pairs of elements of A , i.e., a subset of the Cartesian product $A \times A = A^2$. More generally, a binary relation between two sets A and B is a subset of the Cartesian product $A \times B$.

If $\rho \subset A^2$ is a relation and $(a, b) \in \rho$, we say that the elements a and b are in relation ρ and write $a \rho b$. It is of interest to give a list of important properties that relation $\rho \subset A^2$ may have:

- (a) *reflexive*: for all $a \in A$, it holds that $a \rho a$;
- (b) *symmetric*: for all $a, b \in A$, it holds that if $a \rho b$, then $b \rho a$;
- (c) *transitive*: for all $a, b, c \in A$, it holds that if $a \rho b$ and $b \rho c$, then $a \rho c$;
- (d) *antisymmetric*: for all $a, b \in A$, if $a \rho b$ and $b \rho a$, then $a = b$;
- (e) *asymmetric*: for all $a, b \in A$, if $(a, b) \in \rho$, then $(b, a) \notin \rho$;
- (f) *irreflexive*: for all $a, b \in A$, if $a \rho b$, then $a \neq b$;
- (g) *dichotomous*: for all $a, b \in A$, $a \neq b$, exactly one of $a \rho b$ or $b \rho a$ holds.

An *equivalence relation* is a binary relation that is at the same time a reflexive relation, a symmetric relation, and a transitive relation. Suppose that $\sim$ is an equivalence relation on set A . For every $a \in A$, let us denote

$$C_a = \{x \mid x \in A, a \sim x\}.$$

The set C_a is called the *equivalence class* of the element a . Any two equivalence classes are either equal or disjoint, i.e., for all $a, b \in A$, exactly one of the equalities $C_a = C_b$ or $C_a \cap C_b = \emptyset$ holds.

Orders. Suppose that $\prec$ is a relation on a set A . The relation $\prec$ is a *partial order* on A if it is reflexive, antisymmetric, and transitive. A set with a partial order on it is called a partially ordered set.

The relation $\leq$, defined on A , is a total order if it is a partial order, and for all $a, b \in A$, it holds that $a \leq b$ or $b \leq a$. A total order is also known as a *linear order*. Any linear order $\leq$ defines a strict linear order $<$ as follows: $a < b$ if $a \leq b$ and $a \neq b$. Then, it holds that $a \leq b$ if $a < b$ or $a = b$.

Increasing and strictly increasing functions. Suppose that $\leq$ is a linear order on a set A . A function $f : \{1, 2, \dots, n\} \rightarrow A$ is:

- (a) an *increasing function* if $f(1) \leq f(2) \leq \dots \leq f(n)$;
- (b) a *strictly increasing function* if $f(1) < f(2) < \dots < f(n)$.

1.2 Basic Combinatorial Rules

Definition 1.2.1. A nonempty set A is a *finite set* if there exist a positive integer n and a bijection $f : \{1, 2, \dots, n\} \rightarrow A$. In that case, set A consists of n elements, and we say that A is an *n -set*. The number of elements of set A is denoted by $|A|$. The empty set $\emptyset$ is finite by definition, and $|\emptyset| = 0$. A nonempty set is *infinite* if it is not finite. A *k -subset* of set A is a subset of A which consists of k elements.

Bijection Rule. *Two nonempty finite sets A and B have the same number of elements if and only if there exists a bijection $f : A \rightarrow B$.*

Although the bijection rule is obvious, we formulate it for the following reason. Sometimes one should determine the number of objects with a given property and should recognize the set A of all such objects. If B is a set with the known cardinality k , and there exists a bijection $f : A \rightarrow B$, then set A has the same cardinality k .

Multiplication Rule. *Let A and B be finite sets and $f : A \rightarrow B$ a function such that, for any element $b \in B$, there exist exactly k elements from set A whose image is equal to b . Then,*

$$|A| = k \cdot |B|. \quad (1.2.1)$$

Usually we use this rule to distinguish the cardinality of ordered and non-ordered outcomes in the situation when we choose some elements from a given set.

Sum Rule. *If A is a finite set, and $A = A_1 \cup A_2 \cup \dots \cup A_n$, where $A_i \cap A_j = \emptyset$ for all $i \neq j$, then,*

$$|A| = |A_1| + |A_2| + \dots + |A_n|. \quad (1.2.2)$$

We use the sum rule when considering the combinatorial problem of counting the number of elements of a set A . Sometimes it is natural and easier to partition set A into disjoint subsets (blocks), to determine the number of elements in each block, and to calculate the sum of obtained cardinalities.

Product Rule. *Let $A_1, A_2, \dots, A_n$ be finite sets that consist of $k_1, k_2, \dots, k_n$ elements, respectively. Then, the Cartesian product $A_1 \times A_2 \times \dots \times A_n$ is a $k_1 k_2 \dots k_n$ -set, i.e.*

$$|A_1 \times A_2 \times \dots \times A_n| = |A_1| \cdot |A_2| \cdot \dots \cdot |A_n|. \quad (1.2.3)$$

Particularly, if A is an m -set, then A^n is an m^n -set, i.e., $|A^n| = |A|^n$.

Proof. We shall prove the equality (1.2.3) by induction. For $n = 1$, the equality (1.2.3) becomes $|A_1| = k_1$ and obviously holds. Suppose that the product rule holds for $n - 1$ sets. Let us now consider the sets $A_1, \dots, A_{n-1}, A_n$, such that $|A_i| = k_i$, for $i \in \{1, 2, \dots, n\}$, and $A_n = \{x_1, x_2, \dots, x_{k_n}\}$. By the induction hypothesis we have

$$|A_1 \times A_2 \times \dots \times A_{n-1}| = k_1 k_2 \dots k_{n-1}. \quad (1.2.4)$$

For any $i \in \{1, 2, \dots, k_n\}$ let us denote

$$S_i = \{(a_1, a_2, \dots, a_{n-1}, x_i) \mid a_1 \in A_1, a_2 \in A_2, \dots, a_{n-1} \in A_{n-1}\}. \quad (1.2.5)$$

There is an obvious bijection between S_i and $A_1 \times A_2 \times \dots \times A_{n-1}$, given by $(a_1, a_2, \dots, a_{n-1}, x_i) \mapsto (a_1, a_2, \dots, a_{n-1})$. Hence,

$$|S_i| = k_1 k_2, \dots, k_{n-1}, \quad \text{for any } i \in \{1, 2, \dots, k_n\}. \quad (1.2.6)$$

Note also that sets $S_1, S_2, \dots, S_{k_n}$ are pairwise disjoint, and

$$A_1 \times A_2 \times \dots \times A_n = S_1 \cup S_2 \cup \dots \cup S_{k_n}.$$

Using the sum rule, we obtain that

$$|A_1 \times A_2 \times \dots \times A_n| = |S_1| + |S_2| + \dots + |S_{k_n}| = (k_1 k_2 \dots k_{n-1}) k_n. \quad \square$$

1.3 On the Subject of Combinatorics

First we shall give a few examples of simple combinatorial problems.

Example 1.3.1. Suppose that 7 blue balls, 8 red balls, and 9 green balls should be put into three boxes labeled 1, 2, and 3, so that any box contains at least one ball of each color. How many ways can this arrangement be done?

Solution. First we shall determine how many ways 7 blue balls can be put into three boxes so that any box contains at least one ball. Denote by x , y , and z the number of balls in the first, second, and third boxes, respectively. It is obvious that $x \in \{1, 2, 3, 4, 5\}$. If $x = 1$, then we have five possibilities for y , namely 1, 2, 3, 4, or 5. For any of these possibilities z is uniquely determined. Hence, for $x = 1$ there are five ways to arrange 7 balls into three boxes. Similarly, we obtain four possibilities for $x = 2$, three possibilities for $x = 3$, two possibilities for $x = 4$, and one more possibility for $x = 5$. Finally, by the sum rule, the number of arrangements of 6 blue balls into three boxes is equal to $5 + 4 + 3 + 2 + 1 = 15$.

Similarly we obtain that the number of arrangements of 8 red balls into three boxes (labeled 1, 2, and 3) is equal to $6 + 5 + 4 + 3 + 2 + 1 = 21$. The number of arrangements of 9 green balls is equal to $7 + 6 + 5 + 4 + 3 + 2 + 1 = 28$. Finally, by the product rule we obtain that the number of ways in which all balls can be arranged into three labeled boxes is $15 \cdot 21 \cdot 28 = 8\,820$. $\triangle$

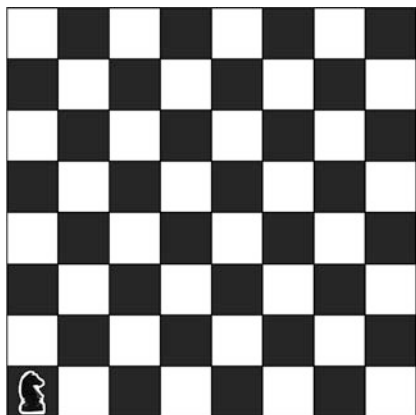


Fig. 1.3.1

28	21	48	9	30	19	46	7
49	10	29	20	47	8	31	18
22	27	64	51	58	55	6	45
11	50	57	54	61	52	17	32
26	23	38	63	56	59	44	5
39	12	25	60	53	62	33	16
24	37	2	41	14	35	4	43
1	40	13	36	3	42	15	34

Fig. 1.3.2

Example 1.3.2. A knight (chess piece) is placed on field $a1$ of a chessboard, see Figure 1.3.1. Is it possible for the knight to make a series of 63 moves (according to chess rules) such that it crosses all the fields of the chessboard and finishes on field $h8$?

Solution. We shall prove that this is not possible. First note that the color of the field where the knight is placed changes after every move. The starting field $a1$ is black. Hence, after any odd number of moves the knight is placed on a white field. Since field $h8$ is also black, it is not possible to place the knight on this field after 63 moves.

Note also that it is possible for the knight to make 63 moves such that it crosses all remaining 63 fields, but without finishing on field $h8$. A sequence of such moves is given in Figure 1.3.2 on page 6, where the starting field $a1$ is labeled 1, and the remaining fields are labeled 2, 3, ..., 64 in the order the knight visited them. $\triangle$

Remark 1.3.3. Note that the problem from Example 1.3.2 can be reformulated the following way. Suppose that the chessboard fields are labeled 1, 2, ..., 64 (not necessarily as on Figure 1.3.2). A series of n moves made by the knight can be determined by the function $f : \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, 64\}$, where $f(k)$ is the name of the field where the knight is placed after the k -th move. The problem is to determine whether there exists a function f with certain properties.

Example 1.3.4. A king (chess piece) is placed on field $a1$ of a chessboard. In any step the king should move to an adjacent field. The allowed directions are up, to the right, or up-right. How many ways can the king reach field $h8$? The answer to this question will be given later on, see Example 2.7.12. $\triangle$

Example 1.3.5. Consider the five-digit positive integers: 12345, 13579, 23367, 11144. Note that the digits in any of these positive integers, considered from left to right, form an increasing sequence. How many five-digit positive integers with this property are there? For the answer to this question, see Example 2.6.6. $\triangle$

The sets $A, B, C, \dots$ that are usually finite, sometimes with a certain structure, and functions $f : A \rightarrow B, g : A \rightarrow C, \dots$ are called *combinatorial configurations*. *Combinatorics* or *combinatorial analysis* is the part of mathematics which deals with problems of the following basic types:

- Proof or disproof of the existence of a combinatorial configuration with certain properties and, particularly, the construction of such configurations;
- Counting of combinatorial configurations;
- Classification of combinatorial configurations;
- Problems of combinatorial optimization; etc.

Elements of combinatorial inference can be found in the first constructions or counting simple configurations such as magic squares or combinations of elements. Combinatorics has developed over the centuries along with

other fields of mathematics. Nowadays combinatorial methods are used in algebra, number theory, geometry, topology, probability theory, mathematical statistics, and many other fields.

Exercises

1.1. A die is thrown five times. The experiment results in a sequence $c_1c_2c_3c_4c_5$, where $c_i \in \{1, 2, \dots, 6\}$ is a positive integer obtained in the i -th throw. How many distinct results are there?

1.2. How many four-digit even positive integers are there without a repetition of the digits?

1.3. How many functions $f : \{1, 2, 3, 4\} \rightarrow \{a, b, c\}$ are there?

1.4. How many *onto* functions $f : \{1, 2, 3, 4\} \rightarrow \{a, b, c\}$ are there?

1.5. How many functions $f : \{1, 2, \dots, n\} \rightarrow \{1, 2\}$ are there?

1.6. How many ten-digit positive integers are there such that all of the following conditions are satisfied:

- (a) each of the digits 0, 1, $\dots$, 9 appears exactly once;
- (b) the first digit is odd;
- (c) five even digits appear in five consecutive positions?

Chapter 2



Arrangements, Permutations, and Combinations

Solving combinatorial problems always requires knowledge of basic combinatorial configurations such as *arrangements*, *permutations*, and *combinations*. All of them are formed from the elements of the finite sets considered, for example, by taking sequences of the elements that belong to some sets or by taking subsets. In this chapter we shall define these combinatorial configurations and provide some examples and exercises.

2.1 Arrangements

Definition 2.1.1. Let A be a finite set. Every element of the set A^k is called a k -*arrangement* of the elements of set A .

Example 2.1.2. Let $A = \{a, b\}$. All 3-arrangements of elements a and b are listed here: aaa , aab , aba , baa , abb , bab , bba , bbb . Hence, the total number of 3-arrangements of the elements of 2-set A is 8. $\triangle$

Theorem 2.1.3. Let A be a set consisting of n elements. The number of k -arrangements of the elements of set A is equal to n^k .

Proof. The theorem follows directly by the product rule. $\square$

Example 2.1.4. Thirteen pairs of football teams will play matches the next Sunday. Peter wants to predict the results of the football matches. The possible predictions for every match are 1, 2, and 0, where: 1 means the victory of the host team, 2 means the victory of the guest team, and 0 means that teams play to a draw. How many different predictions are there?

Answer. Any prediction is a 13-arrangement of the elements 1, 2, and 0. Hence, the total number of predictions is $3^{13} = 1\,594\,323$. $\triangle$

Example 2.1.5. Let us determine the number of 5-digit positive integers without digits 6, 7, 8, and 9 in their decimal representation.

Answer. The number of 5-arrangements of the elements 0, 1, 2, 3, 4, and 5 is equal to 6^5 . The number of such arrangements with 0 in the first position is 6^4 . The number of 5-digit positive integers without digits 6, 7, 8, and 9 in their decimal representation is $6^5 - 6^4 = 6\,480$. $\triangle$

2.2 Arrangements Without Repetitions

Definition 2.2.1. Let A be a set consisting of n elements and $k \leq n$. A k -arrangement without repetitions of the elements of set A is any element of the set A^k whose terms are pairwise different.

Example 2.2.2. Let $A = \{a, b, c, d\}$. All 2-arrangements without repetitions of the elements of set A are the following: $ab, ac, ad, ba, bc, bd, ca, cb, cd, da, db, dc$. The total number of such arrangements is $4 \cdot 3 = 12$. $\triangle$

Theorem 2.2.3. The number of k -arrangements without repetitions of the elements of an n -set is equal to $n(n-1)\dots(n-k+1)$.

Proof. Let $(a_1, a_2, \dots, a_k)$ be a k -arrangement without repetitions of the elements of the n -set A . Then, a_1 can be any of the elements of the n -set A , a_2 can be any of the elements of the $(n-1)$ -set $A \setminus \{a_1\}$, etc. The k -th term a_k can be equal to any of the elements of the $(n-k+1)$ -set $A \setminus \{a_1, a_2, \dots, a_{k-1}\}$. The theorem now follows by the product rule. $\square$

Remark 2.2.4. Note that if $k > n$, then the number of k -arrangements without repetitions of the elements of the n -set is equal to 0.

Example 2.2.5. Twenty applicants apply for four different positions in a company. Let us determine how many ways four applicants can be chosen and employed in these positions.

Let S_1 be the set of all possible choices of four candidates and their arrangement in four positions, and S_2 be the set of all 4-arrangements without repetitions of the elements of set $\{1, 2, \dots, 20\}$. There is an obvious bijection $S_1 \rightarrow S_2$, and consequently we get

$$|S_1| = |S_2| = 20 \cdot 19 \cdot 18 \cdot 17 = 116\,280. \quad \triangle$$

Example 2.2.6. Let us determine the number of 5-digit positive integers without 0's and without the repetition of digits in their decimal representation.

Answer. The number is $9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 = 15\,120$. $\triangle$

2.3 Permutations

Definition 2.3.1. Let A be an n -set. A *permutation* of the elements of set A (or simply a permutation of the set A) is any n -arrangement without repetition of the elements of set A .

It is obvious that any permutation of a finite set A is determined by a bijective function from A to A .

Example 2.3.2. Let $A = \{a, b, c, d\}$. The list of all permutations of set A is given in the following table:

$abcd$,	$abdc$,	$acbd$,	$acdb$,	$adbc$,	$adcb$,
$bacd$,	$badc$,	$bcad$,	$bcda$,	$bdac$,	$bdca$,
$cabd$,	$cadb$,	$cbad$,	$cbda$,	$cdab$,	$cdba$,
$dabc$,	$dacb$,	$dbac$,	$dbca$,	$dcab$,	$dcba$.

The number of permutations of 4-set A is 24. $\triangle$

Generally, the following theorem follows immediately from definition 2.3.1 and Theorem 2.2.3.

Theorem 2.3.3. *The number of permutations of an n -set is $n!$.*

Example 2.3.4. Let us answer the following two questions:

(a) How many ways are there to arrange the digits 1, 2, 3, 4, and 5 in a sequence?

(b) How many ways are there to arrange the digits 1, 2, 3, 4, and 5 in a sequence so that the first two positions are occupied by even digits?

Answer. (a) Using Theorem 2.3.3 we get that the answer is $5! = 120$.

(b) The digits 2 and 4 can be arranged at the first two positions in $2!$ ways, and the digits 1, 3, and 5 can be arranged at the last three positions in $3!$ ways. The number of arrangements that satisfy the given conditions is $2! \cdot 3! = 12$. $\triangle$

Example 2.3.5. Eight pairwise different books should be given to eight students, one book to each of the students. How many ways can this be done?

Answer. There are $8! = 40\,320$ distributions of the books. $\triangle$

2.4 Combinations

Definition 2.4.1. Suppose that A is an n -set, and $k \leq n$. A *k -combination* of the elements of set A is any k -subset of A .

Example 2.4.2. Let $A = \{a, b, c, d\}$. All 3-combinations of the elements of set A are the following: $\{a, b, c\}$, $\{a, b, d\}$, $\{a, c, d\}$, $\{b, c, d\}$. $\triangle$

Theorem 2.4.3. Suppose that A is an n -set, and $k \leq n$. The number of k -combinations of the elements of set A is $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.

Proof. Let S_1 be the set of all k -arrangements without repetition of the elements of set A , and S_2 be the set of all k -combinations of the elements of the same set. By Theorem 2.2.3 we get that $|S_1| = n(n-1) \dots (n-k+1)$. Let us define the function $f : S_1 \rightarrow S_2$ as follows: for each $v = (a_1, a_2, \dots, a_k) \in S_1$, we put

$$f(v) = s = \{a_1, a_2, \dots, a_k\} \in S_2.$$

For every element $s \in S_2$ there are exactly $k!$ elements $v \in S_1$, such that $f(v) = s$. Consequently we get that

$$|S_2| = \frac{|S_1|}{k!} = \frac{n(n-1) \dots (n-k+1)}{k!} = \frac{n!}{k!(n-k)!} = \binom{n}{k}. \quad \square$$

Remark 2.4.4. Suppose that a strict linear order $a_1 < a_2 < \dots < a_n$ is given on the set $A = \{a_1, a_2, \dots, a_n\}$. Then, every k -combination of the elements of set A is uniquely determined by a strictly increasing function $f : \{1, 2, \dots, k\} \rightarrow A$ or, in other words, by a k -tuple $(f(1), f(2), \dots, f(k)) \in A^k$, where $f(1) < f(2) < \dots < f(k)$.

Remark 2.4.5. By definition $\binom{n}{k} = 0$ if $n < k$.

Example 2.4.6. Let us calculate the numbers $\binom{12}{7}$ and $\binom{25}{20}$. We get

$$\binom{12}{7} = \frac{12!}{7!5!} = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} = 792, \quad \binom{25}{20} = 53130. \quad \triangle$$

Example 2.4.7. How many ways can a tourist buy three pairwise different postcards if there are 10 different types of postcards?

Answer. The number of choices is $\binom{10}{3} = 120$. $\triangle$

Example 2.4.8. No three diagonals of a convex octagon intersect at the same point. Let us determine the number of points of intersection of the diagonals of this octagon. (The vertices of the octagon are not considered as the points of intersection of the diagonals.)

Any four vertices of the octagon determine a point of intersection of diagonals. Since no three diagonals intersect at the same point, we get that the number of points of intersection of the diagonals is $\binom{8}{4} = \frac{8 \cdot 7 \cdot 6 \cdot 5}{1 \cdot 2 \cdot 3 \cdot 4} = 70$. $\triangle$

Example 2.4.9. There are 12 boys and 8 girls in a classroom. The teacher needs to choose three boys and two girls to make a team for a mathematical competition. How many ways can this choice be done?

Answer. Three boys can be chosen in $\binom{12}{3} = 220$ ways, and two girls can be chosen in $\binom{8}{2} = 28$ ways. The team can be formed in $\binom{12}{3} \cdot \binom{8}{2} = 220 \cdot 28 = 6160$ ways. $\triangle$

2.5 Arrangements of a Given Type

Definition 2.5.1. Let $A = \{a_1, a_2, \dots, a_m\}$ be a set equipped with the strict linear order $a_1 < a_2 < \dots < a_m$, and let $k_1, k_2, \dots, k_m$ be the nonnegative integers such that

$$n = k_1 + k_2 + \dots + k_m > 0.$$

Every element $v \in A^n$, such that for any $i \in \{1, 2, \dots, m\}$ the element a_i appears in v exactly k_i times, is called an n -arrangement of the elements of set A that has the type $(k_1, k_2, \dots, k_m)$.

Example 2.5.2. Let $A = \{a, b, c, d\}$ and $a < b < c < d$. The 5-arrangements of the elements of set A , whose type is $(3, 2, 0, 0)$, are listed in the following table:

$$\begin{array}{ccccccc} aaabb, & aabba, & abbaa, & bbaaa, & aabab, \\ ababa, & babaa, & abaab, & baaba, & baaab. \end{array} \quad \triangle$$

Example 2.5.3. Let $A = \{a, b, c\}$ and $a < b < c$. Every 3-arrangement of the elements of set A is one of the following 10 types:

$$\begin{array}{cccccc} (3, 0, 0), & (0, 3, 0), & (0, 0, 3), & (2, 1, 0), & (2, 0, 1), \\ (1, 2, 0), & (1, 0, 2), & (0, 1, 2), & (0, 2, 1), & (1, 1, 1). \end{array} \quad \triangle$$

Theorem 2.5.4. Let the conditions of Definition 2.5.1 be satisfied.

(a) The number of n -arrangements of the type $(k_1, k_2, \dots, k_m)$ is

$$\frac{n!}{k_1! k_2! \dots k_m!}.$$

(b) The number of possible types of n -arrangements of the elements of m -set A is equal to

$$\binom{n+m-1}{n}.$$

Proof. (a) Let $B = \{a_1^1, a_1^2, \dots, a_1^{k_1}, \dots, a_m^1, a_m^2, \dots, a_m^{k_m}\}$. Let S_1 be the set of all permutations of set B , and S_2 be the set of those n -arrangements of the elements of set A which have the type $(k_1, k_2, \dots, k_m)$. The set S_1 consists of $(k_1 + k_2 + \dots + k_m)! = n!$ elements. Let us define the function $f: S_1 \rightarrow S_2$ as follows. The map of a permutation $p \in S_1$, is the n -arrangement $v \in S_2$ obtained from p by deleting upper indices. The set S_1 contains exactly $k_1! k_2! \dots k_m!$ permutations of set B , in which, for every $j \in \{1, 2, \dots, m\}$, elements $a_j^1, a_j^2, \dots, a_j^{k_j}$ occupy k_j fixed positions. Hence, for every element $v \in S_2$ there are exactly $k_1! k_2! \dots k_m!$ elements of $u \in S_1$ such that $f(u) = v$, and

$$|S_2| = \frac{|S_1|}{k_1! k_2! \dots k_m!} = \frac{n!}{k_1! k_2! \dots k_m!}.$$

(b) Let $A = \{1, 2, \dots, m\}$ and let S be the set of all $(n + m - 1)$ -arrangements of the elements of set $\{0, 1, 2, \dots, m\}$ which have the following form

$$v = \underbrace{11 \dots 1}_{k_1} \underbrace{022 \dots 2}_{k_2} 0 \dots 0 \underbrace{mm \dots m}_{k_m \text{ times}}.$$

Denote by T the set of all possible types of n -arrangements of the elements of set A . Then, every arrangement $v \in S$ contains exactly $m - 1$ zeros and is uniquely determined by their positions. Hence, $|S| = \binom{m+n-1}{m-1}$. The function $f: S \rightarrow T$, defined by $f(v) = (k_1, k_2, \dots, k_m)$, is a bijection. Consequently, we obtain

$$|T| = |S| = \binom{n+m-1}{m-1} = \binom{n+m-1}{n}. \quad \square$$

Example 2.5.5. How many 7-digit positive integers are there, such that the digit 3 appears three times, and the digits 2 and 3 appear twice in their decimal representation?

The positive integers with the given property are 7-arrangements of the type (3,2,2). By Theorem 2.5.4 there are

$$\frac{7!}{3! 2! 2!} = 210$$

such positive integers. $\triangle$

Example 2.5.6. How many distinct words can be obtained by permuting letters in the word COMBINATORICS?

Letters C, I, and O appear twice, and letters M, B, N, A, T, R, and S appear once in the word COMBINATORICS. Hence, the number of distinct words that can be obtained by permuting letters in this word is

$$\frac{13!}{2! 2! 2!} = 778\,377\,600. \quad \triangle$$

2.6 Combinations with Repetitions Allowed

Definition 2.6.1. Let $A = \{a_1, a_2, \dots, a_m\}$ be a set equipped with the strict linear order $a_1 < a_2 < \dots < a_m$. A n -combination of the elements of set A with repetitions allowed is an n -tuple $(f(1), f(2), \dots, f(n))$, where $f: \{1, 2, \dots, n\} \rightarrow A$ is an increasing function, i.e., a function that satisfies the property $f(1) \leq f(2) \leq \dots \leq f(n)$.

Example 2.6.2. Let $A = \{a, b, c\}$ and $a < b < c$. There are ten different 3-combinations of the elements of set A with repetitions allowed, namely: $aaa, bbb, ccc, aab, aac, abb, acc, bbc, bcc, abc$. $\triangle$

Theorem 2.6.3. The number of n -combinations of the elements of m -set A with repetitions allowed is $\binom{n+m-1}{n}$.

Proof. Let $A = \{a_1, a_2, \dots, a_m\}$ and suppose that $a_1 < a_2 < \dots < a_m$. Let K be the set of n -combinations of the elements of m -set A with repetitions allowed, and let T be the set of possible types of n -arrangements of the elements of set A . Let us define the function $f: T \rightarrow K$, such that for any $(k_1, k_2, \dots, k_m) \in T$,

$$f((k_1, k_2, \dots, k_m)) = \underbrace{a_1 a_1 \dots a_1}_{k_1} \underbrace{a_2 a_2 \dots a_2}_{k_2} \dots \underbrace{a_m a_m \dots a_m}_{k_m \text{ times}} \in K.$$

It is obvious that the function f is a bijection. Using Theorem 2.5.4(b) we obtain that

$$|K| = |T| = \binom{n+m-1}{n}. \quad \square$$

Theorem 2.6.4. The number of n -combinations of the elements of m -set A with repetitions allowed, such that any element $a \in A$ appears at least once in any of these combinations, is equal to $\binom{n-1}{m-1}$.

Proof. Let K be the set of n -combinations of the elements of set $A = \{a_1, a_2, \dots, a_m\}$ with repetitions allowed, such that any element $a \in A$ appears at least once in each of these combinations, and let V be the set of n -arrangements of the elements of set $A \cup \{0\}$. Let us define the function $f: K \rightarrow V$ as follows: for any

$$u = \underbrace{a_1 a_1 \dots a_1}_{k_1} \underbrace{a_2 a_2 \dots a_2}_{k_2} \dots \underbrace{a_m a_m \dots a_m}_{k_m \text{ times}} \in K,$$

we put

$$f(u) = \underbrace{a_1 \dots a_1}_{k_1-1} 0 \underbrace{a_2 \dots a_2}_{k_2-1} 0 \dots 0 \underbrace{a_{m-1} \dots a_{m-1}}_{k_{m-1}-1} 0 \underbrace{a_m \dots a_m}_{k_m \text{ times}}.$$

It is obvious that function f is an injection. Let $f(K) = \{f(u) | u \in K\}$. Note that every arrangement $v \in f(K)$ has zeroes on $m - 1$ positions, and the n -th position is not occupied by 0. Moreover, any $v \in f(K)$ is uniquely determined by the positions of 0's, that is, by an $(m - 1)$ -combination of elements of the set $\{1, 2, \dots, n - 1\}$. Hence,

$$|K| = |f(K)| = \binom{n-1}{m-1}. \quad \square$$

Example 2.6.5. There are 10 types of postcards. How many ways can a tourist buy 3 postcards (some of these postcards may be of the same type)?

Answer. The number of ways to buy 3 postcards is $\binom{10+3-1}{2} = 220$. $\triangle$

Example 2.6.6. How many five-digit positive integers are there, such that the digits, considered from left to right, form an increasing sequence?

Answer. Let S be the set of positive integers with this property. There is a bijection between S and the set of 5-combinations of elements 1, 2, 3, 4, 5, 6, 7, 8, and 9 with repetitions allowed. Using Theorem 2.6.3 we get

$$|S| = \binom{5+9-1}{5} = \binom{13}{5} = 1287. \quad \triangle$$

Remark 2.6.7. Each of the combinatorial configurations previously defined can be uniquely determined by a function $f : X \rightarrow Y$, where:

- (a) X and Y are finite sets (Y may be equal to X);
- (b) the set Y is equipped with a strict linear order;
- (c) the function f may have some of the following properties: f is an injection; f is a surjection; f is an increasing function; f is strictly increasing.

2.7 Some More Examples

For the total number of the combinatorial configurations that are previously defined we often use the following notation:

$$V_0(k, n) = n^k,$$

$$V(k, n) = n(n-1) \dots (n-k+1),$$

$$P(n) = n!,$$

$$K(n, k) = \binom{n}{k},$$

$$V_0(k_1, k_2, \dots, k_m; n) = \frac{n!}{k_1! k_2! \dots k_m!},$$

$$K_0(n, m) = \binom{n+m-1}{n}.$$

Example 2.7.1. Eight elements are chosen from the set $\{1, 2, \dots, 32\}$. We distinguish four cases: choosing with and without replacement, and the order of choosing elements matters, or the order does not matter. In each of these four cases the number of possible choices will be determined.

(a1) *Choosing with replacement, order does not matter.* The result of choosing elements in this case is an 8-arrangement of the elements 1, 2, ..., 32. The number of possible choices is $V_0(8, 32) = 32^8$.

(a2) *Choosing with replacement, order matters.* The result of choosing elements is an 8-arrangement of elements 1, 2, ..., 32 with repetitions allowed. The number of possible choices is

$$K_0(8, 32) = \binom{8 + 32 - 1}{8} = \binom{39}{8} = 61\,523\,748.$$

(b1) *Choosing without replacement, order matters.* The outcome of choosing elements is an 8-arrangement without repetition of elements 1, 2, ..., 32. The number of possible outcomes is $V(8, 32) = 31 \cdot 31 \cdot \dots \cdot 25 = 424\,097\,856\,000$.

(b2) *Choosing without replacement, order doesn't matter.* The outcome is an 8-combination of elements 1, 2, ..., 32. The number of possible outcomes is $K(32, 8) = \binom{32}{8} = 10\,518\,300$. $\triangle$

Example 2.7.2. Let us take a standard deck of 52 cards (26 red and 26 black). We draw cards from the deck, one-by-one and without replacement. There are $P(52) = 52!$ distinct drawings if order matters. There are $(26!)^2$ drawings such that all red cards are chosen in the first 26 drawings. $\triangle$

Example 2.7.3. A box contains four balls numbered 1, 2, 3, and 4. We draw 10 balls from the box with replacement. There are

$$V_0(1, 2, 3, 4; 10) = \frac{10!}{1! 2! 3! 4!} = 12\,600$$

possible drawings if order matters. $\triangle$.

Example 2.7.4. A group consists of two mathematicians and eight economists. They need to form a committee which consists of five members and at least one mathematician among them. How many ways can they do it?

Answer. Using Theorem 2.4.3 and the product and sum rules, we get that the number of ways to form the committee is

$$K(2, 1)K(8, 4) + K(2, 2)K(8, 3) = \binom{2}{1} \binom{8}{4} + \binom{2}{2} \binom{8}{3} = 196. \quad \triangle$$

Example 2.7.5. There are twelve balls numbered 1, 2, $\dots$, 12, and three boxes numbered 1, 2, and 3. How many ways can the balls be put into the boxes such that any box contains 4 balls?

Answer. Any distribution of balls into the boxes is uniquely determined by a 12-arrangement $c_1 c_2 \dots c_{12}$, where, for any $k \in \{1, 2, \dots, 12\}$, c_k is the number associated with the box that contains the k -th ball. And vice versa. The arrangement that corresponds to the case when all three boxes contain 4 balls is of the type (4, 4, 4). The number of distributions of the balls into the boxes that satisfy the given condition is

$$V_0(4, 4, 4; 12) = \frac{12!}{4! 4! 4!} = 34\,650. \quad \triangle$$

Example 2.7.6. How many ways can 12 indistinguishable balls be put into three boxes numbered 1, 2, and 3?

Answer. Let S_1 be the set of all distributions of given balls into boxes numbered 1, 2, 3, and S_2 the set of all types of 12-arrangements of elements 1, 2, and 3. Let $f : S_1 \rightarrow S_2$ be a function defined as follows: for $s \in S_1$ we define $f(s) = (c_1, c_2, c_3) \in S_2$, where, for any $k \in \{1, 2, 3\}$, c_k is the number of balls in the k -th box, related to the distribution s . The function f is a bijective mapping, and by Theorem 2.5.4 (b) we get

$$|S_1| = |S_2| = \binom{12 + 3 - 1}{12} = \binom{14}{12} = \binom{14}{2} = 91. \quad \triangle$$

Example 2.7.7. Given 10 white and ten black balls numbered 1, 2, $\dots$, 10, how many ways can we choose 6 balls such that:

- (a) no two chosen balls have the same number;
- (b) two pairs of chosen balls have the same number?

Answer. (a) Six numbers from the set $\{1, 2, \dots, 10\}$ can be chosen $\binom{10}{6}$ ways. Each of these numbers can appear in two ways on the chosen balls. There are $\binom{10}{6} \cdot 2^6 = 13\,440$ ways of choosing 6 balls, such that no two of them are denoted by the same number.

(b) Two pairs of balls having the same number can be chosen $\binom{10}{2}$ ways. From the remaining 16 balls one can choose two balls numbered differently in $\binom{8}{2} 2^2$ ways. The answer in this case is $\binom{10}{2} \binom{8}{2} 2^2 = 5\,040. \quad \triangle$

Example 2.7.8. Let $m, n \in \mathbb{N}$. Let S_0 be the set of $(m + n)$ -arrangements of elements 0 and 1, which consists of m 0's and n 1's. Denote by S_1 the set which contains arrangements from S_0 without adjacent 0's, and by S_2 the set of arrangements from S_0 with 1's in the first and the last positions. Let us determine $|S_0|$, $|S_1|$ and $|S_2|$.

Answer. (a) Let K be the set of m -combinations of the elements of set $\{1, 2, \dots, m+n\}$ and $f: S_0 \rightarrow K$ be the function determined by the condition: the map of the arrangement $v = c_1 c_2 \dots c_{m+n}$ is that combination from K which contains indices of the terms $c_1, c_2, \dots, c_{m+n}$ that are equal to 0. It is obvious that f is a bijective function. The set K consists of $\binom{m+n}{m}$ elements, and hence $|S_0| = \binom{m+n}{m}$.

(b) In each arrangement that belongs to S_1 , a 0 can be placed at the beginning, between the first and the second 1, $\dots$, between the $(n-1)$ -th and n -th 1, and finally after the n -th 1. Hence, there are $n+1$ positions for the 0's. Moreover, every arrangement from S_1 is uniquely determined by m of those $n+1$ positions, and therefore $|S_1| = \binom{n+1}{m}$.

(c) Similarly as in the previous case we obtain that $|S_2| = \binom{n-1}{m}$. $\triangle$

Example 2.7.9. In statistical physics it is important to consider the distributions of n particles over m quantum states. The Pauli exclusion principle is the quantum mechanical principle that states that two particles cannot occupy the same quantum state simultaneously. Suppose that $n \leq m$. Consider the question: how many distributions of n particles over m states are there? The answer to this question depends on the following assumptions:

- (a) Are the particles distinguishable or not?
- (b) Does the Pauli exclusion principle hold or not?

We shall answer the question in four logically possible cases. Suppose that the states are numbered $1, 2, \dots, m$, and particles are numbered $1, 2, \dots, n$, if they are distinguishable.

Case 1. *Particles are distinguishable and the Pauli exclusion principle does not hold.* Every distribution of particles over states is uniquely determined by the n -arrangement $c_1 c_2 \dots c_n$, where $c_k \in \{1, 2, \dots, m\}$ is the number associated with the state occupied by the k -th particle. The number of distributions in this case is m^n .

Case 2. *Particles are distinguishable and the Pauli exclusion principle holds.* Suppose that $n \leq m$. Every distribution of particles over states is uniquely determined by the n -arrangement without repetition of elements $1, 2, \dots, m$, where $c_k \in \{1, 2, \dots, m\}$ is the number associated with the state occupied by the k -th particle. The number of distributions is $m(m-1)\dots(m-n+1)$.

Case 3. *Particles are indistinguishable and the Pauli exclusion principle holds.* Suppose that $n \leq m$. Every distribution of n particles over m states is uniquely determined by an n -combination $\{c_1, c_2, \dots, c_n\} \subset \{1, 2, \dots, m\}$. The number of distributions in this case is $\binom{m}{n}$.

Case 4. *Particles are indistinguishable and the Pauli exclusion principle does not hold.* Every distribution of n particles over m states is uniquely determined by an n -combination of elements $1, 2, \dots, m$ with repetitions allowed. The number of distributions in this case is $\binom{m+n-1}{n}$.

Remark 2.7.10. The classical *Maxwell-Boltzmann distribution* is the name of the distribution of particles over states, where particles are distinguishable and the Pauli exclusion principle does not hold (Case 1), under the additional assumption that all possible distributions are equally likely. *Fermi-Dirac statistics* describe a distribution of particles over energy states in systems consisting of many identical particles that obey the Pauli exclusion principle (Case 3), under the additional assumption that all possible distributions are equally likely. *Bose-Einstein statistics* correspond to the distribution of indistinguishable particles over states when the Pauli exclusion principle does not hold (Case 4), and under the additional assumption that all possible distributions are equally likely. Among the above-mentioned cases, only Case 2 (particles are distinguishable and the Pauli exclusion principle holds) does not appear in quantum physics.

Example 2.7.11. A square 2×2 consists of four unit squares (fields). The fields are colored red, blue, or yellow.

(a) Suppose that unit fields are also numbered 1, 2, 3, and 4. Then there are $3^4 = 81$ different colorings.

(b) Suppose that two colorings are considered to be the same if the first of them can overlap the second by rotating of the given square 2×2 . Let us determine the number of different colorings under this assumption.

We say that a coloring is of (a, b, c) -type, where $a \geq b \geq c$, if a of the given four fields are colored the same color (red, blue, or yellow), and b and c of the fields are colored the other two colors. Possible types of colorings are $(4, 0, 0)$, $(3, 1, 0)$, $(2, 2, 0)$, and $(2, 1, 1)$. It is easy to see that there are 3, 6, 6, and 9 colorings of these types, respectively. The total number of distinct colorings is $3 + 6 + 6 + 9 = 24$. $\triangle$

Example 2.7.12. A king (chess piece) is placed on field $a1$ of a chessboard. In any step the king may move to an adjacent field. The allowed directions are up, to the right, or up-right. How many ways can the king reach field $h8$?

Answer. Every trajectory of the king from field $a1$ to field $h8$ consists of k moves up, k moves to the right, and $7 - k$ moves up-right, where $k \in \{0, 1, 2, 3, 4, 5, 6, 7\}$. Any such trajectory is uniquely determined by a $(k + 7)$ -arrangement of elements 0, 1, and 2, which has the type $(k, k, 7 - k)$,

where 0 corresponds to a move to the right, 1 corresponds to a move up, and 2 corresponds to a move up-right, and vice versa. The number of ways in which the king can reach field $h8$ starting from field $a1$ is

$$\sum_{k=0}^7 \frac{(k+7)!}{k!k!(7-k)!} = 48\,639. \triangle$$

2.8 A Geometric Method of Counting Arrangements

Certain combinatorial problems can be formulated in terms of counting n -arrangements of elements -1 and 1 for which some additional conditions are satisfied. Every n -arrangement $c_1c_2\ldots c_n$, which consists of terms -1 and 1 , can be associated with a trajectory $Z_0Z_1Z_2\ldots Z_n$, where Z_k is the point with Cartesian coordinates (x_k, y_k) , such that the following equalities hold: $x_0 = 0$, $y_0 = 0$, and, for any $k \in \{1, 2, \ldots, n\}$,

$$x_k = x_{k-1} + 1, \quad y_k = y_{k-1} + c_k.$$

If $c_k = 1$, then $y_k - y_{k-1} = 1$, and $Z_{k-1}Z_k$ is an *increasing part* of the trajectory; if $c_k = -1$, then $y_k - y_{k-1} = -1$, and $Z_{k-1}Z_k$ is a *decreasing part* of the trajectory; Z_0 is the *starting point* and Z_n is the *endpoint* of the trajectory $Z_0Z_1Z_2\ldots Z_n$.

Theorem 2.8.1. (a) Let $M(m, n)$ be the number of trajectories with starting point $(0, 0)$ and endpoint (m, n) , where $m \geq n$, $m, n \in \mathbb{N}$. Then,

$$M(m, n) = \frac{m!}{\left(\frac{m+n}{2}\right)! \left(\frac{m-n}{2}\right)!}, \quad \text{if } m-n \text{ is divisible by } 2;$$

$$M(m, n) = 0, \quad \text{if } m-n \text{ is not divisible by } 2.$$

(b) Let $A(a_1, a_2)$ and $B(b_1, b_2)$ be points whose Cartesian coordinates are integers such that $b_1 > a_1 \geq 0$, $a_2 > 0$, $b_2 \geq 0$, and let $A_1(a_1, -a_2)$ be the point obtained by the reflection of $A(a_1, a_2)$ about the x -axis. Then the following statement holds: the number of trajectories with starting point A and endpoint B that have at least one common point with the x -axis is equal to the number of trajectories with starting point A_1 and endpoint B .

(c) Let $n \in \mathbb{N}$ and let T_1 be the set of trajectories with starting point $(0, 0)$ and endpoint $(2n, 0)$, such that all points of any such trajectory, except the starting and endpoints, have positive y -coordinates. Then,

$$|T_1| = \frac{1}{n} \binom{2n-2}{n-1}.$$

(d) Let $n \in \mathbb{N}$ and let T_2 be the set of trajectories with starting point $(0, 0)$ and endpoint $(2n, 0)$, such that all points of any such trajectory have nonnegative y -coordinates. Then,

$$|T_2| = \frac{1}{n+1} \binom{2n}{n}.$$

(e) Let $n, k \in \mathbb{N}$ and let T_3 be the set of trajectories with starting point $(0, 0)$ and endpoint $(2n, 0)$, and without common points with the line $y = -k$. Then,

$$|T_3| = \binom{2n}{n} - \binom{2n}{n+k}.$$

Proof. (a) Let t be a trajectory with starting point $(0, 0)$ and endpoint (m, n) . Let α be the number of increasing parts of t , and β be the number of decreasing parts of t . Then, we have

$$\alpha + \beta = m, \quad \alpha - \beta = n, \quad \text{i.e.} \quad \alpha = \frac{m+n}{2}, \quad \beta = \frac{m-n}{2}.$$

Since α and β are nonnegative integers, it follows that there is no trajectory with starting point $(0, 0)$ and endpoint (m, n) if one of the integers m and n is even, and another one is odd. If both of the integers m and n are even, or both are odd, i.e., if any of integers $m+n$ and $m-n$ is divisible by 2, then we have

$$M(m, n) = \binom{m}{\alpha} = \binom{m}{\frac{m+n}{2}} = \frac{m!}{\left(\frac{m+n}{2}\right)! \left(\frac{m-n}{2}\right)!}.$$

(b) Let S_1 be the set of trajectories with starting point A and endpoint B that have at least one common point with the x -axis, and let S_2 be the set of trajectories with starting point A_1 and endpoint B , see Figure 2.8.1. Let us define the function $f : S_1 \rightarrow S_2$, where the map $f(t)$ of trajectory $t \in S_1$ is determined as follows. Let us denote by X the common point of trajectory t and the x -axis which has the smallest x -coordinate. Let t_1 be the part of trajectory t with starting point A and endpoint X , and t_2 be the part of t with starting point X and endpoint B . Let t'_1 be the trajectory

obtained by the reflection of t_1 about the x -axis and $f(t) = t'_1 \cup t_2$. The function f is a bijection, and therefore we get $|S_1| = |S_2|$.

(c) Every trajectory from T_2 contains the points $(1, 1)$ and $(2n - 1, 1)$. The problem reduces to counting the number of trajectories with starting point $(1, 1)$ and endpoint $(2n - 1, 1)$, and without common points with the x -axis. The number of all trajectories with starting point $(1, 1)$ and endpoint $(2n - 1, 1)$ is $\binom{2n-2}{n-1}$, because any such trajectory consists of $n - 1$ increasing parts, and $n - 1$ decreasing parts, and all these parts can be arranged in a sequence in all possible ways. The number of trajectories with starting point $(1, 1)$ and endpoint $(2n - 1, 1)$, which have at least one common point with the x -axis, is equal to the number of all trajectories with starting point $(1, -1)$ and endpoint $(2n - 1, 1)$. Every trajectory with starting point $(1, -1)$ and endpoint $(2n - 1, 1)$ has n increasing and $n - 2$ decreasing parts, and hence the number of such trajectories is equal to $\binom{2n-2}{n}$.

Finally, we get

$$|T_1| = \binom{2n-2}{n-1} - \binom{2n-2}{n} = \frac{1}{n} \binom{2n-2}{n-1}.$$

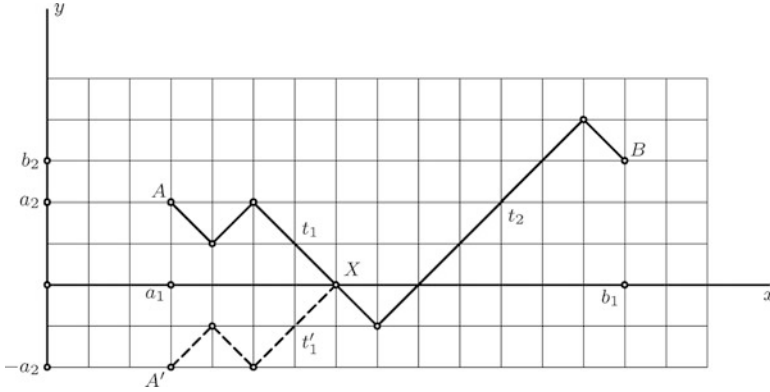


Fig. 2.8.1

(d) The proof is similar to the previous case.

(e) Using similar reasoning as in case (b) we obtain that the number of trajectories with starting point $(0, 0)$ and endpoint $(2n, 0)$, which have at least one common point with the line $y = -k$, is equal to the number of trajectories with starting point $(0, -2k)$ and endpoint $(2n, 0)$, and that is the number $M(2n, 2k)$ of all trajectories with starting point $(0, 0)$ and endpoint $(2n, 2k)$. By the result obtained in case (a), we get

$$|T_3| = M(2n, 0) - M(2n, 2k) = \binom{2n}{n} - \binom{2n}{n+k}. \triangle$$

Example 2.8.2. Suppose that $2n$ persons are in a line in front of a box office. Each of them intends to buy a ticket, and the ticket price is \$5. Suppose also that the first n persons in the line have \$10 notes, and all of the remaining n persons have a \$5 note. There is no money in the cash register at the beginning. How many ways can the line be rearranged, such that no one in the line who has a \$10 note will wait for change?

Answer. By Theorem 2.8.1(d) we obtain that the number of rearrangements of the line, such that the given condition is satisfied, is $\frac{1}{n+1} \binom{2n}{n}$. $\triangle$

Remark 2.8.3. The sequence $(x_n)_{n \geq 1}$ of the positive integers given by

$$x_n = \frac{1}{n+1} \binom{2n}{n},$$

is known as a sequence of *Catalan numbers*. Note that $x_1 = 1$, $x_2 = 2$, $x_3 = 5$, $x_4 = 14$, $x_5 = 42$, ... It can be proved that x_n is an odd integer if and only if n is of the form $n = 2^k - 1$, where k is a positive integer. Catalan numbers often appear as the solution to enumerative combinatorial problems.

2.9 Combinatorial Identities

Theorem 2.9.1. *For every positive integer n the following equality holds:*

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n} = 2^n. \quad (2.9.1)$$

Proof. Let $S = \{1, 2, \dots, n\}$, $\mathcal{P}(S)$ be the set of all subsets of set S , and T be the set of all n -arrangements of elements 0 and 1. Then, T is a 2^n -set. For any $k \in \{0, 1, \dots, n\}$, let A_k be the set of all k -combinations of elements of set S . Then we have

$$|\mathcal{P}(S)| = \sum_{k=0}^n |A_k| = \sum_{k=0}^n \binom{n}{k}.$$

Let us now define the function $f : \mathcal{P}(S) \rightarrow T$ as follows: for any subset $X = \{j_1, j_2, \dots, j_m\}$ of the set S let $f(X)$ be the n -arrangement from T with 1's in the positions $j_1, j_2, \dots, j_m$, and 0's in the remaining positions. The function f is bijective, and therefore we obtain $|\mathcal{P}(S)| = |T|$, i.e., the equality (2.9.1) holds. $\square$

Theorem 2.9.2. *For any positive integer n the following equality holds:*

$$\sum \frac{n!}{k_1! k_2! \cdots k_m!} = m^n, \quad (2.9.2)$$

where the sum on the left-hand side of the equality (2.9.2) runs over all m -tuples $(k_1, k_2, \dots, k_m)$ of nonnegative integers such that $k_1 + k_2 + \cdots + k_m = n$.

Proof. Consider the set $A = \{1, 2, \dots, m\}$. Let V be the set of all n -arrangements of the elements of set A , and $V(k_1, \dots, k_m)$ be the set of those n -arrangements from V that have the type $(k_1, \dots, k_m)$. Since the set V is the union of pairwise disjoint sets $V(k_1, \dots, k_m)$ over all m -tuples $(k_1, \dots, k_m)$ of the nonnegative integers such that $k_1 + \cdots + k_m = n$, the equality (2.9.2) follows from Theorem 2.1.3, Theorem 2.5.4(a), and the sum rule. $\square$

Example 2.9.3. For any positive integer n the following equality holds:

$$\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots = \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \cdots. \quad (2.9.3)$$

Proof. Let $S = \{1, 2, \dots, n\}$ and let $X \subset S$ be a combination of the elements of set X . We say that X is an even combination if $|X| = 2k$, where $k \in \mathbb{N}_0$, and X is an odd combination if $|X| = 2k - 1$, where $k \in \mathbb{N}$. The equality (2.9.3) can be reformulated as follows. *The number of even combinations of an n -set is equal to the number of odd combinations of this set.*

Case 1. Suppose that n is an odd positive integer. The function defined on the set of even combinations of S , such that the map of any even combination is its complement in the set X , is a bijection from the set of even combinations onto the set of odd combinations. Hence, the equality (2.9.3) holds.

Case 2. Suppose that $n = 2k$, where $k \in \mathbb{N}$. Let us introduce the following notation:

- C_1 - the set of all odd combinations of the elements of set $\{1, 2, \dots, 2k - 1\}$;
- C_2 - the set of all even combinations of the elements of set $\{1, 2, \dots, 2k - 1\}$;
- C_3 - the set of all odd combinations of the elements of set $\{1, 2, \dots, 2k\}$, such that any combination from C_3 contains the element $2k$;
- C_4 - the set of all even combinations of the elements of set $\{1, 2, \dots, 2k\}$, such that any combination from C_4 contains the element $2k$.

From the result obtained in Case 1, it follows that $|C_1| = |C_2|$. Note that $C_1 \cup C_3$ is the set of all odd combinations, and $C_2 \cup C_4$ is the set of all even combinations of the elements of set $\{1, 2, \dots, 2k\}$. Moreover, any even combination from C_4 can be obtained by adding $2k$ to an odd combination from C_1 . The mapping from C_1 to C_4 defined this way is a bijection, and hence $|C_4| = |C_1|$. Similarly, any odd combination from C_3 can be obtained by adding $2k$ to an even combination from C_2 . And again it is easy to get $|C_3| = |C_2|$. Finally, we have $|C_1 \cup C_3| = |C_1| + |C_3| = |C_4| + |C_2| = |C_2 \cup C_4|$. This completes the proof of the equality (2.9.3). $\triangle$

Example 2.9.4. Let us prove that for any positive integers m , n , and k the following equalities hold:

$$\sum_{j=1}^m \binom{m}{j} \binom{n-1}{j-1} = \binom{n+m-1}{n}, \quad (2.9.4)$$

$$\sum_{j=0}^n \binom{k-1+j}{j} \binom{m+n-k-1-j}{n-j} = \binom{n+m-1}{n}. \quad (2.9.5)$$

Proof. (a) Let S be the set of all n -combinations of the elements of set $A = \{1, 2, \dots, m\}$ with repetitions allowed. For any $j \in \{1, 2, \dots, m\}$, let S_j be the set of those combinations from S that contains exactly j distinct elements of set A . It follows from Theorem 2.6.3 that $|S| = \binom{n+m-1}{n}$. Using Theorem 2.6.4 we obtain that

$$|S_j| = \binom{m}{j} \binom{n-1}{j-1}, \quad 1 \leq j \leq m.$$

Since $S = S_1 \cup S_2 \cup \dots \cup S_m$, and $S_1, S_2, \dots, S_m$ are pairwise disjoint sets, the equality (2.9.4) follows by the sum rule.

(b) Let A and B be disjoint sets, such that $|A| = k$, $|B| = m - k$, and let S be the set of all n -combinations of the elements of set $A \cup B$ with repetitions allowed. For any $j \in \{0, 1, \dots, n\}$, let $S_j \subset S$ be the set of all n -combinations with repetitions allowed, and with j appearances of elements from A , and $n - j$ appearances of elements from B . Then we have

$$|S| = \binom{n+m-1}{n}, \quad |S_j| = \binom{k+j-1}{j} \binom{n-j+m-k-1}{n-j}.$$

Since $S = S_0 \cup S_1 \cup \dots \cup S_n$, and $S_0, S_1, \dots, S_n$ are pairwise disjoint sets, the equality (2.9.5) follows by the sum rule. $\triangle$

Example 2.9.5. The following identity holds true:

$$\sum_{k=0}^n \binom{2n}{2k} \binom{2k}{k} 2^{2n-2k} = \binom{4n}{2n}. \quad (2.9.6)$$

Suppose that a box contains $2n$ white balls numbered $1, \dots, 2n$, and $2n$ black balls numbered $1, \dots, 2n$. We draw $2n$ balls without replacement. Let S be the set of all possible drawings. Then,

$$|S| = \binom{4n}{2n}. \quad (2.9.7)$$

For any $k \in \{0, 1, \dots, n\}$, let S_k be the set of all drawings with exactly k pairs of balls having the same number. Then we have

$$\begin{aligned} |S_k| &= \binom{2n}{k} \binom{2n-k}{2n-2k} 2^{2n-2k} = \frac{(2n)!}{k! (2n-k)!} \cdot \frac{(2n-k)!}{k! (2n-2k)!} 2^{2n-2k} \\ &= \frac{(2n)!}{(2k)! (2n-2k)!} \cdot \frac{(2k)!}{k! k!} 2^{2n-2k} = \binom{2n}{2k} \binom{2k}{k} 2^{2n-2k}. \end{aligned} \quad (2.9.8)$$

Since $S = S_0 \cup S_1 \cup \dots \cup S_n$, and the sets $S_0, S_1, \dots, S_n$ are pairwise disjoint, it follows by the sum rule that

$$|S| = \sum_{k=0}^n |S_k| = \sum_{k=0}^n \binom{2n}{2k} \binom{2k}{k} 2^{2n-2k}. \quad (2.9.9)$$

The equality (2.9.6) follows from (2.9.7) and (2.9.9). $\triangle$

Exercises

2.1. How many 3-digit positive integers are there, such that three distinct digits appear in the decimal representation of any of them?

2.2. How many 5-digits positive integers are there that are divisible by 5 and without a repetition of digits?

2.3. Twelve basketball teams take part in a tournament. How many possible ways are there for the distribution of gold, silver, and bronze medals?

2.4. There are n light bulbs in a room, each of them with its own switch. How many ways can the room be lighted?

2.5. (a) How many divisors of natural number 2000 are there?

(b) Let $p_1, p_2, \dots, p_m$ be distinct prime numbers, and $k_1, k_2, \dots, k_m$ be positive integers. Determine the number of divisors of the positive integer $n = p_1^{k_1} p_2^{k_2} \dots p_m^{k_m}$.

2.6. Determine all positive integers with an odd number of divisors.

2.7. In a big country no two citizens have the same arrangement of teeth. What is the maximal possible number of citizens in this country?

2.8. Let $n \geq 2$. How many permutations of the set $\{1, 2, \dots, n\}$ are there, such that 1 and 2 are adjacent terms?

2.9. Let $n \geq 2$. How many permutations of the set $\{1, 2, \dots, n\}$ are there, such that 2 is placed after 1, not necessarily at adjacent positions?

2.10. Let $n \geq k + 2$. How many permutations of the set $\{1, 2, \dots, n\}$ are there, such that exactly k elements appear between 1 and 2?

2.11. How many permutations $(a_1, a_2, \dots, a_{3n})$ of the set $\{1, 2, \dots, 3n\}$ are there, such that, for any $k \in \{1, 2, \dots, 3n\}$, the difference $k - a_k$ is divisible by 3?

2.12. How many ways can eight white chess pieces (the king, the queen, two rooks, two knights, and two bishops) be arranged on the first row of a chessboard?

2.13. How many ways can n people sit around a circular table, if the chairs are not labeled?

2.14. Given $n \geq 3$ points in the plane, such that no three of them belong to the same line, how many lines are determined by these points?

2.15. Given $n \geq 3$ points in space, such that no four of them belong to the same plane, how many planes are determined by these points?

2.16. Given a convex n -gon, such that no two lines determined by the vertices of the n -gon are parallel, and no three of them intersect at the same point, determine the number of points of intersection of these lines that are

(a) inside the n -gon;

(b) outside the n -gon.

2.17. A group consists of six mathematicians. How many ways can they form five committees each consisting of three members, such that no two of these committees consists of the same three mathematicians?

2.18. Let us take a deck of cards consisting of 8 red, 8 blue, 8 yellow, and 8 green cards. Cards of the same color are numbered 1, 2, ..., 8. How many ways can 6 cards be chosen such that at least one card of each color is chosen?

2.19. Peter has 12 relatives (five men and seven women), and his wife also has 12 relatives (five women and seven men). They do not have common relatives. They decide to invite 12 guests, six each of their relatives, such that there are six men and six women among the guests. How many ways can they choose 12 guests?

2.20. Let $k \leq n + 1$. How many ways can n 0's and k 1's be arranged in a sequence, such that no two 1's are adjacent?

2.21. Twelve books are arranged in a sequence. How many ways can five books be chosen such that no two of them are adjacent in the sequence?

2.22. Twelve knights are arranged around King's Arthur Round Table. All of them are quarreling with their neighbors. How many ways can King Arthur choose five knights such that no two of them are quarreling?

2.23. How many n -digit positive integers $c_1 c_2 \dots c_n$ are there, such that

$$1 \leq c_1 \leq c_2 \leq \dots \leq c_n \leq 9?$$

2.24. How many 6-digit positive integers are there such that there are exactly three distinct digits in the decimal representation of each of them, and the digit 0 does not appear?

2.25. How many different words can be obtained by permuting the letters of the word LOCOMOTIVE, such that no two O's are adjacent?

2.26. All k -arrangements of elements $a_1, a_2, \dots, a_n$ are written down. Determine how many times the element a_1 is written down.

2.27. A box contains 36 yellow, 27 blue, 18 green, and 9 red balls. Balls of the same color are indistinguishable. How many ways are there to choose 10 balls from the box?

2.28. Determine the number of 0's at the end of the decimal representation of the positive integer $144!$.

2.29. Prove that the product of k consecutive positive integers is divisible by $k!$.

2.30. Let $a_1, a_2, \dots, a_n$ be nonnegative integers. Prove that

$$a_1! a_2! \dots a_n! \leq (a_1 + a_2 + \dots + a_n)!.$$

2.31. Determine the sum of all 4-digit positive integers, such that there are two different even digits and two equal odd digits in the decimal representation of each of them.

2.32. All digits 0, 1, $\dots$, 9 are written down on a sheet of paper. After a 180° rotation, the digits 0, 1, and 8 do not change meaning, the digit 6 becomes 9, and 9 becomes 6. The remaining digits lose their meaning. How many 7-digit positive integers are there, such that they remain the same after the 180° rotation?

2.33. How many 3-combinations $\{x_1, x_2, x_3\}$ of elements 1, 2, $\dots$, $3n$ are there, such that $x_1 + x_2 + x_3$ is divisible by 3?

2.34. Determine the total number of 0's in the decimal representations of the positive integers 1, 2, $\dots$, 10^9 .

2.35. How many positive integers are there in the set $S = \{1, 2, \dots, 10^7\}$, such that the digit 1 does not appear in their decimal representations?

2.36. Determine the sum of all digits that appear in the decimal representations of all positive integers from the set $\{1, 2, \dots, 1\,000\,000\}$.

2.37. How many positive integers less than 10^n are there, such that no two adjacent digits in their decimal representations are equal to each other?

2.38. How many n -arrangements of the elements 0 and 1 are there, such that no two adjacent terms in each of them are both equal to 1?

2.39. How many n -arrangements of the elements 0, 1, $\dots$, n are there with an even number of 0's?

2.40. Let us take 6 red balls, 7 blue balls, and 10 yellow balls. Balls of the same color are numbered and hence distinguishable. How many ways are there to arrange the balls in a sequence such that any red ball is between a blue and a yellow ball, and no blue ball is adjacent to a yellow ball?

2.41. According to the rules, a set of a volleyball game is won by the team that first scores 25 points with a minimum two-point advantage. After each point the result is written down, for example 0 : 1, 0 : 2, 1 : 2, 2 : 2, $\dots$. The first integer always shows points scored by the host team. Suppose that the final result of a set is 25 : n . How many ways can this result be reached,

- (a) if $n \in \{0, 1, 2, \dots, 23\}$ is a fixed integer?
- (b) if we know only that $n \leq 23$?

2.42. Given n indistinguishable balls and m boxes numbered $1, 2, \dots, m$. How many ways can the balls be put into the boxes such that:

- (a) every box contains at least one ball?
- (b) there are exactly k balls in the first box, and every box contains at least one ball?
- (c) there are exactly k balls in the first box?
- (d) for any $k \in \{1, 2, \dots, m\}$, the k -th box contains at least k balls?

2.43. How many ways are there to put n_1 blue balls, n_2 yellow balls, and n_3 red balls into m boxes numbered $1, 2, \dots, m$? Balls of the same color are indistinguishable.

2.44. How many ways are there to arrange $n = n_1 + n_2 + \dots + n_k$ distinguishable balls into k distinguishable boxes, such that, for any $i \in \{1, 2, \dots, k\}$, the i -th box contains exactly n_i balls?

2.45. How many ways are there to put 8 red balls, 9 blue balls, and 10 yellow balls into three boxes numbered 1, 2, and 3, such that any box contains at least one ball of each color?

2.46. Suppose that, for any $i \in \{1, 2, \dots, k\}$, there are $n_i \geq 2s_i$ articles of the i -th kind. How many ways are there to arrange all these articles into two boxes, such that, for any $i \in \{1, 2, \dots, k\}$, each box contains at least s_i articles of the i -th kind?

2.47. How many triples (x_1, x_2, x_3) of positive integers are there such that

- (a) $x_1 x_2 x_3 = 10^3$?
- (b) $x_1 x_2 x_3 = 10^3$ and $x_1 \leq x_2 \leq x_3$?

2.48. Suppose that n points are given at any side of a square. None of these points coincides with a vertex of the square. How many triangles are determined by these $4n$ points?

2.49. Suppose that $n \geq 2$ lines are given in the plane. Let us assume that $k \geq 2$ points are chosen at any of these lines such that the following two conditions are satisfied. (1) None of these points coincides with a point of intersection of the given lines. (2) No three of these points belong to the same line which does not coincide with one of the given lines.

How many triangles are determined by the given points?

2.50. Suppose that 5 points are given in the plane so that the following two conditions are satisfied. (1) No three of these points are collinear. (2) There are no parallel or perpendicular lines among those that are determined by the given points. From each of the given 5 points a perpendicular is dropped to each of the lines determined by the other 5 points. Determine the greatest possible value of the number of points of intersection of these perpendiculars.

2.51. Let $k, m, n \in \mathbb{N}$. Determine the number of k -combinations X of elements $1, 2, \dots, n$ such that $|x - y| > m$ for any $x, y \in X$.

2.52. Suppose that there are $2n + 1$ distinct books and n more books of the same kind. How many ways can we buy n books?

2.53. Let us take n distinct signal flags and k masts arranged in a sequence. A signal is determined by distributing all the given flags on the masts, and by the order of the flags on every mast. Determine the number $A(n, k)$ of all possible signals.

2.54. Let $(p_1, p_2, \dots, p_n)$ be a permutation of the set $\mathbb{N}_n = \{1, 2, \dots, n\}$. A pair (p_i, p_j) is called an inversion if $(i - j)(p_i - p_j) < 0$. Determine the total number of inversions in all permutations of set $\mathbb{N}_n$.

2.55. Determine the greatest possible number of permutations of an n -set A , such that any two elements of A are adjacent in at most one of these permutations.

2.56. Let $k, n, p \in \mathbb{N}$, $k \geq 3$, $n \geq (k + 1)p$, and let $A_1 A_2 \dots A_n$ be a convex n -gon. How many ways can k vertices of the n -gon be chosen, such that there are at least p other vertices of the n -gon between any two of them?

2.57. Determine the greatest possible number of 3-subsets of the set $\mathbb{N}_n = \{1, 2, \dots, n\}$, such that any two of these subsets contain exactly one common element.

2.58. For any positive integer n determine the greatest positive integer $k \in \mathbb{N}_n = \{1, 2, \dots, n\}$ with the following property: there exist k subsets of the set $\mathbb{N}_n$, such that any two of them have a nonempty intersection.

2.59. An international jury consists of n members. Jury documents are stored in a safe box. The safe box has a locks, and each lock has b keys. Determine a and b , and the distribution of the keys among the members of

the jury if the following condition is satisfied: *the safe box can be opened if and only if the number of jury members present is not less than m* , where $m \in \{2, 3, \dots, n-1\}$.

2.60. Let $\mathbb{N}_n = \{1, 2, \dots, n\}$ and $k \in \mathbb{N}$. For any $j \in \{0, 1, \dots, n\}$ determine how many k -tuples $(A_1, A_2, \dots, A_k)$ there are, such that the following conditions are satisfied: $A_1, A_2, \dots, A_k \subset \mathbb{N}_n$, and $|A_1 \cap A_2 \cap \dots \cap A_k| = j$.

2.61. Let S be the set of all $2n$ -arrangements of elements 0 and 1 that have the type (n, n) . For any $v = (c_1, c_2, \dots, c_{2n}) \in S$ we define characteristic $C(v)$ to be the number of pairs (c_i, c_{i+1}) , $i \in \{1, 2, \dots, 2n-1\}$, such that $c_i \neq c_{i+1}$. For any $k \in \{1, 2, \dots, 2n-1\}$, let S_1 be the set of arrangements $v \in S$ such that $C(v) = n - k$, and S_2 be the set of arrangements $v \in S$ such that $C(v) = n + k$. Prove that $|S_1| = |S_2|$.

2.62. How many permutations $(p_1, p_2, \dots, p_n)$ of the set $\{1, 2, \dots, n\}$ are there such that the sum $|p_1 - 1| + |p_2 - 2| + \dots + |p_n - n|$ has the greatest possible value?

2.63. Suppose that $m + n$ persons are in a line in front of a box office. Each of them intends to buy a ticket, and the ticket price is \$5. Suppose that m of the persons in the line have \$5 notes, and all of the remaining n persons have a \$10 note. Assume also that $m \geq n$. How many ways can the line be rearranged, such that no one in the line with a \$10 note will wait for change, if:

- (a) there is no money in the cash register at the beginning?
- (b) there are k \$5 notes in the cash register, where $k < n \leq k + m$?

2.64. How many permutations $i_1 i_2 \dots i_n j_1 j_2 \dots j_n$ of the set $\{1, 2, \dots, 2n\}$ are there such that the following inequalities hold:

$$\begin{aligned} i_1 < j_1, \quad i_2 < j_2, \quad \dots, \quad i_n < j_n, \\ i_1 < i_2 < \dots < i_n, \quad j_1 < j_2 < \dots < j_n? \end{aligned}$$

2.65. Suppose that $2n$ points are located on a circle. How many ways can these $2n$ points be connected into n pairs by n chords without points of intersection inside the circle?

2.66. How many ways can a convex $(n+2)$ -gon be divided into triangles by diagonals that do not intersect inside the polygon?

2.67. Some chess masters and chess grandmasters took part in a chess tournament. All the participants of the tournament played a game against each

other. The number of games played by the masters against the grandmasters was half of the total number of games played in the tournament. Prove that the number of participants in the tournament is a perfect square.

2.68. Consider a group of 30 senators, such that each of them is quarreling with exactly 6 other senators. How many ways can a committee which consists of 3 senators be chosen, such that none of them is quarreling, or all of them are quarreling with each other?

2.69. Let $ABCD$ be a square, such that the Cartesian coordinates of its vertices are given by $A(0, 0)$, $B(n, 0)$, $C(n, n)$, and $D(0, n)$.

(a) How many broken lines L are there, such that the following conditions are satisfied: L starts at point A and ends at the point C , the length of L is equal to $2n$, and the Cartesian coordinates of every vertex of L are positive integers?

(b) The square $ABCD$ is partitioned into n^2 unit squares. How many ways can n chips be arranged into these unit squares, such that every row and every column contains exactly one chip?

(c) Draw a broken line that satisfies the conditions of item (a), and arrange n chips such that the conditions of item (b) are satisfied, and, additionally, all the chips are on the same side of the broken line. How many ways can this be done?

2.70. Some pairs of parentheses are written down in the expression $x_1 : x_2 : \dots : x_n$, and the value of the obtained expression is of the form

$$\frac{x_{i_1} x_{i_2} \dots x_{i_k}}{x_{j_1} x_{j_2} \dots x_{j_{n-k}}},$$

where $i_1 < i_2 < \dots < i_k$, $j_1 < j_2 < \dots < j_{n-k}$. How many distinct expressions of the above form can be obtained this way?

Chapter 3



Binomial and Multinomial Theorems

3.1 The Binomial Theorem

The following equalities hold true for any real numbers x and y :

$$\begin{aligned}(x+y)^2 &= x^2 + 2xy + y^2, \\(x+y)^3 &= x^3 + 3x^2y + 3xy^2 + y^3, \\(x+y)^4 &= x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4.\end{aligned}$$

All these equalities are special cases of the general formula given by the following theorem.

Theorem 3.1.1. *For any positive integer n the following equality holds:*

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k. \quad (3.1.1)$$

Proof. The expression $(x+y)^n$ can be written as follows:

$$(x+y)^n = (x+y)(x+y) \dots (x+y) = \sum c_1 c_2 \dots c_k. \quad (3.1.2)$$

The sum on the right-hand side of (3.1.2) consists of 2^n addends of the form $c_1 c_2 \dots c_n$, where $c_1, c_2, \dots, c_n \in \{x, y\}$. Moreover, for any $k \in \{0, 1, \dots, n\}$, there exist exactly $\binom{n}{k}$ addends $c_1 c_2 \dots c_n$, such that k of factors $c_1, c_2, \dots, c_n$ are equal to y , and the remaining $n-k$ factors are equal to x . Therefore, equality (3.1.1) holds true. $\square$

Equality (3.1.1) is usually referred to as Newton's binomial theorem, and coefficients

$$\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \dots, \binom{n}{n}$$

are called *binomial coefficients of order n* . In the first chapter we proved that $\binom{n}{k}$ is the number of k -combinations of an n -set, and obtained some properties of binomial coefficients. Some other properties of binomial coefficients and certain combinatorial identities will be presented in this chapter. First we give some simple examples.

Example 3.1.2. Using formula (3.1.1) we obtain the following equalities:

$$\begin{aligned} (x+1)^5 &= \binom{5}{0}x^5 + \binom{5}{1}x^4 + \binom{5}{2}x^3 + \binom{5}{3}x^2 + \binom{5}{4}x^1 + \binom{5}{5}x^0 \\ &= x^5 + 5x^4 + 10x^3 + 10x^2 + 5x + 1, \\ (2x+3)^5 &= 32x^5 + 240x^4 + 720x^3 + 1080x^2 + 810x + 243. \triangle \end{aligned}$$

Example 3.1.3. Let us determine the coefficient of x^{10} in the expansion of $(2x+3)^{12}$. It follows from formula (3.1.1) that the term in this expansion which contains x^{10} is equal to $\binom{12}{10}(2x)^{10}3^2$. The coefficient is

$$\binom{12}{2}2^{10}3^2 = 608\,256. \triangle$$

Example 3.1.4. Let $m, n, k \in \mathbb{N}$, $n > k$, and

$$P(x) = (1+x)^k + (1+x)^{k+1} + \dots + (1+x)^n.$$

Let us determine the coefficient c_m of x^m in the expansion of $P(x)$. It follows from the formula for the sum of the terms of a geometric sequence that

$$P(x) = \sum_{j=0}^n (1+x)^j - \sum_{j=0}^{k-1} (1+x)^j = \frac{(1+x)^{n+1} - (1+x)^k}{x}.$$

Now, using formula (3.1.1) it is easy to get coefficient c_m :

$$c_m = \begin{cases} \binom{n+1}{m+1} - \binom{k}{m+1}, & \text{if } m < k, \\ \binom{n+1}{m+1}, & \text{if } k \leq m \leq n, \\ 0, & \text{if } m > n. \triangle \end{cases}$$

Remark 3.1.5. The generalized binomial coefficients $\binom{r}{k}$ are defined for any real number r and $k \in \mathbb{N}_0 = \{0, 1, 2, \dots\}$ as follows:

$$\binom{r}{k} = \frac{r(r-1)(r-2)\dots(r-k+1)}{k!}.$$

For any real number r that is a nonnegative integer the following Newton's binomial theorem holds:

$$(1+x)^r = \sum_{k=0}^{\infty} \binom{r}{k} x^k, \quad |x| < 1.$$

3.2 Properties of Binomial Coefficients

Theorem 3.2.1. For any $n \in \mathbb{N}$, and k and m as indicated, the following equalities hold:

$$(a) \quad \binom{n}{k} = \binom{n}{n-k}, \quad 0 \leq k \leq n, \quad (3.2.1)$$

$$(b) \quad \binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}, \quad 1 \leq k \leq n, \quad (3.2.2)$$

$$(b) \quad \binom{n}{m} \binom{m}{k} = \binom{n}{k} \binom{n-k}{m-k} = \binom{n}{m-k} \binom{n-m+k}{k}, \quad (3.2.3)$$

for $0 \leq k \leq m \leq n$.

Proof. (a) Any k -combination A of the elements of an n -set S uniquely determines the $(n-k)$ -combination $S \setminus A$, and vice versa. Therefore, the number of k -combinations of the elements of set S is equal to the number of $(n-k)$ -combinations of elements of the same set, and equality (3.2.1) follows.

(b) Let $S = \{1, 2, \dots, n\}$, and C_k be the set of k -combinations of the elements of set S . Then $|C_k| = \binom{n}{k}$. Consider the partition $C_k = A \cup B$, where A consists of those k -combinations which do not contain the element n , and B consists of those k -combinations that contain the element n . The number of k -combinations that belong to A is equal to the number of k -combinations of the elements of set $\{1, 2, \dots, n-1\}$, i.e., $\binom{n-1}{k}$. The number of k -combinations that belong to B is equal to the number of $(k-1)$ -combinations of the elements of set $\{1, 2, \dots, n-1\}$, i.e., is $\binom{n-1}{k-1}$. Since $A \cap B = \emptyset$, equality (3.2.2) follows from the equality $C_k = A \cup B$.

(c) It is easy to obtain that each of the products

$$\binom{n}{m} \binom{m}{k}, \quad \binom{n}{k} \binom{n-k}{m-k}, \quad \binom{n}{m-k} \binom{n-m+k}{k}$$

is equal to the number of all pairs (A, B) , such that $A, B \subset S = \{1, 2, \dots, n\}$, $B \subset A$, $|B| = k$ and $|A| = m$. $\square$

Remark 3.2.2. The equalities (3.2.1)–(3.2.3) can be proved by using the definition of the binomial coefficient $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. For example,

$$\begin{aligned} \binom{n-1}{k} + \binom{n-1}{k-1} &= \frac{(n-1)!}{k!(n-1-k)!} + \frac{(n-1)!}{(k-1)!(n-k)!} \\ &= \frac{n!}{k!(n-k)!} \left(\frac{n-k}{n} + \frac{k}{n} \right) = \binom{n}{k}. \end{aligned}$$

Remark 3.2.3. Note that the recurrence equation (3.2.2) and the conditions $\binom{n}{0} = \binom{n}{n} = 1$ uniquely determine all binomial coefficients. They can be given in the following triangular array, usually referred to as Pascal's triangle:

$$\begin{array}{ccccccc} & & & & 1 & & \\ & & & & & & \\ & & 1 & 2 & 1 & & \\ & & & & & & \\ & 1 & 3 & 3 & 1 & & \\ & & & & & & \\ 1 & 4 & 6 & 4 & 1 & & \\ & & & & & & \\ 1 & 5 & 10 & 10 & 5 & 1 & \\ & & & & & & \\ & & & & & & \dots \end{array}$$

The first and the last terms in any row of Pascal's triangle are equal to 1, and every other term is equal to the sum of the two nearest terms in the previous row.

Using the Binomial Theorem and Theorem 3.2.1 we can obtain many other properties of binomial coefficients.

Example 3.2.4. Let us first put $x = y = 1$, and then $x = 1$, $y = -1$ in (3.1.1). We get

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n, \quad (3.2.4)$$

$$\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \dots + (-1)^n \binom{n}{n} = 0. \quad (3.2.5)$$

By summing the left-hand sides of (3.2.4) and (3.2.5), and then by subtracting the left-hand side of (3.2.5) from that of (3.2.4) we obtain that

$$\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \dots = \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \dots = 2^{n-1}. \quad \triangle \quad (3.2.6)$$

Example 3.2.5. By determining the coefficient of x^n in the expansion of

$$(1+x)^n(1+x)^n = (1+x)^{2n},$$

and using equality (3.2.1), it is easy to obtain the following identity

$$\begin{aligned} \binom{n}{0}\binom{n}{n} + \binom{n}{1}\binom{n}{n-1} + \cdots + \binom{n}{n}\binom{n}{0} \\ = \binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2 = \binom{2n}{n}. \quad \triangle \end{aligned}$$

Example 3.2.6. Using the recurrence relation (3.2.2) we get

$$\begin{aligned} \binom{n}{m} &= \binom{n-1}{m} + \binom{n-1}{m-1} = \binom{n-1}{m} + \binom{n-2}{m-1} + \binom{n-2}{m-2} \\ &= \binom{n-1}{m} + \binom{n-2}{m-1} + \binom{n-3}{m-2} + \binom{n-3}{m-3} + \cdots \\ &= \sum_{k=0}^M \binom{n-1-k}{m-k}, \quad \text{where } M = \min\{m, n-1\}. \end{aligned}$$

Example 3.2.7. Vandermonde's identity. If we apply the recurrence relation (3.2.2) first to the binomial coefficient $\binom{n}{m}$, and then to any of the binomial coefficients that appear in the sum, we get Vandermonde's identity:

$$\binom{n}{m} = \sum_{k=0}^r \binom{n-r}{m-k} \binom{r}{k}. \quad (3.2.7)$$

The identity (3.2.7) follows also from the equality

$$(1+x)^n = (1+x)^{n-r} \cdot (1+x)^r,$$

by comparing the coefficients of x^m on the left-hand and the right-hand sides of this equality. $\triangle$

Example 3.2.8. Combinatorial proof of Vandermonde's identity.

Suppose there is a group which consists of $n-r$ men and r women. In how many ways can a committee of m members be formed? The obvious answer is $\binom{n}{m}$. This number can be counted as follows. Suppose that committee consists of $m-k$ men and k women. The number of such committees is $\binom{n-r}{m-k} \binom{r}{k}$. Since $k \in \{0, 1, \dots, r\}$, then equality (3.2.7) follows. $\triangle$

Example 3.2.9. Let us denote

$$S_{nm} = \sum_{k=m}^n (-1)^k \binom{n}{k} \binom{k}{m}, \quad n \geq m \geq 0.$$

For $n = m$ it is obvious that $S_{nn} = (-1)^n$. If $n > m$, then using equality (3.2.5) we obtain

$$\begin{aligned} S_{nm} &= \sum_{k=m}^n (-1)^k \binom{n}{m} \binom{n-m}{k-m} \\ &= \binom{n}{m} \sum_{k=m}^n (-1)^k \binom{n-m}{k-m} = \binom{n}{m} (1-1)^{n-m} = 0. \end{aligned}$$

Finally,

$$S_{nm} = \sum_{k=m}^n (-1)^k \binom{n}{k} \binom{k}{m} = \begin{cases} (-1)^n, & \text{if } n = m, \\ 0, & \text{if } n > m. \end{cases} \quad \triangle \quad (3.2.8)$$

Example 3.2.10. Let $(a_0, a_1, a_2, \dots)$ be a sequence of real numbers and let

$$b_n = \sum_{k=0}^n (-1)^k \binom{n}{k} a_k, \quad n \in \mathbb{N}_0. \quad (3.2.9)$$

By mathematical induction we shall prove that for any $n \in \mathbb{N}_0$,

$$a_n = \sum_{k=0}^n (-1)^k \binom{n}{k} b_k. \quad (3.2.10)$$

It follows from (3.2.9) that $b_0 = a_0$, hence equality (3.2.10) holds for $n = 0$. Suppose now that equality (3.2.10) holds for all positive integers less than n . Using this assumption and equalities (3.2.8) and (3.2.9) we obtain that

$$\begin{aligned} b_n &= \sum_{k=0}^n (-1)^k \binom{n}{k} a_k = \sum_{k=0}^{n-1} (-1)^k \binom{n}{k} \sum_{m=0}^k (-1)^m \binom{k}{m} b_m + (-1)^n a_n \\ &= \sum_{m=0}^{n-1} (-1)^m b_m \left[\sum_{k=m}^{n-1} (-1)^k \binom{n}{k} \binom{k}{m} + (-1)^n \binom{n}{n} \binom{n}{m} \right. \\ &\quad \left. - (-1)^n \binom{n}{n} \binom{n}{m} \right] + (-1)^n a_n = - \sum_{m=0}^{n-1} (-1)^m b_m (-1)^n \binom{n}{m} + (-1)^n a_n \\ &= -(-1)^n \sum_{m=0}^{n-1} (-1)^m \binom{n}{m} b_m + (-1)^n a_n. \end{aligned}$$

Consequently, it follows that

$$a_n = (-1)^n b_n + \sum_{m=0}^{n-1} (-1)^m \binom{n}{m} b_m = \sum_{m=0}^n (-1)^m \binom{n}{m} b_m. \quad \triangle$$

Example 3.2.11. Let us prove that for any positive integer n the following equality holds true:

$$\sum_{k=1}^n (-1)^{k+1} \binom{n}{k} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}. \quad (3.2.11)$$

The proof will be given by mathematical induction. Let us denote $x_n = \sum_{k=1}^n (-1)^{k+1} \binom{n}{k} \frac{1}{k}$. For $n = 1$ it is easy to check that $x_1 = 1$. Suppose that $x_{n-1} = 1 + \frac{1}{2} + \cdots + \frac{1}{n-1}$. Then for the n -th term of the sequence (x_n) we obtain that

$$\begin{aligned} x_n &= \sum_{k=1}^{n-1} \frac{(-1)^{k+1}}{k} \left[\binom{n-1}{k} + \binom{n-1}{k-1} \right] + \frac{(-1)^{n+1}}{n} \\ &= x_{n-1} + \sum_{k=1}^n \frac{(-1)^{k+1}}{n} \binom{n}{k} = x_{n-1} + \frac{1}{n} [1 - (1-1)^n] \\ &= 1 + \frac{1}{2} + \cdots + \frac{1}{n-1} + \frac{1}{n}. \quad \triangle \end{aligned}$$

Example 3.2.12. The next identity follows immediately from equalities (3.2.9), (3.2.10), and (3.2.11):

$$\sum_{k=1}^n (-1)^{k+1} \binom{n}{k} \left(1 + \frac{1}{2} + \cdots + \frac{1}{k} \right) = \frac{1}{n}. \quad \triangle$$

Example 3.2.13. We shall prove that for any positive integer n ,

$$\sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} 2^{n-2k} = \frac{3^n + 1}{2}.$$

Using the binomial theorem we get

$$\left(1 + \frac{1}{2} \right)^n = \sum_{k=0}^n \binom{n}{k} \frac{1}{2^k}, \quad \left(1 - \frac{1}{2} \right)^n = \sum_{k=0}^n (-1)^k \binom{n}{k} \frac{1}{2^k}$$

By summing these two equalities we obtain that

$$\frac{(3/2)^n + (1/2)^n}{2} = \binom{n}{0} \frac{1}{2^0} + \binom{n}{2} \frac{1}{2^2} + \cdots = \sum_{k=0}^{[n/2]} \binom{n}{2k} \frac{1}{2^{2k}}.$$

Multiplication by 2^n yields the desired identity. $\triangle$

3.3 The Multinomial Theorem

Theorem 3.3.1. *Let m and n be positive integers. Then,*

$$(x_1 + x_2 + \cdots + x_m)^n = \sum \frac{n!}{k_1! k_2! \cdots k_m!} x_1^{k_1} x_2^{k_2} \cdots x_m^{k_m}, \quad (3.3.1)$$

where the sum runs over all m -tuples $(k_1, k_2, \dots, k_m)$ of nonnegative integers, such that $k_1 + k_2 + \cdots + k_m = n$.

Proof. The expression on the left-hand side of (3.3.1) is the product of n factors that are equal to $x_1 + x_2 + \cdots + x_m$. By multiplying we obtain that this product is equal to the sum which consists of m^n addends of the form $c_1 c_2 \cdots c_n$, where $c_1, c_2, \dots, c_n \in \{x_1, x_2, \dots, x_m\}$. Let $k_1, k_2, \dots, k_m$ be nonnegative integers such that $k_1 + k_2 + \cdots + k_m = n$. The number of addends $c_1 c_2 \cdots c_n$, such that, for any $j \in \{1, 2, \dots, m\}$, exactly k_j of the numbers $c_1, c_2, \dots, c_n$ are equal to x_j , is equal to the number of n -variations of elements $x_1, x_2, \dots, x_m$, which have the type $(k_1, k_2, \dots, k_m)$, i.e., $\frac{n!}{k_1! k_2! \cdots k_m!}$. Hence, equality (3.3.1) holds true. $\square$

Equality (3.3.1) is usually referred to as the multinomial theorem, and coefficients of the form

$$\frac{n!}{k_1! k_2! \cdots k_m!}$$

are called multinomial coefficients. It is clear that formula (3.1.1) is a particular case of formula (3.3.1), obtained for $m = 2$.

Example 3.3.2. Let us determine the coefficient of x^{15} in the expansion of $(1 - x^2 + x^3)^{20}$.

The number 15 can be written as a sum of addends that are equal to 2 or 3 in the following ways:

$$15 = 2 + 2 + 2 + 2 + 2 + 2 + 2 + 3 = 2 + 2 + 2 + 3 + 3 + 3 = 3 + 3 + 3 + 3 + 3.$$

Using formula (3.3.1) we obtain that the coefficient of x^{15} in the expansion of $(1 - x^2 + x^3)^{20}$ is equal to

$$\frac{20!}{13!6!1!} - \frac{20!}{14!3!3!} + \frac{20!}{15!0!5!} = \binom{20}{7}\binom{7}{1} - \binom{20}{6}\binom{6}{3} + \binom{20}{5}. \triangle$$

Example 3.3.3. Let us prove that for any $n > 1$, the following equality holds true:

$$(1 + x + x^2 + \cdots + x^{n-1})^2 = \sum_{k=0}^{2n-2} (n - |n - k - 1|)x^k.$$

Let c_k be the coefficient of x^k in the expansion of $(1 + x + x^2 + \cdots + x^{n-1})^2$. Then, c_k is equal to the number of pairs (a, b) of integers such that $a + b = k$, $0 \leq a \leq n-1$, $0 \leq b \leq n-1$. For $0 \leq k \leq n-1$, all pairs with this property are the following: $(0, k)$, $(1, k-1)$, $\dots$, $(k, 0)$. The number of such pairs is $c_k = k + 1 = n - |n - k - 1|$. For $n \leq k \leq 2n-2$, the following pairs should not be counted: $(0, k)$, $(1, k-1)$, $\dots$, $(k-n, n)$, $(n, k-n)$, $\dots$, $(k-1, 1)$, $(k, 0)$. In this case we get $c_k = k + 1 - 2(k - n + 1) = n - |n - k - 1|$. $\triangle$

Example 3.3.4. By substituting $x_1 = x_2 = \cdots = x_m = 1$ in (3.3.1) we obtain the equality that was proved in Theorem 2.9.2. $\triangle$

Example 3.3.5. Let us consider the equality:

$$(x_1 + \cdots + x_m)^{n+k} = (x_1 + \cdots + x_m)^n (x_1 + \cdots + x_m)^k. \quad (3.3.2)$$

By determining the coefficients of $x_1^{r_1} x_2^{r_2} \cdots x_m^{r_m}$, where $r_1 + r_2 + \cdots + r_m = n + k$, both on the left-hand and the right-hand side of (3.3.2), we obtain

$$\frac{(n+k)!}{r_1! r_2! \cdots r_m!} = \sum \frac{n!}{s_1! s_2! \cdots s_m!} \cdot \frac{k!}{t_1! t_2! \cdots t_m!}, \quad (3.3.3)$$

where the sum on the right-hand side of (3.3.3) runs over all $2m$ -tuples $(s_1, s_2, \dots, s_m, t_1, t_2, \dots, t_m)$, such that

$$\begin{aligned} s_1 + s_2 + \cdots + s_m &= n, & t_1 + t_2 + \cdots + t_m &= k, \\ s_j + t_j &= r_j \quad \text{for every } j \in \{1, 2, \dots, m\}. \end{aligned} \quad \triangle$$

Exercises

3.1. Determine the coefficient of x^5 in the expansion of $\left(3\sqrt{x} + \frac{1}{2\sqrt[3]{x}}\right)^{20}$.

3.2. Determine the 13-th term in the expansion of $(\sqrt{2} + \sqrt[3]{3})^{15}$.

3.3. How many rational terms are there in the expansion of $(\sqrt{2} + \sqrt[3]{3})^{100}$?

3.4. Determine the sum of all coefficients of powers of x in the expansion of $(3x - 2)^{100}$.

3.5. Prove that for any $n \in \mathbb{N}$, the number $(1 + \sqrt{2})^n + (1 - \sqrt{2})^n$ is an integer.

3.6. Determine $n, k \in \mathbb{N}$ from the following relations:

$$\binom{n}{k} : \binom{n+1}{k} : \binom{n+1}{k+1} = 3 : 4 : 8.$$

3.7. Determine $n, k \in \mathbb{N}$ from the equalities $\binom{n}{k-1} = 2002$, $\binom{n}{k} = 3003$.

3.8. Prove that for any positive integer $n \geq 2$, and any nonnegative real x , the inequality $(1+x)^n \geq 1+nx$ holds true.

3.9. Prove that for any positive integer $n \geq 2$ the following inequalities hold true:

$$2 < \left(1 + \frac{1}{n}\right)^n < 3.$$

3.10. Prove that, for any $n \in \mathbb{N}$, $[(2 + \sqrt{3})^n]$ is an odd positive integer.

3.11. If n is an odd integer greater than 4, then the first n digits in the decimal representation of the number

$$(n + \sqrt{n^2 + 1})^n - [(n + \sqrt{n^2 + 1})^n]$$

are equal to 0. Prove this statement.

Prove the identities formulated in Exercises 3.12–3.30:

$$\mathbf{3.12.} \quad 1 + 14\binom{n}{1} + 36\binom{n}{2} + 24\binom{n}{3} = (n+1)^4 - n^4.$$

$$\mathbf{3.13.} \quad \sum_{k=1}^n \binom{m+k-1}{k} = \sum_{k=1}^m \binom{n+k-1}{k}.$$

$$\mathbf{3.14.} \quad \sum_{k=1}^n \binom{n-1}{k-1} \binom{2n-1}{k}^{-1} = \frac{2}{n+1}.$$

$$3.15. \quad \sum_{k=1}^n \binom{n-1}{k-1} \binom{n+m}{k}^{-1} = \frac{n+m+1}{(m+1)(m+2)}.$$

$$3.16. \quad \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n} = n2^{n-1}.$$

$$3.17. \quad \binom{n}{0} + 2\binom{n}{1} + 3\binom{n}{2} + \cdots + (n+1)\binom{n}{n} = (n+2)2^{n-1}.$$

$$3.18. \quad \binom{n}{0} - 2\binom{n}{1} + 3\binom{n}{2} - \cdots + (-1)^n(n+1)\binom{n}{n} = 0.$$

$$3.19. \quad \binom{n}{0} + \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \cdots + \frac{1}{n+1}\binom{n}{n} = \frac{2^{n+1}-1}{n+1}.$$

$$3.20. \quad \frac{1}{2}\binom{n}{0} + \frac{1}{3}\binom{n}{1} + \frac{1}{4}\binom{n}{2} + \cdots + \frac{1}{n+2}\binom{n}{n} = \frac{2^{n+1}+1}{(n+1)(n+2)}.$$

$$3.21. \quad \binom{n}{0} - \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} - \cdots + (-1)^n \frac{1}{n+1}\binom{n}{n} = \frac{1}{n+1}.$$

$$3.22. \quad \binom{n}{1}^2 + 2\binom{n}{2}^2 + 3\binom{n}{3}^2 + \cdots + n\binom{n}{n}^2 = n\binom{2n-1}{n-1}.$$

$$3.23. \quad \sum_{k=0}^n \frac{(2n)!}{(k!)^2((n-k)!)^2} = \binom{2n}{n}^2.$$

$$3.24. \quad \sum_{k=0}^n \binom{n}{k} \binom{n}{m} \binom{2n}{m+k}^{-1} = \frac{2n+1}{n+1}.$$

$$3.25. \quad \binom{n}{0} + \frac{1}{2}\binom{n+1}{1} + \frac{1}{2^2}\binom{n+2}{2} + \cdots + \frac{1}{2^n}\binom{2n}{n} = 2^n.$$

$$3.26. \quad \sum_{k=1}^n (-1)^k \binom{2n-k}{k} = \begin{cases} 1, & \text{if } n \equiv 0 \pmod{3}, \\ 0, & \text{if } n \equiv 1 \pmod{3}, \\ -1, & \text{if } n \equiv -1 \pmod{3}. \end{cases}$$

$$3.27. \quad \sum_{k=0}^{n-1} \binom{4n}{4k+1} = 2^{4n-2}.$$

$$\mathbf{3.28.} \quad \sum_{k=0}^n (-1)^k \frac{1}{\binom{n}{k}} = \frac{n+1}{n+2} (1 + (-1)^n).$$

$$\mathbf{3.29.} \quad \sum_{k=0}^n (-1)^k \frac{1}{x+k} \binom{n}{k} = \frac{n!}{x(x+1) \cdots (x+n)},$$

where $x \notin \{0, -1, -2, \dots, -n\}$.

$$\mathbf{3.30.} \quad 1 + \sum_{k=0}^{m-1} \binom{m}{k} S_n(k) = (n+1)^m, \text{ where } S_n(k) = \sum_{j=1}^n j^k.$$

3.31. Using de Moivre's formula $(\cos x + i \sin x)^n = \cos nx + i \sin nx$, $n \in \mathbb{N}_0$, and the Binomial Theorem prove the following identities:

$$(a) \quad 1 - 3 \binom{n}{2} + 9 \binom{n}{4} - 27 \binom{n}{6} + \cdots = (-1)^n 2^n \cos \frac{2n\pi}{3},$$

$$(b) \quad \binom{n}{1} - 3 \binom{n}{3} + 9 \binom{n}{5} - \cdots = \frac{1}{\sqrt{3}} (-2)^n \sin \frac{2n\pi}{3}.$$

3.32. Prove the following identities:

$$(a) \quad \binom{n}{0} + \binom{n}{3} + \binom{n}{6} + \cdots = \frac{1}{3} \left(2^n + 2 \cos \frac{2n\pi}{3} \right),$$

$$(b) \quad \binom{n}{1} + \binom{n}{4} + \binom{n}{7} + \cdots = \frac{1}{3} \left(2^n + 2 \cos \frac{(n-2)\pi}{3} \right),$$

$$(c) \quad \binom{n}{2} + \binom{n}{5} + \binom{n}{8} + \cdots = \frac{1}{3} \left(2^n + 2 \cos \frac{(n+2)\pi}{3} \right).$$

3.33. Prove that the following identity holds for all $m, n \in \mathbb{N}$:

$$\binom{n}{0} + \binom{n}{m} + \binom{n}{2m} + \cdots = \frac{2^n}{m} \sum_{k=1}^m \cos^n \frac{k\pi}{m} \cos \frac{nk\pi}{m}.$$

3.34. Determine the square root of the number obtained by inserting k zeros between any two digits of the number 14641.

3.35. A group of 2^n people is moving in the plane starting from point $(0, 0)$. Every day every member of the group crosses a unit of distance. The first day half of them move to the East, and half of them move to the North. Every day every group divides into two equal subgroups, and one subgroup

continues moving to the East, and the other one to the North, etc. Determine the position of the people after n days.

3.36. A group of 2^n people is moving along a line starting from point 0. Half of them are going to the right, and half to the left. After they cross the unit of distance, each group divides into two subgroups, and one of these subgroups continues moving to the right, and the other one to the left, etc. The people stop moving after they cross n units of distances. Determine the position of the people at the moment they stop.

3.37. Let $x_n = 1/n$, $n \in \mathbf{N}_0$. The sequence of the first differences is defined by $r_n^{(1)} = x_{n+1} - x_n$, $n \in \mathbf{N}$. The sequence of the second differences is defined by $r_n^{(2)} = r_{n+1}^{(1)} - r_n^{(1)}$, $n \in \mathbf{N}$, etc. This way we obtain the following double sequence:

$$\begin{array}{ccccccc}
 1 & 1/2 & 1/3 & 1/4 & 1/5 & 1/6 & \dots \\
 -1/2 & -1/6 & -1/12 & -1/20 & -1/30 & \dots & \\
 1/3 & 1/12 & 1/30 & 1/60 & \dots & & \\
 -1/4 & -1/20 & -1/60 & \dots & & & \\
 1/5 & 1/30 & \dots & & & & \\
 -1/6 & \dots & & & & &
 \end{array}$$

Let us consider the table obtained by a rotation of -60° .

$$\begin{array}{cccc}
 & & & 1 \\
 & & -1/2 & 1/2 \\
 & 1/3 & -1/6 & 1/3 \\
 -1/4 & 1/12 & -1/12 & 1/4 \\
 \dots & \dots & \dots & \dots
 \end{array}$$

Then, the following transformations of the table are made: (1) the sign $-$ is removed from the table; (2) then, all the numbers in every row are divided by the first term in their row; (3) then, every number is replaced by its reciprocal. Prove that the obtained table is Pascal's triangle.

3.38. For any $k \in \{0, 1, 2, \dots\}$, let a_k be the coefficient of x^k in the expansion of $(1 + x + x^2)^n$. Prove that:

- (a) $a_0 a_1 - a_1 a_2 + a_2 a_3 - \dots - a_{2n-1} a_{2n} = 0$;
- (b) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + (-1)^{n-1} a_{n-1}^2 + (-1)^n a_n^2 = a_n$;

- (c) $a_0 + a_2 + a_4 + \cdots = \frac{1}{2}(3^n + 1), \quad a_1 + a_3 + a_5 + \cdots = \frac{1}{2}(3^n - 1).$
- (d) $a_k - na_{k-1} + \binom{n}{k}a_{k-2} - \cdots + (-1)^k \binom{n}{k}a_0 = 0, \text{ if } k \equiv 0 \pmod{3}.$

3.39. Determine the coefficient of x^{10} in the expansion of $(1 - x^2 + x^3)^{11}$.

3.40. If p is a prime number, prove that the number $[(2 + \sqrt{5})^p] - 2^{p+1}$ is divisible by p .

3.41. Prove that, for any $n, k \in \mathbb{N}$, $n \geq k$, the greatest common divisor of the binomial coefficients $\binom{n}{k}, \binom{n+1}{k}, \dots, \binom{n+k}{k}$ is equal to 1.

3.42. Determine the greatest common divisor of the binomial coefficients

$$\binom{2n}{1}, \binom{2n}{3}, \binom{2n}{5}, \dots, \binom{2n}{2n-1}.$$

3.43. How many odd positive integers are there among the binomial coefficients of order n ?

3.44. Prove that there is no positive integer n such that the number of even and odd positive integers among the binomial coefficients of order n is the same.

Chapter 4



Inclusion-Exclusion Principle

4.1 The Basic Formula

Counting the number of elements of the union of a few finite sets often appears as part of many combinatorial problems. Let us first consider two simple examples.

Example 4.1.1. Ten students scored good marks on a mathematics exam, 12 students scored good marks on a physics exam, and 7 of them scored good marks on both the mathematics and physics exams. How many students scored good marks on at least one subject exam?

Answer. Let us denote by A and B the sets of students who scored good marks on the math and physics exams, respectively. The question is to determine the number of elements of the set $A \cup B$. Students who scored good marks on both the math and physics exams are counted twice in the sum $|A| + |B|$. Therefore, it follows that

$$|A \cup B| = |A| + |B| - |A \cap B|. \quad (4.1.1)$$

and finally we get $|A \cup B| = 10 + 12 - 7 = 15$. $\triangle$

Example 4.1.2. Let us determine the number of positive integers from the set $S = \{1, 2, \dots, 1000\}$ that are divisible by at least one of the numbers 2, 3, and 5.

Answer. Let us denote by A , B , and C the subsets of elements from S that are divisible by 2, 3, and 5, respectively. We need to determine the number of elements of the set $A \cup B \cup C$. Consider the sum $|A| + |B| + |C|$. Elements

that belong to exactly two of the sets A , B , and C are counted twice in this sum, while elements that belong to all of these sets are counting three times. Consequently we get

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|. \quad (4.1.2)$$

Note that the sets $A \cap B$, $A \cap C$, $B \cap C$, and $A \cap B \cap C$ consist of the numbers from S that are divisible by 6, 10, 15, and 30 respectively. Therefore, $|A| = 500$, $|B| = 333$, $|C| = 200$, $|A \cap B| = 166$, $|A \cap C| = 100$, $|B \cap C| = 66$, and $|A \cap B \cap C| = 33$. Using (4.1.2) we get $|A \cup B \cup C| = 734$. $\triangle$

Theorem 4.1.3. *Let $A_1, A_2, \dots, A_n$ be subsets of a finite set S . Then the following equality holds true:*

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| &= \sum_{i=1}^n |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| \\ &+ \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n-1} |A_1 \cap A_2 \cap \dots \cap A_n|. \end{aligned} \quad (4.1.3)$$

Proof. The number $|A_1 \cup A_2 \cup \dots \cup A_n|$ on the left-hand side of equality (4.1.3) and every addend on the right-hand side can be written as a sum of 1's with corresponding signs $+$ or $-$. Every element $a \in A_1 \cup A_2 \cup \dots \cup A_n$ gives exactly one addend equal to 1 on the left-hand side of (4.1.3). Suppose that exactly r of the sets $A_1, A_2, \dots, A_n$ contain the element a . Then, on the right-hand side this element gives the sum

$$r - \binom{r}{2} + \binom{r}{3} - \dots + (-1)^{r-1} \binom{r}{r} = 1 - (1-1)^r = 1.$$

Consequently, equality (4.1.3) holds. $\square$

Formula (4.1.3) is usually referred to as the inclusion-exclusion principle. We have proved it using simple combinatorial reasoning. This formula can also be proved by mathematical induction and we shall provide this proof as well.

Proof of Theorem 4.1.3 by mathematical induction. For $n = 1$ equality (4.1.3) obviously holds. For $n = 2$ the corresponding formula was proved in Example 4.1.2. Suppose that (4.1.3) holds for some positive integer $n = m \geq 2$. Let $A_1, A_2, \dots, A_m, A_{m+1}$ be arbitrary finite sets. By assumption for sets $A_1, A_2, \dots, A_m$ the following equality holds:

$$\begin{aligned}
|A_1 \cup A_2 \cup \cdots \cup A_m| &= \sum_{j=1}^m |A_j| - \sum_{1 \leq i < j \leq m} |A_i \cap A_j| \\
&+ \sum_{1 \leq i < j < k \leq m} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{m-1} |A_1 \cap A_2 \cap \cdots \cap A_m|. \quad (4.1.4)
\end{aligned}$$

Note that $|(A_1 \cup A_2 \cup \cdots \cup A_m) \cap A_{m+1}| = \left| \bigcup_{j=1}^m (A_j \cap A_{m+1}) \right|$. By assumption for sets $A_1 \cap A_{m+1}, A_2 \cap A_{m+1}, \dots, A_m \cap A_{m+1}$ the following equality holds:

$$\begin{aligned}
|(A_1 \cup A_2 \cup \cdots \cup A_m) \cap A_{m+1}| &= \sum_{j=1}^m |A_j \cap A_{m+1}| \\
&- \sum_{1 \leq i < j \leq m} |A_i \cap A_j \cap A_{m+1}| + \sum_{1 \leq i < j < k \leq m} |A_i \cap A_j \cap A_k \cap A_{m+1}| \\
&- \cdots + (-1)^{m-1} |A_1 \cap A_2 \cap \cdots \cap A_m \cap A_{m+1}|. \quad (4.1.5)
\end{aligned}$$

Using (4.1.4), (4.1.5) and formula (4.1.1) where $A = A_1 \cup A_2 \cup \cdots \cup A_m$ and $B = A_{m+1}$, we obtain

$$\begin{aligned}
|A_1 \cup A_2 \cup \cdots \cup A_m \cup A_{m+1}| &= \\
&= |A_1 \cup A_2 \cup \cdots \cup A_m| + |A_{m+1}| - |(A_1 \cup A_2 \cup \cdots \cup A_m) \cap A_{m+1}| \\
&= \sum_{j=1}^{m+1} |A_j| - \sum_{1 \leq i < j \leq m+1} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq m+1} |A_i \cap A_j \cap A_k| \\
&- \cdots + (-1)^m |A_1 \cap \cdots \cap A_m \cap A_{m+1}|,
\end{aligned}$$

i.e., the inclusion-exclusion principle also holds for $n = m + 1$. $\square$

4.2 The Special Case

It is of interest to consider a special case of formula (4.1.3) where the cardinal number of the intersection of a few sets depends only on the number of sets. More precisely, let $A_1, A_2, \dots, A_n$ be subsets of a finite set S and suppose that the following equalities hold:

$$\begin{aligned}
|S| &= M, \quad |S \setminus (A_1 \cup A_2 \cup \cdots \cup A_n)| = M_0, \\
|A_i| &= M_1 \quad \text{for any } i \in \{1, 2, \dots, n\}, \\
|A_i \cap A_j| &= M_2 \quad \text{for all } i \neq j, \quad i, j \in \{1, 2, \dots, n\}, \\
|A_i \cap A_j \cap A_k| &= M_3 \quad \text{for all } i \neq j \neq k \neq i, \quad i, j, k \in \{1, 2, \dots, n\}, \\
&\dots\dots\dots
\end{aligned}$$

$$|A_1 \cap A_2 \cap \cdots \cap A_n| = M_n.$$

Then,

$$|A_1 \cup A_2 \cup \cdots \cup A_n| = \sum_{i=1}^n (-1)^{i-1} \binom{n}{i} M_i, \quad (4.2.1)$$

$$M_0 = M - \sum_{i=1}^n (-1)^{i-1} \binom{n}{i} M_i. \quad (4.2.2)$$

4.3 Some More Examples

Example 4.3.1. Let us determine how many different words without patterns CC, OO, and II can be obtained by permuting the letters of the word COMBINATORICS.

The total number of arrangements that can be obtained by permuting the letters of the word COMBINATORICS is equal to $\frac{13!}{2!2!2!}$. Let us denote by A_1 , A_2 , and A_3 the sets of arrangements with the patterns CC, OO, and II respectively. Using the inclusion-exclusion principle, i.e., formula (4.2.2), we obtain that the number of words without the specified patterns is equal to

$$\frac{13}{2!2!2!} - |A_1 \cup A_2 \cup A_3| = \frac{13!}{(2!)^3} - 3 \cdot \frac{12!}{(2!)^2} + 3 \cdot \frac{11!}{2!} - 10!. \quad \triangle$$

Example 4.3.2. Let $S = \{1, 2, \dots, n\}$, $F = \{f \mid f : S \rightarrow S\}$, and $F_0 \subset F$ be the set of functions $f : S \rightarrow S$ without fixed points. Let us determine the number of elements of set F_0 .

Let us denote $F_i = \{f \in F \mid f(i) = i\}$, $i \in \{1, 2, \dots, n\}$. Then, for any $k \in \{1, 2, \dots, n\}$, and any k -combination $\{i_1, i_2, \dots, i_k\}$ of the elements of set $\{1, 2, \dots, n\}$ the following equality holds

$$|F_{i_1} \cap F_{i_2} \cap \cdots \cap F_{i_k}| = n^{n-k},$$

This is true because the number of functions $f : S \rightarrow S$, for which the points $i_1, i_2, \dots, i_k$ are fixed, is equal to the number of all functions $g : T \rightarrow S$, where $T = S \setminus \{i_1, i_2, \dots, i_k\}$ and $|T| = n - k$. Using (4.2.2) we obtain that

$$\begin{aligned} |F_0| &= \sum_{k=0}^n (-1)^k \binom{n}{k} n^{n-k} = \sum_{j=n}^0 (-1)^{n-j} \binom{n}{j} n^j \\ &= n^n - \binom{n}{1} n^{n-1} + \binom{n}{2} n^{n-2} - \cdots + (-1)^n. \quad \triangle \end{aligned}$$

Example 4.3.3. Let $X = \{x_1, x_2, \dots, x_n\}$, $Y = \{1, 2, \dots, k\}$, and let F be the set of all surjective functions from X onto Y , i.e.

$$F = \{f: X \rightarrow Y \mid (\forall y \in Y)(\exists x \in X)f(x) = y\}.$$

Let us determine the number of elements of set F . Let us denote

$$F_y = \{f: X \rightarrow Y \mid (\forall x \in X)f(x) \neq y\}, \quad y \in \{1, 2, \dots, k\}.$$

Then, for any $j \in \{1, 2, \dots, k\}$, and any j -combination $\{y_1, y_2, \dots, y_j\}$ of the elements of set $\{1, 2, \dots, k\}$, the following equality holds true

$$|F_{y_1} \cap F_{y_2} \cap \dots \cap F_{y_j}| = (k - j)^n.$$

Using formulae (4.2.1) and (4.2.2) we obtain that

$$\begin{aligned} |F_1 \cup F_2 \cup \dots \cup F_k| &= \sum_{j=1}^k (-1)^{j-1} \binom{k}{j} (k - j)^n, \\ |F| &= k^n - |F(1) \cup F(2) \cup \dots \cup F(k)| \\ &= \sum_{j=0}^k (-1)^j \binom{k}{j} (k - j)^n = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n. \end{aligned}$$

Note that set F is empty for $k > n$. Therefore, for $k > n \geq 1$ the following equality holds true

$$\sum_{j=0}^k (-1)^j \binom{k}{j} (k - j)^n = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n = 0. \quad \triangle$$

In Chapter 2, Section 2.8, we considered a method of counting arrangements using a geometrical method based on counting trajectories. Here we shall prove two more theorems related to trajectories.

Theorem 4.3.4. *Let $m, n, k \in \mathbb{N}$. The number of trajectories with starting point $(0, 0)$ and endpoint $(2n, 0)$ that do not intersect the lines $y = m$ and $y = -k$, is not greater than*

$$\binom{2n}{n} - \binom{2n}{n+m} - \binom{2n}{n+k} + 2 \binom{2n}{n+m+k}.$$

Proof. Let S be the set of all trajectories with starting point $(0, 0)$ and endpoint $(2n, 0)$, S_1 be the set of those trajectories from S that intersect the

line $y = m$, and S_2 be the set of those trajectories from S that intersect the line $y = -k$. For $t \in S_1 \cap S_2$, let us denote

$$\begin{aligned} x_1(t) &= \min\{x \mid (x, m) \in t\}, \\ x_2(t) &= \min\{x \mid (x, -k) \in t\}, \\ T_1 &= \{t \in S_1 \cap S_2 \mid x_1(t) < x_2(t)\}, \\ T_2 &= \{t \in S_1 \cap S_2 \mid x_1(t) > x_2(t)\}. \end{aligned}$$

The sets T_1 and T_2 are disjoint and the equality $S_1 \cap S_2 = T_1 \cup T_2$ holds. Using Theorem 2.8.1 we obtain

$$|S| = \binom{2n}{n}, \quad |S_1| = \binom{2n}{n+m}, \quad |S_2| = \binom{2n}{n+k}.$$

Let $t \in T_1$ and let t_1 be the part of the trajectory t with starting point $(0, 0)$ and endpoint $(x_1(t), m)$, let t_2 be the part of the trajectory t with starting point $(x_1(t), m)$ and endpoint $(x_2(t), -k)$, and let t_3 be the part of the trajectory t with starting point $(x_2(t), -k)$ and endpoint $(2n, 0)$. Now, let t'_3 be the trajectory obtained by the reflection of t_3 across the line $y = -k$, let $t'_2 \cup t'_3$ be the trajectory obtained by the reflection of $t_2 \cup t_3$ across the line $y = m$, and let $t' = t_1 \cup t'_2 \cup t'_3$. The starting point of trajectory t' is $(0, 0)$, and the endpoint is $(2n, 2m + 2k)$. Using Theorem 2.8.1 we obtain

$$|T_1| \leq M(2n, 2m + 2k) = \binom{2n}{n+m+k}.$$

We can similarly prove that $|T_2| \leq M(2n, 2m + 2k)$, and consequently,

$$\begin{aligned} &|S \setminus (S_1 \cup S_2)| \\ &= |S| - |S_1| - |S_2| + |S_1 \cap S_2| = |S| - |S_1| - |S_2| + |T_1| + |T_2| \\ &\leq \binom{2n}{n} - \binom{2n}{n+m} - \binom{2n}{n+k} + 2\binom{2n}{n+m+k}. \quad \square \end{aligned}$$

In Example 2.8.2 we answered the following question: How many $2n$ -arrangements of the elements 0 and 1 are there, such that each of these arrangements consists of n 0's and n 1's, and before each 0 there are more 1's than 0's? The next question is of interest in the theory of statistical quality control. *How many n -arrangements of the elements 0 and 1 are there, such that before each 0 there are at least k times more 1's than 0's?* Before answering this question for $k = 2$ we shall prove the theorem that will be used.

Theorem 4.3.5. *Let $f(A, B)$ be the number of trajectories with starting point A and endpoint B . Let us denote:*

$$\begin{aligned} A_0 &= (0, 0), \quad X = (0, -2), \\ A_{m+n} &= (m+n, m-n), \quad m, n \in \mathbb{N}, \quad m \geq 2n, \\ C_k &= (3k-1, k-1), \quad k = 1, 2, 3, \dots \end{aligned}$$

(a) *Then, for every $k \in \mathbb{N}$, $f(A_0, C_k) = 2f(X, C_k)$.*

(b) *For an arbitrary point Z let us denote by $F_k(Z, A_{m+n})$ the number of trajectories with starting point Z and endpoint A_{m+n} , that contain the point C_k , and contain none of the points $C_1, C_2, \dots, C_{k-1}$. Let $F(Z, A_{m+n})$ be the number of trajectories with starting point Z and endpoint A_{m+n} , that intersect the line $y = (x-2)/3$. Then, we have*

$$\begin{aligned} F_k(A_0, A_{m+n}) &= 2F_k(X, A_{m+n}), \\ F(A_0, A_{m+n}) &= 2F(X, A_{m+n}). \end{aligned}$$

(c) *Let $\alpha(A_0, A_{m+n})$ be the number of trajectories with starting point A_0 and endpoint A_{m+n} , and without points below the line $y = x/3$. Then,*

$$\alpha(A_0, A_{m+n}) = \binom{m+n}{m} - 2\binom{m+n}{m+1}.$$

Proof. First note that, if $m = 2n$, then the point A_{m+n} belongs to the line $y = x/3$, and if $m > 2n$ then this point is above the line $y = x/3$, see Figure 4.3.1 on page 56.

(a) A trajectory with starting point A_0 ends at point C_k , if and only if it consists of $2k-1$ increasing parts and k decreasing parts. A trajectory with starting point X ends at point C_k , if and only if it consists of $2k$ increasing parts and $k-1$ decreasing parts. Using these facts we obtain that

$$\begin{aligned} f(A_0, C_k) &= \binom{3k-1}{k} = \frac{(3k-1)!}{k!(2k-1)!} = \frac{2(3k-1)!}{(k-1)!(2k)!} \\ &= 2\binom{3k-1}{k-1} = 2f(X, C_k). \end{aligned}$$

(b) Using statement (a) and the inclusion-exclusion principle we get

$$\begin{aligned} F_1(A_0, A_{m+n}) &= f(A_0, C_1)f(C_1, A_{m+n}) = 2f(X, C_1)f(C_1, A_{m+n}) \\ &= 2F_1(X, A_{m+n}), \\ F_2(A_0, A_{m+n}) &= f(A_0, C_2)f(C_2, A_{m+n}) - f(A_0, C_1)f(C_1, C_2)f(C_2, A_{m+n}) \\ &= 2f(X, C_2)f(C_2, A_{m+n}) - 2f(X, C_1)f(C_1, C_2)f(C_2, A_{m+n}) \\ &= F_2(X, A_{m+n}), \end{aligned}$$

$$\begin{aligned}
F_3(A_0, A_{m+n}) &= f(A_0, C_3)f(C_3, A_{m+n}) - f(A_0, C_1)f(C_1, C_3)f(C_3, A_{m+n}) \\
&\quad - f(A_0, C_2)f(C_2, C_3)f(C_3, A_{m+n}) \\
&\quad + f(A_0, C_1)f(C_1, C_2)f(C_2, C_3)f(C_3, A_{m+n}) \\
&= 2f(X, C_3)f(C_3, A_{m+n}) - 2f(X, C_1)f(C_1, C_3)f(C_3, A_{m+n}) \\
&\quad - 2f(X, C_2)f(C_2, C_3)f(C_3, A_{m+n}) \\
&\quad + 2f(X, C_1)f(C_1, C_2)f(C_2, C_3)f(C_3, A_{m+n}) \\
&= 2F_3(X, A_{m+n}).
\end{aligned}$$

We can similarly prove that, for any $k \in \mathbb{N}$, the following equality holds:

$$F_k(A_0, A_{m+n}) = 2F_k(X, A_{m+n}).$$

Note that there exists k_0 such that for any $k \geq k_0$, the numbers $F_k(A_0, A_{m+n})$ and $F_k(X, A_{m+n})$ are equal to 0. Any trajectory with starting point A_0 and endpoint A_{m+n} , such that it has common points with the line $y = (x-2)/3$, contains at least one of the points $C_1, C_2, C_3, \dots$. Such a trajectory may intersect the line $y = (x-2)/3$ at some points that do not belong to the set $\{C_1, C_2, C_3, \dots\}$, see Figure 4.3.1. By summing the obtained equalities over all k we get $F(A_0, A_{m+n}) = 2F(X, A_{m+n})$.

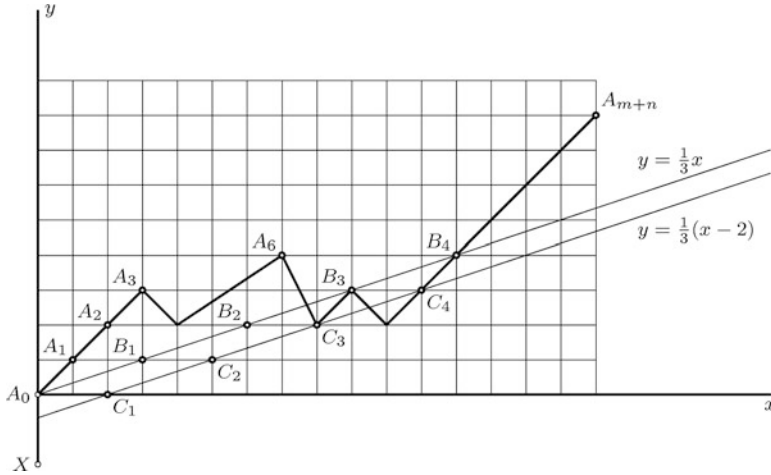


Fig. 4.3.1

(c) Using Theorem 2.8.1(a) we get that the number of all trajectories with starting point A_0 and endpoint A_{m+n} is equal to $\binom{m+n}{m}$.

The number of trajectories with starting point A_0 and endpoint A_{m+n} , that contain points below the line $y = x/3$, is equal to the number of trajectories with starting point A_0 and endpoint A_{m+n} , that have at least one

common point with the line $y = (x - 2)/3$. Using statement (b) we get that this number is equal to $2F(X, A_{m+n})$. All trajectories with starting point X and endpoint A_{m+n} intersect with the line $y = (x - 2)/3$. Hence, $F(X, A_{m+n})$ is the number of all trajectories with starting point X , and end point A_{m+n} , i.e.

$$F(X, A_{m+n}) = \binom{m+n}{m+1}.$$

Now, it is easy to obtain that the number of trajectories with starting point A_0 and endpoint A_{m+n} , such that they do not pass through the points below the line $y = x/3$, is equal to

$$\begin{aligned} \alpha(A_0, A_{m+n}) &= \binom{m+n}{m} - F(A_0, A_{m+n}) = \binom{m+n}{m} - 2F(X, A_{m+n}) \\ &= \binom{m+n}{m} - 2\binom{m+n}{m+1}. \quad \square \end{aligned}$$

Example 4.3.6. Let $A_0A_1A_2 \dots A_s$ be a trajectory with starting point $A_0 = (0, 0)$. Note that none of the points $A_0, A_1, A_2, \dots, A_s$ is below the line $y = x/3$, if and only if, for any $k \in \{1, 2, \dots, s\}$, the following statement holds: the trajectory $A_0A_1A_2 \dots A_k$ has at least two times more increasing parts than decreasing parts. Using this fact and Theorem 4.3.5 we get the following result:

Let $m \geq 2n$, where $m, n \in \mathbb{N}$. The number of arrangements $a_1a_2 \dots a_{m+n}$ consisting of m 0's and n 1's, such that, for every $k \in \{1, 2, \dots, m+n\}$, the number of 1's among $a_1, a_2, \dots, a_k$ is at least twice as large as the number of 0's among them, is equal to

$$\binom{m+n}{m} - 2\binom{m+n}{m+1}. \quad \triangle$$

Example 4.3.7. Let us calculate the sum $\sum 2^{-k}$ over all positive integers k that are not divisible by 2 and 3. Using the inclusion-exclusion principle we obtain that this sum is equal to

$$\begin{aligned} &\sum_{k=1}^{\infty} \left(\frac{1}{2^k} - \frac{1}{2^{2k}} - \frac{1}{2^{3k}} + \frac{1}{2^{6k}} \right) \\ &= \frac{1}{2} \cdot \frac{1}{1 - \frac{1}{2}} - \frac{1}{4} \cdot \frac{1}{1 - \frac{1}{4}} - \frac{1}{8} \cdot \frac{1}{1 - \frac{1}{8}} + \frac{1}{64} \cdot \frac{1}{1 - \frac{1}{64}} = \frac{34}{64}. \quad \triangle \end{aligned}$$

4.4 Generalized Inclusion-Exclusion Principle

Let $S = \{a_1, a_2, \dots, a_m\}$. Suppose that element a_k has weight $w(a_k)$. The weight of set $A \subset S$ is defined as the sum of the weights of its elements. Let $A_1, A_2, \dots, A_n$ be subsets of set S . Let us introduce the following notation:

- $W(A_{j_1}, A_{j_2}, \dots, A_{j_k})$ — the weight of set $A_{j_1} \cap A_{j_2} \cap \dots \cap A_{j_k}$, where $\{j_1, j_2, \dots, j_k\}$ is a k -combination of the elements of set $\{1, 2, \dots, n\}$;
- $W(k) = \sum W(A_{j_1}, A_{j_2}, \dots, A_{j_k})$, where the sum runs over all k -combinations $\{j_1, j_2, \dots, j_k\}$ of the elements of set $\{1, 2, \dots, n\}$;
- $W(0)$ and W — the weights of sets $S \setminus (A_1 \cup A_2 \cup \dots \cup A_n)$ and S , respectively.

Theorem 4.4.1. *The following equality holds true:*

$$W - W(0) = W(1) - W(2) + W(3) - \dots + (-1)^{n-1}W(n). \quad (4.4.1)$$

Remark 4.4.2. Suppose that the weight of each element is equal to 1. In this case formula (4.1.3) is a special case of formula (4.4.1). Both proofs of formula (4.1.3) can be slightly modified in order to cover the more general formula (4.4.1). The case when the sum of the weights of all the elements of set S is equal to 1 is of special interest in probability theory.

Exercises

4.1. If $A_1, A_2, \dots, A_n$ are finite sets, prove that

$$\sum_{i=1}^n |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| \leq |A_1 \cup A_2 \cup \dots \cup A_n| \leq \sum_{i=1}^n |A_i|.$$

4.2. Determine the number of positive integers that are not greater than 10^6 and are not divisible by the numbers: (a) 2, 3, 5; (b) 2, 3, 5, 7.

4.3. (a) How many 6-digit positive integers are there, such that exactly three different digits appear in the decimal representation of each of them?

(b) For every $k \in \{1, 2, \dots, 9\}$ determine how many n -digit positive integers there are, such that exactly k different digits appear in the decimal representation of each of these numbers.

4.4. An international jury consists of nine members from three countries. Each of these countries has three members of the jury.

(a) How many ways can members of the jury be arranged in a row, such that no three adjacent positions are occupied by three members from the same country?

(b) How many ways can members of the jury be arranged in a row, such that no two adjacent positions are occupied by two members from the same country?

4.5. Euler's totient function φ . Let $p_1, p_2, \dots, p_m$ be distinct prime numbers, $k_1, k_2, \dots, k_m$ be positive integers, and $n = p_1^{k_1} p_2^{k_2} \cdots p_m^{k_m}$. Denote by $\varphi(n)$ the number of positive integers less than or equal to n that are relatively prime to n . Prove that

$$\varphi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_m}\right).$$

4.6. Calculate $\varphi(45)$, $\varphi(900)$ and $\varphi(116704)$, where φ is the Euler totient function.

4.7. Let m and n be positive integers. Prove that for the Euler totient function the inequality $\varphi(m)\varphi(n) \leq \varphi(mn)$ holds true. Prove also that if two natural numbers m and n are coprime, then $\varphi(m)\varphi(n) = \varphi(mn)$.

4.8. Let $p_1 = 2, p_2 = 3, p_3 = 5, \dots$ be a sequence of prime numbers. Let us denote by $\pi(n)$ the number of prime numbers that are not greater than n . Prove that

$$\pi(n) = n - 1 + \pi([\sqrt{n}]) + \sum (-1)^k \left\lfloor \frac{n}{p_{j_1} p_{j_2} \cdots p_{j_k}} \right\rfloor,$$

where the sum runs over all nonempty subsets $\{j_1, j_2, \dots, j_k\}$ of the set $\{1, 2, \dots, \pi([\sqrt{n}])\}$.

4.9. Calculate $\pi(120)$, where $\pi : \mathbb{N} \rightarrow \mathbb{N}$ is the function defined in Exercise 4.8.

4.10. The Möbius function $\mu : \mathbb{N} \rightarrow \{-1, 0, 1\}$ is defined by the equalities $\mu(1) = 1$, and

$$\mu(n) = \begin{cases} (-1)^m, & \text{if } k_1 = k_2 = \cdots = k_m = 1, \\ 0, & \text{if } k_j > 1 \text{ for some } j \in \{1, 2, \dots, m\}, \end{cases}$$

for $n = p_1^{k_1} p_2^{k_2} \cdots p_m^{k_m}$, where $p_1, p_2, \dots, p_m$ are distinct prime numbers, and $k_1, k_2, \dots, k_m$ are positive integers. Prove that the equality $\sum \mu(d) = 0$

holds for any positive integer $n > 1$, where the sum runs over all positive divisors d of the natural number n .

4.11. Prove the equality $\varphi(n) = n \sum \frac{\mu(d)}{d}$, where the sum runs over all positive divisors d of the natural number $n > 1$.

4.12. Prove the equality

$$\pi(n) = \pi([\sqrt{n}]) + n - 1 + \sum \mu(d) \left[\frac{n}{d} \right],$$

where the sum runs over all positive divisors d of the natural number n , such that d can be represented as the product of distinct prime numbers that are not greater than $\sqrt{n}$.

4.13. For any positive integer $n \geq 3$ find the smallest positive integer $f(n)$ which has the following property: for any subset $A \subset \{1, 2, \dots, n\}$ consisting of $f(n)$ elements, there exist numbers $x, y, z \in A$, such that every two of them are relatively prime.

4.14. Let $S = \{1, 2, 3, \dots, 280\}$. Find the smallest positive integer n such that any n -subset of set S contains 5 relatively prime numbers.

4.15. How many permutations $a_1 a_2 \dots a_n$ of the set $\{1, 2, \dots, n\}$ are there, such that $a_j \neq j$ for any $j \in \{1, 2, \dots, n\}$?

4.16. Let $1 \leq k \leq n$. How many permutations $a_1 a_2 \dots a_n$ of set $\{1, 2, \dots, n\}$ are there, such that the equality $a_j = j$ holds for exactly k elements $j \in \{1, 2, \dots, n\}$?

4.17. Let $A = \{1, 2, \dots, n\}$ and $1 \leq m < n$. How many permutations $a_1 a_2 \dots a_n$ of set A are there, such that $a_j \neq j$ holds for every $j \in \{1, 2, \dots, m\}$?

4.18. Let $A = \{1, 2, \dots, n\}$ and $1 \leq k \leq m < n$. How many permutations $a_1 a_2 \dots a_n$ of set A are there, such that the equality $a_j = j$ holds for exactly k elements $j \in \{1, 2, \dots, m\}$?

4.19. Let $n \geq 2$ and $A = \{1, 2, \dots, n\}$. How many permutations $a_1 a_2 \dots a_n$ of set A are there, such that $(a_k, a_{k+1}) \neq (j, j+1)$, for any $k \in \{1, \dots, n-1\}$ and any $j \in \{1, 2, \dots, n-1\}$?

4.20. How many ways are there to color the fields of a chessboard 8×8 using 8 colors, such that every color appears in each row, and no two adjacent fields in the same column are of the same color?

4.21. Two physicians, A and B , are to examine the same n patients. Every examination lasts 15 minutes. How many ways are there to arrange the schedule under the condition that all $2n$ examinations are finished during the period of $n/2$ hours?

4.22. How many $2n$ -variations $a_1 a_2 \dots a_{2n}$ of elements $1, 2, \dots, n$ are there, such that any of these elements appears twice, and $a_k \neq a_{k+1}$ for any $k \in \{1, 2, \dots, 2n - 1\}$?

4.23. How many ways can n couples be arranged around a circular table, such that none of the husbands is adjacent to his wife?

4.24. Let $n \geq k$. How many n -arrangements of elements $1, 2, \dots, k$ are there, such that each of these elements appears in every arrangement?

4.25. Let $n, k \in \mathbb{N}$, $c_1, c_2, \dots, c_k \in \mathbb{N}_0$. How many n -combinations of the elements $1, 2, \dots, k$ with repetitions allowed are there, such that, for any $j \in \{1, 2, \dots, k\}$, the element j appears in each n -combination at least c_j times?

4.26. Let $0 \leq k < n \leq m + k$. How many ways can m distinguishable balls be put into n distinguishable boxes, such that there are exactly k empty boxes?

4.27. A die is thrown until every number $1, 2, \dots, 6$ appears. How many n -arrangements of elements $1, 2, \dots, 6$ can be obtained as a result of this experiment?

4.28. Let $A = \{1, 2, \dots, n\}$, $1 \leq m < n$, and let S be the set of all permutations of set A . How many k -arrangements of the elements of set S are there, such that for any permutation $p \in S$ that appears as a term in each of these k -arrangements, there exist exactly m elements $j \in A$ with the property $a_j = j$?

4.29. Let $A = \{1, 2, \dots, n\}$ and let S be the set of all m -combinations of the set A . How many p -arrangements $B_1 B_2 \dots B_p$ of the elements of set S are there, such that every element of set A is contained in at least one of the sets B_s , $s \in \{1, 2, \dots, p\}$?

4.30. Let $S_1, S_2, \dots, S_n$ be finite sets, where $n \in \mathbb{N}$. Prove that

$$|S_1 \cap S_2 \cap \dots \cap S_n| = \sum (-1)^{k+1} |S_{j_1} \cup S_{j_2} \cup \dots \cup S_{j_k}|,$$

where the sum runs over all nonempty subsets $\{j_1, j_2, \dots, j_k\}$ of the set $\{1, 2, \dots, n\}$.

4.31. Let $1 \leq m < n$. Prove the following equality:

$$\sum_{k=0}^n (-1)^k \binom{n}{k} \binom{n-k}{m-k} = 0.$$

4.32. Let $1 \leq m < n$. Prove the equality:

$$\sum_{k=0}^n (-1)^k \binom{n}{k} \binom{m+n-k-1}{m} = 0.$$

4.33. How many ways can k distinguishable balls be put into $m+n$ boxes numbered $1, 2, \dots, m+n$, such that each of the boxes numbered $1, 2, \dots, m$ contains at least one ball?

4.34. Suppose that $3n$ points are chosen on a circle. How many ways are there to draw n mutually disjoint triangles such that the given points are vertices of these triangles?

Chapter 5



Generating Functions

5.1 Definition and Examples

In this chapter we shall introduce one more method for solving combinatorial counting problems that is based on generating functions. We shall also give some examples of the generating functions of certain sequences of positive integers that appear in combinatorial problems.

Definition 5.1.1. The function

$$G(x) = a_0 + a_1x + a_2x^2 + \cdots = \sum_{k=0}^{\infty} a_k x^k, \quad (5.1.1)$$

where the power series on the right-hand side of (5.1.1) converges at some interval is called the *ordinary generating function* of the sequence $(a_n) = (a_0, a_1, a_2, \dots)$.

Example 5.1.2. The function $(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k$ is the ordinary generating function of the binomial coefficients of order n . $\triangle$

Example 5.1.3. The function $\frac{1}{1-x} = 1 + x + x^2 + \cdots = \sum_{k=0}^{\infty} x^k$, $|x| < 1$, is the ordinary generating function of the sequence $(1, 1, 1, \dots)$ whose terms are all equal to 1. $\triangle$

Example 5.1.4. The function

$$\frac{1}{1+x} = 1 - x + x^2 - \cdots = \sum_{k=0}^{\infty} (-1)^k x^k, \quad |x| < 1,$$

is the ordinary generating function of the sequence $(1, -1, 1, -1, \dots)$ whose terms are alternately equal to 1 and -1 . $\triangle$

Example 5.1.5. By mathematical induction we can prove that for every positive integer k the following equality holds

$$\frac{1}{(1-x)^k} = \sum_{n=0}^{\infty} \binom{k+n-1}{n} x^n, \quad |x| < 1. \quad (5.1.2)$$

Equality (5.1.2) can also be proved the following way:

$$\begin{aligned} \frac{1}{(1-x)^k} &= \sum_{n=0}^{\infty} \binom{-k}{n} (-x)^n = \sum_{n=0}^{\infty} \frac{(-k)(-k-1)\dots(-k-n+1)}{n!} (-x)^n \\ &= \sum_{n=0}^{\infty} \frac{k(k+1)\dots(k+n-1)}{n!} x^n = \sum_{n=0}^{\infty} \binom{k+n-1}{n} x^n. \end{aligned}$$

For $k = 2$ equality (5.1.2) becomes

$$\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} (n+1)x^n, \quad |x| < 1, \quad (5.1.3)$$

i.e., $(1-x)^{-2}$ is the ordinary generating function of the sequence of natural numbers $(1, 2, 3, 4, 5, \dots)$. $\triangle$

Example 5.1.6. For every positive integer k the following equality holds

$$\begin{aligned} \frac{1}{(1+x)^k} &= \sum_{n=0}^{\infty} \binom{-k}{n} x^n = \sum_{n=0}^{\infty} (-1)^n \binom{-k}{n} (-x)^n \\ &= \sum_{n=0}^{\infty} (-1)^n \binom{k+n-1}{n} x^n, \quad |x| < 1. \end{aligned}$$

Hence, $(1+x)^{-k}$ is the ordinary generating function of the sequence $c_n = (-1)^n \binom{k+n-1}{n}$, $n = 0, 1, 2, \dots$. For $k = 2$ we obtain that $(1+x)^{-2} = \sum_{n=0}^{\infty} (-1)^n (n+1)x^n$, i.e., $(1+x)^{-2}$ is the ordinary generating function of the sequence $(1, -2, 3, -4, \dots)$. $\triangle$

Example 5.1.7. Consider the equation $x_1 + x_2 + \dots + x_k = n$, where n and k are positive integers. Let us denote by a_n the number of solutions of this equation in the set $\mathbb{N}_0$, and by b_n the number of solutions in the set $\mathbb{N}$.

First, note that the ordinary generating functions of the sequences $(a_0, a_1, a_2, \dots)$ and $(b_0, b_1, b_2, \dots)$ are respectively given by

$$A(x) = (1 + x + x^2 + \dots)^k = \frac{1}{(1-x)^k},$$

$$B(x) = (x + x^2 + x^3 + \dots)^k = \frac{x^k}{(1-x)^k}.$$

From Example 5.1.5 it follows that, for any positive integer k , the following equality holds

$$A(x) = \frac{1}{(1-x)^k} = \sum_{n=0}^{\infty} \binom{k+n-1}{n} x^n, \quad |x| < 1.$$

Consequently we obtain that $a_n = \binom{k+n-1}{n}$, for $n = 0, 1, 2, \dots$

For the generating function $B(x)$ we have

$$\begin{aligned} B(x) &= \frac{x^k}{(1-x)^k} = x^k \sum_{n=0}^{\infty} \binom{k+n-1}{n} x^n = \sum_{n=0}^{\infty} \binom{k+n-1}{n} x^{k+n} \\ &= \sum_{m=k}^{\infty} \binom{m-1}{m-k} x^m = \sum_{n=k}^{\infty} \binom{n-1}{n-k} x^n = \sum_{n=k}^{\infty} \binom{n-1}{k-1} x^n. \end{aligned}$$

Finally, it follows that

$$\begin{aligned} b_n &= \binom{n-1}{k-1}, \quad \text{for } n \geq k; \\ b_n &= 0, \quad \text{for } n \in \{0, 1, \dots, k-1\}. \quad \triangle \end{aligned}$$

5.2 Operations with Generating Functions

Adding. Let $g_1(x) = \sum_{k=0}^{\infty} a_k x^k$ and $g_2(x) = \sum_{k=0}^{\infty} b_k x^k$ be the ordinary generating functions of the sequences (a_n) i (b_n) . Then

$$g(x) = g_1(x) + g_2(x) = \sum_{k=0}^{\infty} (a_k + b_k) x^k,$$

is the ordinary generating function of the sequence $c_k = a_k + b_k$, $k = 0, 1, \dots$

Multiplication by a constant. Let $g(x) = \sum_{k=0}^{\infty} a_k x^k$ be the ordinary generating function of the sequence (a_n) , and $c \in \mathbb{R}$. Then

$$f(x) = cg(x) = \sum_{k=0}^{\infty} ca_k x^k,$$

is the ordinary generating function of the sequence ca_k , $k = 0, 1, 2, \dots$

A linear combination. Let $g_1(x) = \sum_{k=0}^{\infty} a_k x^k$ and $g_2(x) = \sum_{k=0}^{\infty} b_k x^k$ be the ordinary generating functions of the sequences (a_n) and (b_n) . Then

$$g(x) = \alpha g_1(x) + \beta g_2(x) = \sum_{k=0}^{\infty} (\alpha a_k + \beta b_k) x^k,$$

is the ordinary generating function of the sequence $c_k = \alpha a_k + \beta b_k$, $k = 0, 1, 2, \dots$.

Shifting. Let $g(x) = \sum_{k=0}^{\infty} a_k x^k$ be the ordinary generating function of the sequence (a_n) . Then

$$x^n g(x) = \sum_{k=0}^{\infty} a_k x^{n+k} = \sum_{k=n}^{\infty} a_{k-n} x^k,$$

is the ordinary generating function of the sequence $(\underbrace{0, 0, \dots, 0}_{n \text{ times}}, a_0, a_1, a_2, \dots)$. The function

$$\begin{aligned} \frac{g(x) - a_0 - a_1 x - \dots - a_{n-1} x^{n-1}}{x^n} &= x^{-n} \sum_{k=n}^{\infty} a_k x^k \\ &= \sum_{k=n}^{\infty} a_k x^{n-k} = \sum_{k=0}^{\infty} a_{k+n} x^k, \end{aligned}$$

is the ordinary generating function of the sequence $(a_n, a_{n+1}, a_{n+2}, \dots)$.

A change of the variable. Let $g(x) = \sum_{k=0}^{\infty} a_k x^k$ be the ordinary generating function of the sequence (a_n) . Then

$$g(cx) = \sum_{k=0}^{\infty} a_k (cx)^k = \sum_{k=0}^{\infty} c^k a_k x^k,$$

is the ordinary generating function of the sequence $(a_0, ca_1, c^2 a_2, c^3 a_3, \dots)$.

Multiplication. Let $g_1(x) = \sum_{k=0}^{\infty} a_k x^k$ and $g_2(x) = \sum_{k=0}^{\infty} b_k x^k$ be the ordinary generating functions of the sequences (a_n) and (b_n) . Then

$$\begin{aligned} f(x) &= g_1(x)g_2(x) = (a_0 + a_1 x + a_2 x^2 + \dots)(b_0 + b_1 x + b_2 x^2 + \dots) \\ &= a_0 b_0 + (a_0 b_1 + a_1 b_0)x + (a_0 b_2 + a_1 b_1 + a_2 b_0)x^2 + \dots + \\ &\quad + (a_0 b_n + a_1 b_{n-1} + \dots + a_{n-1} b_1 + a_n b_0)x^n + \dots \end{aligned}$$

is the ordinary generating function of the sequence $c_n = \sum_{k=0}^n a_k b_{n-k}$, $n =$

$0, 1, 2, \dots$. In particular, if $g_2(x) = \frac{1}{1-x}$, then

$$g_1(x) \cdot \frac{1}{1-x} = a_0 + (a_0 + a_1)x + (a_0 + a_1 + a_2)x^2 + \dots,$$

is the ordinary generating function of the sequence of partial sums of the sequence (a_n) : $a_0, a_0 + a_1, a_0 + a_1 + a_2, \dots$

Differentiation. Let $g(x) = \sum_{k=0}^{\infty} a_k x^k$ be the ordinary generating function of the sequence (a_n) . Then

$$g'(x) = \sum_{k=0}^{\infty} (k+1)a_{k+1}x^k,$$

is the ordinary generating function of the sequence $(a_1, 2a_2, 3a_3, \dots)$.

Integration. Let $g(x) = \sum_{k=0}^{\infty} a_k x^k$ be the ordinary generating function of the sequence (a_n) . Then

$$\int_0^x g(t) dt = a_0 x + \frac{a_1}{2} x^2 + \frac{a_2}{3} x^3 + \dots = \sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1} = \sum_{k=1}^{\infty} \frac{a_{k-1}}{k} x^k$$

is the ordinary generating function of the sequence $\left(0, a_0, \frac{a_1}{2}, \frac{a_2}{3}, \dots\right)$.

5.3 The Fibonacci Sequence

The Fibonacci sequence $(F_n)_{n \geq 0}$ is defined by $F_0 = 0$, $F_1 = 1$, and

$$F_n = F_{n-1} + F_{n-2} \quad \text{for } n \geq 2.$$

Let us determine the general term of the sequence as a function of n . If we denote by $g(x)$ the generating function of the Fibonacci sequence, then we have

$$\begin{aligned} g(x) &= F_0 + F_1 x + F_2 x^2 + F_3 x^3 + \dots, \\ xg(x) &= F_0 x + F_1 x^2 + F_2 x^3 + F_3 x^4 + \dots, \\ x^2 g(x) &= F_0 x^2 + F_1 x^3 + F_2 x^4 + F_3 x^5 + \dots \end{aligned}$$

As a consequence of previous equalities we get

$$(1-x-x^2)g(x) = F_0 + (F_1 - F_0)x + (F_2 - F_1 - F_0)x^2 + (F_3 - F_2 - F_1)x^3 + \dots = x,$$

and $g(x) = \frac{x}{1-x-x^2}$. Since

$$1-x-x^2=0 \quad \text{for } x_1 = \frac{-1+\sqrt{5}}{2} \quad \text{and } x_2 = \frac{-1-\sqrt{5}}{2},$$

it follows that

$$g(x) = \frac{1}{\sqrt{5}} \left(\frac{1}{1-\alpha_1 x} - \frac{1}{1-\alpha_2 x} \right),$$

where

$$\alpha_1 = \frac{1+\sqrt{5}}{2}, \quad \alpha_2 = \frac{1-\sqrt{5}}{2}.$$

If x is a real number such that $|\alpha_1 x| < 1$ and $|\alpha_2 x| < 1$, then $g(x)$ can be represented in the form

$$\begin{aligned} g(x) &= \frac{1}{\sqrt{5}}(1 + \alpha_1 x + \alpha_1^2 x^2 + \dots) - \frac{1}{\sqrt{5}}(1 + \alpha_2 x + \alpha_2^2 x^2 + \dots) \\ &= \frac{1}{\sqrt{5}}((\alpha_1 - \alpha_2)x + (\alpha_1^2 - \alpha_2^2)x^2 + \dots). \end{aligned}$$

$$\text{Hence, } F_n = \frac{1}{\sqrt{5}}(\alpha_1^n - \alpha_2^n) = \frac{1}{\sqrt{5}} \left\{ \left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right\}. \quad \triangle$$

A number of interesting properties of the Fibonacci sequence are obtained. Here we shall formulate just a few of them.

Example 5.3.1. By the method of mathematical induction it is easy to prove that for every natural number n the following equalities hold:

- (a) $F_1 + F_3 + \dots + F_{2n-1} = F_{2n}$,
- (b) $F_2 + F_4 + \dots + F_{2n} = F_{2n+1} - 1$,
- (c) $F_1^2 + F_2^2 + \dots + F_n^2 = F_n F_{n+1}$.
- (d) $F_{n+1} F_{n-1} - F_n^2 = (-1)^n$,
- (e) $F_1 F_2 + F_2 F_3 + \dots + F_{2n-1} F_{2n} = F_{2n}^2$,
- (f) $F_1 F_2 + F_2 F_3 + \dots + F_{2n} F_{2n+1} = F_{2n+1}^2 - 1$. $\triangle$

Theorem 5.3.2. Every natural number n can be uniquely represented in the form

$$n = F_{k_1} + F_{k_2} + \dots + F_{k_m},$$

where $m \geq 1$, and $k_i - k_{i+1} \geq 2$ for each $i \in \{1, 2, \dots, m-1\}$.

Proof. We shall use mathematical induction. Since $F_1 = 1$, $F_2 = 2$, and $F_3 = 3$, the statement of the theorem obviously holds for $n \in \{1, 2, 3\}$. Suppose that the statement holds for all natural numbers less than n . Let

k be the positive integer such that $F_k \leq n < F_{k+1}$. Then, the difference $n - F_k$ is obviously less than F_{k-1} , and, by the induction hypothesis, it can be represented as a sum Σ of the Fibonacci numbers that satisfy the given condition. Then, $n = F_k + \Sigma$ is the required representation of n .

Now we shall prove the uniqueness of the representation. First, we prove that F_{k_1} is the greatest Fibonacci number not greater than n . Again, let k be the positive integer such that $F_k \leq n < F_{k+1}$. It follows that $F_{k-2} + F_{k-1} \leq n < F_{k+1}$.

Let us suppose that $k_1 < k$, and denote $n_1 = n - F_{k_1}$. Then, $n_1 \geq F_{k-2}$, and $k_2 \leq k - 3$. Since $F_{k-2} > F_{k-3} + F_{k-5} + F_{k-7} + \dots$, it follows that n_1 cannot be represented as a sum of Fibonacci numbers that satisfy the given condition. Hence, the assumption $k_1 < k$ is false, and the uniqueness of k_1 is proved. Similarly we prove that the indices $k_2, k_3, \dots$ can be chosen uniquely. $\square$

5.4 The Recursive Equations

Example 5.4.1. (a) Let $(a_n)_{n \geq 0}$ be the sequence of integers given by $a_0 = 1$, $a_1 = 2$, and

$$a_{n+2} = 3a_{n+1} - 2a_n, \quad n \geq 0. \quad (5.4.1)$$

Let $g(x) = \sum_{n=0}^{\infty} a_n x^n$ be the ordinary generating function of sequence $(a_n)_{n \geq 0}$. Then, for $x \neq 0$, we have

$$\frac{g(x) - a_0}{x} = \sum_{n=0}^{\infty} a_{n+1} x^n, \quad \frac{g(x) - a_0 - a_1 x}{x^2} = \sum_{n=0}^{\infty} a_{n+2} x^n,$$

and consequently

$$\begin{aligned} & \frac{g(x) - a_0 - a_1 x}{x^2} - 3 \cdot \frac{g(x) - a_0}{x} + 2g(x) \\ &= \sum_{n=0}^{\infty} (a_{n+2} - 3a_{n+1} + 2a_n) x^n = 0. \end{aligned}$$

Now, using the fact that $a_0 = 1$ and $a_1 = 2$ it is easy to conclude that

$$g(x) = \frac{1-x}{1-3x+2x^2} = \frac{1}{1-2x} = \sum_{n=0}^{\infty} 2^n x^n, \quad |x| < \frac{1}{2}.$$

Hence, $a_n = 2^n$ for any $n \in \mathbb{N}_0 = \{0, 1, 2, \dots\}$.

(b) Let $(a_n)_{n \geq 0}$ be the sequence of integers given by $a_0 = 2$, $a_2 = 1$, and (5.4.1). As in the previous case we obtain that

$$g(x) = \frac{2-5x}{1-3x+2x^2} = \frac{3}{1-x} - \frac{1}{1-2x}, \quad |x| < \frac{1}{2},$$

is the ordinary generating function of sequence $(a_n)_{n \geq 0}$. It follows that

$$g(x) = 3 \sum_{n=0}^{\infty} x^n - \sum_{n=0}^{\infty} (2x)^n = \sum_{n=0}^{\infty} (3 - 2^n)x^n,$$

and $a_n = 3 - 2^n$ for every $n \in \mathbb{N}_0$. $\triangle$

Example 5.4.2. Let $(a_n)_{n \geq 0}$ be the sequence of real numbers given by $a_0 = 2$, $a_1 = 5$, and $a_{n+2} = 2a_{n+1} - a_n$ for $n \geq 0$. Similarly as in the previous example we obtain that

$$g(x) = \frac{2+x}{(1-x)^2} = \frac{3}{(1-x)^2} - \frac{1}{1-x}, \quad |x| < 1,$$

is the ordinary generating function of sequence $(a_n)_{n \geq 0}$. Using equality (5.1.3) we obtain that

$$g(x) = 3 \sum_{n=0}^{\infty} (n+1)x^n - \sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} (3n+2)x^n,$$

and hence $a_n = 3n+2$ for every $n \in \mathbb{N}_0$. $\triangle$

Example 5.4.3. Let $(a_n)_{n \geq 0}$ be the sequence of real numbers given by $a_0 = 2$, $a_1 = 5$, and $a_{n+2} = 4a_{n+1} - 4a_n$ for $n \geq 0$. For the ordinary generating function $g(x)$ of sequence $(a_n)_{n \geq 0}$ we obtain

$$g(x) = \frac{2-3x}{(1-2x)^2} = \frac{1}{2} \cdot \frac{1}{(1-2x)^2} + \frac{3}{2} \cdot \frac{1}{1-2x}, \quad |x| < \frac{1}{2},$$

and hence $g(x) = \frac{1}{2} \sum_{n=0}^{\infty} (n+1)(2x)^n + \frac{3}{2} \sum_{n=0}^{\infty} (2x)^n = \sum_{n=0}^{\infty} 2^{n-1}(n+4) \cdot x^n$, and $a_n = 2^{n-1}(n+4)$ for every $n \in \mathbb{N}_0$. $\triangle$

For more examples and general cases of the recursive equations see Exercises 5.8–5.13.

5.5 The Catalan Numbers

Example 5.5.1. Suppose that $2n$ points are given on a circle. How many ways can these $2n$ points be connected into n pairs by n chords without points of intersection?

Let us denote by a_n the number to be determined. First note that $a_0 = 1$ and $a_1 = 1$. Denote the given points by $A_1, A_2, \dots, A_{2n}$ in the order of their appearance in a specified direction on the circle. Point A_1 can be

connected with some of the points $A_2, A_4, \dots, A_{2n}$. Hence, for $n \geq 1$, the following equality holds

$$a_n = a_0 a_{n-1} + a_1 a_{n-2} + \dots + a_{k-1} a_{n-k} + \dots + a_{n-1} a_0.$$

Let $g(x)$ be the ordinary generating function of the sequence $(a_0, a_1, a_2, \dots)$. Then we have

$$\begin{aligned} g(x) &= a_0 + a_1 x + a_2 x^2 + \dots, \\ (g(x))^2 &= (a_0 + a_1 x + a_2 x^2 + \dots)(a_0 + a_1 x + a_2 x^2 + \dots) \\ &= a_0 a_0 + (a_0 a_1 + a_1 a_0)x + (a_0 a_2 + a_1 a_1 + a_2 a_0)x^2 + \dots + \\ &\quad + (a_0 a_n + a_1 a_{n-1} + \dots + a_{n-1} a_1 + a_n a_0)x^n + \dots \\ &= a_1 + a_2 x + a_3 x^2 + \dots + a_{n+1} x^n + \dots \end{aligned}$$

Now it is easy to see that $x(g(x))^2 - g(x) + 1 = 0$, and consequently

$$g(x) = \frac{1 \pm \sqrt{1-4x}}{2x}.$$

Since $g(0) = a_0 = 1$ and

$$\lim_{x \rightarrow 0} \frac{1 - \sqrt{1-4x}}{2x} = 1, \quad \lim_{x \rightarrow 0} \left| \frac{1 + \sqrt{1-4x}}{2x} \right| = \infty,$$

it follows that $g(x) = \frac{1 - \sqrt{1-4x}}{2x}$. Using the expansion

$$\begin{aligned} \sqrt{1-4x} &= (1-4x)^{1/2} = \sum_{k=0}^{\infty} \binom{1/2}{k} (-4x)^k \\ &= 1 + \sum_{k=1}^{\infty} \binom{1/2}{k} (-4x)^k \\ &= 1 + \sum_{k=1}^{\infty} \frac{\frac{1}{2}(\frac{1}{2}-1)(\frac{1}{2}-2)\dots(\frac{1}{2}-k+1)}{k!} (-1)^k 4^k x^k \\ &= 1 + \sum_{k=1}^{\infty} \frac{(2k-3)!!(-1)^{k-1}(k-1)!}{2^k k!(k-1)!} (-1)^k 4^k x^k \\ &= 1 - \sum_{k=1}^{\infty} \frac{(2k-3)!!(k-1)!2^k}{k!(k-1)!} x^k \\ &= 1 - 2 \sum_{k=1}^{\infty} \frac{(2k-3)!!(2k-2)!!}{k(k-1)!(k-1)!} x^k \\ &= 1 - 2 \sum_{k=1}^{\infty} \frac{1}{k} \binom{2k-2}{k-1} x^k, \end{aligned}$$

we finally obtain

$$g(x) = \sum_{k=1}^{\infty} \frac{1}{k} \binom{2k-2}{k-1} x^{k-1} = \sum_{n=0}^{\infty} \frac{1}{n+1} \binom{2n}{n} x^n,$$

$$a_n = \frac{1}{n+1} \binom{2n}{n}, \quad n = 0, 1, 2, \dots \triangle$$

5.6 Exponential Generating Functions

Definition 5.6.1. The function $g(x) = \sum_{n=0}^{\infty} \frac{a_n}{n!} x^n$ is called the *exponential generating function* of the sequence $(a_n) = (a_0, a_1, a_2, \dots)$.

Example 5.6.2. Let $f(x) = \sum_{n=0}^{\infty} \frac{a_n}{n!} x^n$ and $g(x) = \sum_{n=0}^{\infty} \frac{b_n}{n!} x^n$ be the exponential generating functions of the sequences (a_n) and (b_n) . Then

$$h(x) = f(x)g(x) = \sum_{n=0}^{\infty} \left\{ \sum_{k=0}^n \binom{n}{k} a_k b_{n-k} \right\} \frac{x^n}{n!},$$

is the exponential generating function of the sequence $c_n = \sum_{k=0}^n \binom{n}{k} a_k b_{n-k}$, $n = 0, 1, 2, \dots$, which is called the *binomial convolution* of the sequences (a_n) and (b_n) . $\triangle$

Exercises

5.1. Determine the ordinary generating functions of the sequences:

- (a) $a_n = n$, $n = 0, 1, 2, \dots$;
- (b) $b_n = n^2$, $n = 0, 1, 2, \dots$;
- (c) $c_n = n(n-1)$, $n = 0, 1, 2, \dots$.

5.2. Determine the exponential generating function of the sequence $a_n = n!$, $n = 0, 1, 2, \dots$.

5.3. Determine the ordinary generating function and the exponential generating function of the sequence $a_n = 2^n + 3^n$, $n = 0, 1, 2, \dots$.

5.4. Determine the ordinary generating function of the harmonic sequence $(H_n)_{n \geq 0}$, that is given by $H_0 = 0$ and $H_n = H_{n-1} + \frac{1}{n}$ for $n \geq 1$.

Remark. The following formula gives the approximation of the general term of the harmonic sequence:

$$H_n = \ln n + \gamma + \frac{1}{2n} - \frac{1}{12n^2} + \frac{1}{120n^4} - \delta,$$

where $0 < \delta < \frac{1}{252n^6}$, and $\gamma = 0.57721566 \dots$ is the Euler constant.

5.5. Determine the ordinary generating function of the sequence $a_n = n\alpha^n$, $n \in \mathbb{N}_0$, where $\alpha \in \mathbb{R}$ is a constant.

5.6. Determine the ordinary generating function of the sequence $(a_n)_{n \geq 0}$, that is given by $a_n = \alpha^n/n!$ for $n \in \mathbb{N}_0$, where $\alpha \in \mathbb{R}$ is a constant.

5.7. Determine the ordinary generating functions of the sequences $(a_n)_{n \geq 0}$ and $(b_n)_{n \geq 0}$, where $a_n = \sin n\alpha$, $b_n = \cos n\alpha$, and $\alpha \in \mathbb{R}$ is a constant.

5.8. Determine the general term of the sequence $(a_n)_{n \geq 0}$ as a function of n , if $a_0 = 1$, $a_1 = 3$ and

(a) $a_n = 2a_{n-1} + 3a_{n-2}$, for $n \geq 2$.

(b) $a_n = 2a_{n-1} - 3a_{n-2}$, for $n \geq 2$.

5.9. Determine the general term of the sequence $(a_n)_{n \geq 0}$ as a function of n , if the terms a_0 and a_1 are given, and $a_n + ba_{n-1} + ca_{n-2} = 0$, for $n \geq 2$. Consider the following two cases:

(a) b and c are real numbers such that the equation $t^2 + bt + c = 0$ has two distinct solutions t_1 and t_2 .

(b) $t^2 + bt + c = (t - t_1)^2$ for some real number t_1 .

5.10. Determine the general solution of the equation

$$f_n = c_1 f_{n-1} + c_2 f_{n-2} + \dots + c_k f_{n-k}, \quad n \geq k,$$

where $c_1, c_2, \dots, c_k$ are real constants such that the characteristic equation $t^k + c_1 t^{k-1} + \dots + c_{k-1} t + c_k = 0$ has k distinct solutions.

5.11. Determine the general term of the sequence $(a_n)_{n \geq 0}$ that is given by $a_0 = 3$, $a_1 = -6$, $a_2 = 22$, $a_3 = -22$, and

$$a_{n+4} = 3a_{n+3} + 6a_{n+2} - 28a_{n+1} + 24a_n, \quad n \geq 0.$$

5.12. Determine the general term of the sequence $(a_n)_{n \geq 0}$ as a function of n , if $a_0 = 1$ and

$$a_n = a_{n-1} + 2a_{n-2} + \dots + na_0 \quad \text{if } n \geq 1.$$

5.13. Determine the general term of the sequence $(a_n)_{n \geq 0}$ as a function of n , if $a_0 = 0$, $a_1 = 1$ and

$$a_n = -2na_{n-1} + \sum_{k=0}^n \binom{n}{k} a_k a_{n-k} \quad \text{if } n \geq 2.$$

5.14. Determine the number of sequences $(a_1, a_2, \dots, a_n)$ whose terms are nonnegative integers such that $a_1 + a_2 + \dots + a_n = n$ and

$$a_1 \leq 1, \quad a_1 + a_2 \leq 2, \quad \dots, \quad a_1 + a_2 + \dots + a_{n-1} \leq n-1?$$

5.15. How many ways can a table $2 \times n$ be filled by the numbers $1, 2, \dots, 2n$, such that the numbers in any row, looking from left to right, form an increasing sequence, and the numbers in any column, from up to down, form an increasing sequence?

Chapter 6



Partitions

In this chapter we shall consider the representation of a given positive integer as a sum of positive integers, as well as the representation of a given finite set in the form of the union of pairwise disjoint sets. These representations will be called the *partitions of positive integers* and the *partitions of finite sets*. We shall be interested in counting the number of partitions that satisfy some additional conditions.

6.1 Partitions of Positive Integers

Definition 6.1.1. A *partition of positive integer n* into k parts is a k -tuple $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_k)$, such that $\alpha_1, \alpha_2, \dots, \alpha_k \in \mathbb{N}$, and

$$\alpha_1 + \alpha_2 + \dots + \alpha_k = n, \quad \alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_k. \quad (6.1.1)$$

Positive integers $\alpha_1, \alpha_2, \dots, \alpha_k$ are called the parts of the partition α .

We shall also use notation $\alpha + \alpha_2 + \dots + \alpha_k$ for the partition $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_k)$. If exactly f_i of the numbers $\alpha_1, \alpha_2, \dots, \alpha_k$ are equal to i , then the partition α will be denoted by $\alpha = (1^{f_1} 2^{f_2} 3^{f_3} \dots)$. In this case, the following equalities hold:

$$f_1 + f_2 + \dots + f_n = k, \quad f_1 + 2f_2 + \dots + nf_n = n. \quad (6.1.2)$$

Example 6.1.2. (a) $5 + 3 + 3 + 1$, $(5, 3, 3, 1)$ and $(1^1 3^2 5^1)$ are different notation of the same partition of the number 12 into 4 parts.

(b) $8 + 8 + 5 + 5 + 5 + 5 + 3 + 3 + 3 + 1 + 1 + 1 + 1 + 1 = (1^5 3^3 5^4 8^2)$ is a partition of the number 50 into 14 parts.

(c) There are 7 different partitions of the number 5. Here is the list of these partitions: $\alpha_1 = (5^1)$, $\alpha_2 = (1^1 4^1)$, $\alpha_3 = (2^1 3^1)$, $\alpha_4 = (1^2 3^1)$, $\alpha_5 = (1^1 2^2)$, $\alpha_6 = (1^3 2^1)$, $\alpha_7 = (1^5)$. $\triangle$

Let $p(n)$ be the number of all partitions of positive integer n . By definition $p(n) = 0$ for every negative integer n , and $p(0) = 1$. It is easy to see that $p(1) = 1$, $p(2) = 2$, $p(3) = 3$, $p(4) = 5$, $p(5) = 7$, $p(6) = 11$, $\dots$. There is a formula for $p(n)$ generally, obtained by Hardy, Ramanujan, and Rademacher. For example, it can be shown that

$$\begin{aligned} p(10) &= 42, & p(20) &= 627, & p(50) &= 204226, \\ p(100) &= 190\,569\,292, & p(200) &= 3\,972\,999\,029\,338. \end{aligned}$$

The function $p(n)$, $n \in \mathbb{N}$, increases very rapidly as n grows. For details see the book by Andrews [1].

Theorem 6.1.3. *Let $n_1, \dots, n_k$ be distinct positive integers. Let us denote by $F(n_1, \dots, n_k; n)$ the number of partitions of a positive integer n into distinct parts, such that every part of any of these partitions belongs to the set $\{n_1, \dots, n_k\}$. Then, the following equality holds*

$$F(n_1, \dots, n_k; n) = F(n_1, \dots, n_{k-1}; n - n_k) + F(n_1, \dots, n_{k-1}; n), \quad (6.1.3)$$

where

$$F(n_1, \dots, n_j; m) = \begin{cases} 0, & \text{if } m < 0, \\ 1, & \text{if } m = 0. \end{cases} \quad (6.1.4)$$

Proof. Let S be the set of all partitions of positive integer n into distinct parts, such that every part of all these partitions belongs to the set $\{n_1, \dots, n_k\}$. Let $S_1 \subset S$ be the set of those partitions from S for which there exists a part equal to n_k , and let $S_2 \subset S$ be the set of those partitions from S for which such a part does not exist. It is obvious that $|S| = F(n_1, \dots, n_k; n)$. Any partition from S_1 is uniquely determined by a partition of the number $n - n_k$ into distinct parts that all belong to the set $\{n_1, \dots, n_{k-1}\}$, and vice versa. Hence,

$$|S_1| = F(n_1, \dots, n_{k-1}; n - n_k).$$

Note also that S_2 is the set of all partitions of n into distinct parts that all belong to the set $\{n_1, \dots, n_{k-1}\}$, and

$$|S_2| = F(n_1, \dots, n_{k-1}; n).$$

Since $S = S_1 \cup S_2$ and $S_1 \cap S_2 = \emptyset$, it follows that $|S| = |S_1| + |S_2|$, i.e., the equality (6.1.3) holds true. $\square$

Corollary 6.1.4. Let $n_1 = 1, n_2 = 2, \dots, n_k = k, F_k(n) = F(1, \dots, k; n)$, and $F_k(0) = 1, F_k(m) = 0$ for $m < 0$. Then,

$$F_k(n) = F_{k-1}(n - k) + F_{k-1}(n). \quad (6.1.5)$$

It is easy to see that $F_1(1) = 1, F_1(n) = 0$ for $n > 1$ and $F_k(n) = F_n(n)$ for $k > n$. Using these equalities and equality (6.1.5) we can calculate the values $F_k(n)$ for any $n, k \in \mathbb{N}$.

Example 6.1.5. Let us calculate some values of the function $F_k(n)$.

(a) $F_2(1) = 1, F_2(2) = 1, F_2(3) = 1, F_2(n) = 0$ for any $n \geq 4$.

(b) $F_3(1) = 1, F_3(2) = 1, F_3(3) = 2, F_3(4) = 1, F_3(5) = 1, F_3(6) = 1$, and $F_3(n) = 0$ for any $n \geq 7$.

(c) $F_4(1) = 1, F_4(2) = 1, F_4(3) = 2, F_4(4) = 2, F_4(5) = 2, F_4(6) = 2, F_4(7) = 2, F_4(8) = 1, F_4(9) = 1, F_4(10) = 1$, and $F_4(n) = 0$ for any $n \geq 11$.

(d) $F_5(8) = F_4(3) + F_4(8) = 2 + 1 = 3$, etc. $\triangle$

Theorem 6.1.6. Let $n_1, \dots, n_k$ be distinct positive integers. Let us denote by $G(n_1, \dots, n_k; n)$ the number of partitions of positive integer n , such that every part of every partition belongs to the set $\{n_1, \dots, n_k\}$. Then, the following equality holds

$$G(n_1, \dots, n_k; n) = G(n_1, \dots, n_k; n - n_k) + G(n_1, \dots, n_{k-1}; n), \quad (6.1.6)$$

where

$$G(n_1, \dots, n_j; m) = \begin{cases} 0, & \text{if } m < 0, \\ 1, & \text{if } m = 0. \end{cases} \quad (6.1.7)$$

Proof. Let S be the set of all partitions of the positive integer n , such that every part of every partition belongs to the set $\{n_1, \dots, n_k\}$. Let $S_1 \subset S$ be the set of those partitions from S for which there exists a part equal to n_k , and let $S_2 \subset S$ be the set of those partitions from S for which such a part does not exist. It is obvious that

$$\begin{aligned} |S| &= G(n_1, \dots, n_k; n), \\ |S_1| &= G(n_1, \dots, n_k; n - n_k), \\ |S_2| &= G(n_1, \dots, n_{k-1}; n), \end{aligned}$$

and the equality (6.1.6) follows from $|S| = |S_1| + |S_2|$. $\triangle$

Corollary 6.1.7. Let $n_1 = 1, n_2 = 2, \dots, n_k = k, G_k(n) = G(1, \dots, k; n)$, and $G_k(0) = 1, G_k(m) = 0$ for $m < 0$. Then, the following equalities hold:

$$G_k(n) = G_{k-1}(n) + G_k(n - k), \quad (6.1.8)$$

$$G_k(n) = G_{k-1}(n) + G_{k-1}(n - k) + G_{k-1}(n - 2k) + \dots \quad (6.1.9)$$

Using Theorem 6.1.6 it is easy to determine the number of partitions of a positive integer into parts that belong to a given set of natural numbers.

Example 6.1.8. Let us determine how many ways the amount of \$50 can be paid by bank notes of \$1, \$2, \$5, \$10, and \$20.

The number to be determined is $G(1, 2, 5, 10, 20; 50)$. It is easy to see that $G(1, n) = 1$, for any integer $n \geq 0$. Using equality (6.1.6) we obtain the following equalities:

$$\begin{aligned}
 G(1, 2; 2k) &= 1 + G(1, 2; 2k - 2) = k + G(1, 2; 0) = k + 1, \quad k \in \mathbb{N}, \\
 G(1, 2; 2k + 1) &= 1 + G(1, 2; 2k - 1) = k + G(1, 2; 1) = k + 1, \quad k \in \mathbb{N}, \\
 G(1, 2, 5; 5) &= G(1, 2, 5; 0) + G(1, 2; 5) = 1 + 3 = 4, \\
 G(1, 2, 5; 10) &= G(1, 2, 5; 5) + G(1, 2; 10) = 4 + 6 = 10, \\
 G(1, 2, 5; 15) &= G(1, 2, 5; 10) + G(1, 2; 15) = 10 + 8 = 18, \\
 G(1, 2, 5; 20) &= G(1, 2, 5; 15) + G(1, 2; 20) = 18 + 11 = 29, \\
 G(1, 2, 5; 25) &= G(1, 2, 5; 20) + G(1, 2; 25) = 29 + 13 = 42, \\
 G(1, 2, 5; 30) &= G(1, 2, 5; 25) + G(1, 2; 30) = 42 + 16 = 58, \\
 G(1, 2, 5; 35) &= G(1, 2, 5; 30) + G(1, 2; 35) = 58 + 18 = 76, \\
 G(1, 2, 5; 40) &= G(1, 2, 5; 35) + G(1, 2; 40) = 76 + 21 = 97, \\
 G(1, 2, 5; 45) &= G(1, 2, 5; 40) + G(1, 2; 45) = 97 + 23 = 120, \\
 G(1, 2, 5; 50) &= G(1, 2, 5; 45) + G(1, 2; 50) = 120 + 26 = 146, \\
 G(1, 2, 5, 10; 10) &= G(1, 2, 5, 10; 0) + G(1, 2, 5; 10) = 1 + 10 = 11, \\
 G(1, 2, 5, 10; 20) &= G(1, 2, 5, 10; 10) + G(1, 2, 5; 20) = 11 + 29 = 40, \\
 G(1, 2, 5, 10; 30) &= G(1, 2, 5, 10; 20) + G(1, 2, 5; 30) = 40 + 58 = 98, \\
 G(1, 2, 5, 10; 40) &= G(1, 2, 5, 10; 30) + G(1, 2, 5; 40) = 98 + 97 = 195, \\
 G(1, 2, 5, 10; 50) &= G(1, 2, 5, 10; 40) + G(1, 2, 5; 50) = 195 + 146 = 341, \\
 G(1, 2, 5, 10, 20; 30) &= G(1, 2, 5, 10; 10) + G(1, 2, 5, 10; 30) = 109, \\
 G(1, 2, 5, 10, 20; 50) &= G(1, 2, 5, 10, 20; 30) + G(1, 2, 5, 10; 50) = 450. \quad \triangle
 \end{aligned}$$

6.2 Ordered Partitions of Positive Integers

Definition 6.2.1. An *ordered partition* of positive integer n into k parts is a solution of the equation $x_1 + x_2 + \cdots + x_k = n$ in the set of positive integers, i.e., a k -tuple $(\alpha_1, \alpha_2, \dots, \alpha_k)$ of positive integers such that their sum is equal to n .

For the ordered partition $(\alpha_1, \alpha_2, \dots, \alpha_k)$ we shall also use the notation $\alpha_1 + \alpha_2 + \cdots + \alpha_k$.

Example 6.2.2. (a) All partitions of the number 3 are: 3, 2 + 1, 1 + 1 + 1. All ordered partitions of the number 3 are: 3, 2 + 1, 1 + 2, 1 + 1 + 1.

(b) All ordered partitions of the number 7 into three parts are: 5 + 1 + 1, 1 + 5 + 1, 1 + 1 + 5, 4 + 2 + 1, 4 + 1 + 2, 2 + 4 + 1, 2 + 1 + 4, 1 + 4 + 2, 1 + 2 + 4, 3 + 3 + 1, 3 + 1 + 3, 1 + 3 + 3, 3 + 2 + 2, 2 + 3 + 2, 2 + 2 + 3. $\triangle$

Theorem 6.2.3. Let $\alpha = (1^{f_1} 2^{f_2} \dots n^{f_n})$ be a partition of positive integer n . Then, there are

$$\frac{(f_1 + f_2 + \dots + f_n)!}{f_1! f_2! \dots f_n!}$$

ordered partitions of n , such that, for any $i \in \{1, 2, \dots, n\}$, there are exactly f_i parts that are equal to i .

Proof. Every ordered partition of positive integer n with f_i parts that are equal to i for any $i \in \{1, 2, \dots, n\}$, is obviously an $(f_1 + f_2 + \dots + f_n)$ -arrangement of the elements 1, 2, $\dots$, n which has the type $(f_1, f_2, \dots, f_n)$. Theorem 6.2.3 now follows from Theorem 2.5.4. $\square$

Theorem 6.2.4. (a) The number of ordered partitions of positive integer n into k parts is equal to $\binom{n-1}{k-1}$.

(b) The number of all partitions of positive integer n is 2^{n-1} .

Proof. (a) Let A be the set of all $(n + k - 1)$ -arrangements of elements 0 and 1, with the following properties:

- exactly n terms of any arrangement from A are equal to 1;
- no two 0's are adjacent;
- the first and the last terms are equal to 1.

The set A contains all $(n + k - 1)$ -arrangements that consist of k series of 1's separated by 0's. There is an obvious bijection between the set A and the set of all ordered partitions of the number n into k parts. Using Example 2.7.8 (c) we obtain that $|A| = \binom{n-1}{k-1}$.

(b) Using the result of part (a) and Theorem 2.9.1 we get that the number of all partitions of n is equal to

$$\binom{n-1}{0} + \binom{n-1}{1} + \dots + \binom{n-1}{n-1} = 2^{n-1}. \quad \square$$

Theorem 6.2.5. Let $n_1, \dots, n_k$ be distinct positive integers. Let us denote by $H(n_1, \dots, n_k; n)$ the number of ordered partitions of positive integer n into parts that belong to the set $\{n_1, \dots, n_k\}$. Then, the following equality holds

$$H(n_1, \dots, n_k; n) = \sum_{j=1}^k H(n_1, \dots, n_k; n - n_j), \quad (6.2.1)$$

where

$$H(n_1, \dots, n_j; m) = \begin{cases} 0, & \text{if } m < 0, \\ 1, & \text{if } m = 0. \end{cases} \quad (6.2.2)$$

Proof. Let S be the set of all ordered partitions of the positive integer n , such that every part of every partition belongs to the set $\{n_1, \dots, n_k\}$. Let S_j be the set of partitions $(\alpha_1, \alpha_2, \dots) \in S$, such that $\alpha_1 = n_j$. Then, set S is the union of the pairwise disjoint sets $S_1, S_2, \dots, S_k$, and consequently we obtain that

$$H(n_1, n_2, \dots, n_k; n) = |S| = \sum_{j=1}^k |S_j| = \sum_{j=1}^k H(n_1, n_2, \dots, n_k; n - n_j). \quad \square$$

Example 6.2.6. The following type of problem appears in Information Theory. Suppose that a message is sent by signals with a length of 1, 2, 3, or 4 units of time. How many different messages can be sent during 10 units of time?

In this example the problem is to determine $H(1, 2, 3, 4; 10)$. Let us denote $H(n) = H(1, 2, 3, 4; n)$ for any $n \in \mathbb{N}$. It is easy to see that $H(1) = 1$, $H(2) = 2$, $H(3) = 4$, and $H(4) = 8$. By Theorem 6.2.5 we get

$$H(n) = H(n-1) + H(n-2) + H(n-3) + H(n-4).$$

By the previous equality it follows that $H(5) = 15$, $H(6) = 29$, $H(7) = 56$, $H(8) = 108$, $H(9) = 208$, and $H(10) = 401$. $\triangle$

6.3 Graphical Representation of Partitions

Definition 6.3.1. *Ferrer's graph* of the partition $\alpha = (\alpha_1, \dots, \alpha_k)$ is the set G_α of points with integer coordinates in the Cartesian plane given by

$$G_\alpha = \{(x, y) \mid -k+1 \leq y \leq 0, 0 \leq x \leq \alpha_{-y+1} - 1\}.$$

Example 6.3.2. Ferrer's graph of the partition $30 = 8+5+5+4+3+2+2+1$ is given in Figure 6.3.1. $\triangle$

Note that the i -th row of Ferrer's graph of the partition $(\alpha_1, \alpha_2, \dots, \alpha_k)$ contains exactly α_i points.

Definition 6.3.3. Let $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_k)$ be a partition of a positive integer and let $m = \alpha_1$. The partition $\beta = (\beta_1, \beta_2, \dots, \beta_m)$ is the *conjugate* of α , if for any $j \in \{1, 2, \dots, m\}$, part β_j is equal to the number of parts of partition α that are not less than j .

Example 6.3.4. Consider the partition $8 + 5 + 5 + 4 + 3 + 2 + 2 + 1$ from Example 6.3.2. The conjugate of this partition is $8 + 7 + 5 + 4 + 3 + 1 + 1 + 1$, see Figure 6.3.2 $\triangle$



Fig. 6.3.1



Fig. 6.3.2

It is obvious that Ferrer's graph of partition β , which is the conjugate of α , can be obtained by a reflection of partition α around the line $y = -x$. This line is called the main diagonal of Ferrer's graph. Consequently, if β is the conjugate of α , then α is the conjugate of β .

Theorem 6.3.5. *The number of partitions of positive integer n into no more than k parts is equal to the number of partitions of n into parts that are not greater than k .*

Proof. Let S_1 be the set of partitions of n into no more than k parts, and S_2 be the set of partitions of n into parts that are not greater than k . It is obvious that if $\alpha \in S_1$, then the conjugate β of partition α is an element of S_2 . The function $f : S_1 \rightarrow S_2$ given by $f(\alpha) = \beta$, where β is the conjugate of α , is a bijection. Hence, $|S_1| = |S_2|$. $\square$

Theorem 6.3.6. *Let $p_0(n)$ be the number of partitions of positive integer n into an even number of distinct parts, and $p_1(n)$ be the number of partitions of positive integer n into an odd number of distinct parts. Then, the following equality holds*

$$p_0(n) - p_1(n) = \begin{cases} (-1)^k, & \text{if } n = \frac{1}{2}k(3k \pm 1), \quad k \in \mathbb{N}, \\ 0, & \text{in all other cases.} \end{cases} \quad (6.3.1)$$

Proof. Let $\alpha = (\alpha_1, \dots, \alpha_k)$ be a partition of positive integer n into distinct parts. Let us denote

$$f_1(\alpha) = \alpha_k, \quad f_2(\alpha) = \max\{j \mid \alpha_j = \alpha_1 - j + 1\}.$$

It is obvious that $f_1(\alpha)$ is the smallest part of partition α , i.e., the number of points in the last row of Ferrer's graph of partition α ; $f_2(\alpha)$ is the number of points of Ferrer's graph of partition α that belong to the line $y = x + 1 - \alpha_1$. For example, if $\alpha = (7, 6, 5, 3, 2)$, then $f_1(\alpha) = 2$, $f_2(\alpha) = 3$, see Figure 6.3.3.

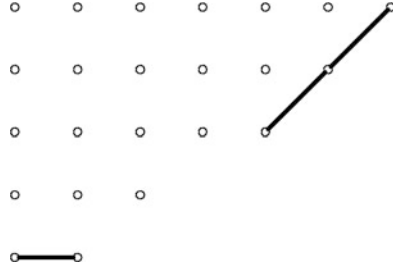


Fig. 6.3.3

Let S be the set of all partitions of n into distinct parts. Let us define transformations T_1 and T_2 that will map some partitions from S into partitions that also belong to S .

(a) Let us consider a partition $\alpha = (\alpha_1, \dots, \alpha_k) \in S$, where $\alpha_1 > \dots > \alpha_k$, and suppose that $f_1(\alpha) \leq f_2(\alpha)$. If $f_1(\alpha) = f_2(\alpha) = k$ does not hold, we define $T_1(\alpha)$ as follows: we remove the smallest part $f_1(\alpha) = \alpha_k$ of partition α , and increase by 1 the $f_1(\alpha)$ greatest parts of α . For example, if $\alpha = (7, 6, 5, 3, 2)$, then $T_1(\alpha) = (8, 7, 5, 3)$, see Figure 6.3.4.

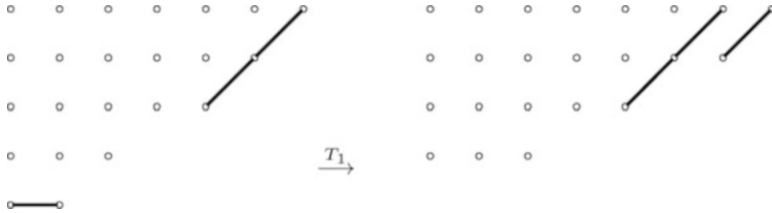


Fig. 6.3.4

If $f_1(\alpha) = f_2(\alpha) = k$ (k is the number of parts of the partition), then the transformation $T_1(\alpha)$ is not defined. Indeed, if we remove the smallest part $\alpha_k = f_1(\alpha) = k$ in this case, then there are only $k - 1$ remaining parts. Hence, in this case we have $\alpha = (2k - 1, 2k - 2, \dots, k + 1, k)$, and the positive integer n is equal to

$$k + (k + 1) + \dots + (2k - 1) = \frac{1}{2}k(3k - 1).$$

(b) Suppose that $f_1(\alpha) > f_2(\alpha)$ for partition $\alpha = (\alpha_1, \dots, \alpha_k)$, where $\alpha_1 > \dots > \alpha_k$. If both equalities $f_2(\alpha) = k$ and $f_1(\alpha) = k + 1$ are not

fulfilled simultaneously, then we define $T_2(\alpha)$ as follows: we decrease by 1 the $f_2(\alpha)$ greatest parts of partition α , and add a new part that is equal to $f_2(\alpha) = \alpha_{k+1}$. For example, if $\alpha = (8, 7, 5, 3)$, then $T_2(\alpha) = (7, 6, 5, 3, 2)$, see Figure 6.3.5.

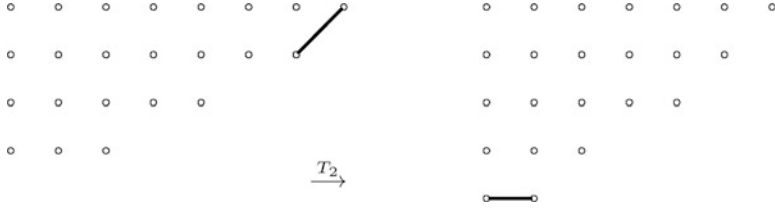


Fig. 6.3.5

If $f_2(\alpha) = k$ and $f_1(\alpha) = k+1$, then the transformation $T_2(\alpha)$ is defined, but the resulting partition has two equal parts. For example, $\alpha = (8, 7, 6, 5)$, $f_1(\alpha) = 5$, $f_2(\alpha) = 4$, $T_2(\alpha) = (7, 6, 5, 4, 4)$, see Figure 6.3.6.

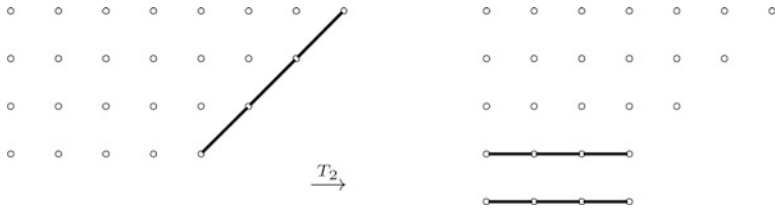


Fig. 6.3.6

Note that in the case $f_2(\alpha) = k$, and $f_1(\alpha) = k+1$, the partition is given by $\alpha = (2k, 2k-1, \dots, k+2, k+1)$, and positive integer n is equal to

$$(k+1) + (k+2) + \dots + 2k = \frac{1}{2}k(3k+1).$$

Note also that after the application of transformation T_1 the number of parts of partition α decreases by 1, and after the application of transformation T_2 , this number increases by 1. Let us consider the following two cases:

Case 1. A positive integer n is not of the form $n = \frac{1}{2}k(3k \pm 1)$, where $k \in \mathbb{N}$. Then, for any partition α of positive integer n into distinct parts, exactly one of the transformations T_1 and T_2 can be applied. This is true because exactly one of the inequalities $f_1(\alpha) \leq f_2(\alpha)$ and $f_1(\alpha) > f_2(\alpha)$ holds. Let S_1 be the set of all partitions of n into an odd number of distinct parts, and let S_2 be the set of all partitions of n into an even number of distinct parts. Let us define the function $T : S_1 \rightarrow S_2$ by

$$T(\alpha) = \begin{cases} T_1(\alpha), & \text{if } f_1(\alpha) \leq f_2(\alpha), \\ T_2(\alpha), & \text{if } f_1(\alpha) > f_2(\alpha). \end{cases} \quad (6.3.2)$$

It is easy to see that T is a bijection, and consequently we obtain that $|S_1| = |S_2|$, i.e., $p_0(n) = p_1(n)$.

Case 2. Let $n = \frac{1}{2}k(3k \pm 1)$, where $k \in \mathbb{N}$ and

$$\alpha_0 = \begin{cases} (2k, 2k-1, \dots, k+2, k+1), & \text{if } n = \frac{1}{2}k(3k+1), \\ (2k-1, 2k-2, \dots, k+1, k), & \text{if } n = \frac{1}{2}k(3k-1). \end{cases} \quad (6.3.3)$$

Let us denote by S_1 (S_2) the set of partitions of positive integer n into an odd (even) number of distinct parts that are not equal to α_0 . Analogously as in Case 1, it follows that $|S_1| = |S_2|$, and consequently we get $p_0(n) = p_1(n) + (-1)^k$. $\square$

Remark 6.3.7. Theorem 6.3.6 is known as the Euler-Legendre theorem. The numbers $\omega_k = \frac{1}{2}k(3k \pm 1)$, $k \in \{\dots, -1, 0, 1, \dots\}$, are called *pentagonal numbers*. If $p(n)$ is the number of all partitions of positive integer n , then the following formula follows from Theorem 6.3.6:

$$p(n) = \sum_{\omega_k \leq n} (-1)^{k-1} \left[p\left(n - \frac{k(3k-1)}{2}\right) + p\left(n - \frac{k(3k+1)}{2}\right) \right], \quad (6.3.4)$$

where $p(0) = 1$, and the sum in (6.3.4) runs over all pentagonal numbers that are not greater than n .

6.4 Partitions of Sets

Definition 6.4.1. A *partition* of a set S into k *blocks* is a collection of sets $\{S_1, S_2, \dots, S_k\}$ such that:

- (a) $S_j \neq \emptyset$, for any $j \in \{1, 2, \dots, k\}$;
- (b) $S_i \cap S_j = \emptyset$, for any $i \neq j$, $i, j \in \{1, 2, \dots, k\}$;
- (c) $S = S_1 \cup S_2 \cup \dots \cup S_k$.

Every partition $\pi = \{S_1, S_2, \dots, S_k\}$ of set S into k blocks generates $k!$ ordered partitions of set S into k blocks. Any ordered partition is a permutation of the blocks of partition π . Every partition $\pi = \{S_1, S_2, \dots, S_k\}$ of an n -set S into k blocks also generates a partition of the positive integer n into k parts that are equal to $|S_1|, |S_2|, \dots, |S_k|$.

Let $\alpha_1, \alpha_2, \dots, \alpha_k$ be positive integers such that $\alpha_1 + \alpha_2 + \dots + \alpha_k = n$, and let us introduce the following notation:

- $B_n(\alpha_1, \alpha_2, \dots, \alpha_k)$ — the number of ordered partitions $(S_1, S_2, \dots, S_k)$ of an n -set S into k blocks, such that for any $j \in \{1, 2, \dots, k\}$, $|S_j| = \alpha_j$;
- B_{nk} — the number of partitions of an n -set into k blocks;
- B_n — the number of all partitions of an n -set.

The numbers B_{nk} and B_n are defined for all $n, k \in \mathbb{N}$. It is obvious that $B_{nk} = 0$ for $k > n$. By definition $B_{00} = B_0 = 1$. The numbers B_{nk} , $n, k \in \mathbb{N}_0$ are called *Stirling numbers of the second kind*. The number B_n is called the *n th Bell number*.

Example 6.4.2. Let us determine the number B_n and all partitions of an n -set, for any $n \in \{1, 2, 3, 4\}$.

(a) $S = \{1\}$, $B_1 = 1$: $\pi_1 = \{\{1\}\}$.

(b) $S = \{1, 2\}$, $B_2 = 2$: $\pi_1 = \{\{1, 2\}\}$, $\pi_2 = \{\{1\}, \{2\}\}$.

(c) $S = \{1, 2, 3\}$, $B_3 = 5$:

$$\begin{aligned}\pi_1 &= \{\{1, 2, 3\}\}, & \pi_2 &= \{\{1, 2\}, \{3\}\}, & \pi_3 &= \{\{1, 3\}, \{2\}\}, \\ \pi_4 &= \{\{2, 3\}, \{1\}\}, & \pi_5 &= \{\{1\}, \{2\}, \{3\}\}.\end{aligned}$$

(d) $S = \{1, 2, 3, 4\}$, $B_4 = 15$:

$$\begin{aligned}\pi_1 &= \{\{1, 2, 3, 4\}\}, & \pi_2 &= \{\{1, 2, 3\}, \{4\}\}, & \pi_3 &= \{\{1, 2, 4\}, \{3\}\}, \\ \pi_4 &= \{\{1, 3, 4\}, \{2\}\}, & \pi_5 &= \{\{2, 3, 4\}, \{1\}\}, & \pi_6 &= \{\{1, 2\}, \{3, 4\}\}, \\ \pi_7 &= \{\{1, 3\}, \{2, 4\}\}, & \pi_8 &= \{\{1, 4\}, \{2, 3\}\}, & \pi_9 &= \{\{1, 2\}, \{3\}, \{4\}\}, \\ \pi_{10} &= \{\{1, 3\}, \{2\}, \{4\}\}, & \pi_{11} &= \{\{1, 4\}, \{2\}, \{3\}\}, \\ \pi_{12} &= \{\{2, 3\}, \{1\}, \{4\}\}, & \pi_{13} &= \{\{2, 4\}, \{1\}, \{3\}\}, \\ \pi_{14} &= \{\{3, 4\}, \{1\}, \{2\}\}, & \pi_{15} &= \{\{1\}, \{2\}, \{3\}, \{4\}\}.\end{aligned} \quad \triangle$$

Theorem 6.4.3. Let $n, k, \alpha_1, \alpha_2, \dots, \alpha_k$ be positive integers such that $\alpha_1 + \alpha_2 + \dots + \alpha_k = n$. Then, the following equality holds:

$$B_n(\alpha_1, \alpha_2, \dots, \alpha_k) = \frac{n!}{\alpha_1! \alpha_2! \dots \alpha_k!}. \quad (6.4.1)$$

Proof. There are $\binom{n}{\alpha_1}$ ways of choosing elements from the set S that will form the block S_1 ; there are $\binom{n - \alpha_1}{\alpha_2}$ ways of choosing elements for the second block of the partition, etc. Hence

$$B_n(\alpha_1, \alpha_2, \dots, \alpha_k) = \binom{n}{\alpha_1} \binom{n - \alpha_1}{\alpha_2} \dots \binom{n - \alpha_1 - \dots - \alpha_{k-1}}{\alpha_k},$$

and equality (6.4.1) follows immediately. $\square$

The second proof of Theorem 6.4.3. Let $\mathcal{X}$ be the set of all ordered partitions $(S_1, S_2, \dots, S_k)$ of the set $S = \{x_1, x_2, \dots, x_n\}$, such that $|S_1| = \alpha_1, \dots, |S_k| = \alpha_k$, and let $\mathcal{Y}$ be the set of all n -arrangements of the elements of set $\{1, 2, \dots, k\}$ that have the type $(\alpha_1, \dots, \alpha_k)$. Let us define the function $f : \mathcal{X} \rightarrow \mathcal{Y}$ as follows

$$f((S_1, S_2, \dots, S_k)) = c_1 c_2 \dots c_k \in \mathcal{Y},$$

where c_j is the index of those of the sets $S_1, S_2, \dots, S_k$ that contains the element $x_j \in S$. It is obvious that function f is a bijection between the sets $\mathcal{X}$ and $\mathcal{Y}$, and hence

$$B_n(\alpha_1, \alpha_2, \dots, \alpha_k) = |\mathcal{X}| = |\mathcal{Y}| = \frac{n!}{\alpha_1! \alpha_2! \dots \alpha_k!}. \quad \square$$

Theorem 6.4.4. *Let $n, k \in \mathbb{N}$ and $n \geq k$. Then, the following equalities hold:*

$$B_{nk} = \frac{1}{k!} \sum_{\alpha_1 + \alpha_2 + \dots + \alpha_k = n} \frac{n!}{\alpha_1! \alpha_2! \dots \alpha_k!}, \quad (6.4.2)$$

$$B_{nk} = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n, \quad (6.4.3)$$

$$B_n = \sum_{k=1}^n B_{nk} = \sum_{k=1}^n \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n. \quad (6.4.4)$$

Proof. Equality (6.4.2) obviously holds.

The number of ordered partitions of an n -set into k blocks is equal to $k! B_{nk}$, because any (unordered) partition into k -blocks generates $k!$ ordered partitions with the same blocks. Every ordered partition $(S_1, S_2, \dots, S_k)$ of the set $S = \{x_1, x_2, \dots, x_n\}$ is uniquely determined by the function $f : S \rightarrow \{1, 2, \dots, k\}$, such that $f(x) = j$, for any $x \in S_j$, and any $j \in \{1, 2, \dots, k\}$, and vice versa. Hence, the number of ordered partitions of the n -set S into k blocks is equal to the number of all surjections from S to $\{1, 2, \dots, k\}$. Using the result given in Example 4.3.3 we obtain that

$$k! B_{nk} = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n,$$

and equality (6.4.3) follows. Then, equality (6.4.4) follows easily from (6.4.3). $\square$

Theorem 6.4.5. *For Bell numbers the following recursive relation holds:*

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k, \quad n \geq 0. \quad (6.4.5)$$

Proof. Let $\mathcal{R}$ be the set of all partitions of the set $S = \{1, 2, \dots, n+1\}$. Then $\mathcal{R} = \mathcal{R}_0 \cup \mathcal{R}_1 \cup \dots \cup \mathcal{R}_n$, where $\mathcal{R}_k$ is the set of partitions of S , such that the element $n+1$ belongs to a block that contains exactly $k+1$ elements, $0 \leq k \leq n$. A block, say S_0 , that contains exactly $k+1$ elements of set S , and element $n+1$ among them, can be chosen in $\binom{n}{k}$ ways. The set $S \setminus S_0$ contains exactly $n-k$ elements, and hence, there are B_{n-k} partitions of this set. Consequently, $|\mathcal{R}_k| = \binom{n}{k} B_{n-k}$, and finally we obtain

$$B_{n+1} = |\mathcal{R}| = \sum_{k=0}^n |\mathcal{R}_k| = \sum_{k=0}^n \binom{n}{k} B_{n-k} = \sum_{k=0}^n \binom{n}{k} B_k. \quad \square$$

Exercises

6.1. Let $n, k \in \mathbb{N}$. How many solutions of the equation $x_1 + x_2 + \dots + x_k = n$ are there: (a) in the set $\mathbb{N}$? (b) in the set $\mathbb{N}_0$?

6.2. Let $n, k \in \mathbb{N}$. How many solutions of the inequality $x_1 + \dots + x_k \leq n$ are there in the set $\mathbb{N}_0$?

6.3. Let $n, k \in \mathbb{N}$ and $c_1, c_2, \dots, c_k$ be integers. How many solutions of the equation $x_1 + x_2 + \dots + x_k = n$ are there, such that $x_1 \geq c_1, x_2 \geq c_2, \dots, x_k \geq c_k$?

6.4. Let $n, k \in \mathbb{N}$, and r and s be integers such that $0 \leq r \leq s$. How many solutions $(x_1, x_2, \dots, x_k)$ of the equation $x_1 + x_2 + \dots + x_k = n$ are there, such that for any $j \in \{1, 2, \dots, k\}$, $r \leq x_j \leq s$ and $x_j \in \mathbb{N}_0$?

6.5. Let $n, k \in \mathbb{N}$ and $k \geq 2$. How many solutions $(x_1, x_2, \dots, x_k)$ of the equation $x_1 + x_2 + \dots + x_k = 2n$ are there in the set $\mathbb{N}_0$, such that $x_1 > x_k$?

6.6. How many n -digit positive integers are there, such that the sum of the digits of any of these numbers is equal to 11?

6.7. How many triplets (x_1, x_2, x_3) of nonnegative integers are there, such that $x_1 + x_2 + x_3 = 100$ and $x_1 \leq x_2 \leq x_3$?

6.8. How many triangles with the circumference 300 are there if the length of every side is a positive integer?

6.9. How many ways can $12n + 5$ indistinguishable balls be put into 4 boxes labeled 1, 2, 3, and 4, such that every box contains at least one ball, and the number of balls in every box is not greater than $6n + 2$?

6.10. Determine the number of partitions of the positive integer 40 into parts that are equal to 5, 10, 15, or 20.

6.11. How many ways can the amount of \$27 be paid by bank notes of:

- (a) \$1 and \$2; (b) \$2 and \$5; (a) \$1, \$2, and \$5?

6.12. How many ways can the amount of \$100 be paid by bank notes of \$1, \$2, \$5, and \$10?

6.13. Prove that the number of all partitions of a positive integer n is equal to the number of partitions of $2n$ into n parts.

6.14. Prove that the number of partitions of a positive integer n into distinct parts is equal to the number of partitions of n into odd parts.

6.15. Let n be a positive integer, and $p(n)$ be the number of partitions of n . For a partition π of n let $q_\pi(n)$ be the number of distinct parts of π . Let $q(n)$ be the sum of numbers $q_\pi(n)$ over all partitions π of n . Prove that:

(a) $q(n) = 1 + p(1) + p(2) + \cdots + p(n - 1)$;

(b) $q(n) < \sqrt{2n} p(n)$.

6.16. Let $\{A_1, A_2, \dots, A_n\}$ and $\{B_1, B_2, \dots, B_n\}$ be two partitions of the set X , such that the following condition is satisfied: if $A_i \cap A_j = \emptyset$ for some $i, j \in \{1, 2, \dots, n\}$, then $|A_i \cup A_j| \geq n$.

(a) Prove that $|X| \geq \frac{1}{2}n^2$.

(b) Can the equality $|X| = \frac{1}{2}n^2$ be fulfilled?

6.17. Let n be a positive integer greater than 3, $k_n = [n(n+1)/6]$, and let X_n be a set that consists of k_n blue elements, k_n red elements, and $\frac{1}{2}n(n+1) - 2k_n$ green elements. Prove that there exists a partition $\{A_1, A_2, \dots, A_n\}$ of the set X_n , such that:

(a) $|A_j| = j$ for any $j \in \{1, 2, \dots, n\}$,

(b) Each of the sets $A_1, A_2, \dots, A_n$ consists of elements of the same color.

6.18. Let m , n , and k be positive integers, such that $m + k \leq n$. Consider two partitions $(A_1, A_2, \dots, A_m)$ and $(B_1, B_2, \dots, B_{m+k})$ of the set

$S = \{1, 2, \dots, n\}$. Let D be a subset of S that consists of elements, denoted by x , with the following property: if $x \in A_i$ and $x \in B_j$, then $|A_i| > |B_j|$. Prove that $|D| > k + 1$.

6.19. Exactly 50 participants of an international conference speak English, exactly 50 participants speak French, and exactly 50 participants speak Spanish. Some of these participants may speak more than one of the three languages. Prove that there is a partition of participants into 5 groups, such that there are exactly 10 English speaking participants, exactly 10 French speaking participants, and exactly 10 Spanish speaking participants in any of these groups.

6.20. Every member of a parliament is quarreling with no more than three other members of the parliament. Prove that the parliament can be divided into two houses such that the following condition is satisfied: every member of the parliament is quarreling with at most one other member in their house.

6.21. Suppose that n lines are given in the plane, so that no two of them are parallel, and no three of them are concurrent. Into how many parts is the plane divided by these n lines?

6.22. Suppose that n planes are given such that no four of them contain the same point, no three of them contain the same line, no two of them are parallel, and every three of them have a common point. Into how many parts is the space divided by these n planes?

6.23. Suppose that n circles are given in the plane, such that any two of them have two common points, and no three of them have a common point. Into how many parts is the plane divided by these n circles?

6.24. Suppose that n spheres are given in space so that no four of them have a common point, no three of them contain a common circle, every three of them have a common point, and every two of them contain a common circle. Into how many parts is the space divided by these n spheres?

6.25. Suppose that a sphere and n planes are given in space, such that each of these planes contains the center of the sphere, and no two of the planes contain the same diameter of the sphere. Into how many parts is the sphere divided by these n planes?

6.26. No three diagonals of a convex n -gon intersect at the same point. Into how many segments are all the diagonals divided by their points of intersection?

6.27. No three diagonals of a convex n -gon intersect at the same point. Into how many parts is the n -gon divided by its diagonals?

6.28. A convex n -gon can be divided into triangles by its diagonals without points of intersection inside the n -gon. There are several such triangulations.

(a) Prove that the number of triangles does not depend on the triangulation. Determine this number.

(b) Prove that the number of diagonals that determine a triangulation does not depend on the triangulation. Determine this number.

Chapter 7



Burnside's Lemma

7.1 Introduction

The simple problem of coloring fields of a square 2×2 using three colors is considered in Example 2.7.11, and the number of nonequivalent colorings is determined by direct counting. At the beginning of this section we shall formulate two more similar problems.

Example 7.1.1. Coloring the vertices of a cube using blue, yellow, and red colors can be done in $3^8 = 6561$ ways. Obviously, we are considering here the case when not all three colors are necessarily used. Two colorings are equivalent (or geometrically equal) if there is a rotation $\mathcal{R}$ of the cube such that the image $\mathcal{R}(v)$ of any vertex v is of the same color as the vertex itself. Let us consider some classes of colorings.

Class $\mathcal{A}$ consists of colorings such that 4 vertices that belong to the same side of the cube are red, and the remaining 4 vertices are blue. All colorings from class $\mathcal{A}$ are obviously equivalent. Class $\mathcal{B}$ consists of colorings such that three vertices are blue, and the remaining 5 vertices are red. Class $\mathcal{C}$ consists of colorings such that each of the three colors is assigned to at least one vertex.

It is obvious that no two colorings from two different classes of $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ are equivalent. An interesting question here is the following. How many nonequivalent colorings are there? $\triangle$

Example 7.1.2. Suppose that k queens are arranged on a chessboard such that no two of them attack each other. New arrangements obtained by isometric transformation of the chessboard (rotation or symmetry) also satisfy

the condition that no two queens attack each other. We shall say that two arrangements of queens are equivalent if one of them can be obtained from the other one using some isometric transformation of the chessboard. The following questions are of interest here:

- (1) What is the maximal value of k ?
- (2) For such maximal value of k , how many nonequivalent arrangements of k queens on a chessboard are there such that no two of them attack each other? $\triangle$

The answer to the question from Example 7.1.1 will be given at the end of this chapter. For the answers to the questions formulated in Example 7.1.2, see Problem 12.46. Necessary notions related to permutations of finite sets will be introduced in the next two sections. A general method for counting nonequivalent configurations will be given in Section 7.4.

7.2 On Permutations

In this section we shall introduce some important notions related to permutations of the set $\mathbb{N}_n = \{1, 2, \dots, n\}$. We start with the notions of *inversion* and the *parity* (*oddness* and *evenness*) of a permutation. An arbitrary permutation of set $\mathbb{N}_n$ will be denoted by $a_1 a_2 \dots a_n$.

Inversion. Consider elements $i, j \in \mathbb{N}_n$. We say that the pair (i, j) is an *inversion* of a permutation $a_1 a_2 \dots a_n$ if $i < j$ and $a_i > a_j$. For example, let us consider the permutation 3142 of the set $\{1, 2, 3, 4\}$. All the inversions of this permutation are $(3, 1)$, $(3, 2)$, and $(4, 2)$.

Parity of a permutation. A permutation $a_1 a_2 \dots a_n$ is *odd* if the number of all its inversions is odd. A permutation $a_1 a_2 \dots a_n$ is *even* if the number of all its inversions is even.

Theorem 7.2.1. *Consider a permutation $a_1 a_2 \dots a_n$. If two elements a_i and a_j exchange positions, then the permutation changes parity.*

Proof. Suppose there are exactly k elements between a and b in the permutation $a_1 a_2 \dots a_n$, where $a, b \in \{a_1, a_2, \dots, a_n\}$. This permutation has the following form

$$a_1 a_2 \dots a_{c_1} c_2 \dots c_k b \dots a_n.$$

Elements a and b can exchange positions the following way. First, element a exchanges positions with elements $c_1, c_2, \dots, c_k$ and b , one after the other, and then element b exchanges positions with elements $c_k, c_{k-1}, \dots, c_1$, one after the other. This way, exactly $2k + 1$ exchanges of the positions of neighboring elements were made. Note that every exchange of the positions of neighboring elements changes the parity of the permutation. Consequently, after $2k + 1$ steps, the obtained permutation is not of the same parity as the starting one, and the proof is completed. $\square$

Every permutation $a_1 a_2 \dots a_n$ of the set $\mathbb{N}_n$ is defined by a bijection $\varphi : \mathbb{N}_n \rightarrow \mathbb{N}_n$, where $\varphi(i) = a_i$ for any $i \in \mathbb{N}_n$, and can be represented in the form

$$\varphi : \begin{pmatrix} 1 & 2 & \dots & n \\ a_1 & a_2 & \dots & a_n \end{pmatrix}. \quad (7.2.1)$$

The *identity permutation* $\varepsilon = 12 \dots n$ of the set $\{1, 2, \dots, n\}$ is determined by $\varepsilon(i) = i$ for all i , and can be represented as

$$\varepsilon : \begin{pmatrix} 1 & 2 & \dots & n \\ 1 & 2 & \dots & n \end{pmatrix}. \quad (7.2.2)$$

Transposition. Exchanging the positions of two elements in a permutation is called a *transposition*. Using Theorem 7.2.1 it is easy to see that every even permutation can be obtained from the identical permutation by an even (and only by an even) number of transpositions, and every odd permutation can be obtained from the identical permutation by an odd (and only by an odd) number of transpositions.

Composition of permutations. If $\varphi : \mathbb{N}_n \rightarrow \mathbb{N}_n$ and $\psi : \mathbb{N}_n \rightarrow \mathbb{N}_n$ are bijections, then the composition $\varphi \circ \psi$, defined by $\varphi \circ \psi(i) = \varphi(\psi(i))$, is also a bijection, i.e., the composition of two permutations of a finite set is again a permutation of this set. If $\varphi : \mathbb{N}_n \rightarrow \mathbb{N}_n$ is a permutation, and k an arbitrary positive integer, then the k -th power of the permutation φ is defined as

$$\varphi^k = \underbrace{\varphi \circ \varphi \circ \dots \circ \varphi}_{k \text{ times}}. \quad (7.2.3)$$

Inverse of a permutation. As permutation φ is a bijection, then it has an inverse φ^{-1} which is also a bijection. If permutation φ is given by (7.2.1), then the inverse φ^{-1} is given by

$$\varphi^{-1} : \begin{pmatrix} a_1 & a_2 & \dots & a_n \\ 1 & 2 & \dots & n \end{pmatrix}. \quad (7.2.4)$$

Example 7.2.2. The permutations $\varphi = 71384526$ and $\psi = 24681357$ of the set $\{1, 2, \dots, 8\}$ can be written in the following form:

$$\varphi : \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 7 & 1 & 3 & 8 & 4 & 5 & 2 & 6 \end{pmatrix}, \quad \psi : \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 4 & 6 & 8 & 1 & 3 & 5 & 7 \end{pmatrix}.$$

Their composition is the permutation

$$\varphi \circ \psi : \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 8 & 5 & 6 & 7 & 3 & 4 & 2 \end{pmatrix},$$

and can also be written as $\varphi \circ \psi = 18567342$. It is easy to check that $\varphi^2 = \varphi \circ \varphi = 27368415$, $\varphi^3 = \varphi \circ \varphi \circ \varphi = 12356874$. The inverses of the permutations φ and ψ are given by

$$\varphi^{-1} : \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 7 & 3 & 5 & 6 & 8 & 1 & 4 \end{pmatrix}, \quad \psi^{-1} : \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 1 & 6 & 2 & 7 & 3 & 8 & 4 \end{pmatrix}.$$

or simply $\varphi^{-1} = 27356814$ and $\psi^{-1} = 51627384$. $\triangle$

Fixed points and the number of moving points. A point $i \in \mathbb{N}_n$ is a fixed point of a permutation $\varphi : \mathbb{N}_n \rightarrow \mathbb{N}_n$ if $\varphi(i) = i$. An important characteristic of permutation φ is the number of moving points, i.e., the number of points $i \in \mathbb{N}_n$ such that $\varphi(i) \neq i$. The number of moving points of permutation φ is denoted by $M(\varphi)$. It is obvious that $M(\varphi)$ can take any of the values from the set $\{0, 1, 2, \dots, n\}$.

Example 7.2.3. Permutation φ from Example 7.2.2 has a unique fixed point 3, and permutation ψ from the same example has no fixed points. Hence, for these two permutations $M(\varphi) = 7$ and $M(\psi) = 8$. $\triangle$

Permutation graph. Let $\varphi : \mathbb{N}_n \rightarrow \mathbb{N}_n$ be a permutation. The graph of permutation φ is a graph whose vertices represent the elements $1, 2, \dots, n$, and whose edges represent pairs $(i, \varphi(i))$, oriented from i toward $\varphi(i)$, for every $i \in \mathbb{N}_n$.

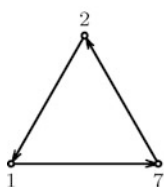


Fig. 7.2.1

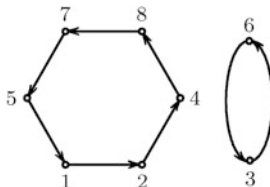
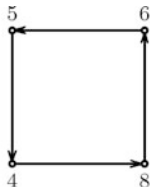


Fig. 7.2.2



Fig. 7.2.3

Example 7.2.4. The graphs of permutations φ and ψ from Example 7.2.2 are given in Figures 7.2.1 and 7.2.2. The graph of the identity permutation $\varepsilon : \mathbb{N}_n \rightarrow \mathbb{N}_n$ is given in Figure 7.2.3. $\triangle$

7.3 Orbits and Cycles

Orbit of an element. Let $\varphi : \mathbb{N}_n \rightarrow \mathbb{N}_n$ be a permutation and let a be an arbitrary element of the set $\mathbb{N}_n$. The sequence

$$a, \varphi(a), \varphi^2(a), \varphi^3(a), \dots \quad (7.3.1)$$

is called the orbit of element a in permutation φ . Since $\mathbb{N}_n$ is a finite set, there are only a finite number of distinct terms in the sequence (7.3.1). Let us denote by $O_\varphi(a)$ the set of all distinct elements that appear in the sequence (7.3.1). The set $O_\varphi(a)$ is sometimes also called the orbit of element a , and the number of its elements $|O_\varphi(a)|$ is called the *length of orbit* $O_\varphi(a)$. It is obvious that the length of the orbit can take any of the values $1, 2, \dots, n$. If $\varphi(a) = a$, then for any positive integer k we have $\varphi^k(a) = a$, and hence $O_\varphi(a) = 1$.

Cyclic permutation. In the special case $O_\varphi(a) = \mathbb{N}_n$, we have $|O_\varphi(a)| = n$, and permutation φ is called a cyclic permutation.

It is easy to prove that for any two elements $a, b \in \mathbb{N}_n$ and any permutation $\varphi : \mathbb{N}_n \rightarrow \mathbb{N}_n$ exactly one of the following equalities holds (see Figures 7.2.1–7.2.3):

$$O_\varphi(a) = O_\varphi(b) \quad \text{or} \quad O_\varphi(a) \cap O_\varphi(b) = \emptyset, \quad (7.3.2)$$

Let us define a relation $\sim$ on set $\mathbb{N}_n$ as follows: $a \sim b$ if the elements a and b belong to the same orbit. Note that $\sim$ is an equivalence relation on set $\mathbb{N}_n$, and every equivalence class coincides with one of the orbits.

Two permutations φ and ψ of set $\mathbb{N}_n$ are *relatively prime* if the sets of their moving points are disjoint.

Example 7.3.1. The following two permutations of the set $\{1, 2, \dots, 6\}$ are relatively prime:

$$\varphi : \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 1 & 4 & 5 & 6 \end{pmatrix} \quad \psi : \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 3 & 5 & 4 & 6 \end{pmatrix}.$$

The compositions $\varphi \circ \psi$ and $\psi \circ \varphi$ of these two permutations are determined as follows:

$$\begin{aligned} \varphi \circ \psi &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 1 & 4 & 5 & 6 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 3 & 5 & 4 & 6 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 1 & 5 & 4 & 6 \end{pmatrix}, \\ \psi \circ \varphi &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 3 & 5 & 4 & 6 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 1 & 4 & 5 & 6 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 1 & 5 & 4 & 6 \end{pmatrix}. \triangle \end{aligned}$$

Note that the permutations φ and ψ from Example 7.3.1 satisfy the equality $\varphi \circ \psi = \psi \circ \varphi$. Generally, the composition of permutations is not commutative, but the following theorem holds.

Theorem 7.3.2. *The composition of two relatively prime permutations is commutative, i.e., does not depend on the order of factors.*

Proof. Let φ and ψ be relatively prime permutations of the set $\mathbb{N}_n$, and a be an arbitrary element of set $\mathbb{N}_n$. Then, element a can be a moving point of at most one of the permutations φ and ψ . Suppose firstly that a is a fixed point for both of φ and ψ . Then we have

$$\varphi \circ \psi(a) = \varphi(\psi(a)) = \varphi(a) = a = \psi(a) = \psi(\varphi(a)) = \psi \circ \varphi(a).$$

Suppose now that a is a moving point of permutation φ , say, and let $\varphi(a) = b$, where $b \neq a$. Then $\varphi(b) \neq b$ (in the opposite case $\varphi(a) = \varphi(b) = b$, we get a contradiction with the assumption that φ is a bijection). Hence, elements a and b are moving points of permutation φ , and, consequently, a and b are fixed points of permutation ψ . It follows that:

$$\varphi \circ \psi(a) = \varphi(\psi(a)) = \varphi(a) = b = \psi(b) = \psi(\varphi(a)) = \psi \circ \varphi(a),$$

and the proof of the equality $\varphi \circ \psi = \psi \circ \varphi$ is completed. $\square$



Fig. 7.3.1

Cycle. Let us consider a permutation $\varphi : \mathbb{N}_n \rightarrow \mathbb{N}_n$. For any element $a \in \mathbb{N}_n$ let us define the function $\tilde{\varphi} : \mathbb{N}_n \rightarrow \mathbb{N}_n$ as follows:

$$\tilde{\varphi}(x) = \begin{cases} \varphi(x), & \text{if } x \in O_\varphi(a), \\ x, & \text{if } x \notin O_\varphi(a) \end{cases} \quad (7.3.3)$$

The function $\tilde{\varphi}$ is obviously a permutation of set $\mathbb{N}_n$, and is called a *cycle*. The graph of such a permutation is given in Figure 7.3.1.

Theorem 7.3.3. *Any permutation of a finite set can be represented as a product (convolution) of cycles.*

Proof. Let $\varphi : \mathbb{N}_n \rightarrow \mathbb{N}_n$ be a permutation. The set $\mathbb{N}_n$ can be represented in the form $\mathbb{N}_n = S_1 \cup S_2 \cup \dots \cup S_k$, where $S_1, S_2, \dots, S_k$ are pairwise

disjoint sets, and each of them coincides with the orbit of an element of $\mathbb{N}_n$. Let us denote

$$\tilde{\varphi}_i(x) = \begin{cases} \varphi(x), & \text{if } x \in S_i, \\ x, & \text{if } x \notin S_i, \end{cases} \quad i = 1, 2, \dots, k. \quad (7.3.4)$$

Then, $\tilde{\varphi}_i$ is a permutation of $\mathbb{N}_n$ with the set of moving points equal to S_i . Since $S_1, S_2, \dots, S_k$ are pairwise disjoint sets, it follows that $\tilde{\varphi}_1, \tilde{\varphi}_2, \dots, \tilde{\varphi}_k$ are relatively prime cycles and $\varphi = \tilde{\varphi}_1 \circ \tilde{\varphi}_2 \circ \dots \circ \tilde{\varphi}_k$. $\square$

Example 7.3.4. Let us consider the permutation of set $\{1, 2, 3, 4, 5, 6, 7\}$ given by

$$\varphi : \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 4 & 1 & 6 & 7 & 2 & 5 \end{pmatrix}.$$

The orbits of the permutation φ are the following: (a) 1, $\varphi(1) = 3$; (b) 2, $\varphi(2) = 4$, $\varphi(4) = 6$; (c) 5, $\varphi(5) = 7$. These orbits determine the sets of moving points $\{1, 3\}$, $\{2, 4, 6\}$, and $\{5, 7\}$ and related permutations $\tilde{\varphi}_1, \tilde{\varphi}_2$, and $\tilde{\varphi}_3$, that are given by

$$\begin{aligned} \tilde{\varphi}_1 &= (1, 3) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 2 & 1 & 4 & 5 & 6 & 7 \end{pmatrix}, \\ \tilde{\varphi}_2 &= (2, 4, 6) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 4 & 3 & 6 & 5 & 2 & 7 \end{pmatrix}, \\ \tilde{\varphi}_3 &= (5, 7) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 2 & 3 & 4 & 7 & 6 & 5 \end{pmatrix}. \end{aligned}$$

Permutation φ can be represented as the product of cycles as follows

$$\varphi = \tilde{\varphi}_1 \circ \tilde{\varphi}_2 \circ \tilde{\varphi}_3 = (1, 3) \circ (2, 4, 6) \circ (5, 7). \quad \triangle$$

Order of a permutation. Let $\varphi : \mathbb{N}_n \rightarrow \mathbb{N}_n$ be a permutation. We shall define the k -th power of permutation φ for any integer k .

If k is a positive integer, then φ^k is given by (7.2.3). Next, $\varphi^0 = \varepsilon$, where ε is the identity permutation, φ^{-1} is the inverse of permutation φ , and for any positive integer k we define

$$\varphi^{-k} = \underbrace{\varphi^{-1} \circ \varphi^{-1} \circ \dots \circ \varphi^{-1}}_{k \text{ times}}. \quad (7.3.5)$$

Now it is easy to prove that for all integers k and l , the following equalities hold:

$$\varphi^k \circ \varphi^l = \varphi^{k+l}, \quad (\varphi^k)^l = \varphi^{kl}.$$

Since the composition of two permutations is again a permutation, we conclude that all functions $\varphi, \varphi^2, \varphi^3, \dots$ are permutations of set $\mathbb{N}_n$. The

number of permutations of set $\mathbb{N}_n$ is finite, and hence there exists two positive integers r and s , such that $r > s$, and $\varphi^r = \varphi^s$. For these r and s we have $\varphi^{r-s} = \varphi^r \circ \varphi^{-s} = \varphi^s \circ \varphi^{-s} = \varphi^{s-s} = \varphi^0 = \varepsilon$. In other words, there exist positive integer k such that

$$\varphi^k = \varepsilon. \quad (7.3.6)$$

The minimal positive integer k for which equality (7.3.6) holds is called the *order of permutation* φ . It is easy to conclude that there exist infinitely many positive integers k , such that the equality $\varphi^k = \varepsilon$ holds, but all of them are divisible by the order of permutation. Indeed, if k is the order of permutation φ , then for any positive integer r we have $\varphi^{rk} = (\varphi^k)^r = \varepsilon^r = \varepsilon$. But for $r \in \{0, 1, 2, \dots\}$ and $s \in \{1, 2, \dots, k-1\}$ the following relations hold

$$\varphi^{rk+s} = \varphi^{rk} \circ \varphi^s = \varepsilon \circ \varphi^s = \varphi^s \neq \varepsilon.$$

The next theorem gives a criterion for determining the order of an arbitrary permutation.

Theorem 7.3.5. *Let $\varphi = \varphi_1 \circ \varphi_2 \circ \dots \circ \varphi_m$ be the representation of some permutation φ as the product of cycles. If the cycles $\varphi_1, \varphi_2, \dots, \varphi_m$ have orders $k_1, k_2, \dots, k_m$, respectively, then the order of permutation φ is equal to the least common multiple of $k_1, k_2, \dots, k_m$.*

Proof. First note that the composition $\varphi_1 \circ \varphi_2 \circ \dots \circ \varphi_m$ of relatively prime permutations maps each of its moving points a to the element $\tilde{a}$, determined by $\varphi_i(a) = \tilde{a}$, where φ_i is a permutation from the set $\{\varphi_1, \varphi_2, \dots, \varphi_m\}$ for which element a is a moving point.

As a consequence we conclude that the composition of a few relatively prime permutations is equal to the identity permutation if and only if every factor in the composition is the identity permutation. Hence, the equality $\varphi^k = \varepsilon$ holds if and only if

$$\varphi_1^k = \varepsilon, \quad \varphi_2^k = \varepsilon, \quad \dots, \quad \varphi_m^k = \varepsilon. \quad (7.3.7)$$

Equalities (7.3.7) hold if and only if positive integer k is divisible by each of the positive integers $k_1, k_2, \dots, k_m$. The smallest value of k for which this is true is equal to LCM($k_1, k_2, \dots, k_m$). $\square$

Example 7.3.6. Consider a permutation of the set $\mathbb{N}_7 = \{1, 2, 3, 4, 5, 6, 7\}$ given by

$$\varphi : \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 4 & 1 & 6 & 7 & 2 & 5 \end{pmatrix}.$$

The permutation φ can be represented as the product of cycles as follows: $\varphi = (1, 3) \circ (2, 4, 6) \circ (5, 7)$. The order of φ is LCM(2, 3, 2) = 6. $\triangle$

7.4 Permutation Groups

We have noticed that any permutation of a finite set S partitions S into equivalence classes, where each of them coincides with the orbit of an element related to this permutation. Usually we consider the set $S = \mathbb{N}_n$, where $n \in \mathbb{N}$. The partition of a finite set into disjoint equivalence classes of elements can also be determined by a set of permutations.

Let G be the set of a few permutations of a finite set S . Let us define a relation $\sim$ on set S as follows: $a \sim b$ holds for $a, b \in S$ if there exists a permutation $\varphi \in G$ such that $\varphi(a) = b$.

Sufficient conditions under which relation $\sim$ is an equivalence relation, i.e., simultaneously a *reflexive* relation, a *symmetric* relation and a *transitive* relation, can be formulated the following way:

- (C1) The identical permutation ε belongs to G .
- (C2) If $\varphi \in G$, then $\varphi^{-1} \in G$.
- (C3) If $\varphi \in G$ and $\psi \in G$, then $\varphi \circ \psi \in G$.

If conditions (C1), (C2), and (C3) are satisfied, then set G is called a *permutation group* of set S . In that case the relation $\sim$ is an equivalence relation and it partitions set S into disjoint *equivalence classes*. Each of these classes is called an *orbit of group G* . The number of elements of an orbit O is called the *length of orbit O* . It is easy to prove the following theorem.

Theorem 7.4.1. *Let G be a permutation group of a finite set S , and $\sim$ be a relation on S defined as follows: $a \sim b$ holds for $a, b \in S$, if there exists a permutation $\varphi \in G$ such that $\varphi(a) = b$. A subset O of set S is an orbit of group G if and only if the following two conditions are satisfied:*

- (a) *For any $\varphi \in G$ and any $a \in O$ the relation $\varphi(a) \in O$ holds.*
- (b) *For any element $b \in O$ there exist a permutation $\varphi \in G$ and an element $a \in O$ such that $b = \varphi(a)$.*

Example 7.4.2. Let S be a finite set, $G = \{\varepsilon = \varphi_0, \varphi_1, \dots, \varphi_{k-1}\}$ be a group of permutations of set S , and $a \in S$. We shall prove that the set $O(a) = \{\varphi_0(a), \varphi_1(a), \dots, \varphi_{k-1}(a)\}$ is an orbit of group G called the *orbit of element a related to group G* . Let us check conditions (a) and (b) of Theorem 7.4.1.

(a) Let us consider a permutation $\varphi_i \in G$ and an element $b = \varphi_j(a) \in O(a)$, where $\varphi_j \in G$. Then we have $\varphi_i \circ \varphi_j \in G$, and $\varphi_i(b) = \varphi_i(\varphi_j(a)) = (\varphi_i \circ \varphi_j)(a) \in O(a)$.

(b) Now, let us consider an element $c = \varphi_i(a) \in O(a)$. For any permutation $\varphi_j \in G$ we have $\varphi_i \circ \varphi_j^{-1} \in G$. Let us denote $b = \varphi_j(a)$. Then, we have $b \in O(a)$ and $c = \varphi_i(a) = (\varphi_i \circ \varphi_j^{-1} \circ \varphi_j)(a) = (\varphi_i \circ \varphi_j^{-1})(b)$. $\triangle$

Example 7.4.3. *Rotation group of a cube.* Let the vertices of a cube be labeled as shown in Figure 7.4.1. Every rotation of the cube is determined by a permutation of the set $\mathbb{N}_8 = \{1, 2, \dots, 8\}$.

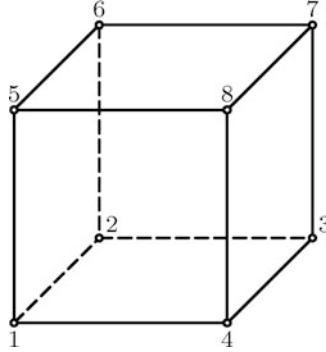


Fig. 7.4.1

(a) There exists a trivial rotation, determined by the identity permutation. This permutation can be written as a product of cycles as follows:

$$(1) \circ (2) \circ (3) \circ (4) \circ (5) \circ (6) \circ (7) \circ (8).$$

(b) There are three nonidentical rotations about each of three lines that connect the centers of two opposite sides of the cube. The related permutations represented as products of cycles are the following:

$$\begin{aligned} &(1, 5, 8, 4) \circ (2, 6, 7, 3), \\ &(1, 8) \circ (5, 4) \circ (2, 7) \circ (6, 3), \\ &(1, 4, 8, 5) \circ (2, 3, 7, 6), \\ &(1, 2, 3, 4) \circ (5, 6, 7, 8), \\ &(1, 3) \circ (2, 4) \circ (5, 7) \circ (6, 8), \\ &(1, 4, 3, 2) \circ (5, 8, 7, 6), \\ &(1, 5, 6, 2) \circ (4, 8, 7, 3), \\ &(1, 6) \circ (5, 2) \circ (4, 7) \circ (8, 3), \\ &(1, 2, 6, 5) \circ (4, 3, 7, 8). \end{aligned}$$

(c) There are two nonidentical rotations about each of the four diagonals of the cube. The related permutations are the following:

$$\begin{aligned} &(1) \circ (2, 5, 4) \circ (3, 6, 8) \circ (7), & (1) \circ (2, 4, 5) \circ (3, 8, 6) \circ (7), \\ &(2) \circ (1, 3, 6) \circ (4, 7, 5) \circ (8), & (2) \circ (1, 6, 3) \circ (4, 5, 7) \circ (8), \end{aligned}$$

$$\begin{aligned} (3) \circ (1, 6, 8) \circ (2, 7, 4) \circ (5), & \quad (3) \circ (1, 8, 6) \circ (2, 4, 7) \circ (5), \\ (4) \circ (1, 3, 8) \circ (2, 7, 5) \circ (6), & \quad (4) \circ (1, 8, 3) \circ (2, 5, 7) \circ (6). \end{aligned}$$

(d) There is exactly one nonidentical rotation about each of six lines that connect the midpoints of the opposite edges of the cube. The related permutations are the following:

$$\begin{aligned} (1, 5) \circ (2, 8) \circ (3, 7) \circ (4, 6), & \quad (1, 2) \circ (3, 5) \circ (4, 6) \circ (7, 8), \\ (1, 7) \circ (2, 3) \circ (4, 6) \circ (5, 8), & \quad (1, 7) \circ (2, 6) \circ (3, 5) \circ (4, 8), \\ (1, 7) \circ (2, 8) \circ (3, 4) \circ (5, 6), & \quad (1, 4) \circ (2, 8) \circ (3, 5) \circ (6, 7). \end{aligned}$$

Hence, the rotation group of a cube consists of 24 permutations of the set $\mathbb{N}_8 = \{1, 2, \dots, 8\}$. Among them, there are: one permutation of the type $[1, 1, 1, 1, 1, 1, 1, 1]$, 6 permutations of the type $[4, 4]$, 9 permutations of the type $[2, 2, 2, 2]$, and 8 permutations of the type $[1, 1, 3, 3]$. Here, k, l , and m in the notation $[k, l, m, \dots]$ represent the number of elements of the cycles of the corresponding permutation. $\triangle$

Theorem 7.4.4. *Let G be a permutation group of the set $\mathbb{N}_n = \{1, 2, \dots, n\}$, a be an arbitrary element of the set S , and $G_a = \{\varphi \mid \varphi \in G, \varphi(a) = a\}$ be the set of those permutations from G for which a is a fixed element. The length of the orbit $O(a)$ is given by*

$$|O(a)| = \frac{|G|}{|G_a|}.$$

Proof. Let $G = \{\varphi_0, \varphi_1, \dots, \varphi_{k-1}\}$ and $G_a = \{\psi_0, \psi_1, \dots, \psi_{s-1}\}$, where $\varphi_0 = \psi_0 = \varepsilon$. Then, there exist permutations $\alpha_0 = \varepsilon, \alpha_1, \dots, \alpha_{l-1}$ in group G , such that the sets

$$\begin{aligned} G_a^{(0)} &= \{\alpha_0 \circ \psi_0, \alpha_0 \circ \psi_1, \dots, \alpha_0 \circ \psi_{s-1}\}, \\ G_a^{(1)} &= \{\alpha_1 \circ \psi_0, \alpha_1 \circ \psi_1, \dots, \alpha_1 \circ \psi_{s-1}\}, \\ &\dots\dots\dots \\ G_a^{(l-1)} &= \{\alpha_{l-1} \circ \psi_0, \alpha_{l-1} \circ \psi_1, \dots, \alpha_{l-1} \circ \psi_{s-1}\}, \end{aligned}$$

are pairwise disjoint and $G = G_a^{(0)} \cup G_a^{(1)} \cup \dots \cup G_a^{(l-1)}$. Permutations $\alpha_1, \alpha_2, \dots, \alpha_{l-1}$ can be chosen such that $\alpha_1 \notin G_a^{(0)}, \alpha_2 \notin G_a^{(0)} \cup G_a^{(1)}$ etc. This follows from the following statement:

If G is a group, and H is a subgroup of group G , then for arbitrary elements $a, b \in G$ exactly one of the following two relations hold:

$$aH = bH \quad \text{or} \quad aH \cap bH = \emptyset.$$

Indeed, if $ax = by$, where $x, y \in H$, then for any element $az \in aH$ we have $az = axx^{-1}z = byx^{-1}z \in bH$.

Suppose now that

$$\begin{aligned} G_a^{(0)} &= \{\psi_0, \psi_1, \dots, \psi_{s-1}\}, \\ G_a^{(1)} &= \{\psi_s, \psi_{s+1}, \dots, \psi_{2s-1}\}, \\ &\dots\dots\dots \\ G_a^{(l-1)} &= \{\psi_{(l-1)s}, \psi_{(l-1)s+1}, \dots, \psi_{ls-1}\}. \end{aligned}$$

Then, for any $i \in \{0, 1, \dots, l-1\}$ we have

$$\psi_{is}(a) = \psi_{is+1}(a) = \dots = \psi_{(i+1)s-1}(a) = \alpha_i(a).$$

Elements $\alpha_0(a)$, $\alpha_1(a)$, $\dots$, $\alpha_{l-1}(a)$ are distinct. Indeed, it follows from $\alpha_i(a) = \alpha_j(a)$ that $a = \alpha_i \circ \alpha_j^{-1}(a)$, i.e., $\alpha_i \circ \alpha_j^{-1} \in G_a$. It follows that α_i and α_j both belong to one of the sets $G_a^{(0)}$, $G_a^{(1)}$, $\dots$, $G_a^{(l-1)}$, and that is a contradiction. Finally, it follows that $|O(a)| = l = |G|/|G_a|$. $\square$

7.5 Burnside's Lemma

Theorem 7.5.1. *Let $G = \{\varphi_0, \varphi_1, \dots, \varphi_{k-1}\}$ be a group of permutations of the set $\mathbb{N}_n = \{1, 2, \dots, n\}$, $B = B(G)$ be the number of orbits of the group G , and $f(\varphi_i)$ be the number of fixed points of the permutation φ_i . Then, the following equality holds:*

$$B(G) = \frac{1}{|G|} [f(\varphi_0) + f(\varphi_1) + \dots + f(\varphi_{k-1})]. \quad (7.5.1)$$

Proof: Let us define the relation $\rho \subset G \times \mathbb{N}_n$ as follows:

$$(\varphi, j) \in \rho \iff \varphi(j) = j. \quad (7.5.2)$$

The graph of relation ρ is given in Figure 7.5.1. We shall determine the number of elements of the set ρ in the following two ways: (1) We shall count points (vertices) of the graph first on the vertical lines, and then calculate the sum. (2) We shall count points of the graph first on the horizontal lines, and then calculate the sum.

Note that there are $f(\varphi_j)$ points of the graph on the vertical line with the abscissa φ_j , and hence

$$|\rho| = f(\varphi_0) + f(\varphi_1) + \dots + f(\varphi_{k-1}) = \sum_{\varphi \in G} f(\varphi). \quad (7.5.3)$$

Let $G_j = \{\varphi \mid \varphi \in G, \varphi(j) = j\}$. There are $|G_j| = |G|/|O(j)|$ points of the graph on the horizontal line with the ordinate j , and hence

$$|\rho| = |G_1| + |G_2| + \dots + |G_n| = \sum_{j=1}^n |G_j|. \quad (7.5.4)$$

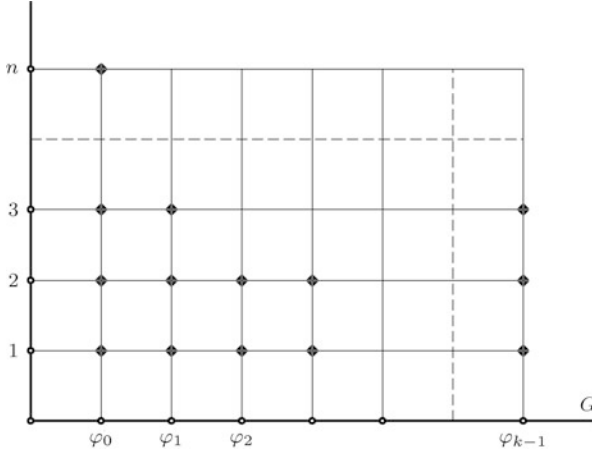


Fig. 7.5.1

If the elements $i, j \in \mathbb{N}_n$ belong to the same orbit, then $O(i) = O(j)$ and $|G_i| = |G| : |O(i)| = |G| : |O(j)| = |G_j|$. Let $O_1, O_2, \dots, O_B$ be all the orbits of group G . Then $\mathbb{N}_n = O_1 \cup O_2 \cup \dots \cup O_B$ and the sets $O_1, O_2, \dots, O_B$ are pairwise disjoint. The following equalities hold:

$$\begin{aligned} \sum_{j=1}^n |G_j| &= \sum_{j \in O_1} |G_j| + \sum_{j \in O_2} |G_j| + \dots + \sum_{j \in O_B} |G_j| \\ &= \sum_{i=1}^B \sum_{j \in O_i} |G_j| = \sum_{i=1}^B \sum_{j \in O_i} \frac{|G|}{|O_i|} = \sum_{i=1}^B |G| = B \cdot |G|. \end{aligned} \quad (7.5.5)$$

Finally, using (7.5.3)–(7.5.5) we obtain that

$$B(G) = \frac{1}{|G|} \sum_{j=1}^n |G_j| = \frac{|\rho|}{|G|} = \frac{1}{|G|} \sum_{\varphi \in G} f(\varphi). \quad \square$$

Example 7.5.2. Suppose that each vertex of a cube is to be colored by one of m colors. Let S be the set of all different colorings. Then, it is obvious that $|S| = m^8$. Two colorings are equivalent if one of them can be obtained from the other one by a rotation of the cube. Let B_m be the number of nonequivalent colorings. We shall determine the formula for B_m , and the exact values of $B_2, B_3, \dots, B_8$.

Let G be the rotation group of a cube, which consists of 24 permutations of the set $\mathbb{N}_8 = \{1, 2, \dots, 8\}$, see Example 7.4.3. The rotation group of a cube generates the permutation group of the set S as follows.

Let $\varphi \in G$ be a rotation of a cube. This rotation maps every element from S to an element of S . It is easy to see that φ is a bijection, i.e., φ is a permutation of S . Let us denote by $\tilde{\varphi}$ the permutation of set S that is determined by the rotation $\varphi \in G$. Let $\tilde{G}$ be the group of all permutations $\tilde{\varphi}$ obtained this way. It is obvious that $|\tilde{G}| = |G|$.

For any fixed positive integer m , let B_m be the number of orbits of the permutation group $\tilde{G}$ of set S . In order to determine B_m it is sufficient to determine the number of moving points of any permutation from $\tilde{G}$ and apply Burnside's lemma.

Any coloring of the vertices of a cube (an element of set S), for example using blue, red, and yellow, is determined by an 8-arrangement of the letters B , R , and Y . The permutations from $\tilde{G}$ transform such arrangements. For example, the permutation $\tilde{\varphi} \in \tilde{G}$, that is determined by the rotation $\varphi = (1, 2, 3, 4) \circ (5, 6, 7, 8) \in G$, transforms the arrangement $BBYBYRRR \in S$ into $BBBYRYRR \in S$. A coloring of the vertices of the cube is a fixed point of the permutation $\tilde{\varphi} \in \tilde{G}$, if and only if all edges in the same cycle of the permutation $\varphi \in G$ are of the same color. Hence we have the conclusion:

If a permutation $\varphi \in G$ is represented as the product of exactly k cycles, and the number of colors is equal to m , then the number of fixed points of the permutation $\tilde{\varphi} \in \tilde{G}$ is equal to m^k .

In Example 7.4.3 any rotation of a cube (which is a permutation of the set of its edges) is represented as the product of cycles. Using these results, Burnside's lemma and the previous conclusion, we obtain that

$$B_m = B(\tilde{G}) = \frac{1}{24} (m^8 + 6 \cdot m^2 + 9 \cdot m^4 + 8 \cdot m^4). \quad (7.5.6)$$

From (7.5.6) it follows that

$$\begin{aligned} B_2 &= 23, & B_3 &= 333, & B_4 &= 2916, & B_5 &= 16\,725, \\ B_6 &= 70\,911, & B_7 &= 241\,913, & B_8 &= 701\,968. \end{aligned} \triangle$$

Exercises

7.1. Prove the following properties of the composition and the inverse of the permutations of a finite set:

- The composition of two even permutations is even.
- The composition of two odd permutations is even.
- The composition of an odd and an even permutation is odd.
- The inverse of every even permutation is even.
- The inverse of every odd permutation is odd.

7.2. Let φ be the permutation of the set $\mathbb{N}_9 = \{1, 2, \dots, 9\}$ given by

$$\varphi: \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 3 & 9 & 5 & 8 & 7 & 2 & 1 & 4 & 6 \end{pmatrix}.$$

Determine the cycles and the order of the permutation φ .

7.3. A *coloring of a regular polyhedron* is a coloring of all its sides by distinct colors. Two colorings are equivalent if there is a rotation of the polyhedron which transforms the first coloring into the second one.

(a) Determine the number of nonequivalent colorings of a tetrahedron that can be obtained using 4 colors.

(b) Determine the number of nonequivalent colorings of a cube that can be obtained using 6 colors.

(c) Determine the number of nonequivalent colorings of an octahedron that can be obtained using 8 colors.

(d) Determine the number of nonequivalent colorings of a dodecahedron that can be obtained using 12 colors.

(e) Determine the number of nonequivalent colorings of an icosahedron that can be obtained using 20 colors.

7.4. A circle is divided into p equal arcs, where p is a prime number. Each of the arcs is to be colored by one of n colors. How many nonequivalent colorings are there? Two colorings are equivalent if one of them can be obtained from the other one by a rotation of the circle about its center.

7.5. Fermat's theorem. Let p be a prime number and n be a positive integer. Prove that $n^p - n$ is divisible by p .

7.6. An n -gon is *starlike* if at least two of its nonadjacent sides intersect each other. Let $p > 2$ be a prime number and suppose that points $A_1, A_2, \dots, A_p$ lie on a circle and divide it into p equal arcs. How many noncongruent starlike p -gons with edges $A_1, A_2, \dots, A_p$ are there?

7.7. Wilson's theorem. Let p be a prime number. Prove that $(p-1)! + 1$ is divisible by p .

7.8. Let S be the set of all n -arrangements of elements 0 and 1, where $n \geq 8$ is a positive integer. The transformation T allows the exchange of the positions of two adjacent triplets of digits in an arrangement from S . For example,

$$d_1 d_2 d_3 d_4 d_5 d_6 d_7 \dots d_n \xrightarrow{T} d_4 d_5 d_6 d_1 d_2 d_3 d_7 \dots d_n.$$

Two arrangements are equivalent if one of them can be obtained from the other one by repeated application of transformation T . How many nonequivalent arrangements are there?

7.9. A deck contains $2n + 1$ cards numbered $1, 2, \dots, 2n + 1$. The starting arrangement is $(1, 2, \dots, 2n + 1)$. The deck can be shuffled according to the following two rules.

(a) A few cards from the beginning of deck can be put at the end of deck, for example, $c_1 c_2 c_3 \dots c_{2n+1} \rightarrow c_3 \dots c_{2n+1} c_1 c_2$.

(b) For a permutation of cards $c_1 c_2 \dots c_{2n+1}$ it is allowed to put simultaneously any of cards $k \in \{1, 2, \dots, n\}$ between cards c_{n+k} and c_{n+k+1} .

How many distinct permutations of cards can be obtained using these two rules?

7.10. How many nonequivalent colorings of the edges of a cube are there if it is allowed to use: (a) two colors? (b) three colors?

7.11. Every vertex of a regular n -gon with center O is labeled $+1$ or -1 . It is allowed to change a labeling by changing the sign of all the edges of a regular k -gon with center O . The case $k = 2$ is allowed; in that case we change the sign of two endpoints of a segment with midpoint O . Two labelings of the vertices of the n -gon are considered equivalent if one of them can be obtained from the other one by repeated application of the given rule.

(a) For $n = 15$ prove that there exists a labeling that is not equivalent to the labeling with all labels equal to $+1$.

(b) For $n = 30$ prove that there exists a labeling that is not equivalent to the labeling with all labels equal to $+1$.

(c) How many nonequivalent labelings of the edges of a regular n -gon are there?



Graph Theory: Part 1

8.1 The Königsberg Bridge Problem

We shall start this chapter with two examples. The first one was formulated in 1736 by Leonard Euler. Now it is known as the Königsberg bridge problem and is usually considered to be the beginning of Graph Theory.

Example 8.1.1. The city of Königsberg (now Kaliningrad) is situated on both sides of the Pregel River. Two islands on the river are connected to each other and the mainland by seven bridges as shown in Figure 8.1.1. The problem is to find out whether there exists a walk through the city that crosses each bridge once and only once. The starting and ending points of the walk need not be the same.

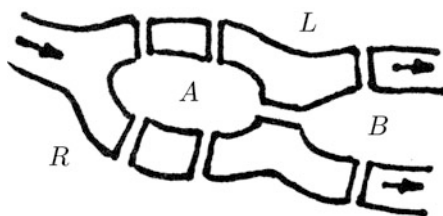


Fig. 8.1.1

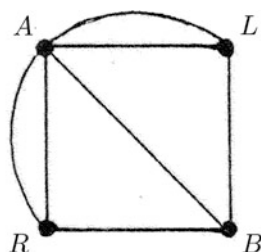


Fig. 8.1.2

In 1736 Euler proved that such a walk is not possible. Let the four land areas (two banks of the river and two islands) be labeled L , R , A , and B , see Figure 8.1.1. In Figure 8.1.2 each land area is replaced by a *vertex*

with the same label, and each bridge is replaced by an *edge* connecting the corresponding vertices. This way a *graph* with four vertices and seven edges is obtained. An equivalent formulation of the problem is the following. Is the obtained graph traversable in such a way that each edge is crossed exactly once?

Let us suppose that the graph is traversable this way and that the starting vertex coincides with the ending vertex. Then each vertex should be incident with an even number of edges. But this is not the case, because every vertex on the graph in Figure 8.1.2 is incident with an odd number of edges. Here we used the following terminology: a vertex X is incident with an edge b , if X is an endpoint of edge b .

Suppose now that the starting vertex does not coincide with the ending vertex. It is obvious that in this case two vertices of the graph should be incident with an odd number of edges, and the remaining two vertices should be incident with an even number of edges. Again, the graph in Figure 8.1.2 does not satisfy this condition. Hence, the problem is unsolvable, i.e., there is no walk in the city of Königsberg that satisfies the given conditions. $\triangle$

Example 8.1.2. Consider now the following modification of the Königsberg bridge problem. Suppose there are two islands on the river, a house H on one of these islands, and a church C on the left bank of the river. Suppose also that there are eight bridges that connect islands and banks of the river as shown in Figure 8.1.3. The problem is formulated as follows. Can a priest reach the house on the island starting the walk from the church and crossing every bridge exactly once? He is not allowed to swim.

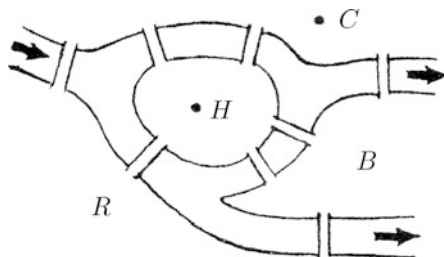


Fig. 8.1.3

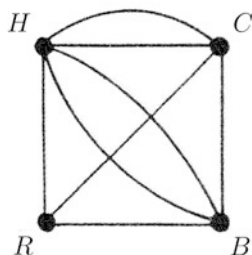


Fig. 8.1.4

In this case we introduce a graph with vertices R - the right bank of the river, C - the left bank with the church, H - the island with the house, and B - the second island, and eight edges that correspond to the bridges, see Figure 8.1.4. If a walk that satisfies the given conditions exists, then each of the vertices C and H should be incident with an odd number of edges, and each of the vertices R and B should be incident with an even number of edges. But this is not true for the graph in Figure 8.1.4, and, hence, a walk that satisfies the given conditions does not exist.

Suppose now that the priest is allowed to walk around the river's source. Then he can reach the house on the island crossing every bridge exactly once as shown in Figure 8.1.5. $\triangle$

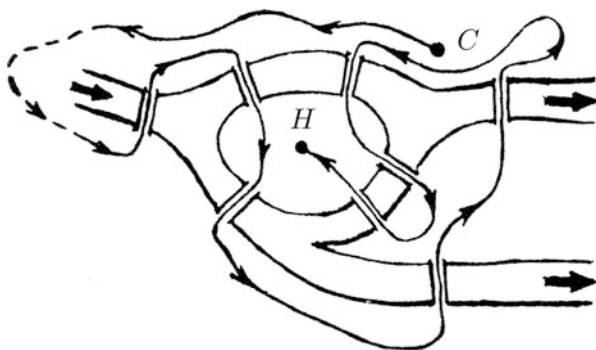


Fig. 8.1.5

8.2 Basic Notions

Vertices and edges. A graph G is determined by a set of its vertices $V(G)$, a set of its edges $E(G)$, and a *relation of incidence* that associates with every edge either two vertices (called its ends), or one vertex (in this case the ends of the edge coincide). A vertex is *even* (*odd*) if it is incident to an even (odd) number of edges. An edge whose ends coincide will be called a *loop*.

Two vertices are *adjacent* if they are connected by an edge. Two edges are *adjacent* if there is a vertex that is incident to both of them. The *degree* of a vertex in a graph is the number of edges incident to it. An *isolated vertex* is a vertex with degree zero. A *leaf vertex* is a vertex with degree one.

An *undirected graph* is a graph in which edges have no orientation. Sometimes there is also a need to consider oriented graphs. A *directed graph* is a graph in which every edge has an orientation, in the sense that one of its ends is the beginning, and another is the end of the edge.

A *null-graph* G_0 is the unique graph with both sets $V(G_0)$ and $E(G_0)$ empty. A *graph without edges* is a graph G for which the set $E(G)$ is empty.

A *multigraph* is a graph in which the same two vertices can be connected by two or more edges. *Multiple edges* are two or more edges that connect the same two vertices. A *simple graph* is an undirected graph in which both multiple edges and loops are disallowed. A *complete graph* is a simple undirected graph in which every pair of distinct vertices is connected by a unique edge.

Example 8.2.1. *A chess tournament.* Suppose that eight chess players, labeled $1, 2, \dots, 8$, attend a chess tournament. They are to play exactly one game against each other. At any given time, a graph can be associated with the tournament, showing which pairs of players have already played a game against each other. The set of vertices is $\{1, 2, \dots, 8\}$. Two vertices i and j are connected with an edge if the players i and j have already played a game against each other. Obviously, before the first tour the corresponding graph is without edges, i.e., all vertices are isolated, see Figure 8.2.1. After the first tour the graph may look as shown in Figure 8.2.2.

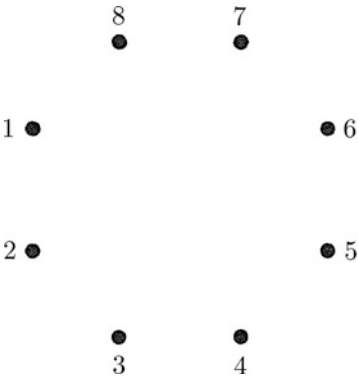


Fig. 8.2.1

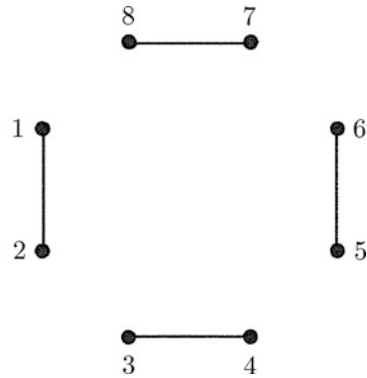


Fig. 8.2.2

After the second tour the graph may look as shown in Figure 8.2.3. When the tournament is finished, then the corresponding graph is a complete graph with 8 vertices, see Figure 8.2.4. All these graphs are simple. $\triangle$

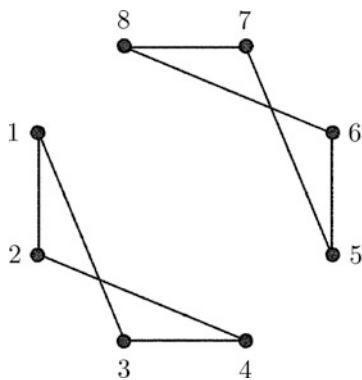


Fig. 8.2.3

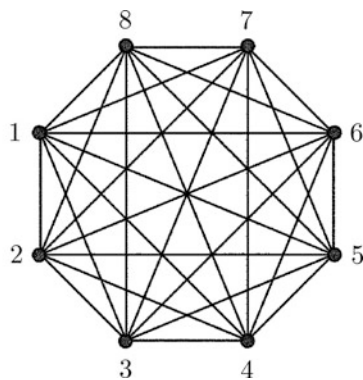


Fig. 8.2.4

Example 8.2.2. Let $V = \{1, 2, \dots, 8\}$ and $f : V \rightarrow V$ be a function defined by

$$f : \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 1 & 5 & 4 & 1 & 6 & 8 \end{pmatrix}.$$

The function f generates the graph G_f with the set of vertices V , and the set of *directed* edges $E = \{(v, f(v)) \mid v \in V\}$, see Figure 8.2.5. Note also that graph G_f is a *directed multigraph* with the multiedge $\{(4, 5), (5, 4)\}$, and with the loop $(8, 8)$.

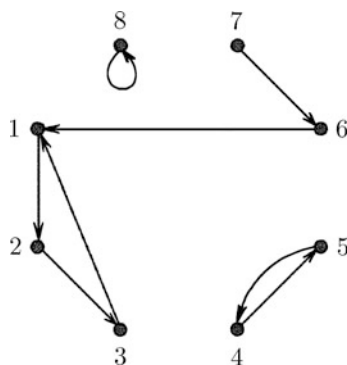


Fig. 8.2.5

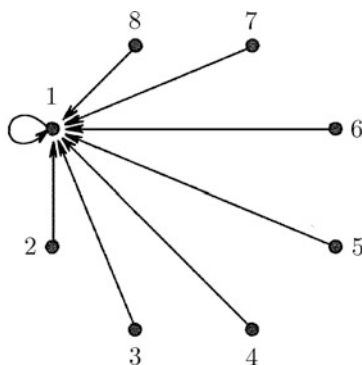


Fig. 8.2.6

Let $g : V \rightarrow V$ be the constant function defined by $g(v) = 1$, for every $v \in V$. The graph G_g of function g is given in Figure 8.2.6. Every vertex from the set $\{2, 3, \dots, 8\}$ is of degree 1, and the vertex 1 is of degree 8.

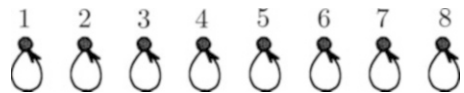


Fig. 8.2.7

Let $i_V : V \rightarrow V$ be the identity function, defined by $i_V(v) = v$ for any $v \in V$. The graph of the function i_V has eight loops, see Figure 8.2.7. $\triangle$

Example 8.2.3. Platonic graphs. A platonic solid is a polyhedron all of whose faces are congruent regular polygons, and such that the same number of faces meet at each vertex. There are five platonic solids: *tetrahedron*, *octahedron*, *hexahedron* (*cube*), *icosahedron*, and *dodecahedron*. We use the notation:

- f - the number of faces of the polyhedron;
- e - the number of edges of the polyhedron;
- v - the number of vertices of the polyhedron;
- n - the number of polygons (faces) meeting at a vertex;
- k - the number of vertices of each polygon (face); this is also the number of edges that bound each polygon;
- k^* - the number of edges meeting at a vertex.

It is well known that the parameters f , e , v , and k^* are uniquely determined by any pair of numbers (n, k) . Combinatorial data for platonic solids are presented in the following table.

Table 8.2.1. Combinatorial data for platonic solids

	f	e	v	n	k	k^*
Tetrahedron	4	6	4	3	3	3
Octahedron	8	12	6	4	3	4
Hexahedron	6	12	8	3	4	3
Icosahedron	20	30	12	5	3	5
Dodecahedron	12	30	20	3	5	3

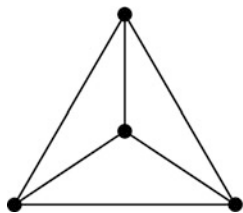


Fig. 8.2.8

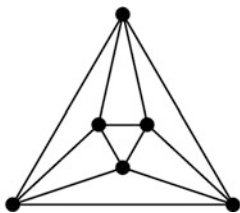


Fig. 8.2.9

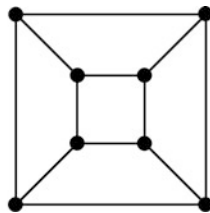


Fig. 8.2.10

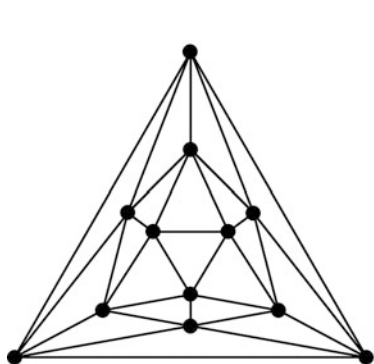


Fig. 8.2.11

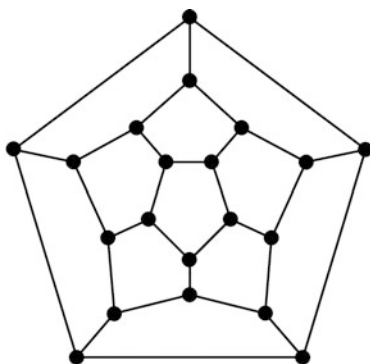


Fig. 8.2.12

The skeleton of a platonic solid (vertices and edges) can be considered as a graph. Therefore, there are *five platonic graphs*, and they are presented on Figures 8.2.8–8.2.12. For every platonic graph all its vertices have the same degree, but only the graph representing a tetrahedron is complete. $\triangle$

8.3 Complement Graphs and Subgraphs

The *complement* or *inverse* of a graph G is a graph H on the same vertices such that two distinct vertices of H are adjacent if and only if they are not adjacent in G .

Example 8.3.1. Consider again a chess tournament with eight participants, labeled 1, 2, $\dots$, 8, as in Example 8.2.1, who are to play exactly one game against each other. Let G be a graph whose edges are determined by the games that have already been played. Then, the complement of graph G is the graph H whose edges are determined by the games remaining till the end of the tournament.

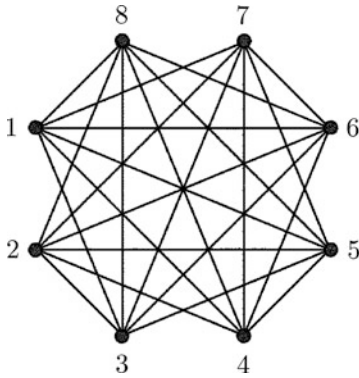


Fig. 8.3.1

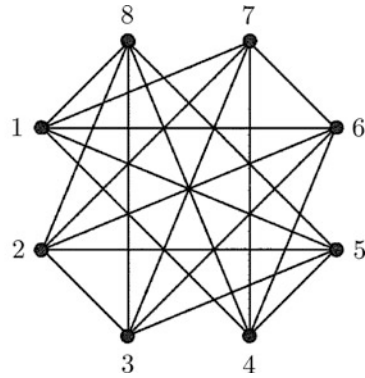


Fig. 8.3.2

The graphs presented in Figures 8.3.1 and 8.3.2 are complements of the graphs given in Figures 8.2.2 and 8.2.3 respectively. It is obvious that the graph without edges and the complete graph (see Figures 8.2.1 and 8.2.4) are complements to each other. $\triangle$

Subgraphs. Let G be a graph, and $V(G)$ and $E(G)$ be related sets of vertices and edges. A graph H is a *subgraph* of graph G if

$$V(H) \subset V(G), \quad E(H) \subset E(G),$$

where $V(H)$ and $E(H)$ are respectively the sets of vertices and edges of graph H . If H is a subgraph of G , then we say that G is a *supergraph* of H .

A *spanning subgraph* of graph G is a graph H with

$$V(H) = V(G), \quad E(H) \subset E(G),$$

It is obvious that if V is an arbitrary set of vertices, then any simple graph with the set of vertices V is a spanning subgraph of the complete graph with the same set of vertices.

Example 8.3.2. A committee consists of six members. Some of them are friends (friendship is a symmetric relation). We shall prove the following statement. *There exist three members of the committee such that every two of them are friends, or there exists three members of the committee such that no two of them are friends.*

Let the members of the committee be labeled 1, 2, 3, 4, 5, and 6. Consider the graph G with the set of vertices $V = \{1, 2, \dots, 6\}$, and where every edge connects two distinct vertices if the corresponding members of the committee are friends. The statement can equivalently be formulated as follows. *There exists a complete subgraph H of the graph G , such that H has exactly three vertices, and H is complete or H is without edges.* It is obvious that exactly one of the following two statements holds.

- (a) Vertex 1 is adjacent to at least three of the other vertices.
- (b) Vertex 1 is not adjacent to at least three of the other vertices.

Suppose that statement (a) holds, and 1 is adjacent to 2, 3, and 4. If two of the vertices 2, 3, and 4 (for example 2 and 3) are adjacent, then subgraph H with the set of vertices $\{1, 2, 3\}$ is complete.

Suppose that statement (b) holds, and 1 is not adjacent to 2, 3, and 4. If two of the vertices 2, 3, and 4 (for example 2 and 3) are not adjacent, then subgraph H with the set of vertices $\{1, 2, 3\}$ is without edges. $\triangle$

8.4 Paths and Connected Graphs

Let G be a graph with the set of vertices V , and the set of edges E . A *walk* in graph G is a finite sequence

$$W = v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n,$$

where $v_0, v_1, \dots, v_n \in V$, and $e_1, e_2, \dots, e_n \in E$, and for any $i \in \{1, 2, \dots, n\}$, the vertices v_{i-1} and v_i are incident to the edge e_i . The vertices v_0 and v_n are called, respectively, the *origin*, and the *terminus* of walk W , while the vertices $v_1, v_2, \dots, v_{n-1}$ are called *internal vertices* of the walk.

The number of edges in the walk is called the *length* of the walk. Origin v_0 and terminus v_n of walk $W = v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n$, need not be distinct. If $v_0 = v_n$, then we say that walk W is *closed*. If $v_0 \neq v_n$, then walk W is *open*. A vertex v_0 can be considered as a *trivial walk*.

A *trail* is a walk $W = v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n$ without repetition in the sequence of its edges $e_1, e_2, \dots, e_n$.

If the vertices $v_0, v_1, \dots, v_n$ of walk $W = v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n$ are distinct, then W is called a *path* from v_0 to v_n .

Example 8.4.1. Let us consider a complete graph with the set of vertices $V = \{v_1, v_2, v_3, v_4, v_5, v_6\}$, see Figure 8.4.1.

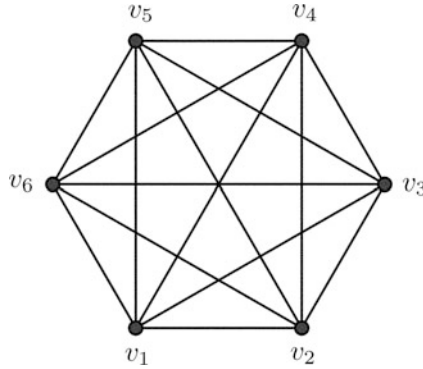


Fig. 8.4.1

For any $i, j \in \{1, 2, 3, 4, 5, 6\}$ let us denote by e_{ij} the edge connecting v_i and v_j . Following is a list of walks, trails, and paths.

$W_1 = v_1 e_{12} v_2 e_{23} v_3 e_{36} v_6 e_{62} v_2 e_{24} v_4 e_{45} v_5$ is an open walk and a trail, but it is not a path (the vertex v_2 is repeated twice).

$W_2 = v_1 e_{12} v_2 e_{25} v_5 e_{51} v_1 e_{12} v_2 e_{23} v_3 e_{31} v_1$ is a closed walk, but it is not a trail (the edge e_{12} is repeated twice in W_2).

$W_3 = v_1 e_{12} v_2 e_{26} v_6 e_{63} v_3 e_{35} v_5 e_{54} v_4$ is a path. $\triangle$

Connected vertices and connected graphs. A vertex u is *connected* to a vertex v in a graph G if there is a path in G from u to v . Graph G is *connected* if every two of its vertices are connected. A graph that is not connected is called *disconnected*.

A *connected component* of an undirected graph G is a subgraph H of the graph G , such that any two vertices of H are connected to each other by paths, and no vertex in H is connected to an additional vertex in the supergraph G .

Example 8.4.2. The graphs presented in Figures 8.2.1, 8.2.2, and 8.2.3 are disconnected. The graphs presented in Figures 8.2.4, 8.2.8–8.2.12, 8.3.1, 8.3.2, and 8.4.1 are connected. $\triangle$

Theorem 8.4.3. Let G be a graph with a set of vertices V . Graph G is disconnected if and only if there exist two nonempty sets V_1 and V_2 , such that $V = V_1 \cup V_2$, $V_1 \cap V_2 = \emptyset$, and there exists no edge in G with end vertices $v_1 \in V_1$ and $v_2 \in V_2$.

Proof. The statement of Theorem 8.4.3 is obvious. $\square$

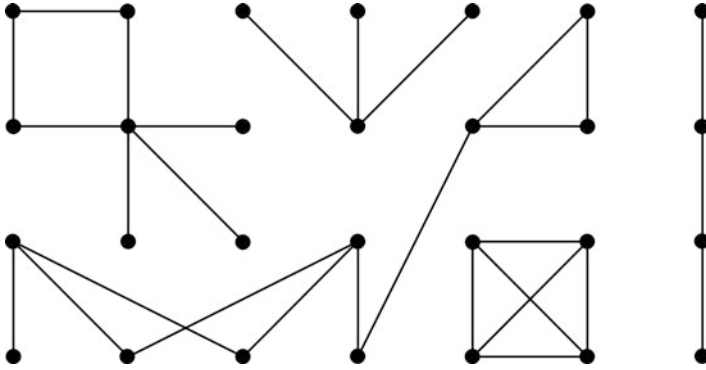


Fig. 8.4.2

Example 8.4.4. The graph in Figure 8.4.2 has five connected components. $\triangle$

Theorem 8.4.5. Suppose that a simple graph G with n vertices has more than $(n-1)(n-2)/2$ edges. Then, G is connected.

Proof. It is sufficient to prove that any disconnected graph with n vertices has at most $(n-1)(n-2)/2$ edges. Let V be the set of vertices of a disconnected graph $\tilde{G}$, and $|V| = n$. Then, there exist two disjoint nonempty subsets V_1 and V_2 of the set V , such that no vertex from V_1 is connected to a vertex from V_2 . Let us denote $k = |V_1|$, where $k \in \{1, 2, \dots, n-1\}$. It follows that the number of edges of the graph $\tilde{G}$ is less than or equal to $\binom{k}{2} + \binom{n-k}{2}$. Note that the inequality

$$\binom{k}{2} + \binom{n-k}{2} \leq \frac{(n-1)(n-2)}{2}$$

is equivalent to $k(n-k) \leq n-1$, and holds for any $k \in \{1, 2, \dots, n-1\}$, with the equality for $k \in \{1, n-1\}$. Hence, the proof is completed. $\square$

8.5 Isomorphic Graphs

Let G be a graph with the set of vertices V , and the set of edges E , and let G' be a graph with the set of vertices V' , and the set of edges E' . The graphs G and G' are *isomorphic* if there exist two bijective functions

$$f : V \rightarrow V', \quad g : E \rightarrow E',$$

such that for each $v \in V$, and each $e \in E$, if v is an endpoint of the edge e , then $f(v)$ is an endpoint of the edge $g(e)$.

Example 8.5.1. Let $V = \{1, 2, 3, 4, 5\}$ be the set of vertices, and $\{i, j\}$ be the edge with incident vertices i and j , for any $i, j \in V$. Let us define G_1 and G_2 to be graphs with the same set of vertices V , and with the sets of edges $E_1 = \{e_1, e_2, e_3, e_4, e_5\}$, and $E_2 = \{\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4, \varepsilon_5\}$, respectively, where:

$$\begin{aligned} e_1 &= \{1, 3\}, \quad e_2 = \{2, 4\}, \quad e_3 = \{3, 5\}, \quad e_4 = \{4, 1\}, \quad e_5 = \{5, 2\}, \\ \varepsilon_1 &= \{1, 2\}, \quad \varepsilon_2 = \{2, 3\}, \quad \varepsilon_3 = \{3, 4\}, \quad \varepsilon_4 = \{4, 5\}, \quad \varepsilon_5 = \{5, 1\}. \end{aligned}$$

Graphs G_1 and G_2 are given in Figures 8.5.1 and 8.5.2, where every edge $\{i, j\}$ is presented as a segment with endpoints i and j .

It is obvious that graphs G_1 and G_2 are complements to each other. We shall prove that these two graphs are also isomorphic.

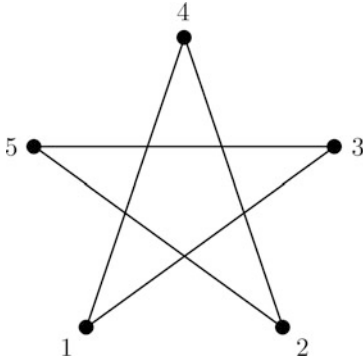


Fig. 8.5.1

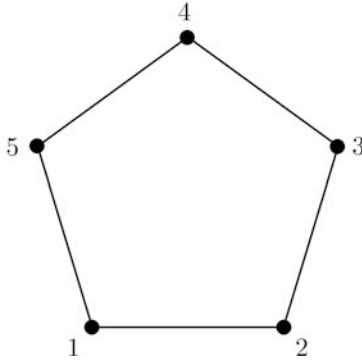


Fig. 8.5.2

Let us define the functions $f : V \rightarrow V$ and $g : E_1 \rightarrow E_2$ as follows:

$$f : \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 1 & 2 \end{pmatrix}, \quad g : \begin{pmatrix} e_1 & e_2 & e_3 & e_4 & e_5 \\ \varepsilon_1 & \varepsilon_2 & \varepsilon_3 & \varepsilon_4 & \varepsilon_5 \end{pmatrix}$$

It is easy to see that for any $i, j \in \{1, 2, 3, 4, 5\}$, if i is an endpoint of the edge e_j , then $f(i)$ is an endpoint of the edge $g(e_j) = \varepsilon_j$. Hence, G_1 and G_2 are isomorphic graphs. $\triangle$

An obvious consequence of the definition is that isomorphic graphs must have the same number of vertices, the same number of edges, the same degrees for corresponding vertices, the same number of connected components, and the same number of loops. In general, it is easier to prove that two graphs are not isomorphic than that they are isomorphic.

8.6 Euler's Graphs

An *Eulerian trail* is a trail in a graph which visits every edge exactly once. A graph with an Eulerian trail is an *Eulerian graph*. An *Eulerian cycle* is an Eulerian trail which starts and ends at the same vertex. The following theorem gives a characterization of graphs that have Eulerian trails. Note that we consider graphs with a finite number of vertices and edges.

Theorem 8.6.1. *An Eulerian cycle exists in a connected graph G if and only if all vertices of G have even degrees.*

Proof. It is obvious that the connectedness of graph G and the evenness of all degrees of its vertices are necessary for the existence of an Eulerian trail. We shall prove that these conditions are also sufficient.

Suppose we begin a walk W_1 at some vertex A , and continue it as far as possible. From any vertex we depart on an edge we have not visited before. Since there are a finite number of edges, the walk must stop after a few steps. Since there are an even number of edges incident to any vertex, there is always an exit from any vertex except A . Hence, the walk must stop at vertex A . If the trail visited all the edges, an Eulerian cycle is obtained. If it did not, and since graph G is connected, there exists a vertex B , lying on W_1 , with an even number of adjacent edges not lying on W_1 . Now we start a new walk W_2 from vertex B , using only edges not lying on W_1 . As before, we can conclude that walk W_2 must stop at the starting vertex B . The longer cyclic trail $W_1 \cup W_2$ can be obtained by following W_1 from A to B , then following W_2 , and finally following the remaining part of W_1 from B to A . If $W_1 \cup W_2$ does not contain all the edges, the cycle can be enlarged the same way till all the edges are visited. $\square$

Similarly we can prove the following theorem.

Theorem 8.6.2. *A connected graph G has an Eulerian trail with the initial vertex $A \in V(G)$, and the ending vertex $B \in V(G)$, if and only if A and B are the only odd vertices of G .*

Proof. If we add a new edge that connects vertices A and B , then all the vertices of the new graph $\tilde{G}$ are even. Hence, $\tilde{G}$ satisfies the conditions of Theorem 8.6.1. Consequently, an Eulerian cycle exists. By removing the edge connecting A and B , we get an Eulerian trail with the given properties. $\square$

Example 8.6.3. The graphs presented in Figures 8.5.1 and 8.5.2 have Eulerian trails. $\triangle$

Example 8.6.4. A complete graph with n vertices has an Eulerian trail if and only if n is odd. $\triangle$

Example 8.6.5. Note that among the Platonic graphs only one associated with a octahedron has all vertices with even degrees. Hence, an Eulerian trail exists for a octahedron, and does not exist for a tetrahedron, hexahedron, icosahedron, and dodecahedron. $\triangle$

8.7 Hamiltonian Graphs

A *Hamiltonian path* or *traceable path* is a path in an undirected or directed graph that visits each vertex exactly once. A *Hamiltonian cycle* is a cycle that visits each vertex exactly once (except for the vertex that is both the start and end, which is visited twice). Let us note that a Hamiltonian cycle, in general, does not visit all the edges.

A graph that contains a Hamiltonian path is called a *traceable graph*. A graph that contains a Hamiltonian cycle is called a *Hamiltonian graph*.

A graph is *Hamiltonian-connected* if for every pair of vertices there is a Hamiltonian path between the two vertices.

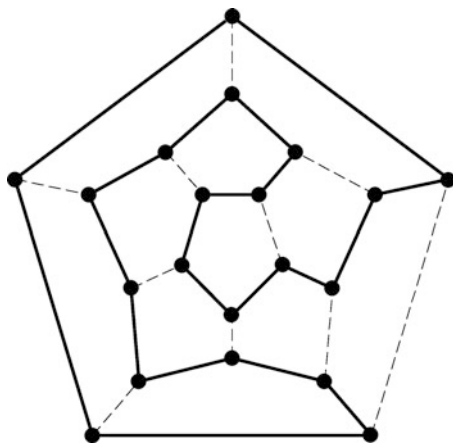


Fig. 8.7.1

Example 8.7.1. The Platonic graph determined by the vertices and edges of a regular dodecahedron is given in Figure 8.2.12. A Hamiltonian cycle of this graph is presented in Figure 8.7.1. $\triangle$

The difference and analogy between Eulerian trails and Hamiltonian cycles is that an Eulerian trail passes through all the edges exactly once, while a Hamiltonian cycle visits all the vertices exactly once. A simple characterization of graphs that have Eulerian trails is given by Theorem 8.6.1. The existence or nonexistence of Hamiltonian cycles is an important problem in graph theory. However, a general characterization of graphs that have

Hamiltonian cycles has not been found so far. The existence of Hamiltonian cycles can be proved in some special cases and with additional conditions. A result of this type is given by the following theorem.

Theorem 8.7.2. (Dirac) *Let G be a simple graph with n vertices. If the degree of any vertex is not less than $n^2/2$, then G is a Hamiltonian graph.*

8.8 Regular Graphs

A *regular graph* is a graph whose vertices all have the same degree. A regular graph with vertices of degree k is called a k -regular graph.

Example 8.8.1. Let G be a complete graph with n vertices. The degree of any vertex is $n - 1$, and hence G is a regular graph. $\triangle$

Example 8.8.2. All five Platonic graphs are regular. Combinatorial data for the Platonic graphs are given in Table 8.2.1, and we conclude that: a tetrahedron is a 3-regular graph; an octahedron is a 4-regular graph; a hexahedron is a 3-regular graph; an icosahedron is a 5-regular graph; a dodecahedron is a 3-regular graph. $\triangle$

Example 8.8.3. *Graph of n -permutations.* Let $G = (V, E)$ be a graph defined as follows. The set of vertices V is the set of all permutations of the set $\mathbb{N}_n = \{1, 2, \dots, n\}$. Two vertices $v_1 = i_1 i_2 \dots i_n$ and $v_2 = j_1 j_2 \dots j_n$ are connected by an edge, if and only if the permutation v_2 can be obtained from the permutation v_1 by the transposition of two elements.

The number of vertices of graph G is $|V| = n!$. The number of edges is $|E| = \frac{n!}{2} \binom{n}{2}$. All vertices of graph G have the same degree $\binom{n}{2}$. Hence, graph G is regular. $\triangle$

A *strongly regular graph* is a regular graph where every adjacent pair of vertices has the same number of neighbors in common, and every nonadjacent pair of vertices has the same number of neighbors in common.

Example 8.8.4. It is easy to see that a cycle graph with 6 vertices is regular, but not strongly regular. $\triangle$

A *regular directed graph* must also satisfy the stronger condition that the *in-degree* and *out-degree* of each vertex are equal to each other. Here, the in-degree of a vertex v is defined as the number of edges incident to v and directed towards v . The out-degree of a vertex v is defined as the number of edges incident to v and directed from v towards the other end vertex.

8.9 Bipartite Graphs

A *bipartite graph* is a graph G whose vertices can be divided into two disjoint nonempty sets V_1 and V_2 , such that every edge connects a vertex in V_1 to one in V_2 . The sets V_1 and V_2 are called the *sides* of bipartite graph G . If a bipartite graph is not connected, it may have more than one bipartition.

Example 8.9.1. A bipartite graph with the sides $V_1 = \{1, 2, 3, 4, 5\}$ and $V_2 = \{A, B, C, D, E\}$ is given in Figure 8.9.1. A possible interpretation of the graph is the following. The vertices from set V_1 represent five workers, and the vertices from set V_2 represent five possible positions. An edge that connects a vertex $x \in V_1$ and a vertex $X \in V_2$ means that the worker x is qualified for the position X . $\triangle$

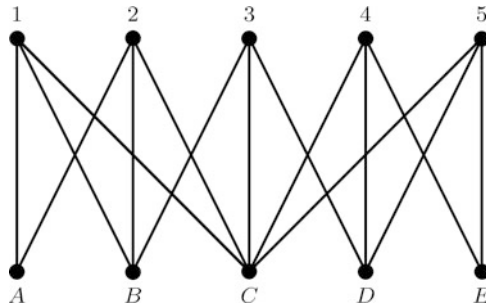


Fig. 8.9.1

Let G be a bipartite graph. If the two subsets (sides) V_1 and V_2 have equal cardinality, then graph G is called a *balanced bipartite graph*. If all vertices on the same side of the bipartition have the same degree, then G is called *biregular*. A simple characterization of bipartite graphs is given by the following theorem.

Theorem 8.9.2. *A graph is bipartite if and only if it does not contain an odd cycle (i.e., a cycle with an odd number of edges).*

Proof. Let G be a graph without odd cycles. It is sufficient to prove that the set of vertices of any connected component of graph G can be partitioned into two nonempty disjoint subsets V_1 and V_2 such that no two vertices from different subsets are connected by an edge. Let us consider a connected component K of graph G , and a vertex v_0 of component K . For any other vertex v of K there is a walk W from v_0 to v . Let us define $v \in V_1$, if walk

W passes through an odd number of edges, and $v \in V_2$ if walk W passes through an even number of edges. We put also $v_0 \in V_2$. Let us note that the subsets V_1 and V_2 are well defined. Suppose, on the contrary, that there are two distinct walks W_1 and W_2 from v_0 to v , such that one of them passes through an even number of edges, while the other one passes through an odd number of edges. Then it is obvious that there exists an odd cycle containing the vertex v_0 . The obtained contradiction completes the proof. $\square$

Let G be a bipartite graph with the set of vertices $V = V_1 \cup V_2$, where $V_1 \cap V_2 = \emptyset$, $V_1 \neq \emptyset$, $V_2 \neq \emptyset$, and such that every vertex in V_1 is connected with all the vertices in V_2 . Graph G is called a *complete bipartite graph on m and n vertices*, and is denoted by $K_{m,n}$.

Example 8.9.3. The total number of edges in the graph $K_{m,n}$ is mn .

Example 8.9.4. The graph presented in Figure 8.9.1 is a bipartite graph, while it is not a complete bipartite graph. $\triangle$

Exercises

8.1. There are at least two bus lines and a set of bus stops in a city, such that the following conditions are satisfied.

- (a) Every two bus stops are connected by a line.
- (b) For every two lines there is exactly one common bus stop.
- (c) On every line there are exactly n bus stops.

How many bus lines are there?

8.2. There are 57 bus lines in a city and the following conditions are satisfied:

- (a) Every two bus stops are connected by a line.
- (b) For any two lines there is exactly one common bus stop.
- (c) On every line there are at least three bus stops.

How many bus stops are there on every line?

8.3. Suppose that n cities are to be connected by a network of one-way roads such that the following three conditions are satisfied:

- (a) For every two cities A and B there is exactly one road from A to B .
- (b) For every two cities A and B there is exactly one city that can be reached directly (by traveling along one road) from both A and B .

(c) For every two cities A and B there is exactly one city from which both A and B can be reached (by traveling along one road).

Find all positive integers $n > 2$ for which such a network of roads can be built.

8.4. There are k cities in a state that are to be connected by a network of phone lines such that the following conditions are satisfied:

- (a) Every phone line connects two cities directly.
- (b) There are exactly $k - 1$ lines.
- (c) Every two cities can be connected not necessarily by a direct line.

Prove that there is a city directly connected to exactly one of the other cities.

8.5. There are $n \geq 2$ cities in a state that are to be connected by a network of phone lines such that the following conditions are satisfied:

- (a) Every phone line connects two cities.
- (b) There are exactly $n - 1$ lines.
- (c) Every two cities can be connected not necessarily by a direct line.

How many ways can such a network of phone lines be made?

8.6. Every city in a state is connected to exactly three other cities by direct air flights. One can fly from each city to any other city with at least one stop. Determine the maximal number of cities in the state.

8.7. A network of roads is built in a state such that there are exactly three roads from any city. A passenger is traveling from city A using one of the three possible roads. When he comes to the next city, he turns to the left, in the next city he turns to the right, then in the next city to the left, etc. Prove that after some time the passenger will come back to city A .

8.8. There are more than two cities in a state. Some cities are connected by simple roads. The road AB is simple if there is no other city on it except A and B . Suppose that no two simple roads intersect. The network of roads is such that *for any three cities A , B , and C the following condition is satisfied: starting from A one can visit B without visiting C* . Prove that a direction can be defined on every simple road such that, for each of the three cities A , B , and C , the previous statement remains valid.

8.9. In a group of n people there are no three people such that every two of them are friends. Friendship is a symmetric relation. Prove that the number of pairs of friends is not greater than $\lfloor n^2/4 \rfloor$.

8.10. At a chess tournament 25 chess players are to play exactly one game against each other. Determine the minimal number of games that should be finished such that for every group of three chess players two of them have already played a game against each other.

8.11. Let us suppose that n mathematicians attend a conference. Some of them are friends with some of the others. Friendship is a symmetric relation. For every two mathematicians who are not friends there are exactly two common friends, and for every two friends there is no common friend. Prove that all the mathematicians have the same number of friends among the conference participants.

Chapter 9



Graph Theory: Part 2

9.1 Trees and Forests

In this section we shall focus on graphs without cycles. A *tree* is a connected graph that has no cycles. A *forest* is a disconnected graph that has no cycles. It is obvious that any connected component of a forest is a tree.

Example 9.1.1. A forest with 5 connected components (trees) is given in Figure 9.1.1. $\triangle$

Example 9.1.2. It follows from Theorem 8.9.2 that every tree is a bipartite graph. The same holds true for any forest. $\triangle$

Let G be a connected graph and let e be an edge in G . Edge e is called a *bridge* if it has the property that after being deleted (removed), the obtained graph becomes disconnected.

It is obvious that a tree with n vertices has $n - 1$ edges. The following theorem (whose proof can be an easy exercise for the reader) gives several equivalent properties that characterize a tree.

Theorem 9.1.3. *Let G be a graph that has exactly n vertices. Then the following statements are equivalent:*

- (a) G is a tree.
- (b) G has no cycles and has exactly $n - 1$ edges.
- (c) G is a connected graph with exactly $n - 1$ edges.

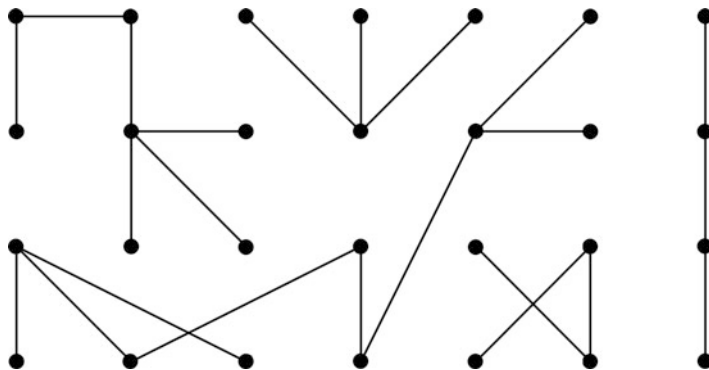


Fig. 9.1.1

- (d) G is a connected graph and every edge is a bridge.
- (e) Any two vertices of G are connected by exactly one simple path.
- (f) Graph G has no cycles, but by adding a new edge it becomes a graph with exactly one cycle.

Graph enumeration. Let G be a graph with n vertices, and let $V(G)$ be the set of its vertices. An *enumeration* of graph G is a bijection $f : \{1, 2, \dots, n\} \rightarrow V(G)$. The pair (G, f) is then called a *labeled graph*. Two labeled graphs are *isomorphic* if there is an isomorphism between G_1 and G_2 that preserves enumeration.

Example 9.1.4. All nonisomorphic labeled graphs on 3 vertices are given in Figure 9.1.2. $\triangle$

A *rooted tree* is a tree in which one vertex has been designated the root. The edges of a rooted tree can be assigned a natural orientation, either away from or toward the root. The obtained structure is then called a *directed rooted tree*. A single vertex e can also be considered a *trivial rooted tree* with root e .

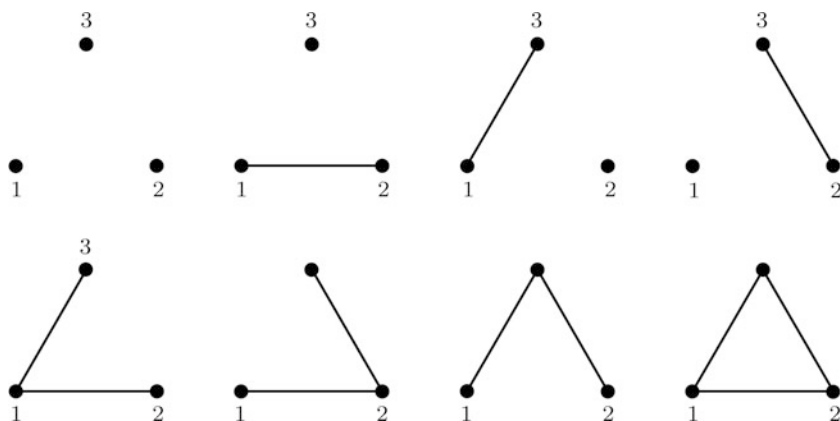


Fig. 9.1.2

Example 9.1.5. A directed rooted tree is presented in Figure 9.1.3. $\triangle$

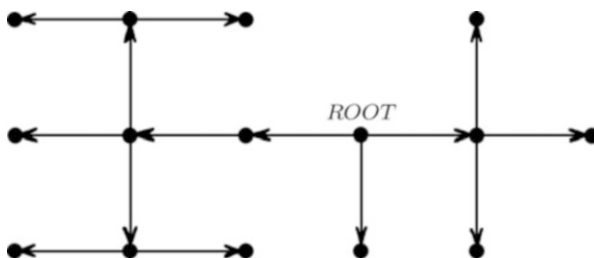


Fig. 9.1.3

It is obvious that the number of labeled graphs with n vertices is $2^{\binom{n}{2}}$. A more interesting result is given by the following theorem.

Theorem 9.1.6. (Cayley 1889) *The number of non-isomorphic labeled trees with $n \geq 2$ vertices is equal to n^{n-2} .*

Proof. Let T_n be the number of non-isomorphic labeled trees with n vertices. Let S be the set of all sequences of directed edges that can be added to an empty graph on n vertices such that a directed tree is formed with the orientation of its edges, for example, away from the root. We shall count the number of elements of set S in two different ways.

One way of counting is to form a sequence $s \in S$ as follows. First we choose one of T_n undirected trees, then choose one of its n vertices that will

be designated the root, and finally choose one of $(n-1)!$ possible sequences of ordered edges. Hence, the number of elements of set S is

$$|S| = T_n n(n-1)! = T_n n!. \quad (9.1.1)$$

Another way to form a sequence $s \in S$ is to consider adding the edges one by one starting with an empty graph on n vertices. There are $n(n-1)$ choices for the first directed edge. Suppose that k directed edges are already added, where $k \in \{1, 2, \dots, n-2\}$. The starting point of the next directed edge can be any vertex, and the endpoint can be any of the vertices except the starting one and the endpoints of the already added edges. Hence, there are $n(n-k-1)$ possibilities for the next directed edge. The total number of sequences of directed edges is

$$|S| = \prod_{k=0}^{n-2} n(n-k-1) = n^{n-1}(n-1)! = n^{n-2}n!. \quad (9.1.2)$$

It follows from (9.1.1) and (9.1.2) that $T_n = n^{n-2}$. $\square$

9.2 Planar Graphs

A *plane graph* is a graph drawn on the plane in such a way that its edges intersect only at their endpoints. A *planar graph* is a graph that is isomorphic to a plane graph. Note that isomorphic graphs are considered to be the same.

Example 9.2.1. The graph presented in Figure 9.2.1 is isomorphic to the plane graph in Figure 9.2.2, and hence, the graph in Figure 9.2.1 is planar. Similarly, the graph in Figure 9.2.3 is isomorphic to the plane graph in Figure 9.2.4, and hence, the graph in Figure 9.2.3 is also planar. $\triangle$

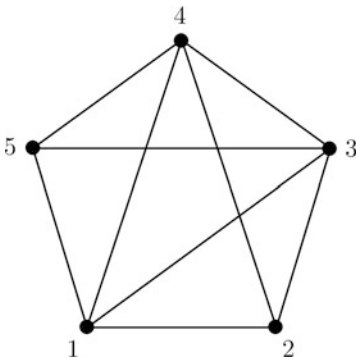


Fig. 9.2.1

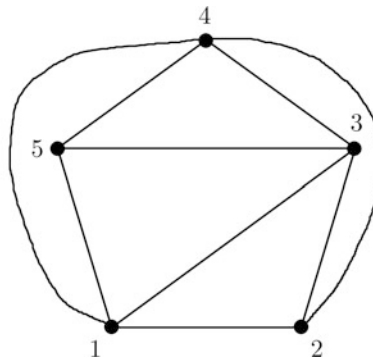


Fig. 9.2.2

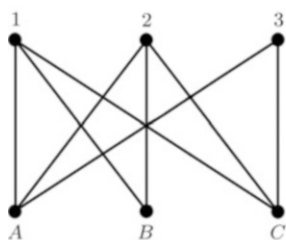


Fig. 9.2.3

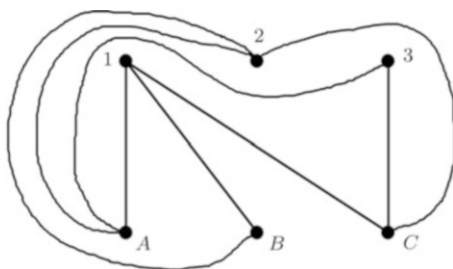


Fig. 9.2.4

Example 9.2.2. Let us consider the graphs presented in Figures 9.2.5 and 9.2.6. The graph in Figure 9.2.5 is obtained from the graph in Figure 9.2.1 by adding the edge $25 := (2, 5)$, and the graph in Figure 9.2.6 is obtained from the graph in Figure 9.2.3 by adding the edge $3B := (3, B)$. Any attempt to draw a plane graph that is isomorphic to the graph in Figure 9.2.5 would be unsuccessful. The same holds for the graph in Figure 9.2.6. $\triangle$

Note that the graph in Figure 9.2.6 is isomorphic to the hexagonal graph presented in Figure 9.2.8. The graphs given in Figures 9.2.5 and 9.2.8 will play an important role in characterizing planar graphs.

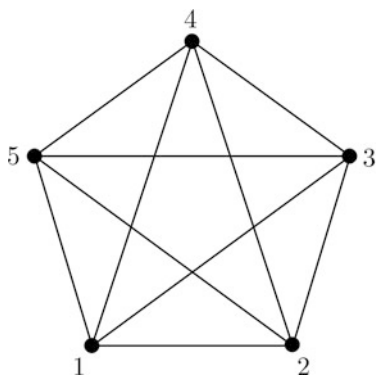


Fig. 9.2.5

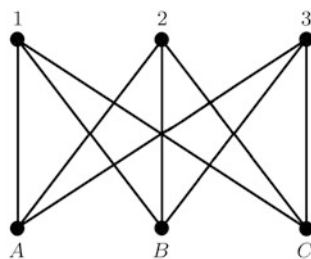


Fig. 9.2.6

In order to prove that the graphs presented in Figures 9.2.5 and 9.2.6 (or 9.2.8) are not planar we need the following result.

Theorem 9.2.3. (Jordan Curve Theorem) *Any continuous closed curve C in the plane divides the plane into an outer part and an inner part. If a point A belongs to the inner part, a point B belongs to the outer part, and a continuous curve L lies in the plane and connects the points A and B , then L intersects C . See Figure 9.2.7.*

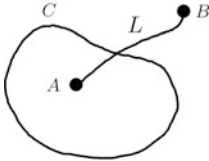


Fig. 9.2.7

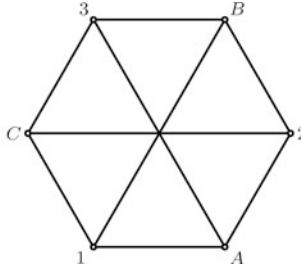


Fig. 9.2.8

Theorem 9.2.4. *The graphs in Figures 9.2.5 and 9.2.6 are not planar.*

Proof. The graph in Figure 9.2.5 is a complete graph with 5 vertices. We suppose that the vertices 1, 2, 3, 4, and 5 form a cycle in this order. Let us also assume that edges 13 and 35 lie inside the pentagon 12345. (The other cases can be considered analogously.) Then it is obvious that the edges 14 and 24 should lie outside the pentagon, see Figure 9.2.2. (If, for example, edge 14 lies inside the pentagon, then edges 14 and 35 intersect each other.) Now, it is not possible to draw edge 25 in such a way that this edge has no point of intersection with any other edge. Indeed, if edge 25 lies inside the pentagon, then 13 and 25 intersect each other, while if edge 25 lies outside the pentagon, then edges 14 and 25 intersect each other. Hence, the graph in Figure 9.2.5 is not planar. We can similarly prove that the graph in Figure 9.2.6 is not planar. $\square$

An important question related to planar graphs is giving their characterization, i.e., finding property $\mathcal{P}$, such that every planar graph has property $\mathcal{P}$, and every nonplanar graph does not have this property.

In order to formulate a criterion for planar graphs we shall first introduce two new notions. Let a graph G be given and suppose that we add a few new vertices on the edges of graph G . This divides some edges into several new edges. Let us denote the new graph by G_1 . This operation transforms graph G into graph G_1 , and is called *expanding* graph G into G_1 , see Figure 9.2.9.

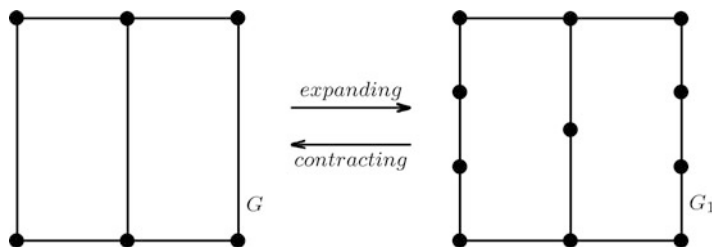


Fig. 9.2.9

Suppose now that a graph G_1 contains a few paths with no additional edges at the intermediate vertices. Let us remove all intermediate vertices on these paths that had become edges this way. Let us denote the obtained graph by G . This operation transforms G_1 into G and is called *contracting* graph G_1 into G , see Figure 9.2.9.

Theorem 9.2.5. (Kuratowski) *A graph G is planar if and only if it does not contain a subgraph that can be contracted to a complete graph on 5 vertices, or to the hexagonal graph given in Figure 9.2.8.*

The proof of Theorem 9.2.5 can be found, for example, in the book by W.T. Tutte [20].

9.3 Euler's Theorem

Euler's polyhedral formula related to the number of vertices, edges, and faces of any polyhedra in space is given by the following theorem.

Theorem 9.3.1. (Leonhard Euler) *Let us denote by v , e , and f the number of vertices, edges, and faces, respectively, of a polyhedra in space. Then,*

$$v + f = e + 2. \quad (9.3.1)$$

Using the data from Table 8.2.1 it is easy to check that formula (9.3.1) holds for five platonic solids. We shall prove this equality in a more general formulation related to planar graphs. Let us first introduce the notion of *stereographic projection*.

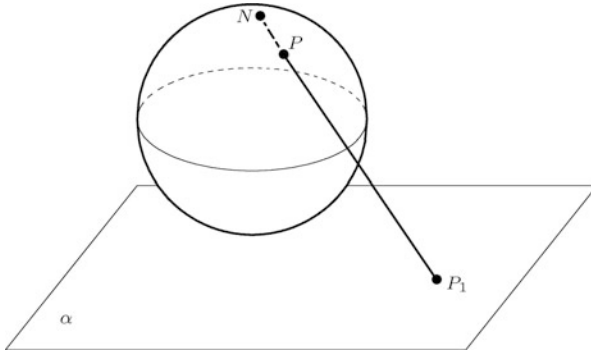


Fig. 9.3.1

Let $\mathbb{S}^2$ be a sphere, $N \in \mathbb{S}^2$ be a fixed point lying on the sphere, and $S \in \mathbb{S}^2$ be the diametrically opposite point. Let α be the plane defined by the conditions $S \in \alpha$ and $NS \perp \alpha$. For any point $P \in \mathbb{S}^2 \setminus \{N\}$, let us denote by $f(P) = P_1$ the point of intersection of the line NP and the plane α , see Figure 9.3.1. The function f is then called the *stereographic projection* of the sphere $\mathbb{S}^2 \setminus \{P\}$ onto plane α . Note that function f is a bijection. It is neither isometric nor area-preserving. But this function is conformal, i.e., it preserves angles. These geometrical properties of the stereographic projection are less important in graph theory.

Note that any graph determined by the skeleton (vertices and edges) of a polyhedra can be spanned on a sphere. Using stereographic projection it is easy to conclude that a graph is planar if and only if it can be spanned on a sphere (without points of intersection of its edges).

Every plane graph divides the plane into polygonal pieces whose edges are not necessarily represented by straight lines. These polygonal pieces, one of which is unbounded, will be called *faces*. Euler's theorem can now be formulated as follows:

Theorem 9.3.2. *If we denote by v , e , and f , the number of vertices, edges, and faces, respectively, of a connected plane graph, then equality (9.3.1) holds.*

Remark. A connected plane graph such that each edge is a side of its face is called a *polygonal graph*. Note also that the connected plane graph in Theorem 9.3.2 is not necessarily isomorphic to the skeleton of a polyhedra or a polygonal graph.

The *proof* of Theorem 9.3.2 is given by induction on the number of faces f . Consider a graph that is a closed cycle with v vertices. Then, we have

$e = v$, and the number of faces is $f = 2$, i.e., there are two polygonal areas, the inner part and the unbounded outer part. Obviously, equality (9.3.1) holds.

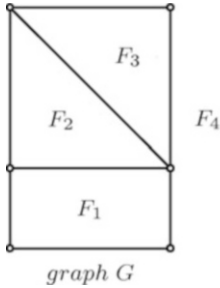


Fig. 9.3.2

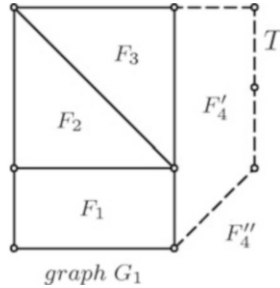


Fig. 9.3.3

Let G be a plane graph with $f \geq 2$ faces. Let us denote by v and e the number of vertices and edges, respectively, of graph G . Suppose that (9.3.1) holds for graph G . Let $V(G)$ be the set of vertices of graph G , and $F_1, F_2, \dots, F_f$ be the faces of G , see Figure 9.3.2. Let us consider a graph G_1 obtained from G by adding a path T contained in one of the faces F_i , and such that the ends of T belong to $V(G)$, see Figure 9.3.3. Let us denote by v_1 , e_1 , and f_1 the number of vertices, edges, and faces, respectively, of graph G_1 . If path T consists of x edges, then it contains $x - 1$ intermediate points, and hence $e_1 = e + x$, $v_1 = v + x - 1$. Path T divides face F_i into two parts F'_i and F''_i , that is $f_1 = f + 1$. Hence,

$$v_1 + f_1 = (v + x - 1) + (f + 1) = (v + f) + x = e + 2 + x = e_1 + 2,$$

i.e., equality (9.3.1) also holds for plane graph G_1 with $f_1 = f + 1$ faces. $\square$

9.4 Dual Graphs

Let G be a polygonal graph and let us denote its faces by $F_1, F_2, \dots$, see Figure 9.4.1. Obviously one of these faces is infinite. Let us define a new graph denoted by G^* , as follows.

We take the set $\{F_1^*, F_2^* \dots\}$ as the set of vertices of graph G^* . Here, F_i^* is an arbitrarily chosen point on the face F_i . Two vertices F_i^* and F_j^* of graph G^* are connected with an edge e_{ij}^* if the faces F_i and F_j of graph G have the common edge e_{ij} . The edge e_{ij}^* should intersect only e_{ij} among the edges of graph G . Moreover, if two faces F_i and F_j of graph G have two or more common sides, then F_i^* and F_j^* are connected by the same number of edges that correspond to the common sides of faces F_i and F_j . Graph G^* is dual to graph G .

Example 9.4.1. Consider the polygonal graph G with four faces and five vertices that is given in Figure 9.4.1. Its dual G^* with four vertices and five faces is given in Figure 9.4.2. Note that the faces F_4 and F_2 have exactly one common side, and hence F_4^* and F_2^* are connected by only one edge. The faces F_4 and F_1 (or F_3) have two common sides, and hence the vertices F_4^* and F_1^* (or F_3^*) are connected by two edges. $\triangle$

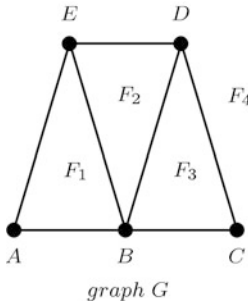


Fig. 9.4.1

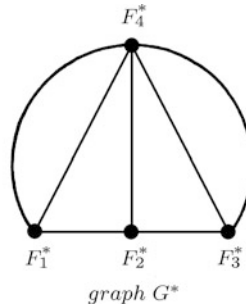


Fig. 9.4.2

Remark 9.4.2. R Note that we have defined a graph that is dual to a polygonal graph, i.e., dual to a plane graph (or to a planar graph already embedded in the plane). A planar graph can be embedded in the plane in different ways, and hence a planar graph may have more than one dual graph. For a polygonal graph its dual is uniquely determined.

We shall now list some properties of dual graphs.

- (a) The dual graph G^* of polygonal graph G is a polygonal graph.
- (b) If G^* is the dual of a plane graph G , then G is the dual of G^* .
- (c) The degree of any vertex of graph G^* is equal to the number of edges of the corresponding face in graph G .
- (d) Let G and G^* be dual graphs. Let v , e , and f be the number of vertices, edges, and faces of graph G , respectively, and v^* , e^* , and f^* be the number of vertices, edges, and faces of graph G^* , respectively. Then, the following equalities hold:

$$e^* = e, \quad v^* = f, \quad f^* = v. \quad (9.4.1)$$

Completely regular graphs. We have already defined a regular graph as a graph whose vertices all have the same degree, see Section 8.8. Let G be a polygonal regular graph, such that the dual graph G^* is also regular. Then, we say that G is *completely regular*.

Let k be the degree of any vertex of a completely regular graph G , and k^* be the degree of any vertex of the dual graph G^* . By definition of the

dual graph it follows that k^* is the number of boundary edges of each face of graph G . It follows also that k is the number of boundary edges of each face of graph G^* .

The only completely regular graphs are the *tetrahedron*, the *octahedron*, the *hexahedron* (the cube), the *icosahedron*, and the *dodecahedron*. Here, we consider the plane embeddings of the graphs of platonic solids, as they are given in Figures 8.2.8–8.2.12.

Moreover, the tetrahedron is its own dual, the octahedron and the cube are duals of each other, while the icosahedron and the dodecahedron are duals of each other, see Exercises 9.1 and 9.2.

9.5 Graph Coloring

A graph coloring is an assignment of colors to some elements of a graph. The conditions that can be imposed here are usually formulated in a very simple way. Following are some examples.

(a) A *vertex coloring* is an assignment of colors to the vertices of a graph in such a way that no two adjacent vertices are of the same color.

(b) An *edge coloring* is an assignment of colors to the edges of an undirected graph G in such a way that no two adjacent edges are of the same color. Note that an edge coloring problem can be reformulated to a problem of coloring the vertices of the graph $L(G)$ obtained from G as follows. Let $E(G) = \{e_1, e_2, \dots, e_n\}$ be the set of edges of G . Then, the vertices of $L(G)$ are denoted by $E_1, E_2, \dots, E_n$. Moreover, two vertices E_i and E_j of graph $L(G)$ are adjacent if and only if e_i and e_j are adjacent edges of graph G . Graph $L(G)$ is then called the *line graph* of G .

(c) A *face coloring of a plane polygonal graph* G is an assignment of colors to the faces of graph G such that no two adjacent faces are of the same color. Note that two faces are adjacent if they share a common edge. Obviously, a face coloring of graph G can be reformulated as a vertex coloring of its dual G^* .

Vertex coloring. Without any additional qualification, a coloring of a graph means a labeling of its vertices with colors such that no two adjacent vertices have the same color. A *k-coloring* is a coloring using no more than k colors.

The *chromatic number* of a graph G is the smallest number of colors needed to color the vertices of graph G . The chromatic number of graph G is denoted by $\chi(G)$.

The *chromatic polynomial* of a graph G is a function $P(G, \cdot)$ that counts the number of colorings of G , i.e., $P(G, k)$ is the number of k -colorings of G .

Example 9.5.1. Let G be a graph with n vertices. It is obvious that $1 \leq \chi(G) \leq n$. If G is a graph without edges, then $\chi(G) = 1$. If G is a complete graph, then $\chi(G) = n$. $\triangle$

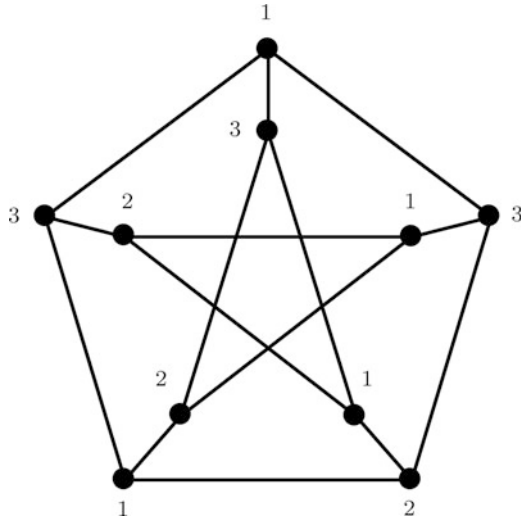


Fig. 9.5.1

Example 9.5.2. The Petersen graph G whose vertices are colored by three colors (denoted 1, 2, and 3) is given in Figure 9.5.1. A coloring with two colors is not possible, and hence $\chi(G) = 3$. $\triangle$

Example 9.5.3. Let C_n be a cycle with n vertices. Then, $\chi(C_n) = 2$ if n is even, while $\chi(C_n) = 3$ if n is odd. $\triangle$

Example 9.5.4. Let K_n be a complete graph with n vertices, T_n be a tree with n vertices, and C_n be a cycle with n vertices. Using simple combinatorial reasoning one can obtain the following equalities:

$$\begin{aligned}\chi(K_n, k) &= k(k-1) \dots (k-n+1), \\ \chi(T_n, k) &= k(k-1)^{n-1}, \\ \chi(C_n, k) &= (k-1)^n + (-1)^n(k-1). \quad \triangle\end{aligned}$$

Theorem 9.5.5. (Brooks' Theorem, 1941) Let G be a connected simple undirected graph G in which the maximum degree of its vertices is $k > 2$. Then, either G is a complete graph with $k+1$ vertices, or it has a k -coloring.

The proof of Theorem 9.5.5 can be found in the book by Tutte [20].

Edge coloring. An edge coloring with k colors is called a k -edge-coloring. The smallest number of colors needed for the edge coloring of a graph G is called the *edge chromatic number* of G , and is denoted by $\chi'(G)$. Let us formulate certain relationships between edge colorability and the maximal degree of a vertex of graph G , denoted by $\Delta = \Delta(G)$.

The inequality $\chi'(G) \geq \Delta(G)$ is obvious because all the edges incident to a vertex with maximal degree should be colored by different colors. Since an edge coloring of graph G is a vertex coloring of its line graph $L(G)$, it is obvious that

$$\chi'(G) = \chi(L(G)).$$

More subtle results are given by the following two theorems.

Theorem 9.5.6. (König's Theorem) *If G is a bipartite graph, then*

$$\chi'(G) = \Delta(G).$$

Theorem 9.5.7. (Vizing's Theorem) *The edge-chromatic number of a graph G is always equal to $\Delta(G)$ or $\Delta(G) + 1$.*

Face Coloring. Let G be a simple polygonal graph. An interesting question in graph theory is formulated as follows. What is the minimal number of colors needed to color the faces of graph G such that no two adjacent faces have the same color? Note that four colors are necessary to color the faces of a tetrahedron. This follows from the fact that any two faces of a tetrahedron are adjacent. Hence, the minimal number of colors needed to color the faces of all polygonal graphs is not less than 4.

Theorem 9.5.8. (Five-Color Theorem) *The faces of any polygonal graph G may be colored using no more than five colors in such a way that no two adjacent faces receive the same color.*

The key steps in the proof of the Five-Color Theorem are the following.

(a) The coloring problem of any polygonal graph can be reduced to that of coloring a 3-regular polygonal graph.

(b) Proof of the statement that every 3-regular graph contains a face bounded by no more than 5 edges.

(c) Graph G can be simplified by excluding faces bounded by two edges, and then excluding faces bounded by three, four, and five edges.

Details of the proof can be found, for example, in the book by Ore [14].

Theorem 9.5.9. (Four-Color Theorem) *The faces of any polygonal graph may be colored using no more than four colors in such a way that no two adjacent faces receive the same color.*

The first formulation of the four color theorem was given in 1852. Unlike the five-color theorem, which has a short elementary proof, the proof of the four-color theorem is significantly harder. There were a number of false proofs and false counterexamples.

The first proof of the four-color theorem was given by Kenneth Appel and Wolfgang Haken in 1976. The proof requires exhaustive use of a computer to check the large number of potential counterexample polygonal graphs.

Exercises

9.1. Cycle rank. Let G be a simple graph with n vertices and k edges, where $k \geq n$. Obviously G contains at least one cycle. What is the smallest number of edges that must be removed such that no cycles remain? This number is called the *cycle rank* of graph G .

9.2. Determine the cycle rank of a complete graph with n vertices.

9.3. Let k be the degree of all the vertices of a completely regular graph G , and k^* be the degree of all the vertices of the dual graph G^* . Prove that

$$(k - 2)(k^* - 2) < 4.$$

9.4. (a) Prove that the *tetrahedron*, the *octahedron*, the *hexahedron* (cube), the *icosahedron*, and the *dodecahedron* are the only completely regular graphs.

(b) Prove that the tetrahedron is its own dual, the octahedron and the cube are duals of each other, while the icosahedron and the dodecahedron are duals of each other.

9.5. Prove that the face coloring problem of any polygonal graph can be reduced to that of coloring a 3-regular polygonal graph.

9.6. Prove that any 3-regular polygonal graph has a face bounded by no more than 5 edges.

Chapter 10



Existence of Combinatorial Configurations

10.1 Magic Squares

A square table $n \times n$ filled with the positive integers $1, 2, \dots, n^2$ is called a *magic square of order n* if the sum of all numbers in each row, the sum of all numbers in each column, and the sum of all numbers in the two main diagonals are equal to each other. This constant sum is called a magic sum. The magic sum of a magic square of order n is

$$\frac{1}{n}(1 + 2 + \dots + n^2) = \frac{1}{n} \cdot \frac{1}{2}n^2(n^2 + 1) = \frac{n(n^2 + 1)}{2}.$$

Example 10.1.1. The unique magic square of order 1 is simply the positive integer 1. It is easy to see that there is no magic square of order 2. The magic squares of order 3 and 4 are given in Figures 10.1.1 and 10.1.2. $\triangle$

8	1	6
3	5	7
4	9	2

Fig. 10.1.1

16	3	2	13
5	10	11	8
9	6	7	12
4	15	14	1

Fig. 10.1.2

It is easy to prove that every magic square of order 3 can be obtained by the rotation and reflection of the square given in Figure 10.1.1. This magic square was constructed in ancient China more than 2000 years ago. A magic square of order 4 (see Figure 10.1.2) is represented on the 1514 engraving *Melancholia* by German Renaissance master Albrecht Dürer. It was proved in the 17th century that a magic square of an arbitrary order greater than 4 can be constructed. The problem of determining the total number of distinct magic squares of a given order can be solved using computers.

We shall prove here how a magic square of an arbitrary order $n > 2$ can be constructed. The construction will be given in the following cases separately: (1) $n = 2m + 1$, (2) $n = 2(2m + 1)$, and (3) $n = 4m$, where $m \in \mathbb{N}$.

Magic square of order $n = 2m + 1$. Let $n = 2m + 1$, where $m \in \mathbb{N}$. We shall label the fields of an $n \times n$ table as it is given in Figure 10.1.3, where the case $n = 5$ is represented. Rows are labeled from the bottom to the top, and columns from the left to the right. A magic square of order n can be constructed as follows. First we put the positive integer 1 in the central field of the n -th row. For $n = 5$, we have $f_{53} = 1$, see Figure 10.1.4. We then arrange the positive integers 2, 3, 4, ... according to the following rules:

(R1) Every next positive integer should be put in the top-right adjacent field, if possible. That means the following: if $f_{ij} = k$, where $1 \leq i \leq 2m$, $1 \leq j \leq 2m$, then we put $f_{i+1,j+1} = k + 1$.

(R2) If a positive integer is placed in one of the first $2m$ fields of the $(2m+1)$ -th row, for example $f_{2m+1,j} = k$ where $1 \leq j \leq 2m$, then we put the next positive integer in the field that is the intersection of the next column and the first row, i.e., $f_{1,j+1} = k + 1$.

(R3) If a positive integer is placed in one of the first $2m$ fields of the last column, for example $f_{i,2m+1} = k$ where $1 \leq i \leq 2m$, then we put the next positive integer in the field that is the intersection of the next row and the first column, i.e., $f_{i+1,1} = k + 1$.

(R4) If the previous rules cannot be applied after the step $f_{ij} = k$ (i.e., the corresponding next field is already occupied), then we put $f_{i-1,j} = k + 1$.

f_{51}	f_{52}	f_{53}	f_{54}	f_{55}
f_{41}	f_{42}	f_{43}	f_{44}	f_{45}
f_{31}	f_{32}	f_{33}	f_{34}	f_{35}
f_{21}	f_{22}	f_{23}	f_{24}	f_{25}
f_{11}	f_{12}	f_{13}	f_{14}	f_{15}

Fig. 10.1.3

17	24	1	8	15
23	5	7	14	16
4	6	13	20	22
10	12	19	21	3
11	18	25	2	9

Fig. 10.1.4

Example 10.1.2. A magic square of order 5, obtained using the rules given above, is represented by the table given in Figure 10.1.4. Let us make the following transformation of this table. We first replace every number f_{ij} by $f_{ij} - 1$, and then represent it in the base 5 number system. The obtained table is given in Figure 10.1.5. Every integer in this table is given as a two-digit base 5 integer, with the first digit possibly equal to 0. Note that each of the digits 0, 1, 2, 3, and 4 appear both in the first and the second positions in every row and every column of this table. It follows from this observation that the square is magic. Similarly we can prove that the square of an arbitrary odd order $n > 2$, obtained by applying rules (R1)–(R4), is also magic. $\triangle$

31	43	00	12	24
42	04	11	23	30
03	10	22	34	41
14	21	33	40	02
20	32	44	01	13

Fig. 10.1.5

I	III
IV	II

Fig. 10.1.6

Magic square of order $n = 2(2m + 1)$. Let us divide a square table $n \times n$ into 4 square tables $(2m + 1) \times (2m + 1)$, and denote the smaller tables by I, II, III, and IV as shown in Figure 10.1.6. Then we fill in the fields of square I with the positive integers $1, 2, \dots, (2m + 1)^2$ according to rules (R1)–(R4). We similarly fill in the fields of square II with the positive integers $(2m + 1)^2 + 1, (2m + 1)^2 + 2, \dots, 2(2m + 1)^2$; then we fill in the fields of square III with the positive integers $2(2m + 1)^2 + 1, 2(2m + 1)^2 + 2, \dots, 3(2m + 1)^2$; finally we fill in the fields of square IV with the positive integers $3(2m + 1)^2 + 1, 3(2m + 1)^2 + 2, \dots, 4(2m + 1)^2$.

The obtained square table for $n = 6$ is given in Figure 10.1.7. This square is not magic, but it will become a magic square after we make the following transformations:

- (T1) Let us mark the first m positive integers in every row except the central row of square I. In the central row of square I we mark the positive integers that belong to the columns labeled $2, 3, \dots, m + 1$. Then we replace all of the marked positive integers by the positive integer that is placed in the corresponding field of square IV, and vice versa.
- (T2) We then replace all positive integers from the last $m - 1$ columns of square III by the positive integers from the corresponding fields of square IV, and vice versa.

8	1	6	26	19	24
3	5	7	21	23	25
4	9	2	22	27	20
35	28	33	17	10	15
30	32	34	12	14	16
31	36	29	13	18	11

Fig. 10.1.7

35	1	6	26	19	24
3	32	7	21	23	25
31	9	2	22	27	20
8	28	33	17	10	15
30	5	34	12	14	16
4	36	29	13	18	11

Fig. 10.1.8

Example 10.1.3. By applying transformations (T1)–(T2) to the square given in Figure 10.1.7 we obtain the square in Figure 10.1.8. It is easy to check that the latter one is a magic square. The square table of order 10 obtained using the rules (R1)–(R4) is given in Figure 10.1.9. The magic square obtained from the square in Figure 10.1.9 by applying transformations (T1)–(T2) is given in Figure 10.1.10. $\triangle$

17	24	1	8	15	67	74	51	58	65
23	5	7	14	16	73	55	57	64	66
4	6	13	20	22	54	56	63	70	72
10	12	19	21	3	60	62	69	71	53
11	18	25	2	9	61	68	75	52	59
92	99	76	83	90	42	49	26	33	40
98	80	82	89	91	48	30	32	39	41
79	81	88	95	97	29	31	38	45	47
85	87	94	96	78	35	37	44	46	28
86	93	100	77	84	36	43	50	27	34

Fig. 10.1.9

Magic square of order $n = 4m$. Let us first explain how we can construct a magic square of order 4. First we fill in the fields of a square 4×4 with the positive integers $1, 2, \dots, 16$ as shown in Figure 10.1.11. Then we mark the integers that are placed on both diagonals, i.e., we mark integers 1, 4, 6, 7, 10, 11, 13, and 16. Then we interchange each of the marked positive integers with the positive integer that is placed symmetrically about the center of the square. The obtained magic square is given in Figure 10.1.12.

A magic square of order $n = 4m$, where $m > 1$, can be constructed as follows. First we place the positive integers $1, 2, \dots, 16m^2$ in the fields of the square $4m \times 4m$ as given in Figure 10.1.13, where the case $n = 8$ is presented. Then we divide this square into squares 4×4 . There are m^2 such squares. We mark the entries in all of them as in the case when we constructed a magic square of order 4. Then we interchange each of the marked positive integers with the positive integer that is placed symmetrically about the center of the square $4m \times 4m$.

92	99	1	8	15	67	74	51	58	40
98	80	7	14	16	73	55	57	64	41
4	81	88	20	22	54	56	63	70	47
85	87	19	21	3	60	62	69	71	28
86	93	25	2	9	61	68	75	52	34
17	24	76	83	90	42	49	26	33	65
23	5	82	89	91	48	30	32	39	66
79	6	13	95	97	29	31	38	45	72
10	12	94	96	78	35	37	44	46	53
11	18	100	77	84	36	43	50	27	59

Fig. 10.1.10

Example 10.1.4. The construction of a magic square of order 8 is given in Figures 10.1.13 and 10.1.14. The marked positive integers that should be interchanged with the positive integers that are placed symmetrically about the center of the square are represented in Figure 10.1.13. The resulting magic square is given in Figure 10.1.14. $\triangle$

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Fig. 10.1.11

16	2	3	13
5	11	10	8
9	7	6	12
4	14	15	1

Fig. 10.1.12

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64

Fig. 10.1.13

64	2	3	61	60	6	7	57
9	55	54	12	13	51	50	16
17	47	46	20	21	43	42	24
40	26	27	37	36	30	31	33
32	34	35	29	28	38	39	25
41	23	22	44	45	19	18	48
49	15	14	52	53	11	10	56
8	58	59	5	4	62	63	1

Fig. 10.1.14

Diabolic squares. A magic square is called *diabolic* or *perfect* if a constant sum appears as the sum of all positive integers contained in any row, any column, any diagonal, and any broken diagonal, see Figures 10.1.15–10.1.17.

The unique magic square of order 3 is not diabolic. It can be proved that for any $m \in \mathbb{N}$, there is no diabolic square of order $n = 2(2m + 1)$.

Example 10.1.5. The diabolic squares of order 4, 5, and 7 are presented in Figures 10.1.15, 10.1.16, and 10.1.17, respectively. For each of these squares the positive integers that belong to one of their broken diagonals are marked. The characteristic sum of these diabolic squares is equal to 34, 65, and 15, respectively. $\triangle$

Remark 10.1.6. All magic squares that can be obtained from a given square by rotation and reflection form an equivalence class of magic squares. There are 8 magic squares of order 3 that all belong to the same equivalence class.

1	14	4	15
8	11	5	10
13	2	16	3
12	7	9	6

Fig. 10.1.15

1	25	19	13	7
14	8	2	21	20
22	16	15	9	3
10	4	23	17	11
18	12	6	5	24

Fig. 10.1.16

There are 7040 magic squares of order 4 that form 880 equivalence classes. It can be proved that 48 of these 880 equivalence classes are diabolic. The number of equivalence classes of magic squares of order 5 is equal to 275 305 224. △

1	32	14	38	20	44	26
45	27	2	33	8	39	21
40	15	46	28	3	34	9
35	10	41	16	47	22	4
23	5	29	11	42	17	48
18	49	24	6	30	12	36
13	37	19	43	25	7	31

Fig. 10.1.17

10.2 Latin Squares

Example 10.2.1. Thirty-six officers problem. The following problem was proposed by Leonard Euler in 1782. There are six regiments consisting of six officers each of six different ranks. Is it possible to arrange these officers in a square 6×6 , such that each row and each column contains one officer of each rank and one of each regiment?

The Euler conjecture was that such an arrangement is not possible. This fact was proved by Gaston Tarry in 1901. The Euler problem has led to the development of important areas in combinatorics, such as Latin squares and design theory. $\triangle$

Definition 10.2.2. A *Latin square* of order n is an $n \times n$ array filled with n distinct elements, such that any of them occurs exactly once in each row and exactly once in each column.

A Latin square is usually denoted by $(a_{ij})_{n \times n}$ (or simply (a_{ij}) if the order is known), where a_{ij} is the intersection of the i -th row and j -th column. Latin squares (a_{ij}) and (b_{ij}) of the same order are *orthogonal* if $(a_{ij}, b_{ij}) \neq (a_{kl}, b_{kl})$ for all indices i, j, k and l such that $(i, j) \neq (k, l)$.

The following theorem is related to the existence of an orthogonal Latin square of a given order.

Theorem 10.2.3. *There is no orthogonal Latin square of order n for $n \in \{2, 6\}$. For any positive integer $n \in \{3, 4, 5, 7, 8, 9, 10, \dots\}$ there exists an orthogonal Latin square of order n .*

Example 10.2.4. Examples of orthogonal Latin squares of order 3, 4, and 5 are given in Figures 10.2.1–10.2.3. Suppose that the orthogonal Latin squares are denoted by (a_{ij}) and (b_{ij}) . Every field of the squares in Figures 10.2.1–10.2.3 contains two integers. The first one is a_{ij} , and the second one is b_{ij} . For example, in Figure 10.2.1 we have $a_{11} = 0, b_{11} = 0$; $a_{12} = 1, b_{12} = 2$, $a_{13} = 2, b_{13} = 1$, etc. $\triangle$

00	12	21
11	20	02
22	01	10

Fig. 10.2.1

00	11	22	33
12	03	30	21
23	32	01	10
31	20	13	02

Fig. 10.2.2

00	12	24	31	43
13	20	32	44	01
21	33	40	02	14
34	41	03	10	22
42	04	11	23	30

Fig. 10.2.3

10.3 System of Distinct Representatives

Definition 10.3.1. Let S be a nonempty set, $\mathbb{P}(S)$ be the set of all subsets of S , $(a_1, a_2, \dots, a_m)$ be an arrangement without repetition of the elements of S , and $(S_1, S_2, \dots, S_m)$ be an arrangement of the elements of $\mathbb{P}(S)$. If $a_k \in S_k$

for any $k \in \{1, 2, \dots, m\}$, then we say that $(a_1, a_2, \dots, a_m)$ is a *system of distinct representatives* of the arrangement of the sets $(S_1, S_2, \dots, S_m)$. The element $a_k \in S_k$ is a *representative* of the set S_k .

We shall use an abbreviation s.d.r. for *system of distinct representatives*, and simply say that $a_1, a_2, \dots, a_m$ is a s.d.r. of the sets $S_1, S_2, \dots, S_m$.

Example 10.3.2. (a) Let us consider the sets $S_1 = \{1, 2\}$, $S_2 = \{1, 2\}$, $S_3 = \{1, 2, 3\}$, and $S_4 = \{3, 4, 5\}$. A s.d.r. of the sets S_1, S_2, S_3 , and S_4 is any of the following arrangements:

$$(1, 2, 3, 4), (1, 2, 3, 5), (2, 1, 3, 4), (2, 1, 3, 5).$$

(b) Let us now consider the sets $S_1 = \{1, 2\}$, $S_2 = \{2, 3\}$, $S_3 = \{1, 3\}$, $S_4 = \{1, 2, 3\}$, and $S_5 = \{3, 4, 5, 6\}$. Note that $S_1 \cup S_2 \cup S_3 \cup S_4 = \{1, 2, 3\}$. It is not possible to choose 4 distinct representatives from a 3-set, and hence, there is no s.d.r. for the sets S_1, S_2, S_3, S_4 , and S_5 . $\triangle$

It is obvious that a *necessary condition* for the sets $S_1, S_2, \dots, S_m$ to have a s.d.r. is that

$$|S_{j_1} \cup S_{j_2} \cup \dots \cup S_{j_k}| \geq k$$

for each $k \in \{1, 2, \dots, m\}$, and each k -combination $\{j_1, j_2, \dots, j_k\}$ of the elements of set $\{1, 2, \dots, m\}$. This condition is also sufficient, as it was proved by Philip Hall in 1935. We shall formulate here a theorem that also gives a bound from above for the number of systems of distinct representatives.

Theorem 10.3.3. (Hall) *Let $S_1, S_2, \dots, S_m$ be sets such that $|S_k| \geq n$ for any $k \in \{1, 2, \dots, m\}$, and that the necessary condition for the existence of a s.d.r. holds. Then, the next statements hold:*

(a) *If $n \leq m$, then $(S_1, S_2, \dots, S_m)$ has at least $n!$ systems of distinct representatives.*

(b) *If $n > m$, then $(S_1, S_2, \dots, S_m)$ has at least $\frac{n!}{(n-m)!}$ systems of distinct representatives.*

Proof by induction on m . For $m = 1$ the theorem obviously holds. Suppose that the theorem holds for all positive integers less than a given positive integer m . We shall prove that the theorem also holds for the positive integer m .

Case 1. Suppose that

$$|S_{j_1} \cup S_{j_2} \cup \dots \cup S_{j_k}| \geq k + 1$$

for all $k \in \{1, 2, \dots, m-1\}$ and every combination $\{j_1, j_2, \dots, j_k\}$ of the elements of set $\{1, 2, \dots, m\}$. Let a_1 be a fixed element of S_1 , and $S'_j = S_j \setminus \{a_1\}$ for $j \in \{2, 3, \dots, m\}$. Then, the arrangement of sets $(S'_2, S'_3, \dots, S'_m)$ satisfies the necessary condition for the existence of a s.d.r. Indeed

$$|S'_{j_1} \cup S'_{j_2} \cup \dots \cup S'_{j_k}| = |(S_{j_1} \cup S_{j_2} \cup \dots \cup S_{j_k}) \setminus \{a_1\}| \geq k + 1 - 1 = k,$$

for $2 \leq j_1 < j_2 < \dots < j_k \leq m$.

If $n \leq m$, i.e., $n-1 \leq m-1$, then by the induction hypothesis it follows that $(S'_2, S'_3, \dots, S'_m)$ has at least $(n-1)!$ systems of distinct representatives. If $(s_2, s_3, \dots, s_m)$ is a s.d.p. for $(S'_2, S'_3, \dots, S'_m)$, then $(a_1, s_2, s_3, \dots, s_m)$ is a s.d.p. for $(S_1, S_2, \dots, S_m)$. Since the element a_1 may be chosen in n ways, we conclude that $(S_1, S_2, \dots, S_m)$ has at least $n \cdot (n-1)! = n!$ systems of distinct representatives.

If $n > m$, i.e., $n-1 > m-1$, then, by the induction hypothesis, it follows that $(S'_2, S'_3, \dots, S'_m)$ has at least $\frac{(n-1)!}{(n-m)!}$ systems of distinct representatives. Similarly as in the case $n \leq m$ we conclude that $(S_1, S_2, \dots, S_m)$ has at least $\frac{n!}{(n-m)!}$ systems of distinct representatives.

Case 2. There is a positive integer $k \in \{1, 2, \dots, m-1\}$, such that the union of some k sets from $\{S_1, S_2, \dots, S_m\}$ consists of exactly k elements. Without loss of generality we can assume that $|S_1 \cup S_2 \cup \dots \cup S_k| = k$. Then, we have

$$n \leq |S_1| \leq |S_1 \cup S_2 \cup \dots \cup S_k| = k < m.$$

By the induction hypothesis it follows that $(S_1, S_2, \dots, S_k)$ has at least $n!$ systems of distinct representatives. Let $(a_1, a_2, \dots, a_k)$ be a s.d.r. and let us denote $S'_j = S_j \setminus \{a_1, a_2, \dots, a_k\}$, for $j \in \{k+1, k+2, \dots, m\}$.

We shall prove that $(S'_{k+1}, S'_{k+2}, \dots, S'_m)$ satisfies the necessary condition for the existence of a s.d.r. Let us suppose, on the contrary, that $|S'_{j_1} \cup S'_{j_2} \cup \dots \cup S'_{j_p}| < p$, for $k+1 \leq j_1 < j_2 < \dots < j_p \leq m$. Then,

$$|S_1 \cup S_2 \cup \dots \cup S_k \cup S_{j_1} \cup S_{j_2} \cup \dots \cup S_{j_p}| < k + p,$$

which contradicts the fact that $(S_1, S_2, \dots, S_m)$ satisfies the necessary condition for the existence of a s.d.r. Hence, $(S'_{k+1}, S'_{k+2}, \dots, S'_m)$ really satisfies the necessary condition. Since $m-k < m$, it follows by the induction hypothesis that $(S'_{k+1}, S'_{k+2}, \dots, S'_m)$ has a s.d.r. Now it is easy to conclude that $(S_1, S_2, \dots, S_m)$ has at least $n!$ systems of distinct representatives. $\square$

Definition 10.3.4. Let $S = \{a_1, a_2, \dots, a_n\}$, and $S_1, S_2, \dots, S_m$ be subsets of S . The matrix $A = (c_{jk})_{m \times n}$, whose elements are given by

$$c_{jk} = \begin{cases} 1, & \text{if } a_k \in S_j, \\ 0, & \text{if } a_k \notin S_j, \end{cases} \quad (10.3.1)$$

is called the *incidence matrix* related to elements $a_1, a_2, \dots, a_n$ and sets $S_1, S_2, \dots, S_m$.

Example 10.3.5. Let us assume that the incidence matrix A related to the elements $a_1, a_2, \dots, a_n$ and the sets $S_1, S_2, \dots, S_m$ has exactly r units in each of its rows and each of its columns, where $1 \leq r \leq m$. Then, there exists a s.d.r. of $(S_1, S_2, \dots, S_m)$.

We shall prove this fact using Hall's theorem. Suppose there is no s.d.r. of $(S_1, S_2, \dots, S_m)$. By Theorem 10.3.3 it follows that there are positive integers $j_1, j_2, \dots, j_p$, such that

$$1 \leq j_1 < j_2 < \dots < j_p \leq m, \quad |S_{j_1} \cup S_{j_2} \cup \dots \cup S_{j_p}| = q < p. \quad (10.3.2)$$

Since every row of the matrix A contains exactly r 1's, it follows that the total number of 1's in the rows labeled $j_1, j_2, \dots, j_p$ is equal to rp . Since every column of A contains exactly r 1's, it follows that each of the elements $a_1, a_2, \dots, a_n$ (and hence each of q elements of the set $S_{j_1} \cup S_{j_2} \cup \dots \cup S_{j_p}$) belongs to exactly r of the sets $S_1, S_2, \dots, S_m$. Hence, the total number of 1's in the rows labeled $j_1, j_2, \dots, j_p$ is not greater than qr . Consequently we get $pr \leq qr$, i.e., $p \leq q$, which contradicts (10.3.2). Hence there is a s.d.r. of $(S_1, S_2, \dots, S_m)$. $\triangle$

Remark 10.3.6. Let $S_1 \cup S_2 \cup \dots \cup S_m = \{a_1, a_2, \dots, a_n\}$. Note that the necessary condition for the existence of a s.d.r. of $(S_1, S_2, \dots, S_m)$ can be formulated as follows: *m units in the incidence matrix related to the elements $a_1, a_2, \dots, a_n$ and the sets $S_1, S_2, \dots, S_m$ can be labeled such that each of m rows contains a labeled unit, and no two of the labeled units belong to the same column.*

Example 10.3.7. Suppose that n persons are to be employed at n positions in a company. Moreover, suppose that any person is qualified to work at exactly r of these n positions, and for any position there are exactly r persons qualified for it, where $r < n$. We shall prove that these n persons can be employed such that every position is occupied by a person qualified for it.

Let $B_1, \dots, B_n$ and $M_1, \dots, M_n$ be the notation for persons and positions, respectively. Let $S_1, \dots, S_n$ be r -subsets of the set $\{B_1, \dots, B_n\}$ defined as follows: $B_k \in S_j$ if and only if the person B_k is qualified to work at the position M_j . Then, the incidence matrix related to the elements $B_1, \dots, B_n$ and the sets $S_1, \dots, S_n$ contains exactly r units in any of its rows and any of its columns. By the result of Example 10.3.5 it follows that there is a s.d.r. of $(S_1, S_2, \dots, S_n)$. In other words, there is a permutation $(k_1, k_2, \dots, k_n)$ of the set $\{1, 2, \dots, n\}$, such that $B_{k_1} \in S_1, B_{k_2} \in S_2, \dots, B_{k_n} \in S_n$. $\triangle$

10.4 The Pigeonhole Principle

The following simple statement is known as the *pigeonhole principle* and may be very useful in proving the existence of some combinatorial configurations.

Theorem 10.4.1. *Let $\{A_1, A_2 \dots A_k\}$ be a partition of an $(nk + 1)$ -set S into k blocks. Then, there exists a block of the partition that contains at least $n + 1$ elements.*

Proof. Suppose, on the contrary, that $|A_j| \leq n$ for any $j \in \{1, 2, \dots, k\}$. Then, $|S| = |A_1| + |A_2| + \dots + |A_k| \leq kn$, and this contradicts the condition $|S| = nk + 1$. $\square$

Example 10.4.2. Suppose that 25 points are given in the plane such that for every three points, we can choose two of them that have a distance of less than 1. We shall prove that there is a circle of radius 1 such that at least 13 of the given 25 points are inside this circle.

Let S be the set of given points, A be an arbitrary point from S , and let c_1 be a circle with center A whose radius is equal to 1. If all points from S are inside c_1 , then the statement is obviously proved. Suppose now that there is a point $B \in S$ such that $d(A, B) \geq 1$, and let c_2 be a circle with center B whose radius is equal to 1. Every point from S lies inside at least one of the circles c_1 and c_2 . Indeed, if for some $C \in S$, $d(A, C) \geq 1$ and $d(B, C) \geq 1$, then the assumption of the problem is not satisfied for the triple of points (A, B, C) . Now by the pigeonhole principle it follows that at least 13 points from S lie inside one of the circles c_1 and c_2 . $\triangle$

Example 10.4.3. Let $S = \{1, 2, \dots, 17\}$ and let $\{A_1, A_2, A_3\}$ be a partition of set S . We shall prove that there are $x, y, z \in S$ and $k \in \{1, 2, 3\}$, such that $x, y, z \in A_k$, and $x = y + z$. By the pigeonhole principle it follows that at least one of the sets A_1 , A_2 , and A_3 contains at least six of the positive integers $1, 2, \dots, 17$. Let us suppose that $a_1, a_2, \dots, a_6 \in A_1$, and $a_1 > a_2 > \dots > a_6$. If for some $k < n$, where $k, n \in \{1, 2, \dots, 6\}$, the difference $a_k - a_n$ belongs to A_1 , the statement is proved. Suppose now that $a_k - a_n \in A_2 \cup A_3$, for all $k < n$, and $k, n \in \{1, 2, \dots, 6\}$. One of the sets A_2 and A_3 (let us suppose it is A_2) contains at least three of the positive integers $a_1 - a_2$, $a_1 - a_3$, $a_1 - a_4$, $a_1 - a_5$, and $a_1 - a_6$. Let us denote these three positive integers by b_1 , b_2 , and b_3 , where $b_1 > b_2 > b_3$. Let us also denote $x = b_1 - b_3$, $y = b_1 - b_2$, and $z = b_2 - b_3$. If at least one of the positive integers x , y , and z belongs to A_2 , the statement obviously holds. In the opposite case we have $x, y, z \in A_3$, and $x = y + z$. $\triangle$

Example 10.4.4. Let us consider an arrangement $(a_1, a_2, \dots, a_{mn+1})$ whose terms are distinct real numbers. We shall prove that at least one of the following two statements holds:

(a) There exist the positive integers $i_1, i_2, \dots, i_{n+1}$, such that

$$i_1 < i_2 < \dots < i_{n+1}, \quad a_{i_1} < a_{i_2} < \dots < a_{i_{n+1}}.$$

(b) There exist the positive integers $j_1, j_2, \dots, j_{m+1}$, such that

$$j_1 < j_2 < \dots < j_{m+1}, \quad a_{j_1} > a_{j_2} > \dots > a_{j_{m+1}}.$$

Let $r(1) = 1$. For $k \in \{2, 3, \dots, mn + 1\}$, let $r = r(k)$ be the greatest positive integer such that there exist the positive integers $j_1, j_2, \dots, j_{r-1}$, satisfying the conditions

$$1 \leq j_1 < j_2 < \dots < j_{r-1} < k, \quad a_{j_1} < a_{j_2} < \dots < a_{j_{r-1}} < a_k.$$

If $a_k < a_j$ for any $j \in \{1, 2, \dots, k-1\}$, then $r(k) = 1$. The positive integer $r(k)$ will be called the *characteristic* of the real number a_k .

Lemma: If $k_1 < k_2$ and $r(k_1) = r(k_2) = r$, then $a_{k_1} > a_{k_2}$.

Proof of the Lemma. Suppose, on the contrary, that $a_{k_1} < a_{k_2}$. Since $r(k_1) = r$, it follows that there are positive integers $j_1, j_2, \dots, j_{r-1}$, such that the following inequalities hold

$$1 \leq j_1 < j_2 < \dots < j_{r-1} < k_1 < k_2, \\ a_{j_1} < a_{j_2} < \dots < a_{j_{r-1}} < a_{k_1} < a_{k_2}.$$

Hence, $r(k_2) \geq r + 1$. This contradicts the condition $r(k_2) = r$, and hence Lemma is proved.

Case 1. At least one of the real numbers $a_1, a_2, \dots, a_{mn+1}$ has the characteristic no less than $n + 1$. Then statement (a) holds.

Case 2. Let $r(k) \in \{1, 2, \dots, n\}$ for every $k \in \{1, 2, \dots, mn + 1\}$. By the pigeonhole principle it follows that at least $m + 1$ of the real numbers $a_1, a_2, \dots, a_{mn+1}$ have the same characteristic, for example, equal to r . Let $a_{j_1}, a_{j_2}, \dots, a_{j_{m+1}}$ be these r real numbers, where $j_1 < \dots < j_{m+1}$. By the Lemma it follows that $a_{j_1} > \dots > a_{j_{m+1}}$, i.e., statement (b) holds. $\triangle$

10.5 Ramsey's Theorem

Example 10.5.1. Let S be an n -set, and let $S = S_1 \cup S_2$, where $S_1 \cap S_2 = \emptyset$. If $n \geq q_1 + q_2 - 1$, where $q_1, q_2 \in \mathbb{N}$, then $|S_1| \geq q_1$ or $|S_2| \geq q_2$. Indeed, if $|S_1| \leq q_1 - 1$ and $|S_2| \leq q_2 - 1$, then

$$|S| = |S_1 \cup S_2| = |S_1| + |S_2| \leq q_1 + q_2 - 2,$$

and this contradicts the fact that S is an n -set. Note that for $n < q_1 + q_2 - 1$, the inequalities $|S_1| \leq q_1 - 1$ and $|S_2| \leq q_2 - 1$, may both be valid, as it is in the case:

$$S_1 = \{1, 2, \dots, q_1 - 1\}, \quad S_2 = \{q_1, q_1 + 1, \dots, q_1 + q_2 - 2\}. \quad \triangle$$

An important generalization of the result of Example 10.5.1 is given by Ramsey's theorem.

Theorem 10.5.2. (Ramsey) *Let r , q_1 , and q_2 be positive integers such that $q_1 \geq r$ and $q_2 \geq r$. Then, there exists the smallest positive integer $R(q_1, q_2; r)$, such that for any positive integer $n \geq R(q_1, q_2; r)$ the following statement holds: If S is an n -set, $\mathbb{P}_r(S)$ is the set of all r -subsets of S , and*

$$P_r(S) = \Phi_1 \cup \Phi_2, \quad \text{where} \quad \Phi_1 \cap \Phi_2 = \emptyset,$$

then for at least one $j \in \{1, 2\}$ there exists a q_j -subset of S , and all of its r -subsets are contained in the collection Φ_j .

Proof by induction on q_1 , q_2 , and r . (a) We shall prove the equalities:

$$R(q_1, q_2; 1) = q_1 + q_2 - 1, \quad q_1 \geq 1, \quad q_2 \geq 1; \quad (10.5.1)$$

$$R(q_1, r; r) = q_1, \quad q_1 \geq r > 1; \quad (10.5.2)$$

$$R(r, q_2; r) = q_2, \quad q_2 \geq r > 1. \quad (10.5.3)$$

The equality (10.5.1) follows from Example 10.5.1. Let us suppose that $n \geq q_1$ and $q_2 = r$. If $\Phi_2 = \emptyset$, then $\Phi_1 = \mathbb{P}_r(S)$, and hence all r -subsets of any q_1 -subset of S belong to Φ_1 . If $\Phi_2 \neq \emptyset$, then any r -set $A \in \Phi_2$ has exactly one r -subset that belongs to Φ_2 . Note that in the case $n < q_1$, $q_2 = r$ and $\Phi_2 = \emptyset$, there is no q_1 -subset of set S (because $|S| = n < q_1$), and there is no r -subset of set S , that belongs to Φ_2 . Hence, (10.5.2) is proved. Analogously one can prove (10.5.3).

(b) Let us now suppose that for $r \geq 2$, $q_1 \geq r + 1$, and $q_2 \geq r + 1$ there exist the positive integers

$$p_1 = R(q_1 - 1, q_2; r), \quad p_2 = R(q_1, q_2 - 1; r), \quad R(p_1, p_2; r - 1),$$

such that the statement of Theorem 10.5.2 holds for any of them. It is sufficient to prove that

$$R(q_1, q_2; r) \leq R(p_1, p_2; r - 1) + 1. \quad (10.5.4)$$

Suppose that the set S contains more than $R(p_1, p_2; r - 1)$ elements. Let $\mathbb{P}_r(S) = \Phi_1 \cup \Phi_2$, where $\Phi_1 \cap \Phi_2 = \emptyset$. Let a_0 be an arbitrary element of set S , $S' = S \setminus \{a_0\}$ and $\mathbb{P}_{r-1}(S') = \Phi'_1 \cup \Phi'_2$, where for any set $A \in \mathbb{P}_{r-1}(S')$,

and any $j \in \{1, 2\}$, $A \in \Phi'_j$ if and only if $A \cup \{a_0\} \in \Phi_j$. Since S' contains at least $R(p_1, p_2; r-1)$ elements, it follows by the induction hypothesis that at least one of the following two statements holds true:

(a) There is a p_1 -subset of set S' , denoted by W , such that all $(r-1)$ -subsets of W belong to the collection Φ'_1 .

(b) There is a p_2 -subset of set S' , such that all its $(r-1)$ -subsets belong to the collection Φ'_2 .

Let us consider the case when statement (a) holds. Since W contains $R(q_1 - 1, q_2; r)$ elements, and all r -subsets of set W belong to $\Phi_1 \cup \Phi_2$, it follows that at least one of the following two statements holds:

(a₁) There is a $(q_1 - 1)$ -subset of set W , denoted by T_1 , such that all r -subsets of T_1 belong to Φ_1 .

(a₂) There is a q_2 -subset of set W , denoted by T_2 , such that all r -subsets of T_2 belong to Φ_2 .

If (a₁) holds, then $T_1 \cup \{a_0\}$ is a q_1 -subset of set S , and all its r -subsets belong to Φ_1 . If (a₂) holds, then T_2 is a q_2 -subset of S , and all its r -subsets belong to Φ_2 . The case when statement (b) holds can be analyzed analogously. We have proved the inequality (10.5.4), and hence the proof of Theorem 10.5.2 is finished. $\square$

Theorem 10.5.3. *Let $r, q_1, q_2, \dots, q_m$ be positive integers such that $r \geq 1$, $m \geq 2$, and $q_j \geq r$ for any $j \in \{1, 2, \dots, m\}$. Then there exists the smallest positive integer $R(q_1, q_2, \dots, q_m; r)$, such that, for any positive integer $n \geq R(q_1, q_2, \dots, q_m; r)$, the following statement holds true:*

If S is an n -set, $\mathbb{P}_r(S)$ is the set of all r -subsets of S , and $\mathbb{P}_r(S) = \Phi_1 \cup \Phi_2 \cup \dots \cup \Phi_m$, where $\Phi_j \cap \Phi_k = \emptyset$ for $k \neq j$ and $k, j \in \{1, 2, \dots, m\}$, then there is a positive integer $j \in \{1, 2, \dots, m\}$ and a q_j -subset of S , such that all its r -subsets belong to Φ_j .

Proof by induction on m . For $m = 2$, Theorem 10.5.3 becomes Theorem 10.5.2. Suppose that Theorem 10.5.3 holds for some positive integer $m - 1 \geq 2$. Let S be an n -set, and $\mathbb{P}_r(S) = \Phi_1 \cup \Phi_2 \cup \dots \cup \Phi_m$, where $\Phi_k \cap \Phi_j = \emptyset$ if $k \neq j$ and $k, j \in \{1, 2, \dots, m\}$. Let us denote $\Phi'_2 = \Phi_2 \cup \Phi_3 \cup \dots \cup \Phi_m$, and $q'_2 = R(q_2, \dots, q_m; r)$. If $n \geq R(q_1, q'_2; r)$, then there is a q_1 -subset of S such that all its r -subsets belong to Φ_1 , or there is a q'_2 -subset of S such that all its r -subsets belong to Φ'_2 . In the second case we conclude by the induction hypothesis that there is a positive integer $j \in \{2, 3, \dots, m\}$ and q_j -subset of S , such that all its r -subsets belong to Φ_j . Consequently, we obtain that $R(q_1, \dots, q_m; r) \geq R(q_1, q'_2; r)$. $\square$

The positive integers $R(q_1, q_2, \dots, q_m; r)$, where $m, q_1, q_2, \dots, q_m, r \in \mathbb{N}$, are called *Ramsey numbers*.

Example 10.5.4. Let every edge of a complete 6-graph be colored red or blue. We shall prove that there exists a monochromatic 3-subgraph, i.e., a 3-subgraph such that all its edges are colored the same color.

Let A_1, A_2, A_3, A_4, A_5 , and A_6 be vertices of the graph. There are five edges that connect A_1 with the other vertices. At least three of them are colored the same color, for example red. Suppose, for example, that the edges A_1A_2, A_1A_3 , and A_1A_4 are colored red. If all the edges of the “triangle” $A_2A_3A_4$ are colored blue, then $A_2A_3A_4$ is a monochromatic 3-subgraph. If at least one of its edges, for example A_2A_3 , is colored red, then $A_1A_2A_3$ is a monochromatic 3-subgraph. $\triangle$

Example 10.5.5. Let all the sides of a pentagon be colored red, and all its diagonals be colored blue. The complete 5-graph obtained this way does not contain a monochromatic 3-subgraph. From this fact and Example 10.5.4 it follows that $R(3, 3; 2) = 6$. $\triangle$

Now we shall formulate two immediate consequences of Theorem 10.5.3 that are related to the edge colorings of a complete n -graph.

Corollary 10.5.6. *Let $q_1 \geq 2, q_2 \geq 2$, and $q_1, q_2 \in \mathbb{N}$. Then, there exists a positive integer $R = R(q_1, q_2, 2)$ that depends only on q_1 and q_2 , such that the following statements hold:*

If $n \geq R$, then for any edge coloring of the complete n -graph in two colors, for example red and blue, there is a red q_1 -subgraph or there is a blue q_2 -subgraph.

If $n < R$, then there exists an edge coloring of the complete n -graph, such that there is no red q_1 -subgraph, and there is no blue q_2 -subgraph.

Corollary 10.5.7. *Let $m, q_1, q_2, \dots, q_m$ be positive integers greater than 1. Then there exists a positive integer $R = R(q_1, q_2, \dots, q_m; 2)$, that depends only on $q_1, q_2, \dots, q_m$, such that the following two statements hold:*

If $n \geq R$, then for any edge coloring of the complete n -graph in m colors, denoted by $1, 2, \dots, m$, there is a positive integer $j \in \{1, 2, \dots, m\}$ for which there exists a monochromatic q_j -subgraph of the j -th color.

If $n < R$, then there exists an edge coloring of the complete n -graph in m colors, such that, for any $j \in \{1, 2, \dots, m\}$, there is no monochromatic q_j -subgraph of the j -th color.

10.6 Arrow's Theorem

Suppose that a society $S = \{1, 2, \dots, n\}$, consisting of n individuals, should choose an option from the set $X = \{A, B, C, \dots\}$, where $|X| = p \geq 3$. Assume that every individual $i \in S$ can compare every two options, and

[illegible]
$$A_1 < A_2 < \cdots < A_p. \quad (10.6.2)$$

Axiom 10.6.2 is also called the *principle of irrelevant options*. The irrelevant options here belong to the set $X \setminus Y$.

Axiom 10.6.3. For any two options $A, B \in X$, there is a system of individual values $<_1, <_2, \dots, <_n$, that implies the social choice $<$, such that $A < B$.

Axiom 10.6.4. There is no individual $i \in S$, such that the individual value $<_i$ coincides with the social choice that follows from every system of individual values containing the individual value $<_i$.

Obviously the meaning of Axiom 10.6.4 can be described as follows. There is no dictator such that his desire is more important than every system of individual values.

Theorem 10.6.5. (Arrow's Impossibility Theorem) *If $|X| \geq 3$, then there is no social choice function that satisfies Axioms 10.6.1–10.6.4.*

Proof. Suppose there is a social choice function F that satisfies Axioms 10.6.1–10.6.3. We need the following definition.

Definition 10.6.6. Let $A, B \in X$ be two distinct options. A set $T \subset S$ is (A, B) -decisive if the condition $A <_i B$ for any $i \in T$ implies $A < B$. We say that a set T is *decisive* if it is (A, B) -decisive for some $A, B \in X$.

Lemma 10.6.7. *The set S is (A, B) -decisive for any two options $A, B \in X$.*

Proof. Without loss of generality we can suppose here that A and B are the only available options (see Axiom 10.6.2). By Axiom 10.6.3 we conclude that there is a system of individual values that implies the social choice such that $A < B$. By Axiom 10.6.1 it follows that $A < B$ remains valid if we change the system of individual values such that $A <_i B$ holds for every $i \in S$. $\square$

Let T_0 be the minimal (in the sense of the inclusion) decisive set. By Axiom 10.6.3 it follows that $T_0 \neq \emptyset$. Then, there exist two options $A_0, B_0 \in X$, such that the set T_0 is (A_0, B_0) -decisive. Let $i_0 \in T_0$.

Lemma 10.6.8. *The equality $T_0 = \{i_0\}$ holds, and for every option $M \neq A_0$, the set T_0 is (A_0, M) -decisive.*

Proof. Let us consider an option $M \in X \setminus \{A_0, B_0\}$, and the following system of individual values:

$$\left. \begin{aligned} A_0 &<_{i_0} B_0 <_{i_0} M, \\ M &<_j A_0 <_j B_0, & \text{for all } j \in T_0 \setminus \{i_0\}, \\ B_0 &<_k M <_k A_0, & \text{for all } k \in X \setminus T_0, \end{aligned} \right\} \quad (10.6.3)$$

The system of individual values (10.6.3) implies the social choice such that:

- (1) $A_0 < B_0$ (a consequence of the fact that T_0 is (A_0, B_0) -decisive);

(2) $B_0 < M$ (in the opposite case, i.e., $M < B_0$, we get that the set $T_0 \setminus \{i_0\}$ is (M, B_0) -decisive, and this contradicts the condition that T_0 is the minimal decisive set).

Since the relation $<$ is a total order of X , it follows that $A_0 < M$. By Axiom 10.6.2 the same conclusion follows from any system of individual values with the same relation between A_0 and M as in (10.6.3). Hence, the set $\{i_0\}$ is (A_0, M) -decisive (note that (10.6.3) implies that i_0 is the only individual from S such that $A_0 <_{i_0} M$). Since T_0 is the minimal decisive set it follows that $T_0 = \{i_0\}$. $\square$

Note that we have also proved the following statement. *The set $\{i_0\}$ is (A_0, M) -decisive for any option $M \neq A_0$.*

Lemma 10.6.9. *The set $T_0 = \{i_0\}$ is (M, N) -decisive for any two distinct options $M, N \in X \setminus \{A_0\}$.*

Proof. Let us consider the following system of individual values:

$$\left. \begin{array}{l} M <_{i_0} A_0 <_{i_0} N, \\ N <_k M <_k A_0, \quad \text{for every } k \neq i_0, \end{array} \right\} \quad (10.6.4)$$

Since $M <_i A_0$ for any $i \in S$, it follows that $M < A_0$. Since $A_0 <_{i_0} N$, and the set $\{i_0\}$ is (A_0, N) -decisive, it follows that $A_0 < N$. Hence, the system of individual values (10.6.4) implies that $M < N$. Now, by Axiom 10.6.2 we conclude that the set $\{i_0\}$ is (M, N) -decisive. $\square$

Lemma 10.6.10. *The set $T_0 = \{i_0\}$ is (M, A_0) -decisive for any $M \neq A_0$.*

Proof. Let us consider two distinct options $M, N \in X \setminus \{A_0\}$, and a system of individual values such that

$$\left. \begin{array}{l} M <_{i_0} N <_{i_0} A_0, \\ N <_k A_0 <_k M, \quad \text{for every } k \neq i_0, \end{array} \right\} \quad (10.6.5)$$

Since $M <_{i_0} N$, it follows by Lemma 10.6.9 that $M < N$. Since $N <_i A_0$ for any i , it follows that $N < A_0$. Hence, the system of individual values (10.6.5) implies that $M < A_0$. By Axiom 10.6.2 we conclude that the set $\{i_0\}$ is (M, A_0) -decisive. $\square$

We proceed now with the proof of Theorem 10.6.5. It follows from Lemma 10.6.8, Lemma 10.6.9, and Lemma 10.6.10 that for any two distinct options $A, B \in X$, the set $\{i_0\}$ is (A, B) -decisive. Hence, i_0 is the dictator, and this conclusion contradicts Axiom 10.6.4. Since Axioms 10.6.1–10.6.3 are satisfied by the assumption, it follows that there is no social choice function that satisfies all four axioms. $\square$

Exercises

Magic Squares

10.1. Give examples of magic squares of the order 7, 10, and 12.

10.2. Let M be a magic square of order n filled with positive integers $1, 2, \dots, n^2$, and M_k be the magic square obtained from M by adding $(k-1)n^2$ to each entry. Let us now replace every entry k in the magic square M by the magic square M_k , and denote the obtained square table by M^* . Prove that M^* is a magic square of order n^2 .

Latin Squares and Rectangles

10.3. A Latin square of order $2n+1$ is filled with positive integers $1, 2, \dots, 2n+1$. If all entries are symmetrically located around the main diagonal, prove that all positive integers $1, 2, \dots, 2n+1$ are represented on this diagonal.

10.4. Give an example of a Latin square of order 2^n .

10.5. The first quadrant of the Cartesian plane is divided into unit squares. Is it possible to fill all unit squares with positive integers, such that every positive integer appears exactly once in every row and every column?

10.6. A rectangle table $m \times n$, with m rows and n columns, where $m < n$, is filled with positive integers $1, 2, \dots, n$, such that every row contains all these positive integers, and no positive integer appears more than once in the same column. Such a table is called a *Latin rectangle*. How many Latin rectangles of the form $2 \times n$ are there?

System of Distinct Representatives

10.7. Determine the number of the systems of distinct representatives of the sets $S_1 = \{2, 3, 4\}$, $S_2 = \{1, 3, 4\}$, $S_3 = \{1, 2, 4\}$ and $S_4 = \{1, 2, 3\}$.

10.8. Let $S_k = \{1, 2, \dots, n\} \setminus \{k\}$, where $k \in \{1, 2, \dots, n\}$. How many systems of distinct representatives of the sets $S_1, S_2, \dots, S_n$ are there?

10.9. Let us consider a Latin rectangle with m rows and n columns, where $m < n$, and whose fields are filled with positive integers $1, 2, \dots, n$. Prove that new $n-m$ rows can be added and filled with positive integers $1, 2, \dots, n$, such that the obtained square table is a Latin square of order n .

10.10. An n -tuple $(x_1, x_2, \dots, x_n)$ is a *common system of distinct representatives* of the collections of sets $\{A_1, A_2, \dots, A_n\}$ and $\{B_1, B_2, \dots, B_n\}$, if there exist two permutations $(i_1, i_2, \dots, i_n)$ and $(j_1, j_2, \dots, j_n)$ of the set $\{1, 2, \dots, n\}$, such that

$$x_m \in A_{i_m} \cap B_{j_m}, \quad \text{for any } m \in \{1, 2, \dots, n\}.$$

Let $\{A_1, \dots, A_n\}$ and $\{B_1, \dots, B_n\}$ be two partitions of a finite set S . The necessary and sufficient condition for the collections $\{A_1, \dots, A_n\}$ and $\{B_1, \dots, B_n\}$ to have a common system of distinct representatives is that, for any $k \in \{1, 2, \dots, n\}$ and any k -tuple $(j_1, j_2, \dots, j_k)$ of distinct positive integers from the set $\{1, 2, \dots, n\}$, the union $A_{j_1} \cup A_{j_2} \cup \dots \cup A_{j_k}$ contains no more than k sets from the collection $\{B_1, B_2, \dots, B_n\}$. Prove this fact.

10.11. Let $\{A_1, \dots, A_n\}$ and $\{B_1, \dots, B_n\}$ be two partitions of a finite set S , such that

$$|A_i| = |B_j|, \quad \text{for all } i, j \in \{1, 2, \dots, n\}.$$

Prove that there is a common system of distinct representatives for the collections $\{A_1, \dots, A_n\}$ and $\{B_1, \dots, B_n\}$.

The Pigeonhole Principle

10.12. Every point with integer coordinates in the Cartesian plane is labeled by a positive integer from the set $\{1, 2, \dots, n\}$. Prove that there is a rectangle whose vertices are all labeled by the same positive integer.

10.13. Prove that among 39 consecutive positive integers there is at least one positive integer with the sum of the digits that is divisible by 11.

10.14. Consider a regular 45-gon. Is it possible to label its vertices by the integers $0, 1, \dots, 9$ such that for any pair (i, j) , $i \neq j$ and $i, j \in \{0, 1, \dots, 9\}$, there is a side of the 45-gon whose endpoints are labeled i and j ?

10.15. A committee had 40 meetings. Exactly 10 members of the committee attended every meeting, but no two members attended the same meeting more than once. Prove that the committee consists of more than 60 members.

10.16. A system has an odd number of planets. All distances between the planets are distinct. From each of these planets an astronomer observes the closest of the other planets. Prove that there is a planet that is not observed.

10.17. Is it possible to label the vertices of a regular 10-gon by the integers $0, 1, \dots, 9$, such that for every side of the 10-gon, the difference of the labels that are assigned to its endpoints belongs to the set $\{3, 4, 5\}$?

10.18. Let $S = \{(i, j) \mid i = 1, 2, \dots, 100; j = 1, 2, \dots, 100\}$. Determine the maximal number of elements of the set $A \subset S$ with the following property: any segment whose endpoints belong to A contains at least one point from the set $S \setminus A$.

10.19. Let $S = \{x_1, x_2, \dots, x_{10}\}$ be a set whose elements are two-digit positive integers. Prove that there are two distinct subsets of set S with the same sum of the elements that belong to them.

10.20. Let $A_1, A_2, \dots, A_{1066}$ be the subsets of a finite set S , such that $|A_k| > |S|/2$ for every $k \in \{1, 2, \dots, 1066\}$. Prove that there exist the elements $x_1, x_2, \dots, x_{10} \in S$, such that each of the sets $A_1, A_2, \dots, A_{1066}$ contains at least one of these elements.

10.21. An international society consists of 330 members from five countries. All members are labeled $1, 2, \dots, 330$. Prove that there are three members from the same country, labeled x, y , and z , such that $x = y + z$, or there are two members from the same country, labeled x and y , such that $x = 2y$.

10.22. Suppose that 20 girls and 20 boys attend a dance party. They are arranged in 20 pairs so that the difference between the heights of the boy and the girl in each pair is less than 10 cm. The pairs are then rearranged so that the tallest boy and the tallest girl form the first pair, the second tallest boy and the second tallest girl form the second pair, etc. Prove that the difference between the heights of the boy and the girl in each of the rearranged pairs is again less than 10 cm.

10.23. Consider $mn + 1$ segments on the real line. Prove that there are $m + 1$ segments among the given segments such that they have a common point, or there are $n + 1$ segments that are pairwise disjoint.

10.24. Let us consider a circle in the Cartesian plane with its center at the origin $(0,0)$ and radius 1990. Let 555 points be given inside the circle such that their coordinates are positive integers, and no three of these points are collinear. Prove that there are two triangles of equal areas whose vertices belong to the set of given points.

10.25. Suppose that 2000 rectangles are given in the first quadrant of the Cartesian plane so that the following conditions are satisfied:

- (a) the origin is the common vertex for all rectangles;
- (b) the sides of all the rectangles are parallel to the coordinate axes, and the length of every side of every rectangle is a positive integer no greater than 100.

Prove that there are at least 40 rectangles among the given rectangles, denoted, for example, by $R_1, R_2, \dots, R_{40}$, so that $R_1 \subset R_2 \subset \dots \subset R_{40}$.

10.26. A square and nine lines are given in the plane. Each of these lines divides the square into two quadrilaterals such that the ratio of their areas is $2 : 3$. Prove that at least three of the given 9 lines have a common point.

10.27. Let c_1 and c_2 be circles of circumference 100. Suppose that 100 points are given on circle c_1 , and a set of arcs with the sum of their lengths that is less than 1 is given on circle c_2 . Prove that circle c_1 can be put on circle c_2 such that all the given points are outside each of the given arcs.

10.28. Twenty unit squares are given inside a square of side 15 cm . Prove that a circle of diameter 2 cm can be put inside the given square of side 15 cm such that it has no common points with any of the given unit squares.

10.29. A circle of diameter 100 and 32 lines are given in the plane α . Prove that there is a circle of diameter 3 which lies inside the given circle of diameter 100, and has no common points with the given lines.

10.30. Given a sequence $c_1 c_2 \dots c_n$, such that $c_1, c_2, \dots, c_n \in \{0, 1, 2, \dots, 9\}$, and $c_1 \neq 0$, prove that there is a positive integer k such that the number 2^k has the sequence $c_1 c_2 \dots c_n$ in the first n positions of its decimal representation.

Ramsey's Theorem

10.31. A group consists of 6 persons. Some of them are friends. Prove that there are 3 persons such that any two of them are friends or there are 3 persons such that no two of them are friends. Friendship is a symmetric relation.

10.32. Let $p \geq 2$ and $q \geq 2$ be positive integers. Prove that: $R(p, 2; 2) = p$, $R(2, q; 2) = q$.

10.33. (a) Let $p \geq 2$ and $q \geq 2$ be positive integers. Prove that

$$R(p+1, q+1; 2) \leq R(p, q+1; 2) + R(p+1, q; 2).$$

(b) If $R(p, q+1; 2)$ and $R(p+1, q; 2)$ are both even, prove that

$$R(p+1, q+1; 2) < R(p, q+1; 2) + R(p+1, q; 2).$$

10.34. Prove that for positive integers p and q the following relation holds:

$$R(p+1, q+1; 2) \leq \binom{p+q}{p}.$$

10.35. Prove that $R(3, 4; 2) = 9$.

10.36. Prove that $R(3, 5; 2) = 14$.

10.37. Every side and every diagonal of a 17-gon is colored blue, yellow, or red. Prove that 3 segments among the sides and diagonals of the 17-gon can be chosen, such that they have the same color and form a triangle.

10.38. Prove that for any positive integer n there is a positive integer $K(n)$ with the following property: if a set S consists of $K(n)$ points in the plane, such that no three of them are collinear, then n points can be chosen from S such that they are vertices of a convex n -gon.

10.39. Schur's theorem. Let k be a positive integer. Prove that there exists the smallest positive integer $S(k)$ such that for any positive integer $n \geq S(k)$ and any partition of the set $\{1, 2, \dots, n\}$ into blocks $A_1, A_2, \dots, A_k$, there is a positive integer $i \in \{1, 2, \dots, k\}$ and $x, y, z \in A_i$, such that $x + y = z$.

10.40. Van der Waerden's theorem. Let m and k be positive integers. Prove that there exists the smallest positive integer $W(m, k)$, such that for any partition of the set $\{1, 2, \dots, W(m, k)\}$ into k blocks, there exists an arithmetic progression consisting of m positive integers, such that all of them belong to the same block of the partition.

Chapter 11



Mathematical Games

11.1 The Nim Game

The nim game is defined as follows. Suppose there are a few piles with a finite number of coins in each of them. Two players alternately take any number of coins from any single one of the piles. The winner is the player who removes the last coin (or the last few coins). Suppose now that two players, A and B , are playing the nim game, and that player A makes the first move. Is there a winning strategy for player A or B ? If the answer to this question is yes, how should the player with a winning strategy play? Let us first consider some examples.

Example 11.1.1. (a) Suppose there are two piles, the first one consisting of 5 and the second consisting of 7 coins. In this case player A has a winning strategy. The first move of this strategy should be that player A removes 2 coins from the second pile. This way, player A makes the number of coins in the second pile equal to the number of coins in the first pile. Now it is obvious that after every move by player B , player A can remove some coins making the number of coins in the first pile equal to that in the second pile (possibly equal to 0). Obviously, this is a winning strategy.

(b) Suppose now that there are two piles, both containing 8 coins. In this case player B has a winning strategy, the same way as A had it in the previous case.

(c) Suppose there are three piles, containing 5, 5, and 8 coins. In this case player A removes all 8 coins from the third pile in the first move. The position with two piles, both containing 5 coins, is lost for player B who makes the next move. $\triangle$

Let $(5, 7)$, $(8, 8)$, and $(5, 5, 8)$ be the notation for the starting positions in Examples 11.1.1 (a)-(c). It follows from previous considerations that $(5, 7)$ and $(5, 5, 8)$ are winning positions for player A (with the first move), while position $(8, 8)$ is lost for player A .

Example 11.1.2. Suppose there are three piles, containing 8, 13, and 15 coins. Let us denote the starting position by $(8, 13, 15)$. Which one of the players A and B has a winning strategy?

In order to formulate a winning strategy (for one of the two players) we need binary representations of the positive integers 8, 13, and 15:

$$8 = 1 \cdot 2^3 + 0 \cdot 2^2 + 0 \cdot 2 + 0 \cdot 1 = 1000_2,$$

$$13 = 8 + 4 + 1 = 1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2 + 1 = 1101_2,$$

$$15 = 8 + 4 + 2 + 1 = 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2 + 1 = 1111_2.$$

Suppose that player A takes 10 coins from the third pile in the first move. The position after that is $(8, 13, 5)$. Let us consider the following tables related to the positions $(8, 13, 15)$ and $(8, 13, 5)$:

8	...	1	0	0	0	8	...	1	0	0	0
13	...	1	1	0	1	13	...	1	1	0	1
15	...	1	1	1	1	5	...	0	1	0	1
		1	0	1	0			0	0	0	0

The first three rows of these tables contain decimal and binary representations of the positive integers that determine the positions $(8, 13, 15)$ and $(8, 13, 5)$. The fourth row is obtained as follows. We consider every column of binary digits. Below the column we write down 0 if this column contains an even number of 1's, and we write down 1 if this column contains an odd number of 1's. The obtained binary representation of a nonnegative integer in the fourth column is called a *characteristic* of the position. Hence, $1010_2 = 10$ is the characteristic of position $(8, 13, 15)$, and 0 is the characteristic of position $(8, 13, 5)$.

The winning strategy is determined by the following two statements:

(A1) *If the characteristic of the position is not equal to 0, the player can play such that the characteristic of the position after their move becomes equal to 0.*

(A2) *If the characteristic of the position is equal to 0 (and there are some coins in the piles), then every subsequent move leads to a position with a characteristic that is not equal to 0.*

Statement (A2) is obvious. In order to prove statement (A1) let us consider the columns of binary digits of the positive integers that determine

the position; mark the first column (looking from left to right) that contains an odd number of 1's; choose one of these 1's and the corresponding positive integer and the related pile of coins; it is obvious that some coins can be removed from this pile such that the digit 1, which was marked, becomes 0, and then, every column of the binary digits contains an even number of 1's.

Note that the characteristic of position (5,13,15) is not equal to 0. Hence, by statements (A1) and (A2) it follows that (5,13,15) is a winning position for player A. They should play the game such that after each move the characteristic of the obtained position becomes equal to 0. $\triangle$

Remark 11.1.3. Suppose there are n piles, containing $k_1, k_2, \dots, k_n$ coins, where $n, k_1, k_2, \dots, k_n$ are arbitrary positive integers. The characteristic of position $(k_1, k_2, \dots, k_n)$ is determined analogously as in the previous example. The strategy based on statements (A1) and (A2) can be applied.

11.2 Golden Ratio in a Mathematical Game

Two players play the following game in which they make moves alternately. There are two piles containing m and n coins, where $m > n$. The player whose turn is to move can take a few coins from one of the piles, such that the number of removed coins is divisible by the number of coins in the other pile. The winner of the game is the player who leaves only one pile of coins after their move. The problem is to determine the values of m and n for which the player that starts the game has a winning strategy.

Let $x = \max \left\{ \frac{m}{n}, \frac{n}{m} \right\}$. We consider m , n , and x as variables that change value after every move. For the starting values m and n we have $m > n$, and hence $x = \frac{m}{n}$. It is obvious that $x \geq 1$ during the game, and after the winning move the variable x becomes equal to ∞ .

Let us determine the real number φ such that $\varphi > 1$ and $\varphi - \frac{1}{\varphi} = 1$. We obtain that $\varphi = \frac{\sqrt{5}+1}{2}$. The real number φ is known as the *golden ratio*, because of the following property: if a and b are real numbers such that $a > b > 0$ and $\frac{a+b}{a} = \frac{a}{b}$, then $\frac{a+b}{a} = \frac{a}{b} = \varphi$.

Let us consider the real numbers

$$\varphi = \frac{\sqrt{5}+1}{2}, \quad \frac{1}{\varphi} = \frac{2}{\sqrt{5}+1} = \frac{\sqrt{5}-1}{2}$$

as points on the real line. The distance between these two points is equal to 1. In order to formulate the winning strategy (for one of the players), note that the following three statements hold:

(A1) If $x = \frac{m}{n} \in \mathbb{N}$, then the player (whose turn it is to move) can take all m coins from one of the piles, and win the game immediately.

(A2) If $x = m/n > \varphi$ and x is not an integer, then the player (whose turn it is to move) can determine the positive integer k and remove kn coins from the pile with m coins, such that the following relations hold

$$\frac{m - kn}{n} = \frac{m}{n} - k \in \left(\frac{\sqrt{5} - 1}{2}, \frac{\sqrt{5} + 1}{2} \right).$$

After this move the fraction x takes a value such that

$$x = \max \left\{ \frac{m - kn}{n}, \frac{n}{m - kn} \right\} \in \left(1, \frac{\sqrt{5} + 1}{2} \right).$$

(A3) If $x = m/n \in (1, \varphi)$, then the player (whose turn it is to move) is forced to take n coins from the pile with m coins. Since

$$\frac{m - n}{n} = \frac{m}{n} - 1 < \frac{\sqrt{5} + 1}{2} - 1 = \frac{\sqrt{5} - 1}{2},$$

it follows that

$$\frac{n}{m - n} > \frac{2}{\sqrt{5} - 1} = \frac{\sqrt{5} + 1}{2},$$

and hence, after this move the variable x takes a value greater than φ .

The following consequences follow from statements (A1)–(A3). If $x < \varphi$, then the player whose turn it is to move is forced to make a move after which variable x becomes greater than φ . If $x > \varphi$, then the player whose turn it is to move, can make the winning move or a move after which variable x becomes less than φ . Hence, we have the following conclusions:

- If $\frac{m}{n} < \frac{\sqrt{5} + 1}{2}$, then the starting position is lost for the player who makes the first move.

- If $\frac{m}{n} > \frac{\sqrt{5} + 1}{2}$, then the starting position is winning for the player who makes the first move. The winning strategy is determined by statements (A1)–(A3).

11.3 Game of Fifteen

Fifteen tiles are put on fifteen fields of a square table 4×4 . One field is unoccupied. It is allowed to move a tile to an adjacent field if it is unoccupied. Two fields (unit squares) are adjacent if they have a common side. Any arrangement of tiles on the fields of the table 4×4 is called a *position*. The starting position is given in Figure 11.3.1. Is it possible to reach the position given in Figure 11.3.2 by a sequence of allowed moves?

Let us denote the fields of the square table 4×4 by positive integers 1, 2, ..., 16 as given in Figure 11.3.3. Then, every position is determined by a permutation $a_1a_2 \dots a_{16}$ of the set $\{1, 2, \dots, 16\}$, where a_k is the notation of the tile that is placed on the field k , if this field is occupied. If the field k is not occupied, we put $a_k = 16$. We say that a position is even (odd) if the corresponding permutation is even (odd).

1	2	3	4
5	6	7	8
9	10	11	12
13	15	14	

Fig. 11.3.1

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

Fig. 11.3.2

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Fig. 11.3.3

Consider a position that is determined by a permutation $a_1a_2 \dots a_{16}$. Suppose that a move is made, and that the tile that was moved is denoted by k . The new position is determined by the permutation that can be obtained from $a_1a_2 \dots a_{16}$ by interchanging terms a_k and 16. By Theorem 7.2.1 it follows that every move transforms an even (odd) position into an odd (even) position.

Let us now consider two positions, p_1 and p_2 , for which the field 16 is not occupied by a tile. If position p_2 can be reached from position p_1 , this can be done after an even number of moves. This is true because the empty field takes the same number of ‘steps’ to the left and to the right, and the same number of ‘steps’ up and down. If position p_2 can be reached from the position p_1 , then both positions are even or both are odd. However, the positions given in Figures 11.3.1 and 11.3.2 are determined by the permutations

$$(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 14, 16),$$
$$(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16),$$

and it is obvious that one of them is even, and the other one is odd. Hence the position in Figure 11.3.2 cannot be reached from the position in Figure 11.3.1.

Remark 11.3.1. Every even (odd) position can be reached if the starting position is even (odd).

11.4 Conway’s Game of Reaching a Level

A finite number of coins are arranged in some points with integer coordinates (x,y) on the Cartesian plane. Only one coin can be placed at one point. Coins may be moved and removed according to the following rules. Let us consider three points A , B , and C with integer coordinates such that the following conditions are satisfied:

(C1) *The points A , B , and C belong to the same horizontal line or to the same vertical line; the point B is the midpoint of the segment AC ; the distance between the points A and C is equal to 2.*

(C2) *There is a coin on points A and B , and there is no coin on point C , see Figures 11.4.1 and 11.4.2.*

Then, it is allowed to move the coin from point A to point C , and remove the coin from point B at the same time.

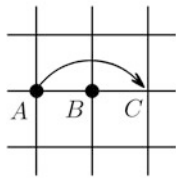


Fig. 11.4.1

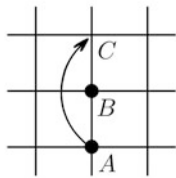


Fig. 11.4.2

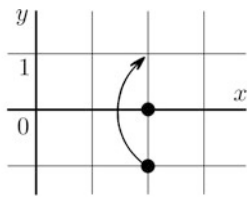


Fig. 11.4.3

If a coin is placed on the point whose coordinates are (x,y) , where $y = k$, we say that this coin is at the level k . Suppose that a finite number of coins are arranged on some points with integer coordinates, and that all these points are on the x -axis or below the x -axis. The player chooses how many coins will be used, and the points where these coins will be placed at the beginning of the game. The goal of the game is to put a coin at the highest possible level.

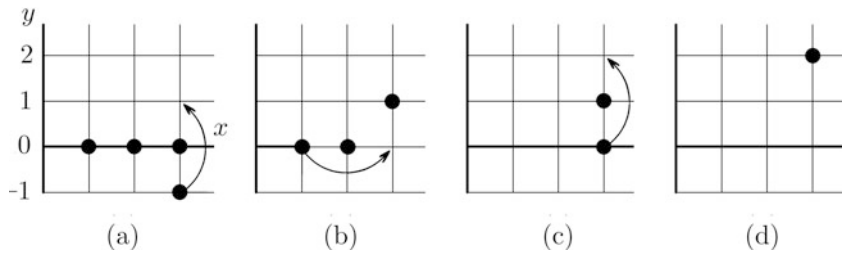


Fig. 11.4.4

Example 11.4.1. Level 1 can be reached as shown in Figure 11.4.3. Two coins are sufficient for the initial configuration. $\triangle$

Example 11.4.2. Four coins, arranged as shown in Figure 11.4.4(a), are sufficient to reach the level 2. The corresponding sequence of three steps (moves) is given in Figure 11.4.4(a)–(d). $\triangle$

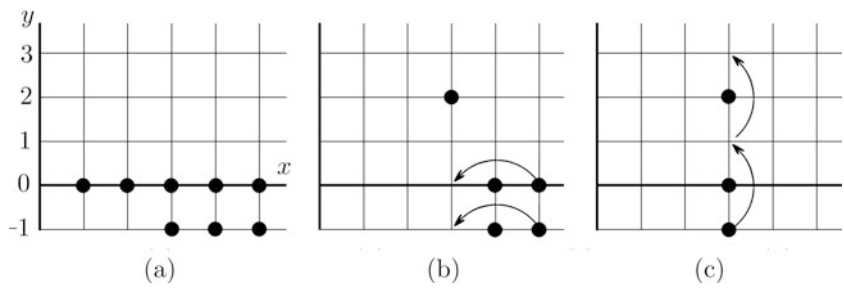


Fig. 11.4.5

Example 11.4.3. Level 3 can be reached using 8 coins after 7 steps. The initial configuration of the coins is given in Figure 11.4.5(a). Four coins can be used to reach level 2 as in Example 11.4.2, see Figure 11.4.5(a)–(b). Using the remaining 5 coins one can reach level 3 as shown in Figure 11.4.5(b)–(c).

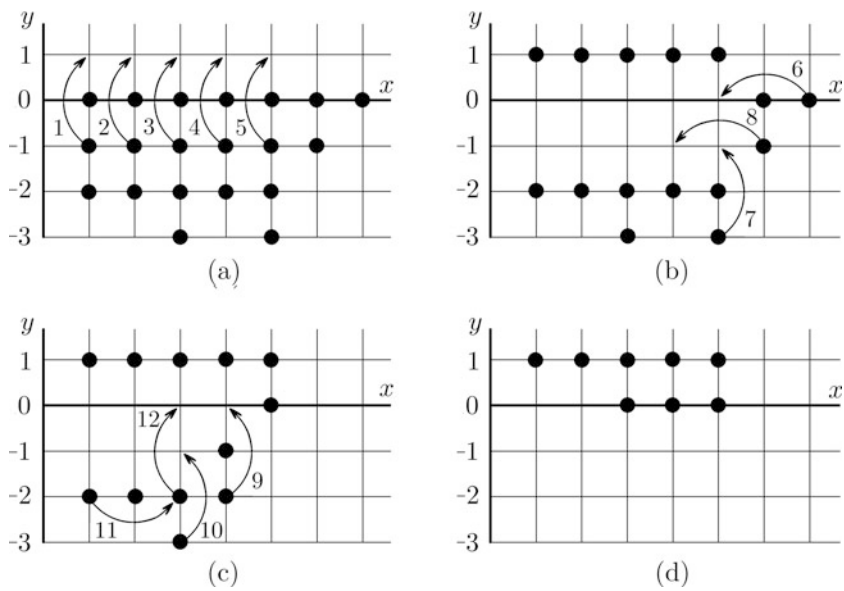


Fig. 11.4.6

Example 11.4.4. How can level 4 be reached? A possible way is presented in Figure 11.4.6(a)–(d). The initial position with 20 coins is presented in

Figure 11.4.6(a). The position with 8 coins obtained after 12 steps is given in Figure 11.4.6(d). Note that this position is the same as the initial position in Example 11.4.2, but placed one level higher. Similarly as in the previous example, one can reach level 4 after 7 additional steps. $\triangle$

Is it possible to reach levels higher than 4? We will show that it is not possible to put a coin at level 5. Then it is obvious that the same is true for any level higher than 4.

Let $A(a_1, a_2)$ and $B(b_1, b_2)$ be two points in the Cartesian plane. The sum $|a_1 - b_1| + |a_2 - b_2|$ is called the coordinate distance between points A and B . In what follows we consider only points with integer coordinates in the Cartesian plane. Let us label all such points as follows. An arbitrary point at level 5 is labeled x^0 . Any other point is labeled x^n , where n is the coordinate distance between this point and x^0 , see Figure 11.4.7.

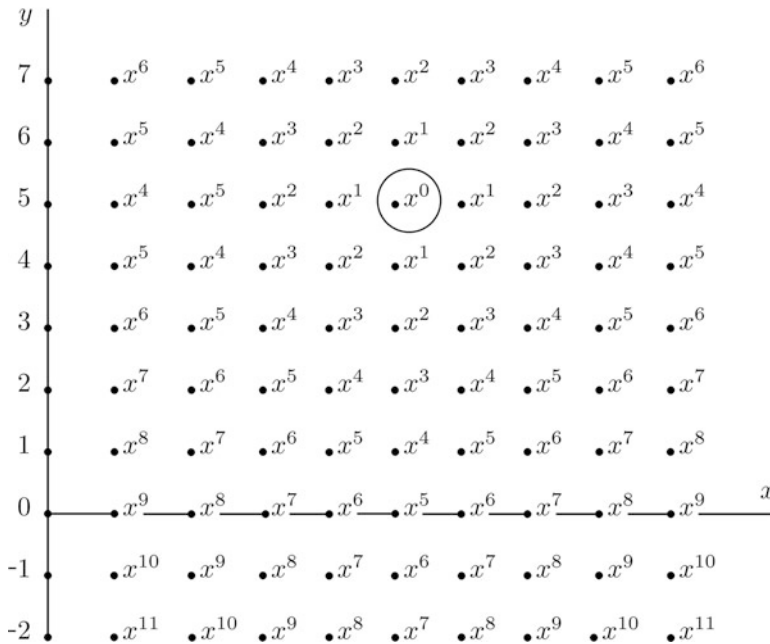


Fig. 11.4.7

Suppose that the coins are placed on some points with integer coordinates (at most one coin on one point). We define the *characteristic* of such a configuration of coins as the sum of the labels of the points occupied by the coins. Let us consider how this characteristic of the configuration of coins changes after a move in the game we are considering.

Case 1. A coin is moved toward point x^0 , and placed on point x^n . In this case two coins are removed from the points labeled x^{n+1} and x^{n+2} . The change of the characteristic is

$$c_1 = x^n - (x^{n+1} + x^{n+2}) = x^n(1 - x - x^2).$$

Case 2. A coin is moved from a point labeled x^n , and placed on a point labeled x^{n+2} . In this case two coins are removed from the points labeled x^n and x^{n+1} , and hence the change of the characteristic is

$$c_2 = x^{n+2} - (x^{n+1} + x^n) = x^n(x^2 - x - 1).$$

Case 3. A coin is moved from a point labeled x^n , and placed on a point labeled x^n . In this case two coins are removed from the points labeled x^n and x^{n-1} . The change of the characteristic is

$$c_3 = x^n - (x^n + x^{n-1}) = -x^{n-1}.$$

Let us now choose x to be equal to $(\sqrt{5}-1)/2$. Then we have $0 < x < 1$, $1 - x - x^2 = 0$, and

$$c_1 = 0, \quad c_2 = x^n(-2x) < 0, \quad c_3 < 0.$$

Hence, if $x = (\sqrt{5}-1)/2$, then the *characteristic of any position does not increase* after any possible move of this game.

Let G_0 be the initial configuration of coins that are all placed on some points with integer coordinates on or below the x -axis. The characteristic of position G_0 is less than

$$\begin{aligned} S &= x^5 + 2(x^6 + x^7 + x^8 + \cdots) + x^6 + 2(x^7 + x^8 + x^9 + \cdots) \\ &\quad + x^7 + 2(x^8 + x^9 + x^{10} + \cdots) + \cdots \\ &= \frac{x^5}{1-x} + 2\left(\frac{x^6}{1-x} + \frac{x^7}{1-x} + \frac{x^8}{1-x} + \cdots\right) \\ &= \frac{x^5}{1-x} + \frac{2x^6}{1-x}(1 + x + x^2 + \cdots) = \frac{x^5}{1-x} + \frac{2x^6}{(1-x)^2} = \frac{x^5}{x^2} + \frac{2x^6}{x^4} \\ &= x^3 + 2x^2 = (1-x)x + 2x^2 = x + x^2 = 1. \end{aligned}$$

It follows that the characteristic of every position obtained from G_0 during the game is less than 1. Suppose that the final position has a coin on point x^0 . The characteristic of such a final position is not less than $x^0 = 1$. This contradicts the fact that the characteristic of any position is less than 1. Hence, it is not possible to place a coin at level 5. This game was invented by John Conway.

11.5 Two More Games

Example 11.5.1. Suppose that n points are given on a circle, where $n \geq 4$. Assume that the points are labeled $1, 2, \dots, n$, one after the other in a chosen direction. Two players, A and B , play the game in which they alternately draw a chord which endpoints belong to the set $\{1, 2, \dots, n\}$. The endpoints of any chord should be of the same parity. It is not allowed for the chords to have points of intersection. Player A starts the game. At the end of the game the winner is the player who has drawn the last chord. Which player has a winning strategy?

We shall prove that player A has a winning strategy if $n = 4k$, $n = 4k+1$, or $n = 4k+3$, where $k \in \mathbb{N}$. Player B has a winning strategy if $n = 4k+2$, where $k \in \mathbb{N}$. Without loss of generality we can assume that the given points are the vertices of a regular n -gon.

Case $n = 4k$, $k \in \mathbb{N}$. Player A draws a chord with endpoints 1 and $2k+1$ as the first move of the game. If player B draws chord ij , then, in the next move, player A draws chord $i'j'$ that is symmetric to ij around the center of the circle.

Case $n = 4k+2$, $k \in \mathbb{N}$. In this case there is no a diameter ij of the circle, such that $i, j \in \{1, 2, \dots, n\}$, where i and j are of the same parity. The winning strategy for player B is to draw chord $i'j'$ that is symmetric to ij around the center of the circle, where ij is the chord drawn by A in the previous move of the game.

Case $n = 4k+1$, $k \in \mathbb{N}$. As the first move, player A draws a chord with endpoints 1 and 3 . In the sequel, the points $4, 5, \dots, 4k+1$ can be chosen as the endpoints of the new chords. There are $4k-2 = 4(k-1)+2$ such points, and, using the strategy of player A in the previous case, now player B will always win the game.

Case $n = 4k+3$, $k \in \mathbb{N}$. As the first move player A draws a chord with endpoints $2k+1$ and $2k+3$. The points $1, 2, \dots, 2k$ and $2k+4, 2k+5, \dots, 4k+3$ can be chosen in the next moves. Without loss of generality we can suppose that the $4k$ points in this order are vertices of a regular $4k$ -gon. The diameters $(1, 2k+4), (2, 2k+5), \dots, (2k, 4k+3)$ are not allowed to be drawn, because the endpoints are not of the same parity. Using a strategy based on symmetry around the center of the circle, player A can win the game. $\triangle$

Example 11.5.2. A coin is placed on the lower-left corner field of a square table 8×8 , see Figure 11.5.1. Two fields of the table are adjacent if they have a common side or a common vertex. Two players, A and B , play a game in which they alternately move a coin to an adjacent field. The allowed

directions are given in Figure 11.5.1. The winner is the player who puts the coin in the top-right corner field. Which player has a winning strategy?

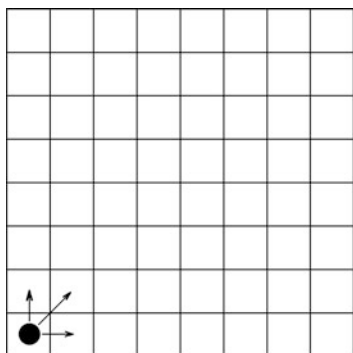


Fig. 11.5.1

-	+	-	+	-	+	-	+
-	-	-	-	-	-	-	-
-	+	-	+	-	+	-	+
-	-	-	-	-	-	-	-
-	+	-	+	-	+	-	+
-	-	-	-	-	-	-	-
-	+	-	+	-	+	-	+
-	-	-	-	-	-	-	-

Fig. 11.5.2

Let the fields (unit squares) of the table 8×8 be labeled $+$ or $-$ as shown in Figure 11.5.2. From any field labeled $+$ the coin necessarily goes to a field labeled $-$. From any field labeled $-$ the coin can be moved to a unit square labeled $+$. The winning strategy for player A is to always put the coin in the allowed field labeled $+$. It is obvious that after a finite number of moves the coin will reach the top-right corner field, and the last move will be made by player A . $\triangle$

Exercises

11.1. Two players, A and B , play the following game. First, player A places a knight (a chess piece) on a field of the chessboard. Then the players move the knight according to the rules of chess. The game is lost by the player who cannot place the knight on a field that was not already visited by the knight. Which player has a winning strategy if the game is played on a board:

- (a) 8×8 ; (b) 9×9 ?

11.2. Two players, A and B , play the following game. First, player A writes down a (decimal) digit d_1 . Then player B writes down a digit d_2 on the left side or on the right side of d_1 . This way the two-digit nonnegative integer d_1d_2 , or d_2d_1 , is obtained. Then the players alternately write down new digits on the left side or on the right side of the previously written positive integer. Prove that player A can play this game such that the positive integer obtained after player B moves is never a perfect square.

11.3. Suppose that n coins are placed on the first n fields of a table $(2n + 1) \times 1$, such that no two coins are placed on the same field. Two players, A and B , play the following game in which they alternately move the coins. Player A starts the game, and always moves an arbitrarily chosen coin to the first free field to the right side of this coin. Player B always moves a coin to the adjacent field to the left side, if this adjacent field is unoccupied. The goal of player A is to put a coin in the last field of the table, and the goal of player B is to prevent this. Which player has the winning strategy?

11.4. The fields of a table $1 \times n$ are labeled $1, 2, \dots, n$. A coin is placed on each of the fields $n - 2, n - 1, n$. Two players play a game in which they alternately move a coin from a field j on an unoccupied field i , such that $i < j$. The game is lost for the player who cannot move a coin anymore. Prove that the player who starts the game has the winning strategy.

11.5. Suppose that 49 coins are placed in a pile. Two players play a game in which they alternately take 1, 2, 3, or 4 coins from the pile. The game lasts until all the coins are taken away. The winner is the player who has taken an even number of coins. Which player, the first or the second, has a winning strategy? Determine the winning strategy.

11.6. Eight white chips are placed on eight fields of the first row of a chessboard 8×8 , one chip on every field. Similarly, eight black chips are placed on eight fields in the last row. Two players, W and B , play a game in which they alternately move the chips. Player W starts the game, and always moves a white chip, while player B always moves a black chip. Each player can move a chip forward or back in the line (column) where this chip belongs, and place it on a free field. It is not allowed to jump over a chip. The game is lost for the player who cannot move a chip anymore. Prove that player B has the winning strategy.

Chapter 12



Elementary Probability

12.1 Discrete Probability Space

In this chapter we shall introduce the basic notions of Probability Theory. In order to avoid a strong measure-theoretical background we shall concentrate on the so-called discrete probability spaces. The reader can find an excellent and exhaustive presentation of Probability Theory in the book by Feller [3].

Probability Theory deals with stochastic experiments that can be repeated many times under approximately the same conditions but with different outcomes. An example is a coin-toss experiment with two possible outcomes: head and tail. In this chapter we shall always use 1 and 0 as notation for these two outcomes. The first step in building this rigorous theory is to introduce the set of possible outcomes (usually called the *sample space*) of an experiment. Such a set is denoted by Ω , and an outcome is denoted by ω . First let us give some examples.

Example 12.1.1. A coin is tossed once. The sample space of this simple experiment is $\Omega = \{0, 1\}$. $\triangle$

Example 12.1.2. A coin is tossed twice. In this case the sample space is $\Omega = \{00, 01, 10, 11\}$. $\triangle$

Example 12.1.3. A coin is tossed three times. The sample space is $\Omega = \{000, 001, 010, 100, 011, 101, 110, 111\}$. $\triangle$

Example 12.1.4. A coin is tossed n times. The sample space is

$$\Omega = \{d_1 d_2 \dots d_n \mid d_k = 0 \text{ or } d_k = 1 \text{ for every } k = 1, 2, \dots, n\}.$$

Obviously, the number of possible outcomes is 2^n , i.e., $|\Omega| = 2^n$. $\triangle$

Example 12.1.5. A coin is tossed until a head (denoted by 1) appears. The sample space is given by $\Omega = \{1, 01, 001, 0001, 00001, \dots\} \cup \{\omega_0\}$ where $\omega_0 = 000\dots$ means that 1 never appears. Note that in this case the sample space is an infinite *countable* set. $\triangle$

Example 12.1.6. A coin is tossed infinitely many times. A possible outcome is, for example, the following 0–1 sequence: $\omega = 0110100010110111\dots$. This outcome can be rewritten in the form $\tilde{\omega} = 0.0110100010110111\dots$, and considered as the binary representation of a real number from the interval $[0, 1]$. Hence, in this case the sample space is $\Omega = [0, 1]$. Note that interval $[0, 1]$ is an infinite *uncountable* set. $\triangle$

In what follows we shall consider only experiments with finite and infinite countable sample spaces. The notation $\Omega = \{\omega_1, \omega_2, \omega_3, \dots\}$ means that the sample space Ω may be an infinite countable set as well as a finite set. The notation $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$ means that the set Ω consists of exactly n elements.

It is worth mentioning that different outcomes of an experiment may share a common property. For example, a coin is tossed 3 times, and the property of the outcomes we are interested in is that the number of obtained 1's (heads) is even. The sample space is given in Example 12.1.3, and the outcomes with the specified property are 000, 011, 101, and 110. The set $A = \{000, 011, 101, 111\}$ is an *event*.

Definition 12.1.7. Let $\Omega = \{\omega_1, \omega_2, \omega_3, \dots\}$ be the sample space of an experiment. Every subset of Ω is an *event*. If $A \subset \Omega$ is an event and $\omega \in A$, we say that outcome ω is *favorable* for event A .

Let the sample space $\Omega = \{\omega_1, \omega_2, \omega_3, \dots\}$ be given, and let $\mathcal{A}$ be the set of all events, i.e., the power set of Ω . Note that $\emptyset$ is called an *impossible* event, and Ω is a *sure* event. Since events are defined as subsets of Ω we can apply set operations to events. If A is an event, then $\bar{A} = \Omega \setminus A$ is the event *complementary* to A . For the *union* and *intersection* of events A and B we shall use notation $A \cup B$, and $A \cap B$ (or simply AB), respectively. Events A and B are *disjoint* if $AB = \emptyset$. Events $A_1, A_2, A_3, \dots$ are *mutually disjoint* if $A_m A_n = \emptyset$ for every $m \neq n$.

Remark 12.1.8. If sample space Ω is an infinite uncountable set, then not every subset of Ω can be considered an event. In such cases we should first choose a collection of events, i.e., a collection $\mathcal{A}$ consisting of subsets of Ω that will be the events. Such a collection $\mathcal{A}$ should be a σ -algebra, i.e., it should satisfy the following three conditions: (1) $\Omega \in \mathcal{A}$; (2) if $A \in \mathcal{A}$, then $\bar{A} \in \mathcal{A}$; (3) if $A_1, A_2, \dots \in \mathcal{A}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.

Suppose that a coin-toss experiment is repeated n times. Let $\alpha_n(1)$ be the number of appearances of the outcome 1. The fraction $\frac{\alpha_n(1)}{n}$ is called the relative frequency of outcome 1 in these n experiments. The relative frequency $\frac{\alpha_n(1)}{n}$ depends on the experiments, but experience teaches us that for large values of n , the following approximate equality holds:

$$\frac{\alpha_n(1)}{n} \approx \frac{1}{2}. \quad (12.1.1)$$

Suppose now that a fair die whose sides are numbered 1, 2, 3, 4, 5, and 6 is thrown n times. Let $\beta_n(6)$ be the number of appearances of number 6 in these n experiments. Then, for large n ,

$$\frac{\beta_n(1)}{n} \approx \frac{1}{6}. \quad (12.1.2)$$

The approximate equalities (12.1.1) and (12.1.2) show us that the possibility of the appearance of an outcome (or an event) in a random experiment can be measured numerically, and lead to the notion of the *probability of an event* that is basic in Probability Theory. First we shall give the so-called *classical definition of probability*.

Definition 12.1.9. Let Ω be the finite sample space of an experiment with equally likely outcomes, and $A \subset \Omega$ be an event. The probability of event A , denoted by $P(A)$, is defined by

$$P(A) = \frac{\text{the number of outcomes of } A}{\text{the number of all possible outcomes}}. \quad (12.1.3)$$

Example 12.1.10. A fair die is tossed twice. The list of possible outcomes is given in the following table:

11	12	13	14	15	16
21	22	23	24	25	26
31	32	33	34	35	36
41	42	43	44	45	46
51	52	53	54	55	56
61	62	63	64	65	66

Here, for example, 13 means that 1 and 3 are obtained in the first and the second tossing, respectively, etc. Let A be the event that the sum of the two obtained number is greater than 6, and B be the event that 5 and 6 do not appear. Using formula (11.1.3) it is easy to obtain that

$$P(A) = \frac{21}{36}, \quad P(B) = \frac{16}{36}, \quad P(AB) = \frac{3}{36}. \quad \triangle$$

Example 12.1.11. The sides of four dice are numbered as follows:

$$\begin{aligned} D_1(1, 1, 1, 5, 5, 5), & \quad D_2(2, 2, 2, 2, 6, 6), \\ D_3(3, 3, 3, 3, 3, 3), & \quad D_4(0, 0, 4, 4, 4, 4). \end{aligned}$$

Peter and Paul play the following game. First, Peter choose a die, and after that Paul choose one of the remaining three dice. Each player throws his die, and the winner is the player who gets the larger number. Which one of the two players is in a better position?

Suppose that Peter and Paul choose dice D_1 and D_2 , respectively. The probability that Paul wins the game is

$$P\{D_1 < D_2\} = \frac{3 \cdot 4 + 6 \cdot 2}{36} = \frac{2}{3}. \quad (12.1.4)$$

Hence, in this case Paul wins the game with a probability of $2/3$. We say that die D_2 is better than die D_1 , and write $D_1 \prec D_2$. Similarly we can compare every two dice, and establish a partial order on the set $\{D_1, D_2, D_3, D_4\}$. Let us now compare dice D_2 and D_3 . We have

$$P\{D_2 < D_3\} = \frac{4 \cdot 6}{36} = \frac{2}{3}. \quad (12.1.5)$$

It follows that D_3 is better than D_2 , i.e., $D_2 \prec D_3$. Note that relations $D_1 \prec D_2$ and $D_2 \prec D_3$ do not imply that $D_1 \prec D_3$; i.e., $\prec$ is not a transitive relation! Indeed, $P\{D_1 < D_3\} = P\{D_1 > D_3\} = 1/2$. Note also that

$$P\{D_3 < D_4\} = \frac{6 \cdot 4}{36} = \frac{2}{3}, \quad (12.1.6)$$

$$P\{D_4 < D_1\} = \frac{2 \cdot 6 + 4 \cdot 3}{36} = \frac{2}{3}. \quad (12.1.7)$$

It follows from (12.1.4)–(12.1.7) that $D_1 \prec D_2 \prec D_3 \prec D_4 \prec D_1$. Hence, Paul is in a better position. He can always choose a die that guarantees he will win the game with a probability of $2/3$. $\triangle$

Example 12.1.12. A group consists of n persons. For the sake of simplicity let us assume that none of them was born on February 29. Let p_n be the probability that there are two persons from the group celebrating their birthdays on the same date. What is the minimal value of n such that $p_n > \frac{1}{2}$ holds?

It is obvious that $0 = p_1 < p_2 < p_3 < \dots < p_{366} = 1$, and $p_n = 1$ for every $n \geq 366$. The probability p_n is given by

$$p_n = 1 - \frac{365 \cdot 364 \cdot 363 \dots (365 - n + 1)}{365^n}$$

It is easy to check that $p_{22} < \frac{1}{2} < p_{23}$, and $p_{55} > 0.99$, $p_{68} > 0.999$. $\triangle$

The classical definition of probability is not appropriate for random experiments with a finite number of possible outcomes that are not equally likely. The following definition gives the probabilistic model for every experiment with finite or infinite countable sample space.

Definition 12.1.13. Let $\Omega = \{\omega_1, \omega_2, \omega_3, \dots\}$ be the sample space of an experiment, and $\mathcal{A}$ be the power set of Ω . Let us assign the probabilities $p_1, p_2, p_3, \dots$ to the outcomes $\omega_1, \omega_2, \omega_3, \dots$, respectively, such that the following two conditions are satisfied:

- (a) $p_n \geq 0$ for every $k \in \mathbb{N}$;
- (b) $\sum_k p_k = 1$.

Real number p_k is called the *probability of outcome* ω_k . If $A \in \mathcal{A}$ is an event, then the *probability of event* A is the sum of the probabilities of all outcomes favorable for event A . If $A = \emptyset$ we define $P(A) = 0$.

Hence, a function $P : \mathcal{A} \rightarrow \mathbb{R}$ is defined. The triple $(\Omega, \mathcal{A}, P)$ is called a *probability space*. Note also that the classical definition of probability is a special case of Definition 12.1.13.

Example 12.1.14. As in Example 12.1.5 let us consider the following experiment. A coin is tossed until a head (denoted by 1) appears. The sample space is given by $\Omega = \{1, 01, 001, 0001, 00001, \dots\} \cup \{\omega_0\}$ where $\omega_0 = 000\dots$ means that 1 never appears. It is natural to assign the following probabilities to the outcomes of this experiment:

$$p(1) = \frac{1}{2}, \quad p(01) = \frac{1}{2^2}, \quad p(001) = \frac{1}{2^3}, \quad p(0001) = \frac{1}{2^4}, \quad \dots$$

For example, if a coin is tossed once, there are two equally likely outcomes, 0 and 1. Hence, $p(1) = 1/2$. If a coin is tossed twice, then there are four equally likely outcomes: 00, 01, 10, and 11. Hence, $p(01) = 1/4$, etc. Note that $\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots = 1$. Condition (b) from Definition 12.1.13 suggests that $p(\omega_0) = 0$, and condition (a) implies that the probability of some outcomes may be equal to 0.

Let A be the event that the experiment finishes after an even number of coin tossing. Then we have

$$P(A) = \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \cdots = \frac{1}{4} \cdot \frac{1}{1 - \frac{1}{4}} = \frac{1}{3}. \triangle$$

Remark 12.1.15. In the probabilistic models of random experiments we assume that the probabilities of the outcomes are *given*. *Probability Theory* does not deal with the question of how to choose (estimate) these probabilities. This problem is the subject of another branch of mathematics, namely *Mathematical Statistics*.

Remark 12.1.16. The following properties of probability follow immediately from Definition 12.1.13:

- A1. $P(\Omega) = 1$;
- A2. $P(A) \geq 0$ for every $A \in \mathcal{A}$;
- A3. (σ -additivity) If $A_1, A_2, A_3, \dots$ are pairwise disjoint events from $\mathcal{A}$, then the following equality holds:

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

In the axiomatic definition of probability these three properties are taken as axioms. The following properties can easily be obtained from Definition 12.1.13 as well.

- B1. $P(\bar{A}) = 1 - P(A)$;
- B2. If $A \subset B$, then $P(A) \leq P(B)$;
- B3. If $AB = \emptyset$, then $P(A \cup B) = P(A) + P(B)$;
- B4. $P(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} P(A_n)$;
- B5. $P(A \cup B) = P(A) + P(B) - P(AB)$;
- B6. Inclusion-exclusion principle:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i A_j) + \sum_{1 \leq i < j < k \leq n} P(A_i A_j A_k) - \cdots + (-1)^n P(A_1 A_2 \dots A_n);$$

B7. If $A_1 \supset A_2 \supset \cdots$ and $\bigcap_{n=1}^{\infty} A_n = \emptyset$, then $\lim_{n \rightarrow \infty} P(A_n) = 0$.

The axiomatic definition of probability was given by A.N. Kolmogorov in 1933. Kolmogorov originally used the properties A1, A2, B3, and B7 as axioms. It can be proved that property A3 is logically equivalent to the conjunction of B3 and B7.

12.2 Conditional Probability and Independence

Definition 12.2.1. Let us consider two events A and B such that $P(A) > 0$. The *conditional probability* of event B under the condition that A occurs is denoted by $P(B|A)$ and given by

$$P(B|A) = \frac{P(AB)}{P(A)}. \quad (12.2.1)$$

Example 12.2.2. Let A and B be the events defined in Example 12.1.10. Then we have $P(B|A) = \frac{3/36}{21/36} = \frac{3}{21} = \frac{1}{7}$. $\triangle$

Theorem 12.2.3. The formula of total probability. Let $A_1, A_2, \dots, A_n$ be events such that the following conditions are satisfied:

- (a) $P(A_i) > 0$ for every $i \in \{1, 2, \dots, n\}$;
- (b) $A_i A_j = \emptyset$ for every $i \neq j$;
- (c) $\Omega = A_1 \cup A_2 \cup \cdots \cup A_n$.

Then, for every event $B \in \mathcal{A}$, its probability is given by

$$P(B) = \sum_{i=1}^n P(A_i)P(B|A_i). \quad (12.2.2)$$

Example 12.2.4. There are two boxes. The first box contains 10 red and 15 blue balls, and the second box contains 8 red and 16 blue balls. We choose a box at random (each box with equal probability $1/2$), and then choose a ball at random from this box (each ball with equal probability).

- (a) What is the probability that the chosen ball is red?
 (b) If the chosen ball is red, what is the probability that it is chosen from the second box?

We shall answer these two questions using the definition of conditional probability and the formula of total probability. Let A_1 be the event that the first box is chosen, A_2 be the event that the second box is chosen, and B be the event that the chosen ball is red.

$$(a) \quad P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) = \frac{1}{2} \cdot \frac{10}{25} + \frac{1}{2} \cdot \frac{8}{24} = \frac{11}{30}.$$

$$(b) \quad P(A_2|B) = \frac{P(A_2)P(B|A_2)}{P(B)} = \frac{\frac{1}{2} \cdot \frac{8}{24}}{\frac{11}{30}} = \frac{5}{11}. \quad \triangle$$

Definition 12.2.5. Events A and B are *independent* if

$$P(AB) = P(A)P(B).$$

Definition 12.2.6. Let $\mathcal{F} = \{A_i | i \in I\}$ be an arbitrary family of events.

(a) The family $\mathcal{F}$ is a collection of *pairwise independent* events if the equality $P(A_i A_j) = P(A_i)P(A_j)$ holds for any two distinct indices $i, j \in I$.

(b) The family $\mathcal{F}$ is a collection of *collectively independent* events if

$$P(A_{i_1} A_{i_2} \dots A_{i_n}) = P(A_{i_1})P(A_{i_2}) \dots P(A_{i_n}).$$

for every finite subset $\{i_1, i_2, \dots, i_n\}$ of the set of indices I .

Events A , B , and C are pairwise independent if $P(AB) = P(A)P(B)$, $P(BC) = P(B)P(C)$, and $P(CA) = P(C)P(A)$. Events A , B , and C are collectively independent if the previous three equalities hold and $P(ABC) = P(A)P(B)P(C)$ holds as well.

A family of collectively independent events is obviously a family of pairwise independent events. The converse statement does not hold. This can be proved by the following simple example.

Example 12.2.7. Let $\Omega = \{00, 01, 10, 11\}$ be the sample space consisting of the outcomes of equal probability $1/4$. Let us consider the following events:

$$A = \{00, 01\}, \quad B = \{00, 10\}, \quad C = \{00, 11\}.$$

Then we have: $P(A) = P(B) = P(C) = 1/2$, $P(AB) = 1/4 = P(A)P(B)$, $P(BC) = 1/4 = P(B)P(C)$, $P(CA) = 1/4 = P(C)P(A)$, and

$$P(ABC) = \frac{1}{4} \neq \frac{1}{8} = P(A)P(B)P(C).$$

Hence, events A , B , and C are pairwise independent, but not collectively independent as a family consisting of three events. $\triangle$

Example 12.2.8. Let A and B be *disjoint* events with positive probabilities. Then, A and B are *dependent* events. Indeed, $AB = \emptyset$, $P(A) > 0$, $P(B) > 0$, implies that $P(AB) = 0 < P(A)P(B)$. $\triangle$

12.3 Discrete Random Variables

Example 12.3.1. Consider the following simple experiment. A coin is tossed three times. The sample space Ω is given in Example 12.1.3. We suppose that all outcomes are of equal probability $1/8$. For every outcome we can count how many 1's (heads) are obtained:

000	001	010	100	011	101	110	111
↓	↓	↓	↓	↓	↓	↓	↓
0	1	1	1	2	2	2	3

Function $X : \Omega \rightarrow \mathbb{R}$ is defined, where $X(000) = 0$, $X(001) = 1$, etc. Such a function is called a *random variable*. Set $\mathcal{S}_X = \{0, 1, 2, 3\}$ is the *set of values* of random variable X . For every value $i \in \mathcal{S}_X$ we can determine the probability $P\{X = i\}$. It is easy to see that

$$P\{X = 0\} = \frac{1}{8}, \quad P\{X = 1\} = \frac{3}{8}, \quad P\{X = 2\} = \frac{3}{8}, \quad P\{X = 3\} = \frac{1}{8}.$$

We say that the *probability distribution of random variable X* has been determined. It can be represented in the following form:

$$X : \begin{pmatrix} 0 & 1 & 2 & 3 \\ \frac{1}{8} & \frac{3}{8} & \frac{3}{8} & \frac{1}{8} \end{pmatrix}. \quad \triangle$$

More generally, let $(\Omega, \mathcal{A}, P)$ be a discrete probability space introduced by Definition 12.1.13. Every function $X : \Omega \rightarrow \mathbb{R}$ is called a *discrete random variable*. Set $\mathcal{S}_X = \{x \mid (\exists \omega \in \Omega) X(\omega) = x\}$ is called the set of values of random variable X . Suppose that $\mathcal{S}_X = \{x_1, x_2, x_3, \dots\}$ and denote $p_k = P\{X = x_k\}$, for every $k = 1, 2, 3, \dots$. The sequence of pairs $(x_1, p_1), (x_2, p_2), (x_3, p_3), \dots$ determines the *probability distribution* of random variable X . It can be written in the following form:

$$X : \begin{pmatrix} x_1 & x_2 & x_3 & \dots \\ p_1 & p_2 & p_3 & \dots \end{pmatrix}. \quad (12.3.1)$$

It is obvious that the condition $p_1 + p_2 + p_3 + \dots = 1$ is satisfied.

Example 12.3.2. Indicator of an event. Let $A \in \mathcal{A}$ be an event. Function $I_A : \Omega \rightarrow \mathbb{R}$ given by

$$I_A(\omega) = \begin{cases} 1, & \text{if } \omega \in A, \\ 0, & \text{if } \omega \notin A, \end{cases} \quad (12.3.2)$$

is called the *indicator* function of event A . Its probability distribution is given by $P\{I_A = 0\} = 1 - p$, $P\{I_A = 1\} = p$, where $p = P(A)$. $\triangle$

Example 12.3.3. Binomial distribution. Consider a sequence of n independent experiments, each of them with two possible outcomes: success (denoted by 1) with probability $p \in (0, 1)$, and failure (denoted by 0) with probability $1 - p$. The sample space Ω is given in Example 12.1.4. For every outcome $\omega = (d_1, d_2, \dots, d_n) \in \{0, 1\}^n$, we define its probability by

$$P(\omega) = p^{d_1+d_2+\dots+d_n}(1-p)^{n-d_1-d_2-\dots-d_n}. \quad (12.3.3)$$

Let us also define the function $S_n : \Omega \rightarrow \mathbb{R}$ by

$$S_n(\omega) = d_1 + d_2 + \dots + d_n. \quad (12.3.4)$$

The set of values of function S_n is $\{0, 1, \dots, n\}$, and it is easy to get

$$P\{S_n = k\} = \binom{n}{k} p^k (1-p)^{n-k}, \text{ for every } k \in \{0, 1, \dots, n\}. \quad (12.3.5)$$

Note that $\sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} = (p + 1 - p)^n = 1$. The distribution given by (12.3.5) is called a *binomial distribution* with the parameters n and p . Random variable S_n is a *binomial random variable*. We use the notation $S_n \in \mathcal{B}(n, p)$. If I_k is the indicator function of success in the k -th experiment, then the binomial random variable S_n can be represented as follows:

$$S_n = I_1 + I_2 + \dots + I_n. \quad (12.3.6)$$

Example 12.3.4. Geometric distribution. Consider again a sequence of independent experiments, each of them with two possible outcomes: success (denoted by 1) with probability $p \in (0, 1)$, and failure (denoted by 0) with probability $1 - p$. Let X be the number of experiments needed to get a success. The probability distribution of random variable X is given by

$$P\{X = k\} = p(1-p)^{k-1}, \quad k \in \{1, 2, 3, \dots\}. \quad (12.3.7)$$

We call it a *geometric distribution* with parameter p , and use notation $X \in \Gamma(p)$. Random variable $Y = X - 1$ represents the number of failures that occur before the first success, and its probability distribution is given by $P\{Y = k\} = p(1-p)^k$, $k \in \{0, 1, 2, 3, \dots\}$. $\triangle$

Example 12.3.5. Negative binomial distribution. Consider a sequence of experiments as in Example 12.3.4. Let X be the number of experiments until r successes occur. The probability distribution of random variable X is given by

$$P\{X = k\} = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k \in \{r, r+1, \dots\}. \quad (12.3.8)$$

This probability distribution is called a *negative binomial distribution* with parameters r and p . Note that random variable X can be represented in the form $X = X_1 + X_2 + \cdots + X_r$, where $X_1, X_2, \dots, X_r$ are independent random variables that have a common geometric distribution with parameter p .

Instead of X , sometimes it is more convenient to consider the random variable $Y = X - r$, i.e., the number of failures that occur before the r -th success. The probability distribution of random variable Y , also called a negative binomial distribution, is given by

$$P\{Y = k\} = \binom{k-r-1}{k} p^r (1-p)^k, \quad k \in \{0, 1, 2, \dots\}. \quad (12.3.9)$$

Example 12.3.6. Poisson distribution. If a random variable X takes values in the set $\{0, 1, 2, \dots\}$, and for some $\lambda > 0$,

$$P\{X = k\} = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k \in \{0, 1, 2, \dots\}, \quad (12.3.10)$$

then we say that X has a *Poisson distribution* with parameter λ . If $p = p(n) = \frac{\lambda}{n}$, then for every fixed $k \in \{0, 1, 2, \dots\}$, the probability $\pi_\lambda(k) = e^{-\lambda} \frac{\lambda^k}{k!}$ can be obtained as the limit value of the binomial probability $P_n(k) = \binom{n}{k} p^k (1-p)^{n-k}$, as $n \rightarrow \infty$. $\triangle$

Definition 12.3.7. Let $\mathcal{S}_X = \{x_1, x_2, \dots\}$ and $\mathcal{S}_Y = \{y_1, y_2, \dots\}$ be the sets of values of the random variables X and Y , respectively. Then, X and Y are *independent* random variables if, for every $x_i \in \mathcal{S}_X$ and $y_j \in \mathcal{S}_Y$,

$$P\{X = x_i, Y = y_j\} = P\{X = x_i\} \cdot P\{Y = y_j\}. \quad (12.3.11)$$

12.4 Mathematical Expectation

In this section we shall introduce two important numerical characteristics of random variables, namely *mathematical expectation* and *variance*.

Definition 12.4.1. Let X be a random variable with the probability distribution

$$X : \begin{pmatrix} x_1 & x_2 & \dots & x_n \\ p_1 & p_2 & \dots & p_n \end{pmatrix}. \quad (12.4.1)$$

The *mathematical expectation* of random variable X is defined by

$$E(X) = \sum_{k=1}^n x_k p_k. \quad (12.4.2)$$

Remark 12.4.2. In this section we shall also consider random variables with a countable set of values. If the probability distribution of random variable X is given by (12.3.1), and $x_k \geq 0$ for every $k \in \{1, 2, 3, \dots\}$, then the mathematical expectation of X is defined by

$$E(X) = \sum_{k=1}^{\infty} x_k p_k. \quad (12.4.3)$$

Note that the sum on the right-hand side of (12.4.3) may be finite or infinite. If the values $x_1, x_2, x_3, \dots$ may be both positive and negative, then the mathematical expectation may be finite and defined by (12.4.3) if the series on the right-hand side of (12.4.3) converges absolutely. Note that the mathematical expectation is not defined for every random variable X .

Theorem 12.4.3. *Let X and Y be random variables defined on a finite sample space $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$. Let p_k be the probability of outcome ω_k , for $k \in \{1, 2, \dots, n\}$. Then, for all $\alpha, \beta \in \mathbb{R}$*

$$E(\alpha X + \beta Y) = \alpha E(X) + \beta E(Y). \quad (12.4.4)$$

For $\alpha = \beta = 1$, equality (12.4.4) becomes $E(X + Y) = E(X) + E(Y)$.

Proof. From the definition of mathematical expectation it follows that

$$\begin{aligned} E(\alpha X + \beta Y) &= \sum_{k=1}^n (\alpha X(\omega_k) + \beta Y(\omega_k)) p_k \\ &= \alpha \sum_{k=1}^n X(\omega_k) p_k + \beta \sum_{k=1}^n Y(\omega_k) p_k = \alpha E(X) + \beta E(Y). \quad \square \end{aligned}$$

The statement of Theorem 12.4.3 holds for all random variables X and Y with finite mathematical expectations.

Example 12.4.4. Consider an indicator function I_A with the probability distribution $P\{I_A = 0\} = 1 - p$, $P\{I_A = 1\} = p$, where $p = P(A)$. The mathematical expectation of I_A is $E(I_A) = 0 \cdot (1 - p) + 1 \cdot p = p$. $\triangle$

Example 12.4.5. Let $S_n \in \mathcal{B}(n, p)$ be a binomial random variable. Since $S_n = I_1 + I_2 + \dots + I_n$, where $I_1, I_2, \dots, I_n$ are indicator functions with parameter p , we obtain that

$$E(S_n) = E\left(\sum_{k=1}^n I_k\right) = \sum_{k=1}^n E(I_k) = np. \quad \triangle$$

Example 12.4.6. Let us determine the mathematical expectation of random variable X with the geometric distribution $\Gamma(p)$ given by (12.3.7). Let us denote $q = 1 - p$, and $S_1 = \sum_{k=1}^{\infty} k q^{k-1}$. Then we have

$$\begin{aligned}
S_1 &= \sum_{k=1}^{\infty} (k-1+1)q^{k-1} = q \sum_{k=1}^{\infty} (k-1)q^{k-2} + \sum_{k=1}^{\infty} q^{k-1} \\
&= qS_1 + \frac{1}{1-q} = qS_1 + \frac{1}{p}.
\end{aligned}$$

It follows that $(1-q)S_1 = \frac{1}{p}$, i.e., $pS_1 = \frac{1}{p}$, and hence

$$E(X) = \sum_{k=1}^{\infty} k(1-p)^{k-1}p = pS_1 = \frac{1}{p}. \quad \triangle$$

Example 12.4.7. Let X be a random variable with the negative geometric distribution given by (12.3.8). Note that X can be represented in the form $X = X_1 + X_2 + \cdots + X_r$, where $X_1, X_2, \dots, X_r$ are random variables with a geometric $\Gamma(p)$ distribution. Using Example 12.4.6 it follows that

$$E(X) = \sum_{i=1}^r E(X_i) = \frac{r}{p}. \quad \triangle$$

A very important property related to the mathematical expectation of the product of independent random variables is given by the following theorem.

Theorem 12.4.8. *If X and Y are independent random variables with finite mathematical expectations, then*

$$E(XY) = E(X)E(Y). \quad (12.4.5)$$

Proof. We shall prove here equality (12.4.5) for independent discrete random variables with finite sets of values. Let $\mathcal{S}_X = \{x_1, x_2, \dots, x_m\}$ and $\mathcal{S}_Y = \{y_1, y_2, \dots, y_n\}$ be the sets of values of two independent random variables X and Y , respectively. Then, we have

$$\begin{aligned}
E(XY) &= \sum_{i=1}^m \sum_{j=1}^n x_i y_j P\{X = x_i, Y = y_j\} = \sum_{i=1}^m \sum_{j=1}^n x_i y_j P\{X = x_i\} P\{Y = y_j\} \\
&= \sum_{i=1}^m x_i P\{X = x_i\} \sum_{j=1}^n y_j P\{Y = y_j\} = E(X)E(Y).
\end{aligned}$$

Note also that the proof remains the same if we assume that X and Y are independent discrete *nonnegative* random variables (with possibly infinite sets of values). $\square$

Definition 12.4.9. Let X be a random variable. For every positive integer n , the n -th *moment* of X is $E(X^n)$ (if this mathematical expectation exists).

The n -th *central moment* of X is $E(X - E(X))^n$.

The second central moment of X is called the *variance* of random variable X and is usually denoted by σ^2 or $\text{var}(X)$.

The positive square root σ of the variance is the *standard deviation*.

The following properties of the variance can easily be proved using the definition and property of mathematical expectation given by (12.4.4):

(a) The variance can be represented in the form

$$\text{var}(X) = E(X - EX)^2 = E(X^2) - (EX)^2. \quad (12.4.6)$$

(b) The variance cannot be negative: $\text{var}(X) = E(X - EX)^2 \geq 0$. Moreover, $\text{var}(X) = 0$ if and only if there exists a real constant c such that $P\{X = c\} = 1$.

(c) For every random variable X with a finite variance, and every real numbers a and b , the following holds: $\text{var}(aX + b) = a^2 \cdot \text{var}(X)$.

Theorem 12.4.10. If X and Y are independent random variables with finite variances, then the following equality holds:

$$\text{var}(X + Y) = \text{var}(X) + \text{var}(Y). \quad (12.4.7)$$

Proof. Using Theorem 12.4.3 we obtain that

$$\begin{aligned} \text{var}(X + Y) &= E(X + Y)^2 - (E(X + Y))^2 \\ &= E(X^2 + 2XY + Y^2) - (EX + EY)^2 \\ &= E(X^2) + 2E(XY) + E(Y^2) - (EX)^2 - 2EX \cdot EY - (EY)^2 \\ &= \text{var}(X) + \text{var}(Y). \quad \square \end{aligned}$$

Example 12.4.11. The statement converse to Theorem 12.4.10 does not hold. Indeed, let X be a random variable that takes values $-2, -1, 1$, and 2 with equal probability $1/4$. Then, X^3 takes values $-8, -1, 1$, and 8 with equal probability $1/4$. It is easy to see that $E(X) = 0$ and $E(X^3) = 0$. If we define $Y = X^2$, then $E(XY) = E(X^3) = 0 = E(X) \cdot E(Y)$. But random variables X and $Y = X^2$ are obviously dependent. Dependence also follows from the fact that $P\{X = 1, Y = 4\} = 0 \neq P\{X = 1\}P\{Y = 4\}$. $\triangle$

Example 12.4.12. Consider an indicator function I_A that takes values: 0 with probability $1 - p$, and 1 with probability $p \in (0, 1)$. Then, from Example 12.4.4 it follows that $E(I_A) = p$. Note that $E(I_A^2) = 0^2 \cdot (1 - p) + 1^2 \cdot p = p$. Hence, it follows that $\text{var}(I_A) = p - p^2 = p(1 - p)$.

Consider now a binomial random variable $S_n \in \mathcal{B}(n, p)$. As stated above it can be represented in the form $S_n = I_1 + I_2 + \cdots + I_n$, where $I_1, I_2, \dots, I_n$ are independent indicator functions, such that each of them takes values 0 and 1 with the probabilities $1 - p$ and p , respectively. Hence, $\text{var}(I_k) = p(1 - p)$ for every $k \in \{1, 2, \dots, n\}$. Using Theorem 12.4.10, we obtain that $\text{var}(S_n) = \text{var}(I_1 + I_2 + \cdots + I_n) = np(1 - p)$. $\triangle$

12.5 Law of Large Numbers

In this section we shall give some explanation of the approximate equalities (12.1.1) and (12.1.2). The quantities $\alpha_n(1)$ and $\beta_n(1)$ that appear there are in fact realizations of the binomial random variables $\mathcal{B}(n, \frac{1}{2})$ and $\mathcal{B}(n, \frac{1}{6})$. For some realizations of the experiments considered, the differences

$$\left| \alpha_n(1) - \frac{1}{2} \right|, \quad \left| \beta_n(1) - \frac{1}{6} \right|$$

may be significantly greater than 0, for example greater than $\varepsilon > 0$. The question is: what is the probability of such an event?

Theorem 12.5.1. Bernoulli's law of large numbers. *Let $S_n \in \mathcal{B}(n, p)$ be a binomial random variable. Then, for every $\varepsilon > 0$ the following inequality holds:*

$$P\left\{\left|\frac{S_n}{n} - p\right| \geq \varepsilon\right\} \leq \frac{p(1-p)}{n\varepsilon^2}. \quad (12.5.1)$$

Note that inequality (12.5.1) implies that $P\{|S_n/n - p| \geq \varepsilon\} \rightarrow 0$ as $n \rightarrow \infty$. This fact, known as the *law of large numbers*, was proved by Jacob Bernoulli (1654–1705), and published firstly in his work *Ars Conjectandi* in 1713. Bernoulli's law of large numbers is the first limit theorem in Probability Theory and one of the basic results of this theory.

We shall prove here a more general result, known as Chebyshev's law of large numbers. For this purpose we need the following lemma.

Lemma 12.5.2. Chebyshev's inequality. *For every random variable X with finite variance σ^2 and every $\varepsilon > 0$ the following inequality holds true:*

$$P\{|X - EX| \geq \varepsilon\} \leq \frac{\sigma^2}{\varepsilon^2}. \quad (12.5.2)$$

Proof. Let X be a discrete random variable with the set of values $\mathcal{S}_X$. Let $\varepsilon > 0$, and consider the following partition of the set $\mathcal{S}_X$:

$$\mathcal{S}_X = \{a_1, a_2, \dots\} \cup \{b_1, b_2, \dots\},$$

such that $|a_i - EX| < \varepsilon$ for all i 's, and $|b_j - EX| \geq \varepsilon$ for all j 's. Then we have

$$\begin{aligned}\sigma^2 &= \sum_i (a_i - EX)^2 P\{X = a_i\} + \sum_j (b_j - EX)^2 P\{X = b_j\} \\ &\geq \sum_i (b_j - EX)^2 P\{X = b_j\} \geq \varepsilon^2 \sum_j P\{X = b_j\} \\ &= \varepsilon^2 P\{|X - EX| \geq \varepsilon\}. \quad \square\end{aligned}$$

Theorem 12.5.3. Chebyshev's law of large numbers:

Let $X_1, X_2, \dots$ be a sequence of independent random variables and $C > 0$ a constant, such that $\text{var}(X_k) \leq C$ for every $k \in \mathbb{N}$. Then for every $\varepsilon > 0$ the following inequality holds:

$$P\left\{\left|\frac{X_1 + \dots + X_n}{n} - E\left(\frac{X_1 + \dots + X_n}{n}\right)\right| \geq \varepsilon\right\} \leq \frac{C}{n\varepsilon^2}. \quad (12.5.3)$$

Proof. Let us denote $X = (X_1 + \dots + X_n)/n$. Using Chebyshev's inequality (12.5.2) and Theorem 12.4.10 we obtain that

$$P\{|X - EX| \geq \varepsilon\} \leq \frac{1}{\varepsilon^2} \text{var}\left(\frac{1}{n} \sum_{k=1}^n X_k\right) = \frac{1}{n^2 \varepsilon^2} \sum_{k=1}^n \text{var}(X_k) \leq \frac{C}{n\varepsilon^2}. \quad \square$$

Since the binomial random variable $S_n \in \mathcal{B}(n, p)$ is a sum of indicator functions, it follows that Theorem 12.5.1 is a special case of Theorem 12.5.3. Note also that we do not assume that random variables $X_1, X_2, \dots$ have a common distribution in Theorem 12.5.3.

Remark 12.5.4. Together with the law of large numbers, the second basic theorem of Probability Theory is the *central limit theorem* whose simplest variant, known as the *integral de Moivre-Laplace theorem*, is related to the binomial random variable. It can be formulated as follows.

If $S_n \in \mathcal{B}(n, p)$, then

$$\sup_{x \in \mathbb{R}} \left| P\left\{\frac{S_n - np}{\sqrt{np(1-p)}} \leq x\right\} - \Phi(x) \right| \leq \frac{p^2 + (1-p)^2}{\sqrt{np(1-p)}}, \quad (12.5.4)$$

where

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt \quad (12.5.5)$$

is the *standard normal distribution function*. Formula (12.5.4) allows us to get the following approximation

$$P\{\alpha \leq S_n \leq \beta\} \approx \Phi\left(\frac{\beta - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{\alpha - np}{\sqrt{np(1-p)}}\right), \quad (12.5.6)$$

for large values n and real numbers α and β , $\alpha < \beta$. The bound of the form $Cn^{-1/2}$ on the right-hand side of inequality (12.5.4) cannot be improved. It follows that n should be greater than 100 in order to get the exact first decimal digit of the probability $P\{\alpha \leq S_n \leq \beta\}$ by the approximation formula (12.5.6).

Exercises

12.1. A coin is tossed three times. What is the probability that at least two heads appear?

12.2. A fair die is thrown three times. Which sum of the obtained numbers has the greater probability: 11 or 12?

12.3. A box contains 5 red, 7 green, and 8 blue balls. Ten balls are chosen simultaneously at random from the box. What is the probability of events A and B defined as follows:

A - no red ball is chosen;

B - there are 3 red, 3 green, and 4 blue balls among the chosen ones?

12.4. A fair die is thrown four times. What is the probability that at least two different numbers appear?

12.5. A box contains 8 red and 8 blue balls numbered 1, 2, ..., 8. Five balls are chosen simultaneously at random from the box. What is the probability that at least two red and at least two blue balls are chosen?

12.6. Suppose that $2n$ tennis players, with A and B among them, are arranged in two groups consisting of n tennis players. What is the probability that A and B are: (a) in the same group; (b) in different groups?

12.7. A fair die is thrown n times. Determine the probability of the events:

(a) the number 1 did not occur;

(b) the number 2 did not occur;

(c) neither of the numbers 1 and 2 occurred;

(d) at least one of the numbers 1 and 2 occurred?

12.8. Positive integers 1, 2, ..., n are arranged in a sequence at random. Let A be the event that number 1 is in the first position, and B be the event that 1 is in the second position. Determine $P(A \cup B)$.

12.9. A fair die is thrown until the sum of the numbers becomes greater than 3.

- (a) Determine the set of possible outcomes.
- (b) What are the natural probabilities of the outcomes?
- (c) What is the probability that the number of throwings of the die is greater than 2?

12.10. A coin is tossed twice. The probabilities of the outcomes 00, 01, 10, and 11 are p^2 , pq , qp , and q^2 , respectively, where $p > 0$, $q > 0$, and $p + q = 1$. What is the probability of the event: (a) $A = \{00, 01\}$, (b) $B = \{10, 11\}$?

12.11. A coin is tossed three times. If at least one head appeared, what is the probability that a tail appeared?

12.12. If A and B are independent events, prove that: (a) A and $\bar{B}$ are independent events; (b) $\bar{A}$ and $\bar{B}$ are independent events;

* * *

In Exercises 12.13–12.20 we consider a sequence of three experiments. For every $i \in \{1, 2, 3\}$, the i -th experiment has two possible outcomes: success with the probability p_i , and failure with the probability $1 - p_i$.

12.13. What is the probability that at least one experiment is successful?

12.14. What is the probability that exactly one experiment is successful?

12.15. What is the probability that at least two experiments are successful?

12.16. What is the probability that at least two consecutive experiments are successful?

12.17. What is the probability that exactly two experiments are successful?

12.18. If exactly one experiment is successful, what is the probability that it is the first experiment?

12.19. If exactly two experiments are successful, what is the probability that the first experiment is unsuccessful?

12.20. If at least two consecutive experiments are successful, what is the probability that exactly two experiments are successful?

* * *

12.21. Twenty questions are offered at an oral exam. The questions are numbered 1, 2, ..., 20. Suppose that three students, Arthur, Bob and Chris, take the exam. Each of them is prepared to give the right answer to 15 questions. The students choose questions one after another, at random and without replacement. Each student chooses only one question, first Arthur, then Bob, and finally Chris. What is the probability that: (a) Arthur will pass the exam; (b) Bob will pass the exam; (c) Chris will pass the exam?

12.22. The set $\{1, 2, \dots, 32\}$ is partitioned into two subset S_1 and S_2 , such that they contain three and five positive integers that are divisible by 4. Suppose that one of the sets S_1 and S_2 is chosen at random, and then, from this set, three numbers are chosen at random without replacement. If the first two natural numbers chosen are both divisible by 4, what is the probability that the third is also divisible by 4?

12.23. A coin is tossed six times. What is the probability that a head appears:

- (a) exactly 3 times;
- (b) more than 3 times?

12.24. A fair die is thrown six times. Determine the probability that the number 6 is obtained:

- (a) exactly 3 times;
- (b) more than 3 times.

12.25. Three fair dice are simultaneously thrown five times. What is the probability that three same numbers are obtained:

- (a) at least once;
- (b) exactly once?

12.26. Consider a sequence of independent experiments with two possible outcomes: success with the probability p and failure with the probability $1 - p$. The experiments are performed until there are 10 successes or 10 failures. What is the probability that exactly 15 trials produce this outcome?

12.27. A coin is tossed 10 times. What is the probability that the number of heads obtained in the first five trials is equal to the number of heads obtained in the last five trials?

12.28. Let $S_n \in \mathcal{B}(n, p)$ be a binomial random variable, $p \in (0, 1)$, and let us denote $P_n(k) = P\{S_n = k\}$.

(a) Determine the values of $k \in \{1, 2, \dots, n\}$ such that the inequality $P_n(k - 1) < P_n(k)$ holds.

(b) Find the conditions under which the sequence $P_n(0)$, $P_n(1)$, $\dots$, $P_n(n)$ is strictly increasing or strictly decreasing.

12.29. Let $S_{2n} \in \mathcal{B}(2n, \frac{1}{2})$ be a binomial random variable, and $P_{2n}(n) = P\{S_{2n} = n\}$. Prove that

$$\frac{1}{2\sqrt{n}} \leq P_{2n}(n) < \frac{1}{\sqrt{2n+1}}.$$

12.30. Consider a sequence of independent experiments with two possible outcomes: success with the probability $p = 0.6$, and failure with the probability $1 - p = 0.4$. What is the minimal number of experiments that should be performed in order to get at least one success with the probability greater than 0.99?

12.31. A fair die is rolled until the side numbered 6 appears. What is the expected number of rolls?

12.32. A fair die is rolled until the same side appears two times consecutively. Let X be the number of rolls.

(a) Find the probability $P\{X = n\}$ for $n \in \{2, 3, 4, \dots\}$.

(b) Determine the mathematical expectation $E(X)$.

12.33. A fair die is rolled until the side numbered 6 appears r times. What is the expected number of rolls?

12.34. Determine $\text{var}(X)$, where X is a random variable with a geometric distribution given by (12.3.7).

12.35. Determine $\text{var}(X)$, where X is a random variable with a negative binomial distribution given by (12.3.8).

12.36. Let $X \in \mathcal{B}(2000, \frac{1}{1000})$ be a binomial random variable with parameters $n = 2000$ and $p = \frac{1}{1000}$. Calculate the binomial probability $P_{2000}(k) = P\{X = k\}$ and the Poisson approximation $\pi_2(k)$ for every $k \in \{0, 1, 2, 3, 4, 5\}$.

12.37. Determine $E(X)$ and $\text{var}(X)$, where X is a random variable with a Poisson $\mathcal{P}(\lambda)$ distribution.

12.38. Let X and Y be independent random variables with Poisson distributions $\mathcal{P}(\lambda)$ and $\mathcal{P}(\mu)$, respectively. Suppose that these two random variables are defined on the same probability space. Determine the distribution of the random variable $X + Y$.

12.39. Symmetric random walk. Let $(X_k)_{k \geq 1}$ be a sequence of independent random variables with the common distribution

$$P\{X_k = -1\} = P\{X_k = 1\} = \frac{1}{2}, \quad k \in \mathbb{N}.$$

Consider the symmetric random walk $S_0, S_1, S_2, \dots$, where $S_0 = 0$, and $S_n = X_1 + X_2 + \dots + X_n$, for every $n \in \mathbb{N}$.

- (a) Determine the probability $u_{2n} = P\{S_{2n} = 0\}$.
- (b) Determine the probability v_{2n} of the first return to 0 at moment $2n$, i.e., $v_{2n} = P\{S_2 \neq 0, \dots, S_{2n-2} \neq 0, S_{2n} = 0\}$.
- (c) Determine the probability $v_2 + v_4 + v_6 + \dots$ of the return to 0.
- (d) Let X be the minimal value of $n \in \{2, 4, 6, \dots\}$ such that $S_{2n} = 0$. Determine the mathematical expectation $E(X)$.

12.40. Random walk. Let $(X_k)_{k \geq 1}$ be a sequence of independent random variables with the common distribution

$$P\{X_k = -1\} = 1 - p, \quad P\{X_k = 1\} = p, \quad k \in \mathbb{N}, \quad .$$

Consider the random walk $S_0, S_1, S_2, \dots$, where $S_0 = 0$, and $S_n = X_1 + X_2 + \dots + X_n$, for every $n \in \mathbb{N}$. What is the probability that the sequence $S_1, S_2, \dots$ will reach the point $n \in \mathbb{N}$, i.e., there exists an index $k \in \mathbb{N}$ such that $S_k = n$?

Chapter 13



Additional Problems

13.1 Basic Combinatorial Configurations

13.1. Let $p_k(n)$ be the number of permutations of the set $\{1, 2, \dots, n\}$ that have exactly k fixed points. Prove the following equalities:

(a) $kp_n(k) = np_{n-1}(k-1)$, where $1 \leq k \leq n$;

(b) $\sum_{k=m}^n \frac{k!}{(k-m)!} p_n(k) = n!$, where $0 \leq m \leq n$.

13.2. Is there a permutation of the set $\mathbb{N}_n = \{1, 2, \dots, n\}$ such that for any two positive integers $i, j \in \mathbb{N}_n$ their arithmetic mean is not equal to a number that is placed between i and j in this permutation?

13.3. Let a finite 0–1 sequence be given such that the following conditions are satisfied:

(a) Any two subsequences which consist of 5 consecutive terms of the given sequence are not equal to each other (the subsequences may overlap partially).

(b) If we add 0 or 1 at the end of the given sequence, then the new sequence does not have property (a) anymore.

Prove that the subsequences consisting of the first four and the last four terms of the starting sequence coincide.

13.4. Prove that there exist 2^{n+1} positive integers such that the following three conditions are satisfied:

(a) Each of these positive integers has only the digits 1 and 2 in its decimal representation.

(b) Each of these positive integers has 2^n digits;

(c) Every two of these 2^{n+1} positive integers have different digits at least in 2^{n-1} positions.

13.5. There are one thousand balls numbered 000, 001, 002, $\dots$, 999 and one hundred boxes numbered 00, 01, 02, $\dots$, 99. It is allowed to put a ball into a box, if the number written on the box can be obtained by erasing a digit of the number written on the ball. Prove the following statements:

(a) All of the balls can be put into 50 boxes.

(b) It is not possible to put all of the balls into 49 boxes.

13.6. Prove that there exists a positive integer with the following property: if we add an arbitrary digit at the end of its decimal representation, then the obtained positive integer is such that a few digits at the beginning repeat in the same order at the end of its decimal representation.

13.7. Suppose that a $(2n+1)$ -arrangement of the elements 0 and 1 is given. Prove that it is possible to delete a term from this arrangement, such that the obtained $2n$ -arrangement has the same number of 0's in even and odd positions.

13.8. Knights from two hostile countries are arranged around a circular table. The number of knights with an enemy on their right-hand side is equal to the number of knights with a friend on their right-hand side. Prove that the total number of knights around the table is divisible by 4.

13.9. Each of the numbers $x_1, x_2, \dots, x_n$ is equal to 1 or -1 . If the equality $x_1x_2 + x_2x_3 + \dots + x_nx_1 = 0$ holds, prove that n is divisible by 4.

13.10. Sperner's lemma. Let $A_1, A_2, \dots, A_n$ be subsets of the set $\mathbb{N}_n = \{1, 2, \dots, n\}$ such that none of them is a subset of another. Prove that

$$k \leq \binom{n}{\lfloor n/2 \rfloor}.$$

13.11. Let $A_1, A_2, \dots, A_k$ be subsets of the set $\{1, 2, \dots, n\}$ such that the following condition is satisfied: if $A_i \subset A_j$, where $i, j \in \{1, 2, \dots, n\}$, then $|A_j \setminus A_i| \leq r - 1$. Prove that the number k is not greater than the sum of the greatest r of the binomial coefficients $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$.

13.12. Let $x_1, x_2, \dots, x_n$ be real numbers such that $x_i > 1$ for any $i \in \{1, 2, \dots, n\}$, and $m = \lfloor n/2 \rfloor$. Prove that for every real number a the following

statement holds: the interval $[a-1, a+1]$ contains no more than $\binom{n}{m}$ sums of the form $\varepsilon_1 x_1 + \varepsilon_2 x_2 + \cdots + \varepsilon_n x_n$, where $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n \in \{-1, 1\}$.

13.13. Let S be a subset of the set of all $2n$ -arrangements of the elements 0 and 1, such that there are no two arrangements $(a_1, a_2, \dots, a_{2n}) \in S$ and $(b_1, b_2, \dots, b_{2n}) \in S$ with all inequalities $a_1 \leq b_1, a_2 \leq b_2, \dots, a_{2n} \leq b_{2n}$ fulfilled. Prove that $|S| \leq \binom{2n}{n}$.

13.14. Suppose that a cube with edges of length 3 is given. What is the minimal number of cuts necessary to cut this cube into 27 unit cubes? It is allowed to cut several parts simultaneously.

13.15. Given a right-angled parallelepiped with edges whose lengths are equal to the positive integers m, n , and p , what is the minimal number of cuts necessary to cut this parallelepiped into mnp unit cubes? It is allowed to cut several parts simultaneously.

13.16. Tennis players numbered $1, 2, 3, \dots, 2^n$ take part in a tournament. If players i and j play against each other, then i always wins if $i + 2 < j$. In each round of the tournament the players are arranged in pairs, and only the winner of each pair continues the competition. Determine the maximal possible number associated with the winner of the tournament.

13.17. How many permutations $(a_1, a_2, \dots, a_n)$ of the set $\mathbb{N}_n = \{1, 2, \dots, n\}$ are there, such that $k - 1 \leq a_k$ for every $k \in \mathbb{N}_n$?

13.18. How many permutations $(a_1, a_2, \dots, a_n)$ of the set $\mathbb{N}_n = \{1, 2, \dots, n\}$ are there, such that $k - 1 \leq a_k \leq k + 1$ for every $k \in \mathbb{N}_n$?

13.19. How many permutations $(a_1, a_2, \dots, a_n)$ of the set $\mathbb{N}_n = \{1, 2, \dots, n\}$ are there, such that the number of elements $j \in \mathbb{N}_n$ satisfying the condition $a_j = \max\{a_1, a_2, \dots, a_j\}$ is equal to k ?

13.20. How many permutations $(a_1, a_2, \dots, a_n)$ of the set $\mathbb{N}_n = \{1, 2, \dots, n\}$ are there, such that the number of elements $j \in \mathbb{N}_n$ satisfying the condition $a_j > j$ is equal to k ?

13.2 Square Tables

13.21. Let us take a square table 50×50 . Some fields (unit squares) of the table are labeled 1, and some of them are labeled -1 . It is allowed that

some fields are not labeled. The sum of all labels is not greater than 100 in absolute value. Prove that there exists a sub-table 25×25 , such that the sum of all labels of the fields of this sub-table is not greater than 25 in absolute value.

13.22. How many ways can a table $m \times n$ be filled with numbers 1 and -1 , such that the product of all the numbers in each column and the product of all the numbers in each row is equal to -1 ?

13.23. All fields of a table $m \times n$ are filled with real numbers. In each step it is allowed to change the sign of all the numbers in the same row or the same column. Prove that repeating steps of this kind produces a table such that the sum of all the numbers in every row and the sum of all the numbers in every column is nonnegative.

13.24. Let $a_1, a_2, \dots, a_m, b_1, b_2, \dots, b_n$ be positive real numbers such that $a_1 + a_2 + \dots + a_m = b_1 + b_2 + \dots + b_n$. Suppose there is no sum of the form $a_1 + a_2 + \dots + a_k$, where $k < m$, that is equal to the sum of the form $b_1 + b_2 + \dots + b_l$, where $l < n$. Prove that it is possible to write $m + n - 1$ positive numbers into $m + n - 1$ fields of a table with m rows and n columns, such that, for each $i = 1, 2, \dots, m$, the sum of all the numbers in the i -th row is equal to a_i , and, for each $j = 1, 2, \dots, n$, the sum of all the numbers in the j -th column is equal to b_j .

13.25. The real numbers from the table

$$\begin{array}{cccc} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{array}$$

satisfy the inequality $\sum_{i=1}^n |a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n| \leq M$, for every choice of numbers $x_i \in \{-1, 1\}$. Prove that $|a_{11}| + |a_{22}| + \dots + |a_{nn}| \leq M$.

13.26. Coins are placed on fields of a table with m rows and n columns, where $n > m$, such that the following two conditions are satisfied:

- (1) Every field contains at most one coin.
- (2) Every column contains at least one coin.

Prove that there exists a coin such that its row contains more coins than its column.

13.27. A *Hadamard matrix* of order n is a matrix with n rows and n columns whose entries are either $+1$ or -1 and whose rows are pairwise orthogonal,

i.e., the sum of the products of the corresponding elements of every two rows is equal to 0. If the order n of a Hadamard matrix is greater than 2, prove that n is divisible by 4.

13.28. Prove that for any positive integer n of the form $n = 2^k$ there exists a Hadamard matrix of order n .

13.29. Give an example of a Hadamard matrix of order 12.

13.30. $3n$ chips are placed on the fields of a square table with $2n$ rows and $2n$ columns. Each field contains no more than one chip. Prove that there exist n rows and n columns such that all $3n$ chips are placed on these rows and columns.

13.31. Prove that $3n + 1$ chips can be arranged on the fields of a square table with $2n$ rows and $2n$ columns (at most one chip on each field) such that the following condition is satisfied: There are no n rows and n columns that contain all $3n + 1$ chips.

13.3 Combinatorics on a Chessboard

A chessboard consists of 64 squares (fields) arranged in eight rows and eight columns. The squares are arranged in two alternating colors (light and dark). The columns of the chessboard are denoted by a, b, c, d, e, f, g , and h , and the rows by $1, 2, \dots, 8$. The chess pieces are: king, queen, rooks, bishops, knights, and pawns. Many interesting combinatorial problems are formulated in terms of the configurations of chess pieces, chips, or positive integers on the chessboard. Sometimes the colors that are associated with the fields are irrelevant. Some problems are related to an *extended* chessboard with n rows and n columns or to an *infinite* chessboard.

13.32. Eight rooks are arranged on a chessboard 8×8 such that every row and every column contains exactly one of them. Prove that the number of rooks on light squares is even.

13.33. The fields of a chessboard 8×8 are numbered 1 to 64 as follows: the fields in the first row from left to right are numbered 1 to 8, in the second row from left to right 9 to 16, etc. Eight rooks are arranged on the chessboard such that no two of them are in the same row or in the same column. Determine the sum of numbers associated with the fields that are occupied by rooks.

13.34. A few fields of a chessboard 8×8 are chosen and a chip is put on each of them. Consider a game with the following moves allowed: each chip can be moved to the adjacent field if this field is unoccupied. Two fields are adjacent if they have a common side. Let us suppose that some moves have been made, and every chip has returned to its starting position crossing every field of the chessboard exactly once. Prove that there was a moment where no chip was at its starting position.

13.35. The fields $a1$ and $h8$ of a chessboard are cut off. Is it possible to cover the remaining part of the chessboard by 31 dominos 2×1 ?

13.36. Twenty-one rectangles 3×1 are placed on a chessboard 8×8 such that only one field of the chessboard is not covered. Determine all the fields of the chessboard that can appear uncovered this way.

Problems 12.37–12.46 are related to chess pieces that are placed on a chessboard. We assume that the chess pieces can move and attack each other according to chess rules.

13.37. How many ways can a black and a white knight be placed on a chessboard, such that they do not attack each other?

13.38. How many ways can twenty chips be placed on the fields of a square table 8×8 , such that the arrangement of the chips remains the same after rotating the table, if the angle of rotation is $\alpha \in \{90^\circ, 180^\circ, 270^\circ\}$?

13.39. Determine the maximal number of rooks that can be placed on a chessboard such that no two of them attack each other.

13.40. How many ways can the maximal number of rooks be placed on a chessboard such that no two of them attack each other?

13.41. Determine the maximal number of bishops that can be placed on a chessboard such that no two of them attack each other.

13.42. Suppose that the maximal number of bishops are placed on a chessboard such that no two of them attack each other. Prove that all the bishops are placed on the margin fields of the chessboard.

13.43. How many ways can the maximal number of bishops be placed on a chessboard such that no two of them attack each other?

13.44. Determine the maximal number of knights that can be placed on a chessboard such that no two of them attack each other.

13.45. How many ways can the maximal number of knights be placed on a chessboard such that no two of them attack each other?

13.46. Determine the maximal number of queens that can be placed on a chessboard such that no two of them attack each other.

13.47. A piece called a *dolphin* is placed on field $a1$ of a chessboard. The dolphin can move to an adjacent field to the right, or up, or to the left-below diagonally. Can the dolphin cross all the fields of the chessboard, such that it is placed on every field exactly once?

13.48. Determine the maximal number of rooks that can be placed on a chessboard $3n \times 3n$, such that each of them is attacked by at most one of the other rooks?

13.49. How many ways are there to put n rooks on a chessboard $n \times n$, such that they attack all the unoccupied fields of the chessboard?

13.50. Is it possible to put five queens on a chessboard 8×8 such that they attack all the unoccupied fields of the chessboard?

13.51. Every field of a chessboard $n \times n$ is to be covered by a red or a blue chip. How many ways can this be done, such that every square 2×2 contains two red and two blue chips?

13.52. Determine the maximal number of chips that can be placed on the fields of a chessboard 8×8 , such that each row, each column and each diagonal contains an even number of chips. At most one chip can be placed on any field.

13.53. A chip is placed on every field of a chessboard 9×9 . Then every chip is moved to an adjacent diagonal field. This way more than one chip can be on the same field, and some fields can remain empty. Determine the minimal number of fields that can remain empty.

13.4 The Counterfeit Coin Problem

In this section we shall give some counterfeit coin problems related to finding a counterfeit coin among the genuine ones. All genuine coins have the same weight. A counterfeit coin looks like a genuine one, but it has a different weight. It is not known whether the counterfeit coin is heavier or lighter than a genuine one. It is allowed to use a balance scale with two pans, and

without weights. Using such a scale every weighing can show only that two things have the same weight, or that one is heavier than the other.

13.54. There are 12 coins, of which only one is counterfeit. It is allowed to make three weighings. How can we find the counterfeit coin and determine whether it is lighter or heavier than the genuine ones?

13.55. There are x_n coins, such that only one of them is counterfeit, plus one more genuine coin that is marked. Determine the maximal value of x_n such that the following statement holds true: *Under the condition that in the first weighing all x_n coins are placed on scale-pans (the additional genuine coin may also be included), it is possible to determine by n weighings which one of the given coins is counterfeit.*

13.56. There are y_n coins, such that only one of them is counterfeit, plus one more genuine coin that is marked. Determine the maximal value of y_n such that the following statement holds: *It is possible to find by n weighings which one of the given coins is counterfeit, and determine whether it is heavier or lighter than the genuine coins.*

13.57. There are z_n coins, such that only one of them is counterfeit. Determine the maximal value of z_n such that the following statement holds: *It is possible to find by n weighings which one of the given coins is counterfeit, and determine whether it is heavier or lighter than the genuine coins.*

13.58. There are u_n coins, such that only one of them is counterfeit, plus one more genuine coin that is marked. Determine the maximal value of u_n such that the following statement holds: *It is possible to find by n weighings which one of the given coins is counterfeit.* It is not important whether the counterfeit coin is heavier or lighter than the genuine coins.

13.59. There are v_n coins, such that only one of them is counterfeit. Determine the maximal value of u_n such that the following statement holds: *It is possible to find by n weighings which one of the given coins is counterfeit.* It is not important whether the counterfeit coin is heavier or lighter than the genuine coins.

13.5 Extremal Problems on Finite Sets

13.60. The number 0 is written on a blackboard. Two players A and B alternately write on the right-hand side of 0 the sign $+$ or $-$ and a numbers from set $S = \{1, 2, \dots, 2001\}$. Player A starts and writes the sign $+$ or $-$

in each of his moves, and player B then writes a number from set S . Each number from S can be written only once. At the end of the game player B gets an amount equal to the absolute value of the final sum written on the blackboard. Determine the maximal amount that B can obtain for sure, independently of the moves of player A .

13.61. Determine the minimal positive integer $n \geq 4$ with the following property: for any n -set S consisting of positive integers, it is possible to choose distinct elements $a, b, c, d \in S$, such that the number $a + b - c - d$ is divisible by 20.

13.62. At the conference of the party of liars and the party of fans of the truth, a presidency of 32 members was chosen. Members of the presidency are arranged to sit on 32 chairs (four rows with eight chairs each, and eight columns with four chairs each).

Members of the party of liars always lie, and the fans of the truth always tell the truth. By definition, B is adjacent to A if B is arranged to sit on a chair which is to the left of A , or to the right of A , or in front of A , or in back of A .

At the coffee break every member of the presidency said that he had a liar and a fan of the truth among his neighbors. Determine the minimal number of liars in the presidency for which the situation described above is possible.

13.63. There are 30 persons arranged around a circular table. Each of them is a liar or a fan of the truth. Every fan of the truth always tells the truth, and every liar sometimes tells the truth and sometimes lies. The number of liars is equal to n . All of these 30 people answered a question regarding who is seated on their right-hand side – a liar or a fan of the truth. Find the maximal value of n with the following property: given 30 answers we can determine a person who is for sure a fan of the truth (independently of the arrangement of people around the table).

13.64. A square table $n \times n$ consists of n^2 unit squares. Determine the maximal value of n for which the following statement holds: it is possible to mark n unit squares such that any rectangle consisting of unit squares of the given square table, and whose area is not less than n , contains at least one marked square.

13.65. Each field of a table 10×19 contains a number from the set $\{0, 1\}$. The sum of all the numbers in each row and each column is determined. Let M be the number of different sums obtained in such a way. What is the maximal value of M ?

13.66. Twenty-five fields are marked on an infinite chessboard. Two fields are disjoint if they do not have a common vertex. Determine the maximal positive integer k such that the following statement holds: *For any choice of marked fields it is possible to choose k of them that are pairwise disjoint.*

13.67. There are n balls in a box. Players, A and B , play the following game. They alternately take balls from the box. Player A starts and in the first move takes at least one, but not all n balls. In each of the subsequent moves, a player takes a few, say k , balls such that k is a divisor of the number of balls that were taken by their rival in the previous move. The winner is the player who takes the last ball (or balls) from the box. Determine the minimal positive integer $n \geq 1998$, such that player B has the winning strategy.

13.68. Let us consider a sequence $c_1 c_2 \dots c_n$, where $c_1, c_2, \dots, c_n \in \{0, 1\}$. Below every pair $c_k c_{k+1}$ we write 0 if $c_k = c_{k+1}$, and 1 if $c_k \neq c_{k+1}$. This way we obtain a second sequence consisting of $n - 1$ terms. A third sequence consisting of $n - 2$ terms is formed from the second by the same rule, etc. Finally we obtain a triangular table. Determine the maximal number of 1's in such a table.

13.69. A family album contains 10 photos. There are three men on each photo: in the middle is a man, to the left of him is his son, and to the right is his brother. No man appears more than once in the middle of a photo. Determine the minimal number of men who can be seen on all the photos.

13.70. A club consists of 30 members who have distinct birthdays. Each of them has the same number of friends among the other club members. A member of the club has the title *senior*, if they are older than most of their friends. Determine the maximal possible number of seniors in the club.

13.71. Let $S = \{1, 2, \dots, 16\}$. Determine the minimal positive integer k , such that there are k partitions of set S into two subsets satisfying the following two conditions:

- (a) $S = A_i \cup B_i$, $|A_i| = |B_i| = 8$, for any $i \in \{1, 2, \dots, k\}$;
- (b) For every two elements $a, b \in S$ there exists $i \in \{1, 2, \dots, k\}$ such that a and b are not in the same subset A_i or B_i .

13.72. Suppose that n teams took part in a football championship, each team playing one match against each of the others. Victory, draw, and defeat in any match are assigned 2, 1, and 0 points, respectively. At the end of the championship there was a team that scored more points than any of the

other teams, but less victories than any of the other teams. Determine the minimal value of n for which such a situation is possible.

13.73. Determine the minimal positive integer k , such that every k -subset of the set $\{1, 2, \dots, 50\}$ contains two elements a and b , such that $a \neq b$ and $a + b$ is a divisor of ab .

13.74. Suppose that n students took an exam. There were four questions with three possible answers to each of them. For every three students there was at least one question to which these three students gave three different answers. Find the minimal value of n for which such a situation is possible.

13.75. Let S be the set of all sequences $(a_1, a_2, \dots, a_7)$, such that $a_1, a_2, \dots, a_7 \in \{0, 1\}$. The distance between sequences $(a_1, a_2, \dots, a_7) \in S$ and $(b_1, b_2, \dots, b_7) \in S$ is defined as $\sum_{i=1}^7 |a_i - b_i|$. Let T be a subset of S , such that the distance between any two elements of T is greater than or equal to 3. Prove that $|T| \geq 16$. Give an example where $|T| = 16$.

13.76. Let k and n be positive integers, such that $k \leq n$, and let S be a set consisting of n real numbers. Let T be the set of all real numbers of the form $x_1 + x_2 + \dots + x_k$, where $x_1, x_2, \dots, x_k \in S$ and $x_i \neq x_j$ for $i \neq j$.

(a) Prove that $|T| \geq k(n - k) + 1$.

(b) Prove that there exists a set S such that $|T| = k(n - k) + 1$.

13.6 Combinatorics at Mathematical Olympiads

In this section we shall give some of the combinatorial problems that were posed at the Balkan Mathematical Olympiads (BMO) and the International Mathematical Olympiads (IMO). The problems at these mathematical competitions are traditionally classified in four areas of elementary mathematics: Algebra (and Analysis), Geometry, Number Theory, and Combinatorics. Notation **BMO-2000** means that the problem was posed at the Balkan Mathematical Olympiad in the year 2000. Similar notation is used for problems posed at the International Mathematical Olympiads. After its formulation the name of the country that contributed the problem is given.

13.77. BMO-1992. For any integer $n \geq 3$ find the smallest natural number $f(n)$ such that for each subset $A \subset \{1, 2, \dots, n\}$ with $f(n)$ elements, there exist elements $x, y, z \in A$ that are pairwise coprime. [Romania]

13.78. BMO-1993. A natural number with decimal representation $a_n \cdot 10^n + a_{n-1} \cdot 10^{n-1} + \dots + a_1 \cdot 10 + a_0$, where $a_0, a_1, \dots, a_n \in \{0, 1, \dots, 9\}$, is

called *monotone* if $a_n \leq a_{n-1} \leq \dots \leq a_1 \leq a_0$. Determine the number of all monotone numbers with at most 1993 digits. [Bulgaria]

13.79. BMO-1994. Let $(a_1, a_2, \dots, a_n)$ be a permutation of the numbers $1, 2, \dots, n$, where $n \geq 2$. Determine the largest possible value of $\sum_{k=1}^{n-1} |a_{k+1} - a_k|$. [Romania]

13.80. BMO-1994. Find the smallest natural number $n > 4$ for which there can exist a set of n people, such that any two people who are acquainted have no common acquaintances, and any two people who are not acquainted have exactly two common acquaintances. Acquaintance is a symmetric relation. [Bulgaria]

13.81. BMO-1995. Let n be a natural number and S be the set of points (x, y) , where $x, y \in \{1, 2, \dots, n\}$. Let T be the set of all squares with vertices in set S . Let us denote by a_k , where $k \geq 0$, the number of (unordered) pairs of points for which there are exactly k squares in T having these two points as vertices. Prove that $a_0 = a_2 + 2a_3$. [Yugoslavia]

13.82. BMO-1996. Prove that there exists a subset A of the set $\{1, 2, \dots, 2^{1996} - 1\}$ with the following properties:

- (a) $1 \in A$ and $2^{1996} - 1 \in A$;
- (b) Every element of the set $A \setminus \{1\}$ is the sum of two (possibly equal) elements of A ;
- (c) Set A contains at most 2012 elements. [Romania]

13.83. BMO-1997. Let $\mathcal{A} = \{A_1, A_2, \dots, A_k\}$ be a collection of subsets of an n -set S . If for any two elements $x, y \in S$ there is a subset $A_i \in \mathcal{A}$ containing exactly one of the two elements x and y , prove that $2^k \geq n$. [Yugoslavia]

13.84. BMO-1998. Consider the finite sequence $\left\lceil \frac{k^2}{1998} \right\rceil$, $k \in \{1, 2, \dots, 1997\}$. How many distinct terms are there in this sequence? [Greece]

13.85. BMO-1999. Let $0 \leq x_0 \leq x_1 \leq x_2 \leq \dots$ be a sequence of nonnegative integers, such that for every $k \geq 0$ the number of terms of the sequence which do not exceed k is finite, say y_k . Prove that, for all positive integers m and n , $\sum_{i=0}^n x_i + \sum_{j=0}^m y_j \geq (n+1)(m+1)$. [Romania]

13.86. BMO-2000. Find the maximal number of rectangles $1 \times 10\sqrt{2}$ that can be cut off from a rectangle 50×90 using cuts parallel to the edges of the big rectangle. [Yugoslavia]

13.87. BMO-2001. A cube of edge 3 is divided into 27 unit cube cells. One of these cells is empty, while in the other cells there are unit cubes which are arbitrarily denoted by $1, 2, \dots, 26$. A *legal move* consists of moving a unit cube into a neighboring cell (two cells are neighboring if they share a face). Does there exist a finite sequence of legal moves after which any two cubes denoted by k and $27 - k$, $k \in \{1, 2, \dots, 13\}$, will exchange their positions?

[Bulgaria]

13.88. BMO-2002. Let $A_1, A_2, \dots, A_n$, where $n \geq 4$ be points on the plane, such that no three of them are collinear. Some pairs of distinct points among them are connected by segments, such that every point is connected to at least three other points. Prove that there exists an integer $k \geq 2$ and distinct points $X_1, X_2, \dots, X_{2k}$ from the set $\{A_1, A_2, \dots, A_n\}$ such that X_i is connected to X_{i+1} , for any $i \in \{1, 2, \dots, 2k\}$, where $X_{2k+1} = X_1$.

[Yugoslavia]

13.89. BMO-2003. Does there exist a set B consisting of 4004 distinct natural numbers, such that for any subset A of the set B containing 2003 elements, the sum of the elements of A is not divisible by 2003?

[FYR Macedonia]

13.90. BMO-2005. Let $n \geq 2$ be an integer, and let S be a subset of $\{1, 2, \dots, n\}$ such that the following conditions are satisfied: (a) S does not contain two coprime elements; (b) S does not contain two elements such that one of them is a divisor of the other.

Determine the maximal possible number of elements of set S . [Romania]

13.91. BMO-2009. A 9×12 rectangle is partitioned into unit squares. The centers of all the unit squares, except for the four corner squares and the eight squares sharing a common side with one of the corner squares, are colored red. Is it possible to label these red centers $C_1, C_2, \dots, C_{96}$ in such a way that the following two conditions are both fulfilled:

- (a) the distances $C_1C_2, C_2C_3, \dots, C_{95}C_{96}, C_{96}C_1$ are all equal to $\sqrt{13}$;
- (b) the closed broken line $C_1C_2 \dots C_{96}C_1$ has a center of symmetry?

[Bulgaria]

* * *

13.92. IMO-1991. Suppose G is a connected graph with k edges. Prove that it is possible to label the edges $1, 2, \dots, k$ in such a way that at each vertex which belongs to two or more edges, the greatest common divisor of the integers labeling these edges is equal to 1.

[USA]

13.93. IMO-1992. Consider nine points in space, no four of which are coplanar. Each pair of points is joined by an edge (i.e., a line segment) and each edge is colored either blue or red or is left uncolored. Find the smallest value of n such that whenever exactly n edges are colored, the set of colored edges necessarily contains a triangle all of whose edges have the same color. [China]

13.94. IMO-1993. On an infinite chessboard a game is played as follows. At the start, n^2 pieces are arranged on the chessboard in an $n \times n$ block of adjoining squares, one piece in each square. A move in the game is a jump in a horizontal or vertical direction over an adjacent occupied square to an unoccupied square immediately beyond. The piece which has been jumped over is removed. Find those values of n for which the game can end with only one piece remaining on the board. [Finland]

13.95. IMO-1993. There are n lamps $L_0, L_1, \dots, L_{n-1}$ in a circle ($n > 1$), where we denote $L_{n+k} = L_k$. A lamp is either on or off at all times. Perform steps $S_0, S_1, \dots$ as follows: at step S_i , if L_{i-1} is lit, switch L_i from on to off or vice versa, otherwise do nothing. Initially all lamps are on. Show that:

- (a) There is a positive integer $M(n)$ such that after $M(n)$ steps all the lamps are on again;
- (b) If $n = 2^k$, we can take $M(n) = n^2 - 1$;
- (v) If $n = 2^k + 1$, we can take $M(n) = n^2 - n + 1$. [Netherlands]

13.96. IMO-1994. For any positive integer k , let $f(k)$ be the number of elements in the set $\{k+1, k+2, \dots, 2k\}$ whose base 2 representation has precisely three 1's.

- (a) Prove that, for each positive integer m , there exists at least one k such that $f(k) = m$.
- (b) Determine all positive integers m for which there exist exactly one k with $f(k) = m$. [Romania]

13.97. IMO-1995. Let p be an odd prime number. How many subsets A of the set $\{1, 2, \dots, 2p\}$ are there, such that the following two conditions hold:

- (a) A has exactly p elements;
- (b) the sum of all elements of A is divisible by p ? [Poland]

13.98. IMO-1997. Let $x_1, x_2, \dots, x_n$ be real numbers satisfying conditions

$$|x_1 + x_2 + \dots + x_n| = 1$$

and $|x_i| \leq \frac{n+1}{2}$ for $i = 1, 2, \dots, n$. Show that there exists a permutation $y_1, y_2, \dots, y_n$ of $x_1, x_2, \dots, x_n$ such that

$$|y_1 + 2y_2 + \cdots + ny_n| \leq \frac{n+1}{2}. \quad [\text{Russia}]$$

13.99. IMO-1997. An $n \times n$ matrix (square array) whose entries come from the set $S = \{1, 2, \dots, 2n-1\}$ is called a *silver* matrix if, for each $i = 1, 2, \dots, n$, the i -th row and the i -th column together contain all the elements of S . Show that:

- (a) there is no silver matrix for $n = 1997$;
- (b) silver matrices exist for infinitely many values of n . [Iran]

13.100. IMO-1997. For each positive integer n , let $f(n)$ denote the number of ways of representing n as a sum of powers of 2 with nonnegative integer exponents. Representations which differ only in the ordering of their summands are considered to be the same. For instance, $f(4) = 4$ because the number 4 can be represented in the following four ways: 4 ; $2+2$; $21+1$; $1+1+1+1$. Prove that, for any integer $n \geq 3$,

$$2^{n^2/4} < f(2^n) < 2^{n^2/2}. \quad [\text{Lithuania}]$$

13.101. IMO-1998. In a competition, there are a contestants and b judges, where $b \geq 3$ is an odd integer. Each judge rates each contestant as either “pass” or “fail.” Suppose k is a number such that, for any two judges, their ratings coincide for at most k contestants. Prove that $\frac{k}{a} \geq \frac{b-1}{2b}$. [India]

13.102. IMO-1999. Consider an $n \times n$ square board, where n is a fixed even positive integer. The board is divided into n^2 unit squares. We say that two different squares on the board are *adjacent* if they have a common side. N unit squares on the board are marked in such a way that every square (marked or unmarked) on the board is adjacent to at least one marked square. Determine the smallest possible value of N . [Belarus]

13.103. IMO-2000. Let $n \geq 2$ be a positive integer and λ a positive real number. Initially, there are n fleas on a horizontal line, not all on the same point. We define a move as choosing two fleas, on some points A and B , with A to be left of B , and letting the flea from A to jump to the point C to the right of B such that $BC/AB = \lambda$. Determine all the values of λ such that, for any point M on the line and any initial position of the n fleas, there exists a sequence of moves that will take them all to a position right of M . [Belarus]

13.104. IMO-2000. A magician has one hundred cards numbered 1 to 100. He puts them into three boxes, a red one, a white one, and a blue one, such

that each box contains at least one card. A member of the audience draws two cards from two different boxes and announces the sum of the numbers on those cards. Given this information, the magician locates the box from which no card has been drawn. How many ways are there to put the cards in the boxes so that this trick works? [Hungary]

13.105. IMO-2001. Twenty-one girls and twenty-one boys took part in a mathematical contest. Each contestant solved at most six problems. For each girl and each boy, at least one problem was solved by both of them. Prove that there was a problem that was solved by at least three girls and at least three boys. [Germany]

13.106. IMO-2001. Let n be an odd integer greater than 1, and let $k_1, k_2, \dots, k_n$ be given integers. For each of the $n!$ permutations $a = (a_1, a_2, \dots, a_n)$ of $1, 2, \dots, n$, let $S(a) = k_1 a_1 + k_2 a_2 + \dots + k_n a_n$. Prove that there are two permutations b and c , $b \neq c$, such that $n!$ is a divisor of $S(b) - S(c)$. [Canada]

13.107. IMO-2002. Let n be positive integer and T be the set of all points (x, y) , where x and y are nonnegative integers such that $x + y < n$. Each element of T is colored red or blue, such that if (x, y) is red and $x' \leq x$, $y' \leq y$, then (x', y') is also red. A set consisting of n blue points from T is called an X -set, if all its points have different x -coordinates; a set consisting of n blue points from T is called a Y -set, if all its points have different y -coordinates. Prove that there are the same number of X -sets and Y -sets. [Colombia]

13.108. IMO-2003. Let A be a subset of the set $S = \{1, 2, \dots, 1\,000\,000\}$, such that A contains exactly 101 elements. Prove that there exist elements $t_1, t_2, \dots, t_{100} \in S$, such that the sets $A_j = \{x + t_j | x \in A\}$, $j \in \{1, \dots, 100\}$, are all pairwise disjoint. [Brazil]

13.109. IMO-2005. In a mathematical competition, in which 6 problems were posed to the participants, every two of these problems were solved by more than $\frac{2}{5}$ of the contestants. Moreover, no contestant solved all 6 problems. Prove that there are at least 2 contestants who solved exactly 5 problems. [Romania]

13.110. IMO-2006. Let P be a regular 2006-gon. A diagonal of P is called *good* if its endpoints divide the boundary of P into two parts, each composed of an odd number of sides of P . The sides of P are also called *good*. Suppose P has been dissected into triangles by 2003 diagonals, no two of which have

a common point in the interior of P . Find the maximum number of isosceles triangles having two good sides that could appear in such a configuration.

[Serbia]

13.111. IMO-2007. In a mathematical competition some competitors are friends. Friendship is always mutual. Call a group of competitors a *clique* if each two of them are friends. The number of members of a clique is called its *size*. Given that, in this competition, the largest size of a clique is even, prove that the competitors can be arranged in two rooms such that the largest size of the clique contained in one room is the same as the largest size of the clique contained in the other room.

[Russia]

13.112. IMO-2008. Let n and k be positive integers with $k \geq n$ and $k - n$ an even number. Let $2n$ lamps labeled $1, 2, \dots, 2n$ be given, each of which can be either on or off. Initially all the lamps are off. We consider sequences of steps: at each step one of the lamps is switched, from on to off, or from off to on. Let N be the number of such sequences consisting of k steps and resulting in the state where lamps 1 through n are all on, and lamps $n + 1$ through $2n$ are all off. Let M be the number of such sequences consisting of k steps, resulting in the state where lamps 1 through n are all on, and lamps $n + 1$ through $2n$ are all off, but where none of the lamps $n + 1$ through $2n$ is ever switched on. Determine the ratio N/M .

[France]

13.113. IMO-2009. Let $a_1, a_2, \dots, a_n$ be distinct positive integers and let M be a set of $n - 1$ positive integers not containing $s = a_1 + a_2 + \dots + a_n$. A grasshopper is to jump along the real axis, starting at point 0 and making n jumps to the right with lengths $a_1, a_2, \dots, a_n$ in some order. Prove that the order can be chosen in such a way that the grasshopper never lands on any point in M .

[Russia]

13.114. IMO-2010. In each of six boxes B_i , $1 \leq i \leq 6$, there is initially one coin. Two types of operations are allowed:

Type 1: Choose a nonempty box B_j with $1 \leq i \leq 5$. Remove one coin from B_j and add two coins to B_{j+1} .

Type 2: Choose a nonempty box B_k with $1 \leq i \leq 4$. Remove one coin from B_k and exchange the contents of (possibly empty) boxes B_{k+1} and B_{k+2} .

Determine whether there is a finite sequence of such operations that results in B_1, B_2, B_3, B_4 , and B_5 being empty and box B_6 containing exactly $2010^{2010^{2010}}$ coins.

[Netherlands]

13.115. IMO-2011. Let $n > 0$ be an integer. We are given a balance scale and n weights of weight $2^0, 2^1, \dots, 2^{n-1}$. We are to place each of the n

weights on the balance, one after another, in such a way that the right pan is never heavier than the left pan. At each step we choose one of the weights that has not yet been placed on the balance, and place it in either the left pan or the right pan, until all of the weights have been placed. Determine the number of ways in which this can be done. [Iran]

13.116. IMO-2012. The *liar's guessing game* is a game played between two players A and B . The rules of the game depend on two positive integers k and n which are known to both players.

At the start of the game A chooses integers x and N with $1 \leq x \leq N$. Player A keeps x secret, and truthfully tells N to player B . Player B now tries to obtain information about x by asking player A questions as follows: each question of B specifies an arbitrary set S of positive integers (possibly one specified in some previous question), and asks A whether x belongs to S . Player B may ask as many such questions as they wish. After each question, player A must immediately answer it with YES or NO, but is allowed to lie as many times as they want; the only restriction is that, among $k + 1$ consecutive answers, at least one answer must be truthful. After B has asked as many questions as they want, they must specify a set X of at most n positive integers. If $x \in X$, then B wins; otherwise, they lose. Prove that:

(a) If $n \geq 2^k$, then B can guarantee a win.

(b) For all sufficiently large k , there exists an integer $n \geq 1.99^k$ such that B cannot guarantee a win. [Canada]

13.117. IMO-2013. Let $n \geq 3$ be an integer, and consider a circle with $n+1$ equally spaced points marked on it. Consider all labelings of these points with the numbers $0, 1, \dots, n$ such that each label is used exactly once; two such labelings are considered to be the same if one can be obtained from the other by a rotation of the circle. A labeling is called *beautiful* if, for any four labels $a < b < c < d$ with $a + d = b + c$, the chord joining the points labeled a and b does not intersect the chord joining the points labeled b and c . Let M be the number of beautiful labelings, and let N be the number of ordered pairs (x, y) of positive integers such that $x + y \leq n$ and $\gcd(x, y) = 1$. Prove that $M = N + 1$. [Russia]

13.118. IMO-2014. Let $n \geq 2$ be an integer. Consider an $n \times n$ chessboard consisting of n^2 unit squares. A configuration of n rooks on this board is *peaceful* if every row and every column contains exactly one rook. Find the greatest positive integer k such that for each peaceful configuration of n rooks there is a $k \times k$ square which does not contain a rook on any of its k^2 unit squares. [Croatia]

13.119. IMO-2015. The sequence $a_1, a_2, \dots$ of integers satisfies the following conditions:

- (a) $1 \leq a_j \leq 2015$ for all $j \geq 1$;
- (b) $k + a_k \neq l + a_l$ for all $1 \leq k < l$.

Prove that there exist two positive integers b and N such that

$$\left| \sum_{j=m+1}^n (a_j - b) \right| \leq 1007^2$$

for all integers m and n satisfying $n > m \geq N$.

[Australia]

13.120. IMO-2016. Find all positive integers n for which each cell of an $n \times n$ table can be filled with one of the letters I, M, and O in such a way that:

- in each row and each column, one-third of the entries are I, one-third are M, and one-third are O, and
- in any diagonal, if the number of entries on the diagonal is a multiple of three, then one-third of the entries are I, one-third are M, and one-third are O.

[Australia]

13.121. IMO-2017. An integer $n \geq 2$ is given. A collection of $n(n+1)$ soccer players, no two of whom are of the same height, stand in a row. The coach wants to remove $n(n-1)$ players from this row leaving a new row of $2n$ players in which the following conditions hold:

- (1) no one stands between the two tallest players;
- (2) no one stands between the third and the fourth tallest players;
-
- (n) no one stands between the two shortest players.

Prove that this is always possible.

[Russia]

13.7 Elementary Probability

13.122. A fair die is rolled successively. Let X be the number of rolls necessary to obtain all six sides. Determine $P\{X = n\}$, for $n \in \{6, 7, \dots\}$, and calculate the mathematical expectation $E(X)$.

13.123. Let a probability space $(\Omega, \mathcal{A}, P)$ be given, and let $A, H_1, \dots, H_n \in \mathcal{A}$ be events such that the following conditions are satisfied:

- (a) $H_1 \cup H_2 \cup \dots \cup H_n = \Omega$, $H_i H_j = \emptyset$ for $i \neq j$;
- (b) $P(AH_i) > 0$ for every $i \in \{1, 2, \dots, n\}$;
- (c) Event A does not depend on events $H_1, H_2, \dots, H_n$.

Then for every event $B \in \mathcal{A}$ the following equality holds:

$$P(B|A) = \sum_{i=1}^n P(H_i)P(B|AH_i).$$

13.124. Consider a sequence of independent trials with two possible outcomes: A with probability p , and B with probability q , where $p + q = 1$. Determine the probability that a series (i.e., a sequence of consecutive occurrences) of A 's of length m will occur before a series of B 's of length n .

13.125. Prove that for every two events A and B the following inequality holds:

$$|P(AB) - P(A)P(B)| \leq \frac{1}{4}.$$

Find the conditions under which $|P(AB) - P(A)P(B)| = 1/4$ holds.

13.126. Let X_1 and X_2 be independent random variables, such that each of them takes the values -1 and 1 with equal probability $1/2$, and X_3 be a random variable such that $X_1X_2X_3 = 1$. Prove that the random variables X_1 , X_2 , and X_3 are pairwise independent, but dependent if they are considered as a collection of three random variables.

13.127. A fair die is rolled successively. Let X and Y denote, respectively, the number of rolls necessary to obtain an even number and 1.

- (a) Calculate $P\{X = m, Y = n\}$, where m and n are positive integers.
- (b) Find the probability $P\{X > Y\}$.

13.128. Positive integers $1, 2, \dots, n$ are arranged in a sequence $a_1a_2\dots a_n$ at random (all permutations are of equal probability $1/n!$). Let X be the number of k 's such that $a_k = k$. Determine $E(X)$.

13.129. A box contains k red and l blue balls, and next to the box there are m red and m blue balls. A ball is chosen at random from the box, and then the chosen ball and the m balls of the same color are put into the box.

(a) Let X be the number of red balls in the box at the end of the experiment. Determine $E(X)$.

(b) Suppose that a ball is chosen from the box in the new situation. What is probability that the chosen ball is red?

13.130. Let X be a random variable with finite variance. Determine the real constant c for which $E(X - c)^2$ takes the minimal value.

13.131. A random variable X takes the values $x_1, x_2, \dots, x_m$ with probability $p_1, p_2, \dots, p_m$, respectively, where $a \leq x_1 < x_2 < \dots < x_m \leq b$ and $p_1 + p_2 + \dots + p_m = 1$. Prove that

$$\text{var}(X) \leq \frac{1}{4}(b - a)^2.$$

Chapter 14



Solutions

14.1 Solutions for Chapter 1

1.1. 6^5 . 1.2. 4500. 1.3. 3^4 . 1.4. 36. 1.5. 2^n . 1.6. 72 000.

14.2 Solutions for Chapter 2

2.1. Digit c_1 of the three-digit number $c_1c_2c_3$ can be chosen arbitrarily from the set $\{1, 2, \dots, 9\}$. There are 9 possible choices of digit c_2 such that $c_2 \neq c_1$, and there are 8 possible choices of digit c_3 such that $c_3 \neq c_1$ and $c_3 \neq c_2$. By the product rule it follows that the number of positive integers with the given properties is equal to $9 \cdot 9 \cdot 8 = 648$.

2.2. 512. **2.3.** 1320. **2.4.** 2^n . **2.5.** (a) 20, (b) $(k_1+1)(k_2+1)\dots(k_m+1)$.

2.6. A positive integer n has an odd number of divisors if and only if n is a perfect square.

2.7. Every arrangement of teeth uniquely determines the 32-variation of elements 0 and 1. Hence, the maximal possible number of citizens is 2^{32} .

2.8. The number of permutations of the set $\{1, 2, \dots, n\}$ in which elements 1 and 2 are adjacent, and 1 is placed before 2, is equal to the number of permutations of the set $\{b, 3, \dots, n\}$, where b is notation for 12, i.e., $(n-1)!$. The number of permutations of the set $\{1, 2, \dots, n\}$, in which elements 1

and 2 are adjacent, and 1 comes after 2 is the same. Therefore, the number of permutations of the set $\{1, 2, \dots, n\}$ with adjacent elements 1 and 2 is $2(n-1)!$.

2.9. The number of permutations of the set $\{1, 2, \dots, n\}$ in which element 2 is placed after element 1 (not necessarily in the adjacent position) is equal to the number of permutations in which 2 is placed before 1. Since the total number of permutations of the n -set is equal to $n!$, it follows that the number we are asking for is equal to $\frac{1}{2}n!$.

2.10. The number of permutations of the set $\{1, 2, \dots, n\}$ such that element 1 is placed at position i , and element 2 is placed at position j , where $i \neq j$, and $i, j \in \{1, 2, \dots, n\}$, is equal to the number of permutations of a set consisting of $n-2$ elements, i.e., $(n-2)!$. The list of pairs of positions that can be occupied by 1 and 2 is the following: $(1, k+2), (2, k+3), \dots, (n-k-1, n)$. Hence, there are $n-k-1$ such pairs. Elements 1 and 2 can occupy any of these pairs in 2 ways. Hence, the number of permutations that satisfy the given conditions is $2(n-k-1)(n-2)!$.

2.11. $(n!)^3$. **2.12.** $\frac{8!}{2!2!2!} = 5040$. **2.13.** $(n-1)!$. **2.14.** $\binom{n}{2}$. **2.15.** $\binom{n}{3}$.

2.16. (a) The answer is $\binom{n}{4}$ because there is a bijection between the set of 4-tuples of vertices of the convex n -gon and the set of the points of intersection of the lines considered that are inside the n -gon.

$$(b) \frac{1}{2} \binom{n}{2} \left[\binom{n}{2} - 1 \right] - \binom{n}{4} - n \binom{n-1}{2} = 2 \binom{n}{4}.$$

2.17. The number of three-member committees is $\binom{6}{3} = 20$, and five of them can be chosen in $\binom{20}{5} = 15\,504$ ways.

2.18. $4\binom{8}{3}8^3 + \binom{4}{2}\binom{8}{2}8^2 = 415\,744$. **2.19.** $\sum_{k=0}^5 \left[\binom{5}{k} \binom{7}{6-k} \right]^2 = 267\,148$.

2.20. The units can be put in k positions of the following $n+1$ positions: before the first zero, between the first and the second zeroes, between the second and the third zeroes, $\dots$, and finally after the n -th zero. Therefore, the number of sequences consisting of n zeroes and k units such that no two units are adjacent is equal to $\binom{n+1}{k}$.

2.21. The number of choices of k books from a sequence consisting of n books, such that no two adjacent books are chosen, is equal to the number of n -variations of the elements 0 and 1, consisting of $n-k$ zeroes and k ones,

and such that no two ones are adjacent. Using the result from Exercise 2.20 we get that this number is equal to $\binom{n-k+1}{k}$. For $n = 12$ and $k = 5$ the number of choices is $\binom{12-5+1}{5} = 56$.

2.22. Suppose that n is the number of knights arranged around the table, every knight is quarreling with his neighbors, and King Arthur wishes to choose k knights so that no two of them are quarreling. Let L be the first knight, and S be the set of all k -combinations of knights, such that no two knights from the same combination are quarreling. Let S_1 be the set of those combinations from S that contain L , and S_2 the set of those combinations from S that does not contain L . Note that any combination from S_1 does not contain the neighbors of L , and the combinations from S_2 may contain one or both of his neighbors. Similarly as in Exercise 2.21 we get

$$|S_1| = \binom{n-3-(k-1)+1}{k-1}, \quad |S_2| = \binom{n-1-k+1}{k}.$$

The number of choices of k knights (such that the given conditions are satisfied) is equal to $\binom{n-k-1}{k-1} + \binom{n-k}{k}$. For $n = 12$ and $k = 5$ the number of choices is 36.

2.23. The number of n -digit positive integers $c_1 c_2 \dots c_n$, such that $1 \leq c_1 \leq c_2 \leq \dots \leq c_n \leq 9$, is equal to $(n+8)$ -variations of elements 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, that contain exactly eight 0's, and such that all 1's are placed before the first 0, all 2's are placed between the first and the second 0, $\dots$, and finally all 9's are placed after the last 0. Every such variation is determined by the positions of the 0's, and therefore their number is $\binom{n+8}{8}$.

2.24. *Answer:* $\binom{9}{3} \left[3 \frac{6!}{1!1!4!} + 6 \frac{6!}{1!2!3!} + \frac{6!}{2!2!2!} \right] = 45\,360.$

2.25. The letters L, C, M, T, I, V, and E can be arranged in a sequence in $7!$ ways. Three letters O can be added so that no two of them are adjacent in $\binom{8}{3}$ ways, see Exercise 2.20. By the product rule the number of different words that can be obtained is $7! \binom{8}{3} = 282\,240$.

2.26. The total number of appearances of all elements $a_1, a_2, \dots, a_n$ in all k -arrangements is kn^k . Hence, every element appears kn^{k-1} times.

2.27. The number of 13-arrangements of the elements Y , B , G , R , and s , that have the form $Y \dots Y s B \dots B s G \dots G s R \dots R$, is $\binom{13}{3}$. Here, letters Y , B , G , and R are notation for yellow, blue, green, and red balls, respectively, and letter s separates blocks of letters of the same kind. The subsequences $Y \dots Y$, $B \dots B$, $G \dots G$ may consist of 0, 1, ..., 10 terms, while the subsequence $R \dots R$ consists of no more than 9 (possibly 0) terms. Hence, the number of ways is $\binom{13}{3} - 1 = 285$.

2.28. Let us represent $144!$ in the canonical form $144! = 2^{k_1} 3^{k_2} 5^{k_3} \dots$. Note that $k_3 = \left\lfloor \frac{144}{5} \right\rfloor + \left\lfloor \frac{144}{25} \right\rfloor + \left\lfloor \frac{144}{125} \right\rfloor = 28 + 5 + 1 = 34$, and $k_1 > k_3$. Every zero at the end of the decimal representation of $144!$ is obtained as the product $2 \cdot 5$. Since $k_1 > k_3$, the number of zeroes is 34.

2.29.
$$\frac{n(n+1) \cdots (n+k-1)}{k!} = \binom{n+k-1}{k}.$$

2.30. Note that $\frac{(a_1 + a_2 + \cdots + a_n)!}{a_1! a_2! \cdots a_n!}$ is the number of $(a_1 + a_2 + \cdots + a_n)$ -arrangements of the elements $1, 2, \dots, n$, that have the type $(a_1, a_2, \dots, a_n)$.

2.31. Let us introduce the following notation:

X – the set of positive integers whose sum we are interested in;

S – the sum of positive integers from set X ;

S_1 – the sum of integers from X without 0's in their representation;

S_2 – the sum of integers from X that have 0's in their representation;

$S_1(a, b)$ – the sum of integers from X with even digits $a \neq 0$ and $b \neq 0$;

$S_1(a, b, n)$ – the sum of positive integers from X with even digits $a \neq 0$ and $b \neq 0$, and an odd digit n ;

$S_2(a)$ – the sum of positive integers from X with even digits 0 and $a \neq 0$;

$S_2(a, n)$ – the sum of positive integers from X with even digits 0 and $a \neq 0$, and an odd digit n .

All positive integers from X with even digits $a \neq 0$ and $b \neq 0$, and an odd digit n are the following: $abnn$, $anbn$, $annb$, $bann$, $bnan$, $bnnn$, $nanb$, $nban$, $nanb$, $nbna$, $nnab$, $nnba$. Their sum is $S_1(a, b, n) = 1111(3a + 3b + 6n) = 3333(a + b) + 6666n$. Consequently we obtain

$$\begin{aligned} S_1(a, b) &= 5 \cdot 3333(a + b) + 6666(1 + 3 + 5 + 7 + 9) \\ &= 16665(a + b) + 166650, \end{aligned}$$

$$S_1 = 16665(6 + 8 + 10 + 10 + 12 + 14) + 6 \cdot 166650 = 1999800.$$

The list of all positive integers from X with even digits 0 and $a \neq 0$, and an odd digit n is: $a0nn$, $an0n$, $ann0$, $na0n$, $n0an$, $nan0$, $n0na$, $nnn0$.

Their sum is $S_2(a, n) = 3222a + 6444n$, and consequently we get

$$\begin{aligned} S_2(a) &= 5 \cdot 3222a + 6444(1 + 3 + 5 + 7 + 9) = 16\,110a + 161\,100, \\ S_2 &= 16110(2 + 4 + 6 + 8) + 4 \cdot 161\,100 = 966\,600. \end{aligned}$$

Finally, $S = S_1 + S_2 = 2\,966\,400$.

2.32. If a seven-digit positive integer $c_1c_2c_3c_4c_5c_6c_7$ does not change after the 180° rotation, then $c_4 \in \{0, 1, 8\}$, $c_1 \in \{1, 6, 8, 9\}$, $c_2, c_3 \in \{0, 1, 6, 8, 9\}$, and the digits c_5 , c_6 , and c_7 are uniquely determined by the digits c_3 , c_2 , and c_1 . By the product rule it follows that the number of seven-digit positive integers that remain the same after the 180° rotation is $4 \cdot 5 \cdot 5 \cdot 3 = 300$.

2.33. Answer: $3 \binom{n}{3} + n^3$.

2.34. The number of k -digit positive integers is $10^k - 10^{k-1} = 9 \cdot 10^{k-1}$. The total number of digits in their decimal representation is $9k \cdot 10^{k-1}$. The number of 0's in the decimal representation of all k -digit positive integers is $(9k \cdot 10^{k-1} - 9 \cdot 10^{k-1})/10 = 9(k-1)10^{k-2}$. The total number of 0's in the decimal representation of all positive integers from the set $\{1, 2, \dots, 10^9\}$ is

$$\sum_{k=1}^9 9(k-1)10^{k-2} + 9 = 9 \cdot 87\,654\,321 + 9 = 788\,888\,898.$$

2.35. Let us consider the sequence 0000000, 0000001, 0000002, $\dots$, 9999999. The number of terms in this sequence without the digit 1 in their decimal representation is 9^7 , and the number of terms with the digit 1 is $10^7 - 9^7$. In the sequence $1, 2, \dots, 10^7$ there are $9^7 - 1 = 4\,782\,968$ positive integers without the digit 1, and $5\,217\,032$ positive integers with the digit 1 in their decimal representation.

2.36. Let us consider the following pairs of nonnegative integers: $(0, 999\,999)$, $(1, 999\,998)$, $(2, 999\,997)$, $\dots$, $(499\,998, 500\,001)$, $(499\,999, 500\,000)$. The sum of digits that appear in each of these pairs is $6 \cdot 9 = 54$. The sum of digits that appear in the decimal representation of all positive integers from the set $\{1, 2, \dots, 1\,000\,000\}$ is $500\,000 \cdot 54 + 1 = 27\,000\,001$.

2.37. The number of k -digit positive integers without equal digits in adjacent positions is 9^k (the first digit can be any of nine digits not equal to zero, and every next digit, different from the previous one, can also be chosen in nine ways). The number of positive integers that are not

greater than 10^n and without equal digits in adjacent positions is equal to $9 + 9^2 + \cdots + 9^n = \frac{9}{8}(9^n - 1)$.

2.38. *The first solution.* The number of units in an n -arrangement of the elements 0 and 1 without the pattern 11 can take any of the values 0, 1, 2, ..., k , where

$$k = \begin{cases} n/2, & \text{if } n \text{ is even,} \\ (n+1)/2, & \text{if } n \text{ is odd,} \end{cases}$$

By the sum rule and Exercise 2.20 we get that the number of n -arrangements of the elements 0 and 1 without the pattern 11 is equal to

$$1 + \binom{n}{1} + \binom{n-1}{2} + \cdots + \binom{n-k+1}{k}.$$

The second solution. For any positive integer n let S_n be the set of n -arrangements of the elements 0 and 1 without the pattern 11, and let us denote $f_n = |S_n|$. Then, $f_1 = 2$, $f_2 = 3$. For $n \geq 3$ let X_n be the subset of S_n consisting of those n -arrangements that have 1 in the n -th position, and let Y_n be the subset of S_n consisting of those n -arrangements that have 0 in the n -th position. Every n -arrangement from X_n has 0 in the $(n-1)$ -th position, while the n -arrangements from set Y_n may have 0 or 1 in the $(n-1)$ -th position. The function $g: X_n \rightarrow S_{n-2}$ given by

$$g(c_1 c_2 \cdots c_{n-2} 01) = c_1 c_2 \cdots c_{n-2}, \quad c_1, c_2, \dots, c_{n-2} \in \{0, 1\},$$

is a bijection, and therefore $|X_n| = |S_{n-2}| = f_{n-2}$. Analogously we get $|Y_n| = f_{n-1}$. Since $S_n = X_n \cup Y_n$ and $X_n \cap Y_n = \emptyset$, it follows that $f_n = f_{n-1} + f_{n-2}$, i.e., (f_n) is the Fibonacci sequence with the initial terms $f_1 = 2$, $f_2 = 3$.

2.39. Let S_n be the set of n -arrangements of the elements 0, 1, 2, ..., k , with an even number of zeroes, $a_n = |S_n|$, and let $A_0^{(n)}$, $A_1^{(n)}$, ..., $A_k^{(n)}$ be subsets of S_n consisting of n -arrangements that have 0, 1, ..., k in the first position, respectively. Then,

$$|A_1^{(n)}| = |A_2^{(n)}| = \cdots = |A_k^{(n)}| = a_{n-1}, \quad |A_0^{(n)}| = (k+1)^{n-1} - a_{n-1},$$

and hence, for $n \geq 2$ we get $a_n = (k-1)a_{n-1} + (k+1)^{n-1}$. Note that $a_1 = k$. By mathematical induction it is easy to prove that for any $n \in \mathbb{N}$,

$$a_n = \frac{1}{2} [(k+1)^n + (k-1)^n].$$

2.40. Let **R** be the notation for a red ball, and **B** and **Y** be notation for a few blue balls, or a few yellow balls, respectively. Note that 6 red balls, 7 blue balls, and 10 yellow balls are arranged in a sequence so that the given conditions are satisfied if and only if the sequence is of the form **YRBRYYRBRYYRBRYY**, or **BRYRBRYYRBRYYR**. The number of arrangements that satisfy the given conditions is $6!7!(10)! \left[\binom{6}{3} \binom{9}{2} + \binom{6}{2} \binom{9}{3} \right]$.

2.41. (a) For a fixed n the result $25 : n$ can be reached in $\binom{24+n}{n}$ ways.

(b) The host team can reach 25 points with at least a two-point margin in $\binom{24}{24} + \binom{25}{24} + \cdots + \binom{47}{24}$ ways.

2.42. (a) $\binom{n-1}{m-1}$, (b) $\binom{n-k-1}{m-1}$, (c) $\binom{n-k+m-2}{m-2}$,
(d) $\binom{n-1-2-\cdots-(m-1)}{m-1}$.

Hint. Consider the arrangements of the elements 0 and 1 that have the form $11 \cdots 1011 \cdots 1011 \cdots$, where each block of 1's represents balls from the same box.

2.43. $\binom{n_1+m-1}{m-1} \binom{n_2+m-1}{m-1} \binom{n_3+m-1}{m-1}$.

2.44. $\binom{n_1+\cdots+n_k}{n_1} \binom{n_2+\cdots+n_k}{n_2} \cdots \binom{n_k}{n_k} = \frac{(n_1+n_2+\cdots+n_k)!}{n_1!n_2!\cdots n_k!}$.

2.45. $\binom{7}{2} \binom{8}{2} \binom{9}{2} = 21\,168$. **2.46.** $\prod_{i=1}^k (n_i - 2s_i + 1)$.

2.47. (a) Answer: 100. (b) Answer: 18. *Hint.* Consider the cases: $x_1 = 1$, $x_1 = 2$, $x_1 = 4$, $x_1 = 5$, $x_1 = 8$, and $x_1 = 10$.

2.48. The number of triangles whose vertices belong to three different sides of the square is $4(n-1)^3$. The number of triangles with two vertices on the same side of the square is $4\binom{n-1}{2}(3n-3)$. The total number of triangles is $2(n-1)^2(5n-8)$.

2.49. $\binom{n}{3}k^3 + \binom{n}{1}\binom{k}{2}(n-1)k$.

2.50. Let A_1, A_2, A_3, A_4 , and A_5 be the given points. Six perpendicular lines are dropped from each of these points. The perpendicular lines dropped from A_1 and A_2 intersect at $6 \cdot 6 - 3 = 33$ points (the perpendicular lines dropped from A_1 and A_2 to any of the lines A_3A_4, A_4A_5 , and A_5A_1 do not intersect). Two points among five can be chosen in 10 ways. Therefore, the number of points of intersection of all the dropped perpendicular lines is not greater than 330. Any three of the points A_1, A_2, A_3, A_4 , and A_5 determine a triangle whose altitudes intersect at the same point. There are 10 such triangles, and hence the total number of points of intersection of the dropped perpendiculars is $330 - 2 \cdot 10 = 310$.

2.51. Let S_1 be the set of all k -combinations of the elements of set $\{1, \dots, n\}$, such that $|x - y| > m$ for any $x, y \in X$, where $x \neq y$ and $X \in S_1$. Let S_2 be the set of all k -combinations of the set $\{1, 2, \dots, n - (k - 1)m\}$. Let us consider the function $f: S_1 \rightarrow S_2$, defined as follows. For

$$X = \{x_1, x_2, \dots, x_k\} \in S_1, \quad x_1 < x_2 < \dots < x_k,$$

we define $f(X) = Y = \{y_1, y_2, \dots, y_k\} \in S_2$, where $y_j = x_j - (j - 1)m$. The function f is a bijection between S_1 and S_2 , and therefore

$$|S_1| = |S_2| = \binom{n - (k - 1)m}{k}.$$

2.52. The number of choices of n books, such that exactly k books are chosen from the set of n books of the same kind, is $\binom{2n+1}{n-k}$. Since k can take any value from the set $\{0, 1, \dots, n\}$, it follows that the total number of choices is

$$\binom{2n+1}{n} + \binom{2n+1}{n-1} + \dots + \binom{2n+1}{0} = \frac{1}{2} \sum_{k=0}^{2n+1} \binom{2n+1}{k} = 2^{2n}.$$

2.53. Let $\{c_1, c_2, \dots, c_n\}$ be the set of all signal flags, and let S be the set of $(n + k - 1)$ -arrangements of the elements $0, c_1, c_2, \dots, c_n$, such that any element c_j appears exactly once, and 0 appears exactly $k - 1$ times. There is an obvious bijection between the set of all possible signals and the set S . Hence, the total number of signals is

$$|S| = \frac{(n + k - 1)!}{1! 1! \dots 1! (k - 1)!} = n! \binom{n + k - 1}{k - 1}.$$

2.54. The number of permutations of the set S in which the elements $j, k \in S$ form an inversion is equal to the number of permutations of the set

S in which elements $j, k \in S$ do not form an inversion. The total number of inversions in all permutations is $\binom{n}{2} \frac{n!}{2}$.

2.55. Let $S = \{1, 2, \dots, n\}$, and Π be the set of permutations of the set S such that any two elements from S are adjacent in at most one of the permutations from Π . Let us denote $m = |\Pi|$. Note that any permutation of S has $n-1$ pairs of adjacent elements, and the number of two-element subsets of S is $n(n-1)/2$. Therefore, $m(n-1) \leq \frac{1}{2}n(n-1)$, i.e., $m \leq [n/2]$. For $n = 8$, equality is attained if $\Pi = \{12837465, 23148576, 34251687, 45362718\}$. A similar example can be given for any positive integer n .

2.56. Let S be the set of all convex k -gons that should be counted, and whose vertices belong to the set $\{A_1, A_2, \dots, A_n\}$. Let us consider the partition $S = S_0 \cup S_1$, $S_0 \cap S_1 = \emptyset$, where S_1 is the set of those k -gons from S with one vertex belonging to the set $\{A_1, A_2, \dots, A_p\}$. If A_1 is a vertex of a k -gon from S_1 , then none of the points $A_2, A_3, \dots, A_{p+1}, A_n, A_{n-1}, \dots, A_{n-p+1}$ is a vertex of this k -gon. By Exercise 2.51 it follows that $k-1$ vertices can be chosen from the set $\{A_{p+2}, A_{p+3}, \dots, A_{n-p}\}$ in $\binom{n-2p-1-(k-2)p}{k-1}$ ways, such that these vertices together with A_1 form a convex k -gon from S_1 . Therefore, $|S_1| = p \binom{n-kp-1}{k-1}$. Similarly, $|S_0| = \binom{n-p-(k-1)p}{k}$, and hence

$$|S| = p \binom{n-kp-1}{k-1} + \binom{n-kp}{k}.$$

2.57. Let S be a set of 3-combinations of elements of the set $\{1, 2, \dots, n\}$, such that every two sets from S have exactly one common element. Let us denote $x_n = |S|$, and consider the following three cases.

(a) No three of the sets from S have a common element. Suppose that $\{1, 2, 3\} \in S$. Since any other set from S contains exactly one of the elements 1, 2, and 3, it follows that $x_n \leq 1 + 3 \cdot 1 = 4$.

(b) There are three sets $A_1, A_2, A_3 \in S$ that have a common element, and no four sets from S have a common element. Then, $n \geq |A_1 \cup A_2 \cup A_3| = 7$. Suppose that $\{1, 2, 3\} \in S$. Since each of the elements 1, 2, and 3 is contained in at most two of the other sets from S , it follows that $x_n \leq 1 + 3 \cdot 2 = 7$.

(c) There are 4 sets $A_1, A_2, A_3, A_4 \in S$ that have a common element. Then, $|A_1 \cup A_2 \cup A_3 \cup A_4| = 9$, and the common element of the sets A_1, A_2, A_3 , and A_4 belongs to each of the sets from S (in the opposite case there exists a set from S that has at least 4 elements). By this fact it follows that $1 + 2x_n \leq n$, i.e., $x_n \leq [(n-1)/2]$. Let us determine the maximal possible value of x_n . We shall denote it by x_n^* .

Obviously we have $x_1^* = x_2^* = 0$, $x_3^* = x_4^* = 1$, $x_5^* = 2$. If $n = 6$, then no 3 sets from S have a common element (in the opposite case we easily get that $n \geq 7$), and therefore $x_6 \leq 4$. The subsets $\{1, 2, 3\}$, $\{3, 4, 5\}$, $\{5, 6, 1\}$, $\{2, 4, 6\}$ of the set $\{1, 2, 3, 4, 5, 6\}$ show that $x_6 = 4$.

Let $n \in \{7, 8, \dots, 16\}$. In case (c) we have $x_n \leq [(16 - 1)/2] = 7$. In cases (a) and (b) the inequality $x_n \leq 7$ also holds. Subsets $\{1, 2, 3\}$, $\{3, 4, 5\}$, $\{5, 6, 1\}$, $\{2, 4, 6\}$, $\{1, 4, 7\}$, $\{2, 5, 7\}$, and $\{3, 6, 7\}$ of the set $\{1, 2, \dots, n\}$, where $7 \leq n \leq 16$, show that $x_7^* = x_8^* = \dots = x_{16}^* = 7$. For $n \geq 17$ the equality $x_n = [(n - 1)/2]$ holds. This fact follows from the following example of the subsets of set $\{1, 2, \dots, n\}$:

$$\begin{aligned} &\{1, 2, 2k - 1\}, \{3, 4, 2k - 1\}, \dots, \{2k - 3, 2k - 2, 2k - 1\}, \quad \text{if } n = 2k, \\ &\{1, 2, 2k + 1\}, \{3, 4, 2k + 1\}, \dots, \{2k - 1, 2k, 2k + 1\}, \quad \text{if } n = 2k + 1. \end{aligned}$$

2.58. There are 2^{n-1} subsets of the set $\mathbb{N}_n = \{1, 2, \dots, n\}$ that contain the element 1. From this fact it follows that $k \geq 2^{n-1}$. Let Φ be a family of subsets of set $\mathbb{N}_n$, such that every two sets from Φ have a nonempty intersection. If $A \in \Phi$, then obviously $S \setminus A \notin \Phi$. Therefore, $k \leq 2^{n-1}$, and finally it follows that $k = 2^{n-1}$.

2.59. The condition from the formulation of the problem is satisfied if and only if for any group of $m - 1$ members of the jury there exists a lock L such that none of these $m - 1$ members has the key to lock L , and all of the remaining $n - m + 1$ members of the jury have the key to lock L . Hence the number of locks is $\binom{n}{m-1}$, and the number of keys is

$$(n - m + 1) \binom{n}{m-1} = m \binom{n}{m}.$$

Every member of the jury should have $\binom{n}{m} \frac{m}{n} = \binom{n-1}{m-1}$ keys.

2.60. Let us introduce the following notation: x_n is the number of k -tuples $(A_1, A_2, \dots, A_k)$ such that $A_1, A_2, \dots, A_k \in S_n$ and $|A_1 \cap A_2 \cap \dots \cap A_k| \neq \emptyset$, and $x_{n,j}$ is the number of k -tuples $(A_1, A_2, \dots, A_k)$ such that $A_1, A_2, \dots, A_k \in S_n$, and $|A_1 \cap A_2 \cap \dots \cap A_k| = j$. Then, $x_1 = 1$, and $x_{n+1} = (2^k - 1)x_n + 2^{kn}$ for $n \geq 1$. If we denote $a = 2^k - 1$ and $b = 2^k$, then

$$\begin{aligned} x_{n+1} &= ax_n + b^n = a(ax_{n-1} + b^{n-1}) + b^n = \dots \\ &= a^n x_1 + a^{n-1}b + \dots + ab^{n-1} + b^n = \frac{b^{n+1} - a^{n+1}}{b - a}. \end{aligned}$$

Hence, $x_n = \frac{b^n - a^n}{b - a} = 2^{kn} - (2^k - 1)^n$. Consequently, it follows that

$$x_{n,0} = 2^{kn} - x_n = (2^k - 1)^n, \quad x_{n,n} = 1,$$

$$x_{n+1,j} = (2^k - 1)x_{n,j} + x_{n,j-1}, \quad j = 1, 2, \dots, n.$$

Let us denote $a_{n,j} = (2^k - 1)^{j-n} x_{n,j}$, for $j \in \{0, 1, \dots, n\}$, $n \in \mathbb{N}$. Then, we get $a_{n,0} = a_{n,n} = 1$, $a_{n+1,j} = a_{n,j} + a_{n,j-1}$, for $j \in \{1, 2, \dots, n\}$, $n \in \mathbb{N}$. The last equalities uniquely determine all terms of the sequence $a_{n,j}$, where $j \in \{0, 1, 2, \dots, n\}$, $n \in \mathbb{N}$. Note that the binomial coefficients satisfy these equalities, see Chapter 3, and hence $a_{n,j} = \binom{n}{j}$, $x_{n,j} = \binom{n}{j} (2^k - 1)^{n-j}$.

2.61. Every arrangement from S consists of blocks of 1's and blocks of 0's. Every arrangement with the characteristic $2m + 1$ has $m + 1$ blocks of 1's and $m + 1$ blocks of 0's. The number of such arrangements is $2 \binom{n-1}{m}^2$, because a sequence consisting of n equal digits can be partitioned into m blocks in $\binom{n-1}{m}$ ways. Every arrangement with the characteristic $2m$ has $m + 1$ blocks of one of the digits 0 or 1, and m blocks of the other digit. The number of such arrangements is $2 \binom{n-1}{m-1} \binom{n-1}{m}$. Note that positive integers $n - k$ and $n + k$ have the same parity. It is easy to see that

$$2 \binom{n-1}{\frac{n-k-1}{2}}^2 = 2 \binom{n-1}{\frac{n+k-1}{2}}^2.$$

The statement of Exercise 2.61 follows from the last equality if $n - k$ and $n + k$ are both even positive integers. Similarly, if $n - k$ and $n + k$ are both odd, then the statement follows from the equality

$$2 \binom{n-1}{\frac{n-k}{2} - 1} \binom{n-1}{\frac{n-k}{2}} = 2 \binom{n-1}{\frac{n+k}{2} - 1} \binom{n-1}{\frac{n+k}{2}}.$$

2.62. The sum $|p_1 - 1| + |p_2 - 2| + \dots + |p_n - n|$ is equal to the sum of integers

$$1, 1, 2, 2, 3, 3, \dots, n, n, \quad (1)$$

where n of them are taken with the sign “+,” and n of them with the sign “−.” This sum has a maximal value if the first n terms of the sequence (1) are taken with the sign “−.” The maximal value of the sum is $(n^2 - 1)/2$, if n is odd, while the maximal value is $n^2/2$, if n is even.

(a) Case $n = 2k$. The maximal value is attained if and only if

$$\begin{aligned} \{p_1, p_2, \dots, p_k\} &= \{k + 1, k + 2, \dots, 2k\}, \\ \{p_{k+1}, p_{k+2}, \dots, p_{2k}\} &= \{1, 2, \dots, k\}. \end{aligned}$$

Therefore, the number of permutations $(p_1, p_2, \dots, p_n)$ that satisfy this condition is $(k!)^2$.

(b) Case $n = 2k + 1$. The maximal value is attained if and only if

$$\begin{aligned}\{p_1, p_2, \dots, p_{k+1}\} &= \{k+1, k+2, \dots, 2k+1\}, \\ \{p_{k+2}, p_{k+3}, \dots, p_{2k+1}\} &= \{1, 2, \dots, k\},\end{aligned}$$

or there exists a positive integer $x \in \{1, 2, \dots, k\}$, such that $p_{k+1} = x$ and

$$\begin{aligned}\{p_1, p_2, \dots, p_k\} &= \{k+2, k+3, \dots, 2k+1\}, \\ \{p_{k+2}, p_{k+3}, \dots, p_{2k+1}\} &= \{1, 2, \dots, k+1\} \setminus \{x\}.\end{aligned}$$

Hence, the number of permutations for which the maximal value is attained is equal to $k!(k+1)! + k(k!)^2 = (2k+1)(k!)^2$.

2.63. For $m = n$, the results are given by Theorem 2.8.1(d)–(e). For $m \geq n$ the corresponding results can be obtained similarly as in Theorem 2.8.1.

$$(a) \binom{m+n}{m} - \binom{m+n}{m+1}; \quad (b) \binom{m+n}{m} - \binom{m+n}{n-k-1}.$$

2.64. Let S_1 be the set of $2n$ -arrangements of the elements 0 and 1 that have the type (n, n) , and such that before any 1 there are more 0's than 1's. Let S_2 be the set of permutations of the set $\{1, 2, \dots, 2n\}$, denoted by $i_1 i_2 \dots i_n j_1 j_2 \dots j_n$, such that the following inequalities hold:

$$i_1 < j_1, \quad i_2 < j_2, \dots, \quad i_n < j_n, \quad (1)$$

$$i_1 < i_2 < \dots < i_n, \quad j_1 < j_2 < \dots < j_n. \quad (2)$$

For any arrangement $v \in S_1$ let us define $f(v) = i_1 i_2 \dots i_n j_1 j_2 \dots j_n$, where $i_1, i_2, \dots, i_n$ ($i_1 < i_2 < \dots < i_n$) are the positions of the 0's in arrangement v , and $j_1, j_2, \dots, j_n$ ($j_1 < j_2 < \dots < j_n$) are the positions of the 1's in arrangement v . Then, $i_k < j_k$ for any $k \in \{1, 2, \dots, n\}$. [Suppose, on the contrary, that $j_k < i_k$ for some k . Then, before the 1 in the j_k -th position of arrangement v , there are $k-1$ 1's and no more than $k-1$ 0's, and this contradicts the assumption that $v \in S_1$.] Therefore $f(v) \in S_2$, i.e., the function $f: S_1 \rightarrow S_2$ is well defined. Obviously, f is a *one-to-one* function. Let us prove that f is an *onto* function. Let $p = i_1 i_2 \dots i_n j_1 j_2 \dots j_n$ be a permutation of the set $\{1, 2, \dots, 2n\}$, such that the inequalities (1) and (2) hold, and let v be the $2n$ -arrangement of the elements 0 and 1, such that 0's are placed in the positions $i_1, i_2, \dots, i_n$, and 1's are placed in the positions $j_1, j_2, \dots, j_n$. Since $i_1 < i_2 < \dots < i_k < j_k$ for any $k \in \{1, 2, \dots, n\}$, it follows that $v \in S_1$ and $f(v) = p$. Hence, function f is a bijection. From the

result of Example 2.8.2 it follows that

$$|S_2| = |S_1| = \frac{1}{n+1} \binom{2n}{n}.$$

2.65. Let us introduce the following definition: *the pairs of positive integers (x, y) and (z, u) , where $x < y$ and $z < u$, separate each other if*

$$x < z < y < u \quad \text{or} \quad z < x < u < y.$$

Let us denote the given points by $1, 2, \dots, 2n$ in the order of their appearance on the circle. The chord determined by the points i and j , where $i < j$, is denoted by ij . Point i is the beginning, and j is the endpoint of chord ij . We also say that i and j are both endpoints of chord ij . The next three statements hold:

(a) *The necessary and sufficient condition that chords ij and rs intersect is that the pairs of positive integers (i, j) and (r, s) separate each other.*

(b) Let $i_1j_1, i_2j_2, \dots, i_nj_n$, where $j_1 < j_2 < \dots < j_n$, be n chords without points of intersection, such that the set of endpoints of these chords coincides with the set of the given $2n$ points. Then, $j_k \geq 2k$ for every $k \in \{1, 2, \dots, n\}$.

(c) If $j_1, j_2, \dots, j_n \in \{1, 2, \dots, 2n\}$, and $j_k \geq 2k$ for any $k \in \{1, 2, \dots, n\}$, then there exists exactly one k -tuple of chords $(i_1j_1, i_2j_2, \dots, i_nj_n)$ without points of intersection, and such that these chords connect points $1, 2, \dots, 2n$ in pairs.

Statement (a) is obvious.

Proof of statement (b): For any $k \in \{1, 2, \dots, n\}$ the following inequalities hold

$$\begin{aligned} j_k &> j_{k-1} > \dots > j_2 > j_1, \\ j_k &> i_k, \quad j_{k-1} > i_{k-1}, \quad \dots, \quad j_2 > i_2, \quad j_1 > i_1, \end{aligned}$$

and therefore it follows that $j_k \geq 2k$.

Proof of statement (c): Let us suppose $j_1, j_2, \dots, j_n \in \{1, 2, \dots, 2n\}$, and $j_k \geq 2k$ for any $k \in \{1, 2, \dots, n\}$. Let $i_1 = j_1 - 1$, and

$$i_k = \max \{1, 2, \dots, j_k - 1\} \setminus \{i_1, j_1, i_2, j_2, \dots, i_{k-1}, j_{k-1}\},$$

for any $k \in \{1, 2, \dots, n\}$. Then, i_k is well defined because $\{1, 2, \dots, j_{k-1}\}$ consists of $2k - 2$ elements. The points i_k and j_k determine two arcs of the circle. Note that the points $i_1, j_1, i_2, j_2, \dots, i_{k-1}, j_{k-1}$ all belong to one of these two arcs, and the points $i_{k+1}, j_{k+1}, \dots, i_n, j_n$ all belong to the

other arc. Now it is easy to conclude that chords $i_1j_1, i_2j_2, \dots, i_nj_n$ do not intersect. Suppose that for some k the next inequality holds:

$$i_k < \max\{1, 2, \dots, j_{k-1}\} \setminus \{i_1, j_1, \dots, i_{k-1}, j_{k-1}\} = M.$$

Since $j_{k-1} < M < j_k$, it follows that M cannot be the endpoint of a chord. Point M cannot be the beginning of a chord (in the opposite case this chord intersects chord i_kj_k). This finishes the proof of statement (c).

Let S_1 be the set of $2n$ -arrangements of the elements 0 and 1 that have the type (n, n) , and such that, before every 1, there are more 0's than 1's. Let S_2 be the set of all connections of the points $1, 2, \dots, 2n$ into pairs by chords without points of intersection inside the circle. Every such connection by chords $i_1j_1, i_2j_2, \dots, i_nj_n$ determines a $2n$ -arrangement of elements 0 and 1, with 0's in the positions $i_1, i_2, \dots, i_n$, and 1's in the remaining positions. And vice versa. By statements (b) and (c), and Example 2.8.2 it follows that $|S_2| = |S_1| = \frac{1}{n+1} \binom{2n}{n}$.

Remark. Let $x_0 = 1$, and x_n be the number of connections of the given $2n$ points into n pairs by chords without points of intersection. Then, $x_n = x_0x_{n-1} + x_1x_{n-2} + \dots + x_{n-1}x_0$ for every $n \geq 1$.

2.66. In what follows, a *partition* will always be a partition of the polygon into triangles by diagonals without points of intersection inside the polygon. First, note that any diagonal of a convex polygon divides this polygon into two convex polygons. For each $n \in \mathbb{N}$, let y_n be the number of partitions of a convex $(n+2)$ -gon. Then, $y_1 = 1, y_2 = 2$. Let $A_1A_2 \dots A_nA_{n+1}A_{n+2}$ be a convex $(n+2)$ -gon. The number of partitions of this $(n+2)$ -gon, such that $A_1A_2A_3$ is one of the obtained triangles, is equal to y_{n-1} , i.e., the number of partitions of the convex $(n+1)$ -gon $A_1A_3A_4 \dots A_{n+2}$. The number of partitions of $A_1A_2 \dots A_nA_{n+1}A_{n+2}$, such that $A_1A_2A_{n+2}$ is one of the obtained triangles, is also equal to y_{n-1} .

Let $k \in \{3, 4, \dots, n\}$. The number of partitions of the $(n+2)$ -gon $A_1A_2 \dots A_{n+2}$, such that $A_1A_2A_{k+1}$ is one of the obtained triangles, is equal to $y_{k-2}y_{n-k+1}$. Here, y_{k-2} is the number of partitions of the k -gon $A_2A_3 \dots A_{k+1}$, and y_{n-k+1} is the number of partitions of the $(n-k+3)$ -gon $A_1A_{k+1}A_{k+2} \dots A_{n+2}$. By the sum rule it follows that

$$y_n = y_{n-1} + y_1y_{n-2} + y_2y_{n-3} + \dots + y_{n-2}y_1 + y_{n-1}.$$

Therefore, the sequence (y_n) satisfies the same recursive relation as the sequence (x_n) from Exercise 2.65. Since $x_1 = y_1 = 1$, it follows that $y_n = x_n = \frac{1}{n+1} \binom{2n}{n}$ for every $n \in \mathbb{N}$.

2.67. Let x be the number of chess masters and y be the number of grand-masters participating in the tournament. Then

$$\frac{x(x-1)}{2} + \frac{y(y-1)}{2} = xy,$$

and therefore $x + y = (x - y)^2$.

2.68. Let us say a committee is *good* if no two of its members are quarreling or every two of them are quarreling. Otherwise, a committee is *bad*. Let x be the number of good committees, and y be the number of bad committees. Then, $x + y = \binom{30}{3} = 4060$. Suppose that every senator makes a list of possible committees containing them as a member. Every such list contains $\binom{23}{2} + \binom{6}{2} = 268$ committees. Note that every good committee is listed on exactly three senator's lists, and every bad committee is listed on exactly one of them. Hence, $3x + y = 30 \cdot 268 = 8040$. It follows $x = 1990$.

2.69. (a) $\binom{2n}{n}$, (b) $n!$, (c) *Answer:* $2 \cdot (2n - 1)!!$. *Hint.* Let x_n be the number of ways in which a broken line can be drawn and n chips be arranged such that the conditions of items (a) and (b) are satisfied, and additionally all the chips are placed between the broken line and the x -axis. Note that $x_1 = 1$, and prove that $x_n = (2n - 1)x_{n-1}$ for $n > 1$.

2.70. *Answer:* 2^{n-2} . *Hint.* Prove that x_1 is always placed in the nominator of the fraction, x_2 is always placed in the denominator, and every of the numbers $x_3, \dots, x_n$ can be placed in the nominator as well as in the denominator.

14.3 Solutions for Chapter 3

$$\mathbf{3.1.} \quad \binom{20}{14} 3^{14} 2^{-6} \quad \mathbf{3.2.} \quad \binom{15}{12} (\sqrt{2})^3 (\sqrt[3]{3})^{12} = 73\,710\sqrt{2}. \quad \mathbf{3.3.} \quad 17.$$

3.4. The sum of all coefficients is the value of the polynomial for $x = 1$ and is equal to 1.

3.5. All terms that contain $\sqrt{2}$ vanish.

3.6. $n = 7$, $k = 2$. **3.7.** $n = 14$, $k = 6$.

3.8. By the binomial theorem it follows that

$$(1 + x)^n = 1 + nx + \sum_{k=2}^n \binom{n}{k} x^k \geq 1 + nx, \quad \text{if } x \geq 0.$$

3.9. *Hint.* Use the Bernoulli inequality, the binomial theorem and the fact that for $k \in \{2, 3, \dots, n\}$ the inequalities $\binom{n}{k} \frac{1}{n^k} < \frac{1}{k!} < \frac{1}{2^{k-1}}$ hold.

3.10. From the binomial theorem it follows that

$$(2 + \sqrt{3})^n + (2 - \sqrt{3})^n = 2 \left[2^n + \binom{n}{2} 2^{n-2} 3 + \binom{n}{4} 2^{n-4} 3^2 + \dots \right].$$

Therefore, $(2 + \sqrt{3})^n + (2 - \sqrt{3})^n$ is an even positive integer. Since $0 < (2 - \sqrt{3})^n < 1$, it follows that $[(2 + \sqrt{3})^n] = (2 + \sqrt{3})^n + (2 - \sqrt{3})^n - 1$, i.e., $[(2 + \sqrt{3})^n]$ is an odd positive integer.

3.11. For every positive integer n it follows by the binomial theorem that

$$(n + \sqrt{n^2 + 1})^n - (\sqrt{n^2 + 1} - n)^n = x_n \in \mathbb{N}.$$

For $n \geq 5$ we obtain

$$\begin{aligned} y_n &= (\sqrt{n^2 + 1} - n)^n = \frac{1}{(\sqrt{n^2 + 1} + n)^n} \\ &\leq \frac{1}{(\sqrt{26} + 5)^n} < 10^{-n} = 0.\underbrace{00 \dots 01}_{n-1}. \end{aligned}$$

Since $(n + \sqrt{n^2 + 1})^n = x_n + y_n$, the statement follows.

3.12. $1 + 14\binom{n}{1} + 36\binom{n}{2} + 24\binom{n}{3} = 4n^3 + 6n^2 + 4n + 1 = (n + 1)^4 - n^4.$

3.13. Note that

$$\begin{aligned} &\binom{m}{0} + \binom{m}{1} + \binom{m+1}{2} + \binom{m+2}{3} + \dots + \binom{m+n-1}{n} \\ &= \binom{m+1}{1} + \binom{m+1}{2} + \binom{m+2}{3} + \dots + \binom{m+n-1}{n} \\ &= \binom{m+2}{2} + \binom{m+2}{3} + \dots + \binom{m+n-1}{n} = \dots = \binom{m+n}{n} \end{aligned}$$

Therefore, $\sum_{k=1}^n \binom{m+k-1}{k} = \binom{m+n}{n} - 1$. Similarly we obtain that

$$\sum_{k=1}^m \binom{n+k-1}{k} = \binom{m+n}{n} - 1.$$

3.14. The identity follows from the equalities:

$$\begin{aligned}
 \sum_{k=1}^n \binom{n-1}{k-1} \binom{2n-1}{k}^{-1} &= \frac{(n-1)!}{(2n-1)!} \sum_{k=1}^n \frac{k(2n-k-1)!}{(n-k)!} \\
 &= \frac{2n}{2n-1} \binom{2n-2}{n-1}^{-1} \sum_{k=1}^n \binom{2n-k-1}{n-1} - \binom{2n-1}{n-1}^{-1} \sum_{k=1}^n \binom{2n-k}{n} \\
 &= \frac{2n}{2n-1} \binom{2n-2}{n-1}^{-1} \binom{2n-1}{n-1} - \binom{2n-1}{n-1}^{-1} \binom{2n}{n-1} = \frac{2}{n+1}.
 \end{aligned}$$

3.15. The identity follows from the equalities:

$$\begin{aligned}
 \sum_{k=1}^n \binom{n-1}{k-1} \binom{n+m}{k}^{-1} &= \frac{(n+m+1)(n-1)!m!}{(n+m)!} \sum_{k=1}^n \binom{n+m-k}{n-k} \\
 &\quad - \frac{(n-1)!(m+1)!}{(n+m)!} \sum_{k=1}^n \binom{n+m-k+1}{n-k} \\
 &= \frac{(n+m+1)(n-1)!m!}{(n+m)!} \binom{n+m}{n-1} - \frac{(n-1)!(m+1)!}{(n+m)!} \binom{n+m+1}{n-1} \\
 &= \frac{n+m+1}{(m+1)(m+2)}.
 \end{aligned}$$

$$\mathbf{3.16.} \quad \sum_{k=1}^n k \binom{n}{k} = n \sum_{k=1}^n \binom{n-1}{k-1} = n \sum_{j=1}^{n-1} \binom{n-1}{j} = n2^{n-1}.$$

$$\mathbf{3.17.} \quad \sum_{k=0}^n (k+1) \binom{n}{k} = \sum_{k=1}^n k \binom{n}{k} + \sum_{k=0}^n \binom{n}{k} = n2^{n-1} + 2^n = (n+2)2^{n-1}.$$

$$\mathbf{3.18.} \quad \sum_{k=0}^n (-1)^k (k+1) \binom{n}{k} = \sum_{k=1}^n (-1)^k k \binom{n}{k} = n \sum_{k=1}^{n-1} (-1)^k \binom{n-1}{k-1} = 0.$$

$$\mathbf{3.19.} \quad \sum_{k=0}^n \frac{1}{k+1} \binom{n}{k} = \frac{1}{n+1} \sum_{k=0}^n \binom{n+1}{k+1} = \frac{1}{n+1} \sum_{k=1}^{n+1} \binom{n+1}{k} = \frac{2^{n+1} - 1}{n+1}.$$

3.20. From the properties of binomial coefficients and Exercise 3.16 it follows:

$$\begin{aligned}
\sum_{k=0}^n \frac{1}{k+2} \binom{n}{k} &= \frac{1}{(n+1)(n+2)} \sum_{k=0}^n (k+1) \binom{n+2}{k+2} \\
&= \frac{1}{(n+1)(n+2)} \left[\sum_{k=-1}^n (k+2) \binom{n+2}{k+2} - \sum_{k=-1}^n \binom{n+2}{k+2} \right] \\
&= \frac{1}{(n+1)(n+2)} \left[\sum_{k=1}^{n+2} k \binom{n+2}{k} - \sum_{k=1}^{n+2} \binom{n+2}{k} \right] \\
&= \frac{(n+2)2^{n+1} - 2^{n+2} + 1}{(n+1)(n+2)} = \frac{n2^{n+1} + 1}{(n+1)(n+2)}.
\end{aligned}$$

3.21. The identity follows from the equalities:

$$\begin{aligned}
\sum_{k=0}^n (-1)^k \frac{1}{k+1} \binom{n}{k} &= \frac{1}{(n+1)} \sum_{k=0}^n (-1)^k \binom{n+1}{k+1} \\
&= \frac{1}{n+1} \left[1 - \sum_{k=0}^{n+1} (-1)^k \binom{n+1}{k} \right] = \frac{1}{n+1}.
\end{aligned}$$

3.22. *Hint.* Consider the coefficients of x^{n-1} on both sides of the equality

$$n(1+x)^{2n-1} = \sum_{k=0}^n \binom{n}{k} x^k \sum_{j=1}^n j \binom{n}{j} x^{n-j}.$$

$$\mathbf{3.23.} \quad \sum_{k=0}^n \frac{(2n)!}{(k!)^2((n-k)!)^2} = \frac{(2n)!}{n!n!} \sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}^2.$$

3.24. The identity follows from the equalities:

$$\begin{aligned}
\sum_{k=0}^n \binom{n}{k} \binom{n}{m} \binom{2n}{n+m}^{-1} &= \frac{n!}{(2n)!} \binom{n}{m} \sum_{k=0}^n \frac{(k+m)!(2n-k-m)!}{k!(n-k)!} \\
&= \frac{(n!)^2}{(2n)!} \sum_{k=0}^n \binom{k+m}{k} \binom{2n-k-m}{n-m} = \frac{(n!)^2}{(2n)!} \binom{2n+1}{n+1} = \frac{2n+1}{n+1}.
\end{aligned}$$

3.25. Let us denote $x_n = \sum_{k=0}^n \frac{1}{2^k} \binom{n+k}{k}$, for $k \in \mathbb{N}$. Then, $x_1 = 2$ and

$$x_{n+1} = \sum_{k=0}^{n+1} \frac{1}{2^{k+1}} \binom{n+1+k}{k} = \sum_{k=0}^{n+1} \frac{1}{2^{k+1}} \binom{n+k}{k} + \sum_{k=1}^{n+1} \frac{1}{2^k} \binom{n+k}{k-1}$$

$$\begin{aligned}
&= x_n + \frac{1}{2^{n+1}} \binom{2n+1}{n+1} + \frac{1}{2} \sum_{k=1}^{n+2} \frac{1}{2^{k-1}} \binom{n+k}{k-1} - \frac{1}{2^{n+2}} \binom{2n+2}{n+1} \\
&= x_n + \frac{1}{2} x_{n+1},
\end{aligned}$$

i.e., $x_{n+1} = 2x_n$ for $n \geq 1$. It follows that $x_n = 2^n$.

3.26. *Hint.* Let $S_n = \sum_{k=0}^n (-1)^k \binom{2n-k}{k}$. Prove that $S_1 = 0$, $S_2 = -1$, $S_3 = 1$, and $S_{n+1} = -(S_n + S_{n-1})$ for $n \geq 2$.

3.27. The identity follows from the equalities:

$$\begin{aligned}
\sum_{k=0}^{n-1} \binom{4n}{4k+1} &= \frac{1}{2} \sum_{k=0}^{n-1} \binom{4n}{4k+1} + \frac{1}{2} \sum_{k=0}^{n-1} \binom{4n}{4(n-1-k)+1} \\
&= \frac{1}{2} \sum_{k=0}^{2n-1} \binom{4n}{2k+1} = \frac{1}{2} \sum_{k=0}^{2n-1} \binom{4n-1}{2k+1} + \frac{1}{2} \sum_{k=0}^{2n-1} \binom{4n-1}{2k} \\
&= \frac{1}{2} \sum_{k=0}^{4n-1} \binom{4n-1}{k} = \frac{1}{2} (1+1)^{4n-1} = 2^{4n-2}.
\end{aligned}$$

3.28. Let us denote

$$A(n, k) = (-1)^k \binom{n}{k}^{-1} = (-1)^k \frac{(n-k)! k!}{n!}, \quad 0 \leq k \leq n, \quad n \in \mathbb{N}.$$

Then, $A(n+1, k+1) - A(n+1, k) = -\frac{n+2}{n+1} A(n, k)$ for $0 \leq k \leq n$, and consequently we obtain that

$$\sum_{k=1}^n A(n, k) = -\frac{n+1}{n+2} (A(n+1, n+1) - A(n+1, 0)) = (1 + (-1)^n) \frac{n+1}{n+2}.$$

3.29. Let us denote

$$S(n, x) = \sum_{k=0}^n (-1)^k \frac{1}{x+k} \binom{n}{k}, \quad \text{for } x \notin \{0, -1, -2, \dots, -n\}, \quad n \in \mathbb{N}.$$

By the method of mathematical induction we shall prove that, for any $n \in \mathbb{N}$,

$$S(n, x) = \frac{n!}{x(x+1) \cdots (x+n)}.$$

For $n = 1$ and $x \notin \{0, 1\}$ we get $S(1, x) = \frac{1}{x(x+1)}$. Let us assume that for some $n \in \mathbb{N}$ and every $x \notin \{0, -1, \dots, -n+1\}$ the following equality holds:

$$S(n-1, x) = \frac{(n-1)!}{x(x+1) \cdots (x+n-1)}.$$

Then for $x \in \{0, -1, \dots, -n\}$ we obtain that

$$\begin{aligned} S(n, x) &= \frac{1}{x} + \sum_{k=1}^{n-1} (-1)^k \frac{1}{x+k} \left[\binom{n-1}{k} + \binom{n-1}{k-1} \right] + (-1)^n \frac{1}{x+n} \\ &= \sum_{k=0}^{n-1} (-1)^k \frac{1}{x+k} \binom{n-1}{k} + \sum_{k=1}^n \frac{1}{x+k} \binom{n-1}{k-1} \\ &= S(n-1, x) - S(n-1, x+1) \\ &= \frac{(n-1)!}{(x+1)(x+2) \cdots (x+n-1)} \left(\frac{1}{x} - \frac{1}{x+n} \right) \\ &= \frac{n!}{x(x+1) \cdots (x+n)}. \end{aligned}$$

3.30. The identity follows by adding the equalities

$$(j+1)^m = 1 + \binom{m}{1}j + \cdots + \binom{m}{m-1}j^{m-1} + j^m, \quad j = 1, 2, \dots, n.$$

3.31. Let $z = -1/2 + i\sqrt{3}/2 = \cos(2\pi/3) + i\sin(2\pi/3)$. Using the binomial theorem we get

$$z^n = \frac{(-1)^n}{2^n} \left\{ \left[1 - 3\binom{n}{2} + 9\binom{n}{4} - \cdots \right] - i\sqrt{3} \left[\binom{n}{1} - 3\binom{n}{3} + \cdots \right] \right\}.$$

On the other hand, by de Moivre's formula it follows that

$$z^n = \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)^n = \cos \frac{2n\pi}{3} + i \sin \frac{2n\pi}{3}.$$

The identities (a) and (b) follow by equalizing the real and imaginary parts of z^n on the left-hand side and the right-hand side of the last equality.

3.32. (a) Let $\varepsilon = \cos(2\pi/3) + i \sin(2\pi/3)$. Then $1 + \varepsilon + \varepsilon^2 = 0$,

$$\begin{aligned} 1 + \varepsilon &= -\varepsilon^2 = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}, \\ 1 + \varepsilon^2 &= -\varepsilon = \cos \frac{\pi}{3} - i \sin \frac{\pi}{3}, \end{aligned}$$

and by de Moivre's formula it follows that

$$2^n + (1 + \varepsilon)^n + (1 + \varepsilon^2)^n = 2^n + 2 \cos \frac{n\pi}{3}. \quad (1)$$

Note that

$$1 + \varepsilon^k + \varepsilon^{2k} = \begin{cases} 0, & \text{if } k \text{ is not divisible by } 3, \\ 3, & \text{if } k \text{ is divisible by } 3, \end{cases}$$

and therefore

$$\begin{aligned} 2^n + (1 + \varepsilon)^n + (1 + \varepsilon^2)^n &= \sum_{k=0}^n \binom{n}{k} (1 + \varepsilon^k + \varepsilon^{2k}) \\ &= 3 \left[\binom{n}{0} + \binom{n}{3} + \binom{n}{6} + \cdots \right]. \end{aligned} \quad (2)$$

Using (1) and (2) we obtain that $\binom{n}{0} + \binom{n}{3} + \binom{n}{6} + \cdots = \frac{1}{3} \left(2^n + 2 \cos \frac{n\pi}{3} \right)$.

(b)–(c): *Hint.* Consider the sums $2^n + \varepsilon(1 + \varepsilon)^n + \varepsilon^2(1 + \varepsilon^2)^n$ and $2^n + \varepsilon^2(1 + \varepsilon)^n + \varepsilon(1 + \varepsilon^2)^n$.

3.33. Let $\varepsilon_k = \cos \frac{2k\pi}{m} + i \sin \frac{2k\pi}{m}$, $k = 0, 1, 2, \dots, m-1$. By the binomial theorem it follows that

$$S_n = \sum_{k=1}^m (1 + \varepsilon_k)^n = \sum_{j=0}^n \binom{n}{j} (\varepsilon_1^j + \varepsilon_2^j + \cdots + \varepsilon_m^j).$$

If j is divisible by m , then $\varepsilon_1^j + \varepsilon_2^j + \cdots + \varepsilon_m^j = \varepsilon_1^j + \varepsilon_1^{2j} + \cdots + \varepsilon_1^{mj} = m$. If j is not divisible by m , then $\varepsilon_1^j \neq 1$, and therefore

$$\varepsilon_1^j + \varepsilon_2^j + \cdots + \varepsilon_m^j = \varepsilon_1^j + \varepsilon_1^{2j} + \cdots + \varepsilon_1^{mj} = \varepsilon_1^j \frac{1 - \varepsilon_1^{mj}}{1 - \varepsilon_1^j} = 0.$$

Consequently we get

$$S_n = m \left[\binom{n}{0} + \binom{n}{m} + \binom{n}{2m} + \cdots \right]. \quad (1)$$

Note that $1 + \varepsilon_k = 1 + \cos \frac{2k\pi}{m} + i \sin \frac{2k\pi}{m} = 2 \cos \frac{k\pi}{m} \left(\cos \frac{k\pi}{m} + i \sin \frac{k\pi}{m} \right)$, and by de Moivre's formula it follows that

$$S_n = 2^n \sum_{k=1}^m \cos^n \frac{k\pi}{m} \left(\cos \frac{nk\pi}{m} + i \sin \frac{nk\pi}{m} \right). \quad (2)$$

The identity that should be proved follows from (1) and (2).

3.34. The obtained positive integer is $n = (10^{k+1} + 1)^4$, and therefore $\sqrt{a} = (10^{k+1} + 1)^2$.

3.35. The path passed by every passenger after n days is a broken line consisting of n unit segments. Hence, every passenger stopped at one of the following n points: $B_k = (k, n - k)$, where $k \in \{0, 1, \dots, n\}$. All these points belong to the line $y = -x + n$. Let S be the set of all broken lines of length n with starting point $(0, 0)$ and endpoint in one of the points $B_0, B_1, \dots, B_n$, and whose vertices all have integer coordinates. There is a bijection between S and the set of all n -arrangements of the elements 0 and 1, and hence $|S| = 2^n$. Since there are 2^n passengers, exactly one of them passed along every one of the broken lines. Note also that there is a bijection between the set $S_k \subset S$ of broken lines with the endpoint $B_k = (k, n - k)$ and the set of all n -arrangements of elements 0 and 1 with exactly k zeroes. It follows that $|S_k| = \binom{n}{k}$. Therefore, after n days there are $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$ passengers at points $B_0, B_1, \dots, B_n$, respectively.

3.36. At the end of the walk there are $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n-1}, \binom{n}{n}$ persons at points $-\frac{n}{2}, -\frac{n}{2} + 1, \dots, -\frac{1}{2}, \frac{1}{2}$, respectively.

3.37. The terms of the given sequence and the sequences of differences can be represented in the form

$$\begin{array}{ccccccc} 1 & \frac{1}{2\binom{1}{1}} & \frac{1}{3\binom{2}{2}} & \frac{1}{4\binom{3}{3}} & \cdots \\ & \frac{1}{2\binom{1}{0}} & \frac{1}{3\binom{2}{1}} & \frac{1}{4\binom{3}{2}} & \cdots \\ & & \frac{1}{3\binom{2}{0}} & \frac{1}{4\binom{3}{1}} & \cdots \\ & & & \frac{1}{4\binom{3}{0}} & \cdots \end{array}$$

This is true because the equality

$$\frac{1}{n\binom{n-1}{k}} - \frac{1}{(n+1)\binom{n}{k+1}} = \frac{1}{(n+1)\binom{n}{k}},$$

holds for $0 \leq k \leq n$. The statement of the exercise now follows easily.

3.38. Let us replace x by $1/y$ in the equality

$$(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \cdots + a_{2n}x^{2n}, \quad (1)$$

and multiply both sides of (1) by y^{2n} . Equality (1) becomes

$$(y^2 + y + 1)^n = a_0y^{2n} + a_1y^{2n-1} + \cdots + a_{2n} = a_0 + a_1y + \cdots + a_{2n}y^{2n}. \quad (2)$$

(a) If we replace x by $-x$ in (1), we get

$$(1 - x + x^2)^n = a_0 - a_1x + a_2x^2 - \cdots + a_{2n}x^{2n}. \quad (3)$$

By multiplying the polynomials on the left-hand sides and the right-hand sides of (1) and (2) it follows that

$$(1 + x^2 + x^4)^n = \sum_{k=0}^{4n} (-1)^k (a_0a_k - a_1a_{k-1} + \cdots + (-1)^k a_k a_0) x^k \quad (4)$$

where $a_k = 0$, for $k \in \{2n+1, 2n+2, \dots, 4n\}$. By equalizing the coefficients of x^{2n-1} on the left-hand side and the right-hand side of (4), it follows that

$$0 = a_0a_{2n-1} - a_1a_{2n-2} + \cdots - a_{2n-1}a_0 = a_0a_1 - a_1a_2 + \cdots - a_{2n-1}a_{2n}.$$

(b) *Hint.* Consider the coefficients of x^{2n} in the equality (4).

(c) *Hint.* Consider the equalities (1) and (3) for $x = 1$.

(d) *Hint.* Multiply (1) by $(1-x)^n$ and equalize the coefficients of x in the obtained polynomials.

3.39. The positive integer 10 can be represented as the sum of addends equal to 2 or 3 in two ways: $10 = 2 + 2 + 2 + 2 + 2 = 2 + 2 + 3 + 3$. It follows that the coefficient of x^{10} in the expansion of the polynomial $(1 - x^2 + x^3)^{11}$ is $-\binom{11}{5} + \binom{11}{2}\binom{9}{2} = 1518$.

3.40. By the binomial theorem it follows that $(\sqrt{5} + 2)^p - (\sqrt{5} - 2)^p$ is a positive integer. Since $0 < (\sqrt{5} - 2)^p < 1$, it follows that

$$\left[(\sqrt{5} + 2)^p \right] = (\sqrt{5} + 2)^p - (\sqrt{5} - 2)^p = 2 \left[2^p + \binom{p}{2} c_2 + \cdots + \binom{p}{p-1} c_{p-1} \right],$$

where $c_2, c_4, \dots, c_{p-1} \in \mathbb{N}$. Hence

$$\left[(\sqrt{5} + 2)^p\right] - 2^{p+1} = 2c_2 \binom{p}{2} + 2c_4 \binom{p}{4} + \dots + 2c_{p-1} \binom{p}{p-1}.$$

Now, it is sufficient to prove that $\binom{p}{k}$ is divisible by p , if p is a prime positive integer and $k \in \{1, 2, \dots, p-1\}$. This statement follows from the facts that $\frac{p(p-1) \cdots (p-k+1)}{k!}$ is a positive integer, and $k!$ is relatively prime to p for every $k \in \{1, 2, \dots, p-1\}$.

3.41. Let d be the greatest common divisor of $\binom{n}{k}, \binom{n+1}{k}, \dots, \binom{n+k}{k}$. Then, d is a divisor of the difference of any two neighboring terms of this sequence, i.e., d is a divisor of any of the positive integers $\binom{n-1}{k-1}, \binom{n}{k-1}, \dots, \binom{n+k-1}{k-1}$. By repeating this operation k times it follows that d is a divisor of $\binom{n-k}{0} = 1$. Hence $d = 1$.

3.42. *Answer.* The greatest common divisor is 2^{k+1} , where k is the exponent of 2 in the canonical representation $n = 2^k 3^l \dots$.

3.43. *Answer.* The number of odd positive integers among the binomial coefficients of order n is equal to 2^{k+1} , where k is the number of 1's in the binary representation of n .

3.44. *Hint.* Use the result of Exercise 3.43.

14.4 Solutions for Chapter 4

4.1. The inequalities are an immediate consequence of the inclusion-exclusion principle.

4.2. (a) Let A , B , and C be sets of positive integers that are not greater than 10^6 , and are divisible by 2, 3, and 5, respectively. The positive integer we are to determine is

$$\begin{aligned} & 10^6 - |A \cup B \cup C| \\ &= 10^6 - |A| - |B| - |C| + |A \cap B| + |A \cap C| + |B \cap C| - |A \cap B \cap C| \\ &= 10^6 - 500\,000 - 333\,333 - 200\,000 + 166\,666 + 100\,000 + 66\,666 \\ &\quad - 33\,333 = 266\,666. \end{aligned}$$

(b) Similarly as in the previous case we get the answer 228571.

4.3. (a) Let $c_1 \neq 0$, $c_2 \neq 0$, $c_3 \neq 0$ be three distinct digits. The number of 6-digit positive integers whose decimal representation can be written by

the digits c_1 , c_2 , and c_3 is 3^6 . The number of 6-digit positive integers whose decimal representation can be written by no more than two of the digits c_1 , c_2 , and c_3 is equal to $\binom{3}{2}2^6$, while the number of 6-digit positive integers that can be written by one of these digits is 3. The number of 6-digit positive integers with a decimal representation containing exactly three digits from the set $\{1, 2, \dots, 9\}$ is

$$\binom{9}{3} \left\{ 3^6 - \binom{3}{2}2^6 + \binom{3}{1}1^6 \right\} = 84 \cdot 540 = 45\,360.$$

The number of 6-digit positive integers with a decimal representation containing 0 and exactly two more digits from the set $\{1, 2, \dots, 9\}$ is

$$\binom{9}{2} 2 \left(3^5 - \binom{2}{1}2^5 + 1^5 \right) = 36 \cdot 2 \cdot 180 = 12\,960.$$

Hence, the number of 6-digit positive integers with a decimal representation containing exactly three distinct digits is $45\,360 + 12\,960 = 58\,320$.

(b) The number of n -digit positive integers with exactly k distinct digits in their decimal representation is

$$\binom{9}{k} \sum_{j=0}^{k-1} (-1)^j \binom{k}{j} (k-j)^n + \binom{9}{k-1} (k-1) \sum_{j=0}^{k-1} (-1)^j \binom{k-1}{j} (k-j)^{n-1}.$$

4.4. (a) Let A , B , and C be the jury members' countries. Nine members of the jury can be arranged in a row in $9!$ ways. The number of permutations in which three members from country A occupy three adjacent positions is $3!7!$. The number of permutations in which members from country A occupy three adjacent positions, and members from country B also occupy three adjacent positions is $(3!)^25!$.

The number of permutations in which three members from each country occupy three adjacent positions is $(3!)^4$. It follows by the inclusion-exclusion principle that the number of permutations that satisfy the required condition is $9! - 3 \cdot 3!7! + 3(3!)^25! - (3!)^4 = 283\,824$.

(b) We shall determine the number of permutations in which one can find gathered together:

- (1) 2 members of the jury from the same country;
- (2) 2 members from the same country, and 2 members from another country (not necessarily all 4 members together);
- (3) 2 members from each country;
- (4) 3 members from the same country;

(5) 3 members from the same country, and again three members from another country;

(6) 3 members from each country;

(7) 3 members from the same country, and 2 from another one;

(8) 3 members from the same country, and 2 members from another two countries;

(9) 3 members from two countries, and 2 members from a third one.

These numbers are the following, respectively:

$$\begin{aligned} & 3 \binom{3}{2} 2! 8!, \quad \binom{3}{2}^3 (2!)^2 7!, \quad \binom{3}{2}^3 (2!)^3 6!, \quad 3 \cdot 3! 7!, \quad \binom{3}{2} (3!)^2 5!, \\ & (3!)^4, \quad 3 \cdot 2 \binom{3}{2} 3! 2! 6!, \quad 3 \binom{3}{2}^2 3! (2!)^2 5!, \quad 3 \binom{3}{2} (3!)^2 2! 4!. \end{aligned}$$

By the inclusion-exclusion principle it follows that the number of permutation with the required property is

$$\begin{aligned} & 9! - 9 \cdot 2! 8! + 27(2!)^2 7! - 27(2!)^3 6! + 3 \cdot 3! 7! + 3(3!)^2 5! + (3!)^4 \\ & - 18 \cdot 3! 2! 6! + 27 \cdot 3! (2!)^2 5! - 9(3!)^2 2! 4! = 37584. \end{aligned}$$

4.5. Let $S = \{1, 2, \dots, n\}$. For any $j \in \{1, 2, \dots, m\}$, let A_j be the set of positive integers from S that are divisible by p_j . By the inclusion-exclusion principle it follows that

$$\begin{aligned} \varphi(n) &= |S \setminus (A_1 \cup A_2 \cup \dots \cup A_m)| \\ &= n - \sum_{j=1}^m \frac{n}{p_j} + \sum_{1 \leq i < j \leq m} \frac{n}{p_i p_j} - \sum_{1 \leq i < j < k \leq m} \frac{n}{p_i p_j p_k} + \dots \\ &= n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_m}\right). \end{aligned}$$

4.6. Using the formula for $\varphi(n)$ we get:

$$\begin{aligned} \varphi(45) &= \varphi(3^2 \cdot 5) = 45 \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right) = 24, \\ \varphi(900) &= \varphi(2^2 3^2 5^2) = 900 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right) = 240, \\ \varphi(116704) &= \varphi(2^5 \cdot 7 \cdot 521) = 116704 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{7}\right) \left(1 - \frac{1}{521}\right) = 49920. \end{aligned}$$

4.7. Let $p_1, p_2, \dots, p_k, q_1, q_2, \dots, q_s$ be all prime divisors of the positive integer m , and $p_1, p_2, \dots, p_k, r_1, r_2, \dots, r_t$ be all prime divisors of the positive integer n , where $q_i \neq r_j$ for every $i \in \{1, 2, \dots, s\}$ and $j \in \{1, 2, \dots, t\}$. Then, the primes $p_1, p_2, \dots, p_k, q_1, q_2, \dots, q_s, r_1, r_2, \dots, r_t$ are all prime divisors of the positive integer mn , and hence it follows that

$$\begin{aligned}\varphi(m)\varphi(n) &= mn \prod_{j=1}^k \left(1 - \frac{1}{p_j}\right)^2 \prod_{j=1}^s \left(1 - \frac{1}{q_j}\right) \prod_{j=1}^t \left(1 - \frac{1}{r_j}\right) \\ &\leq mn \prod_{j=1}^k \left(1 - \frac{1}{p_j}\right) \prod_{j=1}^s \left(1 - \frac{1}{q_j}\right) \prod_{j=1}^t \left(1 - \frac{1}{r_j}\right) = \varphi(mn).\end{aligned}$$

4.8. Let $S = \{2, 3, \dots, n\}$, and let $\{p_1, p_2, \dots, p_m\}$ be the set of all primes not greater than $\sqrt{n}$. For any $j \in \{1, 2, \dots, m\}$, let A_j be the set of positive integers from S that are divisible by p_j . Then, for $1 \leq j_1 < j_2 < \dots < j_k \leq m$, we have

$$|A_{j_1} \cap A_{j_2} \cap \dots \cap A_{j_k}| = \left\lfloor \frac{n}{p_{j_1} p_{j_2} \dots p_{j_k}} \right\rfloor.$$

By the inclusion-exclusion principle we obtain that the number of primes that are greater than $\sqrt{n}$ but not greater than n is given by

$$\begin{aligned}\pi(n) - \pi([\sqrt{n}]) &= |S \setminus (A_1 \cup A_2 \cup \dots \cup A_m)| \\ &= n - 1 + \sum (-1)^k \left\lfloor \frac{n}{p_{j_1} p_{j_2} \dots p_{j_k}} \right\rfloor,\end{aligned}$$

where the sum runs over all $j_1, j_2, \dots, j_k$ such that $\emptyset \neq \{j_1, j_2, \dots, j_k\} \subset \{1, 2, \dots, m\}$.

4.9. Since $[\sqrt{120}] = 10$, and 2, 3, 5, and 7 are all primes not greater than 10, it follows that

$$\begin{aligned}\pi(120) &= 119 + \pi(10) - \left\lfloor \frac{120}{2} \right\rfloor - \left\lfloor \frac{120}{3} \right\rfloor - \left\lfloor \frac{120}{5} \right\rfloor - \left\lfloor \frac{120}{7} \right\rfloor \\ &\quad + \left\lfloor \frac{120}{2 \cdot 3} \right\rfloor + \left\lfloor \frac{120}{2 \cdot 5} \right\rfloor + \left\lfloor \frac{120}{2 \cdot 7} \right\rfloor + \left\lfloor \frac{120}{3 \cdot 5} \right\rfloor + \left\lfloor \frac{120}{3 \cdot 7} \right\rfloor + \left\lfloor \frac{120}{5 \cdot 7} \right\rfloor \\ &\quad - \left\lfloor \frac{120}{2 \cdot 3 \cdot 5} \right\rfloor - \left\lfloor \frac{120}{2 \cdot 3 \cdot 7} \right\rfloor - \left\lfloor \frac{120}{2 \cdot 5 \cdot 7} \right\rfloor - \left\lfloor \frac{120}{3 \cdot 5 \cdot 7} \right\rfloor \\ &\quad + \left\lfloor \frac{120}{2 \cdot 3 \cdot 5 \cdot 7} \right\rfloor = 30.\end{aligned}$$

4.10. Let $n > 1$. Note that $\mu(d) \in \{-1, 0, 1\}$ for any positive divisor d of the positive integer n . The number of positive divisors d such that $\mu(d) = (-1)^r$, where $r \in \{0, 1, 2, \dots, m\}$, is equal to $\binom{m}{r}$. The sum of all $\mu(d)$'s is

$$1 - \binom{m}{1} + \binom{m}{2} - \dots + (-1)^m \binom{m}{m} = (1 - 1)^m = 0.$$

4.11. This is another form of the formula for $\varphi(n)$ from Exercise 4.5.

4.12. This is another form of the formula from Exercise 4.8.

4.13. We shall prove that for any positive integer n , the following equality holds:

$$f(n) = \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor + 1. \quad (1)$$

Let A be the set of even positive integers that are not greater than n , and B be the set of positive integers that are not greater than n and are divisible by 3. The set $A \cup B$ consists of $\left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor$ elements. Any 3-subset of the set $A \cup B$ contains two even positive integers or two positive integers that are divisible by 3. Hence, it follows that

$$f(n) \geq \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor + 1. \quad (2)$$

In order to prove equality (1) it is now sufficient to prove that

$$f(n) \leq \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor + 1. \quad (3)$$

First, we shall prove the next lemma. *Let k be a positive integer, and $C = \{k, k+1, k+2, k+3, k+4, k+5\}$. For any 5-subset A of set C there are 3 positive integers from A that are relatively prime.*

Indeed, there exists an odd positive integer x , such that $x, x+2, x+4 \in C$, and $x, x+2, x+4$ are relatively prime. Let $y \in \{x+1, x+3\}$ be a positive integer that is not divisible by 3. Then, $x, x+2, x+4$, and y are relatively prime. Note that if we choose 5 elements from set C , then among them there are at least three of the positive integers $x, x+2, x+4$, and y , i.e., there are 3 positive integers that are relatively prime.

We shall prove the inequality (3) by “induction from n to $n+6$.”

First, we check inequality (3) for $n \in \{3, 4, 5, 6, 7, 8\}$. Since 1, 2, and 3 are relatively prime, it follows that $f(3) \leq 3$ and $f(4) \leq 4$, i.e., inequality (3) holds for $n = 3$ and $n = 4$.

Let us prove that $f(5) \leq 4 = [5/2] + [5/3] - [5/6] + 1$. If we choose 1 and three of the positive integers 2, 3, 4, and 5, then we have 1, 2, and 3 or 1, 4, and 5 among the chosen positive integers, i.e., there are three relatively prime positive integers among the chosen ones. If we choose 2, 3, 4, and 5, then 3, 4, and 5 are relatively prime. The inequality $f(6) \leq 5 = [6/2] + [6/3] - [6/6] + 1$ follows from the above lemma.

Let us prove that $f(7) \leq 5 = [7/2] + [7/3] - [7/6] + 1$. Indeed, if we choose five of the positive integers 2, 3, ..., 7, then from the lemma proved above, it follows that three positive integers among the chosen ones are relatively prime. If we choose 1, and four of the positive integers 2, 3, 4, 5, 6, and 7, then there are two consecutive positive integers among the chosen ones, say x and $x + 1$, where $x \neq 1$. It is obvious that 1, x and $x + 1$ are relatively prime.

Now we shall prove that $f(8) \leq 6 = [8/2] + [8/3] - [8/6] + 1$. If we choose five of the positive integers 3, 4, 5, 6, 7, and 8, then from the above lemma it follows that three of them are relatively prime. If we choose 1, 2, and four of the positive integers 3, 4, 5, 6, 7, and 8, then there are two consecutive positive integers among the chosen ones, say x and $x + 1$, such that $x \geq 3$. Then, 1, x , and $x + 1$ are relatively prime.

Induction step. Let us suppose that inequality (3) holds for a positive integer $n \geq 3$. We shall prove that this inequality also holds for the positive integer $n + 6$. Let us denote $g(n) = [n/2] + [n/3] - [n/6] + 1$. It is easy to check that $g(n + 6) = g(n) + 4$. Let A be a subset of the set $\{1, 2, \dots, n + 6\}$, such that $|A| = g(n + 6)$. If

$$|A \cap \{n + 1, n + 2, \dots, n + 6\}| \geq 5,$$

then, from the lemma proved above, it follows that A contains three relatively prime positive integers. In the opposite case, set A contains at least $g(n + 6) - 4 = g(n)$ elements from the set $\{1, 2, \dots, n\}$, and, by the induction hypothesis, it follows that there are three relatively prime positive integers contained in set A . Hence, inequality (3) holds for every positive integer $n \geq 3$.

4.14. Let A_p be the set of positive integers from S that are divisible by p . By the inclusion-exclusion principle it follows that

$$\begin{aligned} |A_2 \cup A_3 \cup A_5 \cup A_7| &= |A_2| + |A_3| + |A_5| + |A_7| - |A_6| - |A_{10}| - |A_{14}| \\ &\quad - |A_{15}| - |A_{21}| - |A_{35}| + |A_{30}| + |A_{42}| + |A_{70}| + |A_{105}| - |A_{210}| \\ &= 140 + 93 + 56 + 40 - 46 - 28 - 20 - 18 - 13 - 8 + 9 + 6 + 4 + 2 - 1 \\ &= 216. \end{aligned}$$

By the pigeonhole principle it follows that any 5-subset of the set $A_2 \cup A_3 \cup A_5 \cup A_7$ contains at least two elements from one of the sets A_2 , A_3 , A_5 , and A_7 , and these two positive integers are not relatively prime. Hence, $n > 216$. Let us prove that $n = 217$.

Note that the set $A_2 \cup A_3 \cup A_5 \cup A_7$ contains the primes 2, 3, 5, 7, and 212 composite positive integers. The set $S \setminus (A_2 \cup A_3 \cup A_5 \cup A_7)$ contains exactly 8 composite positive integers, namely: 11^2 , $11 \cdot 13$, $11 \cdot 17$, $11 \cdot 19$, $11 \cdot 23$, 13^2 , $13 \cdot 17$ and $13 \cdot 19$. Hence, set S contains exactly $212 + 8 = 220$ composite positive integers, the first positive integer 1, and 59 primes.

Let A be a subset of set S , such that $|A| = 217$, and suppose that A does not contain 5 positive integers that are relatively prime. Then, set A contains at most four primes, and at least 213 composite positive integers. Since set S contains exactly 220 composite positive integers, it follows that the set $S \setminus A$ contains at most 7 composite positive integers. Hence, set $S \setminus A$ has a nonempty intersection with at most 7 of the following eight 5-sets:

$$\begin{aligned} &\{2 \cdot 23, 3 \cdot 19, 5 \cdot 17, 7 \cdot 13, 11 \cdot 11\}, & \{2 \cdot 29, 3 \cdot 23, 5 \cdot 19, 7 \cdot 17, 11 \cdot 13\}, \\ &\{2 \cdot 31, 3 \cdot 29, 5 \cdot 23, 7 \cdot 19, 11 \cdot 17\}, & \{2 \cdot 37, 3 \cdot 31, 5 \cdot 29, 7 \cdot 23, 11 \cdot 19\}, \\ &\{2 \cdot 41, 3 \cdot 37, 5 \cdot 31, 7 \cdot 29, 11 \cdot 23\}, & \{2 \cdot 43, 3 \cdot 41, 5 \cdot 37, 7 \cdot 31, 13 \cdot 17\}, \\ &\{2 \cdot 47, 3 \cdot 43, 5 \cdot 41, 7 \cdot 37, 13 \cdot 19\}, & \{2 \cdot 2, 3 \cdot 3, 5 \cdot 5, 7 \cdot 7, 13 \cdot 13\}. \end{aligned}$$

It follows that at least one of these eight sets is a subset of set A . Denote by X this subset of A . Then, X is a 5-set, and its elements are relatively prime. Hence, we have proved that $n = 217$.

4.15. Let S be the set of all permutations $(a_1, \dots, a_n)$ of the set $\{1, \dots, n\}$. For every $j \in \{1, 2, \dots, n\}$, let A_j be the set of permutations of set $\{1, \dots, n\}$, such that $a_j = j$. Then,

$$|A_{j_1} \cap A_{j_2} \cap \dots \cap A_{j_k}| = (n - k)!,$$

for $1 \leq j_1 < j_2 < \dots < j_k \leq n$. By the inclusion-exclusion principle it follows that the number of permutations $a_1 a_2 \dots a_n$ of the set $\{1, 2, \dots, n\}$, such that $a_j \neq j$ for every j , is equal to

$$\begin{aligned} x_n &= |S \setminus (A_1 \cup A_2 \cup \dots \cup A_n)| \\ &= n! - \binom{n}{1}(n-1)! + \binom{n}{2}(n-2)! - \dots + (-1)^n \binom{n}{n} \\ &= n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \dots + (-1)^n \frac{1}{n!} \right). \end{aligned}$$

Second solution. Let S_n be the set of permutations $(a_1, a_2, \dots, a_n)$ of the set $\{1, 2, \dots, n\}$, such that $a_j \neq j$ for every j , and let $x_n = |S_n|$. For any

$k \in \{2, 3, \dots, n\}$, where $n > 2$, let us denote

$$\begin{aligned} A_{n,k} &= \{x \mid x = a_1 a_2 \dots a_n \in S_n, a_k = 1\}, \\ B_{n,k} &= \{x \mid x = a_1 a_2 \dots a_n \in S_n, a_k = 1, a_1 = k\}, \\ C_{n,k} &= \{x \mid x = a_1 a_2 \dots a_n \in S_n, a_k = 1, a_1 \neq k\}. \end{aligned}$$

Then, the sets $B_{n,k}$ and $C_{n,k}$ are disjoint, and $A_{n,k} = B_{n,k} \cup C_{n,k}$. The sets $A_{n,2}, A_{n,3}, \dots, A_{n,n}$, are pairwise disjoint, and $S_n = A_{n,2} \cup A_{n,3} \cup \dots \cup A_{n,n}$. For the sequence $(x_n)_{n \geq 1}$ we obtain $x_1 = 0$, $x_2 = 1$, and

$$\begin{aligned} |B_{n,k}| &= x_{n-2}, \quad |C_{n,k}| = x_{n-1}, \quad |A_{n,k}| = x_{n-1} + x_{n-2}, \\ x_n &= |S_n| = \sum_{k=2}^n |A_{n,k}| = (n-1)(x_{n-1} + x_{n-2}), \quad \text{for } n > 2. \end{aligned}$$

Hence, the sequence $(x_n)_{n \geq 1}$ is determined by the initial terms $x_1 = 0$, $x_2 = 1$, and the recursive relation $x_n = (n-1)(x_{n-1} + x_{n-2})$, for $n > 2$.

4.16. Note that k elements of the set $\{1, 2, \dots, n\}$, that will be placed in the same positions as in permutation $(1, 2, \dots, n)$, can be chosen in $\binom{n}{k}$ ways. The remaining $n - k$ elements can be arranged in the remaining $n - k$ positions, such that none of them occupies the same position as in $(1, 2, \dots, n)$, in x_{n-k} ways (x_{n-k} is determined in Exercise 4.15). Hence, the number of permutations that satisfy the given condition is

$$\binom{n}{k} x_{n-k} = \frac{n!}{k!} \left(1 - \frac{1}{1!} + \frac{1}{2!} - \dots + (-1)^n \frac{1}{(n-k)!} \right).$$

4.17. Let $x_{n,m}$ be the number of permutations of the set $\{1, 2, \dots, n\}$ that satisfy the given condition. Similarly as in the previous two exercises we obtain that

$$x_{n,m} = n! - \binom{m}{1} (n-1)! + \binom{m}{2} (n-2)! - \dots + (-1)^m \binom{m}{m} (n-m)!.$$

4.18. Answer: $\binom{m}{k} x_{n-k, m-k}$, where $x_{n,m}$ is determined in Exercise 4.17.

4.19. Let S be the set of all permutations of the set $\{1, 2, \dots, n\}$. For any $j \in \{1, 2, \dots, n-1\}$, let A_j be the set of permutations from S , such that the element $j+1$ comes immediately after the element j . Then,

$$|A_j| = (n-1)!, \quad |A_i \cap A_j| = (n-2)!, \quad |A_i \cap A_j \cap A_k| = (n-3)!, \dots,$$

and by the inclusion-exclusion principle we obtain that

$$\begin{aligned} |S \setminus (A_1 \cup A_2 \cup \cdots \cup A_{n-1})| &= n! + \sum_{k=1}^{n-1} (-1)^k \binom{n-1}{k} (n-k)! \\ &= (n-1)! \left(n - \frac{n-1}{1!} + \frac{n-2}{2!} - \cdots + (-1)^{n-1} \frac{1}{(n-1)!} \right). \end{aligned}$$

4.20. The first row can be colored in $8!$ ways. Using Exercise 4.15 we obtain that every subsequent row can be colored in

$$8! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + \frac{1}{8!} \right) = 14833$$

ways. The number of colorings, such that the given conditions are satisfied, is $(8!)^8 (14833)^7$.

4.21. Answer: $(n!)^2 \left(1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + (-1)^n \frac{1}{n!} \right)$.

4.22. Let S be the set of all $2n$ -arrangements of the elements $1, 2, \dots, n$, in which each of these elements appears twice. For any $i \in \{1, 2, \dots, n\}$, let A_i be the set of $2n$ -arrangements from S in which the pattern ii appears.

Let us suppose that $1 \leq j_1 < j_2 < \cdots < j_k \leq n$, and let X be the set of $(2n-k)$ -arrangements of the elements $1, 2, \dots, n$, in which each of the elements $j_1, j_2, \dots, j_k$ appears once, and all of the remaining $n-k$ elements appear twice. Obviously there is a bijection between the sets X and $A_{j_1} \cap A_{j_2} \cap \cdots \cap A_{j_k}$. Hence,

$$|A_{j_1} \cap A_{j_2} \cap \cdots \cap A_{j_k}| = \frac{(2n-k)!}{2^{n-k}}.$$

By the inclusion-exclusion principle it follows that

$$|S \setminus (A_1 \cup A_2 \cup \cdots \cup A_n)| = \frac{(2n)!}{2^n} + \sum_{k=1}^n (-1)^k \binom{n}{k} \frac{(2n-k)!}{2^{n-k}}.$$

4.23. Let us suppose that there are $2n$ chairs around the table. Then, n wives can be arranged in n chairs (with a free chair between any two of them) in $(n-1)!$ ways. Let us consider a *fixed arrangement of the wives*, and let them be labeled $1, 2, \dots, n$ in the order as they appear around the table in a chosen direction. Suppose that every husband has the same label as his wife. Suppose that any free chair has two labels, equal to the labels of the wives who occupy the neighboring chairs.

The husbands are to be arranged in free chairs. First, we shall determine the number of arrangements of husbands in free chairs, such that at least k husbands occupy a chair that is adjacent to the chair occupied by his wife. Let us choose a husband to be seated in the chair adjacent to his wife. This husband can be chosen in n ways, and arranged in 2 chairs. Suppose that the husband labeled n is seated in the chair labeled n and 1. Let us remove the label n from the chair that is placed between wives labeled $n-1$ and n . The remaining $n-1$ husbands should be arranged such that at least $k-1$ of them are neighbors with their wives. In other words, $k-1$ distinct positive integers should be chosen from the sequence

$$1, 2, 2, 3, 3, \dots, n-2, n-2, n-1, n-1, \quad (1)$$

such that none of the pairs $(1, 2), (2, 3), \dots, (n-2, n-1)$ is chosen. The last condition related to the pairs is required because a free chair can be occupied by only one person. The question can be reformulated as follows. Choose $k-1$ positive integers from sequence (1) such that no two of them are adjacent.

This can be done in $\binom{2n-k-1}{k-1}$ ways, see Chapter 2, Exercise 2.22. Let us consider a fixed k -combination of chosen husbands who are arranged to be neighbors to their wives. This k -combination can be obtained in k ways (because any of these k husbands can be chosen first). Note that there are $(n-k)!$ possibilities to arrange the remaining $n-k$ husbands. Using the product rule, and taking into account the multiplicity of any k -tuple, we get the conclusion. If an arrangement of wives is fixed, then the number of arrangements of husbands, such that at least k of them are neighbors to their wives, is equal to

$$2n \binom{2n-k-1}{k-1} \frac{1}{k} (n-k)! = \frac{2n}{2n-k} \binom{2n-k}{k} (n-k)!.$$

By the inclusion-exclusion principle we get the following conclusion: if an arrangement of wives is fixed, then the number of arrangements of husbands, such that none of them is a neighbor to his wife, is equal to

$$x_n := \sum_{k=0}^n (-1)^k \frac{2n}{2n-k} \binom{2n-k}{k} (n-k)!.$$

Since the wives can be arranged in $(n-1)!$ ways, it follows that the total number of arrangements that satisfy the required condition is $(n-1)!x_n$.

Remark: It is easy to check that $x_2 = 0$, $x_3 = 1$, $x_4 = 2$, and

$$x_n = nx_{n-1} + \frac{n}{n-2}x_{n-2} + (-1)^{n+1} \frac{4}{n-2}, \quad n \geq 4.$$

4.24. Let S be the set of all n -arrangements of the elements $1, 2, \dots, k$. For any $j \in \{1, 2, \dots, k\}$, let A_j be the set of n -arrangements $a_1 a_2 \dots a_n \in S$, such that $a_i \neq j$ for any i . Then, $|S| = k^n$. For $1 \leq j_1 < j_2 < \dots < j_r \leq k$, we have $|A_{j_1} \cap A_{j_2} \cap \dots \cap A_{j_r}| = (k-r)^n$. By the inclusion-exclusion principle it follows that

$$|S \setminus (A_1 \cup A_2 \cup \dots \cup A_k)| = k^n - \sum_{r=1}^k (-1)^r \binom{k}{r} (k-r)^n.$$

4.25. Let S be the set of all n -combinations of the elements $1, 2, \dots, k$ with repetitions allowed. For any $j \in \{1, 2, \dots, k\}$, let A_j be the set of n -combinations from S in which the element j appears more than c_j times.

By Theorem 2.6.3 it follows that $|S| = \binom{n+k-1}{n}$. Suppose that $1 \leq j_1 < j_2 < \dots < j_r \leq k$. Then, $A_{j_1} \cap A_{j_2} \cap \dots \cap A_{j_r}$ is the set of n -combinations of the elements $1, 2, \dots, k$, with repetitions allowed, and such that the elements $j_1, j_2, \dots, j_r$ appear at least $c_{j_1} + 1, c_{j_2} + 1, \dots, c_{j_r} + 1$ times, respectively. It follows that the number of elements of the set $A_{j_1} \cap A_{j_2} \cap \dots \cap A_{j_r}$ is equal to the number of $(n - \sum_{i=1}^r (c_{j_i} + 1))$ -combinations of the elements $1, 2, \dots, k$, with repetitions allowed, i.e.

$$|A_{j_1} \cap A_{j_2} \cap \dots \cap A_{j_r}| = \binom{n - \sum_{i=1}^r c_{j_i} - r + k - 1}{k-1}.$$

The number of elements of the set $|S \setminus (A_1 \cup A_2 \cup \dots \cup A_k)|$, to be determined, is equal to

$$\binom{n+k-1}{n} + \sum (-1)^r \binom{n - \sum_{i=1}^r c_{j_i} - r + k - 1}{k-1},$$

where the sum runs over all $\{j_1, j_2, \dots, j_r\}$, such that $\emptyset \neq \{j_1, j_2, \dots, j_r\} \subset \{1, 2, \dots, k\}$.

4.26. The number of arrangements with exactly k empty boxes is equal to

$$\sum_{j=k}^n (-1)^{j-k} \binom{j}{k} \binom{n}{j} (n-j)^m.$$

4.27. Let S be the set of all $(n-1)$ -arrangements of the elements of the set $X = \{1, 2, 3, 4, 5\}$. For any $j \in X$, let A_j be the set of $(n-1)$ -arrangements from S in which the element j does not appear. Then, $|S| = 5^{n-1}$, and, for $1 \leq j_1 < j_2 < \dots < j_r \leq 5$, we have $|A_{j_1} \cap A_{j_2} \cap \dots \cap A_{j_r}| = (5-r)^{n-1}$.

By the inclusion-exclusion principle it follows that the number of $(n-1)$ -arrangements of the elements 1, 2, 3, 4, 5, in which each of these elements appears at least once, is

$$|S \setminus (A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5)| = 5^{n-1} - 5 \cdot 4^{n-1} + 10 \cdot 3^{n-1} - 10 \cdot 2^{n-1} + 5.$$

Obviously this is also the number of n -arrangements that can appear as the result of the experiment with the digit 6 in the last position. Since the experiment can end by any of the digits 1, 2, 3, 4, 5, and 6, it follows that the total number of possible results of the experiment is

$$6(5^{n-1} - 5 \cdot 4^{n-1} + 10 \cdot 3^{n-1} - 10 \cdot 2^{n-1} + 5).$$

4.28. The number of k -arrangements of the form $P_1 P_2 \dots P_k$, where $P_1, P_2, \dots, P_k$ are permutations of the set $A = \{1, 2, \dots, n\}$ with r fixed points and arbitrarily arranged remaining $n-r$ elements, is $\left[\binom{n}{r} (n-r)! \right]^k$. By the inclusion-exclusion principle it follows that the number of k -arrangements of the form $P_1 P_2 \dots P_k$, where $P_1, P_2, \dots, P_k$ are permutations of set A , with exactly m fixed points, is equal to

$$\sum_{r=m}^n (-1)^{r-m} \binom{r}{m} \left[\binom{n}{r} (n-r)! \right]^k.$$

4.29. Let X be the set of all p -arrangements of the elements of set S . Since $|S| = \binom{n}{m}$, it follows that $|X| = \left(\binom{n}{m} \right)^p$. For any $j \in \{1, 2, \dots, n\}$, let A_j be the set of p -arrangements $K_1 K_2 \dots K_p \in X$, with the property that none of the m -combinations $K_1, K_2, \dots, K_p$ contain the element j . Note that, for $1 \leq j_1 < j_2 < \dots < j_r \leq n$, $A_{j_1} \cap A_{j_2} \cap \dots \cap A_{j_r}$ is the set of p -arrangements of the form $K_1 K_2 \dots K_p$, where $K_1, \dots, K_p$ are m -combinations of the set $\{1, 2, \dots, n\} \setminus \{j_1, j_2, \dots, j_r\}$. Hence, it follows that

$$|A_{j_1} \cap A_{j_2} \cap \dots \cap A_{j_r}| = \binom{n-r}{m}^p,$$

By the inclusion-exclusion principle it follows that the number of p -arrangements of set S that satisfy the given condition is equal to

$$|X \setminus (A_1 \cup A_2 \cup \dots \cup A_n)| = \sum_{r=0}^{n-m} (-1)^r \binom{n}{r} \binom{n-r}{m}^p.$$

4.30. If X is a finite set, and if the positive integer $|X|$ is represented as a sum of 1's, then any element of X contributes exactly one 1 in this sum. Let us now consider the positive integers

$$|S_1 \cap S_2 \cap \dots \cap S_n| \quad |S_{j_1} \cup S_{j_2} \cup \dots \cup S_{j_k}|,$$

where $\emptyset \neq \{j_1, j_2, \dots, j_k\} \subset \{1, 2, \dots, n\}$, and represent each of them as the sum of 1's. Let a be an arbitrary element of the set $S_1 \cup S_2 \cup \dots \cup S_n$.

Case 1. All of the sets $S_1, S_2, \dots, S_n$ contain the element a . Then, element a contributes exactly one 1 in the representation of the positive integer $|S_1 \cap S_2 \cap \dots \cap S_n|$ as the sum of 1's. Let us now consider the sum $Z = \sum (-1)^{k+1} |S_{j_1} \cup S_{j_2} \cup \dots \cup S_{j_k}|$, where each of the positive integers $|S_{j_1} \cup S_{j_2} \cup \dots \cup S_{j_k}|$ is represented as a sum of 1's. Element a contributes the following sum in the representation of Z :

$$\sum_{k=1}^n (-1)^{k+1} \binom{n}{k} = 1 - \sum_{k=0}^n (-1)^k \binom{n}{k} = 1 - (1-1)^n = 1.$$

Case 2. Suppose that exactly m of the sets $S_1, S_2, \dots, S_n$ contain the element a , where $0 \leq m < n$. Element a contributes no 1 in the representation of the positive integer $|S_1 \cap S_2 \cap \dots \cap S_n|$, and contributes the following sum in the representation of Z :

$$\begin{aligned} \sum_{k=1}^n (-1)^{k+1} \left[\binom{n}{k} - \binom{n-m}{k} \right] &= \sum_{k=1}^n (-1)^{k+1} \binom{n}{k} \\ &\quad - \sum_{k=1}^{n-m} (-1)^{k+1} \binom{n-m}{k} = 1 - 1 = 0. \end{aligned}$$

The statement of the exercise now follows easily.

4.31. Let S be the set of m -combinations of the elements $1, 2, \dots, n$. For any $j \in \{1, 2, \dots, n\}$, let A_j be the set of m -combinations from S that contain the element j . Then we have $S = A_1 \cup A_2 \cup \dots \cup A_n$, $|S| = \binom{n}{m}$, and

$$|A_{j_1} \cap A_{j_2} \cap \dots \cap A_{j_k}| = \binom{n-k}{m-k},$$

where $1 \leq j_1 < j_2 < \dots < j_r \leq n$, $k \leq m$. By the inclusion-exclusion principle it follows that

$$0 = |S \setminus (A_1 \cup A_2 \cup \dots \cup A_n)| = \sum_{k=0}^n (-1)^k \binom{n}{k} \binom{n-k}{m-k}.$$

4.32. Let S be the set of m -combinations of elements of the set $\{1, 2, \dots, n\}$, with repetitions allowed. For any $j \in \{1, 2, \dots, n\}$ let A_j be the set of m -combinations from S that do not contain element j . Then we have $|S| =$

$\binom{m+n-1}{m}$, and $|A_{j_1} \cap A_{j_2} \cap \cdots \cap A_{j_k}| = \binom{m+n-k-1}{m}$, where $1 \leq j_1 < j_2 < \cdots < j_k \leq n$, and $k \leq m$. Note that $A_1 \cup A_2 \cup \cdots \cup A_n$ is the set of m -combinations of the elements $1, 2, \dots, n$, that do not contain at least one of these elements. Since $m < n$, it follows that $A_1 \cup A_2 \cup \cdots \cup A_n = S$, and, by the inclusion-exclusion principle we obtain that

$$0 = |S \setminus (A_1 \cup A_2 \cup \cdots \cup A_n)| = \sum_{k=0}^n (-1)^k \binom{n}{k} \binom{m+n-k-1}{m}.$$

4.33. Answer: $(m+n)^k - \binom{m}{1}(m+n-1)^k + \binom{m}{2}(m+n-2)^k + \cdots + (-1)^m n^k$.

4.34. Let $1, 2, \dots, 3n$ be notation for the points in the order they appear on the circle. Let $\{i_1, j_1, k_1\}, \{i_2, j_2, k_2\}, \dots, \{i_n, j_n, k_n\}$ be a partition of the set $\{1, 2, \dots, 3n\}$ into n 3-sets. Without loss of generality we can suppose that $i_s < j_s < k_s$ for any $s \in \{1, 2, \dots, n\}$, and $k_1 < k_2 < \cdots < k_n$. By proceeding similarly as in Chapter 2, Exercise 2.65, we can derive the following statement:

The necessary and sufficient condition for the triangles $i_1 j_1 k_1, i_2 j_2 k_2, \dots, i_n j_n k_n$ to be pairwise disjoint is that $k_s \geq 3s$ for any $s \in \{1, 2, \dots, n\}$. In that case, the triangles $i_1 j_1 k_1, i_2 j_2 k_2, \dots, i_n j_n k_n$ are uniquely determined by the points $k_1, k_2, \dots, k_n$.

Using this statement and Example 4.3.6 it is easy to see that the number we are interested in is equal to

$$\binom{2n+n}{2n} - 2 \binom{2n+n}{2n+1} = \frac{1}{2n+1} \binom{3n}{n} = \frac{1}{3n+1} \binom{3n+1}{n}.$$

14.5 Solutions for Chapter 5

5.1. (a) $g_1(x) = \frac{x}{(1-x)^2}$, $|x| < 1$. (b) $g_2(x) = \frac{x^2+x}{(1-x)^3}$, $|x| < 1$.

(c) $g(x) = g_2(x) - g_1(x) = \frac{2x^2}{(1-x)^3}$, $|x| < 1$.

5.2. $g(x) = \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} n! \cdot \frac{x^n}{n!}$, $|x| < 1$.

5.3. $g(x) = \frac{1}{1-2x} + \frac{1}{1-3x}$, $|x| < \frac{1}{3}$; $f_e(x) = e^{2x} + e^{3x}$.

$$\mathbf{5.4.} \quad g(x) = \frac{1}{1-x} \ln \frac{1}{1-x} = \sum_{n=0}^{\infty} H_n x^n.$$

5.5. Using equality (5.1.3) we obtain that the ordinary generating function of sequence $a_n = n\alpha^n$, $n \in \mathbb{N}_0$, is

$$\begin{aligned} g(x) &= \sum_{n=0}^{\infty} n\alpha^n x^n = \alpha x \sum_{n=1}^{\infty} n(\alpha x)^{n-1} = \alpha x \sum_{n=0}^{\infty} (n+1)(\alpha x)^n \\ &= \frac{\alpha x}{(1-\alpha x)^2}, \quad |\alpha x| < 1. \end{aligned}$$

$$\mathbf{5.6.} \quad g(x) = \sum_{n=0}^{\infty} \frac{\alpha^n}{n!} x^n = \sum_{n=0}^{\infty} \frac{(\alpha x)^n}{n!} = e^{\alpha x}.$$

5.7. Let us denote $z(\alpha) = \cos \alpha + i \sin \alpha$ and

$$G(x, \alpha) = \sum_{n=0}^{\infty} z^n(\alpha) x^n = \sum_{n=0}^{\infty} (xz(\alpha))^n = \sum_{n=0}^{\infty} x^n \cos n\alpha + i \sum_{n=0}^{\infty} x^n \sin n\alpha.$$

Then we have

$$G(x, \alpha) = \frac{1}{1 - xz(\alpha)} = \frac{1}{1 - x(\cos \alpha + i \sin \alpha)}$$

Let $g_s(x)$ and $g_c(x)$ be the ordinary generating functions of the sequences $(a_n)_{n \geq 0}$ and $(b_n)_{n \geq 0}$, respectively. Now it is easy to obtain that

$$\begin{aligned} g_s(x) &= \frac{1}{2i} (G(x, \alpha) - G(x, -\alpha)) = \frac{x \sin \alpha}{1 - 2x \cos \alpha + x^2}, \\ g_c(x) &= \frac{1}{2} (G(x, \alpha) + G(x, -\alpha)) = \frac{1 - x \cos \alpha}{1 - 2x \cos \alpha + x^2}. \end{aligned}$$

5.8. (a) The ordinary generating function of the sequence $(a_n)_{n \geq 0}$ is $g(x) = \frac{1}{1-3x} = \sum_{n=0}^{\infty} (3x)^n$, $|x| < \frac{1}{3}$. It follows that $a_n = 3^n$ for every $n \geq 0$.

(b) The ordinary generating function of the sequence (a_n) is given by $g(x) = \frac{1+x}{1-2x+3x^2}$. It follows that

$$a_n = \frac{1-i\sqrt{2}}{2}(1+i\sqrt{2})^n + \frac{1+i\sqrt{2}}{2}(1-i\sqrt{2})^n, \quad \text{for every } n \geq 0.$$

5.9. (a) $a_n = At_1^n + Bt_t^n$, for every $n \in \mathbb{N}_0$, where constants A and B are determined by the initial conditions $A+B=a_0$ and $At_1+Bt_2=a_1$.

(b) $a_n = (A + Bn)t_1^n$, for every $n \in \mathbb{N}_0$, where constants A and B are determined by the initial conditions $A = a_0$ and $(A + B)t_1 = a_1$.

5.10. Let $t_1, t_2, \dots, t_k$ be the solutions of characteristic equation $t^k + c_1 t^{k-1} + \dots + c_{k-1} t + c_k = 0$. The general solution of recursive equation $f_n = c_1 f_{n-1} + c_2 f_{n-2} + \dots + c_k f_{n-k}$ is of the form

$$f_n = A_1 t_1^n + A_2 t_2^n + \dots + A_k t_k^n,$$

where $A_1, A_2, \dots, A_k$ are real constants that can be determined from the given values of the first k terms of sequence $(f_n)_{n \geq 0}$.

5.11. Note that the characteristic equation is $t^4 - 3t^3 - 6t^2 + 28t - 24 = (t - 2)^3(t + 3) = 0$. The general term of sequence $(a_n)_{n \geq 0}$ is of the form $a_n = (A + Bn + Cn^2) \cdot 2^n + D \cdot (-3)^n$. From the initial conditions $a_0 = 3$, $a_1 = -6$, $a_2 = 22$, and $a_3 = -22$ it follows that $A = 1$, $B = -2$, $C = 1$, and $D = 2$. Hence, $a_n = (n - 1)^2 \cdot 2^n + 2 \cdot (-3)^n$ for every $n \in \mathbb{N}_0$.

5.12. The ordinary generating function of the sequence $(a_n)_{n \geq 0}$ is given by

$$g(x) = 1 + \frac{x}{1 - 3x + x^2},$$

It follows that $a_0 = 1$, and $a_n = F_{2n}$ for any $n \geq 1$, where $(F_n)_{n \geq 0}$ is the Fibonacci sequence given by $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for every $n \geq 2$.

5.13. The ordinary generating function of the sequence $(a_n)_{n \geq 0}$ is given by

$$g(x) = \frac{1 + 2x - \sqrt{1 + 4x^2}}{2}.$$

It follows that $a_0 = 1$, $a_{2n+1} = 0$ for every $n \geq 0$, and

$$a_{2n} = (-1)^n \frac{(2n)!}{n} \binom{2n-2}{n-1} \quad \text{for every } n \geq 1.$$

5.14. Answer: $\frac{1}{n+1} \binom{2n}{n}.$

5.15. Answer: $\frac{1}{n+1} \binom{2n}{n}.$

14.6 Solutions for Chapter 6

6.1. (a) See Theorem 6.2.4(a).

(b) Let S_1 be the set of solutions of the equation $x_1 + x_2 + \cdots + x_k = n$ in the set $\mathbb{N}_0$, and let S_2 be the set of $(n+k-1)$ -arrangements of elements 0 and 1, with exactly $k-1$ 1's in each of them. Obviously, there is a bijection between S_1 and S_2 . Since any arrangement from S_2 is determined by the positions of the 1's, it follows that

$$|S_1| = |S_2| = \binom{n+k-1}{k-1}.$$

6.2. Let S be the number of k -tuples $(x_1, x_2, \dots, x_k)$ such that $x_1 + x_2 + \cdots + x_k \leq n$, and $x_1, x_2, \dots, x_k \in \mathbb{N}_0$. Let S^* be the set of solutions of the equation $x_1 + x_2 + \cdots + x_k + x_{k+1} = n$ in the set $\mathbb{N}_0$. Obviously, there is a bijection between S and S^* . From Exercise 6.1(b) it follows that

$$|S| = \binom{n+k}{k}.$$

6.3. Let us denote $y_i = x_i - c_i$, for $i \in \{1, 2, \dots, k\}$. We are interested in the number of solutions of the equation $y_1 + y_2 + \cdots + y_k = n - c_1 - c_2 - \cdots - c_k$, where $y_1, y_2, \dots, y_k \in \mathbb{N}_0$. From Exercise 6.1(b) it follows that this number is equal to

$$\binom{n - c_1 - \cdots - c_k + k - 1}{k - 1}.$$

6.4. Answer. The number of solutions is the coefficient of x^n in the expansion of the polynomial $(x^r + x^{r+1} + \cdots + x^s)^k$.

6.5. The number of solutions of the equation $x_1 + x_2 + \cdots + x_k = 2n$ in the set $\mathbb{N}_0$ is $\binom{2n+k-1}{k-1}$. The number of solutions of this equation such that $x_1 = x_k = j$, where $j \in \{0, 1, 2, \dots, n\}$, and $x_2, x_3, \dots, x_{k-1} \in \mathbb{N}_0$, is equal to $\binom{2n-2j+k-3}{k-3}$. The number of solutions in the set $\mathbb{N}_0$, such that $x_1 > x_k$, is equal to

$$\frac{1}{2} \left[\binom{2n+k-1}{k-1} - \sum_{j=0}^n \binom{2n-2j+k-3}{k-3} \right].$$

6.6. Let S be the set of solutions of the equation $x_1 + x_2 + \cdots + x_n = 11$, such that $x_1, x_2, \dots, x_n \in \mathbb{N}_0$. Let A , B , and C be, respectively, the sets of those solutions from S that satisfy the conditions: (a) $x_1 = 0$; (b) $x_i = 11$ for some $i \in \{1, 2, \dots, n\}$; (c) $x_i = 10$ for some $i \in \{1, 2, \dots, n\}$. Then we have

$$|S| = \binom{n+10}{11}, |A| = \binom{n+9}{11}, |B| = n, |C| = n(n-1), |A \cap B| = n-1,$$

$|B \cap C| = 0$, $|C \cap A| = (n-1)(n-2)$. The number of n -digit positive integers with the sum of digits equal to 11 is

$$\begin{aligned} |S \setminus (A \cup B \cup C)| &= |S| - |A| - |B| - |C| + |A \cap B| + |B \cap C| \\ &\quad + |C \cap A| - |A \cap B \cap C| = \binom{n+9}{10} - 2n + 1. \end{aligned}$$

6.7. Let m be a positive integer. It is easy to prove that the number of pairs (a, b) of nonnegative integers satisfying the conditions $a \leq b$ and $a + b = m$ is equal to $[m/2] + 1$, where $[x]$ is the greatest integer that is not greater than x . Let S be the set of solutions of the equation $x_1 + x_2 + x_3 = 100$ in the set $\mathbf{N}_0$, such that $x_1 \leq x_2 \leq x_3$. For any $k \in \{0, 1, 2, \dots, 33\}$, let A_k be the set of solutions of the equation $x_1 + x_2 + x_3 = 100$, such that $k = x_1 \leq x_2 \leq x_3$. It is obvious that $S = A_0 \cup A_1 \cup \dots \cup A_{33}$, where $A_0, A_1, \dots, A_{33}$ are pairwise disjoint sets. Let us determine $|A_k|$. If we denote $a = x_2 - k$ and $b = x_3 - k$, then $|A_k|$ is the number of pairs (a, b) of nonnegative integers such that $a + b = 100 - 3k$, and $0 \leq a \leq b$, i.e., $|A_k| = \left\lceil \frac{100 - 3k}{2} \right\rceil + 1$. Hence,

$$|S| = \sum_{k=0}^{33} |A_k| = 34 + \sum_{j=0}^{33} \left\lceil \frac{4j+1}{2} \right\rceil = 884.$$

6.8. The number of triplets (x, y, z) of positive integers such that $x_1 + x_2 + x_3 = 300$, $x < y + z$, $y < z + x$, and $z < x + y$ is equal to the number of triplets (x, y, z) of positive integers such that $z = 300 - x - y$, $x < 150$, $y < 150$, and $x + y > 150$. This number, denoted by s , is also the number of points (x, y) , with the integer coordinates satisfying the conditions $x < 150$, $y < 150$, and $x + y > 150$. Hence, $s = 1 + 2 + 3 + \dots + 148 = 11026$. Let X and Y be the number of triangles with three distinct sides, and exactly two equal sides, respectively, and the circumference 300, whose sides are positive integers. Then, we have $11026 = 6X + 3Y + 1 = 6(X + Y + 1) - 3(Y + 1) - 2$, and the number of triangles that satisfy the given condition is

$$X + Y + 1 = \frac{11026}{6} + \frac{Y + 1}{2} + \frac{1}{3}.$$

Note that $Y + 1$ is the number of pairs (x, y) of positive integers such that $2x + y = 300$ and $2x > y$, i.e., the number of positive integers x such that $75 < x < 150$. Hence, $Y + 1 = 74$. Finally, it follows that $X + Y + 1 = 1875$.

6.9. In this exercise we always consider equations in the set of positive integers. The number of solutions of the equation

$$x_1 + x_2 + x_3 + x_4 = 12n + 5 \quad (1)$$

is $\binom{12n+4}{3}$. The number of solutions of the equation $x_2 + x_3 + x_4 = 12n - k + 5$ is $\binom{12n+4-k}{3}$. It follows that the number of solutions of equation (1), such that $x_1 > 6n + 2$, is $\sum_{k=6n+3}^{12n+2} \binom{12n-k+4}{2} = \binom{6n+2}{3}$. Hence, the number of solutions of equation (1), such that none of x_1, x_2, x_3, x_4 is greater than $6n + 2$, is equal to $\binom{12n+4}{3} - 4\binom{6n+2}{3}$. The equation $x_3 + x_4 = 12n - 2k + 5$ has exactly $12n - 2k + 4$ solutions. Now it follows that the equation

$$2x_1 + x_3 + x_4 = 12n + 5 \quad (2)$$

has exactly $\sum_{k=1}^{6n+1} (12n - 2k + 4) = 2\binom{6n+2}{2}$ solutions. The equation $x_1 + x_4 = 12n + 5 - 2k$ has exactly $6n - k + 2$ solutions, and the equation $2x_1 + x_4 = 12n + 5 - 2k - 1$ has exactly $6n - k + 1$ solutions. Now it follows that the number of solutions of equation (2), such that $x_3 = k > 6n + 2$, is

$$\sum_{k=3n+2}^{6n+1} (6n - k + 2) + \sum_{k=3n+1}^{6n} (6n - k + 1) = 2\binom{3n+1}{2}.$$

The number of solutions of equation (1), such that $x_1 = x_2$, and none of x_1, x_2, x_3, x_4 is greater than $6n + 2$, is equal to $2\binom{6n+2}{2} - 4\binom{3n+1}{2}$. The number of solutions of the equation $3x_1 + x_4 = 12n + 5$ is $4n + 1$, and the number of solutions of this equation, such that $x_4 > 6n + 2$, is $2n$. Hence, the number of solutions of the equation $3x_1 + x_4 = 12n + 5$, such that $x_1 \leq 6n + 2$ and $x_4 \leq 6n + 2$, is $2n + 1$. By the inclusion-exclusion principle it follows that the number of solutions of equation (1), such that x_1, x_2, x_3 , and x_4 are distinct positive integers, is equal to

$$\begin{aligned} & \left[\binom{12n+4}{3} - 4\binom{6n+3}{3} \right] - \binom{4}{2} \left[2\binom{6n+2}{2} - 4\binom{3n+1}{2} \right] \\ & + 2\binom{4}{3}(2n+1) = 12n(12n^2 + 3n - 1). \end{aligned}$$

The number of solutions of equation (1), such that there are exactly three distinct positive integers among x_1, x_2, x_3, x_4 , is equal to

$$\binom{4}{2} \left[2\binom{6n+2}{2} - 4\binom{3n+1}{2} \right] - 12(2n+1) = 12n(9n+4).$$

The number of solutions of equation (1), such that there are exactly two distinct positive integers among x_1, x_2, x_3, x_4 , is equal to $4(2n+1)$. The positive integer we are interested in is

$$\begin{aligned} & \frac{12n(12n^2+3n-1)}{24} + \frac{12n(9n+4)}{12} + \frac{4(2n+1)}{4} \\ &= \frac{1}{2}(n+1)(12n^2+9n+2). \end{aligned}$$

6.10. We need to determine $H(5, 10, 15, 20; 60)$, that is defined by Theorem 6.2.5. Note that $H(5, 10, 15, 20; 60) = H(1, 2, 3, 4; 12) = H(12)$. It was proved that $H(7) = 56$, $H(8) = 108$, $H(9) = 208$, $H(10) = 401$, and $H(n) = H(n-1) + H(n-2) + H(n-3) + H(n-4)$, see Example 6.2.6. Hence, $H(11) = 773$, and $H(12) = 1490$.

6.11. (a) $G(1, 2; 27) = 14$; (b) $G(2, 5; 27) = 3$; (c) $G(1, 2, 5; 27) = 34$.

6.12. From Theorem 6.1.6 it follows that $G(1, 2, 5, 10; 100) = 2156$.

6.13. Let S_1 be the set of partitions of the positive integer $2n$ into n parts, and S_2 be the set of all partitions of the positive integer n . For every $(\alpha_1, \alpha_2, \dots, \alpha_n) \in S_1$, such that $\alpha_1 \geq \dots \geq \alpha_k > 1 = \alpha_{k+1} = \dots = \alpha_n$, let us define $f((\alpha_1, \alpha_2, \dots, \alpha_n)) = (\alpha_1 - 1, \alpha_2 - 1, \dots, \alpha_k - 1) \in S_2$. The function $f: S_1 \rightarrow S_2$ is a bijection. Hence, $|S_1| = |S_2|$.

6.14. Let S_1 be the set of partitions of the positive integer n into odd parts, and S_2 be the set of partitions of the positive integer n into distinct parts. Let $\alpha \in S_1$ be the partition given by $n = \alpha_1 + \alpha_1 + \dots + \alpha_2 + \alpha_2 + \dots = \sum_{j=1}^s k_j \alpha_j$, where $\alpha_1, \alpha_2, \dots, \alpha_s$ are distinct odd integers. Let

$$k_j = 2^{c_{j1}} + 2^{c_{j2}} + \dots, \quad c_{j1} > c_{j2} > \dots, \quad j \in \{1, 2, \dots, s\},$$

be the binary representation of $k_1, k_2, \dots, k_s$. Let $f(\alpha)$ be the partition of n given by $n = 2^{c_{11}}\alpha_1 + 2^{c_{12}}\alpha_1 + \dots + 2^{c_{21}}\alpha_2 + 2^{c_{22}}\alpha_2 + \dots$. The parts of this partition, i.e., $2^{c_{11}}\alpha_1, 2^{c_{12}}\alpha_1, \dots, 2^{c_{21}}\alpha_2, 2^{c_{22}}\alpha_2, \dots$, are distinct positive integers. Hence, $f(\alpha) \in S_2$. Obviously, the function $f: S_1 \rightarrow S_2$ is an injection. We shall prove that this function is also a surjection. Let

$$n = n_1 + n_2 + \dots + n_r, \quad n_1 > n_2 > \dots > n_r, \quad (1)$$

be a partition of n . The parts of this partition can be represented in the form $n_1 = 2^{c_1}\alpha_1, n_2 = 2^{c_2}\alpha_2, \dots, n_r = 2^{c_r}\alpha_r$, where $\alpha_1, \alpha_2, \dots, \alpha_r$ are odd positive integers. For any $j \in \{1, 2, \dots, r\}$, let us replace $n_j = 2^{c_j}\alpha_j$ in

equality (1) by the sum of 2^{c_j} addends that are equal to α_j each. This way we obtain a partition α of the positive integer n into odd parts, such that $f(\alpha)$ is given by (1).

6.15. (a) Let S_1 be the set of all partitions of the positive integers $1, \dots, n-1$, and S_2 be the set of all partitions of the positive integer n . Let $k = \alpha_1 + \alpha_2 + \dots + \alpha_s$ be a partition of a positive integer $k \in \{1, 2, \dots, n-1\}$. Let $f(\alpha_1, \alpha_2, \dots, \alpha_s)$ be the partition of n given by $n = \alpha_1 + \alpha_2 + \dots + \alpha_s + (n-k)$. This way we have defined the function $f: S_1 \rightarrow S_2$. Let $\pi \in S_2$ be the partition of n given by $n = \beta_1 + \beta_2 + \dots + \beta_r$, $r \geq 2$. The number of partitions $\alpha \in S_1$, such that $f(\alpha) = \pi$, is equal to the number of distinct positive integers among $\beta_1, \beta_2, \dots, \beta_r$, that is exactly $q_\pi(n)$. For the trivial partition π_0 , which is given by $n = n$, there is no partition $\alpha \in S_1$, such that $f(\alpha) = \pi_0$. Hence, it follows that

$$|S_1| = p(1) + p(2) + \dots + p(n-1) = \sum_{\pi} q_{\pi}(n) - 1,$$

where $\sum_{\pi} q_{\pi}(n)$ is notation for the sum of all positive integers $q_{\pi}(n)$ over all partitions π of the positive integer n . Since this sum is also equal to $q(n)$, it follows that $q(n) = 1 + p(1) + p(2) + \dots + p(n-1)$.

(b) Let $\alpha_1, \alpha_2, \dots, \alpha_k$ be distinct parts of a partition π of a positive integer n , and suppose that $\alpha_1 > \alpha_2 > \dots > \alpha_k > 0$. Then,

$$\alpha_1 + \alpha_2 + \dots + \alpha_k \leq n, \quad \alpha_k \geq 1, \quad \alpha_{k-1} \geq 2, \quad \dots, \quad \alpha_2 \geq k-1, \quad \alpha_1 \geq k,$$

and consequently it follows that

$$n \geq \sum_{j=1}^k \alpha_k \geq \sum_{j=1}^k j = \frac{1}{2}k(k+1) > \frac{1}{2}k^2,$$

and $q_{\pi}(n) = k \leq \sqrt{2n}$. The last inequality holds for any partition π of the positive integer n . By taking the sum of such inequalities over all partitions of n , we get $q(n) < \sqrt{2n}p(n)$.

6.16. Let us denote $m = \min\{|A_1|, |A_2|, \dots, |A_n|, |B_1|, |B_2|, \dots, |B_n|\}$ and suppose, for example, $|A_1| = m$. Since the sets $B_1, B_2, \dots, B_n$ are pairwise disjoint, it follows that at most m of these sets have a nonempty intersection with the set A_1 . Let us suppose, for example,

$$\begin{aligned} A_1 \cap B_j &\neq \emptyset, & \text{for each } j \in \{1, 2, \dots, k\}, \\ A_1 \cap B_j &= \emptyset, & \text{for each } j \in \{k+1, k+2, \dots, n\}, \end{aligned}$$

where $k \leq m$. Then, $|B_j| \geq m$ for any $j \in \{1, 2, \dots, k\}$, and $|B_j| \geq n-m$ for any $j \in \{k+1, k+2, \dots, n\}$. If $m < n/2$, then using the inequality $k \leq m$,

we obtain that

$$\begin{aligned} |X| &= \sum_{k=1}^n |B_k| \geq km + (n-k)(n-m) = n(n-m) - k(n-2m) \\ &\geq n(n-m) - m(n-2m) = \frac{1}{2}n^2 + \frac{1}{2}(n-2m)^2 \geq \frac{1}{2}n^2. \end{aligned}$$

If $m \geq n/2$, then $|X| = \sum_{k=1}^n |B_k| \geq nm \geq \frac{1}{2}n^2$.

(b) The equality is possible, and this fact follows from the following example. Let n be an even positive integer, and let $A_1, A_2, \dots, A_n$ be pairwise disjoint sets, such that $|A_i| = n/2$ for any i . Let us denote $X = A_1 \cup A_2 \cup \dots \cup A_n$, and $B_j = A_j$ for any $j \in \{1, 2, \dots, n\}$. Then, all the conditions are satisfied, and $|X| = n^2/2$.

6.17. We shall use notation A_j^b , A_j^r , or A_j^g for a block A_j of a partition of the set X_n , if $|A_j| = j$, and all elements of A_j are blue, red, or green, respectively. For any $n \in \{4, 5, 6, 7, 8, 9\}$, we give a partition of the set X_n , such that conditions (a) and (b) are satisfied.

$$\begin{aligned} n = 4, \quad k_n = 3 : & \quad A_1^p, A_2^p, A_3^c, A_4^z; \\ n = 5, \quad k_n = 5 : & \quad A_1^p, A_2^c, A_3^c, A_4^p, A_5^z; \\ n = 6, \quad k_n = 7 : & \quad A_1^p, A_2^c, A_3^z, A_4^z, A_5^c, A_6^p; \\ n = 7, \quad k_n = 9 : & \quad A_1^z, A_2^p, A_3^c, A_4^z, A_5^z, A_6^c, A_7^5; \\ n = 8, \quad k_n = 12 : & \quad A_1^z, A_2^z, A_3^z, A_4^p, A_5^c, A_6^z, A_7^z, A_8^c, A_9^c; \\ n = 9, \quad k_n = 15 : & \quad A_1^p, A_2^p, A_3^p, A_4^p, A_5^p, A_6^c, A_7^z, A_8^z, A_9^c. \end{aligned}$$

The proof that the statement holds for any positive integer $n \geq 4$ will be given by the method of mathematical induction. Let us suppose that $n \geq 10$, and assume that the statement holds for all positive integers $4, 5, \dots, n-1$, particularly for $n-6$. Note that

$$\begin{aligned} |X_n| - |X_{n-6}| &= \frac{1}{2}n(n+1) - \frac{1}{2}(n-6)(n-5) = 6n - 15 = 3(2n-5), \\ k_n - k_{n-6} &= 2n - 5. \end{aligned}$$

Let $\{A_1, A_2, \dots, A_{n-6}\}$ be a partition of the set X_{n-6} satisfying conditions (a) and (b). It is obvious that

$$\{A_1, A_2, \dots, A_{n-6}, A_{n-5}^b, A_{n-4}^r, A_{n-3}^g, A_{n-2}^g, A_{n-1}^r, A_n^b\}$$

is a partition of set X_n that also satisfies the conditions (a) and (b).

6.18. Let $\{A_1, A_2, \dots, A_m\}$ and $\{B_1, B_2, \dots, B_{m+k}\}$ be two partitions of set S . For every $i \in S$, exactly one of the blocks $A_1, A_2, \dots, A_m$ contains i .

Let x_i be the number of elements of this block. Similarly, exactly one of the blocks $B_1, B_2, \dots, B_{m+k}$ contains i . Let y_i be the number of elements of the block B_s that contains i . For every $i \in \{1, 2, \dots, m\}$, there are exactly $|A_i|$ positive integers among $x_1, x_2, \dots, x_n$ that are equal to $|A_i|$. Hence,

$$\sum_{i=1}^n \frac{1}{x_i} = m, \quad \text{and analogously,} \quad \sum_{i=1}^n \frac{1}{y_i} = m + k. \quad (1)$$

It follows from equalities (1) that

$$\sum_{i=1}^n \left(\frac{1}{y_i} - \frac{1}{x_i} \right) = k. \quad (2)$$

All addends on the left-hand side of equality (2) are less than 1. Hence, at least $k + 1$ of these addends are greater than 0. It remains to note that $1/y_i - 1/x_i > 0$ if and only if $x_i > y_i$, and the proof is finished.

6.19. Let S_E be the set of participants who speak only English, S_{EF} be the set of participants who speak English and French and do not speak Spanish, and S_{EFS} be the set of participants who speak all three languages. Similarly we define the sets S_F , S_S , S_{ES} , and S_{FS} . The following lemma holds.

A few participants can be chosen, such that exactly two of them speak English, exactly two of them speak French, and exactly two of them speak Spanish. Some of the chosen participants may speak more than one language.

Case 1. If each of the sets S_{EF} , S_{ES} , and S_{FS} is nonempty, then we choose an element from each of them, and form a 3-set this way.

Case 2. If, for example, $S_{EF} = \emptyset$, and $S_{ES} \neq \emptyset$, $S_{FS} \neq \emptyset$, then we choose an element from each of the sets S_E , S_F , S_{ES} , and S_{FS} .

Case 3. Let us suppose, for example, that $S_{EF} = S_{ES} = \emptyset$ and $S_{FS} \neq \emptyset$. If $S_{EFS} \neq \emptyset$, then we choose an element from each of the sets S_{EFS} , S_{FS} , and S_E . If $S_{EFS} = \emptyset$ and $|S_{FS}| \geq 2$, then we choose two elements from each of the sets S_{FS} and S_E . If $S_{EFS} = \emptyset$ and $|S_{FS}| = 1$, then we choose two elements from S_E , and an element from each of the sets S_{FS} and S_F , S_S .

Case 4. Let us suppose that $S_{EF} = S_{ES} = S_{FS} = \emptyset$. If $|S_{EFS}| \geq 2$, then we choose two elements from the set S_{EFS} . If $|S_{EFS}| = 1$, then we choose an element from each of the sets S_{EFS} , S_E , S_F , and S_S . If $|S_{EFS}| = 0$, then we choose two elements from each of the sets S_E , S_F , and S_S . It is easy to see that such choices are possible.

By proceeding the same way, we can form five disjoint subsets $A_i \subset X$, $i \in \{1, 2, 3, 4, 5\}$, where X is the set of all participants, such that every A_i contains exactly two English speaking participants, exactly two French

speaking participants, and exactly two Spanish speaking participants. The union $A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5$ satisfies the required conditions.

6.20. Let us divide the parliament into two houses arbitrarily. If there is a member of the parliament, denoted by A , with two enemies in the same house, then we move A into the other house. This way the total number of pairs of members of parliament, who are enemies and belong to the same house, decreases. By repeating this operation we can reach the required structure of the houses of parliament.

6.21. Suppose that the given lines divide the plane into x_n parts. Then we have $x_1 = 2$, and $x_n = x_{n-1} + n$ for $n \geq 2$. Indeed, let $l_1, l_2, \dots, l_n$ be the given lines. The lines $l_1, l_2, \dots, l_{n-1}$ divide the plane into x_{n-1} parts and have $n-1$ points of intersection with the line l_n . These $n-1$ points divide the line l_n into n segments (two of them are unbounded), and each of these segments divides one of x_{n-1} parts of the plane into two new parts.

Now it is easy to obtain that $x_n = 2 + 2 + 3 + 4 + \dots + n = (n^2 + n + 2)/2$.

6.22. Let x_n be the number of parts into which space is divided by the given planes. Similarly as in the Exercise 6.21 we obtain that $x_1 = 2$, and $x_n = x_{n-1} + (n^2 - n + 2)/2$ for $n \geq 2$. Consequently we obtain that

$$x_n = 2 + \frac{1}{2} \sum_{k=2}^n (k^2 - k + 2) = (n+1)(n^2 - n + 6)/6.$$

6.23. Let (x_n) be the number of parts into which the plane is divided by the given circles. Then, $x_1 = 2$, and $x_n = x_{n-1} + 2n - 2$ for $n \geq 2$. It follows that $x_n = 2 + \sum_{k=2}^n (2k - 2) = n^2 - n + 2$.

6.24. *Hint.* First prove the following statement. Suppose that k circles are given on the sphere such that every two of them have a common point, and no three of them have a common point. Then, these circles divide the sphere into $k^2 - k + 2$ parts. *Answer.* The number of parts into which space is divided by the given spheres is $2 + \sum_{k=2}^n (k^2 - k + 2) = \frac{n}{3}(n^2 + 5)$.

6.25. *Answer.* $2 + \sum_{k=2}^n (2k - 2) = n^2 - n + 2$.

6.26. Let $A_1 A_2 \dots A_n$ be the given convex n -gon. For any $3 \leq k \leq n-1$, the diagonal $A_1 A_k$ has $(k-2)(n-k)$ points of intersection with the other

diagonals. These points divide the diagonal A_1A_k into $(k-2)(n-k)+1$ parts. The total number of obtained segments is

$$\frac{n}{2} \sum_{k=3}^{n-1} [(k-2)(n-k)+1] = \frac{1}{12} n(n-3)(n^2-3n+8).$$

6.27. The number of diagonals of the n -gon is $n(n-3)/2$. Let S be the set of points of intersections of the diagonals of the given convex n -gon. There is a bijection between S and the set of 4-combinations of vertices of the given n -gon. Hence $|S_1| = \binom{n}{4}$. Let us draw all the diagonals, one after the other. Suppose that some diagonals are already drawn, and we draw a new diagonal d . If diagonal d has $k \geq 0$ points of intersection with the diagonals previously drawn, then we obtain $k+1$ new parts of the n -gon. It follows that the n -gon is divided into $1 + \frac{n(n-3)}{2} + \binom{n}{4}$ parts.

6.28. *Answer:* (a) $n-2$; (b) $n-3$.

14.7 Solutions for Chapter 7

7.1. (a)–(c): A *transposition* is an exchange of two elements in a permutation. Use the following facts: Every transposition changes the parity of a permutation. An even (odd) permutation is a composition of an even (odd) number of transpositions.

(d)–(e): Use the fact that $\varphi \circ \varphi^{-1} = \varepsilon$.

7.2. $\varphi = (1, 3, 5, 7) \circ (2, 9, 6) \circ (4, 8)$. The order of the permutation φ is 12.

7.3. (a) The number of colorings is $4! = 24$. Every class of equivalent colorings is a set consisting of 12 elements. Hence, the number of nonequivalent colorings is equal to 2.

$$(b) \frac{6!}{6 \cdot 4} = 30, \quad (c) \frac{8!}{8 \cdot 3} = 1680, \quad (d) \frac{12!}{12 \cdot 5} = 7983360, \quad (e) \frac{20!}{60}.$$

7.4. Since there are n colors for any arc, it follows that there are n^p possible colorings of all p arcs. We shall consider all equivalence classes. If all arcs are colored the same color, then this coloring is unique in its class. There are n such classes.

We shall prove that any coloring with at least two colors used belongs to a class consisting of exactly p elements (colorings). *All colorings from this class can be obtained from the initial one by rotation about the center of the circle, where the angles of rotation are $0, \frac{2\pi}{p}, 2\frac{2\pi}{p}, \dots, (p-1)\frac{2\pi}{p}$.*

Let c_0 be the initial coloring of the arcs with at least two colors used, O be the center of the circle, and $\mathcal{R}_{O,\alpha}$ be the rotation about O with the angle of rotation α . For the proof of the above statement it is sufficient to check that $\mathcal{R}_{O,2k\pi/p}(c_0) \neq \mathcal{R}_{O,2m\pi/p}(c_0)$, where $0 \leq k < m \leq p-1$. Suppose, on the contrary, that $\mathcal{R}_{O,2k\pi/p}(c_0) = \mathcal{R}_{O,2m\pi/p}(c_0)$. Then, we have $\mathcal{R}_{O,(m-k)2\pi/p}(c_0) = c_0$, and consequently

$$\mathcal{R}_{O,i(m-k)2\pi/p}(c_0) = c_0 \quad \text{for any } i \in \{0, 1, \dots, p-1\}.$$

It is easy to see that the first arc runs over all the other arcs after the rotations $\mathcal{R}_{O,i(m-k)2\pi/p}$, $i \in \{0, 1, \dots, p-1\}$. Hence, the number of nonequivalent colorings, i.e., the number of classes, is $n + (n^p - n)/p$.

Remark: Suppose that a circle is divided into p equal arcs, where p is not a prime. Let us consider a coloring of arcs with at least two colors used. Then the number of elements of a class of equivalent colorings does not have to be equal to p . Let us consider the case $p = 8$. The circle is divided into 8 equal arcs. If the arcs are alternately red and blue, then the related class of equivalent colorings consists of two elements. If seven arcs are red, and the remaining arc is blue, then the related class of equivalent colorings consists of 8 elements.

7.5. If p is a prime, then, by Exercise 7.4, it follows that $(n^p - n)/p$ is a positive integer, i.e., $n^p - n$ is divisible by p .

7.6. We say that a starlike p -gon is regular if all its sides are congruent to each other, and all its angles are congruent to each other. It is obvious that the shorter of two arcs of the circle determined by two adjacent vertices of a starlike p -gon contains the same number of points from the set $\{A_1, A_2, \dots, A_p\}$, i.e., this number does not depend on the choices of two adjacent vertices. Note also that this number belongs to the set $\{1, 2, \dots, (p-3)/2\}$. Let us consider the partition of the set of all starlike p -gons with the vertices $A_1, A_2, \dots, A_p$ into disjoint classes of congruent p -gons. Then, the following statements hold (prove that):

(a) The number of p -gons with vertices $A_1, A_2, \dots, A_p$ is $(p-1)!/2$, and the number of starlike p -gons with the same vertices is $(p-1)!/2 - 1$.

(b) The number of regular starlike p -gons is $(p-3)/2$.

(c) Any regular starlike p -gon is the unique element in its class of congruent starlike p -gons.

(d) Any starlike non-regular p -gon belongs to the class that consists of exactly p congruent p -gons. All starlike p -gons from this class can be obtained from any of them by the rotations $\mathcal{R}_{O,i2\pi/p}$, $i \in \{0, 1, \dots, p-1\}$, where O is the center of the circle.

(e) The number of noncongruent starlike p -gons, i.e., the number of classes of congruent starlike p -gons, is

$$\frac{p-3}{2} + \frac{1}{p} \left[\frac{1}{2}(p-1)! - 1 - \frac{p-3}{2} \right] = \frac{1}{2} \left[p-4 + \frac{(p-1)! + 1}{p} \right].$$

7.7. If $p = 2$, then $(p-1)! + 1 = 2$, and the statement obviously holds. If $p \geq 3$, then the statement follows from Exercise 7.6.

7.8. If we exchange two adjacent triplets of digits, then the number of 0's (and 1's) in the positions that are congruent to each other modulo 3 remains the same. The necessary and sufficient condition for two arrangements from S to be equivalent is that the arrangements have the same number of 0's in the positions congruent to each other modulo 3. Prove this statement for $n = 8$ and then use the method of mathematical induction.

Let x_n be the number of nonequivalent arrangements, where $n \geq 8$. The sequence $(x_n)_{n \geq 8}$ is given by:

$$\begin{aligned} x_{3k} &= (k+1)^3, & k &= 3, 4, 5, \dots; \\ x_{3k+1} &= (k+2)(k+1)^2, & k &= 3, 4, 5, \dots; \\ x_{3k+2} &= (k+2)^2(k+1), & k &= 2, 3, 4, \dots \end{aligned}$$

7.9. Let us consider all permutations with 1 in the first position that can be obtained from the identical permutation by repeated application of operations (a) and (b). The element 2 can occupy the following positions in these permutations:

$$2 \pmod{2n+1}, \quad 4 \pmod{2n+1}, \quad 8 \pmod{2n+1}, \quad \dots$$

It follows that the number of distinct permutations that can be obtained this way is equal to $(2n+1)K$, where K is the number of distinct remainders obtained after dividing the positive integers $2, 4, 8, \dots, 2^k, \dots$ by $2n+1$.

7.10. Similarly as in Example 7.5.2 we get that the number of nonequivalent edge colorings of the cube using m colors is equal to

$$D_m = \frac{1}{24} = (m^{12} + 6m^7 + 3m^6 + 8m^4 + 6m^3).$$

For $m = 2$ and $m = 3$, we obtain that $D_2 = 218$ and $D_3 = 22815$.

7.11. A *simple polygon* is any regular p -gon whose vertices belong to the set of vertices of the given n -gon, and where p is a prime number less than or equal to n . Let X be the set of all simple polygons (including 2-gons). Let $\mathcal{P}$ be the set of all subsets of X . Let L_1 be the labeling with all labels equal

to $+1$. Any labeling that is equivalent to L_1 can be obtained by successive changing of the labels of all the vertices of the simple polygons. Hence, any labeling equivalent to L_1 is determined by a subset of X .

(a) If $n = 15$, then there are 8 simple polygons, namely 5 triangles, and 3 pentagons. There is a bijection between the set of all labelings and the set $\mathcal{P}$. Hence, the number of labelings that are equivalent to L_1 is 2^8 . Since the total number of labelings is 2^{15} , it follows that some labelings are not equivalent to L_1 .

(b) Suppose that $n = 30$. The number of simple polygons (including 2-gons) is $15 + 10 + 6 = 31$. Let Δ be a simple triangle, and Δ' be the triangle obtained from Δ by reflection about the point O . Let us first change the labels of the vertices of triangle Δ , and then change the labels of all 2-gons that have no common vertex with Δ' . The resulting labeling is the same as that obtained by changing the labels of all the vertices of triangle Δ' . Hence, we can exclude triangle Δ' from consideration. A similar remark holds for simple pentagons. It follows that all the labelings can be obtained using $15 + \frac{10}{2} + \frac{6}{2} = 23$ simple polygons. The number of labelings that are equivalent to L_1 is 2^{23} . The total number of labelings is 2^{30} , and hence some labelings are not equivalent to L_1 .

(c) *Answer.* Let $n = p_1 p_2 \dots p_m$, where $p_1, p_2, \dots, p_m$ are distinct prime numbers. There is a bijection between the set of nonequivalent labelings of the vertices of the n -gon and the set

$$\{(r_1, r_2, \dots, r_m) \mid 1 \leq r_j \leq p_j - 1, \text{ za } j = 1, 2, \dots, m\}.$$

It follows that the number of nonequivalent labelings, i.e., the number of classes of equivalence, is $2^{(p_1-1)(p_2-1)\dots(p_m-1)}$. If $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_m^{\alpha_m}$ is the canonical representation of positive integer n , then the number of nonequivalent labelings is $2^{\varphi(n)}$, where φ is Euler's totient function.

14.8 Solutions for Chapter 8

8.1. We shall prove that exactly n bus lines pass through every bus stop. Let l_1 be an arbitrarily chosen bus line, and $A_1, A_2, \dots, A_n$ be all the bus stops on bus line l_1 . Let B be a bus stop that is not on line l_1 . The existence of such a bus stop follows from the given conditions. It follows from condition (b) that every bus line that passes through B has exactly one bus stop on line l_1 . It follows from condition (a) that for any $k \in \{1, 2, \dots, n\}$, there is a bus line that connects B and A_k . Hence, there are exactly n bus lines that pass through B . Let us consider a bus stop A_k on bus line l_1 , and suppose that $A_j \neq A_k$. Let l_2 be the bus line that connects B and A_j . It follows from condition (b) that A_k is not on bus line l_2 (because l_2 passes through

n bus stops). Analogously we prove that there are exactly n bus lines that pass through A_k . Now it is easy to conclude that, for any $j \in \{1, 2, \dots, n\}$, there are exactly $n - 1$ bus lines different from l_1 that pass through A_j . From condition (b) all these bus lines are different from each other. It follows that the total number of bus lines is $n(n - 1) + 1$.

8.2. Let l_1 be an arbitrary bus line, and suppose that there are exactly n bus stops on line l_1 , where $n \geq 3$. Let B be a bus stop that is not on line l_1 . The same way as in Exercise 8.1 we conclude that there are exactly n lines that pass through B . Let l_2 be a bus line different from l_1 . It follows from condition (b) that there is exactly one bus stop common to lines l_1 and l_2 . Let us denote this bus stop by A . Let B_1 and B_2 be bus stops on lines l_1 and l_2 , respectively, such that $B_1 \neq A$ and $B_2 \neq A$. It follows from condition (a) that there is a line l_3 , such that l_3 passes through B_1 and B_2 . It follows from condition (b) that $l_3 \neq l_1$ and $l_3 \neq l_2$. It follows from condition (c) that there is a bus stop X on line l_3 , such that $X \neq B_1$, and $X \neq B_2$. Since X is not on line l_1 , there are exactly n lines that pass through X . Every line through X has exactly one bus stop on line l_2 , and for every bus stop on line l_2 there is a line that connects it with X . Hence, there are exactly n bus stops on line l_2 too. Since every line has exactly n bus stops, it follows from Exercise 8.1 that the total number of bus stops is $n(n - 1) + 1$. Hence, $n(n - 1) + 1 = 57$, i.e., $n = 8$.

8.3. Suppose there is a network of roads that connect cities $A_1, A_2, \dots, A_n$, such that conditions (a), (b), and (c) are satisfied. Let S be the set of all roads, and $A_i A_j$ be a one-way road with starting point A_i and endpoint A_j . For every $k \in \{1, 2, \dots, n\}$ let us denote $F_k = \{A_j \mid A_k A_j \in S\}$ and $T_k = \{A_j \mid A_j A_k \in S\}$. For every $A_m \in F_1$, it follows from condition (c) that there is a uniquely determined city A_t , such that $A_t A_m \in S$, $A_t A_1 \in S$, and $A_t \in T_1$. Hence, $|F_1| \leq |T_1|$. For every $A_m \in T_1$, it follows from condition (b) that there is a uniquely determined A_t , such that $A_m A_t \in S$, $A_1 A_t \in S$, and $A_t \in F_1$. Hence, $|T_1| \leq |F_1|$. Since $|F_1| \leq |T_1|$ and $|T_1| \leq |F_1|$, we conclude that $|F_1| = |T_1|$.

From condition (a) it follows that $n = 1 + |F_1| + |T_1| = 1 + 2|F_1|$, i.e., n is an odd positive integer. Hence, $n = 2r + 1$, where $r \in \mathbb{N}$. Then, $|F_1| = \dots = |F_n| = |T_1| = \dots = |T_n| = r$. Let us denote

$$X = \{(i, j) \mid i, j = 1, 2, \dots, 2r + 1, i \neq j\},$$

$$X_k = \{(i, j) \mid i \neq j, A_k A_i \in S, A_k A_j \in S\}.$$

It follows from condition (c) that, for every pair (i, j) , where $i \neq j$ and $i, j \in \{1, 2, \dots, n\}$, there is a uniquely determined $k \in \{1, 2, \dots, n\}$, such that $(i, j) \in X_k$. Hence, the sets $X_1, X_2, \dots, X_{2r+1}$ are pairwise disjoint, and $X = X_1 \cup X_2 \cup \dots \cup X_{2r+1}$. Since $|F_k| = r$ (i.e., exactly r cities can

be reached from city A_k by direct roads), it follows that $|X_k| = r(r-1)$. Obviously $|X| = (2r+1)2r$, and hence

$$(2r+1)2r = |X| = \sum_{j=1}^{2r+1} |X_j| = r(r-1)(2r+1). \quad (1)$$

Equality (1) implies that $r = 3$, i.e., $n = 7$. It remains to find an example of roads that connect cities $A_1, A_2, \dots, A_7$, and satisfy conditions (a), (b), and (c). Such an example is the following network consisting of 21 roads:

$$\begin{aligned} &12, 14, 16, 23, 24, 27, 31, 34, 35, 45, 46, 47, \\ &51, 52, 57, 62, 63, 65, 71, 73, 76. \end{aligned}$$

We have used the notation ij instead of A_iA_j . It is easy to check that conditions (a)–(c) are satisfied.

8.4. Let the cities be denoted by $A_1, A_2, \dots, A_k$, and let $x_1, x_2, \dots, x_k$ be direct lines from these cities, respectively. It follows from condition (b) that $x_1 + x_2 + \dots + x_k = 2(k-1)$. It is easy to conclude now that there is a positive integer $j \in \{1, 2, \dots, k\}$, such that $x_j < 2$. From condition (c) it follows that $x_j \geq 1$, and hence $x_j = 1$.

8.5. A k -network is a set of k cities and $k-1$ lines that satisfy conditions (a) and (c). Let $S = \{1, 2, \dots, n\}$ be a set consisting of n cities, S_1 be the set of all possible n -networks connecting these cities, and S_2 the set of $(n-2)$ -arrangements of the elements $1, 2, \dots, n$. Let us define function $f: S_1 \rightarrow S_2$ as follows. Let $s \in S_1$ be an arbitrarily chosen n -network. By the same arguments as in Exercise 8.4 it follows that there is a city directly connected to exactly one of the other cities. Let i_1 be the first such city in the sequence $1, 2, \dots, n$, and let j_1 be the city directly connected to i_1 . If we exclude city i_1 and the line that connects i_1 and j_1 , then the remaining cities and lines form an $(n-1)$ -network. Let i_2 be the first city in the sequence $1, 2, \dots, n$ without i_1 , that is connected directly to exactly one of the other cities in this $(n-1)$ -network. Such a city exists from Exercise 8.4. Let us denote by j_2 the city directly connected to i_2 . Then, we exclude city i_2 and the line that connects i_2 and j_2 , and obtain an $(n-2)$ -network. Then, we continue with this procedure. After $n-2$ steps in total, there will be $n-2$ excluded cities, denoted by $i_1, i_2, \dots, i_{n-2}$, and connected directly to $j_1, j_2, \dots, j_{n-2}$, respectively. The remaining two cities, one of which is j_2 , are also connected by a direct line, and form a 2-network. Let us define $f(s) = j_1j_2 \dots j_{n-2} \in S_2$. The function $f: S_1 \rightarrow S_2$ is obviously *one-to-one*. Let us prove that f is also an *onto* function. Note that the arrangement $j_1j_2 \dots j_{n-2} \in S_2$ is the image of the network consisting of the lines that connect the following cities:

$$\begin{array}{ll} j_1 & \text{and } i_1 = \min S \setminus \{j_1, j_2, \dots, j_{n-2}\}, \\ j_2 & \text{and } i_2 = \min S \setminus \{i_1, j_2, j_3, \dots, j_{n-2}\}, \\ j_3 & \text{and } i_3 = \min S \setminus \{i_1, i_2, j_3, \dots, j_{n-2}\}, \\ & \dots\dots\dots \\ j_{n-2} & \text{and } i_{n-2} = \min S \setminus \{i_1, i_2, \dots, i_{n-3}, j_{n-2}\} = S', \\ j_{n-2} & \text{and the unique city from } S'. \end{array}$$

The given set of lines is an n -network from S_1 . This is true because the first line connects two cities, and each of the next $n - 2$ lines includes a new city in the network or connects two networks. We have proved that the function $f: S_1 \rightarrow S_2$ is a bijection, and hence $|S_1| = |S_2| = n^{n-2}$.

8.6. Let the cities be denoted by $1, 2, \dots, n$, where n is unknown. Let us suppose that city 1 is connected directly to cities 2, 3, and 4. The number of cities that can be reached from 1, over 2, 3, or 4 with one additional flight, is at most 6. Hence, there are no more than $1 + 3 + 6 = 10$ cities in the state. Ten cities can be connected by airlines that satisfy the given condition as shown in Figure 14.8.1.

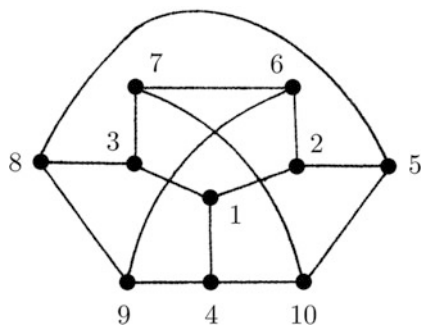


Fig. 14.8.1

8.7. Suppose that the passenger starts the trip at moment 0. Let us denote the moments of changing direction by $1, 2, 3, \dots$. A *direct path* is a road that connects two cities, but does not pass through a third city. Let n be the total number of direct paths. After a long time the passenger will take $4n + 1$ direct paths. It follows that the passenger will take some direct path that connects two cities, say A and B , at least 5 times. We also conclude that the passenger will take this path in a given direction, say from A toward

B , at least 3 times. The passenger will choose the same direction in B two times, for example, they will turn to the right two times. Suppose that this happens at moments i and j , where $i < j$. We conclude that the passenger will turn to the left at moments $i - 1$ and $j - 1$, they will turn to the right at moments $i - 2$ and $j - 2$, etc. It follows that the road that the passenger takes from moment 0 to moment i is the same as the road that they take from moment $j - i$ to moment j . Hence, the passenger will be in their own city at moment $j - i$.

8.8. Let A and B be two cities that are connected by a simple road AB . We shall prove that, in addition to simple road AB , there is a road that connects A and B with at least one more city on it. Let us consider the third city C . Then, there is a road L_1 from A to C such that B is not on L_1 , and there is a road L_2 from B to C such that A is not on L_2 . Each of the roads L_1 and L_2 may consist of more simple roads. Let us consider road L_1 from A to C , and the first city on this road that is also on road L_2 . This first city on $L_1 \cap L_2$, denoted by C^* , may be C , but not necessarily. Consider the part of road L_1 from A to C^* and the part of road L_2 from C^* to B . The union of these two parts is the road L_{AB} from A to B , and is not a simple road. Let us direct L_{AB} from A toward B . Let us also direct the simple road AB from B toward A . Let M_1 be the union of the directed roads L_{AB} and AB . From every city on road M_1 we can reach any other city on this road by moving in the allowed direction. If all the cities are on road M_1 , the proof is finished.

In the opposite case there is a city X that is not on road M_1 , and is connected to a city, say Y , on road M_1 by a simple road. By considering the cities on road M_1 that are different from Y , it is easy to see that there is a road L_{XZ} , from X to Z , such that Z is on road M_1 , $Z \neq Y$, and road L_{XZ} does not contain the simple road XY or any simple road which is a part of M_1 . Let us direct the road XY from Y to X , and the road L_{XZ} from X to Z . The network $M_2 = M_1 \cup XY \cup L_{XZ}$ satisfies the following condition: from every city in network M_2 we can reach any other city in this network by moving always in the allowed direction. If there is a city that is not included in network M_2 , then we can extend it similarly.

8.9. Let x_n be the maximal number of pairs of friends. By the method of mathematical induction “from k to $k + 2$ ” we shall prove that $x_n \leq \lceil n^2/4 \rceil$ for any $n \in \mathbb{N}$. It is easy to check that $x_1 = 0 = \lceil 1^2/4 \rceil$, $x_2 = 1 = \lceil 2^2/4 \rceil$, and $x_3 = 2 = \lceil 3^2/4 \rceil$. Let us suppose that $x_k \leq \lceil k^2/4 \rceil$ for some $k \in \mathbb{N}$. Let S be the set consisting of $k + 2$ persons satisfying the given conditions. Suppose that $A, B \in S$ are friends. Every $X \in S$ is friends with at most one of the persons A and B . It follows that

$$x_{k+2} \leq 1 + k + x_k \leq 1 + k + \left\lceil \frac{1}{4}k^2 \right\rceil = \left\lceil \frac{1}{4}k^2 + k + 1 \right\rceil = \left\lceil \frac{1}{4}(k+2)^2 \right\rceil.$$

Hence, $x_n \leq [n^2/4]$ for every $n \in \mathbb{N}$.

Let $n = 2m$, and let the set consisting of $2m$ persons be partitioned into two disjoint blocks consisting of m persons. Let us suppose that two persons are friends if and only if they belong to different blocks. Then, there are no 3 persons such that any two of them are friends, and the number of pairs (A, B) , such that A and B are friends, is

$$m^2 = \frac{1}{4}(2m)^2 = \frac{1}{4}n^2 = \left\lfloor \frac{1}{4}n^2 \right\rfloor.$$

Let $n = 2m - 1$ and let the set consisting of $2m - 1$ persons be partitioned into two blocks consisting of m and $m - 1$ persons. Again, suppose that two persons are friends if and only if they belong to different blocks. Then, there are no 3 persons such that any two of them are friends, and the number of pairs (A, B) , such that A and B are friends, is $m(m - 1) = 4m(m - 1)/4 = [(4m^2 - 4m + 1)/4] = [n^2/4]$. Finally, we get $x_n = [n^2/4]$ for any $n \in \mathbb{N}$.

8.10. Let us divide the chess players into two groups consisting of 12 and 13 members, and suppose that any two players have already played a game against each other if and only if they belong to the same group. The number of finished games is $\binom{12}{2} + \binom{13}{2} = 144$. For any three chess players there are two of them who belong to the same group, and hence these two players have already played a game against each other.

Let us now suppose that the number of finished games is $n \leq 144$, and, among any three of the chess players, there are two of them who have already played a game against each other. There is a player, say A , who played no more than 11 games. Let S_1 be the set of players who have already played against A , and S_2 be the set of players who have not played against A . Then, we have $|S_1| = k \leq 11$ and $|S_2| = 24 - k$. All players from S_2 played $\binom{24-k}{2}$ games against each other. Since $\binom{23}{2} > 144$ and $n \leq 144$, it follows that $k > 1$. Suppose that m pairs of players from S_1 have not played against each other, where $0 \leq m \leq \binom{k}{2}$. At least one chess player from every such pair (X, Y) has already played a game against B , where B is a fixed chess player from S_2 (in the opposite case consider the chess players X , Y , and B). The number of such pairs is not less than $m(24 - k)/(k - 1)$, because any player from S_1 can be a member of at most $k - 1$ pairs mentioned. Hence,

$$\begin{aligned} n &\geq \binom{24-k}{2} + k + \binom{k}{2} - m + \frac{m(24-k)}{k-1} \\ &\geq \binom{24-k}{2} + k + \binom{k}{2} = (k-11)(k-12) + 144 \geq 144. \end{aligned}$$

8.11. (a) Let A and B be friends. Let us suppose that $A_1, A_2, \dots, A_n, B$ are all friends of A . Then no two of $A_1, A_2, \dots, A_n, B$ are friends. Since A_1 and B are not friends, it follows that they have two common friends, say A and B_1 . Then B_1 and A are not friends, and hence the set of their common friends is $\{A_1, B\}$. It follows that none of $A_2, A_3, \dots, A_n$ is a friend of B_1 . Analogously we prove that there is $B_2 \neq B_1$, such that B_2 is a common friend of A_2 and B , etc. It follows that there are n mathematicians, $B_1 \neq A$, $B_2 \neq A$, $\dots$, and $B_n \neq A$, such that they are friends of B . It follows that the number of friends of B is not less than the number of friends of A . Analogously, the number of friends of A is not less than the number of friends of B . Finally we conclude that mathematicians A and B have the same number of friends.

(b) If X and Y are not friends, then they have a common friend, say Z . By the previous considerations it follows that they both have the same number of friends as Z .

14.9 Solutions for Chapter 9

9.1. Answer: $k - n + 1$. **9.2.** Answer: $\binom{n}{2} - n + 1 = \frac{(n-1) - (n-2)}{2}$.

9.3. Let v , e and f be the number of vertices, edges, and faces respectively, of the complete regular graph G , and v^* be the number of vertices of the dual graph G^* . Then we have $v^* = f$, and

$$kv = 2e. \quad (1)$$

$$k^*v^* = k^*f = 2e. \quad (2)$$

It follows from (1) and (2) that $e = \frac{kv}{2}$, and $f = \frac{2e}{k^*} = \frac{kv}{k^*}$. By replacing these values for e and f in Euler's formula, it follows that

$$v + \frac{kv}{k^*} - \frac{kv}{2} = 2. \quad (3)$$

Equality (3) is equivalent to $v(2k + 2k^* - kk^*) = 4k^*$, implying that $2k + 2k^* - kk^* > 0$, or equivalently $(k-2)(k^*-2) < 4$.

9.4. (a) Let us consider the inequality $(k-2)(k^*-2) < 4$, and suppose that $k-2 > 0$ and $k^*-2 > 0$. It follows that $k-2, k^*-2 \in \{1, 2, 3\}$. All solutions (k, k^*) , and related values of v , e , and f are given in Table 8.2.1. The case where $k-2$ and k^*-2 are not both positive leads to trivial solutions.

(b) The result follows from the data given in Table 8.2.1.

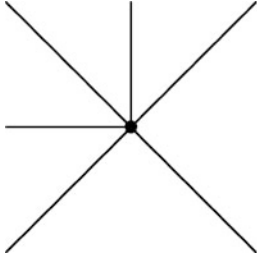


Fig. 14.9.1

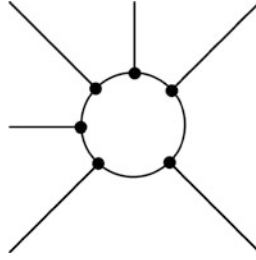


Fig. 14.9.2

9.5. Every vertex A of degree 2 can be removed, and two edges that are incident to A can be replaced by one edge. Every vertex of degree $k > 3$ can be removed and replaced by a new face with k vertices of order 3. Figures 14.9.1 and 14.9.2 show how this transformation works. If the new graph can be face-colored with no more than n colors, this can also be done for the original graph. After coloring the new graph we can simply contract every new face into the point.

9.6. Let G be a 3-regular polygonal graph. Let v , e , and f be the number of vertices, edges, and faces respectively, of graph G . For any $k \geq 2$, let f_k be the number of faces with k vertices (and k edges). Some f_k 's are equal to 0. Then the following equalities hold:

$$f = \sum_{k \geq 2} f_k, \quad v = \frac{1}{3} \sum_{k \geq 2} k f_k, \quad e = \frac{1}{2} \sum_{k \geq 2} k f_k. \quad (1)$$

Euler's formula and the equalities (1) imply that

$$\begin{aligned} 2 = v + f - e &= \sum_{k \geq 2} f_k + \frac{1}{3} \sum_{k \geq 2} k f_k - \frac{1}{2} \sum_{k \geq 2} k f_k \\ &= \frac{1}{3} f_2 + \frac{1}{4} f_3 + \frac{1}{6} f_4 + \frac{1}{12} f_5 + \sum_{k \geq 7} \left(1 - \frac{k}{6}\right) f_k. \end{aligned} \quad (2)$$

Since $1 - \frac{k}{6} < 0$ for any $k \geq 7$, it follows from equality (2) that at least one of the integers f_2 , f_3 , f_4 , and f_5 is greater than 0.

14.10 Solutions for Chapter 10

10.1. *Hint:* Use the algorithms from Chapter 10, Section 10.1.

10.2. The sum of positive integers in every row, every column, and every diagonal of the magic square M of order n^2 is $S = n(n^2 + 1)/2$. The sum of the positive integers in every row, every column, and every diagonal of the square table M^* is

$$nS + (nS - n)n^2 = \frac{n^2(n^4 + 1)}{2} = \frac{1}{n^2}(1 + 2 + \cdots + n^4).$$

It follows that M^* is a magic square.

10.3. Let L be a Latin square of order $2n + 1$. Each of the positive integers $1, 2, \dots, 2n + 1$ appears $2n + 1$ times in L . Since the arrangement of the entries is symmetric with respect to the main diagonal of L , it follows that each entry appears in an even number of fields that are not on the diagonal. Hence, all of the positive integers $1, 2, \dots, 2n + 1$ appear on the diagonal.

10.4. A Latin square of the order 2^n can be constructed inductively. Let A_n be a Latin square of the order 2^n . It is shown in Figure 14.10.1 how a Latin square A_{n+1} of the order 2^{n+1} can be obtained from the Latin square A_n . A Latin square of the order $8 = 2^3$ is given in Figure 14.10.2.

$A_n + 2^n$	A_n
A_n	$A_n + 2^n$

Fig. 14.10.1

8	7	6	5	4	3	2	1
7	8	5	6	3	4	1	2
6	5	8	7	2	1	4	3
5	6	7	8	1	2	3	4
4	3	2	1	8	7	6	5
3	4	1	2	7	8	5	6
2	1	4	3	6	5	8	7
1	2	3	4	5	6	7	8

Fig. 14.10.2

10.5. The answer is positive. The arrangement of positive integers in the unit squares can be obtained as follows. First we put 1 into the square $[0, 1] \times [0, 1]$. Then we construct successively Latin squares of the order 2, 4, 8, $\dots$ according to the algorithm from Exercise 10.4.

10.6. Answer: $(n!)^2 \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!} \right)$.

10.7. The number of systems of distinct representatives of the sets S_1, S_2, S_3 , and S_4 is 9. All systems are the following: 2143, 2341, 2413, 3142, 3412, 3421, 4123, 4312, and 4321.

10.8. Any system of distinct representatives of the sets $S_1, S_2, \dots, S_n$ is a permutation $a_1 a_2 \dots a_n$ of the set $\{1, 2, \dots, n\}$, such that, $a_j \neq j$ for any j . The number of such permutations is determined in Chapter 4, Exercise 4.15.

10.9. For every $k \in \{1, 2, \dots, n\}$, let $S_k \subset \{1, 2, \dots, n\}$ be the set of numbers that do not appear in the k -th column of the Latin rectangle. Let us consider the incidence matrix of the elements $1, 2, \dots, n$ related to the sets $S_1, S_2, \dots, S_n$. There are exactly $n - m$ units in every row and every column of this matrix. From Example 10.3.5 there is a s.d.r. $(x_1, x_2, \dots, x_n)$ for the sets $(S_1, S_2, \dots, S_n)$. If we extend the given Latin rectangle $m \times n$ by adding $(x_1, x_2, \dots, x_n)$ as the $(m+1)$ -st row, then the obtained table $(m+1) \times n$ is a Latin rectangle as well. If $m+1 = n$, the obtained table is a Latin square of the order n . If $m+1 < n$, we continue by adding new rows as before.

10.10. (a) We shall prove that the given condition is necessary for the existence of a common system of distinct representatives. Let $(x_1, x_2, \dots, x_n)$ be a common system of distinct representatives of collections $\{A_1, A_2, \dots, A_n\}$ and $\{B_1, B_2, \dots, B_n\}$. Let us suppose that for some $k \in \{1, 2, \dots, n\}$ and $1 \leq i_1 < i_2 < \dots < i_k \leq n$, the set

$$A_{i_1} \cup A_{i_2} \cup \dots \cup A_{i_k} \quad (1)$$

contains at least $k+1$ of the sets $B_1, B_2, \dots, B_n$. Then set (1) contains at least $k+1$ of the elements $x_1, x_2, \dots, x_n$. Hence, the number of elements $x_1, x_2, \dots, x_n$ that are not contained in set (1) is less than $n - k$. Among them it is not possible to find distinct representatives for the remaining $n - k$ sets from the collection $\{A_1, A_2, \dots, A_n\}$, and this conclusion contradicts the assumption that $(x_1, x_2, \dots, x_n)$ is a common s.d.r. Hence, the given condition is necessary.

(b) Let us suppose that the given condition holds. We shall use the following notation: $X = \{A_1, A_2, \dots, A_n\}$, and $X_j = \{A_i \mid A_i \cap B_j \neq \emptyset\}$, $j = 1, 2, \dots, n$. Let us prove that the n -arrangement $(X_1, X_2, \dots, X_n)$ of the subsets of set X satisfies the necessary condition (and by Theorem 10.3.3 is a sufficient condition for the existence of a s.d.r. as well). Let us suppose, on the contrary, that, for example,

$$X_1 \cup X_2 \cup \dots \cup X_k \cup X_{k+1} = \{A_{i_1}, A_{i_2}, \dots, A_{i_k}\}. \quad (2)$$

Since every of the blocks $A_1, A_2, \dots, A_n$ that is not contained in set (2) has an empty intersection with any of the sets $B_1, B_2, \dots, B_k, B_{k+1}$, it follows that $B_1 \cup B_2 \cup \dots \cup B_k \cup B_{k+1} \subset A_{i_1} \cup A_{i_2} \cup \dots \cup A_{i_k}$, and this conclusion contradicts the given condition. Hence, by Theorem 10.3.3 there is an n -tuple $(A_{i_1}, A_{i_2}, \dots, A_{i_n})$ which is a s.d.r. for $(X_1, X_2, \dots, X_n)$. Then, $A_{i_m} \in X_m$ for every $m \in \{1, 2, \dots, n\}$, i.e., $A_{i_m} \cap B_m \neq \emptyset$. Let us choose $x_m \in$

$A_{i_m} \cap B_m$. The obtained n -tuple $(x_1, x_2, \dots, x_n)$ is a common s.d.r. for the given collections.

10.11. The statement follows from Exercise 10.10.

10.12. Let us denote $S = \{(i, j) \mid i = 0, 1, 2, \dots, n^{n+1}, j = 0, 1, 2, \dots, n\}$. Each of the lines $x = i$, where $i \in \{0, 1, \dots, n^{n+1}\}$, contains $n+1$ points from S . The number of such lines is $n^{n+1} + 1$. The number of different labelings of $n+1$ points using the labels $1, 2, \dots, n$ is n^{n+1} . By the pigeonhole principle it follows that there are integers $x_1, x_2 \in \{0, 1, 2, \dots, n^{n+1}\}$, such that $x_1 < x_2$, and sequences of points $(x_1, 0), (x_1, 1), \dots, (x_1, n)$, and $(x_2, 0), (x_2, 1), \dots, (x_2, n)$, are both labeled by the same $(n+1)$ -arrangement of positive integers $1, 2, \dots, n$. Again by the pigeonhole principle it follows that there are two points in the first sequence, say (x_1, y_1) and (x_1, y_2) , with $0 \leq y_1 < y_2 \leq n$, that are labeled the same positive integer. Then points $(x_1, y_1), (x_2, y_1), (x_2, y_2)$, and (x_1, y_2) are the vertices of a rectangle, and they are all labeled the same positive integer.

10.13. Let n be a positive integer. There are two positive integers in the set $\{n, n+1, \dots, n+19\}$ with the digit 0 in the last position of their decimal representation. One of these two positive integers, denoted by a , has a digit different from 9 in the second-to-last position of its decimal representation. Note that $a + 19 \leq n + 19 + 19 = n + 38$. Let S be the sum of the digits of a , and let us consider the sum of the digits of any of the positive integers $a, a+1, \dots, a+19$. Each of the positive integers $S, S+1$, and $S+10$ appears among these sums, and hence one of them is divisible by 11.

10.14. If such a labeling exists, then each of the digits $0, 1, 2, \dots, 9$ is used at least 5 times as a label. Since $45 < 5 \cdot 10$, such a labeling does not exist.

10.15. Let S be the set of all members of the committee. For every meeting there were $\binom{10}{2} = 45$ pairs $\{A, B\}$ such that A and B both attended this meeting. For all meetings there were $45 \cdot 40 = 1800$ such pairs. Let us suppose that $|S| \leq 60$. Then the total number of pairs $\{A, B\}$, such that $A \in S$ and $B \in S$, is not greater than $\binom{60}{2} = 1770$. Since $1770 < 1800$, the last conclusion contradicts the assumption that any two members of the committee attended the same meeting at most once. Hence, $|S| > 60$.

10.16. Let $2k+1$ be the number of planets, $k \in \mathbb{N}$. Let A and B be planets with the minimal distance between them. Then the astronomer from A observes planet B , and the astronomer from B observes planet A . We shall prove the statement by induction on k . For $k=1$, there is only one more planet in addition to A and B and neither of the astronomers observes it. Let us suppose that the statement holds for any system consisting of $2(k-1)+1 = 2k-1$ planets, and satisfying the conditions.

Let us now consider a system consisting of $2k+1$ planets, and satisfying the conditions. A and B are planets with the minimal distance between them. Let us denote the remaining $2k-1$ planets by $A_1, A_2, \dots, A_{2k-1}$.

Case 1. At least one of the astronomers located on $A_1, \dots, A_{2k-1}$ observes A or B . It follows that the remaining $2k-1$ planets are observed by at most $2k-2$ astronomers. Hence, there is a planet such that none of the astronomers observes it.

Case 2. None of the astronomers located on $A_1, \dots, A_{2k-1}$ observes A or B . The set consisting of $A_1, \dots, A_{2k-1}$ satisfies the given condition, and by the induction hypothesis there is a planet that is not observed.

10.17. No two of the integers from the set $\{0, 1, 2, 8, 9\}$ can be used as the labels of two adjacent vertices of the 10-gon. It follows that the vertices labeled 0, 1, 2, 8, 9 are separated by the remaining 5 vertices of the 10-gon. None of these five vertices can be labeled 7. It follows that there is no labeling that satisfies the given condition.

10.18. Let us denote $S_{kr} = \{(2k-1, 2r-1), (2k-1, 2r), (2k, 2r-1), (2k, 2r)\}$. The set S is the union of the pairwise disjoint sets S_{kr} , $k, r \in \{1, 2, \dots, 50\}$. It is obvious that the set A with the given property may contain at most one element from any of the sets S_{kr} , $k, r \in \{1, 2, \dots, 50\}$. The set

$$A_0 = \{(2k-1, 2r-1) \mid k = 1, \dots, 50, r = 1, \dots, 50\}$$

contains exactly one point from any of those sets and has the given property. Since $|A_0| = 2500$, it follows that the maximal number of elements of a set with the given property is 2500.

10.19. The sum of elements of any nonempty subset of the set S is not greater than $90 + 91 + 92 + \dots + 99 = 945$, and not less than 10. Hence, any such sum belongs to the set $V = \{10, 11, \dots, 945\}$, consisting of 936 elements. Since the set S has $2^{10} - 1 = 1023$ distinct nonempty subsets, it follows from the pigeonhole principle that there are two distinct subsets $A, B \subset S$, such that the sum of elements contained in A is equal to the sum of elements contained in B . Obviously, neither of the sets A and B is a subset of the other one. If $A \cap B = \emptyset$, the proof is finished. If $A \cap B \neq \emptyset$, then the sets $A \setminus B$ and $B \setminus A$ satisfy the required condition.

10.20. Since $|A_k| > |S|/2$ for any $k \in \{1, 2, \dots, 1066\}$, it follows that $|A_1| + |A_2| + \dots + |A_{1066}| > 533|S|$. By the pigeonhole principle we conclude that there is an element $x_1 \in S$ contained in at least 534 of the sets $A_1, A_2, \dots, A_{1066}$. Let us denote these 534 sets by $B_{533}, B_{534}, \dots, B_{1066}$. Similarly, we conclude that there are elements $x_2, x_3, \dots, x_{10}$, such that

$$\begin{aligned}
x_2 &\in B_{266} \cap B_{267} \cap \cdots \cap B_{532}, & x_3 &\in B_{133} \cap B_{134} \cap \cdots \cap B_{265}, \\
x_4 &\in B_{66} \cap B_{67} \cap \cdots \cap B_{132}, & x_5 &\in B_{33} \cap B_{34} \cap \cdots \cap B_{65}, \\
x_6 &\in B_{16} \cap B_{17} \cap \cdots \cap B_{32}, & x_7 &\in B_8 \cap B_9 \cap \cdots \cap B_{15}, \\
x_8 &\in B_4 \cap B_5 \cap B_6 \cap B_7, & x_9 &\in B_2 \cap B_3, \quad x_{10} \in B_1,
\end{aligned}$$

where $(B_1, B_2, \dots, B_{1066})$ is a permutation of the set $\{A_1, A_2, \dots, A_{1066}\}$. Then, each of the sets $A_1, A_2, \dots, A_{1066}$ contains at least one of the elements $x_1, x_2, \dots, x_{10}$.

10.21. Let us assume that the given statement does not hold. Since $65 \cdot 5 = 325 < 330$, there are 66 members of the society from the same country, denoted by A . Let $a_1, a_2, \dots, a_{66}$ be their labels in increasing order. It is obvious that $a_2 - a_1, a_3 - a_1, \dots, a_{66} - a_1$ are then the labels of 65 members of the society from the remaining four countries. Now we conclude that 17 of these 65 members are from the same country, denoted by B . Let $b_1, b_2, \dots, b_{17}$ be their labels in increasing order. Then $b_2 - b_1, b_3 - b_1, \dots, b_{17} - b_1$ are the labels of 16 members of the society from the remaining three countries. (Note that, for every $i \in \{2, 3, \dots, 17\}$, there are $j, k \in \{2, 3, \dots, 66\}$, such that $j < k$, and $b_i - b_1 = (a_k - a_1) - (a_j - a_1) = a_k - a_j$. Hence, the member labeled $b_i - b_1$ does not come from country A . It is obvious that none of these 16 members is from country B .) Five of these 16 members come from the country $C \notin \{A, B\}$. Let c_1, c_2, c_3, c_4 , and c_5 be their labels in increasing order. Then $c_2 - c_1, c_3 - c_1, c_4 - c_1$, and $c_5 - c_1$ are the labels of 4 members from the remaining two countries. (It is obvious that none of these 4 members comes from country C . Prove that none of them comes from A or B as well!) Two of these four members come from the same country $D \notin \{A, B, C\}$. Let d_1 and d_2 be their labels, where $d_1 < d_2$.

Consider the member labeled $d_2 - d_1$. Any assumption about the country they come from contradicts the initial assumption.

10.22. Let $a_1, a_2, \dots, a_{20}$ be the heights of the boys, and $b_1, b_2, \dots, b_{20}$ be the heights of the girls, where $a_1 > a_2 > \cdots > a_{20}$ and $b_1 > b_2 > \cdots > b_{20}$. Let us suppose that there is a positive integer $k \in \{1, 2, \dots, 20\}$, such that $a_k - b_k \geq 10$. Let k_0 be the smallest positive integer with this property. In the first arrangement of pairs, all boys with heights $a_1, a_2, \dots, a_k$ can dance only with girls whose heights are $b_1, b_2, \dots, b_{k-1}$. This conclusion contradicts the assumption that there are 20 pairs satisfying the given condition.

10.23. Let $[a_1, b_1], [a_2, b_2], \dots, [a_{mn+1}, b_{mn+1}]$ be the given intervals, and $D_1 = \{b_1, b_2, \dots, b_{mn+1}\}$. Some elements of set D_1 may be equal to each other. Let us denote $x_1 = \min D_1$, and let I_1 be one of the given intervals with the right endpoint x_1 . Let D_2 be the set of right endpoints of the given

intervals that do not contain the point x_1 . Let us denote $x_2 = \min D_2$, and let I_2 be one of the given intervals with right endpoint x_2 . Then $x_1 < x_2$, and the intervals I_1 and I_2 are disjoint. Let D_3 be the set of right endpoints of the given intervals that contain no points from the set $\{x_1, x_2\}$. Let us denote $x_3 = \min D_3$, and let I_3 be one of the given intervals with right endpoint x_3 . Then, $x_1 < x_2 < x_3$, and the intervals I_1 , I_2 , and I_3 are pairwise disjoint. If we proceed this way, we will obtain real numbers $x_1, x_2, \dots, x_k$ and intervals $I_1, I_2, \dots, I_k$, such that the following conditions are satisfied:

- (a) $x_1 < x_2 < \dots < x_k$;
- (b) For any $j \in \{1, 2, \dots, k\}$, x_j is the right endpoint of I_j ;
- (c) The intervals $I_1, I_2, \dots, I_k$ are pairwise disjoint;
- (d) Any of $mn + 1$ given intervals contains a real number from the set $\{x_1, x_2, \dots, x_k\}$.

If $k \geq n + 1$, the proof of the statement is finished. If $k \leq n$, then from the pigeonhole principle and condition (d) we conclude that some of the real numbers $x_1, x_2, \dots, x_k$ belong to at least $m + 1$ of the given intervals.

10.24. It is easy to prove that the area of any triangle whose vertices have integer coordinates is an integer or half of an integer. Since the area of the given circle is $1990^2\pi$, it follows that the area of any triangle whose vertices belong to the given set of 555 points belongs to the set consisting of $[2\pi \cdot 1990^2] + 1$ distinct real numbers. The number of triangles whose vertices belong to the set of given points is $\binom{555}{3} > [2\pi \cdot 1990^2] + 1$. By the pigeonhole principle it follows that two of them have the same area.

10.25. Each of the given rectangles is determined by a pair (a, b) , where a is the length, and b is the altitude. Let us consider the following 50 sequences of rectangles (not all of these rectangles are given):

1. $(1, 1), (1, 2), \dots, (1, 100), (2, 100), (3, 100), \dots, (100, 100),$
2. $(2, 2), (2, 3), \dots, (2, 99), (3, 99), (4, 99), \dots, (99, 99),$
3. $(3, 3), (3, 4), \dots, (3, 98), (4, 98), (5, 98), \dots, (98, 98),$
.....
49. $(49, 49), (49, 50), (49, 51), (49, 52), (50, 52), (51, 52), (52, 52),$
50. $(50, 50), (50, 51), (51, 51).$

All the given rectangles appear in this table, and none of them appears twice. All of the sequences in the table are increasing with respect to the relation $\subset$. Since $2000 > 50 \cdot 39$, it follows by the pigeonhole principle that 40 of the given rectangles belong to the same sequence.

10.26. Let $ABCD$ be the given square, and A_1, B_1, C_1 , and D_1 be the midpoints of the segments AB, BC, CD , and DA , respectively. There are two points, denoted by X and Y , such that each of them divides the segment A_1C_1 into two parts with the ratio of their lengths $2/3$. Analogously, there are two points, denoted by U and V , such that each of them divides the segment B_1D_1 into two parts with the ratio of their lengths $2/3$. Any line that divides the square $ABCD$ into two quadrilaterals with the ratio of their areas $2/3$ must contain one of the points X, Y, U , and V . (Prove this statement!) It follows by the pigeonhole principle that at least three of the given lines contain one of these four points.

10.27. Suppose that circle c_2 is fixed. Let us put circle c_1 on circle c_2 , and rotate c_1 around the center. Suppose that the fixed point $A \in c_1$ leaves a trail when some of the given points on c_1 belong to some of the given arcs on c_2 . After a 360° rotation the total length of the trail is less than $100 \cdot 1 = 100$. Hence, there is a point $X \in c_2$, that does not belong to the trail. At the moment when A overlaps X , none of the given points is inside one of the given arcs.

10.28. Let $\mathcal{K}$ be the given square with side 15. Let us denote the point of intersection of its diagonals by O , and call it the center of the square. Let $\mathcal{K}_1$ be a square with side 13, with center O , and sides that are parallel to the sides of $\mathcal{K}$. Let $\mathcal{S}$ be a square with side 1 that lies inside square $\mathcal{K}$. Let $\mathcal{S}^*$ be the set of all points A , such that $d(A, \mathcal{S}) \leq 1$. Note that $\mathcal{S}^*$ consists of 5 unit squares and four quarters of the unit disc. Hence, $\text{area}(\mathcal{S}^*) = 5 + \pi$. The center of the circle of radius 1 that we are trying to find must be inside square $\mathcal{K}_1$, and outside each of the 20 figures that are congruent to $\mathcal{S}^*$. Hence, it is sufficient to prove that the sum of the areas of these 20 figures is less than the area of square $\mathcal{K}_1$. It is easy to check that the inequality $20(5 + \pi) < 13^2$ really holds.

10.29. Let k be the given circle of diameter 100, and k_1 be the concentric circle with diameter 97. Let $l_i, i \in \{1, 2, \dots, 32\}$, be the given lines. Let l'_i and l''_i be two distinct lines that are parallel to $l_i, 1 \leq i \leq 32$, and such that $d(l_i, l'_i) = 1.5$ and $d(l_i, l''_i) = 1.5$, where $d(l_i, l'_i)$ is notation for the distance between lines l_i and l'_i , see Figure 14.10.3. The part of the plane α between two parallel lines will be called a *strip*. Let S_i be the strip between lines l'_i and l''_i . The center of a circle of diameter 3 which has the given property (if such a circle exists) is an inner point of circle k_1 and must not belong to any of the strips $S_1, S_2, \dots, S_{32}$. Now, it is sufficient to prove that strips $S_1, S_2, \dots, S_{32}$ do not cover the whole disc D_1 bounded by circle k_1 . Without loss of generality we can assume that each of the lines l'_i and $l''_i, i \in \{1, 2, \dots, 32\}$, has a common point with circle k_1 .

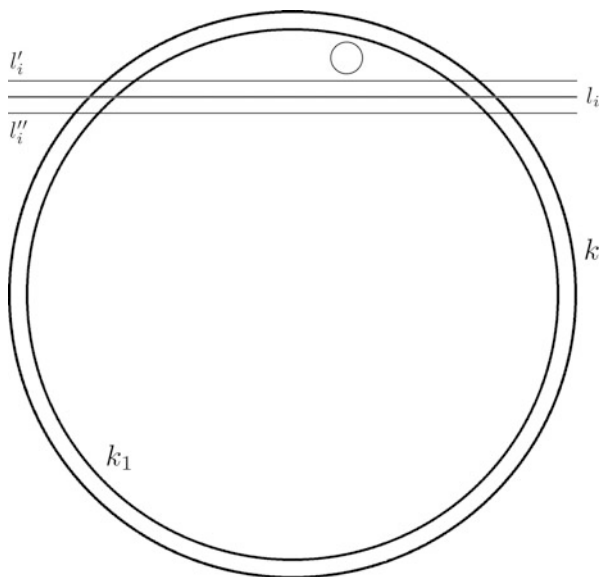


Fig. 14.10.3

Let $\mathcal{S}_1$ be the sphere with the great circle k_1 . The planes α'_i and α''_i that contain lines l'_i and l''_i , respectively, such that $\alpha'_i \perp \alpha$ and $\alpha''_i \perp \alpha$, cut a part of sphere $\mathcal{S}_1$ that is called a *spherical zone*. Let $\mathcal{Z}_i$ be the spherical zone cut by the planes α'_i and α''_i . The surface area of the spherical zone (which excludes the bases) is given by $2\pi Rh$, where R is the radius of the sphere, and h is the distance between the parallel planes that cut the zone. Note that the surface area of the spherical zone depends on the distance between the parallel planes, but does not depend on the distances of these planes from the center of the sphere. The distance between the parallel planes that cut the zone is called the height of the zone. Note that the sum of the surface area of the spherical zones $\mathcal{Z}_1, \dots, \mathcal{Z}_{32}$ is $32\pi \cdot 97 \cdot 3 = 96 \cdot 97\pi$. Since the surface area of sphere $\mathcal{S}_1$ is $97^2\pi$, it follows that there is a point $X \in \mathcal{S}_1$, such that $X \notin \mathcal{Z}_1 \cup \dots \cup \mathcal{Z}_{32}$. Let A be the orthogonal projection of point X onto plane α . Circle k_0 with center A and radius 1.5 lies inside circle k_1 and does not have common points with lines $l_1, \dots, l_{32}$.

10.30. It is necessary and sufficient to prove that there are positive integers m and k , such that

$$\overline{d_1 d_2 \dots d_n \underbrace{00 \dots 0}_m} \leq 2^k \leq \overline{d_1 d_2 \dots d_n \underbrace{99 \dots 9}_m}. \quad (1)$$

Inequalities (1) are equivalent to the following:

$$\log_2 \overline{d_1 d_2 \dots d_n} + m \leq k \log_2 2 < \log_2 (\overline{d_1 d_2 \dots d_n} + 1) + m. \quad (2)$$

Let us introduce the following notation:

$$\begin{aligned} x &= \log \overline{d_1 d_2 \dots d_n}, & y &= \log (\overline{d_1 d_2 \dots d_n} + 1), \\ I_m &= [x + m, y + m), & m &= 1, 2, \dots \end{aligned}$$

We shall prove that there are positive integers k and m , such that $k \log_2 2 \in S_m$. Let $\mathcal{K}$ be a circle of circumference 1, and A be an arbitrarily chosen point on it. Let $f : [0, +\infty) \rightarrow \mathcal{K}$ be the function that winds the half-line $[0, +\infty)$ around circle $\mathcal{K}$, such that $f(k) = A$ for any $k \in \{0, 1, 2, \dots\}$. Let us denote $\mathcal{L} = f(I_1)$. The length of the arc $\mathcal{L}$ is $y - x$, and $\mathcal{L} = f(I_m)$ for every $m \in \mathbb{N}$.

It remains to prove that there is a positive integer $k \geq \log_2 x / \log_2 2$, such that $f(k \log_2 2) \in \mathcal{L}$. Let r be a positive integer such that $1/r < y - x$. Let us divide circle $\mathcal{K}$ into arcs $\mathcal{L}_1, \mathcal{L}_2, \dots, \mathcal{L}_r$ of length $1/r$, and consider the sequence $k \log_2 2$, where $k \geq \log_2 x / \log_2 2$. By the pigeonhole principle it is easy to conclude that there are positive integers $k_1, k_2 \geq \log_2 x / \log_2 2$ and a positive integer $i \in \{1, 2, \dots, r\}$, such that $f(k_1 \log_2 2) \in \mathcal{L}_i$ and $f(k_2 \log_2 2) \in \mathcal{L}_i$. The arc distance between the points $f(k_1 \log_2 2)$ and $f(k_2 \log_2 2)$ is less than $y - x$. Let us denote $d = k_2 - k_1$, and consider the sequence of points $A_j = f((k_1 + jd) \log_2 2)$, $j = 0, 1, 2, \dots$. The arc distance between the adjacent points in this sequence is less than $y - x$, and hence some of these points belong to $\mathcal{L}$.

10.31. Compare this Exercise with Example 10.5.4.

10.32. Let us suppose that every edge of the complete p -graph G_p is colored blue or red. If all the edges are blue, then G is a blue p -graph. If there is a red edge, then this edge is a red 2-subgraph. On the other hand, if all the edges of the complete $(p-1)$ -graph G_{p-1} are blue, then G_{p-1} does not contain a blue p -subgraph, or a red 2-subgraph. Hence, $R(p, 2; 2) = p$.

10.33. (a) By equality (10.5.1) and inequality (10.5.4) it follows that:

$$\begin{aligned} R(p+1, q+1; 2) &\leq R(R(p, q+1; 2), R(p+1, q; 2); 1)+1 \\ &= R(p, q+1; 2) + R(p+1, q; 2). \end{aligned}$$

(b) Suppose that $R(p, q+1; 2) = 2a$, $R(p+1, q; 2) = 2b$, where a and b are positive integers, and let $n = 2a + 2b - 1$. Let us consider the complete n -graph G whose edges are colored blue or red. Let A be a vertex of the graph G , n_1 be the number of blue edges incident to A , and n_2 be the

number of red edges incident to A . Let G_1 be the complete n_1 -subgraph whose vertices are connected to A by blue edges, and let G_2 be the complete n_2 -subgraph whose vertices are connected to A by red edges. Then, we have $n - 1 = n_1 + n_2 = R(p, q + 1; 2) + R(p + 1, q; 2) - 2$. Let us consider the following three cases:

Case 1. $n_1 \geq R(p, q + 1; 2)$, $n_2 < R(p + 1, q; 2)$. By definition of the number $R(p, q + 1; 2)$, the graph G_1 has a blue p -subgraph (by adding vertex A with blue edges incident to A we obtain a blue $(p + 1)$ -subgraph), or G_1 has a red $(q + 1)$ -subgraph.

Case 2. $n_1 < R(p, q + 1; 2)$, $n_2 \geq R(p + 1, q; 2)$. In this case G_2 has a blue $(p + 1)$ -subgraph or a red q -subgraph (in the latter case we obtain a red $(q + 1)$ -subgraph by adding vertex A with red edges incident to A).

Case 3. $n_1 = 2a - 1$, $n_2 = 2b - 1$.

In this case we consider the other vertices of graph G . If there is a vertex different from A that satisfies the conditions of Case 1 or Case 2, then we get the same conclusions as above.

It remains to prove that there is a vertex of graph G , such that the conditions of Case 3 are not satisfied. If all the vertices satisfy these conditions, then the number of blue edges of graph G is $(2a + 2b - 1)(2a - 1)/2$, i.e., we obtain a contradiction. Hence, we have proved that $R(p + 1, q + 1; 2) \leq 2a + 2b - 1 < R(p, q + 1; 2) + R(p + 1, q; 2)$.

10.34. We shall prove the inequality by the method of mathematical induction on $p + q$. Let us first consider the case $p + q = 4$, i.e., $p = 2$, $q = 2$. It follows from Example 10.5.5 that $R(3, 3; 2) = 6 = \binom{4}{2} = \binom{2+2}{2}$. Let us now suppose that

$$R(p, q + 1; 2) \leq \binom{p + q - 1}{p - 1}, \quad R(p + 1, q; 2) \leq \binom{p + q + 1}{p}.$$

Using these two inequalities and the inequality that was proved in Exercise 10.33, we obtain that

$$\begin{aligned} R(p + 1, q + 1; 2) &\leq R(p, q + 1; 2) + R(p + 1, q; 2) \\ &\leq \binom{p + q - 1}{p - 1} + \binom{p + q - 1}{p} = \binom{p + q}{p}. \end{aligned}$$

10.35. Let us consider a complete 8-graph whose vertices are labeled 1, 2, 3, 4, 5, 6, 7, and 8, with edge coloring such that the edges 15, 26, 37, 48, 13, 35, 57, and 71 are blue, and the remaining edges are red. It is easy to check that there is no blue 3-subgraph, and there is no red 4-subgraph.

Hence, $R(3, 4; 2) > 8$. On the other hand, it follows from Exercise 10.34, Exercise 10.33, and Example 10.5.4 that $R(3, 4; 2) < R(3, 3; 2) + R(2, 4; 2) = 6 + 4 = 10$. Hence, $R(3, 4; 2) = 9$.

10.36. Let G be a complete 13-graph whose vertices are labeled $0, 1, \dots, 12$. Consider an edge coloring defined as follows. An edge (i, j) is blue if and only if $|i - j| \in \{2, 3\}$. The remaining edges are red. Then, there is no blue 3-subgraph, and there is no red 5-subgraph. Hence, $R(3, 5; 2) > 13$. Using Exercises 10.32, 10.33, and 10.34 we obtain that $R(3, 5; 2) \leq R(2, 5; 2) + R(3, 4; 2) = 5 + 9 = 14$. Hence, $R(3, 5; 2) = 14$.

10.37. Let us consider a complete 17-graph whose edges are colored blue, yellow, or red. Let A be a vertex of the graph. There are 16 edges incident to A . At least 6 of these edges are of the same color. Suppose that 6 edges incident to A are blue, and let $B_1, B_2, \dots, B_6$ be vertices connected to A by these blue edges. If there is a blue edge $B_i B_j$, then $AB_j B_k$ is a blue 3-graph (a blue triangle). In the opposite case all the edges of the complete graph with vertices B_1, B_2, B_3, B_4, B_5 , and B_6 are yellow or red. Let us consider the edges $B_1 B_2, B_1 B_3, B_1 B_4, B_1 B_5$, and $B_1 B_6$. At least three of them are of the same color, for example, yellow. Suppose that edges $B_1 B_2, B_1 B_3$, and $B_1 B_4$ are yellow. If $B_2 B_3 B_4$ is a red triangle, the proof is finished. If, for example, the edge $B_2 B_3$ is yellow, then the triangle $B_1 B_2 B_3$ is yellow.

10.38. (a) Since a triangle is a convex figure, it follows that $K(3) = 3$.

(b) We shall prove that $K(4) = 5$, i.e., if A, B, C, D , and E are points in the plane such that no three of them are collinear, then there are 4 points among them that are vertices of a convex quadrilateral. If the convex hull of the set $\{A, B, C, D, E\}$ is a pentagon or a quadrilateral, the proof is finished. Suppose that this convex hull is a triangle, for example, ABC . Then, D and E are the inner points of the triangle ABC . Suppose also that the line DE meets the sides AB and BC , and does not meet the side AC . Then, the points A, C, D , and E are the vertices of a convex quadrilateral.

(c) Lemma: Suppose that $A_1, A_2, \dots, A_n$ are points in the plane such that any four of them are the vertices of a convex quadrilateral. Then $A_1, A_2, \dots, A_n$ are the vertices of a convex n -gon.

Proof of the Lemma. Let us suppose, on the contrary, that the convex hull of the set $\{A_1, A_2, \dots, A_n\}$ is, for example, a k -gon with vertices $A_1, A_2, \dots, A_k$, where $k < n$. Then point A_n is an inner point of one of the triangles $A_1 A_2 A_3, A_1 A_3 A_4, \dots, A_1 A_{k-1} A_k$. Suppose, for example, that A_n is an inner point of the triangle $A_1 A_2 A_3$. Then, the points A_1, A_2, A_3 , and A_n are not vertices of a convex quadrilateral, and this conclusion contradicts the assumption of the Lemma.

(d) Let n be a positive integer greater than 4, and S be a set consisting of at least $R(n, 5; 4)$ points in the plane, such that no three of them are collinear. Let Φ_1 be the set of all 4-combinations of points from S that are vertices of a convex quadrilateral, and Φ_2 be the set of the remaining 4-combinations of points from S . By Ramsey's Theorem it follows that there are n points from S such that any four of them are the vertices of a convex quadrilateral, or there are 5 points from S such that no four of them are the vertices of a convex quadrilateral. It follows from (b) that the second possibility should be eliminated. Hence, there are n points from S such that any four of them are the vertices of a convex quadrilateral. Now, using the above Lemma we obtain that these n points are the vertices of a convex n -gon. We have also proved that $K(n) \leq R(n, 5; 4)$.

10.39. Let $n \geq R(\overbrace{3, 3, \dots, 3}^k; k)$. Let $A_1 \cup A_2 \cup \dots \cup A_k$ be a partition of the set $\{1, 2, \dots, n\}$ into k blocks, let Φ be the set of all 2-combinations of the elements $1, 2, \dots, n$, and let $\Phi = \Phi_1 \cup \Phi_2 \cup \dots \cup \Phi_k$ be a partition of the set Φ into k blocks defined as follows: for any $j \in \{1, 2, \dots, k\}$,

$$\{a, b\} \in \Phi_j \text{ if and only if } |a - b| \in A_j.$$

By Ramsey's Theorem there is a set $\{a_1, a_2, a_3\} \subset \{1, 2, \dots, n\}$ and a positive integer $j \in \{1, 2, \dots, k\}$ such that: $\{a_1, a_2\} \in \Phi_j$, $\{a_2, a_3\} \in \Phi_j$, $\{a_3, a_1\} \in \Phi_j$. Without loss of generality we can assume that $a_1 < a_2 < a_3$. Let us denote $x = a_2 - a_1$, $y = a_3 - a_2$, and $z = a_3 - a_1$. Then, the positive integers x , y , and z belong to the set A_j , and $x + y = z$.

14.11 Solutions for Chapter 11

11.1. (a) Let the fields of the chessboard 8×8 be labeled as shown in Figure 14.11.1. There are 32 pairs of fields labeled the same number. If player A puts the knight on the field labeled k , then, after that, the winning strategy for player B is to put the knight on the other field with the same label.

(b) Let the fields of the chessboard 9×9 be labeled as shown in Figure 14.11.2. There are 40 pairs of fields labeled the same number, and the field in the lower left corner is without a label. As a first move, player A should put the knight on the field without a label, and then use the strategy of player B from the previous case.

11.2. Note that the following three simple statements hold:

- (a) The last digit of a perfect square does not belong to the set $\{7, 8\}$.
- (b) A two-digit positive integer $\overline{d_1 d_2}$ with the first digit $d_1 = 7$ is not a perfect square.

14	11	24	5	15	10	23	4
25	5	15	10	24	4	16	9
11	14	32	26	29	28	3	23
6	25	29	27	31	26	9	16
13	12	19	32	28	30	22	3
20	6	13	30	27	31	17	8
12	19	1	21	7	18	2	22
1	20	7	18	2	21	8	17

Fig. 14.11.1

2	10	23	17	3	11	24	18	4
23	16	3	10	24	17	4	11	25
9	2	29	33	38	36	30	5	18
16	22	38	35	30	33	39	25	12
1	9	32	29	39	31	36	19	5
22	15	35	37	32	40	34	12	26
8	1	28	40	34	37	31	6	19
15	21	8	27	14	20	7	26	13
	28	14	21	7	27	13	20	6

Fig. 14.11.2

(c) If $n \geq 10$ is a positive integer, then $(n+1)^2 - n^2 > 20$.

The winning strategy for player A is the following. First, A writes down the digit 7. Then player A chooses every subsequent digit as follows. Let us suppose that $d_1 d_2 \dots d_{2k}$, where $k \geq 1$, is the positive integer written down. Consider the positive integers $d_1 d_2 \dots d_{2k} 70$, $d_1 d_2 \dots d_{2k} 71$, $d_1 d_2 \dots d_{2k} 72$, $\dots$, $d_1 d_2 \dots d_{2k} 89$. At most one of these positive integers is a perfect square. If there is a perfect square among them, and its second last digit is 7 (8), then player A writes down the digit 8 (7) at the end of $d_1 d_2 \dots d_{2k}$. If there is no perfect square among these 20 positive integers, then player A can choose any of the digits 7 or 8, and write it down after d_{2k} in the representation $d_1 d_2 \dots d_{2k}$.

11.3. Let us suppose that the fields are labeled $1, 2, \dots, 2n+1$ from left to right. We shall use the following notation for the moves of the game: (n, k) means a coin is moved from field n to field k . Player B has the winning strategy. If A plays $(k, k+1)$, then B plays $(k+1, k)$. Let us suppose that A plays $(k, k+n)$, where $n > 1$. In this case the fields $k+1, k+2, \dots, k+n-1$ are occupied by coins, and player B plays $(k+1, k)$. It is easy to see that the position obtained after every move by player B has the following property. Field 1 is occupied, and there are no two unoccupied fields followed by a field occupied by a coin. Now we have the conclusions. There are at most $n-1$ unoccupied fields among the first $2n-1$ fields. All n coins are placed on the first $2n-1$ fields. In the next move, player A cannot put a coin on field $2n+1$.

11.4. Let us consider the pairs $(2, 3), (4, 5), (6, 7), \dots$. If n is even, then the last pair in the sequence is $(n-2, n-1)$, and positive integers 1 and n

are not included in the pairs. If n is odd, then the last pair is $(n - 1, n)$. Let A and B be the first and second player, respectively. In the first move, player A moves a coin from field n (from field $n - 2$) to field 1 if n is even (if n is odd). After the first move by player A there are an even number of unoccupied fields between occupied field 1 and a pair of occupied fields. The winning strategy for A is then the following. If B moves a coin to field k , where k is even, then A moves a coin to field $k + 1$. It remains to note that the game can last a finite number of moves.

11.5. *Answer:* Losing positions for the player who plays the next move are the following: (1) The player has an even number of coins, and there are $6k + 1$ coins on the table, where k is a nonnegative integer. (2) The player has an odd number of coins, and there are $6k - 1$ or $6k$ coins on the table. There are $49 = 6 \cdot 8 + 1$ coins on the table at the initial position. Hence, the initial position means the player who starts the game loses.

11.6. Player B has the winning strategy. Let us consider a partition of the table into four parts, such that each part consists of two columns. After every move by player W (in a column of one of the four parts), player B moves a chip in the other column of the same part of the table. If W moves his chip k fields forward, then B also moves his chip k fields forward. If W moves his chip k fields back, then B moves his chip k fields forward. After every move by player B the distance between the black and the white chips in both columns is the same. The initial position is given in Figure 14.11.3. A possible position reached after two moves by both players is given in Figure 14.11.4. Since player B always moves his chips forward, the game will finish after a finite number of moves with all white chips in the first row, and all black chips in the second row.

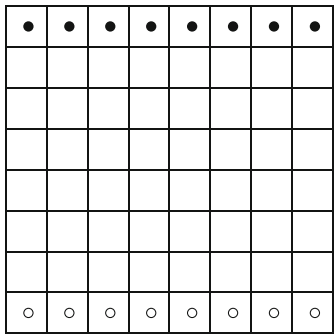


Fig. 14.11.3

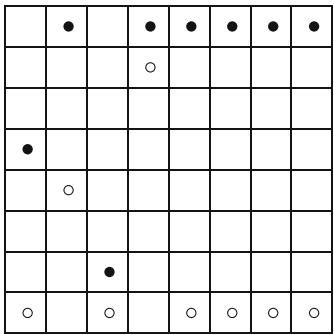


Fig. 14.11.4

14.12 Solutions for Chapter 12

12.1. The sample space consists of the following equally likely outcomes: 000, 001, 010, 100, 011, 101, 110, and 111. We are interested in the probability of event $A = \{011, 101, 110, 111\}$. Hence, $P(A) = \frac{4}{8} = \frac{1}{2}$.

12.2. There are 216 outcomes of equal probability $1/216$. The partitions of the natural numbers 11 and 12 into three parts that belong to the set $\{1, 2, 3, 4, 5, 6\}$ are given by:

$$\begin{aligned} 11 &= 6 + 4 + 1 = 6 + 3 + 2 = 5 + 5 + 1 \\ &= 5 + 4 + 2 = 5 + 3 + 3 = 4 + 4 + 3, \\ 12 &= 6 + 5 + 1 = 6 + 4 + 2 = 6 + 3 + 3 \\ &= 5 + 5 + 2 = 5 + 4 + 3 = 4 + 4 + 4. \end{aligned}$$

A partition into three distinct parts can be obtained in 6 ways. Any of the partitions mentioned above with two distinct parts can be obtained in 3 ways. There is only one way to obtain the partition $4 + 4 + 4$. Hence, the sums 11 and 12 occur with the following probability:

$$\begin{aligned} P(11) &= \frac{6 + 6 + 3 + 6 + 3 + 3}{216} = \frac{27}{216}, \\ P(12) &= \frac{6 + 6 + 3 + 3 + 6 + 1}{216} = \frac{25}{216}. \end{aligned}$$

The probability of the sum 11 is greater than the probability of the sum 12.

12.3. Answer: $P(A) = \binom{15}{10} \binom{20}{10}^{-1}$, $P(B) = \binom{5}{3} \binom{7}{3} \binom{8}{4} \binom{20}{10}^{-1}$.

12.4. Answer: $1 - \frac{6}{6^4} = \frac{215}{216}$. **12.5.** Answer: $2 \binom{8}{2} \binom{8}{3} \binom{16}{5}^{-1}$.

12.6. Answer: (a) $\frac{n-1}{2n-1}$, (b) $\frac{n}{2n-1}$.

12.7. Let A be the event that 1 did not occur, and B be the event that 2 did not occur. Then we have:

$$\begin{aligned} \text{(a) } P(A) &= \frac{5^n}{6^n}, \quad \text{(b) } P(B) = \frac{5^n}{6^n}, \quad \text{(c) } P(AB) = \frac{4^n}{6^n}, \\ \text{(d) } P(A \cup B) &= \frac{2 \cdot 5^n - 4^n}{6^n}. \end{aligned}$$

12.8. $P(A \cup B) = P(A) + P(B) - P(AB) = \frac{2(n-1)! - (n-2)!}{n!} = \frac{2n-3}{n(n-1)}$.

12.9. (a) The sample space consists of the following four groups of outcomes: the first group: 4, 5, and 6; the second group: 31, 32, 33, 34, 35, 36, 22, 23,

24, 25, 26, 13, 14, 15, and 16; the third group: 112, 113, 114, 115, 116, 121, 122, 123, 124, 125, 126, 211, 212, 213, 214, 215, and 216; the fourth group: 1111, 1112, 1113, 1114, 1115, and 1116.

(b) Outcomes from the same group are of equal probability, namely, $1/6$ in the first group, $1/36$ in the second group, $1/216$ in the third group, and $1/1296$ in the fourth group. The sum of the probabilities of all the outcomes is $\frac{3}{6} + \frac{15}{36} + \frac{17}{216} + \frac{6}{1296} = 1$.

$$(c) \frac{17}{216} + \frac{6}{1296} = \frac{18}{216} = \frac{1}{12}.$$

12.10. Answer: (a) $P(A) = p^2 + pq = p$, (b) $P(B) = qp + q^2 = q$.

12.11. Let A be the event that a head (denoted by 1) appeared, and B be the event that a tail (denoted by 0) appeared. The sample space is given in Example 12.1.3. The events A and AB are given by

$$\begin{aligned} A &= \{001, 010, 100, 011, 101, 110, 111\}, \\ AB &= \{001, 010, 100, 011, 101, 110\}. \end{aligned}$$

Hence, $P(B|A) = P(AB)/P(A) = 6/7$.

12.12. (a) The independence of A and B implies that $P(AB) = P(A)P(B)$. Note that $P(A\bar{B}) + P(AB) = P(A\bar{B} \cup AB) = P(A)$. Hence

$$\begin{aligned} P(A\bar{B}) &= P(A) - P(AB) = P(A) - P(A)P(B) \\ &= P(A)\{1 - P(B)\} = P(A)P(\bar{B}). \end{aligned}$$

(b) If we start with the independent events A and $\bar{B}$, then similarly as in the previous case we get that $\bar{A}$ and $\bar{B}$ are independent.

12.13. Answer: $1 - (1 - p_1)(1 - p_2)(1 - p_3)$.

12.14. Answer: $p_1(1 - p_2)(1 - p_3) + (1 - p_1)p_2(1 - p_3) + (1 - p_1)(1 - p_2)p_3$.

12.15. Answer: $p_1p_2(1 - p_3) + p_1(1 - p_2)p_3 + (1 - p_1)p_2p_3 + p_1p_2p_3$.

12.16. Answer: $p_1p_2(1 - p_3) + (1 - p_1)p_2p_3 + p_1p_2p_3$.

12.17. Answer: $p_1p_2(1 - p_3) + p_1(1 - p_2)p_3 + (1 - p_1)p_2p_3$.

12.18. The possible outcomes are 111, 110, 101, 011, 100, 010, 001, and 000. Let A be the event that exactly one experiment is successful, and B be the event that the first experiment is successful. Then we have $A = \{100, 010, 001\}$, $B = \{111, 110, 101, 100\}$, $AB = \{100\}$, and hence

$$P(B|A) = \frac{p_1(1 - p_2)(1 - p_3)}{p_1(1 - p_2)(1 - p_3) + (1 - p_1)p_2(1 - p_3) + (1 - p_1)(1 - p_2)p_3}.$$

$$\mathbf{12.19.} \text{ Answer: } \frac{(1-p_1)p_2p_3}{p_1p_2(1-p_3) + p_1(1-p_2)p_3 + (1-p_1)p_2p_3}.$$

$$\mathbf{12.20.} \text{ Answer: } \frac{p_1p_2(1-p_3) + (1-p_1)p_2p_3}{p_1p_2(1-p_3) + (1-p_1)p_2p_3 + p_1p_2p_3}.$$

12.21. The probability that Arthur will pass the exam is obviously $\frac{15}{20}$. Bob and Chris have the same probability to pass the exam. This fact can be obtained using the formula of total probability.

The same conclusion can be obtained as follows. Suppose that 20 students take the exam under the same rules (every student is prepared to answer 15 questions correctly). When choosing the questions, the students arrange them in a random permutation $q_1q_2\ldots q_{20}$. The probability that a good question is in the i -th position is $\frac{15}{20} = \frac{3}{4}$.

12.22. Let A_i be the event that the set S_i is chosen, $i \in \{1, 2\}$, B be the event that the first two chosen natural numbers are both divisible by 4, and C be the event that the third chosen number is also divisible by 4. We are interested in $P(C|B)$. First, we calculate $P(B)$ and $P(BC)$.

$$P(B) = \frac{1}{2} \cdot \frac{3}{16} \cdot \frac{2}{15} + \frac{1}{2} \cdot \frac{5}{16} \cdot \frac{4}{15} = \frac{13}{240},$$

$$P(BC) = \frac{1}{2} \cdot \frac{3}{16} \cdot \frac{2}{15} \cdot \frac{1}{14} + \frac{1}{2} \cdot \frac{5}{16} \cdot \frac{4}{15} \cdot \frac{3}{14} = \frac{11}{1120}.$$

$$\text{It follows that } P(C|B) = \frac{P(BC)}{P(B)} = \frac{33}{182}.$$

$$\mathbf{12.23.} \text{ (a) } \binom{6}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^3 = 0.3125; \text{ (b) } \sum_{k=4}^6 \binom{6}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{6-k} = \frac{1}{2}.$$

$$\mathbf{12.24.} \text{ (a) } \binom{6}{3} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^3 \approx 0.054; \text{ (b) } \sum_{k=4}^6 \binom{6}{k} \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{6-k} \approx 0.0087.$$

12.25. If three fair dice are thrown once, then the probability that three same numbers are obtained is $\frac{6}{216} = \frac{1}{36}$. Now it is easy to get the answers:

$$\text{(a) } 1 - \left(\frac{35}{36}\right)^5 \approx 0.131; \text{ (b) } 5 \cdot \frac{1}{36} \left(\frac{35}{36}\right)^4 \approx 0.124.$$

$$\mathbf{12.26.} \text{ Answer: } \binom{14}{9} p^{10}(1-p)^5 + \binom{14}{9} p^5(1-p)^{10}.$$

12.27. Note that the probability that the number of heads in the first five trials and the number of heads in the last five trials are both equal to $k \in \{0, 1, 2, 3, 4, 5\}$, is $\binom{5}{k}^2 \left(\frac{1}{2}\right)^{10}$. Hence, the answer to the question is

$$\begin{aligned} & \left\{ \binom{5}{0}^2 + \binom{5}{1}^2 + \binom{5}{2}^2 + \binom{5}{3}^2 + \binom{5}{4}^2 + \binom{5}{5}^2 \right\} \left(\frac{1}{2}\right)^{10} \\ &= \binom{10}{5} \left(\frac{1}{2}\right)^{10} = \frac{63}{2^8} = \frac{63}{128} \approx 0.246. \end{aligned}$$

12.28. (a) Let us denote $q = 1 - p$. It is easy to prove that

$$P_n(k) - P_n(k-1) = P_n(k-1) \frac{(n+1)p - k}{kq}.$$

The inequality $P_n(k-1) < P_n(k)$ holds if and only if $k < (n+1)p$. Let us denote $k_0 = [(n+1)p]$. It is worth mentioning the following facts. If $(n+1)p$ is not a positive integer, then $P_n(k_0)$ is the maximal term in the sequence $P_n(0), P_n(1), \dots, P_n(n)$. If $(n+1)p$ is a positive integer, then $P_n(k_0-1) = P_n(k_0)$ are two maximal terms in this sequence.

(b) If $0 < p < 1/(n+1)$, then $P_n(0) > P_n(1) > \dots > P_n(n)$.

If $n/(n+1) < p < 1$, then $P_n(0) < P_n(1) < \dots < P_n(n)$.

12.29. We shall give the proof by the method of mathematical induction. For $n = 1$ the inequality $\frac{1}{2} \leq \frac{1}{2} < \frac{1}{\sqrt{3}}$ holds. Suppose that, for some $n \in \mathbb{N}$,

$$\frac{1}{2\sqrt{n}} \leq P_{2n}(n) < \frac{1}{\sqrt{2n+1}}.$$

Note that

$$P_{2n+2}(n+1) = \binom{2n+2}{n+1} \frac{1}{2^{2n+2}} = \binom{2n}{n} \frac{1}{2^{2n}} \cdot \frac{2n+1}{2n+2} = P_{2n}(n) \cdot \frac{2n+1}{2n+2}.$$

By the last equality and the induction hypothesis it follows that

$$\begin{aligned} P_{2n+2}(n+1) &< \frac{1}{\sqrt{2n+1}} \cdot \frac{2n+1}{2n+2} = \frac{\sqrt{2n+1}}{2n+2} < \frac{1}{\sqrt{2n+3}}, \\ P_{2n+2}(n+1) &\geq \frac{1}{2\sqrt{n}} \cdot \frac{2n+1}{2n+2} > \frac{1}{2\sqrt{n+1}}. \end{aligned}$$

Note that the equality $\frac{1}{2\sqrt{n}} = P_{2n}(n)$ holds only for $n = 1$.

12.30. The probability that at least one out of n independent experiments is successful is $1 - (1-p)^n$. From the condition $1 - 0.4^n > 0.99$ it follows that $n \geq 6$. Hence, the minimal number of experiments is 6.

12.31. Let X be the number of rolls necessary to obtain a 6. Then, X has a geometric distribution with the parameter $p = 1/6$, and from Example 12.4.6 it follows that $E(X) = 6$.

12.32. *Answer:* (a) $P\{X = n\} = \left(\frac{5}{6}\right)^{n-2} \frac{1}{6}$; (b) $E(X) = 7$.

12.33. Let X be the number of rolls necessary to obtain a six r times. Then, X has a negative binomial distribution with the parameters r and $p = 1/6$. From Example 12.4.7 it follows that $E(X) = 6r$.

12.34. *Answer:* $\text{var}(X) = (1 - p)p^{-2}$.

12.35. *Answer:* $\text{var}(X) = r(1 - p)p^{-2}$. *Hint.* Use Theorem 12.4.10, Exercise 12.34, and the fact that a negative binomial random variable is the sum of the geometric random variables.

12.36. The values are given in the following table:

k	0	1	2	3	4	5
$P_{2000}(k)$	0.1366	0.2728	0.2724	0.1813	0.0904	0.0361
$\pi_2(k)$	0.1353	0.2707	0.2707	0.1804	0.0902	0.0361

12.37. *Answer:* $E(X) = \lambda$; $\text{var}(X) = \lambda$.

12.38. For every $n \in \{0, 1, 2, \dots\}$ we obtain that

$$\begin{aligned}
 P\{X + Y = n\} &= \sum_{k=0}^n P\{X = k, Y = n - k\} = \sum_{k=0}^n e^{-\lambda} \frac{\lambda^k}{k!} \cdot e^{-\mu} \frac{\mu^{n-k}}{(n-k)!} \\
 &= \frac{e^{-(\lambda+\mu)}}{n!} \sum_{k=0}^n \binom{n}{k} \lambda^k \mu^{n-k} = e^{-(\lambda+\mu)} \frac{(\lambda + \mu)^n}{n!}.
 \end{aligned}$$

Hence, the random variable $X + Y$ has a Poisson $\mathcal{P}(\lambda + \mu)$ distribution.

12.39. (a) It is obvious that $u_{2n} = \binom{2n}{n} \frac{1}{2^{2n}}$. By the Stirling formula $n! \sim n^n e^{-n} \sqrt{2\pi n}$, $n \rightarrow \infty$, it follows that $u_{2n} \sim \frac{1}{\sqrt{\pi n}}$ as $n \rightarrow \infty$.

(b) By considering the events $\{S_1 > 0, S_2 > 0, \dots, S_{2n} = 2k\}$, for $k \in \{1, 2, \dots, n\}$, and using the method from Section 2.6, it is easy to obtain

$$\begin{aligned}
P\{S_2 \neq 0, S_4 \neq 0, \dots, S_{2n} \neq 0\} &= 2 \sum_{k=1}^n P\{S_1 > 0, \dots, S_{2n-1} > 0, S_{2n} = 2k\} \\
&= 2 \sum_{k=1}^n \left[\binom{2n-1}{n+k-1} - \binom{2n-1}{n+k} \right] \frac{1}{2^{2n}} \\
&= 2 \binom{2n-1}{n} \frac{1}{2^{2n}} = \binom{2n}{n} \frac{1}{2^{2n}} = u_{2n}.
\end{aligned}$$

Using the last equality, and assuming that $u_0 = 1$, we obtain

$$v_{2n} = u_{2n-2} - u_{2n} = \frac{1}{2n-1} \binom{2n}{n} \frac{1}{2^{2n}} \sim \frac{1}{2\sqrt{\pi n^{3/2}}}, \quad n \rightarrow \infty.$$

$$(c) \quad \sum_{n=1}^{\infty} v_{2n} = \sum_{n=1}^{\infty} (u_{2n-2} - u_{2n}) = 1.$$

$$(d) \quad E(X) = \sum_{n=1}^{\infty} 2n v_{2n} \sim \sum_{n=1}^{\infty} \frac{1}{\sqrt{\pi n}} = +\infty.$$

12.40. Let p_n be the probability that point (level) n will be reached. Since the random variables $X_1, X_2, X_3, \dots$ (steps in the random walk) are independent, it follows that $p_n = p_1^n$. Since the first step can be either -1 or 1 , it follows that $p_1 = p \cdot 1 + (1-p)p_1^2$, and consequently $p_1 = 1$ or $p_1 = \frac{p}{1-p}$.

Let us consider the probability p_1 as a function of the probability p , i.e., $p_1 = p_1(p)$, $0 \leq p \leq 1$. We can prove that p_1 is a continuous function of p . (This can be done by representing p_1 in the form $p_1 = v_1 + v_3 + v_5 + \dots$, where v_{2k-1} is the probability that level 1 will be reached for the first time after the $(2k-1)$ -th step, and calculating v_{2k-1} for every $k \in \mathbb{N}$. This way we obtain a power series that defines a continuous function.)

Since $\frac{p}{1-p} > 1$ for $p > \frac{1}{2}$, it follows that $p_1 = 1$ for $p > \frac{1}{2}$. Note also that $p_1(0) = 0$. Since p_1 is a continuous function of p , it follows that $p_1 = \frac{p}{1-p}$ for $0 \leq p \leq \frac{1}{2}$. Finally, we obtain that

$$p_n = \begin{cases} [p/(1-p)]^n, & \text{if } 0 \leq p \leq 1/2, \\ 1, & \text{if } 1/2 \leq p \leq 1. \end{cases}$$

14.13 Solutions for Chapter 13

13.1. (a) Let S_k be the set of all pairs (P, j) , where P is a permutation of elements $1, 2, \dots, n$ with exactly k fixed points, and j is a fixed point of the permutation P . Obviously, $|S_k| = kp_n(k)$.

Let us now consider an arbitrary element $j \in \{1, 2, \dots, n\}$. If j is a fixed point of a permutation P with exactly k fixed points, then there are $k - 1$ fixed points of this permutation in the set $\{1, 2, \dots, n\} \setminus \{j\}$. Hence, element j is the second component of exactly $p_{n-1}(k - 1)$ pairs from S_k . Since j is one of n elements, it follows that $|S_k| = np_{n-1}(k - 1)$. Finally we obtain that $|S_k| = kp_n(k) = np_{n-1}(k - 1)$.

(b) By equality (a), where $1 \leq m \leq k \leq n$, it follows that:

$$\begin{aligned} \frac{k!}{(k-m)!} p_n(k) &= k(k-1) \dots (k-m+1) p_n(k) \\ &= n(k-1) \dots (k-m+1) p_{n-1}(k-1) = \dots = \\ &= n(n-1) \dots (n-m+1) p_{n-m}(k-m) \\ &= \frac{n!}{(n-m)!} p_{n-m}(k-m). \end{aligned}$$

By adding the equalities obtained for $k = m, m+1, \dots, n$, we get

$$\sum_{k=m}^n \frac{k!}{(k-m)!} p_n(k) = \frac{n!}{(n-m)!} \sum_{k=m}^n p_{n-m}(k-m) = n!.$$

Note that we have used the equality $\sum_{k=m}^n p_{n-m}(k-m) = (n-m)!$, which holds true, because $(n-m)!$ is the number of permutations of $n-m$ elements, and the sum on the left-hand side is obviously equal to the sum of the number of permutations of $n-m$ elements with exactly $0, 1, \dots, n-m$ fixed points.

13.2. Answer. There is a permutation with the given property. *Proof* by the method of mathematical induction. For $n = 1$ the statement obviously holds. Let us suppose that for some $n \in \mathbb{N}$ there is a permutation $P = a_1 a_2 \dots a_n$ of the elements $1, 2, \dots, n$, that satisfies the given condition. We shall define a permutation $Q = b_1 b_2 \dots b_{2n}$ of the elements $1, 2, \dots, 2n$, that satisfies the given condition as follows:

$$\begin{aligned} b_i &= 2a_i - 1, & i &\in \{1, 2, \dots, n\}, \\ b_{n+i} &= 2a_i, & i &\in \{1, 2, \dots, n\}. \end{aligned}$$

Permutation Q really satisfies the given condition. This condition is satisfied for two odd or two even terms because the relation $\frac{p+q}{2} \neq r$ implies that $\frac{(2p-1) + (2q-1)}{2} \neq 2r-1$ and $\frac{2p+2q}{2} \neq 2r$. For an even and an odd term of permutation Q the condition is satisfied because their mean is not an integer. Hence, the statement is proved for the positive integers $1, 2, 4, 8, \dots$. If $n = 2^k + r$, where $1 \leq r < 2^k$, then we first construct a permutation

of the elements $1, 2, \dots, 2^{n+1}$ that satisfies the given condition, and then remove the terms $n+1, n+2, \dots, 2^{k+1}$. The obtained permutation of the elements $1, 2, \dots, n$ also satisfies the given condition.

13.3. Suppose that the given 0–1 sequence has $abcd$ in the last four positions. Then, $abcd0$ and $abcd1$ are its subsequences. In the opposite case we can add 0 or 1 at the end of the sequence and obtain a new sequence without identical 5-subsequences, and this contradicts the condition (b). Hence, the subsequence $abcd$ appears three times in the given sequence. The digits 0 and 1 can appear once before such a subsequence. Hence, the given sequence has $abcd$ in the first four positions.

13.4. We shall prove the statement by induction on n . Note that for $n = 1$, the positive integers 11, 12, 21, and 22 satisfy the given condition. Let us suppose that, for a positive integer n , there is a set S_n , such that $|S_n| \geq 2^{n+1}$, all elements of S_n are 2^n -digit positive integers, and the digits of any two elements of S_n differ in at least 2^{n-1} positions. For any $x \in S_n$, let us denote by $\bar{x}$ the positive integer obtained from x the following way: all digits of x that are equal to 1 are replaced by 2, and all 2's are replaced by 1's. Let S_{n+1} be the set of all positive integers of the form xx or $x\bar{x}$, where $x \in S_n$, and xx and $x\bar{x}$ are the positive integers obtained by concatenation. Then we have $|S_{n+1}| \geq 2^{n+2}$, all elements of S_{n+1} are 2^{n+1} -digit positive integers, and the digits of any two elements of S_{n+1} differ in at least 2^n positions.

13.5. (a) The number of boxes labeled by integers that have only the digits 0, 1, 2, 3, and 4 is equal to 25. Similarly, the number of boxes labeled by integers that have only the digits 5, 6, 7, 8, and 9 is 25. All the balls can be put into these $25 + 25 = 50$ boxes.

(b) Suppose that all the balls are in the boxes and the given condition is satisfied. Let x_j be the number of boxes that have a label with the first digit j , and contain at least one ball, where $j \in \{0, 1, \dots, 9\}$. Let us denote $N = x_0 + x_1 + \dots + x_9$. Without loss of generality we can assume that x_1 is the minimum among $x_0, x_1, \dots, x_9$. Let us consider the boxes 10, 11, $\dots$, 19, and suppose that exactly k of them contain at least one ball. We can assume that boxes $\overline{1c}$, where $c \in \{0, 1, \dots, k-1\}$ are not empty. None of these k boxes contains balls labeled $\overline{1c_2c_3}$, where $c_2, c_3 \in \{k, k+1, \dots, 9\}$. The ball $\overline{1c_2c_3}$, where $c_2 \geq k, c_3 \geq k$, is contained in the box labeled $\overline{c_2c_3}$. The number of nonempty boxes labeled $\overline{c_2c_3}$, where $c_2 \geq k$ and $c_3 \geq k$, is $(10-k)^2$. The number of nonempty boxes that have labels with the first digit from the set $\{0, 1, 2, \dots, k-1\}$ is not less than k^2 . Hence, $N \geq (10-k)^2 + k^2 \geq 50$.

13.6. A positive integer with the given property can be obtained as follows. Let us first write down the digit 0. Then we write the digit 1 on both the left

and the right side of 0, and get the sequence 101. Then we write the digit 2 at the beginning of sequence 101, between every two consecutive digits, and at the end of this sequence. The obtained positive integer is 2120212. Then we write the digit 3 at the beginning of the sequence, between every two consecutive digits, and at the end of the sequence. The obtained positive integer is 323132303231323. Then we continue to write the digits 4, 5, ..., 9. Finally we get

$$9897989698979895 \dots 909 \dots 5989798969897989. \quad (1)$$

There are $1 + 2 + 2^2 + \dots + 2^9 = 1023$ digits in sequence (1). For every $k \in \{0, 1, \dots, 9\}$ the following statement holds: if we add the digit k at the end of (1), then the subsequence consisting of the first 2^{9-k} digits of the obtained sequence coincides with the subsequence consisting of the last 2^{9-k} digits.

13.7. We shall prove the statement by mathematical induction on n . For $n = 1$ the statement holds, because we shall remove 0 or 1 from every 3-arrangement of the elements 0 and 1, such that the obtained 2-arrangement consists of two 0's or two 1's. Suppose that the statement holds for a positive integer n , and consider a $(2n + 3)$ -arrangement of the elements 0 and 1.

Case 1. There are two adjacent digits in this $(2n + 3)$ -arrangement that are both equal to 0 or 1. Let us remove these two digits. From the obtained $(2n + 1)$ -arrangement we can remove one more digit such that the new $2n$ -arrangement has the required property. Then we put back two digits that were removed firstly, and put them in the same positions. The obtained $(2n + 2)$ -arrangement has the required property.

Case 2. No two adjacent digits in the $(2n + 3)$ -arrangement are equal to each other. The first and the last digits are both equal to 0 or 1. Now we can remove these two digits and repeat the procedure from Case 1.

13.8. Let us consider a partition of the knights into several blocks determined as follows. All knights from the same block come from the same country, and occupy several adjacent places around the table. The knights that belong to two adjacent blocks come from different countries. Let n be the number of blocks. It is obvious that the blocks alternately consist of knights from different countries, and hence n is an even positive integer i.e., $n = 2k$, where $k \in \mathbb{N}$. Let $m_1, m_2, \dots, m_n$ be the number of knights in these blocks. The number of knights that have an enemy on the right-hand side is equal to n . The number of knights that have a friend on the right-hand side is $(m_1 - 1) + (m_2 - 1) + \dots + (m_n - 1)$. It follows that $(m_1 - 1) + (m_2 - 1) + \dots + (m_n - 1) = n$, i.e., $m_1 + m_2 + \dots + m_n = 2n = 4k$.

13.9. Let us consider a regular n -gon and label its vertices by $x_1, x_2, \dots, x_n$ consecutively in the chosen direction. We say that a vertex is good if

its label is the same as the label of the next vertex. A vertex is bad if it is not good. Since $x_1x_2 + x_2x_3 + \cdots + x_nx_1 = 0$, it follows that the number of good vertices is the same as the number of bad vertices. Similarly as in Exercise 13.8 we conclude that n is divisible by 4.

13.10. For each $j \in \{1, 2, \dots, k\}$, let us denote $n_j = |A_j|$, and let S_j be the set of permutations of the set $\mathbb{N}_n$, in which the first n_j positions are occupied by the elements of set A_j . Then, $|S_j| = n_j!(n - n_j)!$. Since none of the sets $A_1, A_2, \dots, A_k$ is a subset of the others, it follows that $S_1, S_2, \dots, S_k$ are pairwise disjoint sets. Using this fact, and the fact that $S_1 \cup S_2 \cup \cdots \cup S_k$ is a subset of the set of all permutations of the elements $1, 2, \dots, n$, we obtain that

$$n! \geq |S_1 \cup S_2 \cup \cdots \cup S_k| = \sum_{j=1}^k |S_j| = \sum_{j=1}^k n_j!(n - n_j)! \quad (1)$$

It is easy to see that (1) implies the inequality

$$\sum_{j=1}^k \frac{1}{\binom{n}{n_j}} \leq 1.$$

Since $\binom{n}{[n/2]}$ is the greatest of the binomial coefficients $\binom{n}{0}, \dots, \binom{n}{n}$, it follows that

$$k \leq \sum_{j=1}^k \frac{1}{\binom{n}{n_j}} \binom{n}{[n/2]} \leq \binom{n}{[n/2]},$$

and the proof of the given inequality is completed. Note that the inequality becomes an equality if we choose $A_1, A_2, \dots$ to be all $[n/2]$ -combinations of the elements of set $\{1, 2, \dots, n\}$.

13.11. As in Exercise 13.10, let us denote $n_j = |A_j|$, where $j \in \{1, 2, \dots, k\}$, and let S_j be the set of permutations of the set $\mathbb{N}_n$, in which the first n_j positions are occupied by the elements of set A_j . The imposed condition on $A_1, A_2, \dots, A_k$ implies that any permutation of $\mathbb{N}_n$ belongs to at most r of the sets $S_1, S_2, \dots, S_k$. Hence,

$$r n! \geq |S_1 \cup S_2 \cup \cdots \cup S_k| = \sum_{j=1}^k |S_j| = \sum_{j=1}^k n_j!(n - n_j)!,$$

i.e., $\sum_{j=1}^k \binom{n}{n_j}^{-1} \leq r$. Let us consider the set

$$\left\{ \binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n} \right\} = \{a_1, a_2, \dots, a_{n+1}\},$$

and, without loss of generality, suppose that $a_1 \geq a_2 \geq \cdots \geq a_{n+1}$. Then,

$$\binom{k-1}{r-1} \left(\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_k} \right) = \sum^* \left(\frac{1}{a_{j_1}} + \frac{1}{a_{j_2}} + \cdots + \frac{1}{a_{j_r}} \right),$$

where the sum $\sum^*$ runs over all r -combinations $\{j_1, j_2, \dots, j_r\} \subset \{1, 2, \dots, k\}$. Note also that for all nonnegative real numbers $x_1, x_2, \dots, x_r, y_1, y_2, \dots, y_r$, we have that $r \sum_{j=1}^r x_j y_j \leq \sum_{j=1}^r x_j \sum_{j=1}^r y_j$. By the last two relations it follows that

$$\begin{aligned} k &= \binom{k-1}{r-1}^{-1} \binom{k}{r} r \leq \binom{k-1}{r-1}^{-1} \sum^* \left(\frac{a_1}{a_{j_1}} + \frac{a_2}{a_{j_2}} + \cdots + \frac{a_r}{a_{j_r}} \right) \\ &\leq \binom{k-1}{r-1}^{-1} \sum^* \left(\frac{1}{a_{j_1}} + \frac{1}{a_{j_2}} + \cdots + \frac{1}{a_{j_r}} \right) \frac{a_1 + a_2 + \cdots + a_r}{r} \\ &= \frac{a_1 + a_2 + \cdots + a_r}{r} \left(\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_k} \right) \\ &\leq \frac{a_1 + a_2 + \cdots + a_r}{r} \cdot r = \sum_{j=1}^r a_j, \end{aligned}$$

and the proof is completed.

Remark. For every positive integer n we can choose the subsets $A_1, A_2, \dots, A_k$ of the set $\mathbb{N}_n = \{1, 2, \dots, n\}$, that satisfy the imposed conditions, and such that k is equal to the sum of the greatest r of the binomial coefficients $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$. If $n = 2m$ and $r = 2t + 1$, $t \in \mathbb{N}$, then we can choose k to be equal to the number of subsets of set $\mathbb{N}_n$ with no less than $m - t$ elements and no more than $m + t$ elements, and $A_1, A_2, \dots, A_k$ to be all such subsets. There are similar examples for the other values of n and r .

13.12. Every sum of the form $\varepsilon_1 x_1 + \varepsilon_2 x_2 + \cdots + \varepsilon_n x_n$ determines a subset A of the set $\{1, 2, \dots, n\}$, as follows: $k \in A$ if and only if $\varepsilon_k = 1$. Let A and B be the subsets determined respectively by the sums

$$s_1 = \varepsilon_1 x_1 + \varepsilon_2 x_2 + \cdots + \varepsilon_n x_n, \quad s_2 = \varepsilon'_1 x_1 + \varepsilon'_2 x_2 + \cdots + \varepsilon'_n x_n,$$

where $\varepsilon'_j \neq \varepsilon_j$ for at least one $j \in \{1, 2, \dots, n\}$. If $A \subset B$, then $s_2 - s_1 = \sum_{k \in B \setminus A} 2x_k$, and the imposed condition implies that $s_2 - s_1 > 2$. Hence, if $A \subset B$, then there is no segment of length 2 that contains both s_1 and s_2 . In other words, if s_1 and s_2 both belong to a segment of length 2, neither of sets A and B is a subset of the other. By Exercise 13.10 it follows that at most $\binom{n}{m}$ sums of the given form belong to $[a - 1, a + 1]$.

13.13. The statement follows from Exercise 13.10.

13.14. Six cuts are necessary to get the central unit cube. It is obvious that six cuts are also sufficient to get all 27 unit cubes.

13.15. Let a, b, c be nonnegative integers such that

$$2^{a-1} < m \leq 2^a, \quad 2^{b-1} < n \leq 2^b, \quad 2^{c-1} < p \leq 2^c. \quad (1)$$

It is easy to see that $a + b + c$ cuts are sufficient to get mnp unit cubes. Let us prove that this number of cuts is also necessary. A *characteristic* of the parallelepiped $m \times n \times p$ is the sum $a + b + c$, where a, b, c are determined by inequalities (1). The characteristic of the unit cube is equal to 0. Now it is sufficient to note that if we cut any parallelepiped, then the characteristic of at least one of the two obtained parts decreases by at most 1.

13.16. *Hint.* Let A be the player labeled 1, B be the player that eliminated A , C be the player that eliminated B , etc. Let us consider the sequence of games $G_1 G_2 \dots G_n$ determined as follows. The first few terms of this sequence are the games played by A , then come the games played by B after the elimination of A , then come the games played by C after the elimination of B , etc. Let $L_1 L_2 \dots L_n$ be the sequence of positive integers defined as follows: L_k is the label of the winner of the game G_k . It follows from the imposed conditions that $L_1 \leq 3$, $L_2 \leq 5$, $\dots$, $L_n \leq 2n + 1$. Prove that L_n cannot take the value $2n + 1$, but can be equal to $2n$.

13.17. Let x_n be the number of permutations with the given property. Then, $x_1 = 1$ and $x_n = 2x_{n-1}$ for $n \geq 2$. Hence, $x_n = 2^{n-1}$ for any $n \in \mathbb{N}$. The recurrence relation is obtained using the fact that $a_n = n$ or $a_n = n - 1$ for any permutation with the given property.

13.18. Let y_n be the number of permutations with the given property. Then, $y_1 = 1$, $y_2 = 2$, and $y_n = y_{n-1} + y_{n-2}$ for $n \geq 3$. Hence, $y_n = F_{n+1}$ is the $(n + 1)$ -st Fibonacci number. The recurrence relation is obtained using the fact that $a_n = n$ or $a_n = n - 1$ for any permutation with the given property.

13.19. Let x_{nk} be the number of permutations with the given property. The numbers x_{nk} , where n is a positive integer and $1 \leq k \leq n$, are determined by $x_{n,1} = (n - 1)!$, $x_{n,n} = 1$, and $x_{nk} = (n - 1)x_{n-1,k} + x_{n-1,k-1}$ for $1 < k < n$. The last recurrence relation is obtained by considering the cases where $a_n = n$ and $a_n < n$.

13.20. Let y_{nk} be the number of permutations with the given property. The integers y_{nk} , where n is a positive integer and $0 \leq k \leq n$, are determined by $y_{n0} = 1$, $y_{nn} = 0$, and $y_{nk} = (k + 1)y_{n-1,k} + (n - k)y_{n-1,k-1}$ for $0 < k < n$.

13.21. Let us partition the given square table $\tau = 50 \times 50$ into four square tables 25×25 . Let $\Sigma_1, \Sigma_2, \Sigma_3, \Sigma_4$ be the sum of the entries in these four square tables 25×25 . If $\Sigma_i \geq 0$ for any $i \in \{1, 2, 3, 4\}$, then $\Sigma_i \leq 25$ for some $i \in \{1, 2, 3, 4\}$. Similarly, if $\Sigma_i \leq 0$ for any $i \in \{1, 2, 3, 4\}$, then $\Sigma_i \geq -25$ for some $i \in \{1, 2, 3, 4\}$.

It remains to consider the case where some of the sums $\Sigma_1, \Sigma_2, \Sigma_3, \Sigma_4$ are positive, and some of them are negative. In this case there are two square tables 25×25 , denoted by τ_1 and τ_2 , that have a *common side*, and the related sums Σ_i and Σ_j such that $\Sigma_i < 0$ and $\Sigma_j > 0$. Without loss of generality we can assume that τ_1 is determined as the intersection of the first 25 rows and the first 25 columns of the square table τ , while τ_2 is determined as the intersection of the first 25 rows and the last 25 columns of τ . Let us now consider the sequence of 26 square tables $\tau_1 = T_1, T_2, T_3, \dots, T_{25}, T_{26} = \tau_2$, such that, for any $i \in \{1, 2, \dots, 26\}$, the table T_i is the intersection of the first 25 rows of τ and 25 columns labeled $i, i+1, \dots, i+24$ in the square table τ . Let S_i be the sum of the entries of the square table T_i . Then, $S_1 S_{26} < 0$, and, for any $i \in \{1, 2, \dots, 25\}$, $|S_{i+1} - S_i| \leq 50$. It follows that there is $i_0 \in \{1, 2, \dots, 25\}$ such that $S_{i_0} S_{i_0+1} < 0$. Since $|S_{i_0+1} - S_{i_0}| \leq 50$, it follows that at least one of the inequalities $|S_{i_0}| \leq 25$ and $|S_{i_0+1}| \leq 25$ holds true.

13.22. Let $x_{ij} \in \{-1, 1\}$ be the entry placed into the unit square that is the intersection of the i -row and j -th column of the table $m \times n$. The necessary and sufficient conditions for the product of integers in any row and the product of integers in any column to be equal to -1 are the following equalities:

$$x_{in} = - \prod_{j=1}^{n-1} x_{ij}, \quad i = 1, 2, \dots, m-1; \quad (1)$$

$$x_{mj} = - \prod_{i=1}^{m-1} x_{ij}, \quad j = 1, 2, \dots, n-1. \quad (2)$$

The necessary and sufficient conditions for the product of the integers in the m -th row and the product of the integers in the n -th column to be equal to -1 are the following equalities:

$$x_{mn} = - \prod_{j=1}^{n-1} x_{mj} = - \prod_{j=1}^{n-1} \left(- \prod_{i=1}^{m-1} x_{ij} \right) = (-1)^n \prod_{j=1}^{n-1} \prod_{i=1}^{m-1} x_{ij}, \quad (3)$$

$$x_{mn} = - \prod_{i=1}^{m-1} x_{in} = - \prod_{i=1}^{m-1} \left(- \prod_{j=1}^{n-1} x_{ij} \right) = (-1)^m \prod_{i=1}^{m-1} \prod_{j=1}^{n-1} x_{ij}. \quad (4)$$

Both equalities (3) and (4) are possible if and only if $(-1)^m = (-1)^n$, i.e., if and only if the positive integers m and n are of the same parity.

Let us now consider a table with m rows and n columns, where m and n are of the same parity. Let us choose the integers $x_{ij} \in \{-1, 1\}$, where $i \in \{1, 2, \dots, m-1\}$ and $j \in \{1, 2, \dots, n-1\}$. The integers that should be placed in the fields of the last row and the last column are then determined by equalities (1) and (2). Note that x_{mn} is uniquely determined by (3) and (4). Now it is obvious that the number of ways that the table can be filled by the integers -1 and 1 , such that the given conditions are satisfied, is equal to $2^{(m-1)(n-1)}$.

13.23. The number of distinct tables that can be obtained by successive applications of the operation allowed is not greater than 2^{mn} , because any entry can appear as $+x_{ij}$ or $-x_{ij}$. For any of these tables consider the sum of all real numbers from its fields, and let T be the table for which this sum takes the maximal value. Let us denote this maximal sum by Σ . The sum of real numbers placed in any row (column) of table T is not negative. Indeed, if the sum of the real numbers that are placed, for example, in the first row of the table T is less than 0, then we can change the sign of all the real numbers in the first row and obtain a new table T' , such that the corresponding sum Σ' is greater than Σ .

13.24. Let AB be a segment of length $a_1 + a_2 + \dots + a_m$. Let us consider two partitions of this segment into smaller segments: $AB = \mathcal{D}_1 \cup \mathcal{D}_2 \cup \dots \cup \mathcal{D}_m$, such that a_i is the length of segment $\mathcal{D}_i$, and $AB = \mathcal{E}_1 \cup \mathcal{E}_2 \cup \dots \cup \mathcal{E}_n$, such that b_i is the length of segment $\mathcal{E}_i$. The total number of points that determine these two partitions is $m + n - 2$, and they divide the segments AB into $m + n - 1$ parts. Each of these parts is of the form $\mathcal{D}_i \cap \mathcal{E}_j$, for some $i \in \{1, 2, \dots, m\}$, $j \in \{1, 2, \dots, n\}$. Let us fill the unit square that belongs to i -th row and j -th column by a real number that is the length of $\mathcal{D}_i \cap \mathcal{E}_j$. The obtained table satisfies the given condition.

13.25. Let X be the set of n -arrangements $x = (x_1, x_2, \dots, x_n)$ of the set $\{-1, 1\}$. Then, $\sum_{i=1}^n |a_{i1}x_1 + \dots + a_{in}x_n| \leq M$ for any $x \in X$. Since set X consists of 2^n elements, it follows that

$$\sum_{x \in X} \sum_{i=1}^n |a_{i1}x_1 + \dots + a_{in}x_n| \leq 2^n M. \quad (1)$$

Inequality (1) can be written in the equivalent form:

$$\sum_{i=1}^n \left(\frac{1}{2^n} \sum_{x \in X} |a_{i1}x_1 + \dots + a_{in}x_n| \right) \leq M. \quad (2)$$

Note that the sum $S_i := \sum_{x \in X} |a_{i1}x_1 + \cdots + a_{in}x_n|$ can be represented as the sum of 2^{n-1} summands of the form $|A + a_{ii}| + |A - a_{ii}|$, where A is of the form $A = a_{i1}x_1 + \cdots + a_{i,i-1}x_{i-1} + a_{i,i+1}x_{i+1} + \cdots + a_{in}x_n$. Note that

$$|A + a_{ii}| + |A - a_{ii}| \geq |(A + a_{ii}) - (A - a_{ii})| = 2|a_{ii}|. \quad (3)$$

By (3) it follows that $S_i \geq 2^{n-1} \cdot 2|a_{ii}| = 2^n|a_{ii}|$, and hence

$$|a_{11}| + \cdots + |a_{nn}| = \sum_{i=1}^n \left(\frac{1}{2^n} 2^n |a_{ii}| \right) \leq \sum_{i=1}^n \frac{1}{2^n} S_i \leq M.$$

13.26. Let T be the given table. Let us replace every coin in table T by the real number $1/p$, where p is the number of coins in the related column, and denote the obtained table by T_1 . Similarly if we replace every coin in table T by the real number $1/q$, where q is the number of coins in the related row, then we obtain a new table which is denoted by T_2 . Note that the sum of real numbers in every column of table T_1 is equal to 1, and hence the sum of all the real numbers in table T_1 is n . The sum of the real numbers in every row of table T_2 is equal to 0 or 1. Hence the sum of all the real numbers in table T_2 is not greater than m . Since $n > m$, it follows that there is a coin C that is replaced by a greater real number in T_1 than in T_2 . Let us suppose that C belongs to the i -th row and the j -th column of table T . Then, the i -th row contains more coins than the j -th column.

13.27. It is easy to check that there are Hadamard matrices for $n = 1$ and $n = 2$, and there is no Hadamard matrix for $n = 3$. Let us consider the case $n \geq 4$. If we multiply all elements of a column of a Hadamard matrix by -1 , then we obtain a Hadamard matrix as well. Hence, it is sufficient to consider the Hadamard matrices with all entries in the first row equal to 1. Let H be such a Hadamard matrix. For any column of H , let us consider the pair of elements in the second and third rows. This pair has one of the following forms:

$$\begin{array}{|c|} \hline 1 \\ \hline 1 \\ \hline \end{array} \quad \begin{array}{|c|} \hline 1 \\ \hline -1 \\ \hline \end{array} \quad \begin{array}{|c|} \hline -1 \\ \hline 1 \\ \hline \end{array} \quad \begin{array}{|c|} \hline -1 \\ \hline -1 \\ \hline \end{array}$$

Let x , y , z , and t be the number of columns with the above forms of the pair of elements in the second and the third rows, respectively. Then, $x+y+z+t = n$, $x-y+z-t = 0$, $x+y-z-t = 0$, and $x-y-z+t = 0$. By summing these equalities we obtain that $4x = n$.

13.28. A Hadamard matrix of order 2 is given in Figure 14.13.1. Figure 14.13.2 shows how a Hadamard matrix of order $2n$ can be obtained from

1	1
1	-1

Fig. 14.13.1

A_n	A_n
A_n	$-A_n$

Fig. 14.13.2

a Hadamard matrix A_n of order n . Note that $-A_n$ is the matrix obtained from A_n by multiplying all its elements by -1 .

13.29. A Hadamard matrix of order 12 is given in Figure 14.13.3. Any “+” sign in this table should be considered as “+1,” and any “-” sign as “-1.”

+	+	+	+	+	+	+	+	+	+	+	+
+	+	+	+	+	+	-	-	-	-	-	-
+	+	+	-	-	-	+	+	+	-	-	-
+	+	-	+	-	-	+	-	-	+	+	-
+	+	-	-	+	-	-	+	-	-	+	+
+	+	-	-	-	+	-	-	+	+	-	+
-	+	+	+	-	-	-	+	-	+	-	+
-	+	+	-	+	-	-	-	+	+	+	-
-	+	+	-	-	+	+	-	-	-	+	+
+	-	+	+	-	-	-	-	+	-	+	+
+	-	+	-	+	-	-	+	-	+	-	+
+	-	+	-	-	+	+	-	-	+	+	-

Fig. 14.13.3

13.30. Let us label n rows containing the maximal number of chips in the sum. We shall prove that the labeled rows contain no less than $2n$ chips. Suppose, on the contrary, that there are less than $2n$ chips in the labeled rows. It follows that there is a labeled row containing at most one chip, and there is an unlabeled row containing more than one chip. This conclusion obviously contradicts the fact that the labeled rows contain the maximal number of chips.

Hence, the labeled rows really contain no less than $2n$ chips. It follows that the unlabeled rows contain at most n remaining chips. Now we can label n columns such that they contain all the remaining chips.

•	•						
		•	•				
			•	•			
				•	•		
•					•		
						•	
							•

Fig. 14.13.4

1	2	1	2	1	2	1	2
3	4	3	4	3	4	3	4
1	2	1	2	1	2	1	2
3	4	3	4	3	4	3	4
1	2	1	2	1	2	1	2
3	4	3	4	3	4	3	4
1	2	1	2	1	2	1	2
3	4	3	4	3	4	3	4

Fig. 14.13.5

13.31. The square table 8×8 that is given in Figure 14.13.4 contains 13 chips on its fields. The chips are arranged such that no four rows and four columns contain all 13 chips. A similar example can be constructed for any square table $2n \times 2n$.

13.32. Let us label all the fields of a square table 8×8 as given in Figure 14.13.5. Since there is exactly 1 rook in every row, it follows that all the fields that are labeled 1 and 2 contain 4 rooks. By similar reasoning we conclude that the fields labeled 2 and 4 contain 4 rooks. It follows that the number of rooks that are placed on fields labeled 1 is equal to the number of rooks that are placed on fields labeled 4. Hence the number of rooks that are placed on fields labeled 1 and 4 is even. It remains to note that all the white fields of the chessboard 8×8 are labeled 1 and 4 in Fig 14.13.5.

13.33. For every $k, n \in \{1, 2, \dots, 8\}$, the field that belong to the k -th row and the n -th column is labeled $8(k - 1) + n$. Now it is easy to conclude that the sum of the eight positive integers used to label the fields that are occupied by rooks is equal to

$$8(0 + 1 + 2 + \dots + 7) + (1 + 2 + \dots + 8) = 260.$$

13.34. Let X be the chip that first returned to its initial position, and suppose that it happened in the n -th move. Let us consider the position after the $(n - 1)$ -st move. Till this moment none of the chips returned to its initial position, and every chip left its initial position. The previous conclusion holds because X visited all the fields.

13.35. A chessboard 8×8 has 32 white and 32 black fields. Suppose that two white fields (the ends of a diagonal) are cut off. The remaining part of the chessboard has 30 white and 32 black fields. A domino 2×1 always covers a

white and a black field. It follows that 31 dominos cannot be arranged such that the remaining part of the chessboard is covered.

1	2	0	1	2	0	1	2
0	1	2	0	1	2	0	1
2	0	1	2	0	1	2	0
1	2	0	1	2	0	1	2
0	1	2	0	1	2	0	1
2	0	1	2	0	1	2	0
1	2	0	1	2	0	1	2
0	1	2	0	1	2	0	1

Fig. 14.13.6

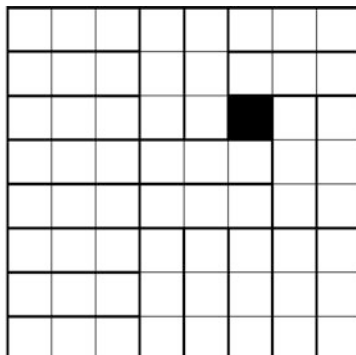


Fig. 14.13.7

13.36. All fields of a chessboard 8×8 are filled with integers 0, 1, and 2 as shown in Figure 14.13.6. There are 21 0's, 22 1's, and 21 2's. Every trimino covers each of the digits 0, 1, and 2. It follows that the uncovered field is filled by 1. After rotations of 90° , 180° , and 270° this field overlaps a field that is also filled by 1. Only four fields satisfy this condition: $c3$, $f3$, $c6$, and $f6$. The position with the uncovered field $f6$ is given in Figure 14.13.7.

13.37. All the fields of a chessboard are labeled as shown in Figure 14.13.8. The label of every field F is the number of fields into which a knight can jump from field F . It follows that the number of arrangements of two knights such that they do not attack each other is $4 \cdot 61 + 8 \cdot 60 + 20 \cdot 59 + 16 \cdot 57 + 16 \cdot 55 = 3796$.

13.38. $\binom{16}{5} = 4368$. **13.39.** 8 rooks. **13.40.** $8!$ ways.

13.41. Let us consider the 15 *diagonals* of a chessboard 8×8 that are drawn in Figure 14.13.9. The two bishops (chess pieces) that are placed on the chessboard such that they do not attack each other should be on distinct diagonals. The bishops that are placed on fields $a1$ and $h8$ ($a8$ and $h1$) attack each other. It follows that the maximal number of bishops that can be placed on the chessboard such that they do not attack each other is not greater than 14. An arrangement of 14 bishops such that no two of them attack each other is given in Figure 14.13.9.

Remark. The maximal number of bishops that can be placed on a chessboard $n \times n$ such that no two of them attack each other is $2n - 2$.

2	3	4	4	4	4	3	2
3	4	6	6	6	6	4	3
4	6	8	8	8	8	6	4
4	6	8	8	8	8	6	4
4	6	8	8	8	8	6	4
4	6	8	8	8	8	6	4
3	4	6	6	6	6	4	3
2	3	4	4	4	4	3	2

Fig. 14.13.8

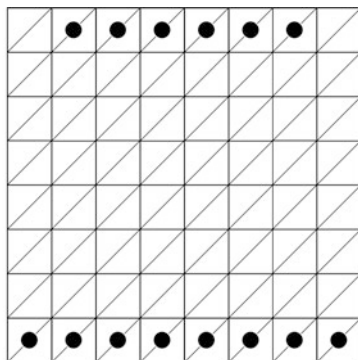


Fig. 14.13.9

13.42. We shall prove the statement for a chessboard $n \times n$. The maximal number of bishops that can be placed on it such that no two of them attack each other is $2n - 2$, see Exercise 13.41. A *margin field* of a chessboard is a field with at least one side that is not a common side of two fields. Each of the four *corner fields* has two such sides. The corner fields are also margin fields.

Every bishop that is placed on a margin field, see Figure 14.13.10, attacks n fields including the field where it is placed. Every bishop that is placed on an inner field attacks more than n fields (including the field where it is placed). Suppose that $2n - 2$ bishops are put on a chessboard $n \times n$ such that no two of them attack each other. Then, any field occupied by a bishop and any corner field can be attacked by at most one bishop. The number of such fields is greater than or equal to $2n - 2 + 2 = 2n$, because at least two corner fields are not occupied by a bishop. The number of the remaining fields is less than or equal to $n^2 - 2n$, and each of them can be attacked by two bishops. The number of pairs (B, F) where B is a bishop, and F is a field attacked by this bishop, is less than or equal to

$$2(n^2 - 2n) + 2n = (2n - 2)n.$$

If all the bishops are placed on margin fields, then the number of pairs (B, F) such that B attacks F is $(2n - 2)n$. If there is a bishop that is placed on an inner field, then the number of pairs (B, F) such that B attacks F is greater than $(2n - 2)n$, and this contradicts the above conclusion. Hence, all the bishops are placed on margin fields.

13.43. Let us consider a margin (but not a corner) field. This field belongs to two small diagonals. There are two more small diagonals that contain the end fields of the previously mentioned two small diagonals. The end fields of these four diagonals are all margin fields, see Figure 14.13.10. Note that these four margin fields are uniquely determined by the initial choice of one of them, and hence, there are $n - 2$ such 4-tuples of margin fields. Suppose that $2n - 2$ bishops are placed on the chessboard such that no two of them attack each other. Then, at most two of the four margin fields that are presented in Figure 14.13.10 can be occupied by bishops. Note also that there are four ways to put two bishops on the corner fields, such that these two fields are not on the same diagonal. Now it is easy to conclude that the number of arrangements of $2n - 2$ bishops on a chessboard $n \times n$, such that no two of them attack each other, is $4 \cdot 2^{n-2} = 2^n$.

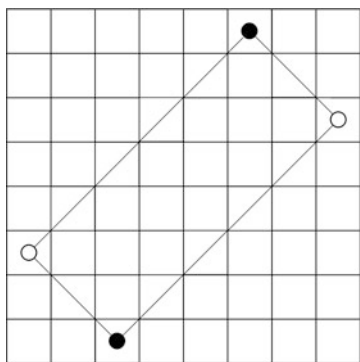


Fig. 14.13.10

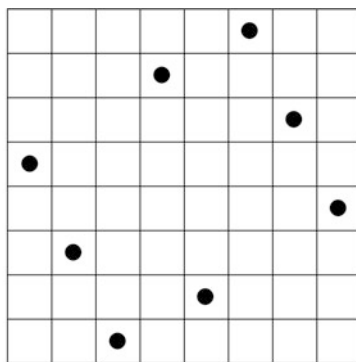


Fig. 14.13.11

13.44. Note that no more than four knights can be placed on a board 4×2 such that no two of them attack each other. This statement holds because any knight on the board 4×2 attacks exactly one of the remaining fields. A chessboard 8×8 can be partitioned into 8 boards 4×2 . It follows that we can put no more than 32 knights on the chessboard 8×8 , such that no two of them attack each other. Note that if we put 32 knights on the 32 black fields of the chessboard 8×8 , then no two of them attack each other.

13.45. There are two arrangements of 32 knights on the chessboard 8×8 such that no two of them attack each other. All the knights should be put on fields of the same color.

13.46. Any queen on a chessboard 8×8 attacks all the other fields in the same column. From this fact we conclude that no more than 8 queens can be placed on the chessboard such that no two of them attack each other. An arrangement of 8 queens with this property is given in Figure 14.13.11.

Remark. There are 92 distinct arrangements of 8 queens on the chessboard such that no two of them attack each other. Moreover, there are 12 basic arrangements with the following properties: (1) Each of the other arrangements can be obtained from the basic one by a rotation of the chessboard, or by symmetry about an axis. (2) None of the basic arrangements can be obtained from another one by rotation or by symmetry.

A collection of basic arrangements is not uniquely determined. The following list represents a set of 12 basic arrangements.

- 1) a2, b6, c8, d3, e1, f4, g7, h5;

2) a2, b7, c5, d8, e1, f4, g6, h3;

3) a3, b5, c8, d4, e1, f7, g2, h6;

4) a4, b7, c3, d8, e2, f5, g1, h6;

5) a6, b3, c1, d8, e4, f2, g7, h5;

6) a7, b2, c6, d3, e1, f4, g8, h6;
- 7) a2, b6, c1, d7, e4, f8, g3, h5;

8) a3, b5, c2, d8, e6, f4, g7, h1;

9) a4, b6, c8, d3, e1, f7, g5, h2;

10) a5, b3, c1, d7, e2, f8, g6, h4;

11) a6, b3, c1, d8, e5, f2, g4, h7;

12) a8, b4, c1, d3, e6, f2, g7, h5.

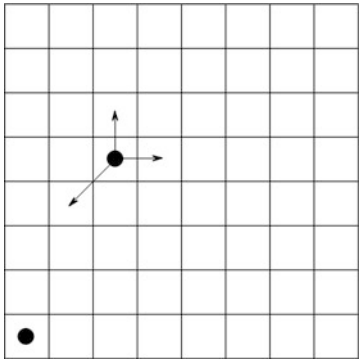


Fig. 14.13.12

1	2	0	1	2	0	1	2
0	1	2	0	1	2	0	1
2	0	1	2	0	1	2	0
1	2	0	1	2	0	1	2
0	1	2	0	1	2	0	1
2	0	1	2	0	1	2	0
1	2	0	1	2	0	1	2
0	1	2	0	1	2	0	1

Fig. 14.13.13

13.47. The initial position of the dolphin and the allowed moves are given in Figure 14.13.12. All the fields of the chessboard are labeled as shown in Figure 14.13.13. Let us consider a sequence $l_0 l_1 l_2 l_3 \dots$ consisting of the digits 0, 1, and 2, that is defined as follows. The first term is $l_0 = 0$ (the label of the field a1). For any $k \in \mathbb{N}$, the term l_k is the label of the field where the dolphin is placed after the k -th move. Every possible trajectory of the dolphin's moves produces the sequence 012012012...

Let us suppose that there is a trajectory of the dolphin's moves such that it visits every field exactly once. The corresponding sequence 012012012... consists of 64 terms, such that 22 of them are equal to 0,

21 of them are equal to 1, and 21 of them are equal to 2. Note that there are 21 0's in the table given in Figure 14.13.13. Hence, there is no trajectory of the dolphin's moves that satisfies the given condition.

13.48. Suppose that a few rooks are placed on a chessboard $3n \times 3n$, such that each of them is attacked by at most one of the remaining rooks. Let x be the number of pairs consisting of two rooks attacking each other, and y be the number of rooks such that none of them is attacked by the other rooks. Any two rooks that attack each other also attack all the fields in three lines (two horizontal and one vertical line, or one horizontal and two vertical lines). Every rook that does not attack the other rooks, attacks all the fields on a vertical and on a horizontal line. The number of lines that are *covered* by all the rooks is $3x + 2y$. Since the total number of (horizontal and vertical) lines is $6n$, it follows that $3x + 2y \leq 6n$. The number of rooks that are on the chessboard is $2x + y$. Now we conclude that

$$2x + y \leq 2x + \frac{4}{3}y = \frac{2}{3}(3x + 2y) \leq 4n.$$

Note that $4n$ rooks can be arranged on the chessboard $3n \times 3n$, such that each of them is attacked by at most one of the remaining rooks. Such an arrangement is given in Figure 14.13.14 in the case $n = 3$, i.e., $3n = 9$. A similar example can be given for any $n \in \mathbb{N}$.

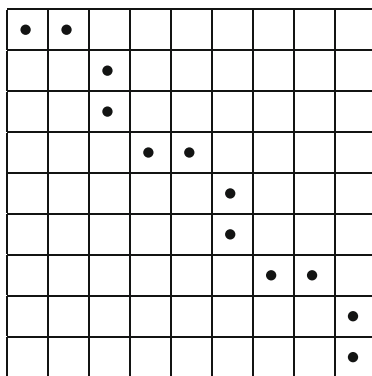


Fig. 14.13.14

13.49. Suppose that n rooks are placed on a chessboard $n \times n$. A necessary and sufficient condition for these n rooks to attack all the fields of the chessboard is that at least one of the following two statements holds:

(A) At least one rook is placed on every horizontal line.

(B) At least one rook is placed on every vertical line.

It is obvious that the above condition is sufficient. Let us prove that the condition is necessary. Suppose, on the contrary, that neither of statements (A) and (B) holds. Then there exist a horizontal line H , and a vertical line V , without rooks on their fields. It follows that the field $H \cap V$ is not attacked, and this conclusion contradicts the initial assumption. Hence, the condition is necessary.

The number of arrangements of n rooks on a chessboard $n \times n$, such that (A) holds, is n^n . Analogously, the number of arrangements of n rooks, such that (B) holds, is also n^n . The number of arrangements of n rooks, such that both (A) and (B) hold, is $n!$. Now it is easy to conclude that the number of arrangements of n rooks on the chessboard $n \times n$, such that at least one of the conditions (A) and (B) is satisfied, is $2n^n - n!$.

13.50. For example, the queens can be placed on fields $c6$, $d3$, $e5$, $f7$, and $g4$.

13.51. *Answer.* $2^{n+1} - 2 = (2^n - 2) + 2^n$. *Hint.* Consider the following two cases. (a) Two chips of the same color are put on two adjacent fields in the first row. In this case every arrangement of chips in the first row uniquely determines the arrangement of chips on the whole chessboard. (b) The chips are alternately red and blue (or blue and red) in the first row. In this case the arrangement of chips in the first column uniquely determines the arrangement of chips on the whole chessboard.

13.52. A chessboard 8×8 has 16 diagonals with an odd number of fields. It follows that the maximal number of chips that can be put on the chessboard such that the given conditions are satisfied is less than or equal to $64 - 16 = 48$. If we put 48 chips on all the fields except the fields that belong to two great diagonals, then the conditions are satisfied.

13.53. *Result:* 9. *Hint.* Let the columns be labeled 1, 2, ..., 9 from left to right. Use the fact that the difference between the number of fields in columns with an odd label and the number of fields in columns with an even label is equal to 9.

13.54. Let the coins be labeled $1, 2, \dots, 12$. Let $\tau\{i, j, k, \dots\}$ be notation for the weight of the set of coins $\{i, j, k, \dots\}$. In the first weighing we shall compare the weight of the sets $\{1, 2, 3, 4\}$ and $\{5, 6, 7, 8\}$. Let us consider the following cases.

(a) $\tau\{1, 2, 3, 4\} = \tau\{5, 6, 7, 8\}$. Then the counterfeit coin belongs to the set $\{9, 10, 11, 12\}$. In the second weighing we compare the sets $\{1, 2, 3\}$ and $\{9, 10, 11\}$.

(a1) If $\tau\{1, 2, 3\} = \tau\{9, 10, 11\}$, then the counterfeit coin is labeled 12. In the third weighing we compare coins 1 and 12, and find out whether coin 12 is lighter or heavier than the genuine coins.

(a2) If $\tau\{1, 2, 3\} < \tau\{9, 10, 11\}$, the counterfeit coin belongs to the set $\{9, 10, 11\}$, and is heavier than the genuine coins. In the third weighing we compare coins 9 and 10. If $\tau\{9\} = \tau\{10\}$, then the counterfeit coin is labeled 11. If, for example $\tau\{9\} < \tau\{10\}$, then the coin labeled 10 is the counterfeit.

(a3) The case $\tau\{1, 2, 3\} > \tau\{9, 10, 11\}$ is analogous to case (a2).

(b) $\tau\{1, 2, 3, 4\} < \tau\{5, 6, 7, 8\}$. The counterfeit coin belongs to the set $\{1, 2, \dots, 8\}$. In the second weighing we compare the sets $\{1, 2, 5\}$ and $\{3, 4, 6\}$. Moreover, we assume that coins 1, 2, and 6 are in the same pan as in the first weighing.

(b1) If $\tau\{1, 2, 5\} = \tau\{3, 4, 6\}$, then the counterfeit coin belongs to the set $\{7, 8\}$. In the third weighing we compare the coins labeled 7 and 8. The heavier one is counterfeit.

(b2) If $\tau\{1, 2, 5\} < \tau\{3, 4, 6\}$, then the counterfeit coin is still in the same pan as in the first weighing. There are two possibilities: the counterfeit coin is labeled 6 and is heavier than the genuine coins, or it belongs to the set $\{1, 2\}$ and is lighter than the genuine coins. In the third weighing we compare coins 1 and 2. If $\tau\{1\} = \tau\{2\}$, then the counterfeit coin is labeled 6. If, for example, $\tau\{1\} < \tau\{2\}$, the counterfeit coin is labeled 1.

(b3) If $\tau\{1, 2, 5\} > \tau\{3, 4, 6\}$, then the counterfeit coin is still on the balance scale, but not in the same pan as in the first weighing. There are two possibilities: the counterfeit coin is labeled 5 and is heavier than the genuine coins, or it belongs to the set $\{3, 4\}$ and is lighter than the genuine coins. In the third weighing we compare coins 3 and 4. If $\tau\{3\} = \tau\{4\}$, then the counterfeit coin is labeled 5. If, for example, $\tau\{3\} < \tau\{4\}$, the counterfeit coin is labeled 3.

(c) The case $\tau\{1, 2, 3, 4\} > \tau\{5, 6, 7, 8\}$ is analogous to case (b).

13.55. *Hint.* Let $\mathcal{G}_0$ be the set of given coins. If $x_n = 2k$ consider the partition $\mathcal{G}_0 = S \cup T$, such that $|S| = |T| = k$. If $x_n = 2k - 1$, let S be a subset of $\mathcal{G}_0$ such that $|S| = k$, and $T = (\mathcal{G}_0 \setminus S) \cup \{g_0\}$, where g_0 is the marked genuine coin. Compare the sets S and T in the first weighing. Then consider the partitions $S = S_1 \cup S_2 \cup S_3$, $T = T_1 \cup T_2 \cup T_3$ with (approximately) the same number of elements in the blocks of the partitions. In the second weighing compare the sets $S_1 \cup T_1$ and $S_2 \cup T_2$. If necessary used the marked genuine coin, and prove that $x_n \leq 3x_{n-1}$. Since $x_1 = 1$ and $x_n = 2$, it follows that $x_n \leq 3^{n-1}$. Then prove that $x_n = 3^{n-1}$.

In order to prove the last equality consider the set $\mathcal{G}_0$ consisting of 3^{n-1} coins, such that only one of them is counterfeit. Suppose that one more genuine coin is given. Then consider the partition $\mathcal{G}_0 = S \cup T$, such that

$$|S| = (3^{n-1} + 1)/2, \quad |T| = (3^{n-1} - 1)/2.$$

For the second weighing consider the sets S_1, S_2, S_3, T_1, T_2 , and T_3 such that

$$\begin{aligned} |S_1| &= (3^{n-2} + 1)/2, & |S_2| &= (3^{n-2} + 1)/2, & |S_3| &= (3^{n-2} - 1)/2, \\ |T_1| &= (3^{n-2} - 1)/2, & |T_2| &= (3^{n-2} - 1)/2, & |T_3| &= (3^{n-2} + 1)/2, \end{aligned}$$

and use the marked genuine coin when necessary.

13.56. *Hint.* Prove that $y_1 = 1$, $y_2 = 4$, and $y_n = y_{n-1} + x_n = y_{n-1} + 3^{n-1}$ for $n \geq 2$. It follows from these equalities that $y_n = (3^n - 1)/2$. Suppose that a set $\mathcal{G}_0$ consisting of $(3^n - 1)/2$ coins is given, such that only one of these coins is counterfeit, and there is one more genuine coin. Let S be a set consisting of $(3^n - 1)/2$ coins from set $\mathcal{G}_0$ and the marked genuine coin, and T be a set consisting of $(3^{n-1} + 1)/2$ coins from set $\mathcal{G}_0$. The first weighing: put the coins from set S in one pan of the balance scale, and the coins from set T in the other pan. Then consider the cases $\tau(S) = \tau(T)$, $\tau(S) < \tau(T)$, and $\tau(S) > \tau(T)$.

13.57. *Answer:* $z_n = (3^n - 3)/2$. *Hint.* For $z_n = (3^n - 3)/2$, put $(3^{n-1} - 1)/2$ coins in one pan, and the same number of coins in the other pan.

13.58. *Hint.* Prove that $u_1 = 2$, $u_2 = 5$, and $u_n = u_{n-1} + 3^{n-1}$ for $n \geq 2$, and hence $u_n = (3^n + 1)/2$. Let $\mathcal{G}_0$ be a set consisting of u_n coins such that one of them is counterfeit. Suppose that one more genuine coin g_0 is given. In the first weighing put $(3^{n-1} + 1)/2$ coins from $\mathcal{G}_0$ in one pan, and $(3^{n-1} - 1)/2$ coins from $\mathcal{G}_0$ and the coin g_0 in the other pan.

13.59. *Result.* $v_n = u_n - 1 = (3^n - 1)/2$. In the first weighing put $(3^{n-1} - 1)/2$ coins in both pans.

13.60. The strategy of player A that prevents player B from obtaining a sum greater than 2001 is the following. Player A starts with the sign $-$. Let S_n be the sum obtained after the n -th move of player B . If $S_n < 0$, then A continues with the sign $+$. If $S_n \geq 0$, then A writes the sign $-$.

Now we formulate the strategy of player B that provides him a sum of 2001. Let us denote

$$S_1 = \{1, 4, 5, 8, \dots, 4k - 3, 4k, \dots, 1997, 2000\};$$

$$S_2 = \{2, 3, 6, 7, \dots, 4k - 2, 4k - 1, \dots, 1998, 1999\}.$$

After every $+$, player B writes an element from set S_1 if this set is not exhausted. After any $-$, player B writes an element from set S_2 . The positive integer 2001 should be written down when one of the signs $+$ or $-$ appears the 1001-st time.

13.61. *Answer.* $n = 9$. *Hint.* Let $n = 8$ and $S = \{0, 20, 40, 1, 2, 4, 7, 12\}$. The set of remainders obtained after dividing the integers from S by 20 is $\{0, 0, 0, 1, 2, 4, 7, 12\}$. Using these remainders prove that set S does not have the given property.

For a set S consisting of n integers, where $n \geq 9$, consider the remainders obtained after dividing the integers from S by 20, and the remainders obtained after dividing integers of the form $a + b$ by 20, where $a, b \in S$.

13.62. Let us consider a partition of the set of chairs (occupied by members of the presidency) into 8 blocks as shown in Figure 14.13.15. If the number of liars in the presidency is less than 8, then there is a block of the partition mentioned above such that all the chairs in this block are occupied by fans of truth. But in every block there is a member such that all their neighbors are from the same block, and this contradicts the statement given by all the members of presidency. Hence, there are at least 8 liars in the presidency.

An arrangement of 8 liars and 24 fans of the truth in the presidency, such that each of them has a liar and a fan of the truth among their neighbors, is given in Figure 14.13.16.

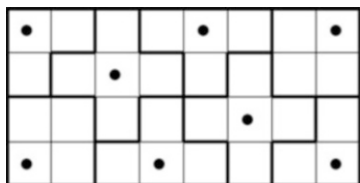


Fig. 14.13.15

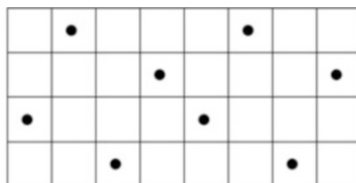


Fig. 14.13.16

13.63. *Answer:* 8; **13.64.** *Answer:* 7; **13.65.** *Answer:* 19; **13.66.** *Answer:* 7.

13.67. Player B has the winning strategy if and only if $n = 2^k$, where $k \in \mathbb{N}$. The smallest positive integer of the form 2^n , that belongs to the set $\{1998, 1999, \dots\}$, is 2048.

(a) Let $n = 2^k$, $k \geq 1$. Suppose that player A takes $2^{k_1}(2l_1 + 1)$ balls in the first move, where $k_1 \geq 0$, $l_1 \geq 0$. Then player B takes 2^{k_1} balls. The number of remaining balls is of the form $2l_2 \cdot 2^{k_1}$, i.e., an even multiple of 2^{k_1} , and also an even multiple of any power of 2 with an exponent that is less than k_1 . After any subsequent move by player A , player B should take the same number of balls as player A .

(b) Let $n = 2^k(2l + 1)$, where $l \geq 1$. In this case player A has the winning strategy. Player A should take 2^k balls in the first move, and then apply the strategy of player B from the previous case.

13.68. Let x_n be the maximal number of 1's in a triangular array with n terms in the first row. We shall prove by induction on n that

$$x_n \leq \left\lfloor \frac{n(n+1)+1}{3} \right\rfloor. \quad (1)$$

For $n \in \{1, 2, 3\}$, it is easy to see that $x_1 = 1$, $x_2 = 2$, and $x_3 = 4$, and inequality (1) holds. Let us suppose that (1) holds for some positive integer n , and prove that it holds for the positive integer $n + 3$ as well.

Lemma. *The first three rows of a triangular array with $n + 3$ terms in the first row, contain at least $n + 2$ zeroes.*

Proof of the Lemma. Let us consider the first three rows:

$$\begin{array}{cccccccccccc} a_1 & a_2 & a_3 & a_4 & \dots & a_n & a_{n+1} & a_{n+2} & a_{n+3} \\ b_1 & b_2 & b_3 & b_4 & \dots & b_n & b_{n+1} & b_{n+2} \\ c_1 & c_2 & c_3 & c_4 & \dots & c_n & c_{n+1} \end{array}$$

Any of the triplets (a_i, b_i, c_i) , where $i \in \{1, 2, \dots, n + 1\}$, will be called a *diagonal*. If any of these $n + 1$ diagonals contains a 0, the statement of the Lemma holds, because at least one of the terms a_{n+2} , a_{n+3} , b_{n+2} is equal to 0. If $a_i = b_i = c_i = 1$ for some $i \in \{1, 2, \dots, n + 1\}$, then $a_{i+1} = b_{i+1} = 0$. Hence, there are at least $n + 2$ 0's in the first three rows. It follows that there are no more than $2n + 4$ 1's in the first three rows.

By using the Lemma and the induction hypothesis we obtain that

$$x_{n+3} \leq 2n + 4 + \left\lfloor \frac{n(n+1)+1}{3} \right\rfloor = \left\lfloor \frac{(n+3)(n+4)+1}{3} \right\rfloor,$$

and the proof of inequality (1) is completed. The existence of a triangular array with n terms in the first row and exactly $[(n^2 + n + 1)/3]$ 1's in the whole array can be proved by examples. The first row in the corresponding array is given for $n = 10$, $n = 11$, $n = 12$ as follows:

$n = 12$: (1,1,0,1,1,0,1,1,0,1,1,0).

$n = 11$: (0,1,1,0,1,1,0,1,1,0,1);

$n = 10$: (1,0,1,1,0,1,1,0,1,1);

A similar example can be constructed for any $n \in \mathbb{N}$. Hence, the maximal number of 1's in a triangular array with n terms in the first row is $[(n^2 + n + 1)/3]$.

13.69. Answer: 16; **13.70.** Answer: 25; **13.71.** Answer: 4; **13.72.** Answer: 6; **13.73.** Answer: 39; **13.74.** Answer: 9.

13.75. Let T_0 be a set whose elements are the following sequences:

(0,0,0,0,0,0), (1,1,1,1,1,1),

$s_1 = (1, 1, 0, 1, 0, 0)$ and 6 cyclic permutations obtained from s_1 ,

$s_2 = (0, 0, 1, 0, 1, 1)$ and 6 cyclic permutations obtained from s_2 .

Then, T_0 is a set consisting of 16 sequences whose terms belong to the set $\{0, 1\}$, and such that the distance between any two sequences from T_0 is not less than 3.

13.76. *Hint.* (a) Use the method of mathematical induction on n .

(b) The equality $|T| = k(n - k) + 1$ holds if $S = \{1, 2, \dots, n\}$.

13.77. We shall prove that the following equality holds for any positive integer n :

$$f(n) = \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor + 1. \quad (1)$$

Let A be the set of even positive integers that are not greater than n , and B be the set of positive integers divisible by 3 that are not greater than n . The set $A \cup B$ consists of $\left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor$ elements. Every 3-subset of set $A \cup B$ contains two even positive integers or two positive integers that are divisible by 3. Hence,

$$f(n) \geq \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor + 1. \quad (2)$$

Now it is sufficient to prove that the following inequality also holds:

$$f(n) \leq \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor + 1. \quad (3)$$

Lemma. Let k be a positive integer, and suppose that A_0 is a 5-subset of the set $C = \{k, k + 1, k + 2, k + 3, k + 4, k + 5\}$. Then there exist three elements of set A_0 that are pairwise coprime.

Proof of the Lemma. It is easy to see that there exists an odd positive integer x such that $x, x+2, x+4 \in C$, and $x, x+2, x+4$ are pairwise coprime. If $y \in \{x+1, x+3\}$ is a positive integer that is not divisible by 3, then the positive integers $x, x+2, x+4, y$ are pairwise coprime. Note that at least three of the positive integers $x, x+2, x+4, y$ belong to set A_0 .

Now we shall prove inequality (3) by induction “from n to $n+6$.” First, we shall check this inequality for $n \in \{3, 4, 5, 6, 7, 8\}$.

Since 1, 2, 3 are pairwise coprime, it follows that $f(3) \leq 3$ and $f(4) \leq 4$, i.e., inequality (3) holds for $n = 3$ and $n = 4$.

Now we check the inequality $f(5) \leq 4 = [5/2] + [5/3] - [5/6] + 1$. If we choose 1 and three of the positive integers 2, 3, 4, and 5, then one of the 3-sets $\{1, 2, 3\}$ and $\{1, 4, 5\}$ consists of the chosen positive integers that are pairwise coprime. If we choose the positive integers 2, 3, 4, and 5, then 3, 4, and 5 are pairwise coprime.

The inequality $f(6) \leq 5 = [6/2] + [6/3] - [6/6] + 1$ follows from the Lemma.

Let us prove the inequality $f(7) \leq 5 = [7/2] + [7/3] - [7/6] + 1$. If we choose five of the positive integers 3, 4, 5, 6, 7, and 8, then by the Lemma it follows that three of the chosen positive integers are pairwise coprime. If we choose 1 and four of the positive integers 2, 3, 4, 5, 6, and 7, then there is a positive integer k such that the positive integers 1, k , and $k+1$ are all chosen and (obviously) pairwise coprime.

Now we prove the inequality $f(8) \leq 6 = [8/2] + [8/3] - [8/6] + 1$. If we choose five of the positive integers 3, 4, 5, 6, 7, and 8, then we again apply the above Lemma. If we choose 1, 2, and four of the positive integers 3, 4, 5, 6, 7, and 8, then there is a positive integer $k \in \{3, 4, 5, 6, 7\}$, such that 1, k , and $k+1$ are all chosen and pairwise coprime.

Let us now suppose that inequality (3) holds for a positive integer $n \geq 3$. We shall prove that this inequality then holds for $n+6$ as well. For any $n \geq 3$, let us denote $g(n) = [n/2] + [n/3] - [n/6] + 1$. It is easy to prove that $g(n+6) = g(n) + 4$. Let A be a subset of the set $\{1, 2, \dots, n+6\}$ such that $|A| = g(n+6)$. If $|A \cap \{n+1, n+2, \dots, n+6\}| \geq 5$, then by the Lemma it follows that A contains three elements that are pairwise coprime. In the opposite case set A contains at least $g(n+6) - 4 = g(n)$ elements of the set $\{1, 2, \dots, n\}$. By the induction hypothesis we can choose three of them such that they are pairwise coprime. Hence, (3) holds for every $n \geq 3$.

13.78. Let A be the set of all monotone numbers with no more than 1993 digits, and B be the set of all sequences $d_1 d_2 \dots d_{1993}$ that have the form

$$\underbrace{00\dots 0}_{11} \underbrace{11\dots 1}_{22} \underbrace{22\dots 2}_{\dots} \underbrace{99\dots 9}_{\dots},$$

contain 1993 terms, and at least one digit is not equal to 0. Not all the digits

should appear in a sequence. There is an obvious bijection between sets A and B , and hence

$$|A| = |B| = \binom{1993 + 10 - 1}{10 - 1} - 1 = \binom{2002}{9} - 1.$$

13.79. For any permutation $(a_1, a_2, \dots, a_n)$ of the set $\{1, 2, \dots, n\}$ let us denote

$$S(a_1, a_2, \dots, a_n) = \sum_{k=1}^{n-1} |a_{k+1} - a_k| + |a_1 - a_n|. \quad (1)$$

Sum (1) can be represented in the form

$$S(a_1, a_2, \dots, a_n) = \sum_{k=1}^n \varepsilon_k (a_{k+1} - a_k) = \sum_{k=1}^n (\varepsilon_{k-1} - \varepsilon_k) a_k, \quad (2)$$

where $\varepsilon_k \in \{-1, 1\}$, and $a_{n+1} = a_1$, $\varepsilon_0 = \varepsilon_n$. It follows from (2) that

$$S(a_1, a_2, \dots, a_n) = \sum_{k=1}^n k b_k, \quad (3)$$

where $b_k \in \{-2, 0, 2\}$ and $b_1 + b_2 + \dots + b_n = 0$. The number of b_k 's that are equal to -2 is the same as the number of b_k 's that are equal to $+2$. Hence, the sum $S(a_1, a_2, \dots, a_n)$ can be written as

$$S(a_1, a_2, \dots, a_n) = 2(x_1 + x_2 + \dots + x_m) - 2(y_1 + y_2 + \dots + y_m), \quad (4)$$

where $x_1, x_2, \dots, x_m, y_1, y_2, \dots, y_m$ are distinct positive integers from the set $\{1, 2, \dots, n\}$. The maximal value of sum (4) is $2m(n - m)$ and is attained for $m = \lfloor n/2 \rfloor$, and

$$\begin{aligned} \{x_1, x_2, \dots, x_m\} &= \{n, n-1, \dots, n-m+1\}, \\ \{y_1, y_2, \dots, y_m\} &= \{1, 2, \dots, m\}. \end{aligned}$$

Let us now consider the permutation $(a_1, a_2, \dots, a_n)$ of the set $\{1, 2, \dots, n\}$ that is defined as follows: $a_{2k-1} = n - m + k$, $a_{2k} = k$ for $k \leq \lfloor n/2 \rfloor$, and, if n is an odd positive integer, $a_n = m + 1$. For this permutation we have $S(a_1, a_2, \dots, a_n) = 2m(n - m)$, and $a_1 - a_n = 1$. Hence, the largest possible value of the sum $S(a_1, a_2, \dots, a_n) - |a_1 - a_n|$ is $2 \lfloor \frac{n}{2} \rfloor (n - \lfloor \frac{n}{2} \rfloor) - 1$.

13.80. Let a be a person from the set S that satisfies the given conditions. Suppose that a has exactly r acquaintances, and denote them by $a_1, \dots, a_r$. It is obvious that, for any $1 \leq i < j \leq r$, the person a_i is not acquainted with a_j . Moreover, a_i and a_j have exactly two common acquaintances, a

and $a_{ij} \neq a$. Hence, the set $\{a_{ij} : 1 \leq i < j \leq r\}$ is in fact the set of persons who are not acquainted with a . It follows that

$$n = 1 + r + \binom{r}{2}, \quad r = \frac{\sqrt{8n-7}-1}{2}.$$

Note that r depends only on n , and is the same for any person $a \in S$. For $r = 2$, we have $n = 4$ and this is not the case we are interested in. Let us consider the case $r \geq 3$. The acquaintances of person a_{12} are a_1, a_2 , and $r - 2$ more persons from the set $\{a_{ij} : 3 \leq i < j \leq r\}$. It follows that $r - 2 \leq \binom{r-2}{2}$, i.e., $r \geq 5$ and $n \geq 16$. Consider the set

$$S = \{a, a_1, a_2, a_3, a_4, a_5, a_{12}, a_{13}, a_{14}, a_{15}, a_{23}, a_{24}, a_{25}, a_{34}, a_{35}, a_{45}\}$$

consisting of 16 persons with the following acquaintances:

1. (a, a_i) , where $1 \leq i \leq 5$;
2. (a_i, a_{ij}) and (a_j, a_{ij}) , where $1 \leq i < j \leq 5$;
3. (a_{ij}, a_{kl}) , where $1 \leq i < j \leq 5, 1 \leq k < l \leq 5, \{i, j\} \cap \{k, l\} = \emptyset$.

All the conditions are satisfied. Hence, the smallest natural number n , for which there exists a set that satisfies the given conditions, is $n = 16$.

13.81. The number of (unordered) pairs of elements of set S is

$$a_0 + a_1 + a_2 + a_3 = \binom{n^2}{2} = \frac{(n-1)n^2(n+1)}{2}.$$

Let T_k be the set of squares from T whose sides are equal to k and parallel to the coordinate axes. Then, $|T_k| = (n-k)^2$. Every square $\mathcal{K}$ from set T_k contains $k-1$ squares whose vertices belong to the sides of $\mathcal{K}$, and whose sides are not parallel to the coordinate axes. It follows that

$$|T| = \sum_{k=1}^{n-1} k(n-k)^2 = \frac{(n-1)n^2(n+1)}{12}.$$

On the other hand, since any square from T gives 6 pairs of points from S , we obtain that $|T| = (a_0 + 2a_2 + 3a_3)/6$. Hence,

$$a_0 + 2a_2 + 3a_3 = \frac{(n-1)n^2(n+1)}{12} = a_0 + a_1 + a_2 + a_3,$$

and consequently $a_0 + a_2 = 2a_3$.

13.82. Let us consider the sequence: $a_k = 2^{n+k} - 2^k$ for $k \in \{0, 1, \dots, n\}$, and $a_{n+1} = 2^{2n} - 1$. Note that, $a_k = 2a_{k-1}$ for any $k \in \{1, 2, \dots, n\}$, and $a_{n+1} = a_1 + a_n$. Hence, if a set S has property (b), and $a_0 = 2^n - 1 \in S$, then the set $S \cup \{a_1, a_2, \dots, a_n, a_{n+1}\}$ has property (b) as well.

Let us denote $A_n = \{2^{n+1} - 2, 2^{n+2} - 2^2, \dots, 2^{2n} - 2^n, 2^{2n} - 1\}$. Then the set $B = \{1\} \cup \bigcup_{i=1}^8 A_{2^i-1}$ has property (b), and $2^{2^8} - 1 \in B$. Note that the number of elements of B is $1 + (1+1) + (2+1) + (2^2+1) + \dots + (2^8+1) = 264$. The set

$$C = B \cup \{2^k(2^{256} - 1) \mid 1 \leq k \leq 243\} \cup \\ \cup \{2^{499} - 2^{115}, 2^{499} - 2^{51}, 2^{499} - 2^{19}, 2^{499} - 2^3, 2^{499} - 2, 2^{499} - 1\}$$

also has property (b) following from these facts: $2^{256} = 1 \in B$, $2^{k+1}(2^{256} - 1) = 2 \cdot 2^k(2^{256} - 1)$ and

$$\begin{aligned} 2^{499} - 2^{115} &= (2^{499} - 2^{243}) + (2^{243} - 2^{115}), & 2^{243} - 2^{115} &\in A_{128}, \\ 2^{499} - 2^{51} &= (2^{499} - 2^{115}) + (2^{115} - 2^{51}), & 2^{115} - 2^{51} &\in A_{64}, \\ 2^{499} - 2^{19} &= (2^{499} - 2^{51}) + (2^{51} - 2^{19}), & 2^{51} - 2^{19} &\in A_{32}, \\ 2^{499} - 2^3 &= (2^{499} - 2^{19}) + (2^{19} - 2^3), & 2^{19} - 2^3 &\in A_{16}, \\ 2^{499} - 2 &= (2^{499} - 2^3) + (2^3 - 2), & 2^3 - 2 &\in A_2, \\ 2^{499} - 1 &= (2^{499} - 2) + (2^2 - 1), & 2^2 - 1 &\in A_1. \end{aligned}$$

Note that $|C| = 264 + 243 + 6 = 513$. Finally, set $A = C \cup A_{499} \cup A_{998}$ has property (b), contains the elements 1 and $2^{1996} - 1$ ($1996 = 2 \cdot 998$), and the number of elements of set A is $513 + (499 + 1) + (998 + 1) = 2012$.

13.83. Suppose that $S = \{x_1, x_2, \dots, x_n\}$. Let $M = (m_{ij})_{1 \leq i \leq n, 1 \leq j \leq k}$ be a matrix whose terms are defined as follows:

$$m_{ij} = \begin{cases} 1, & \text{if } x_i \in A_j, \\ 0, & \text{if } x_i \notin A_j. \end{cases}$$

The number of rows of matrix M is equal to $|S| = n$. It follows from the given conditions that all rows of matrix M are distinct. Hence, the number of elements of set S is not greater than the number of k -arrangements of the elements 0 and 1, i.e., $n \leq 2^k$.

13.84. If a and b are real numbers such that $a - b > 1$, then $[a] \neq [b]$. Since $\frac{(k+1)^2}{1998} - \frac{k^2}{1998} = \frac{2k+1}{1998} > 1$ for $k \geq 999$, it follows that all the terms of the sequence $[k^2/1998]$, $k = 1000, 1001, \dots, 1997$, are distinct positive integers. For $k < 999$, we obtain that $\frac{(k+1)^2}{1998} - \frac{k^2}{1998} = \frac{2k+1}{1998} < 1$, and

hence the sequence $[k^2/1998]$, $k = 1, 2, \dots, 999$, contains all the integers from $[1^2/1998] = 0$ to $[999^2/1998] = 499$. The number of distinct terms in the given sequence is $500 + 1997 - 999 = 1498$.

13.85. Case 1. If $x_0 = x_1 = \dots = x_n = 0$, then $y_j \geq n + 1$ for any $j \in \mathbb{N}_0$, and hence the following inequality obviously holds:

$$\sum_{i=0}^n x_i + \sum_{j=0}^m y_j \geq (n+1)(m+1). \quad (1)$$

Case 2. Suppose that $x_0 = x_1 = \dots = x_{k-1} = 0$ and $x_k = l$, for some $k < n$ and $l \in \mathbb{N}$. Let $(x'_n)_{n \geq 0}$ be the sequence defined by

$$x'_n = \begin{cases} 0, & \text{if } n = k, \\ x_n, & \text{if } n \neq k. \end{cases}$$

Suppose that the sequence $(y'_n)_{n \geq 0}$ is obtained from the sequence $(x'_n)_{n \geq 0}$ the same way as the sequence $(y_n)_{n \geq 0}$ is obtained from the sequence $(x_n)_{n \geq 0}$.

Then we have $y'_n = \begin{cases} y_n + 1, & \text{if } n \leq l \\ y_n, & \text{if } n > l \end{cases}$, and hence

$$\sum_{i=0}^n x_i + \sum_{j=0}^m y_j \geq \sum_{i=0}^n x'_i + \sum_{j=0}^m y'_j. \quad (2)$$

Inequality (2) holds because the sum $\sum_{i=1}^n x_i$ decreases for l , and the sum $\sum_{i=1}^m y_i$ increases for $\min\{l, m\} \leq l$. If we continue to apply the above transformation of the sequences $(y'_n)_{n \geq 0}$ and $(x'_n)_{n \geq 0}$, we will reach the sequence $(x^*_n)_{n \geq 0}$ such that $x^*_0 = x^*_1 = \dots = x^*_n = 0$, and $(y^*_n)_{n \geq 0}$, for which inequality (1) holds. Hence this inequality holds for the initial sequences.

13.86. Let us consider the rectangle $ABCD$, such that the Cartesian coordinates of the vertices are determined as follows: $A(0, 0)$, $B(90, 0)$, $C(90, 50)$, and $D(0, 50)$. Let us introduce two more points: $E(60\sqrt{2}, 0)$ and $F(60\sqrt{2}, 50)$. Consider 49 horizontal lines given by $y = 1, y = 2, \dots, y = 49$, and 6 vertical lines determined by $x = 10\sqrt{2}, x = 20\sqrt{2}, \dots, x = 60\sqrt{2}$. These lines cut off $6 \cdot 50 = 300$ rectangles $1 \times 10\sqrt{2}$ from rectangle $ABCD$. The sides of the remaining rectangle $EBCF$ are given by

$$EB = 90 - 60\sqrt{2} \approx 5.15, \quad BC = 50.$$

Since $3 \cdot 10\sqrt{2} < 50 < 4 \cdot 10\sqrt{2}$, it follows that $15 (= 5 \cdot 3)$ more rectangles of the form $1 \times 10\sqrt{2}$ can be cut off from rectangle $EBCF$. Hence, it is possible to cut off 315 rectangles from rectangle $ABCD$, such that the sides of these 315 rectangles are parallel to the coordinate axes.

Now we prove that there is no way to cut off more than 315 rectangles $1 \times 10\sqrt{2}$ from rectangle $ABCD$, such that their sides are parallel to the coordinate axes. Let us consider the lines

$$x + y = k \cdot 10\sqrt{2}, \quad k \in \{1, 2, \dots, 9\},$$

and the segments $s_1, s_2, \dots, s_9$ determined as the intersection of these lines with rectangle $ABCD$. The sum of the length of these segments is $570\sqrt{2} - 360$ (prove that!). Any rectangle of the form $1 \times 10\sqrt{2}$ that is contained in $ABCD$ and has sides parallel to the coordinate axes, contains part of the segments $s_1, \dots, s_9$ of the total length $\sqrt{2}$ (prove that!). Since $316\sqrt{2} > 570\sqrt{2} - 360$, it follows that there is no a way to cut off 316 rectangles of the given form such that their sides are parallel to the coordinate axes.

13.87. If we denote the empty cell by 0, then the starting position is determined by the permutation $p = (0, 1, 2, \dots, 25, 26)$. The final position (if it can be reached) is determined by the permutation $q = (0, 26, 25, \dots, 2, 1)$. Note that p is an even permutation, while q is an odd permutation. Every legal move is a transposition of two terms and changes the parity of the permutation. Hence, if the final position can be reached, we conclude that this can be done after an odd number of moves.

Suppose that the position determined by permutation q can be reached, and consider a sequence of moves that leads to the final position. Let us consider also the trajectory of the empty cell and the steps that the empty cell makes. Let l, r, f, b, u , and d be the number of these steps to the left, to the right, forward, backward, up, and down, respectively. Since the starting position of the empty cell is the same as its final position, it follows that $l = r$, $f = b$, and $u = d$. Hence, the final position is reached after an even number of legal moves, i.e., permutation q is obtained from permutation p after an even number of transpositions. This conclusion contradicts the previous one. We finally conclude that the position determined by permutation q cannot be reached.

13.88. We consider a graph G with the set of vertices $\{A_1, A_2, \dots, A_n\}$, such that the degree of any vertex is greater than 2. The aim is to prove that there is a cycle with an even number of vertices. Let m be the largest positive integer with the following property (denoted by) $\mathcal{P}$: there is a path $X_1X_2 \dots X_m$, such that $X_1, X_2, \dots, X_m$ are distinct vertices of graph G , and X_iX_{i+1} is an edge of graph G for each $i \in \{1, 2, \dots, m-1\}$. Since the degree of vertex X_1 is greater than 2, there are at least two more edges incident to X_1 , in addition to X_1X_2 . Let X_1Y and X_1Z be the edges incident to X_1 , such that $Y \neq X_2$ and $Z \neq X_2$. Then, $\{Y, Z\} \subset \{X_3, X_4, \dots, X_m\}$. Indeed, if, for example, $Y \notin \{X_3, X_4, \dots, X_m\}$, then the path $YX_1X_2 \dots X_m$ shows that $m+1$ has property $\mathcal{P}$ as well. This contradicts the assumption that m is

the largest positive integer with property $\mathcal{P}$. Hence, $Y = X_i$ and $Z = X_j$ for some distinct $i, j \in \{3, 4, \dots, m\}$. If i (or j) is an even positive integer, then $X_1 X_2 \dots X_{i-1} X_i X_1$ (or $X_1 X_2 \dots X_{j-1} X_j X_1$) is a cycle with an even number of vertices. If both i and j are odd, and $i < j$, then $X_1 X_i X_{i+1} \dots X_j X_1$ is a cycle with an even number of vertices.

13.89. Let $C = \{2003k \mid 1 \leq k \leq 2002\}$ and $D = \{2003k+1 \mid 1 \leq k \leq 2002\}$. Let A be a subset of the set $B = C \cup D$, such that $|A| = 2003$. Let us suppose that $|A \cap D| = k$, where $1 \leq k \leq 2002$. Then $|A \cap C| = 2003 - k$. The remainder that the sum of the elements of set A gives after division by 2003 is $(2003 - k) \cdot 0 + k \cdot 1 = k \in \{1, 2, \dots, 2002\}$. Hence, set $B = C \cup D$ has the given property.

13.90. Let $a, b \in \{1, 2, \dots, n\}$. If a and b are coprime, or one of them is a divisor of the other, then we say that $\{a, b\}$ is a *forbidden subset* of set S .

Let S be a subset of the set $\{1, 2, \dots, n\}$ without forbidden subsets. Let s be the smallest element of set S . Then $2s \notin S$. If $s \leq n/2$, let us define $S' = (S \setminus \{s\}) \cup \{2s\}$. Then, set S' does not contain forbidden subsets as well. Hence, without loss of generality we can assume that all the elements of set S are greater than $n/2$. Note that any two consecutive positive integers are coprime. Hence, if a is a positive integer such that $n/2 < a$, then at most one of the integers a and $a+1$ may belong to set S . By considering the cases $n = 4k, n = 4k+1$, where $k \geq 1$, and $n = 4k+2, n = 4k+3$, where $k \geq 0$, it is easy to conclude that $|S| \leq [(n+2)/2]$. Set $S = \{2k \mid n/2 < 2k \leq n\}$ has exactly $[(n+2)/4]$ elements, and does not contain forbidden subsets.

13.91. Consider the rectangle $OXVY$ whose vertices are determined by the Cartesian coordinates as follows: $O(0,0)$, $X(12,0)$, $V(12,9)$, and $Y(0,9)$. This rectangle is partitioned into $12 \cdot 9 = 108$ unit squares that are arranged in 9 rows and 12 columns. Let the rows and columns be labeled as shown in Figure 14.13.17. We shall use the notation S_{ij} for the unit square that is the intersection of the i -th row and the j -th column. We shall say that a square S_{ij} is *even* (*odd*) if $i+j$ is an even (odd) positive integer.

The solutions of the equation $x^2 + y^2 = 13$ in the set of positive integers are pairs $(2,3)$ and $(3,2)$. Let us assume that it is possible to label red centers $C_1, C_2, \dots, C_{96}$, such that conditions (a) and (b) are satisfied.

Then, for any $i \in \{1, 2, \dots, 96\}$, the following statement holds. If C_i and C_{i+1} , where $C_{97} = C_1$, are the centers of the squares S_{i_1, j_1} and S_{i_2, j_2} , then $\{|i_1 - i_2|, |j_1 - j_2|\} = \{(2,3), (3,2)\}$. The closed broken line $C_1 C_2 \dots C_{96} C_1$ consists of 96 segments, and the length of each segment is $\sqrt{13}$. The center of symmetry is the point C with Cartesian coordinates $(6, 9/2)$. Let A be the center of the square $S_{2,2}$, and B be the center of the square $S_{11,8}$, see

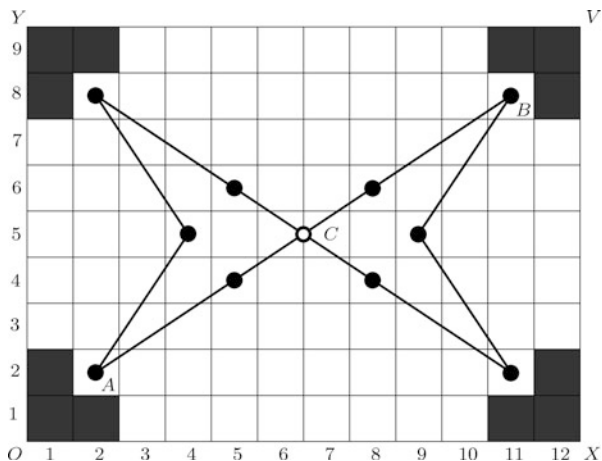


Fig. 14.13.17

Figure 14.13.17. Points A and B divide the broken line $C_1C_2 \dots C_{96}C_1$ into two parts, say $\mathcal{L}_1$ and $\mathcal{L}_2$. We consider the symmetry $\mathcal{S}_c$ around the point C as a function $\mathcal{S}_c : \mathbb{R}^2 \rightarrow \mathbb{R}^2$. Note that $\mathcal{S}_c(A) = B$. Since the broken line is symmetrical around the center C , there are two possibilities.

Case 1. $\mathcal{S}_c(\mathcal{L}_1) = \mathcal{L}_2$. In this case each of the broken lines $\mathcal{L}_1$ and $\mathcal{L}_2$ consists of 48 segments. Suppose that $\mathcal{L}_1 = C_0^*C_1^* \dots C_{47}^*C_{48}^*$, where $C_0^* = A$ and $C_{48}^* = B$. Let S_i^* be a unit square with center C_i^* , where $i \in \{0, 1, \dots, 48\}$. Note that $S_0^* = S_{2,2}$ is an even square, and the squares $S_0^*, S_1^*, S_2^*, \dots$ are alternately even and odd. Hence, S_{48}^* should be an even square. On the other hand, $S_{48}^* = S_{11,8}$ is an odd square which contradicts the previous conclusion.

Case 2. $\mathcal{S}_c(\mathcal{L}_1) = \mathcal{L}_1$ and $\mathcal{S}_c(\mathcal{L}_2) = \mathcal{L}_2$. The assumption that each of the broken lines $\mathcal{L}_1$ and $\mathcal{L}_2$ consists of an even number of segments leads to a contradiction as in Case 1. Suppose that both of $\mathcal{L}_1$ and $\mathcal{L}_2$ consist of an odd number of segments. It follows that, for any $i \in \{1, 2\}$, the broken line $\mathcal{L}_i$ contains a segment J_i such that $\mathcal{S}_c(J_i) = J_i$. There are two segments with this property: the segment J_1 that connects the centers of the squares $S_{5,4}$ and $S_{8,6}$, and the segment J_2 that connects the centers of the squares $S_{5,6}$ and $S_{8,4}$. Note also that each of the red points A, B , the center of $S_{11,2}$, and the center of $S_{2,8}$, can be connected with exactly two other red centers. Let us consider 8 segments that are incident to these 4 red centers, and the segments J_1 and J_2 . They form the closed broken line that is given in Figure 14.13.17. This broken line does not visit any of the red centers, and again we have reached a contradiction!

Hence, it is not possible to label the centers $C_1, C_2, \dots, C_{96}$ such that conditions (a) and (b) are satisfied.

13.92. Let v_0 be an arbitrary vertex, and $E_1 = v_0v_1v_2 \dots v_{k_1}$ be an Eulerian trail that cannot be continued beyond the vertex v_{k_1} (we say that this trail is *maximal*). Let us label the edges of this trail as follows: $1 = v_0v_1, 2 = v_1v_2, \dots, k_1 = v_{k_1-1}v_{k_1}$. If the trail E_1 contains all the edges of graph G , then $k_1 = k$, and all edges are labeled. In the opposite case there is an edge that is incident to one of the vertices $v_0, v_1, \dots, v_{k_1-2}$ (graph G is connected!), and does not belong to trail E_1 . Let $k_1 + 1$ be the label of this edge, and v_i be the vertex from the set $\{v_0, v_1, \dots, v_{k_1-2}\}$ that is incident to $k_1 + 1$. Let E_2 be an Eulerian trail with the following properties: (a) E_2 starts with the vertex v_i , and $k_1 + 1$ is its first edge; (b) E_2 does not contain any edge that belongs to E_1 ; (c) E_2 cannot be extended beyond its end with an edge that does not belong to E_1 . Let $k_1 + 1, k_1 + 2, \dots$, be labels of the edges that belong to E_2 in the order as they appear on the trail. If $E_1 \cup E_2$ does not contain all the edges of graph G we continue by introducing new Eulerian trails and labeling the remaining edges.

The labeling obtained at the end of this process has the given property. Indeed, let v be an arbitrary vertex incident to at least two edges, and let $GCD(v)$ be the greatest common divisor of the integers that are used as the labels of all edges incident to v . If $v = v_0$, then $GCD(v_0) = 1$ because edge 1 is incident to v_0 . Suppose now that $v \neq v_0$. Let m be the minimal positive integer that is used as the label of an edge incident to v . Then, $m + 1$ is the label of an edge incident to v as well. It follows that $GCD(v) = 1$.

13.93. Let S be the set consisting of the given 9 points. There are $\binom{9}{2} = 36$ segments determined by these points. Suppose that 33 segments are colored (blue or red), and only three of them are left uncolored. It is obvious that we can choose 3 points from S , such that any uncolored segment is incident to a chosen point. All segments determined by the remaining 6 points from S are colored blue or red. From Example 10.5.4 it follows that, among these 6 points, we can choose 3 points that are connected by segments of the same color. The example given in Figure 14.13.18 shows that 32 segments can be colored blue or red, and 4 segments be left uncolored, such that there is no triangle with all sides colored the same color. (In Figure 14.13.18 the dashed segments are, for example, red, and all the other segments are blue.)

13.94. For $n = 1$ the game is trivially finished. For $n = 2$ the game can end with only one piece remaining on the board as shown in Figure 14.13.19.

Now let us consider the following positions and sequence of moves.

The combination of moves presented in Figure 14.13.20 allows us to remove three pieces from the fields that form a rectangle 3×1 (or 1×3). One

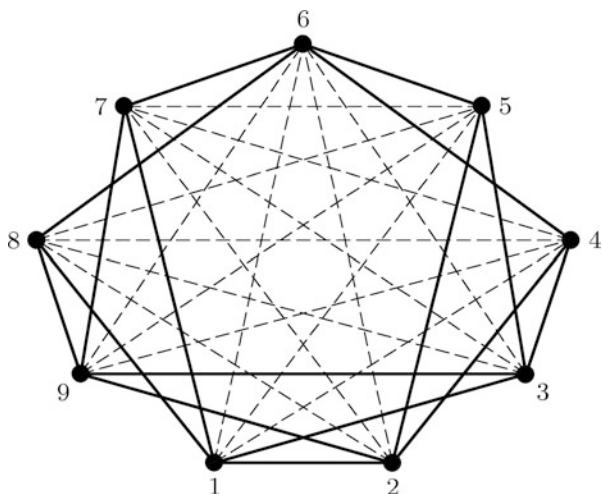


Fig. 14.13.18

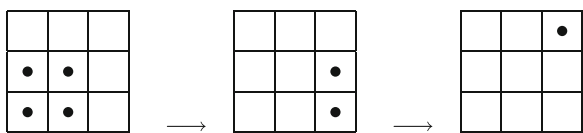


Fig. 14.13.19

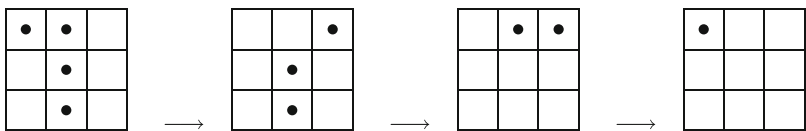


Fig. 14.13.20

additional piece and an unoccupied field are also used in this combination. Using this combination any square $(n + 3) \times (n + 3)$, where $n \geq 4$, can be reduced to a square $n \times n$, see Figure 14.13.21. From the previous consideration we conclude that the game can end with only one piece remaining on the board for any positive integer that is not divisible by 3.

Let us assume that all fields of a square table $3k \times 3k$ are filled by pieces. This square table $3k \times 3k$ is considered to be part of an infinite chessboard.

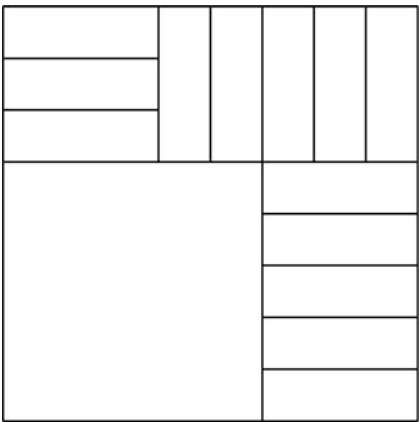


Fig. 14.13.21

1	2	3	1	2	3	1	2	3	1
2	3	1	2	3	1	2	3	1	2
3	1	2•	3•	1•	2•	3•	1•	2	3
1	2	3•	1•	2•	3•	1•	2•	3	1
2	3	1•	2•	3•	1•	2•	3•	1	2
3	1	2•	3•	1•	2•	3•	1•	2	3
1	2	3•	1•	2•	3•	1•	2•	3	1
2	3	1•	2•	3•	1•	2•	3•	1	2
3	1	2	3	1	2	3	1	2	3
1	2	3	1	2	3	1	2	3	1

Fig. 14.13.22

Suppose that the game can end with only one piece on the board. Let us label the field that contains a piece at the end of the game by 1, and then label all the fields of the infinite chessboard as presented in Figure 14.13.22. A *position* is any arrangement of a finite number of pieces on the chessboard. A *characteristic* of a position is the sum of the labels of the fields occupied by pieces. Let us consider how a move changes the characteristic of a position. Suppose that a piece placed on the field labeled 1 jumps over the field labeled

2, and reaches the field labeled 3. The change of the characteristic is $-1-2+3=0$. The list of all possible changes of the characteristic is the following:

$$\begin{aligned} -1-2+3 &= 0, & -2-3+1 &= -4, & -3-1+2 &= -2, \\ -1-3+2 &= -2, & -2-1+3 &= 0, & -3-2+1 &= -4. \end{aligned}$$

The characteristic of the starting position is $18k^2$, and the characteristic of the final position is 1. Since the characteristic remains the same or decreases for an even positive integer after every move, it follows that the position with characteristic 1 cannot be reached.

13.95. Any state of the lamps $L_0, L_1, L_2, \dots, L_{n-1}$ is determined by the vector $v = (v_1, v_1, \dots, v_{n-1})$, where

$$v_i = \begin{cases} 1, & \text{if lamp } L_i \text{ is on,} \\ 0, & \text{if lamp } L_i \text{ is off.} \end{cases}$$

The initial state is given by the vector $e = (1, 1, \dots, 1)$. Note that

$$S_i((v_0, v_1, \dots, v_{n-1})) = (v_0, v_1, \dots, v_{i-1}, v_{i-1} + v_i, v_{i+1}, \dots, v_{n-1}),$$

where $+$ is the sum modulo 2, and hence $v_{i-1} + v_i \in \{0, 1\}$.

(a) It is sufficient to prove that $S_k S_{k-1} \dots S_1 S_0 e = e$ for some $k \in \mathbb{N}$. Let T be the operation defined by $T((v_0, v_1, \dots, v_{n-1})) = (v_1, \dots, v_{n-1}, v_0)$. Then, $S_i = T^{-i} S_0 T^i$, and

$$\begin{aligned} S_k S_{k-1} \dots S_1 S_0 &= (T^{-k} S_0 T^k)(T^{-k+1} S_0 T^{k-1})(T^{-2} S_0 T^2)(T^{-1} S_0 T^1) S_0 \\ &= T^{-k} S_0 (T S_0)^k = T^{-k-1} (T S_0)^{k+1}. \end{aligned}$$

Note that the condition $S_k S_{k-1} \dots S_1 S_0 e = e$ is equivalent to $(T S_0)^{k+1} e = e$. Let us consider the sequence $e, T S_0 e, (T S_0)^2 e, (T S_0)^3 e, \dots$. Since there are 2^n possible states, it follows that there exist the positive integers i and j , such that $i < j$, and $(T S_0)^i e = (T S_0)^j e$. For $k = j - i - 1$ we obtain that $(T S_0)^{k+1} e = (T S_0)^{j-i} e = e$.

(b) For $v = (v_0, v_1, \dots, v_{n-1})$, let us define the polynomial

$$P_v(x) = v_{n-1}x^{n-1} + v_0x^{n-2} + \dots + v_{n-4}x^2 + v_{n-3}x + v_{n-2}.$$

Since $T S_0 v = T(v_{n-1} + v_0, v_1, \dots, v_{n-1}) = (v_1, v_2, \dots, v_{n-1}, v_{n-1} + v_0)$, it follows that

$$Q_v(x) := P_{T S_0 v}(x) = (v_{n-1} + v_0)x^{n-1} + v_1x^{n-2} + \dots + v_{n-3}x^2 + v_{n-2}x + v_{n-1}.$$

Note that $Q_v(x) \equiv x P_v(x) \pmod{x^n - x^{n-1} - 1}$. It follows that the equality $(T S_0)^k e = e$ is equivalent to $x^k \equiv 1 \pmod{x^n - x^{n-1} - 1}$. For $n = 2^k$ we obtain that

$$x^{n^2} \equiv (x^n)^n \equiv (x^{n-1} + 1)^n \equiv x^{n^2-n} + 1, \quad (1)$$

where $\equiv$ is always congruence modulo $x^n - x^{n-1} - 1$. Equality (1) holds because all the binomial coefficients $\binom{2^k}{1}, \binom{2^k}{2}, \dots, \binom{2^k}{2^k-1}$ are even positive integers, i.e., equal to 0 modulo 2. It follows from (1) that

$$1 \equiv x^{n^2} - x^{n^2-n} \equiv x^{n^2-n}(x^n - 1) \equiv x^{n^2-n} \cdot x^{n-1} \equiv x^{n^2-1}.$$

Hence, after $n^2 - 1$ steps all the lamps will be on again.

(c) Let $n = 2^k + 1$. Then similarly as in the previous case we obtain that

$$x^{n^2-1} \equiv (x^{n+1})^{n-1} \equiv [x(x^{n-1} + 1)]^{n-1} \equiv (x^n + x)^{n-1} \equiv x^{n(n-1)} + x^{n-1},$$

It follows that $x^{n^2-1} - x^{n(n-1)} \equiv x^{n-1}$, i.e., $x^{n^2-n}(x^{n-1} - 1) \equiv x^{n-1}$. Hence $x^{n^2-n} \cdot x^n \equiv x^{n-1}$, and finally $x^{n^2-n+1} \equiv 1$. The conclusion is that after $n^2 - n + 1$ steps all the lamps will be on again.

13.96. (a) Let $g(k)$ be the number of elements of the set $\{1, 2, \dots, k\}$ that have exactly three 1's in the base 2 representation. It is obvious that $(g(k))_{k \geq 1}$ is a nondecreasing sequence, and $f(2k) = g(2k) - g(k)$. Hence,

$$\begin{aligned} f(k+1) - f(k) &= (g(2k+2) - g(k+1)) - (g(2k) - g(k)) \\ &= (g(2k+2) - g(2k)) - (g(k+1) - g(k)). \end{aligned}$$

Since $2k+2$ and $k+1$ have the same number of 1's in the base 2 representation, it follows that $f(k+1) - f(k) = 1$ if $2k+1$ has precisely three 1's in the base 2 representation, and $f(k+1) - f(k) = 0$ in all other cases. From the previous fact we conclude that the sequence $(f(k))_{k \geq 1}$ contains all positive integers. Since $g(2^n) = g(2^n - 1) = \binom{n}{3}$, we conclude that

$$f(2^n) = g(2^{n+1}) - g(2^n) = \binom{n+1}{3} - \binom{n}{3} = \binom{n}{2}.$$

Hence, the sequence $(f(k))_{k \geq 1}$ is unbounded from above, and for any positive integer m , there is at least one positive integer k , such that $f(k) = m$.

(b) Let m be a positive integer. Let us assume that there is precisely one positive integer k such that $f(k) = m$. The above assumption holds if and only if $f(k+1) - f(k) = 1$ and $f(k) - f(k-1) = 1$. The first condition holds if and only if $2k+1$ has exactly three 1's in the base 2 representation, i.e., if and only if k has exactly two 1's in the base 2 representation. The same conclusion holds for $k-1$. This is possible if and only if $k-1$ has at least three digits in the base 2 representation, and, moreover, the first and the last digits are equal to 1, while all the remaining digits are equal to 0. It follows that $k = 2^n + 2$ for some $n \geq 2$, and

$$f(2^n + 2) = g(2^{n+1} + 4) - g(2^n + 2) = 1 + g(2^{n+1}) - g(2^n) = 1 + \binom{n}{2}.$$

Hence, equation $f(k) = m$ has a unique solution if and only if $m = 1 + \binom{n}{2}$ for some $n \geq 2$.

13.97. Without loss of generality we can consider the set $S = \{0, 1, \dots, 2p-1\}$ instead of $\{1, 2, \dots, 2p\}$. Let $S = K \cup L$ be a partition of set S , where $K = \{0, 1, \dots, p-1\}$ and $L = \{p, p+1, \dots, 2p-1\}$. For each $A \subset S$ let $|A|$ be the number of elements of set A , and $\sigma(A)$ be the sum of its elements. Let S_p be the collection of all p -subsets of set S , and $f : S_p \rightarrow S_p$ be the function defined by

$$f(A) = \{a+1 \mid a \in A \cap K\} \cup \{A \cap L\},$$

where $+$ is the sum modulo p . It is obvious that $f(K) = K$ and $f(L) = L$, i.e., K and L are fixed points of function f . Let us consider an element $A \in S_p \setminus \{K, L\}$, and denote $k = |A \cap K|$. Then, we have $1 \leq k \leq p-1$. Since $\sigma(f^i(A)) = \sigma(A) + ik$, for any $i \in \{0, 1, \dots, p-1\}$, it follows that

$$\{\sigma(f^i(A)) \mid i = 0, 1, \dots, p-1\} = \{0, 1, \dots, p-1\}.$$

Hence, the sets $A, f(A), f^2(A), \dots, f^{p-1}(A)$ are distinct, and only one of them has a sum of elements that is divisible by p . Let us consider the orbits of function f that do not contain fixed points K and L . The number of elements of any such orbit is equal to p , and only one set from the orbit has a sum of elements that is divisible by p . Therefore, it follows that the number of subsets of set $\{1, 2, \dots, 2p\}$ that satisfy the given conditions is $\frac{1}{p} [\binom{2p}{p} - 2] + 2$.

13.98. Let $\pi = (y_1, y_2, \dots, y_n)$ be an arbitrary permutation of the real numbers $x_1, x_2, \dots, x_n$, and $d = (n+1)/2$. Let us define $S(\pi) = y_1 + 2y_2 + \dots + ny_n$, and denote $\pi_0 = (x_1, x_2, \dots, x_n)$ and $\pi_0^* = (x_n, x_{n-1}, \dots, x_1)$.

Case 1. $|S(\pi_0)| \leq d$ or $|S(\pi_0^*)| \leq d$. The statement obviously holds.

Case 2. $|S(\pi_0)| > d$ and $|S(\pi_0^*)| > d$. In this case we obtain that

$$\begin{aligned} S(\pi_0) + S(\pi_0^*) &= (x_1 + 2x_2 + \dots + nx_n) + (x_n + 2x_{n-1} + \dots + nx_1) \\ &= (n+1)(x_1 + 2x_2 + \dots + nx_n), \end{aligned}$$

and since $|x_1 + x_2 + \dots + x_n| = 1$, it follows that $|S(\pi_0) + S(\pi_0^*)| = n+1 = 2d$. Now we conclude that one of the real numbers $S(\pi_0)$ and $S(\pi_0^*)$ is greater than d , while the other one is less than $-d$.

There is a sequence of permutations $\pi_0, \pi_1, \dots, \pi_m = \pi_0^*$, such that, for each $i \in \{1, 2, \dots, m\}$, the permutation π_i can be obtained from π_{i-1} by the transposition of two adjacent elements of π_{i-1} . If $\pi_i = (y_1, y_2, \dots, y_n)$ and $\pi_{i-1} = (z_1, z_2, \dots, z_n)$, then there is $k \in \{1, 2, \dots, n-1\}$, such that

$z_k = y_{k+1}$, $z_{k+1} = y_k$, and $z_i = y_i$ for any $i \notin \{k, k+1\}$. Since $|x_i| \leq d$ for any i , it follows that

$$\begin{aligned} S(\pi_i) + S(\pi_{i-1}) &= |(z_1 + 2z_2 + \cdots + nz_n) - (y_1 + 2y_2 + \cdots + ny_n)| \\ &= |kz_k + (k+1)z_{k+1} - ky_k - (k+1)y_{k+1}| \\ &= |y_k - y_{k+1}| \leq |y_k| + |y_{k+1}| \leq 2d. \end{aligned}$$

Hence, the difference of any two consecutive terms of the sequence $S(\pi_0)$, $S(\pi_1)$, $\dots$, $S(\pi_m)$ is less than or equal to $2d$. Since the real numbers $S(\pi_0)$ and $S(\pi_m) = S(\pi_0^*)$ lie on different sides of the interval $[-d, d]$, it follows that $S(\pi_k) \in [-d, d]$ for some $k \in \{1, 2, \dots, m-1\}$, i.e., $|S(\pi_k)| \leq d$.

13.99. (a) Suppose that for $n > 1$ there exists a silver matrix $A = [a_{ij}]_{n \times n}$. The union of the i -th row and i -th column will be called the i -th *cross*. Every cross of a silver matrix contains every element of set S exactly once. Since only n elements appear on the main diagonal, it follows that some elements of the set $S = \{1, 2, \dots, 2n-1\}$ do not appear on this diagonal. Let x be such an element. Element x appears in each cross exactly once. If $x = a_{ij}$, where $i \neq j$, then x belongs to the i -th cross and also to the j -th cross. We say that these crosses are x -linked. This implies that all n crosses are partitioned into pairs of x -linked crosses. Therefore n is an even positive integer. This implies that for $n = 1997$ there is no silver matrix.

(b) We shall prove that for any $n = 2^k$, where $k \in \mathbb{N}$, there is a silver matrix $A = [a_{ij}]_{n \times n}$. For $n = 2$, a silver matrix is given by $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$. Now it is sufficient to prove the following statement: if there is a silver matrix $A = [a_{ij}]_{n \times n}$, then there is a silver matrix $D = [d_{ij}]_{2n \times 2n}$. Let $A = [a_{ij}]_{n \times n}$ be a silver matrix, and B be the matrix given by

$$D = \begin{bmatrix} A & B \\ C & A \end{bmatrix},$$

where matrices B and C are defined as follows. Let $(2n, 2n+1, \dots, 3n-1)$ be the first row of matrix B . The subsequent rows of matrix B are cyclic permutations of the first row. That means the second row of matrix B is $(3n-1, 2n, 2n+1, \dots, 3n-2)$, the third row is $(3n-2, 3n-1, 2n, \dots, 3n-3)$, etc. Similarly, the first row of matrix C is $(3n, 3n+1, \dots, 4n-1)$, and the subsequent rows are its cyclic permutations. Now we shall prove that D really is a silver matrix of order $2n$. Indeed, if $1 \leq i \leq n$, then the i -th cross of matrix D contains the i -th cross of matrix A from the upper-left corner, the i -th row of matrix B , and the i -th column of matrix C . Therefore, it follows that the i -th cross of matrix D contains all the elements from the set $\{1, 2, \dots, 4n-1\}$. If $n+1 \leq i \leq 2n$, then the i -th cross of matrix D contains the $(i-n)$ -th cross of matrix A from the lower-right corner, the

$(i - n)$ -th column of matrix B , and the $(i - n)$ -th row of matrix C . Hence, in this case, the i -th cross of matrix D contains all the elements from the set $\{1, 2, \dots, 4n - 1\}$ as well.

13.100. The partition of a positive integer into a sum of powers of 2 (the order of summands does not matter) will be called in short a *partition*, and only this kind of partition will be considered in what follows.

(a) Every partition of the positive integer $2k + 1$ contains a summand of the form $1 = 2^0$. By deleting the summand 1 we obtain a partition of the positive integer $2k$. Similarly, by adding the summand 1 to a partition of $2k$, we obtain a partition of $2k + 1$. Therefore, it follows that

$$f(2k + 1) = f(2k). \quad (1)$$

Let S be the set of all partitions of $2k$, S_1 be the subset of S that consists of partitions that contain the summand 1, and S_2 be the subset of S that consists of partitions without the summand 1. There is a bijection between S_1 and the set of all partitions of $2k - 1$. (Indeed, by removing the summand 1 from a partition that belong to S_1 , we obtain a partition of $2k - 1$. By adding the summand 1 to a partition of $2k - 1$, we obtain a partition that belongs to S_1 .) Similarly, there is a bijection between S_2 and the set of all partitions of k . (Indeed, if $a_1 + a_2 + \dots + a_i$ is a partition from S_2 , then $a_1/2 + a_2/2 + \dots + a_i/2$ is a partition of k . If $b_1 + b_2 + \dots + b_j$ is a partition of k , then $2b_1 + 2b_2 + \dots + 2b_j$ is a partition from S_2 .) Therefore,

$$f(2k) = f(2k - 1) + f(k). \quad (2)$$

If we define $f(0) = 1$ and $f(-1) = 0$, then formulae (1) and (2) hold for every $k \geq 0$. It follows from (1) and (2) that $f(0) = f(1) < f(2) = f(3) < f(4) = f(5) < f(6) = \dots$, i.e., $(f(n))_{n \geq 0}$ is an increasing sequence, and

$$f(2k) - f(2k - 2) = f(k), \quad \text{for } k \in \mathbb{N}. \quad (3)$$

By adding the expressions on the left-hand side and on the right-hand side of equalities (3) for $k \in \{1, 2, \dots, n\}$ we obtain

$$f(2n) = f(0) + f(1) + \dots + f(n). \quad (4)$$

Since $f(0) = f(1) = 1$, and $(f(n))$ is an increasing sequence, we obtain for $n \geq 2$ that

$$\begin{aligned} f(2n) &= 2 + (f(2) + f(3) + \dots + f(n)) \leq 2 + (n - 1)f(n) \\ &\leq f(n) + (n - 1)f(n) = nf(n). \end{aligned} \quad (5)$$

Using equality (5) it follows that

$$\begin{aligned} f(2^n) &= 2^{n-1} f(2^{n-1}) \leq 2^{n-1} \cdot 2^{n-2} f(2^{n-2}) \leq \dots \leq \\ &\leq 2^{n-1} \cdot 2^{n-2} \dots 2^1 \cdot 2^0 \cdot f(2) = 2 \cdot 2^{n(n-1)/2} < 2^{n^2/2}, \quad \text{for } n \geq 3. \end{aligned}$$

(b) Suppose that a and b are two integers of the same parity, and such that $b \geq a \geq 0$. We shall prove the inequality

$$f(b+1) - f(b) \geq f(a+1) - f(a). \quad (6)$$

Indeed, if a and b are both even, then (1) implies that $f(b+1) - f(b) = f(a+1) - f(a) = 0$. If $b = 2b_1 - 1$ and $a = 2a_1 - 1$, where a_1 and b_1 are positive integers such that $b_1 \geq a_1$, then using (2) and the fact that $(f(k))_{k \geq 0}$ is an increasing sequence, it follows that

$$\begin{aligned} f(b+1) - f(b) &= f(2b_1) - f(2b_1 - 1) = f(b_1) \geq f(a_1) \\ &= f(2a_1) - f(2a_1 - 1) = f(a+1) - f(a). \end{aligned}$$

Let r be an even positive integer, and k be a positive integer, such that $r \geq k \geq 1$. If we put $a = r - i$ and $b = r + i$, $i = 0, 1, \dots, k-1$, in inequality (6), and sum the obtained inequalities, we get

$$f(r+k) - f(r) \geq f(r+1) - f(r-k+1). \quad (7)$$

Since r is even, it follows that $f(r+1) = f(r)$, and (7) implies that

$$f(r+k) + f(r-k+1) \geq 2f(r), \quad \text{for } k = 1, 2, \dots, r. \quad (8)$$

By summing the inequalities (8) and using equality (3), it follows that

$$2rf(r) \leq f(1) + f(2) + \dots + f(2r) = f(4r) - 1.$$

Therefore $f(4r) > 2rf(r)$ for any even positive integer r . For $r = 2^{m-2}$ we obtain that

$$f(2^m) > 2^{m-1} f(2^{m-2}). \quad (9)$$

Note that r^{m-2} is even for $m > 2$, and that inequality (9) also holds for $m = 2$. Now, let us consider a positive integer $n \geq 2$ and a positive integer l such that $2l \leq n$. If we put $m = n, n-1, \dots, n-2l+2$ in (9), we get

$$\begin{aligned} f(2^n) &> 2^{n-1} f(2^{n-2}) > 2^{n-1} \cdot 2^{n-3} f(2^{n-4}) > \dots > \\ &> 2^{(n-1)+(n-3)+\dots+(n-2l+1)} f(2^{n-2l}) = 2^{l(n-l)} f(2^{n-2l}). \end{aligned} \quad (10)$$

If n is even, and $l = n/2$, then (10) implies that $f(2^n) > 2^{n^2/4} \cdot f(2^0) = 2^{n^2/4}$. If n is odd, and $l = (n-1)/2$, then (10) implies that

$$f(2^n) > 2^{(n^2-1)/4} \cdot f(2^1) = 2^{(n^2-1)/4} \cdot 2 > 2^{n^2/4}.$$

13.101. Since the number of pairs of judges is equal to $\binom{b}{2}$, and the judges from any pair have at most k equal rates, it follows that the total number of pairs of equal rates is at most $k\binom{b}{2}$. Let x_i and y_i be the number of judges that rate the i -th contestant as “pass” and “fail,” respectively, where $1 \leq i \leq a$. Then, $x_i + y_i = b$ for any i . The number of pairs of judges that rate contestant i equally is $f(2k) = g(2k) - g(k)$. Hence,

$$\begin{aligned} \binom{x_i}{2} + \binom{y_i}{2} &= \frac{1}{2}(x_i^2 + y_i^2 - x_i - y_i) \geq \frac{1}{2} \left(\frac{1}{2}(x_i + y_i)^2 - b \right) \\ &= \frac{1}{4}(b^2 - 2b) = \frac{1}{4}((b-1)^2 - 1). \end{aligned}$$

Since $b \geq 3$ is odd, it follows that $\binom{x_i}{2} + \binom{y_i}{2} \geq \frac{1}{4}(b-1)^2$ as well. Since the total number of pairs of equal rates is $\sum_{i=1}^a \left(\binom{x_i}{2} + \binom{y_i}{2} \right)$, it follows that

$$k\binom{b}{2} \geq \sum_{i=1}^a \left(\binom{x_i}{2} + \binom{y_i}{2} \right) \geq \frac{a(b-1)^2}{4}. \quad (1)$$

Inequality (1) implies that $\frac{k}{a} \geq \frac{b-1}{2b}$.

Remark. If $a = b = 2r + 1$ and $k = r$, where $r \in \mathbb{N}$, then $\frac{k}{a} = \frac{b-1}{2b}$ holds.

If b is even, then the inequality $\frac{k}{a} \geq \frac{b-2}{2b-2}$ can be proved a similar way.

13.102. Let the fields of the given square board $n \times n$ be colored as the fields of a chessboard, see Figure 14.13.23. Let $N = x_n$ be the smallest possible number of marked squares such that the given condition is satisfied. Let w_n be the smallest number of marked white squares, such that any black square has at least one adjacent and marked white square. Similarly, let b_n be the smallest number of marked black squares, such that any white square has at least one adjacent and marked black square. Since $n = 2k$ for some $k \in \mathbb{N}$, it follows that $w_n = b_n$, and $x_n = w_n + b_n$.

Let the white diagonals that are “parallel” to the great black diagonal be labeled $1, 2, \dots, 2k$. The case $k = 5$ is presented in Figure 14.13.23. Consider the diagonals labeled $2, 4, \dots, 2k$. Let us mark $2, 4, \dots, 3, 1$ fields in these diagonals, respectively, as presented in Figure 14.13.23. Every black square is adjacent to exactly one marked white square. If $k = 2l - 1$ for some $l \in \mathbb{N}$, then the number of marked white squares in this example is

$$(2 + 4 + \dots + (k-1)) + (k + \dots + 3 + 1) = \frac{k(k+1)}{2}.$$

Similarly, if $k = 2l$ for some $l \in \mathbb{N}$, then the number of marked white squares in this example is $(2 + 4 + \cdots + k) + ((k - 1) + \cdots + 3 + 1) = k(k + 1)/2$.

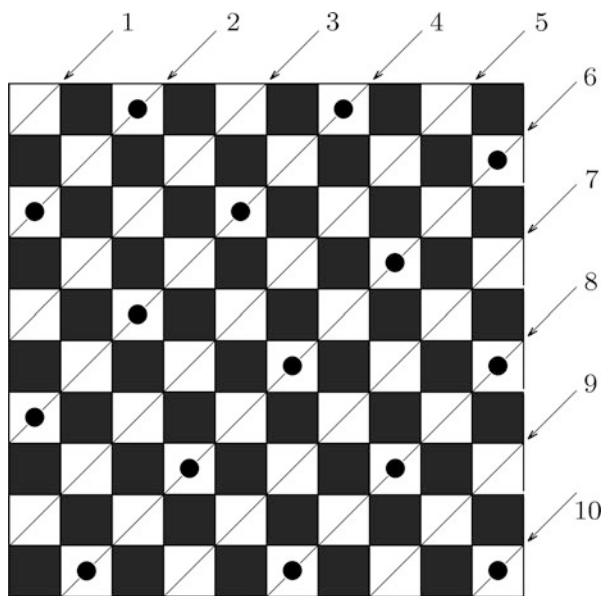


Fig. 14.13.23

Therefore, it follows that

$$w_n \leq \frac{k(k + 1)}{2}. \tag{1}$$

In the given example there is no black square that is adjacent to more than one marked white square. It follows that it is necessary to mark at least $k(k + 1)/2$ black squares. Hence,

$$b_n \geq \frac{k(k + 1)}{2}. \tag{2}$$

Since $w_n = b_n$, it follows from (1) and (2) that $w_n = b_n = k(k + 1)/2$. Hence, $x_n = w_n + b_n = k(k + 1)$.

13.103. The answer is $\lambda \geq 1/(n - 1)$. We shall consider separately the cases $\lambda \geq 1/(n - 1)$ and $\lambda < 1/(n - 1)$.

Case $\lambda \geq 1/(n - 1)$. Every point on the real line (the position of a flea) is determined by a real number (the coordinate of this point). Let us consider the following strategy. For *every move* we choose A to be the point

with the minimal coordinate that is occupied by a flea (or more than one flea). Moreover, we choose B to be the point with the maximal coordinate that is occupied by a flea. The flea from A jumps over point B to point C such that $BC/AB = \lambda$. After a few moves of this type all the fleas will be on distinct points, and, without loss of generality we can consider this position to be the starting one. Let $\delta_k > 0$ be the minimal distance between two fleas after k moves, and d_k be the maximal distance between two fleas after k moves. Note that δ_0 and d_0 are related to the starting position. Obviously $d_k \geq (n-1)\delta_k$ for any $k \geq 0$.

After the $(k+1)$ -st move a new distance between two fleas appears. Note that this distance is λd_k . If this is the minimal distance between two fleas in the new position, then $\lambda d_k = \delta_{k+1}$. In the opposite case $\delta_{k+1} \geq \delta_k$, and

$$\frac{\delta_{k+1}}{\delta_k} \geq \min \left\{ 1, \frac{\lambda d_k}{\delta_k} \right\} \geq \min \{1, (n-1)\lambda\}.$$

Since $\lambda \geq 1/(n-1)$, it follows that $\delta_{k+1} \geq \delta_k$ for any $k \geq 0$. Let x_k be the minimal coordinate that is occupied by a flea after k moves. It is obvious that $x_k \geq x_{k-1} + \delta_{k-1}$, for any $k \geq 1$. It follows that

$$x_k > x_0 + \delta_0 + \delta_1 + \cdots + \delta_{k-1} \geq x_0 + k\delta_0.$$

If $k > (M - x_0)/\delta_0$, then $x_k > M$, i.e., after k moves all the fleas are in positions to the right of M .

Case $\lambda < 1/(n-1)$. We shall prove that for any initial position of the fleas there is a point M on the real line such that the fleas cannot jump over it. Let s_k be the sum of real numbers that represent the positions of the fleas after the k -th move, and M_k be the maximum of these real numbers. It is obvious that $s_k \leq nM_k$. It is sufficient to prove that the sequence $(M_k)_{k \geq 0}$ is bounded from above.

In the $(k+1)$ -st move a flea jumps from point A over point B to point C . Suppose that these three points are determined by the real numbers a , b , and c , respectively. Then, $s_{k+1} = s_k + c - a$, and $c - b = \lambda(b - a)$. It follows that $c - a = (1 + \lambda)(b - a)$, and $\lambda(c - a) = (1 + \lambda)(c - b)$. Hence,

$$s_{k+1} - s_k = c - a = \frac{1 + \lambda}{\lambda} \cdot (c - b).$$

If $c > M_k$, then obviously $M_{k+1} = c$. Since point b is occupied by a flea after the k -th move, we conclude that $b \leq M_k$. Therefore,

$$s_{k+1} - s_k = \frac{1 + \lambda}{\lambda} \cdot (c - b) \geq \frac{1 + \lambda}{\lambda} \cdot (M_{k+1} - M_k). \quad (1)$$

If $c \leq M_k$, then $M_{k+1} = M_k$, $s_{k+1} - s_k = c - a > 0$, and (1) holds as well. Let $(z_k)_{k \geq 0}$ be the sequence of real numbers defined by

$$z_k = \frac{1 + \lambda}{\lambda} \cdot M_k - s_k, \quad k = 0, 1, 2, \dots$$

Using (1) we conclude that $z_{k+1} - z_k \leq 0$ for any $k \geq 0$. Hence, $(z_k)_{k \geq 0}$ is a decreasing sequence, and $z_k \leq 0$ for any $k \geq 0$.

Since $\lambda < 1/(n-1)$, it follows that $1 + \lambda > n\lambda$. Let us denote $\mu = \frac{1+\lambda}{\lambda} - n$. Then $\mu > 0$, and for any $k \geq 0$, we obtain that

$$z_k = (n + \mu)M_k - s_k = \mu M_k + (nM_k - s_k) \geq \mu M_k.$$

Therefore, $M_k \leq z_k/\mu \leq z_0/\mu$ for any $k \geq 0$, i.e., the sequence $(M_k)_{k \geq 0}$ is bounded from above by a constant that depends on the number of fleas n , λ and the initial position, but does not depend on the moves.

13.104. Hint. Consider the following two cases: (a) There are three cards labeled k , $k+1$, and $k+2$ such that each of the given boxes contains one of these cards. (b) There are no three cards labeled k , $k+1$, and $k+2$ such that each of the boxes contains one of these cards.

Answer. There are 12 ways to put the cards into the boxes such that the trick works. Six ways are determined as follows. The cards are partitioned into the sets $\{3k \mid 1 \leq k \leq 33\}$, $\{3k+1 \mid 0 \leq k \leq 33\}$, $\{3k+2 \mid 0 \leq k \leq 32\}$, and each box contains one of these sets of cards. Another six ways are determined as follows. The cards are partitioned into the sets $\{1\}$, $\{2, 3, \dots, 99\}$, $\{100\}$, and each box contains one of these sets of cards.

13.105. Let us introduce the following notation: G – the set of girls, B – the set of boys, P – the set of problems, $P(g)$ – the set of problems solved by the girl $g \in G$, $P(b)$ – the set of problems solved by the boy $b \in B$, $G(p)$ – the set of girls who have solved the problem $p \in P$, and $B(p)$ – the set of boys who have solved the problem $p \in P$. Let us label the problems red or blue as follows. If $|G(p)| \leq 2$, then problem $p \in P$ is labeled red; if $|G(p)| > 2$, then problem $p \in P$ is labeled blue.

Assumption. Suppose that the statement of the problem does not hold, i.e., for every $p \in P$, we have $|G(p)| \leq 2$ or $|B(p)| \leq 2$.

Consider a square table 21×21 , where the rows are related to the girls, and the columns are related to the boys. For every $g_i \in G$ and $b_j \in B$, where $i, j \in \{1, 2, \dots, 21\}$, let the field (g_i, b_j) be labeled red or blue according to the following rule. Choose an element $p \in P(g_i) \cap P(b_j)$, and assign the color of the problem p to the field (g_i, b_j) . By the pigeonhole principle it follows that there are 221 fields colored the same color. It follows that there is a row with 11 blue fields, or there is a column with 11 red fields.

Case 1. The row related to the girl g_i has 11 blue fields. For each of these 11 fields the next statement holds: the problem whose color was assigned to this field is solved by at most two boys. It follows that there are at least $\lceil 11/2 \rceil + 1 = 6$ problems solved by g_i . Hence, girl g_i has solved exactly 6 problems. But then, for at most 12 boys there is a problem solved by a boy and girl g_i . This contradicts the imposed condition.

Case 2. There is a column with 11 red fields. In this case we reach a contradiction analogously.

Therefore, there is a problem $p \in P$, such that $|G(p)| \geq 3$ and $|B(p)| \geq 3$.

13.106. Let $\mathcal{P}$ be the set of all permutations of the set $\{1, 2, \dots, n\}$. Suppose that there is no permutation with the given property. Then we obtain all remainders $0, 1, \dots, n! - 1$ after dividing $S(a)$ by $n!$ for all $a \in \mathcal{P}$. It follows that

$$\sum_{a \in \mathcal{P}} \equiv 0 + 1 + \dots + (n! - 1) \equiv \frac{(n! - 1)!}{2} \equiv \frac{n!}{2} \pmod{n!}. \quad (1)$$

On the other hand, the coefficient of k_i in $\sum_{a \in \mathcal{P}} S(a)$ is $(n-1)! \cdot (1+2+\dots+n)$ for any $i \in \{1, 2, \dots, n\}$. Using the fact that n is odd we obtain that

$$\sum_{a \in \mathcal{P}} S(a) \equiv \frac{n+1}{2} \cdot (k_1 + k_2 + \dots + k_n) \cdot n! \equiv 0 \pmod{n!}. \quad (2)$$

Since (2) contradicts (1), it follows that there are two distinct permutations $b, c \in \mathcal{P}$, such that $n!$ is a divisor of $S(b) - S(c)$.

13.107. Let a_i be the number of blue points with the x -coordinate equal to i , and b_i be the number of blue points with the y -coordinate equal to i . Let r be the number of red points. Then $r \in \{0, 1, \dots, n(n+1)/2\}$. Note that the number of X -sets is $a_0 a_1 \dots a_{n-1}$, while the number of Y -sets is $b_0 b_1 \dots b_{n-1}$. By the method of mathematical induction on the number of red points r we shall prove that

$$a_0 a_1 \dots a_{n-1} = b_0 b_1 \dots b_{n-1}. \quad (1)$$

Moreover, $(a_0, a_1, \dots, a_{n-1})$ is a permutation of $b_0, b_1, \dots, b_{n-1}$.

A *configuration* is the set T of the given points with any coloring that satisfies the given condition. If all points are blue, i.e., if $r = 0$, equality (1) obviously holds. Suppose that equality (1) holds for all configurations such that the number of red points belongs to the set $\{0, 1, \dots, r-1\}$. Consider a configuration $\mathcal{C}$ with r red points. Let (x, y) be a red point with the greatest value of the sum $x + y$. It is easy to see that $a_x = a_y = n - 1 - x - y$. Note that for any i , the numbers a_i and b_i depend on the configuration, i.e., $a_i = a_i(\mathcal{C})$, $b_i = b_i(\mathcal{C})$.

Now, let us change the color of the point (x, y) from red to blue. Then, the obtained configuration $\mathcal{C}^*$ has $r - 1$ red points. Moreover, it is easy to see that the configurations $\mathcal{C}$ and $\mathcal{C}^*$ satisfy the following equalities:

$$a_x(\mathcal{C}^*) = a_x(\mathcal{C}) + 1 = n - x - y = a_y(\mathcal{C}^*) = a_y(\mathcal{C}) + 1, \quad (2)$$

$$a_i(\mathcal{C}^*) = a_i(\mathcal{C}), \quad b_i(\mathcal{C}^*) = b_i(\mathcal{C}), \quad \text{for any } i \neq x. \quad (3)$$

By the induction hypothesis equality (1) holds for $\mathcal{C}^*$. Using (2) and (3) we obtain that (1) holds for $\mathcal{C}$ as well.

13.108. Let us consider the set $D = \{x - y \mid x, y \in A\}$. Then, $|D| \leq 101 \cdot 100 + 1$. Suppose that $t_i, t_j \in S$. The sets $A + t_i$ and $A + t_j$ are disjoint if and only if $t_i - t_j \notin D$. Hence, it is sufficient to choose elements $t_1, t_2, \dots, t_{100} \in S$, such that $t_i - t_j \notin D$ for all $i, j \in \{1, 2, \dots, 100\}$, and $i \neq j$. First we choose an arbitrary element $t_1 \in S$. Then we choose an element $t_2 \in S \setminus (D + t_1)$. It is obvious that $t_2 - t_1 \notin D$. Suppose that the elements $t_1, t_2, \dots, t_k \in S$ are chosen such that $t_i - t_j \notin D$, for all $i, j \in \{1, 2, \dots, k\}$ and $i \neq j$, where $1 \leq k \leq 99$. Note that

$$\left| \bigcup_{i=1}^k (D + t_i) \right| \leq \sum_{i=1}^k |D + t_i| \leq 99 \cdot (101 \cdot 100 + 1) = 999\,999 < 10^6 = |S|.$$

Hence, we can choose t_{k+1} from the set $S \setminus (\cup_{i=1}^k (D + t_i))$. Since $t_{k+1} \notin D + t_i$ for every $i \in \{1, 2, \dots, k\}$, it follows that $t_{k+1} - t_i \notin D$.

13.109. Let n_i be the number of participants that solved exactly i problems, where $0 \leq i \leq 6$. By the given condition $a_6 = 0$. The total number of contestants is $n := \sum_{i=0}^6 n_i$. For all $i, j \in \{1, 2, 3, 4, 5, 6\}$, $i \neq j$, let n_{ij} be the number of participants that solved both the i -th and the j -th problems. The total number of pairs of problems solved by the same contestant is

$$S := \sum_{i=0}^6 \binom{i}{2} n_i = n_2 + 3n_3 + 6n_4 + 10n_5. \quad (1)$$

Since every two of the problems were solved by more than $2/5$ of the contestants, it follows that

$$S = \sum_{i < j} n_{ij} \geq \binom{6}{2} \cdot \frac{2n+1}{5} = 6n + 3 = 6 \sum_{i=0}^6 n_i + 3. \quad (2)$$

It follows from (1) and (2) that

$$4n_5 - 3 \geq 6n_0 + 6n_1 + 5n_2 + 3n_3. \quad (3)$$

Obviously $n_5 > 0$. It is sufficient to prove that n_5 cannot be equal to 1. Suppose, on the contrary, that $n_5 = 1$. Then, inequality (3) implies that $n_0 = n_1 = n_2 = n_3 = 0$. Using (2) and (1) we obtain that $n_4 = n - 1$, and $S = 6(n - 1) + 10 = 6n + 4$.

Case 1. $2n + 1$ is not divisible by 5. Then, using (2) we obtain that $6n + 4 = S \geq \binom{6}{2} \cdot \frac{2n+2}{5} = 6n + 6$, and this is an obvious contradiction.

Case 2. $\frac{2n+1}{5} = k \in \mathbb{N}$. Since $S = 6n + 4 = 15k + 1$, it follows from (2) that not all n_{ij} 's are equal to k , and only one of them is equal to $k + 1$. Let C be the contestant that solved exactly 5 problems. Without loss of generality we can assume that C solved namely the 1st, 2nd, 3rd, 4th, and 5th problems, and that $n_{1i} = k$ for $2 \leq i \leq 6$. If x is the number of contestants that solved the 6th problem, and y is the number of contestants that solved four problems including the first one, then $n_{16} + n_{26} + n_{36} + n_{46} + n_{56} = 3x$ (because every contestant that solved the 6th problem contributes 3 to the previous sum), while $n_{12} + n_{13} + n_{14} + n_{15} + n_{16} = 3y + 4$ (because every contestant that solved the 1st problem contributes 3 to the last sum, except one that contributes 4).

Now we conclude that $3x = n_{16} + n_{26} + n_{36} + n_{46} + n_{56} \in \{5k, 5k + 1\}$, i.e., $3x \in \{2n + 1, 2n + 2\}$. It follows that n is not divisible by 3. Since the pair (i, j) for which $n_{ij} = k + 1$ does not participate in the sum $n_{12} + n_{13} + n_{14} + n_{15} + n_{16} = 3y + 4$, it follows that $5k = 3y + 4$, i.e., $2n + 1 = 3y + 4$. It follows that n is divisible by 3, and this contradicts the previous conclusion.

Therefore, we finally obtain that $n_5 \geq 2$.

13.110. Let τ be a triangulation of the n -gon P . Consider an isosceles triangle ABC with two good sides that appear in the triangulation τ , and suppose that $AB = AC$. Obviously, the good sides are AB and AC , and BC is not a good one. Points A and B divide the boundary of P into two broken lines: the part $\mathcal{L}_1$ that does not contain point C , and the part $\mathcal{L}_2$ that contains point C . Similarly, points A and C divide the boundary of P into two broken lines: the part $\mathcal{L}_3$ that does not contain point B , and the part $\mathcal{L}_4$ that contains point B . Then, each of the broken lines $\mathcal{L}_1$ and $\mathcal{L}_3$ contains a side of P that is not a good side of a triangle of the triangulation τ . These two sides cannot appear in the same context for any other triangle of the triangulation τ . It follows that the number of good triangles of the triangulation τ is not greater than $2006/2 = 1003$.

Consider a triangulation of the regular 2006-gon $A_1A_2 \dots A_{2006}$ that is determined as follows. First we draw the diagonals $A_1A_3, A_3A_5, \dots, A_{2003}A_{2005}, A_{2005}A_1$. In order to obtain a triangulation we draw the remaining diagonals arbitrarily. Any triangulation obtained this way has 1003 isosceles triangles having two good sides.

13.111. Let $2n$ be the greatest size of a clique. Consider a clique M of the size $2n$ (there may be more than one clique of this size). An M -competitor is a competitor that belongs to clique M . Let us arrange all the competitors in two rooms, denoted A and B , such that all M -competitors are arranged in room A , and all the remaining competitors are arranged in room B . Let $s(A) = 2n$ and $s(B)$ be the greatest size of a clique in rooms A and B , respectively. Then, $s(B) \leq 2n$. If $s(B) = 2n$, then the proof is finished. Suppose that $s(B) < 2n$. If we move an M -competitor from room A into room B , then size $s(A)$ decreases by 1, and size $s(B)$ increases by 1, or remains the same. The difference $s(A) - s(B)$ changes by 1 or 2. Let us continue to move M -competitors, one by one, from room A into room B . After a few steps we have $s(B) = s(A)$, and the proof is then finished, or $s(B) = s(A) + 1$. Let us consider the second possibility, and let k be the value of $S(B) = S(A) + 1$ at this moment.

Case 1. *There is an M -competitor c in room B , and there is a clique C of greatest size $k + 1$ in room B such that $c \notin C$.* Let us move competitor c back to room A . We obtain an arrangement of competitors, such that $s(B) = k + 1$ and $s(A) = k + 1$.

Case 2. *Every M -competitor in room B belongs to every clique of greatest size $k + 1$ in this room.* The number of M -competitors in room B is $2n - k$. It follows that $k + 1 \geq 2n - k$. Since $k + 1$ and k are of different parity, it follows that $k + 1 > 2n - k$. Hence, every clique in room B of size $k + 1$ contains a member that is not an M -competitor.

Choose a clique of size $k + 1$ in room B , and move its member that is not an M -competitor to room A . Then choose one of the remaining cliques of size $k + 1$ and move its member that is not an M -competitor to room A , etc. Obviously, after a finite number of steps we will reach an arrangement of the competitors such that $s(B) = k$.

Now we prove that $s(A)$ is still equal to k at the end of this process. Let us suppose, on the contrary, that there is a clique C_0 in room A , such that its size is greater than or equal to $k + 1$. Let M_B be the set of $2n - k$ M -competitors that have been moved to room B . Obviously, the set $C_0 \cup M_B$ consists of more than $2n$ competitors. Moreover, $C_0 \cup M_B$ is a clique, because every element of set C_0 is an M -competitor, or came from clique C^* in room B , such that $M_B \subset C^*$. This contradicts the given condition that $2n$ is the greatest size of a clique in the set of all competitors. Hence, the situation considered in Case 2 is not possible.

13.112. Let $\mathfrak{N}$ be a set of sequences consisting of k steps, and resulting in the state where lamps 1 through n are all on, and lamps $n + 1$ through $2n$ are all off. Let $\mathfrak{M} \subset \mathfrak{N}$ be the subset that consists of sequences of steps such that none of the lamps $n + 1$ through $2n$ is ever switched on. Obviously,

$|\mathfrak{N}| = N$, and $|\mathfrak{M}| = M$. Note also that $\mathfrak{M}$ is a nonempty set. For example, the next sequence belongs to $\mathfrak{M}$: one switches the first lamp $k - n + 1$ times, and then switches the lamps $2, 3, \dots, n$, each of them once.

Let us define a function $f : \mathfrak{N} \rightarrow \mathfrak{M}$ as follows. For every sequence $(\mathbf{n}_1, \dots, \mathbf{n}_k) \in \mathfrak{N}$, let us define $f((\mathbf{n}_1, \dots, \mathbf{n}_k)) = (\mathbf{m}_1, \dots, \mathbf{m}_k) \in \mathfrak{M}$ the following way. If $\mathbf{n}_i$ switches lamp l , where $1 \leq l \leq n$, then $\mathbf{m}_i$ switches lamp l as well. If $\mathbf{n}_i$ switches lamp l , where $n + 1 \leq l \leq 2n$, then $\mathbf{m}_i$ switches lamp $l - n$. It is obvious that every $(\mathbf{m}_1, \dots, \mathbf{m}_k) \in \mathfrak{M}$ is a fixed point of function f , i.e., $f((\mathbf{m}_1, \dots, \mathbf{m}_k)) = (\mathbf{m}_1, \dots, \mathbf{m}_k)$.

Let us consider a sequence $(\mathbf{m}_1, \dots, \mathbf{m}_k) \in \mathfrak{M}$. This sequence of steps switched the i -th lamp $2a_i + 1$ times, where $1 \leq i \leq n$, and never switched on any of the lamps $n + 1$ through $2n$. Note that $\sum_{i=1}^n (2a_i + 1) = k$ is the number of steps. Instead of switching the i -th lamp $2a_i + 1$ times, we could switch it $2a_1 + 1 - 2l$ times and switch the $(i + n)$ -th lamp $2l$ times, for $l \in \{0, 1, \dots, a_i\}$, and the resulting state would be the same. The number of ways in which this can be done is

$$\binom{2a_i + 1}{0} + \binom{2a_i + 1}{2} + \dots + \binom{2a_i + 1}{2a_i} = 2^{2a_i}.$$

Now it is easy to conclude that the number of sequences $(\mathbf{n}_1, \dots, \mathbf{n}_k) \in \mathfrak{N}$, such that $f((\mathbf{n}_1, \dots, \mathbf{n}_k)) = (\mathbf{m}_1, \dots, \mathbf{m}_k) \in \mathfrak{M}$ is

$$\prod_{i=1}^n 2^{2a_i} = 2^{2(a_1 + a_2 + \dots + a_n)} = 2^{k-n}.$$

Therefore, $N/M = 2^{k-n}$.

13.113. We shall prove the statement by the method of mathematical induction on n . For $n = 1$ the statement trivially holds. Let $n \geq 2$ be given. Suppose that the statement holds for all $k < n$ (this is the induction hypothesis). Let us denote $d = \min M$. Without loss of generality we can suppose that $a_1 < a_2 < \dots < a_n$.

Case 1: $a_n > d$. If $a_n \notin M$, then the interval (a_n, s) contains at most $n - 2$ elements of set M . The first jump of the grasshopper is from point 0 to point a_n . By the induction hypothesis we then conclude that the grasshopper can reach point s after $n - 1$ additional jumps of lengths $a_1, \dots, a_{n-1}$ in some order, but never landing on a point in M .

If $a_n \in M$ then $|(0, a_n] \cap M| \geq 2$. Let us consider the sets $\{a_n\}$ and $\{a_i, a_i + a_n\}$, for all $i \in \{1, 2, \dots, n - 1\}$. All these sets are pairwise disjoint, and their union consists of $2n - 1$ elements. Since $a_n \in M$, and $|M| = n - 1$, it follows that there exists $i \in \{1, 2, \dots, n - 1\}$, such that $M \cap \{a_i, a_i + a_n\} = \emptyset$. Then, $|(a_i + a_n, s) \cap M| \leq n - 3$. The first jump of the grasshopper is from

point 0 to point a_i , and the second jump is from point a_i to point $a_i + a_n$. By the induction hypothesis we then conclude that the grasshopper can reach point s after $n - 2$ additional jumps of lengths from $\{a_1, \dots, a_{n-1}\} \setminus \{a_i\}$, but never landing on a point in M .

Case 2: $a_n \leq d$. Note that $|M \setminus \{d\}| = n - 2$. The first jump is from point 0 to point a_n . By the induction hypothesis it follows that the grasshopper can reach point s after $n - 1$ additional jumps of lengths $a_1, \dots, a_{n-1}$ in some order, but never landing on a point from $M \setminus \{d\}$. If the grasshopper has not landed on point d , the proof is finished. In the opposite case we have $d = a_n + \sum_{i=1}^m a_{k_i}$ for some $m \in \{1, 2, \dots, n - 2\}$, where $(a_{k_1}, a_{k_2}, \dots, a_{k_{m+1}})$ is a permutation of the set $\{a_1, a_2, \dots, a_{n-1}\}$. Now we prove that n jumps of lengths $a_1, \dots, a_n$ in the order

$$a_{k_1}, a_{k_2}, \dots, a_{k_{m+1}}, a_n, a_{k_{m+2}}, \dots, a_{k_{n-1}}$$

fulfill all the imposed conditions. Indeed, since $\sum_{i=1}^{m+1} a_{k_i} < \sum_{i=1}^m a_{k_i} + a_n = d$, we conclude that the grasshopper did not land on a point in M till the $(k + 1)$ -st jump. Note also that for every $l \geq m + 1$,

$$\sum_{i=1}^l a_{k_i} + a_n \geq \sum_{i=1}^{m+1} a_{k_i} + a_n = d + a_{k_{m+1}} > d,$$

and $\sum_{i=1}^l a_{k_i} + a_n \notin M$ as stated before.

13.114. Notation for k boxes containing $n_1, \dots, n_k$ coins is $(n_1, \dots, n_k)$.

Let the starting state of the boxes be $(a, 0, 0)$, where a is a positive integer. We shall prove that by applying the allowed operations we can reach the state $(a - k, 2^k, 0)$, where $1 \leq k \leq a$. The proof will be given by the method of mathematical induction on k . For $k = 1$, note that a Type 1 operation leads to the state $(a - 1, 2^1, 0)$. Suppose that the statement holds for some $k \in \{1, 2, \dots, a - 1\}$. Consider the state $(a - k, 2^k, 0)$. By applying a Type 1 operation on the second and third boxes 2^k times repeatedly, we reach the state $(a - k, 0, 2^{k+1})$. A Type 2 operation then leads to the state $(a - k - 1, 2^{k+1}, 0)$. A *Type 3 operation* is defined as a sequence of operations that transform the state $(a, 0, 0)$ into the state $(0, 2^a, 0)$.

Let $p_1 = 2$, and $p_{n+1} = 2^{p_n}$ for $n \geq 1$. By applying Type 1, Type 2, and Type 3 operations we can transform the state $(a, 0, 0)$ to the state $(a - k, p_k, 0, 0)$, where $1 \leq k \leq a$. Proof by induction on k . For $k = 1$ note that a Type 1 operation leads to the state $(a - 1, 2, 0, 0)$. Suppose that the statement holds for some $k \in \{1, 2, \dots, a - 1\}$. Let us consider the state $(a_k - k, p_k, 0, 0)$. By applying a Type 3 operation on the last three boxes we can reach the state $(a - k, 0, 2^{p_k}, 0) = (a - k, 0, p_{k+1}, 0)$. A Type 2 operation,

applied then to the first three boxes, leads to the state $(a - k - 1, p_{k+1}, 0, 0)$. A *Type 4 operation* is defined as a sequence of operations that transform the state $(a, 0, 0, 0)$ into the state $(0, p_a, 0, 0)$.

A Type 1 operation, applied for $j = 5$ to the state $(1, 1, 1, 1, 1, 1)$, transforms it to the state $(1, 1, 1, 1, 0, 3)$. A Type 2 operation applied then successively for $k = 4, k = 3, k = 2$, and $k = 1$ to this state transforms it to the state $(0, 3, 0, 0, 0, 0)$. Then, a Type 4 operation, applied to boxes 2, 3, 4, and 5, leads to the state $(0, 0, p_3, 0, 0, 0)$. A Type 4 operation, applied to boxes 3, 4, 5, and 6, leads then to the state $(0, 0, 0, p_{p_3}, 0, 0)$. Let us denote $4r = 2010^{2010^{2010}} = q$. Since $2010 < 2^{11}$ and $15 < p_{13}$, it follows that

$$q < 2^{11 \cdot 2010^{2010}} < 2^{2010^{2011}} < 2^{2^{11} \cdot 2011} < 2^{2^{2^4} \cdot 2^{11}} = 2^{2^{2^{15}}} = 2^{2^{2^{p_{13}}}} = p_{16} = p_{p_3}.$$

Starting from the last state $(0, 0, 0, p_{p_3}, 0, 0)$, and applying a Type 2 operation $p_{16} - r$ times repeatedly for $k = 4$, we reach the state $(0, 0, 0, r, 0, 0)$. A Type 1 operation applied then r times for $j = 4$ leads to the state $(0, 0, 0, 0, 2r, 0)$. Finally, a Type 1 operation applied $2r$ times repeatedly for $j = 5$ gives the state $(0, 0, 0, 0, 0, 4r)$, i.e., $(0, 0, 0, 0, 0, q)$. Therefore, the answer to the given question is positive.

13.115. Let a_n be the number of ways in which the weights can be placed on the balance scale such that the given condition is satisfied. Obviously, $a_1 = 1$. Note that $2^k > 2^0 + 2^1 + \dots + 2^{k-1}$ for any positive integer k . Using this fact we conclude that any $n - 1$ of the given weights can be placed on the balance such that the given condition is satisfied in a_{n-1} ways. If weight 1 is first placed on the balance, then it should be in the left pan. If a weight heavier than 1 is first placed on the balance, then weight 1 can be placed arbitrarily in the left or in the right pan. It follows that $a_n = (2n - 1)a_{n-1}$ for $n \geq 1$. Now it is easy to conclude that $a_n = (2n - 1) \cdot \dots \cdot 3 \cdot a_1 = (2n - 1)!!$.

13.116. (a) Suppose that $n \geq 2^k$. Let $X = \{a_0, a_1, \dots, a_{a^k-1}, a_{2^k}\}$ be a subset of the set $\{1, 2, \dots, N\}$. (Do not confuse N with n .) It is sufficient to prove that, under the given conditions, player B can determine an element of set X that is not equal to x . Let S_i be the set that B chooses at the i -th step, and r_i be the answer given by player A , where

$$r_i = \begin{cases} 1, & \text{if } A \text{ says } x \in S_i, \\ 0, & \text{if } A \text{ says } x \notin S_i. \end{cases}$$

Player B chooses the set S_i to be equal to $\{a_{2^k}\}$ till the first appearance of the answer $x \in S_i$ ($r_i = 1$), but at most $k + 1$ times.

Case 1. Suppose that $r_1 = r_2 = \cdots = r_{k+1} = 0$. In this case we have $S_i = \{a_{2^k}\}$ for any $i \in \{1, 2, \dots, k+1\}$. Since at least one of these $k+1$ answers must be truthful, it follows that the relation $x \notin \{a_{2^k}\}$ holds. In other words $a_{2^k} \neq x$.

Case 2. $r_1 = \cdots = r_{i_*-1} = 0$, $r_{i_*} = 1$, for some $i_* \leq k$. In this case player B chooses the sets S_{i_*+i} , $i \in \{1, 2, \dots, k\}$ as follows.

For any $j \in \{0, 1, \dots, 2^k - 1\}$, let $j = 2^k c_k + 2^{k-1} c_{k-1} + \cdots + 2c_1 + c_0$, with $c_0, \dots, c_k \in \{0, 1\}$, be its base 2 representation. For every $i \in \{1, 2, \dots, k\}$ the set S_{i_*+i} is defined by

$$S_{i_*+i} = \left\{ a_j \mid j \text{ has the base 2 representation } \sum_{l=0}^{k-1} 2^l c_l \text{ with } c_{i-1} = 0 \right\}.$$

Using $r_{i_*+1}, r_{i_*+2}, \dots, r_{i_*+k}$, define a real number $s = \sum_{i=1}^k 2^{i-1} r_{i_*+i}$, and consider the element a_s of set X . Then, $s \neq 2^k$, i.e., $a_s \neq a_{2^k}$. If we assume that $a_s = x$, then all answers given in steps i_* , $i_* + 1$, $\dots$, $i_* + k$ are not truthful. This contradicts the given conditions, and hence, $a_s \neq x$.

(b) Player A can apply the following strategy. First, player A chooses a set $Y = \{y_1, y_2, \dots, y_{n+1}\} \subset \{1, 2, \dots, N\}$, and assigns the weight $w_i = c^{-(k+1)}$ to any element $y_i \in Y$, where $c > 0$. The weight of set Y is defined as $W_0 = w_1 + w_2 + \cdots + w_{n+1} = (n+1)c^{-(k+1)}$. Player A will choose answers to the question *whether x belongs to S* , and change the weight of the elements y_i , $1 \leq i \leq n+1$, as follows.

The answer to the question *whether x belongs to S* is a lie related to element y_i if the answer is YES and $y_i \notin S$, or the answer is NO and $y_i \in S$. Let us define $g_i(t) = 1$ if the answer at the t -th step is a lie related to element y_i . Otherwise, $g_i(t) = 0$. The weight of element y_i after the t -th step is $c^{-(k+1-m_i)}$, where m_i is the number of untruthful consecutive answers related to y_i , assuming that only the last steps are counted, i.e., $g_i(t) = g_i(t-1) = \cdots = g_i(t-m_i+1) = 1$ and $g_i(t-m_i) = 0$. The weight of set Y after the t -th step is defined as $W_t = \sum_{i=1}^{n+1} c^{-(k+1-m_i)}$.

Suppose that B chooses a set S at the $(t+1)$ -st step. Let us consider the partition of set Y that is defined by $Y = P \cup Q$, where $P = Y \cap S$ and $Q = Y \setminus S$. Then, we have

$$W_{t+1} = \sum_{y_i \in P} \frac{1}{c^{k+1}} + c \cdot \sum_{y_i \in Q} \frac{1}{c^{k+1-m_i}} \quad \text{if the answer at the step } t+1 \text{ is YES,}$$

$$W_{t+1} = \sum_{y_i \in Q} \frac{1}{c^{k+1}} + c \cdot \sum_{y_i \in P} \frac{1}{c^{k+1-m_i}} \quad \text{if the answer at the step } t+1 \text{ is NO.}$$

Player A chooses the answer such that the obtained value of W_{t+1} is the smaller one. Hence, player A can guarantee that

$$\begin{aligned} W_{t+1} &\leq \frac{1}{2} \left\{ \sum_{y_i \in P} \frac{1}{c^{k+1}} + c \cdot \sum_{y_i \in Q} \frac{1}{c^{k+1-m_i}} + \sum_{y_i \in Q} \frac{1}{c^{k+1}} + c \cdot \sum_{y_i \in P} \frac{1}{c^{k+1-m_i}} \right\} \\ &= \frac{1}{2} (W_0 + cW_t) = \frac{1}{2} \cdot \left(\frac{n+1}{c^{k+1}} + cW_t \right). \end{aligned} \quad (1)$$

Let us choose $c = 1.995$. Since $\frac{1.99}{1.995} < 1$, we can choose k such that $\left(\frac{1.99}{1.995}\right)^k < 0.0025c = 0.0025 \cdot 1.995$. Then we have $0.005 \cdot 1.995^{k+1} > 2 \cdot 1.99^k$. Therefore, $0.005 \cdot 1.995^{k+1} - 1.99^k > 1.99^k$. It follows that there is a positive integer n such that $n \geq 1.99^k$ and $n+1 < 0.005 \cdot 1.995^{k+1} = 0.005 \cdot c^{k+1}$.

We prove by induction that $W_t < 1$ for every nonnegative integer t . Obviously $W_0 = (n+1)c^{-(k+1)} < 0.005 < 1$. Let us suppose that $W_t < 1$ for some t . Then, using (1) we obtain that $W_{t+1} < 1$.

Therefore, player A can guarantee that W_t is always less than 1. It follows that $m_i < k+1$ for any $i \in \{1, 2, \dots, n+1\}$ at any moment t . Indeed, if $m_i = k+1$ for some i at the moment t , then $W_t = \sum_{i=1}^{n+1} c^{-(k+1-m_i)} > 1$.

13.117. Let a and b be two distinct integers from the set $\{0, 1, \dots, n\}$. By definition, an (a, b) -labeling is a beautiful labeling if a , 0 , and b are the labels of three consecutive points that appear in this order clockwise on the circle. Hence a and b are adjacent to 0 . We shall prove that (a, b) -labeling exists if and only if $\gcd(a, b) = 1$ and $a + b > n$. Moreover, if (a, b) -labeling exists, it is uniquely determined. Notation for the chord joining the points labeled x and y is $[x, y]$.

If $a + b \leq n$, then the chords $[0, a+b]$ and $[a, b]$ intersect each other, and hence, (a, b) -labeling does not exist.

Suppose that $a + b > n$, and let L be an (a, b) -labeling. For a given positive integer $c \in \{1, 2, \dots, n\}$ let us define $z_0 = c$, and, for $k \geq 0$,

$$z_{k+1} = \begin{cases} z_k + a, & \text{if } z_k \leq n - a, \\ z_k - b, & \text{if } z_k > n - a \text{ and } z_k + k \geq b, \\ z_k + a - b, & \text{in all other cases.} \end{cases}$$

The sequence $(z_k)_{k \geq 0}$ is well defined. Note that, for any $k \geq 1$, at least one of the equalities $z_{k+1} + 0 = z_k + a$, $z_{k+1} + b = z_k + 0$, and $z_{k+1} + b = z_k + a$ holds. Points 0 , z_k , and z_{k+1} appear in this order clockwise on the circle. If $0 \notin \{z_1, z_2, \dots, z_m\}$, points $z_1, z_2, \dots, z_m$ appear in this order clockwise on the circle (z_i and z_{i+1} are not necessarily adjacent points).

• Suppose that $\gcd(a, b) > 1$, and take $c = 1$. Then, 0 does not appear in the sequence $(z_k)_{k \geq 0}$, and this is a contradiction.

- Suppose that $\gcd(a, b) = 1$, and take $c = 0$. Note that

$$z_{k+1} \equiv \begin{cases} z_k + a & (\text{mod } a + b), \text{ if } z_k + a \leq n, \\ z_k + 2a & (\text{mod } a + b), \text{ if } z_k + a > n. \end{cases}$$

Consider the sequence $a, 2a, 3a, \dots \pmod{a+b}$. If we remove all the terms of this sequence that are greater than n , the obtained sequence is $(z_k)_{k \geq 1}$. Therefore, $z_1, z_2, \dots, z_n$ is a permutation of the set $\{1, 2, \dots, n\}$, and labeling L is uniquely determined..

It is easy to prove that the number of pairs (a, b) such that $\gcd(a, b) = 1$ and $a + b > n$ is equal to $N + 1$.

13.118. Let a peaceful configuration be given. Consider a rectangle $k \times n$ such that the rook from the first column is contained in it. Since any row contains exactly one rook, this rectangle contains exactly k rooks. Then remove the first column. The remaining rectangle R is of the form $k \times (n-1)$, and contains $k-1$ rooks. Suppose that $k^2 < n$. Then, rectangle R contains at least k pairwise disjoint squares $k \times k$. At least one of them does not contain a rook. Let $n = k^2$, where $k > 1$. We give a peaceful configuration, such that any square $k \times k$ contains at least one rook. Let (a, b) be the cell that belongs to the intersection of the $(a+1)$ -st row and the $(b+1)$ -st column. Let us arrange the rooks on the cells $(ik+j, jk+i)$, for any $i, j \in \{0, 1, \dots, k-1\}$. See Figure 14.13.24, where the case $n = 16$ is presented. Since every integer from the set $\{0, 1, \dots, n-1\}$ can be represented in the form $ik+j$ in a unique way, it follows that this arrangement of rooks is a peaceful one.

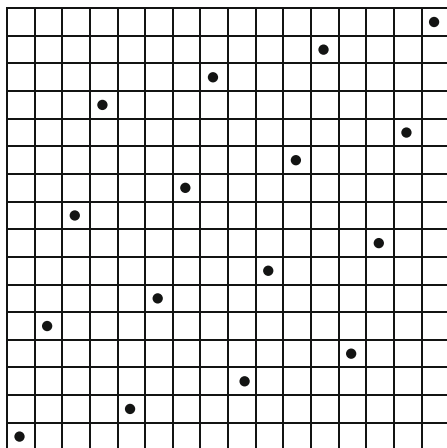


Fig. 14.13.24

It remains to prove that, in this example, any square $k \times k$ contains a rook. Let us consider a square S of the form $k \times k$ with the lower-left corner cell (a, b) , where $a, b \in \{0, 1, \dots, k^2 - k + 1\}$. There is a cell (x, y) with $x \geq a$ and $y \geq b$, such that a rook is placed on it. Let us consider such a cell $A(x, y)$ with the minimal value of the sum $x + y$. Suppose that $A \notin S$. Without loss of generality we can assume that $x \geq a + k$. Each of the cells $B(x - k + 1, y + k - 1)$, $C(x - k, y - 1)$, and $D(x - 2k + 1, y + k - 2)$ (if it is inside square S) contains a rook in the previous example. From the minimality condition for cell A , it follows that $x - 2k + 1 < a$ and $y - 1 < b$, i.e., $x \leq a + 2k - 2$ and $y = b$. It follows that square S contains cell B . The case $n < k^2$ can now be analyzed easily.

The greatest positive integer k with the given property is $k = \lfloor \sqrt{n - 1} \rfloor$.

13.119. Let us define $c_n = n + a_n$ for each $n \in \mathbb{N}$. All the terms of the sequence $(c_n)_{n \geq 1}$ are distinct, and $n + 1 \leq c_n \leq n + 2015$ for each $n \in \mathbb{N}$. Let us consider the set $M = \mathbb{N} \setminus \{c_n \mid n \in \mathbb{N}\}$. Note that $1 \in M$. For every $n \in \mathbb{N}$, the set $\{1, 2, \dots, n + 2015\}$ contains all the terms $c_1, c_2, \dots, c_n$. Hence, the set $\{1, 2, \dots, n + 2015\}$ contains at most 2015 elements of set M . It follows that $1 \leq |M| \leq 2015$. We shall prove that $b = |M|$ and $N = \max M$ satisfy the given conditions.

Suppose that $k \geq N$. The set $M_k = M \cup \{c_1, c_2, \dots, c_k\}$ is a subset of $\{1, 2, \dots, k + 2015\}$, and contains the integers $1, 2, \dots, k + 1$. Hence,

$$M_k = \{1, 2, \dots, k + 1\} \cup \{k + i + 1 \mid i \in T_k\}$$

for some $(b - 1)$ -set $T_k \subset \{1, 2, \dots, 2014\}$. Let $S(X)$ be notation for the sum of the elements of set X . Then, we have $S(M_k) = S(M) + \sum_{i=1}^k (i + a_i)$, and, on the other hand, $S(M_k) = \sum_{i=1}^k i + (k + 1)b + S(T_k)$. It follows that $S(T_k) = S(M) - b + \sum_{i=1}^k (a_i - b)$. Therefore, for $n > m \geq N$,

$$S(T_n) - S(T_m) = \sum_{i=m+1}^n (a_i - b). \quad (1)$$

Note also that, for any $k \geq N$,

$$1 + 2 + \dots + (b - 1) \leq S(T_k) \leq 2014 + 2013 + \dots + (2016 - b). \quad (2)$$

It follows from (1) and (2) that $|S(T_n) - S(T_m)| \leq (b - 1)(2015 - b) \leq ((b - 1) + (2015 - b))^2 / 4 = 1007^2$.

13.120. An example for $n = 9$ is given in Figure 14.13.25. Using a square table 9×9 it is easy to obtain a square table $9k \times 9k$ satisfying the conditions.

I	I	I	M	M	M	O	O	O
M	M	M	O	O	O	I	I	I
O	O	O	I	I	I	M	M	M
I	I	I	M	M	M	O	O	O
M	M	M	O	O	O	I	I	I
O	O	O	I	I	I	M	M	M
I	I	I	M	M	M	O	O	O
M	M	M	O	O	O	I	I	I
O	O	O	I	I	I	M	M	M

Fig. 14.13.25

Suppose now that a square table $n \times n$ is filled such that the given conditions are satisfied. Obviously, n is divisible by 3, i.e., $n = 3k$ for some $k \in \mathbb{N}$. For every $i, j \in \{1, 2, 3\}$, let a_{ij} be the total number of the letter M in the cells (x, y) , where $x \equiv i \pmod{3}$, and $y \equiv j \pmod{3}$. The given conditions imply that

$$a_{11} + a_{22} + a_{33} = a_{21} + a_{22} + a_{23} = a_{31} + a_{22} + a_{31} = k^2, \quad (1)$$

$$a_{11} + a_{21} + a_{31} = a_{31} + a_{23} + a_{33} = k^2. \quad (2)$$

Let us multiply equalities (2) by -1 , and add the obtained two equalities and equalities (1). We get that $3a_{22} = k^2$. It follows that 3 is a divisor of k , and 9 is a divisor of n .

13.121. Let us divide the players into n pairwise disjoint groups consisting of $n + 1$ players, such that all the players from each group stand in the $n + 1$ consecutive positions in the row. Let P_i and Q_i be the tallest and the second tallest players in the i -th group. Let Q_k be the tallest one among the players $Q_1, \dots, Q_n$. The coach removes players $P_1, \dots, P_{k-1}, P_{k+1}, \dots, P_n$ and all players from the k -th group except P_k and Q_k . Players P_k and Q_k are the tallest and the second tallest in the remaining row, and no one stands between them. Consider now the remaining row ignoring P_k and Q_k . It consists of $n - 1$ groups with n players each. By repeating the previous procedure the coach can obtain a row that satisfies the given conditions.

13.122. Suppose that a fair die is successively rolled until all six sides are obtained. A possible outcome of this experiment is of the form $\omega = c_1 c_2 \dots c_n$, where $c_1, c_2, \dots, c_n \in S_0 := \{1, 2, 3, 4, 5, 6\}$, such that exactly five elements of the set S_0 appear among $c_1, c_2, \dots, c_{n-1}$, and c_n is the sixth number. The probability of the outcome $\omega = c_1 c_2 \dots c_n$ is $p(\omega) = 1/6^n$.

Let n be a fixed natural number from the set $\{6, 7, 8, \dots\}$, and A_n be the event that each of the natural numbers 1, 2, 3, 4, and 5 appears in the first $n - 1$ rolls, and the number 6 does not appear.

Let S be the set of $(n - 1)$ -arrangements of the elements 1, 2, 3, 4, and 5, and S_k be the set of $(n - 1)$ -arrangements of elements of the set $\{1, 2, 3, 4, 5\} \setminus \{k\}$. Then we have $A_n = S \setminus (S_1 \cup S_2 \cup S_3 \cup S_4 \cup S_5)$.

Set S consists of 5^{n-1} elements. Each of the sets S_1, S_2, S_3, S_4 , and S_5 consists of 4^{n-1} elements, the intersection of every two of them consists of 3^{n-1} elements, etc. By the inclusion-exclusion principle it follows that

$$\begin{aligned} |A_n| &= |S \setminus (S_1 \cup S_2 \cup S_3 \cup S_4 \cup S_5)| \\ &= 5^{n-1} - 5 \cdot 4^{n-1} + 10 \cdot 3^{n-1} - 10 \cdot 2^{n-1} + 5. \end{aligned}$$

The probability of the event A_n is

$$P(A_n) = \left(\frac{5}{6}\right)^{n-1} - 5\left(\frac{4}{6}\right)^{n-1} + 10\left(\frac{3}{6}\right)^{n-1} - 10\left(\frac{2}{6}\right)^{n-1} + 5\left(\frac{1}{6}\right)^{n-1}.$$

The probability that exactly five distinct numbers appear in the first $n - 1$ rolls of a fair die is $6P(A_n)$. The probability that the missed number appears in the n -th roll is $1/6$. Hence, $P\{X = n\} = \frac{1}{6} \cdot 6 \cdot P(A_n) = P(A_n)$.

Note that the mathematical expectation $E(X) = \sum_{n=6}^{\infty} nP(A_n)$ can be calculated by summing the geometric series. It can be obtained also the following way. The random variable X can be represented in the form $X = \sum_{i=1}^6 X_i$, where $X_1 = 1$, and, for every $i \in \{2, 3, 4, 5, 6\}$, X_i is a random variable with geometric $\Gamma\left(\frac{6-i+1}{6}\right)$ distribution and mathematical expectation $E(X_i) = \frac{6}{6-i+1}$. Hence, $E(X) = 1 + \frac{6}{5} + \frac{6}{4} + \frac{6}{3} + \frac{6}{2} + \frac{6}{1} = 14.7$.

13.123. Using the given conditions and the properties of probability and conditional probability, it follows that

$$\begin{aligned} P(B|A) &= \frac{P(BA)}{P(A)} = \frac{1}{P(A)} P\left(\bigcup_{i=1}^n BAH_i\right) = \sum_{i=1}^n \frac{P(BAH_i)}{P(A)} \\ &= \sum_{i=1}^n \frac{P(H_i)P(BAH_i)}{P(A)P(H_i)} = \sum_{i=1}^n P(H_i) \frac{P(BAH_i)}{P(AH_i)} = \sum_{i=1}^n P(H_i)P(B|AH_i). \end{aligned}$$

13.124. Let A_k and B_k be the notation of the outcomes A and B in the k -th experiment, and let C be the event that a series of A 's of length m occurs before a series of B 's of length n . Let us denote $x = P(C \mid A_1)$, $y = P(C \mid B_1)$, and consider the following tables:

Table 1. The first partition of Ω

H_i	$P(H_i)$	$P(C \mid A_1 H_i)$
B_2	q	y
$A_2 B_3$	pq	y
$A_2 A_3 B_4$	$p^2 q$	y
$\dots$	$\dots$	$\dots$
$A_2 A_3 \dots A_{m-1} B_m$	$p^{m-2} q$	y
$A_2 A_3 \dots A_{m-1} A_m$	p^{m-1}	1

Table 2. The second partition of Ω

H_i	$P(H_i)$	$P(C \mid B_1 H_i)$
A_2	p	x
$B_2 A_3$	qp	x
$B_2 B_3 A_4$	$q^2 p$	x
$\dots$	$\dots$	$\dots$
$B_2 B_3 \dots B_{n-1} A_n$	$q^{n-2} p$	x
$B_2 B_3 \dots B_{n-1} B_n$	q^{n-1}	0

Note that A_1 and the events H_i from the first column of Table 1 satisfy the conditions of Exercise 13.123, and hence

$$\begin{aligned}
 x &= P(C \mid A_1) = \sum P(H_i) P(C \mid A_1 H_i) \\
 &= p^{m-1} + \sum_{k=0}^{m-2} p^k q y = p^{m-1} + (1 - p^{m-1})y.
 \end{aligned}$$

Similarly, using the events from Table 2 it follows that

$$y = P(C \mid B_1) = \sum_{k=0}^{n-2} q^k p x = (1 - q^{n-1})x.$$

From the previous two equations it follows that

$$x = \frac{p^{m-1}}{1 - (1 - p^{m-1})(1 - q^{n-1})}, \quad y = \frac{p^{m-1}(1 - q^{n-1})}{1 - (1 - p^{m-1})(1 - q^{n-1})}.$$

Now by the formula of total probability we obtain that

$$\begin{aligned} P(C) &= P(A_1)P(C | A_1) + P(B_1)P(C | B_1) \\ &= px + qy = \frac{p^{m-1}(1 - q^n)}{1 - (1 - p^{m-1})(1 - q^{n-1})}. \end{aligned}$$

Let C_1 be the event that a series of B 's of length n occurs before a series of A 's of length m . Analogously we get

$$P(C_1) = \frac{q^{n-1}(1 - p^m)}{1 - (1 - p^{m-1})(1 - q^{n-1})}.$$

It is easy to check that $P(C) + P(C_1) = 1$.

13.125. Let us denote $x = P(A \setminus B)$, $y = P(B \setminus A)$, $z = P(AB)$, and $u = 1 - x - y - z = P(\Omega \setminus (A \cup B))$. Then, $P(A) = x + z$, $P(B) = y + z$ and $P(AB) - P(A)P(B) = z - (x + z)(y + z) = z(1 - x - y - z) - xy = uz - xy$.

Since $x, y, z, u \in [0, 1]$ and $x + y + z + u = 1$, it follows that

$$uz - xy \leq uz \leq \left(\frac{u+z}{2}\right)^2 \leq \left(\frac{1}{2}\right)^2 = \frac{1}{4}.$$

Analogously we obtain $xy - uz \leq 1/4$, i.e., $-1/4 \leq uz - xy$, and hence the inequality $|P(AB) - P(A)P(B)| \leq 1/4$ holds. From the previous consideration it follows that the equality $|P(AB) - P(A)P(B)| = 1/4$ holds if and only if $u = z = 1/2$, $x = y = 0$ or $u = z = 0$, $x = y = 1/2$.

13.126. Let us determine the distribution of random variable X_3 .

$$P\{X_3 = 1\} = P\{X_1 = 1, X_2 = 1\} + P\{X_1 = -1, X_2 = -1\} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

Analogously we get $P\{X_3 = -1\} = 1/2$. Hence, random variable X_3 has the same probability distribution as random variables X_1 and X_2 . Since the relation between X_1 , X_2 , and X_3 is symmetric, and X_1 and X_2 are independent, it follows that X_1 , X_2 , and X_3 are pairwise independent. It is obvious that X_1 , X_2 , and X_3 are dependent if these three random variables are considered together as a collection.

13.127. (a) For every natural number n , it holds that $P\{X = n, Y = n\} = 0$. For all natural numbers m and k we obtain that

$$P\{X = m, Y = m + k\} = \left(\frac{2}{6}\right)^{m-1} \frac{3}{6} \left(\frac{5}{6}\right)^{k-1} \frac{1}{6} = \frac{1}{12} \left(\frac{1}{3}\right)^{m-1} \left(\frac{5}{6}\right)^{k-1}.$$

For all natural numbers k and n we obtain that

$$P\{X = n + k, Y = n\} = \left(\frac{2}{6}\right)^{n-1} \frac{1}{6} \left(\frac{3}{6}\right)^{k-1} \frac{3}{6} = \frac{1}{12} \left(\frac{1}{3}\right)^{n-1} \left(\frac{1}{2}\right)^{k-1}.$$

(b) Now it is easy to get

$$\begin{aligned} P\{X > Y\} &= \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} P\{X = n + k, Y = n\} = \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{12} \left(\frac{1}{3}\right)^{n-1} \left(\frac{1}{2}\right)^{k-1} \\ &= \frac{1}{12} \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^{n-1} \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^{k-1} = \frac{1}{12} \cdot \frac{1}{1 - \frac{1}{3}} \cdot \frac{1}{1 - \frac{1}{2}} = \frac{1}{4}. \end{aligned}$$

The same result can be obtained in a simple way as follows. Note that $\{X > Y\}$ is the event that 1 is the first number from the set $\{1, 2, 4, 6\}$ to occur in a sequence of die rolls. Hence, $P\{X > Y\} = 1/4$.

13.128. Let I_k be the indicator function of the event $\{a_k = k\}$. Note that $P\{I_k = 1\} = 1/n$, and $P\{I_k = 0\} = 1 - 1/n$, and $E(I_k) = 1/n$, for every $k \in \{1, 2, \dots, n\}$. Since $X = I_1 + I_2 + \dots + I_n$, it follows that $E(X) = n \cdot \frac{1}{n} = 1$.

Note that indicators $I_1, I_2, \dots, I_n$ are not independent, i.e., X is not a binomial random variable. For example, if indicators $I_1, I_2, \dots, I_{n-1}$ take the value 1, then I_n takes the value 1 with probability 1.

13.129. (a) The number of red balls added to the box is a random variable Y , with distribution $P\{Y = 0\} = \frac{l}{k+l}$, $P\{Y = m\} = \frac{k}{k+l}$. Hence, it follows that $E(X) = k + E(Y) = k + \frac{mk}{k+l}$.

(b) Let B be the event that the second chosen ball is red, A_1 be the event that the second chosen ball was in the box at the beginning, and A_2 be the event that the second chosen ball was next to the box at the beginning. From the formula of total probability it follows that

$$\begin{aligned} P(B) &= P(A_1)P(B|A_1) + P(A_2)P(B|A_2) \\ &= \frac{k+l}{k+l+m} \cdot \frac{k}{k+l} + \frac{m}{k+l+m} \cdot \frac{k}{k+l} = \frac{k}{k+l}. \end{aligned}$$

13.130. Note that

$$\begin{aligned} E(X - c)^2 &= E(X - E(X) + E(X) - c)^2 \\ &= E(X - E(X))^2 + 2(E(X) - c) \cdot E(X - E(X)) + (E(X) - c)^2 \\ &= D(X) + (E(X) - c)^2 \geq D(X). \end{aligned}$$

The minimal value of $E(X - c)^2$ is $\text{var}(X)$ and is attained for $c = E(X)$.

13.131. The following relations hold:

$$\begin{aligned}\text{var}(X) &= \text{var}\left(X - \frac{a+b}{2}\right) \leq E\left\{\left(X - \frac{a+b}{2}\right)^2\right\} = \sum_{k=1}^m \left(x_k - \frac{a+b}{2}\right)^2 p_k \\ &\leq \sum_{k=1}^m \left(\frac{b-a}{2}\right)^2 p_k = \left(\frac{b-a}{2}\right)^2 \sum_{k=1}^m p_k = \left(\frac{b-a}{2}\right)^2.\end{aligned}$$

The equality is attained if $P\{X = a\} = P\{X = b\} = 1/2$.

Bibliography

- [1] G.E. Andrews, *The Theory of Partitions*, Addison-Wesley Publishing Company, London, 1976.
- [2] W.W.R. Ball, H.S.M. Coxeter, *Mathematical Recreations and Essays*, University of Toronto Press, 1974.
- [3] W. Feller, *An Introduction to Probability Theory and its Applications, Volume I*, 3rd ed. John Willey & Sons, New York, Chichester, Brisbane, Toronto, 1970.
- [4] L.R. Graham, D.E. Knuth and O. Patashnik, *Concrete Mathematics*, Addison-Wesley Publishing Company, New York, 1989.
- [5] M. Hall, *Combinatorial Theory*, Blaisdel, Waltham, 1976.
- [6] F. Harary, *Graph Theory*, Addison-Wesley Publishing Company, 1969.
- [7] R. Honsberger, *Mathematical Gems II*, Mathematical Association of America, 1976.
- [8] R. Honsberger, *Mathematical Morsels*, Mathematical Association of America, 1978.
- [9] V. Janković, Z. Kadelburg, P. Mladenović, *International and Balkan Mathematical Olympiads 1984–1995*, Mathematical Society of Serbia, Belgrade, 1996. [in Serbian]
- [10] Z. Kadelburg, P. Mladenović, *Yugoslav Mathematical Olympiads 1960–1989*, Mathematical Society of Serbia, Belgrade, 1990. [in Serbian]
- [11] Dj. Krtinić, P. Mladenović, *International and Balkan Mathematical Olympiads 1986–2006*, Mathematical Society of Serbia, Belgrade, 2007. [in Serbian]

- [12] Dj. Krtinić, *Mathematical Olympiads 2007–2011*, Mathematical Society of Serbia, Belgrade, 2012. [in Serbian]
- [13] W. Lipski, *Kombinatorika dla programistów*, Wydawnictwa naukowo-techniczne, Warszawa, 1982.
- [14] O. Ore, *Graphs and Their Uses* (revised and updated by R.J. Wilson), Mathematical Association of America, 1990.
- [15] J. Riordan, *An Introduction to Combinatorial Analysis*, John Wiley, New York, 1958
- [16] J. Riordan, *Combinatorial Identities*, John Wiley, New York, 1968.
- [17] H.J. Ryser, *Combinatorial Mathematics*, The Mathematical Association of America, 1963.
- [18] I. Tomescu, *Problems in Combinatorics and Graph Theory*, (Translated by R.A. Melter), Wiley, 1985.
- [19] R. Tošić, *Combinatorics*, University of Novi Sad, Novi Sad, 1999. [in Serbian]
- [20] W.T. Tutte, *Graph Theory* (Encyclopedia of Mathematics and Its Applications), Addison-Wesley Publishing Company, 1984.
- [21] R.J. Wilson, *Introduction to Graph Theory*, 4th ed. Addison-Wesley Longman Limited, 1996.
- [22] Н.Б. Васильев, А.А. Егоров, *Задачи всесоюзных математических олимпиад*, Наука, Москва, 1988. [N.B. Vasil'ev, A.A. Egorov, *All-Soviet Mathematical Olympiads*]
- [23] Н.Я. Виленкин, *Комбинаторика*, Наука, Москва, 1969. [N.Ya. Vilenkin, *Combinatorics*]
- [24] Н.Н. Воробьев, *Числа Фибоначчи* (издание четвертое), Наука, Москва, 1978. [N.N. Vorob'ev, *Fibonacci Numbers*, 4th ed.]
- [25] Е.Я. Гик, *Математика на шахматной доске*, Наука, Москва, 1976. [E.Ya. Gik, *Mathematics on a Chessboard*]
- [26] Л.А. Калужнин, В.И. Суцанский, *Преобразования и перестановки*, Наука, Москва, 1985. [L.A. Kaluzhnin, V.I. Sushchanskii, *Transformations and Permutations*]
- [27] *Комбинаторный анализ. Задачи и упражнения*. Под. ред. Рыбников, К.А., Наука, Москва, 1982. [*Combinatorial Analysis, Problems and Exercises*, Ed. K.A. Rybnikov]

- [28] Й. Кюршак, Д. Нейкомм, Д. Хайош, Я. Шурани, *Венгерские математические олимпиады* (перевод с венгерского), Издательство Мир, Москва, 1976. [Kyrurshak et all. *Hungarian Mathematical Olympiads*]
- [29] А.А. Леман, *Сборник задач московских математических олимпиад*, Просвещение, Москва, 1965. [A.A. Leman, *Moscow Mathematical Olympiad*]
- [30] В.Н. Сачков, *Введение в комбинаторные методы дискретной математики*, Наука, Москва 1982. [V.N. Sachkov, *An Introduction to Combinatorial Methods of Discrete Mathematics*]
- [31] А.М. Яглом, И.М. Яглом, *Неэлементарные задачи в элементарном изложении*, Гос. изд. технико-теоретической литературы, Москва, 1954. [A.M. Yaglom, I.A. Yaglom, *Challenging Mathematical Problems with Elementary Solutions*]
- [32] А.М. Яглом, И.М. Яглом, *Вероятность и информация*, Наука, Москва, 1973. [A.M. Yaglom, I.A. Yaglom, *Probability and Information*]
- [33] *Квант*. 1971, No. 1; 1973, No. 3, 10; 1974, No. 2; 1978, No. 7; 1979, No. 1, 12; 1981, No. 12; 1984, No. 9; 1987, No. 2, 10; 1988, No. 1, 3, 8; 1989, No. 2, 3; 1990, No. 1, 2, 3, 8, 9, 11; 1991, No. 1, 7. [Russian journal: *Kvant*]
- [34] *Тангента*. 1996, No. 5; 1997, No. 10; 1998, No. 13; 1999, No. 17; 2000, No. 21; 2012, No. 70; 2013, No. 74; 2014, No. 78; 2016, No. 83; 2016, No. 85. [Serbian journal: *Tangent*]

Further Reading

- [35] B. Bollobás, *Random Graphs*, 2nd ed. Cambridge University Press, Cambridge, 2001.
- [36] M. Bóna, *Combinatorics of Permutations*, Chapman & Hall/CRC, Boca Raton, London, New York, Washington, 2004.
- [37] F. Chung, R. Graham, *Erdős on Graphs - His Legacy of Unsolved Problems*, A. K. Peters, Wellesley, Massachusetts, 1998.
- [38] В.Н. Сачков, *Вероятностные методы в комбинаторном анализе*, Наука, Москва 1978. [V.N. Sachkov, *Probabilistic Methods in Combinatorial Analysis*]
- [39] T. Tao, V.H. Vu, *Additive Combinatorics*, Cambridge University Press, Cambridge, 2006.

Index

A

- Arrangements, 9
 - of a given type, 13
 - without repetitions, 10
- Arrow's impossibility theorem, 158

B

- Bell numbers, 85
- Bernoulli's law of large numbers, 191
- Bijection, 2
 - rule, 4
- Binary relation, 3
- Binomial coefficients of order n , 36
 - distribution, 186
 - random variable, 186
 - theorem, 35
- Bipartite graph, 122
- Bose-Einstein statistics, 20
- Brooks' theorem, 138
- Burnside's lemma, 102

C

- Cartesian product, 2
- Catalan numbers, 24, 70
- Chebyshev's inequality, 191
 - law of large numbers, 192
- Choosing with replacement, 17
 - without replacement, 17

- Chromatic number, 137
 - polynomial, 138
- Classical definition of probability, 179
- Collectively independent events, 184
- Combinations, 11
 - with repetition allowed, 15
- Combinatorial identities, 24
- Complement graphs, 113
- Complete graph, 110
- Completely regular graph, 136
- Composition of functions, 2
 - of permutations, 93
- Conditional probability, 183
- Connected component, 116
 - graph, 116
 - vertices, 116
- Contracted graph, 133
- Conway's game, 170
- Counterfeit coin problem, 205
- Cycle, 96
 - rank, 140
- Cyclic permutation, 95

D

- Degree of a vertex, 109
- de Moivre formula, 46

de Moivre-Laplace theorem, 192

Directed graph, 110

Discrete probability space, 181
 random variable, 185

Dual graphs, 135

E

Equivalence relation, 3
 class, 4

Eulerian cycle, 119
 graph, 119
 trail, 119

Euler-Legendre theorem, 84

Euler's theorem, 133

Euler's totient function, 59

Expanded graph, 132

Exponential generating function, 72

F

Fermat's theorem, 105

Fermi-Dirac statistics, 20

Ferrer's graph, 80

Fibonacci sequence, 67

Five-color theorem, 139

Fixed point, 2

Forest, 127

Formula of total probability, 183

Four-color theorem, 140

G

Game of fifteen, 168

Geometric distribution, 186

Golden ratio, 167

Graph, 109

 coloring, 137
 of n -permutations, 121

H

Hadamard matrix, 202

Hall's theorem, 149

Hamiltonian cycle, 120
 graph, 120
 trail, 120

I

Identity permutation, 93

Incidence matrix, 151
 relation, 109

Inclusion-exclusion principle, 50
 generalized, 58

Increasing function, 4

Independent events, 184
 random variables, 187

Indicator function, 186

Inverse of a permutation, 93

Isolated vertex, 109

Isomorphic graphs, 117, 128

J

Jordan curve theorem, 132

K

k -arrangement, 9

k -coloring, 137

k -combination, 11

Königsberg bridge problem, 107

König's theorem, 139

L

Labeled graph, 128

Latin rectangle, 160

Latin square of order n , 148

Leaf vertex, 109

Length of orbit, 99

Loop, 109

M

Magic square of order n , 141

Mathematical expectation, 187

Maxwell-Boltzmann distribution, 20

Möbius function, 59

Moving point, 2

Multigraph, 110

Multinomial theorem, 42

Multiplication rule, 4

N

Negative binomial distribution, 187
Nim game, 165
Normal distribution function, 192
 n -set, 4

O

Orbit of an element, 95
Ordered partition, 78
Order of a permutation, 97
Ordinary generating function, 63
Orthogonal Latin squares, 148

P

Pairwise independent events, 184
Parity of a permutation, 92
Partial order, 4
Partition of positive integer n , 75
Partitions of sets, 84
Path, 115
Pauli exclusion principle, 19
Pentagonal numbers, 84
Permutation, 11
 graph, 94
 group, 99
Petersen graph, 138
Pigeonhole principle, 152
Planar graph, 130
Plane graph, 130
Platonic graphs, 112
Poisson distribution, 187
Probability distribution, 185
Product rule, 5

R

Ramsey numbers, 155
Ramsey's theorem, 154
Random walk, 197
Recursive equations, 69
Regular graph, 121
Relations, 3

Relatively prime permutations, 95

Rooted tree, 128

Rotation group of a cube, 100

S

Schur's theorem, 164
Silver matrix, 213
Simple graph, 110
Social choice function, 157
Sperner's lemma, 200
Standard deviation, 190
Stereographic projection, 133
Stirling number of the 2nd kind, 85
Strictly increasing function, 4
Strongly regular graph, 121
Subgraph, 114
Sum rule, 5
Supergraph, 114
Symmetric random walk, 197
System of distinct representatives, 149
System of individual values, 157

T

Thirty-six officers problem, 147
Trail, 115
Trajectories, 21, 53, 55
Transposition, 93
Tree, 127

U

Undirected graph, 110

V

Vandermonde's identity, 39
Van der Waerden's theorem, 164
Variance, 190
Vizing's theorem, 139

W

Walk in a graph, 115
Wilson's theorem, 105